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Surprises and Counterexamples in Real Function Theory

A. R. Rajwade
A. K. Bhandari

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Texts and Readings in Mathematics

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Preface

Our aim in this book is to consider a variety of intriguing, surprising and appealing topics, and nonroutine proofs of the usual results of real function theory. The reader is expected to have done a first course in real analysis (or advanced calculus), since the book assumes a knowledge of continuity and differentiability of functions, Rolle's theorem, the mean value theorem, Taylor expansion and Riemann integration. However, no sophisticated knowledge of analysis is required and a student at the masters or advanced undergraduate level should have no difficulty in going through the book.

Though this book has some part of the title common with the book "Counterexamples in Analysis" by Gelbaum and Olmstead, it is totally different in nature and contents. Some examples and counterexamples (fifteen or twenty) in our book are essentially the same as given in the book by Gelbaum and Olmstead, but otherwise the intersection is small.

This book contains a number of surprising and unexpected results. It is meant to be a reference book and is expected to be a book to which one turns for finding answers to curiosities which one comes across while studying or teaching elementary analysis. *For example: We know that continuous functions defined on an interval satisfy the intermediate value property. Are there functions which are not continuous but have this property in every interval?(Example 4.1.2) If a one to one onto function is continuous at a point, is its inverse also continuous at the image of that point?(Example 2.4.17) Most would believe that " if* $y = f(x)$ *is a function defined on* $[a, b]$ *and* $c \in (a, b)$ *is any point, then the tangent to its graph* Γ_f *exists at the point* $(c, f(c))$ *if and only if* f *is differentiable at* c*"– see §5.2 for a negative answer. Where does one find easily accessible details of everywhere continuous, nowhere differentiable functions?(Chapter 3)*

Chapter 1 of the book gives an introduction to algebraic, irrational and transcendental numbers. It has several results about the numbers e and π and contains a detailed account of the construction of the curious Cantor ternary set.

In Chapter 2, we consider functions with extraordinary properties. For example, an increasing function $f : [0, 1] \to [0, 1]$, $f(0) = 0$, $f(1) = 1$, the length of whose graph is equal to 2. Another example studied is a function defined on the entire real line that is differentiable at each point but is monotone in no interval.

Chapter 3 discusses, in detail, functions that are continuous at each point but differentiable at no point.

Chapters 4 and 5 include the intermediate value property, periodic func-

tions, properties of derivatives, Rolle's theorem, Taylor's theorem, L'Hospital's rule, points of inflexion, tangents to curves etc. The geometric interpretation of the second and the third derivatives and some intricate aspects of Riemann integration are also included in this chapter.

Chapter 6 discusses sequences, series and Euler's constant γ. The restricted harmonic series is a beautiful topic included here. The surprising rearrangements of alternating harmonic series, leading to Riemann's theorem are also discussed in this Chapter. Some number theoretic aspects are also treated.

In Chapter 7, the infinite exponential $x^{x^{x^{\cdot^{\cdot^{\cdot}}}}}$ with its peculiar range of convergence is studied in detail. We have included an analytic proof of its convergence as well as a very revealing graphical proof.

Appendix I deals with Stirling's formula, a specialized topic of Schwartz's differentiability and some curious properties of Cauchy's functional equation.

Some exercises, conforming to the spirit and style of the book, are included at the end of each chapter. Hints and/or full solutions for the exercises are provided in Appendix II.

References for most of the material in the book are given at the end. Certainly, many other interesting topics could have easily found a place in the book, but there are limitations of time and space.

We hope that the book will be useful to students and teachers alike.

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Chapter 1

Introduction to the real line $\mathbb{R}$ and some of its subsets

§1.1. The real number system

The system of real numbers has evolved as a result of a process of successive extensions of the system of natural numbers (i.e., the positive whole numbers). We shall denote by $\mathbb{N}$, $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$ and $\mathbb{C}$, respectively, the sets of natural numbers, integers, rational numbers, real numbers and complex numbers. The sets $\mathbb{Q}$, $\mathbb{R}$ and $\mathbb{C}$ form a field with respect to the usual operations of addition and multiplication. The fields $\mathbb{Q}$ and $\mathbb{R}$ are ordered fields:

Definition 1.1.1 An ordered field is a field $\mathbf{F}$ that contains a subset $\mathcal{P}$ such that
(i) $\mathcal{P}$ is closed with respect to addition; that is,

$$x \in \mathcal{P}\ ,\ y \in \mathcal{P} \implies x + y \in \mathcal{P}.$$

(ii) $\mathcal{P}$ is closed with respect to multiplication, that is,

$$x \in \mathcal{P}\ ,\ y \in \mathcal{P} \implies x.y \in \mathcal{P}.$$

(iii) For all $x \in \mathbf{F}$, exactly one of the following three statements is true:

$$x \in \mathcal{P}\ ;\ x = 0\ ;\ -x \in \mathcal{P},$$

where 0 is the additive identity of the field $\mathbf{F}$.

A member x of $\mathbf{F}$ is called positive if and only if $x \in \mathcal{P}$ and is called negative if and only if $-x \in \mathcal{P}$. Inequalities in an ordered field are defined by: $x < y$ if and only if $y - x \in \mathcal{P}$ and $x \leq y$ if and only if $y - x \in \mathcal{P}$ or $x = y$. If $\mathbf{F}$ is an ordered field and if $x \in \mathbf{F}$, then $\mid x \mid$, called the absolute value of x, is defined to be x in case $x \geq 0$ and to be $-x$ in case $x < 0$.

Suppose that $\mathbf{F}$ is an ordered field. Let $u \in \mathbf{F}$ and $A \subseteq \mathbf{F}$. If $x \leq u$ for every $x \in A$, then u is called an upper bound of A. A non empty subset A of $\mathbf{F}$ is called bounded above in $\mathbf{F}$ if and only if there exits an element of $\mathbf{F}$ which is an upper bound of A. If s is an upper bound of A and if s is less than or equal to every other upper bound of A, then s is called the least upper bound or supremum of A, denoted by $\sup A$. Similarly one defines the notion

of being bounded below, lower bound and the greatest lower bound or infimum of a non-empty set A. The infimum of A is denoted by $\inf A$.

Definition 1.1.2. A complete ordered field is an ordered field $\mathbf{F}$ in which a least upper bound exists for every non-empty subset of $\mathbf{F}$ which is bounded above in $\mathbf{F}$.

From any of the well-known constructions of the real number system, it follows that *the set $\mathbb{R}$ of real numbers is a complete ordered field.*

Let $\mathbf{F}$ be an ordered field. A sequence in $\mathbf{F}$ is a function with values in $\mathbf{F}$ whose domain is the set of natural numbers $\mathbb{N}$. Its values are denoted by a_n and the sequence itself by $\{a_n\}$. A sequence $\{a_n\}$ is said to converge and to have a limit a if and only if for every ϵ in the set $\mathcal{P}$ of positive numbers, there exists $N \in \mathbb{N}$ such that $\mid a_n - a \mid < \epsilon$ for all $n > N$. A sequence that is not convergent is said to be a divergent sequence. A sequence $\{a_n\}$, where the terms a_n are members of an ordered field $\mathbf{F}$, is called a Cauchy sequence if for every $\epsilon \in \mathcal{P}$, there exists $N \in \mathbb{N}$ such that $\mid a_m - a_n \mid < \epsilon$ for all $m, n > N$. It follows that every convergent sequence is a Cauchy sequence. The completeness property of $\mathbb{R}$ is equivalent to the fact that every Cauchy sequence in $\mathbb{R}$ is convergent. Not all fields can be ordered in the sense of Definition 1.1.1. We have:

Example 1.1.3 The field $\mathbb{C}$ can not be ordered, i.e., it possesses no subset $\mathcal{P}$ satisfying the properties of Definition 1.1.1. Indeed, assume that there does exist such a subset $\mathcal{P}$ of $\mathbb{C}$. Consider the complex number ι. Since $\iota \neq 0$, there are two possibilities. The first is that $\iota \in \mathcal{P}$, in which case $\iota^2 = -1 \in \mathcal{P}$, whence $\iota^4 = 1 \in \mathcal{P}$. Since ι^2 and ι^4 are additive inverses of each other, it is impossible for both of them to be in $\mathcal{P}$. We thus obtain a contradiction. The other alternative is that $-\iota \in \mathcal{P}$, in which case $(-\iota)^2 = -1 \in \mathcal{P}$, whence $(-\iota)^4 = 1 \in \mathcal{P}$, and we arrive at the same contradiction.

It can be shown that the fields $\mathbb{Q}$ as well as $\mathbb{R}$ admit only one ordering (see, for example [61]). The set $\mathcal{P}$ of positive elements of $\mathbb{R}$ is the set of squares of the elements of $\mathbb{R}$. The set of positive elements of $\mathbb{Q}$ is $\mathbb{Q} \cap \mathcal{P}$.

However, there are fields which can be ordered in more than one ways.

Example 1.1.4. Let m be any positive integer which is not a perfect square. Let $\mathbf{F} = \{a + b\sqrt{m} \mid a, b \in \mathbb{Q}\}$. It is easy to see that $\mathbf{F}$ is a field under the usual operations of addition and multiplication of real numbers. Let $\mathcal{P}$ be the set of usual positive elements of $\mathbb{R}$ (i.e., squares) and let $\mathcal{P}' = \mathcal{P} \cap \mathbf{F}$. Then $\mathcal{P}'$ serves as a subset of positive elements of $\mathbf{F}$ according to Definition 1.1.1. A second way in which $\mathbf{F}$ is an ordered field is provided by the subset $\mathcal{P}''$ defined by

$$a + b\sqrt{m} \in \mathcal{P}'' \iff a - b\sqrt{m} \in \mathcal{P} ;$$

that $\mathcal{P}''$ satisfies the three requirements of Definition 1.1.1 can be easily verified.

Example 1.1.5. The ordered field $\mathbb{Q}$ of rational numbers is not complete (i.e., does not satisfy the requirements of Definition 1.1.2). Let

$$A = \{r \in \mathbb{Q} \mid r > 0 \, , \, r^2 < 2\}.$$

The set A is non empty ($1 \in A$) and is bounded above by 2. Let us assume that $\mathbb{Q}$ is complete. Then there must be a positive rational number c that is the supremum of A. Since there is no rational number whose square is equal to 2, either $c^2 < 2$ or $c^2 > 2$. Assume first that $c^2 < 2$ and let d be the positive number $d = \frac{1}{2}\min\{\frac{2-c^2}{(c+1)^2}\ ,\ 1\}$. Then $c+d$ is a positive rational number greater than c whose square is less than 2, i.e., $(c+d)^2 < c^2 + d(c+1)^2 < 2$; but then $c+d \in A$, whereas c is an upper bound of A. Assuming $c^2 > 2$, let d be the positive number $d = \frac{c^2-2}{2(c+1)^2}$. Then $c-d$ is a positive rational number less than c whose square is greater than 2, i.e., $(c-d)^2 > c^2 - d(c+1)^2 > 2$. Since $c-d$ is an upper bound of A which is less than the least upper bound c, we arrive at a contradiction.

Recall that the set $\mathbb{Q}$ of rational numbers is dense in $\mathbb{R}$ (in the usual distance topology). We close this section by recording a somewhat surprising result, which will be needed later.

Theorem 1.1.6. *Let θ be an irrational number. Then the set of numbers of the form $m + n\theta$, $m, n \in \mathbb{Z}$, is dense in $\mathbb{R}$.*

Proof. Let $\epsilon > 0$ be an arbitrary real number. We shall first find m, n such that $0 < m + n\theta < \epsilon$. Choose N such that $\frac{1}{N} < \epsilon$ and then for each $n = 0, 1, 2, \ldots, N$, choose m $(= m(n)$, i.e., m depends on $n)$ such that $0 < m + n\theta < 1$ (for example, $m = -[n\,\theta]$ will do, where for a real number x, $[x]$ denotes the largest integer less than or equal to x). Then we have $N+1$ distinct numbers $m(n) + n\,\theta$, $0 \le n \le N$, in $[0, 1]$ and therefore there exist two such numbers which are at a distance less than $\frac{1}{N}$ apart (box principle), say, $|\,(m_1 + n_1\,\theta) - (m_2 + n_2\,\theta)\,| < \frac{1}{N} < \epsilon$, i.e., $|\,(m_1 - m_2) - (n_1 - n_2)\,\theta\,| < \epsilon$. Thus, we find a number of the form $m + n\,\theta$ such that $|\,m + n\,\theta\,| < \epsilon$. By changing signs we get a number $0 < m + n\,\theta < \epsilon$. Now, given any real number r, find an integer k such that $k(m + n\,\theta) < r < (k+1)(m + n\,\theta)$. We thus get a number of the required form which is as close to r as we want.

§1.2. Irrational and transcendental numbers

A real number α is called an *algebraic number*, if α is a root of a nonconstant polynomial with rational coefficients. Let $\mathbb{A}$ denote the set of all such real numbers. In the usual notations, the inclusions

$$\mathbb{N} \subseteq \mathbb{Z} \subseteq \mathbb{Q} \subseteq \mathbb{A} \subseteq \mathbb{R}$$

give rise to the following disjoint unions:

$$\mathbb{R} = \mathbb{Q} \cup (\mathbb{R} \setminus \mathbb{Q}) = \mathbb{Q} \cup \mathbb{I} \ ; \ \mathbb{R} = \mathbb{A} \cup (\mathbb{R} \setminus \mathbb{A}) = \mathbb{A} \cup \mathbb{T}\,,$$

where, $\mathbb{I} = \mathbb{R} \setminus \mathbb{Q}$ and $\mathbb{T} = \mathbb{R} \setminus \mathbb{A}$. As $\mathbb{Q} \subseteq \mathbb{A}$, it follows that $\mathbb{T} \subseteq \mathbb{I}$.

Definition 1.2.1. Elements of $\mathbb{I}$ are called irrational numbers while those of $\mathbb{T}$ are called transcendental numbers.

Perhaps it is very surprising that while on one hand it is not immediately clear that $\mathbb{T}$ is non empty, on the other hand, it turns out that (in a well defined sense) almost all real numbers are transcendental. Not only that, it is hard to identify individual numbers as being transcendental and such identifications, as have been made are mathematical epics. According to a Cambridge story, G.H. Hardy was prepared to resign his chair in favour of anyone who proved that the Euler's constant

$$\gamma = \lim_{n\to\infty}(1 + \frac{1}{2} + \cdots + \frac{1}{n} - \log n)$$

was irrational, let alone transcendental ! It was in 1873 that Hermite proved that

$$e = 1 + \frac{1}{1!} + \frac{1}{2!} + \cdots$$

is transcendental and in 1882, Lindemann did the same for π; the status of γ is still undecided.

It will give some idea of the difficulty of transcendence questions if we note that the transcendence of $2^{\sqrt{2}}$ was the seventh in the list of the famous 23 unsolved problems that David Hilbert presented, at the International Congress of 1900 in Paris, as signposts for twentieth century Mathematics. Not only that but speaking informally at a seminar in Göttingen, twenty years later, Hilbert declared that none of the audience would live to see a solution of this problem. As it happened, Hilbert was wrong : the problem was settled some twelve years later by Gelfond and Schneider, while several questions that Hilbert believed to be easier are still unanswered.

A beautiful result that settles the transcendence of $2^{\sqrt{2}}$ is the following:

Theorem 1.2.2.(Gelfond-Schneider, 1934) *If $\alpha \neq 0$ or 1, $\alpha \in \mathbb{A}$, $\beta \in \mathbb{I}\cap\mathbb{A}$, then $\alpha^{\beta} \in \mathbb{T}$.*

For a self-contained proof of this result the reader is referred to [44]. Taking $\alpha = 2$, $\beta = \sqrt{2}$ shows that $2^{\sqrt{2}}$ is transcendental.

In 1851, Liouville was the first to exhibit a class of transcendental numbers, viz. numbers of the form $a_1/10 + a_2/10^{2!} + a_3/10^{3!} + \cdots$, $0 \leq a_i \leq 9$, with infinitely many a_i's nonzero. This class is uncountable whereas the set $\mathbb{A}$ can be shown to be countable. This already shows that *most numbers are transcendental.*

Theorem 1.2.3. (Liouville, 1851) *The real number*

$$L = \sum_{i=1}^{\infty} \frac{1}{10^{i!}} = \frac{1}{10} + \frac{1}{10^2} + \frac{1}{10^6} + \frac{1}{10^{24}} + \cdots = 0.1100010\ldots$$

is transcendental.

Proof. Let α be an algebraic number. Then it satisfies a polynomial $P(x)$ with integral coeffcients. Suppose that the degree of $P(x)$ is n. We first claim

that then α can not be the limit of a sequence of distinct rational numbers p_i/q_i satisfying

$$| p_i/q_i - \alpha | < k/q_i^{n+1} , \qquad (*)$$

where k is a fixed number. For otherwise, the mean value theorem gives $P(p_i/q_i) - P(\alpha) = (p_i/q_i - \alpha)P'(\xi_i)$ for some ξ_i between p_i/q_i and α, i.e.,

$$\begin{aligned} | P(p_i/q_i) | &= | p_i/q_i - \alpha || P'(\xi_i) | && (\text{since } P(\alpha) = 0) \\ &< (k/q_i^{n+1}) | P'(\xi_i) | && (\text{by}(*)) \\ &< (k/q_i^{n+1})(| P'(\alpha) | + 1) , \end{aligned}$$

if i is large enough, since $\xi_i \to \alpha$, (as $i \to \infty, p_i/q_i \to \alpha$) and P' is a polynomial, so continuous. This gives $| q_i^n P(p_i/q_i) | < (k/q_i)(| P'(\alpha) + 1 |)$, if i is large. Here the left hand side $q_i^n P(p_i/q_i) \in \mathbb{Z}$, since $P(p_i/q_i)$ is a polynomial in p_i/q_i of degree n with integer coefficients, while the right hand side tends to 0 as i tends to infinity, i.e., the absolute value of the right hand side is less than 1 (note that p_i/q_i being distinct implies that $q_i \to \infty$ as $i \to \infty$). It follows that the left hand side is equal to 0 for all $i \geq J$, say, i.e., $P(p_i/q_i) = 0$ for $i \geq J$, which is a contradiction since P is a polynomial of degree n and so has no more than n zeros. This proves the claim.

Now going back to the number L, we have $L = 1/10+1/10^2+0/10^3+0/10^4+0/10^5+1/10^6+0/10^7+\cdots$, so that its decimal expansion equals 0.1100010.... as stated in the theorem, with the i^{th} nonzero decimal digit equal to 1 in the $i!^{\text{th}}$ place. If we truncate this decimal expansion after the i^{th} nonzero digit, we obtain a rational approximation $p_i/10^{i!}$, which differs from L by less than $2/10^{i!+1}$. Indeed, we have

$$L = \underbrace{(1/10+1/10^2+0/10^3+0/10^4+0/10^5+1/10^6+\cdots+1/10^{i!})}_{\textit{truncated at this digit}} + 0/10^{i!+1} + \cdots$$

and we call the quantity in brackets p_i/q_i, which is equal to $p_i/10^{i!}$ (common denominator equal to $10^{i!}$), and then

$$\begin{aligned} L - p_i/q_i &= 0/10^{i!+1} + \cdots + 1/10^{(i+1)!} + \cdots \\ &= 0 + \cdots + 1/10^{(i+1)!} + \cdots \\ &= (1/10^{(i+1)!})(1 + 1/10^{i+2} + 1/10^{(i+3)(i+2)} + \cdots) \\ &< (1/10^{(i+1)!}) \cdot 2 \quad (\text{the sum being a sub-geometric progression}) \\ &= 2/(10^{i!})^{i+1} \\ &\leq 2/(10^{i!})^{n+1} , \end{aligned}$$

for any $n \geq i$; and so, since the q_i are all distinct ($q_i = 10^{i!}$), the claim above yields that L can not satisfy a polynomial equation of degree n with integer coefficients and hence it follows that L is not algebraic.

For the general Liouville number $L = \sum a_i/10^{i!}$, $0 \leq a_i \leq 9$, a slight modification of the above argument shows L to be transcendental. Alternatively, one could use binary expansion and write L in the form $L = \sum a_i/2^{i!}$ $(a_i = 0, 1)$ and then, exactly as in Theorem 1.2.3, L may be shown to be transcendental. This was the first encounter of a class of transcendental numbers (in 1851). The Gelfond-Schneider theorem mentioned above guarantees another such class of transcendental numbers. To check specific numbers for transcendence is an extremely difficult job. Results that help in this direction are the following:

Theorem 1.2.4. *If β is a positive number such that 2^β, 3^β, 5^β, 7^β, 11^β, ... are all integers, i.e., if p^β is an integer for every prime p then β itself is an integer.*

For a proof see [41].

The hypothesis of the above theorem can be weakened and indeed a deeper argument given in [41] to prove Theorem 1.2.4 yields the following result.

Theorem 1.2.5.(Siegel) *If β is a positive real number such that 2^β, 3^β, 5^β are integers, then β is an integer.*

It is an open question whether the hypothesis of Theorem 1.2.5 can be reduced to requiring only that 2^β and 3^β be integers. If that were possible, as is conjectured, the result would be the best possible, since $2^{\log 3/\log 2} = 3$ is an integer but $\log 3/\log 2$ is irrational (for, if it is equal to p/q, then $q \log 3 = p \log 2$, or $3^q = 2^p$, which is impossible). Theorem 1.2.5 tells us that if β is irrational, at least one of 2^β, 3^β, 5^β is not an integer.

The following far-reaching extension of Theorem 1.2.2 was proved by Baker (see [6]) in 1966.

Theorem 1.2.6. *If $\alpha_1, \ldots, \alpha_n$ are algebraic numbers, $\alpha_i \neq 0, 1$, and if $\beta_1, \ldots, \beta_n$ are algebraic numbers which are different from 0 and 1 and are linearly independent over $\mathbb{Q}$, then $\alpha_1^{\beta_1}\alpha_2^{\beta_2} \ldots \alpha_n^{\beta_n}$ is transcendental.*

The case $n = 1$ is Theorem 1.2.2. As an example, it follows that the number $2^{\sqrt{2}}.3^{\sqrt{3}}.5^{\sqrt{5}}$ is transcendental.

Example 1.2.7. Can a rational or an irrational number raised to a rational or an irrational power be rational or irrational? All the eight possibilities are as follows: (i) $(irrational)^{irrational} = rational$: Observe that $(\sqrt{2}^{\sqrt{2}})^{\sqrt{2}} = (\sqrt{2})^{\sqrt{2}.\sqrt{2}} = 2$ and that $\sqrt{2}^{\sqrt{2}}$ is irrational, being the square root of Hilbert's number $2^{\sqrt{2}}$ (Theorem 1.2.2). It was actually proved to be transcendental by Kuzmin in 1930.
(ii) $(irrational)^{irrational} = irrational$: Yes; $(\sqrt{2})^{\sqrt{2}}$, as above.
(iii) $(rational)^{irrational} = irrational$: Yes; $2^{\sqrt{2}}$ is irrational, by Theorem 1.2.2.
(iv) $(rational)^{irrational} = rational$: Yes; $2^{\log_2 3} = 3$. However, a rational number raised to an algebraic irrational power is irrational, by Theorem 1.2.2.

(v) $(irrational)^{rational} = rational$: Yes; $(\sqrt{2})^2 = 2$.
(vi) $(irrational)^{rational} = irrational$:Yes; trivially, $(\sqrt{2})^1 = \sqrt{2}$.
(vii) $(rational)^{rational} = rational$: Yes; $2^2 = 4$.
(viii) $(rational)^{rational} = irrational$:Yes; $2^{\frac{1}{2}} = \sqrt{2}$ is irrational.

The reader is referred to [52] for more such results.

Example 1.2.8. The number $\xi = \sum_{i=1}^{\infty} \frac{1}{c_i}$ is transcendental, where $c_0 = 0$, $c_{i+1} = 2^{c_i}$.

To see this, observe that c_i are rapidly increasing powers of 2. Write out a few and then check (by induction) that if $k \geq 1$, then $k \leq c_{k-1}$ and that $c_{k-1}^2 \leq 2^{c_{k-1}}$. It follows that $kc_{k-1} \leq c_{k-1}^2 \leq 2^{c_{k-1}} = c_k$, i.e., $kc_{k-1} \leq c_k$ and so $2^{kc_{k-1}} \leq 2^{c_k} = c_{k+1}$, or $(2^{c_{k-1}})^k \leq c_{k+1}$, or $c_k^k \leq c_{k+1}$ (note that $x^y - y^x \leq 0$ for (x, y) in regions IV and II of the plane (see Chapter 7); hence $c_k^2 - 2^{c_k} \leq 0$, since $(c_k, 2)$ is in the region IV).

Now, let $\xi_k = \sum_{i=1}^{k} \frac{1}{c_i}$. Then $\xi = p_k/c_k$, where $(p_k, c_k) = 1$ (for $\xi_k =$ (a sum of even integers+1)$/c_k$). Now suppose ξ is algebraic of degree n $(n > 1)$, then

$$\xi - \frac{p_k}{c_k} = \sum_{i \geq k+1} c_i = \frac{1}{c_{k+1}} + \left\{ \frac{1}{c_{k+2}} + \cdots \right\} < \frac{2}{c_{k+1}} \leq \frac{2}{c_k^k} ,$$

because, $\{\frac{1}{c_{k+2}} + \cdots\} \leq \frac{1}{c_{k+2}}$ + terms of a geometric progression, all of the form $1/2^r$, from c_{k+2} onward $\leq 2/c_{k+2} < 1/c_{k+1}$ and $c_k^k \leq c_{k+1}$, by above.

It follows that

$$0 \leq \lim_{k \to \infty} c_k^{n+1}(\xi - p_k/c_k) \leq \lim_{k \to \infty} c_k^{n+1}.(2/c_k^k) = 0 ,$$

and so $| c_k^{n+1}(\xi - p_k/c_k) | < 1$, if k is large, which gives that $| p_k/c_k - \xi | < 1/c_k^{n+1}$ and since p_k/c_k are distinct, this contradicts the claim in the beginning of the proof of Liouville's theorem.

Finally, there remains the case $n = 1$, i.e., ξ is rational, say, $\xi = p/q$ (p, q integers). Choose k such that $c_k > q$. Then $0 < (\xi - \xi_k)qc_k = (p/q - p_k/c_k)qc_k = pc_k - p_k q$, which is an integer.

However, $(\xi - \xi_k)qc_k = qc_k(\sum_{i \geq k+1} 1/c_i) < qc_k.(2/c_{k+1}) \leq 2q/c_k^{k-1} < 2/c_k^{k-2}$, since $c_k^k \leq c_{k+1}$ and k is so chosen that $c_k > q$; and it follows that for $k \geq 3$, $(\xi - \xi_k)qc_k < 1$, contradicting the above observation that it is an integer.

There are many outstanding problems regarding the irrationality of numbers. In particular, many irrational numbers can be explicitly exhibited, e.g., $\sqrt{n}$ is irrational for all positive integers n which are not perfect squares. Also $\sqrt{2} + \sqrt{3}$, $\sqrt{2} + \sqrt{3} + \sqrt{6}$ etc. are irrational. In what follows, we give a couple of such interesting results, which yield irrationality of some classes of numbers. The proofs of these results require familiarity with the basic concepts of field extensions.

Theorem 1.2.9. *Let $a_1, \ldots, a_n$ be positive integers, none perfect squares and*

coprime in pairs. Then the 2^n *algebraic numbers* $(a_1^{\varepsilon_1}a_2^{\varepsilon_2}\dots a_n^{\varepsilon_n})^{\frac{1}{2}}$, $\varepsilon_i = 0, 1$, *are linearly independent over* $\mathbb{Q}$.

Proof. We use induction on n. For $n = 1$, the result is trivial. So suppose the result is true for $n-1$ and we prove it for n.. Define a tower of fields: $K_0 = \mathbb{Q}$, $K_1 = \mathbb{Q}(\sqrt{a_1})$, $K_2 = K_1(\sqrt{a_2})$, $\dots$, $K_n = K_{n-1}(\sqrt{a_n})$. We first prove that each K_i is of degree 2 over K_{i-1}. To see this we again use induction. For $n = 1$, clearly $[K_1 : K_0] = 2$. So, suppose the result is true for $n-1$ and we are to show that $[K_n : K_{n-1}] = 2$. If not, then $K_n = K_{n-1}$, i.e., $\sqrt{a_n} \in K_{n-1} = K_{n-2}(\sqrt{a_{n-1}})$ and so $\sqrt{a_n} = \beta + \gamma\sqrt{a_{n-1}}$, $\beta, \gamma \in K_{n-2}$, or $a_n = \beta^2 + \gamma^2 a_{n-1} + 2\beta\gamma\sqrt{a_{n-1}}$. Here $\sqrt{a_{n-1}} \notin K_{n-2}$, by induction hypothesis, so $\beta\gamma = 0$. If $\gamma = 0$, then $\sqrt{a_n} = \beta \in K_{n-2}$, which contradicts the induction hypothesis for $n-1$ numbers $a_1, a_2, \dots, a_{n-2}, a_n$. If $\beta = 0$, then $\sqrt{a_n} = \gamma\sqrt{a_{n-1}}$ and so $\sqrt{(a_{n-1}a_n)} = a_{n-1}\gamma \in K_{n-2}$, which again contradicts the induction hypothesis for $n-1$ numbers $a_1, \dots, a_{n-2}, a_{n-1}a_n$. This completes the proof of the observation and hence $[K_n : \mathbb{Q}] = 2^n$.

By induction hypothesis, the set $\{(a_1^{\varepsilon_1}\dots a_{n-1}^{\varepsilon_{n-1}})^{1/2} \mid \varepsilon_i = 0, 1\}$ is linearly independent over $\mathbb{Q}$ and therefore constitutes a basis of K_{n-1} over $\mathbb{Q}$, as $[K_{n-1} : \mathbb{Q}] = 2^{n-1}$, by above.

Let $E \subseteq K \subseteq L$ be a tower of fields. Let $\{\alpha_1, \dots, \alpha_s\}$ be a basis of K over E and $\{\beta_1, \dots, \beta_t\}$ be a basis of L over K. Then it is easy to see that $\{\alpha_i\beta_j \mid 1 \le i \le s, 1 \le j \le t\}$ is a basis of L over E.

Now, $\{(a_1^{\varepsilon_1}\dots a_{n-1}^{\varepsilon_{n-1}})^{1/2} \mid \varepsilon_i = 0, 1\}$ is a basis of K_{n-1} over $\mathbb{Q}$ and $\{1, \sqrt{a_n}\}$ is a basis of K_n over K_{n-1}, the result follows.

Example 1.2.10. As an application of the above result it follows that an expression like $7\sqrt{19}/4 - 3\sqrt{7} + 8\sqrt{6}/5$ is irrational.

While we are at it, we prove the following very useful:

Theorem 1.2.11. *Let* m *be a square-free integer greater than* 1 *and let* $\alpha, \beta, \gamma \in \mathbb{Q}(\sqrt{m}) = \{a + b\sqrt{m} \mid a, b \in \mathbb{Q}\}$. *A necessary and sufficient condition that there exist* $u, v, w \in \mathbb{Q}$ *such that*

$$u\sqrt{\alpha} + v\sqrt{\beta} + w\sqrt{\gamma} = 0 \qquad (1)$$

is that there exist $\lambda, \mu, \nu \in \mathbb{Q}(\sqrt{m})$ *such that*

$$\alpha : \beta : \gamma = \lambda^2 : \mu^2 : \nu^2. \qquad (2)$$

Proof. Write $\alpha = a_1 + a_2\sqrt{m}$, $\beta = b_1 + b_2\sqrt{m}$, $\gamma = c_1 + c_2\sqrt{m}$, $a_1, a_2, b_1, b_2, c_1, c_2 \in \mathbb{Q}$. Then (1) $\Rightarrow (u\sqrt{\alpha} + v\sqrt{\beta})^2 = (-w\sqrt{\gamma})^2 \Rightarrow u^2\alpha + v^2\beta + 2uv\sqrt{\alpha\beta} = w^2\gamma \Rightarrow \sqrt{\alpha\beta} \in \mathbb{Q}(\sqrt{m})$, i.e., that $\alpha\beta$ is a square in $\mathbb{Q}(\sqrt{m})$, say, $\alpha\beta = 1/\nu^2$ and similarly $\beta\gamma = 1/\lambda^2$, $\gamma\alpha = 1/\mu^2$, with $\lambda, \mu, \nu \in \mathbb{Q}(\sqrt{m})$. On dividing, it follows that $\alpha/\beta = \lambda^2/\mu^2$ and $\alpha/\gamma = \lambda^2/\nu^2$, which together give (2).

Conversely, let (2) hold. Then $\sqrt{\alpha} : \sqrt{\beta} : \sqrt{\gamma} = \lambda : \mu : \nu$, i.e., say, $\sqrt{\alpha} = c\lambda, \sqrt{\beta} = c\mu, \sqrt{\gamma} = c\nu$, where $c \in \mathbb{Q}(\sqrt{m})$. Then $\sqrt{\alpha} = c(a_1 + a_2\sqrt{m})$, $\sqrt{\beta} = c(b_1 + b_2\sqrt{m})$, $\sqrt{\gamma} = c(c_1 + c_2\sqrt{m})$ and we want to solve for u, v, w (not all zero) in $\mathbb{Q}$, with $u\sqrt{\alpha} + v\sqrt{\beta} + w\sqrt{\gamma} = 0$, i.e., $u(a_1 + a_2\sqrt{m}) + v(b_1 + b_2\sqrt{m}) +$

$w(c_1 + c_2\sqrt{m}) = 0$, i.e., we want to solve the equations $ua_1 + vb_1 + wc_1 = 0$, $ua_2 + vb_2 + wc_2 = 0$, simultaneously for u, v, w, nontrivially, in $\mathbb{Q}$. As these are two equations in three variables, there exist a nontrivial solution u, v, w in $\mathbb{Q}$, as required.

Observe that when (2) is satisfied, (1) becomes $u\lambda + v\mu + w\nu = 0$.

As an application, we have the following:

Example 1.2.12. $59\sqrt{(90 - 14\sqrt{7})}+4\sqrt{(4555 + 1721\sqrt{7})} = 145\sqrt{(26 + 2\sqrt{7})}$.

Indeed, let $m = 7$ and let $\alpha = 90-14\sqrt{7}$, $\beta = 4555+1721\sqrt{7}$, $\gamma = 26+2\sqrt{7}$. We need to verify that there exist u, v, w (given respectively as 59, 4, -145) such that $u\sqrt{\alpha} + v\sqrt{\beta} + w\sqrt{\gamma} = 0$ (i.e., (1)). It is enough to verify (2), i.e., that $\alpha : \beta : \gamma = \lambda^2 : \mu^2 : \nu^2$. We have $\alpha/\gamma = (90 - 14\sqrt{7})(26 + 2\sqrt{7}) = (317 - 68\sqrt{7})/81 = ((17 - 2\sqrt{7})/9)^2$, and $\beta/\gamma = (4555+1721\sqrt{7})/(26+2\sqrt{7}) = (47168+17818\sqrt{7})/324 = ((151+59\sqrt{7})/18)^2$, so $\alpha = ((17-2\sqrt{7})/9)^2 \cdot \gamma$, $\beta = ((151 + 59\sqrt{7})/18)^2 \cdot \gamma$ and of course $\gamma = 1^2 \cdot \gamma$. These give us the ratios $\lambda^2 : \mu^2 : \nu^2$ and proceeding as in the proof of Theorem 1.2.11, one checks easily that $u = 59, v = 4, w = -145$ is a solution of the system of equations

$$\frac{17}{9}u + \frac{151}{18}v + w = 0$$
$$\frac{-2}{9}u + \frac{59}{18}v = 0$$

Remark 1.2.13. From elementary field theory it follows that if K/F and L/K are algebraic extensions, then so is L/F. From this, it is easy to deduce that the set of all algebraic real numbers is a subfield of $\mathbb{R}$, containing $\mathbb{Q}$, i.e., the sum, difference, product and quotient of two algebraic numbers is algebraic.

If θ is a rational number, it follows that $\cos \pi\theta$ and $\sin \pi\theta$ and hence $\cot \pi\theta = \frac{\cos \pi\theta}{\sin \pi\theta}$ are algebraic numbers (indeed, for $\theta = p/q$, $(\cos \frac{\pi p}{q} + \iota \sin \frac{\pi p}{q})^q = (e^{\iota \pi p/q})^q = e^{\iota \pi p} = \pm 1$, yields polynomial equations with rational coefficients satisfied by $\cos \pi\theta$ and $\sin \pi\theta$). As an application, we have:

Example 1.2.14. The number $g(\theta) = \sum_{n=1}^{\infty} \theta/(n(n + \theta))$ is transcendental (and hence irrational) for infinitely many rational numbers $\theta \in (0, 1)$.

Proof (sketch). Assuming the partial fraction expansion of $\pi \cot \pi\theta$, i.e.,

$$\pi \cot \pi\theta = \sum_{n=2}^{\infty} \frac{2\theta}{\theta^2 - n^2} \quad ,$$

the proof proceeds as follows:

$$g(\theta) = 1 - 1/(1+\theta) + \sum_{n=2}^{\infty}(1/n - 1/(n+\theta)) \,, \tag{1}$$

$$\begin{aligned} g(1-\theta) &= \sum_{n=1}^{\infty}(1/n - 1/(n+1-\theta)) \\ &= (1 - 1/(2-\theta)) + (1/2 - 1/(3-\theta)) + \cdots \\ &= 1 + \sum_{n=2}^{\infty}(1/n - 1/(n-\theta)) \,. \end{aligned} \tag{2}$$

(1) and (2) imply that

$$\begin{aligned} g(1-\theta) - g(\theta) &= \frac{1}{1+\theta} + \sum_{n=2}^{\infty}(2\,\theta/(\theta^2 - n^2)) \\ &= \frac{1}{1-\theta} + \sum_{n=1}^{\infty}(2\,\theta/(\theta^2 - n^2)) \\ &= \frac{1}{1-\theta} + \pi\cot\pi\theta \quad . \end{aligned}$$

Since for θ rational, $\cot\pi\theta$ is algebraic, whence $\pi\cot\pi\theta$ is transcendental (as π is transcendental, see next section), therefore, at least one of $g(1-\theta)$ and $g(\theta)$ is transcendental.

§1.3. The numbers e and π

In this section we investigate two special real numbers e and π, their irrationality and transcendence. We have tried to collect interesting facts about them at one place. We begin with the following:

Theorem 1.3.1. *(i) The series* $1 + 1/1! + 1/2! + \cdots$ *is convergent. Its limit is called e. Moreover* $2 < e < 3$.
(ii) The number e is irrational.
(iii) Let $a_n = (1 + 1/n)^{n+1}$, $b_n = (1 - 1/n)^n$, $c_n = (1 + 1/n)^n$. *Then* c_n *increases,* $a_n > c_n$ *for all* n *and* $\lim_{n\to\infty} a_n$, $\lim_{n\to\infty} c_n$ *both exist and are equal to e. Further,* $\lim_{n\to\infty} b_n$ *also exists and equals* $1/e$.

Proof. (i) The partial sums $S_n = 1 + 1/1! + 1/2! + \cdots + 1/n! < 1 + 1 + 1/2 + 1/2^2 + \cdots + 1/2^{n-1} = 3$, are bounded above by 3 and below by 2 and are clearly increasing as n increases and so tend to a limit e say, as required.

(ii) We have $0 < e - S_n$ and that

$$\begin{aligned} e - S_n &= 1/(n+1)! + 1/(n+2)! + \cdots \\ &= (1/(n+1)!)(1 + 1/(n+2) + 1/(n+3)(n+2) + \cdots) \\ &< (1/(n+1)!)(1 + 1/(n+1) + 1/(n+1)^2 + \cdots) \\ &= (1/(n+1)!)(1/(1 - 1/(n+1))) \\ &= 1/n! \cdot n \,. \end{aligned}$$

Now if $e = p/q$, say, with $1 \leq p, q$, $(p, q) = 1$, then $0 < p/q - S_n < 1/n! \cdot n$. In this taking $n = q$ we get $0 < p/q - S_q < 1/q! \,. q$, and multiplying by $q!$, this becomes $0 < (q-1)! \cdot p - q! \cdot S_q < 1/q \leq 1$. Here $(q-1)! \cdot p$ and $q! \cdot S_q$ are both integers and so their difference ρ say, is an integer, i.e., $0 < \rho < 1$, which is not possible.

(iii) We first prove that $c_n = (1 + 1/n)^n \to e$ as $n \to \infty$. Using the binomial theorem

$$\begin{aligned} (1+1/n)^n = 1 &+ n(1/n) + (n(n-1)/2!)(1/n^2) + (n(n-1)(n-2)/3!)(1/n^3) + \cdots \\ = 1 &+ 1 + (1/2!)(1 - 1/n) + (1/3!)(1 - 1/n)(1 - 2/n) + \cdots + \\ &+ (1/n!)(1 - 1/n)(1 - 2/n) \cdots (1 - (n-1)/n) \qquad (*) \\ < 1 &+ 1 + 1/2! + 1/3! + \cdots + 1/n! \\ < e & \end{aligned}$$

Further, each term on the right hand side of (*) increases as n increases and the number of terms increases too as n increases, i.e., $(1 + 1/n)^n$ increases as n increases, and it is bounded above by e and so tends to a limit $\eta \leq e$. However, $(1+1/n)^n \geq 1+1+(1/2!)(1-1/n)+(1/m!)(1-1/n)(1-2/n)\ldots(1-(m-1)/n)$, (by (*) again), if $n \geq m$. Now keeping m fixed and letting $n \to \infty$, we get that $\eta \geq 1 + 1 + 1/2! + \cdots + 1/m!$ and this is true for all m. Letting $m \to \infty$, we get $\eta \geq e$.

Short proofs of the other parts are as follows. The arithmetic mean of the $n+2$ numbers 1, $n/(n+1)$, $n/(n+1)$, ... , $n/(n+1)$ is greater than their geometric mean. This gives $(1+n)/(n+2) > ((n/(n+1))^{n+1})^{1/(n+2)}$, i.e., on taking reciprocals $1+1/(n+1) < (1+1/n)^{(n+1)/(n+2)}$, i.e., $(1+1/(n+1))^{(n+2)} < (1+1/n)^{(n+1)}$, showing that a_n decreases as n increases.

Next, $b_{n+1} = (1 - 1/(n+1))^{(n+1)} = (n/(n+1))^{(n+1)} = 1/a_n$; hence b_n increases, as required.

Finally, the arithmetic mean of the $n+1$ numbers 1, $1+1/n$, $1+1/n$, ... , $1+1/n$ is greater than their geometric mean. Hence $(1 + n(1 + 1/n))/(n+1) > ((1+1/n)^n)^{1/(n+1)}$, i.e., $((1+1/(n+1))^{(n+1)} > (1+1/n)^n$, showing that c_n increases as n increases. Since $c_n \to e$ (already proved), so $a_n = c_n(1+1/n) \to e$ too. Finally, $1/b_n = 1/a_{n-1} \to 1/e$ as required.

Remark 1.3.2. The value of e has been calculated with great accuracy (see [63], page 101):

$$2.71828182\ldots < e < 2.71828184\ldots$$

We next consider the number π. A reference to an excellent account of π, its computation and all other developments through the ages, is the exhaustive article " The Ubiquitous π ", by Dario Castellanos [22].

With the help of computers, π has now been calculated to thousands of decimal places:

$$\pi = \mathbf{3.14159265358979323846}\ldots .$$

Many *mnemonics* have been composed to remember and write down the value of π, the number of letters in each word representing successive digits of π. A well known mnemonic is: *How I want a drink, alcoholic of course, after the heavy lectures involving quantum mechanics.*(15 digits).

One of the most basic result in which π appears is the sum of the following infinite series:

Theorem 1.3.3. $1 + 1/2^2 + 1/3^2 + 1/4^2 + \cdots = \pi^2/6.$

We shall present here a simple proof given in [78]. First we prove the following simple

Lemma 1.3.4. $\sum_{k=1}^{m} \cot^2(k\pi/(2m+1)) = m(2m-1)/3$.

Proof. By equating the imaginary parts in the formula

$$\cos n\,\theta + \iota \sin n\,\theta = (\cos\theta + \iota \sin\theta)^n = \sin^n\theta\,(\cot\theta + \iota)^n$$
$$= \sin^n\theta \sum_{k=0}^{n} \binom{n}{k} \cdot \iota^k \cot^{n-k}\theta\ ,$$

we obtain the identity

$$\sin n\theta = \sin^n\theta \left[\binom{n}{1} \cot^{n-1}\theta - \binom{n}{3} \cot^{n-3}\theta + \binom{n}{5} \cot^{n-5}\theta - \cdots \right],$$

where $0 < \theta < \pi/2$. Take $n = 2m+1$ and write this in the form $\sin(2m+1)\theta = \sin^{2m+1}\theta \cdot P_m(\cot^2\theta)$, where $P_m(x) = \binom{2m+1}{1}x^m - \binom{2m+1}{3}x^{m-1} + \binom{2m+1}{5}x^{m-2} - \cdots + \binom{2m+1}{2m+1}$ is a polynomial of degree m. Since $\sin\theta \neq 0$, if $0 < \theta < \pi/2$, it follows from above that $P_m(\cot^2\theta) = 0$ if and only if $(2m+1)\theta = k\pi$, $k \in \mathbb{Z}$. Therefore, $P_m(x)$ vanishes at the m distinct points $x_k = \cot^2(\pi k/(2m+1))$, for $k = 1, 2, \ldots, m$. Since the degree of $P_m(x)$ is m, these are all the zeros of $P_m(x)$ and their sum is $-(-\binom{2m+1}{3}/\binom{2m+1}{1})$, which proves the lemma.

Proof of Theorem 1.3.3. Start with the inequality $\sin x < x < \tan x$ for $0 < x < \pi/2$. Take reciprocals and square to obtain $\cot^2 x < 1/x^2 < 1 + \cot^2 x$. Now put $x = k\pi/(2m+1)$, where k, m are integers, $1 \leq k \leq m$, and sum from $k = 1$ to $k = m$ to get

$$\sum_{k=1}^{m} \cot^2(k\pi/(2m+1)) < ((2m+1)^2/\pi^2) \sum_{k=1}^{m} 1/k^2 < m + \sum_{k=1}^{m} \cot^2(k\pi/(2m+1)).$$

Putting in the value of the sums, using the lemma, we get

$$m(2m-1)/3 < ((2m+1)^2/\pi^2)\sum_{k=1}^{m} 1/k^2 < m + m(2m-1)/3\,,$$

which, on dividing throughout by $(2m+1)^2/\pi^2$ and then letting $m \to \infty$, proves the theorem.

One of the most beautiful expressions for π, given by John Wallis in 1655, called Wallis' product, is the following:

Theorem 1.3.5.

$$\frac{\pi}{2} = \lim_{m\to\infty}\left(\frac{2\cdot 2\cdot 4\cdot 4\cdot 6\cdots(2m)\cdot(2m)}{1\cdot 3\cdot 3\cdot 5\cdot 5\cdots(2m-1)\cdot(2m+1)}\right).$$

Proof. Let $I_n = \int_0^{\pi/2} \sin^n\theta\, dx$. Integration by parts easily gives $I_n = ((n-1)/n)I_{n-2}$ $(n > 1)$. Since $I_0 = \pi/2$, $I_1 = 1$, it follows that

$$I_{2m} = ((2m-1)/2m)\cdot((2m-3)/(2m-2))\cdots(1/2).(\pi/2) \qquad (1)$$

$$I_{2m+1} = (2m/(2m+1))\cdot((2m-2)/(2m-1))\cdots(2/3).1 \qquad (2)$$

Now in the range $0 \le \theta \le \pi/2$, we have $0 \le \sin\theta \le 1$ (in fact < 1 if $\theta < \pi/2$) and so $\sin^n\theta > \sin\theta.\sin^n\theta = \sin^{n+1}\theta$ for all $\theta \in (0, \pi/2)$. It follows that (see Figure 1.1) $0 < I_{2m+1} < I_{2m} < I_{2m-1}$.

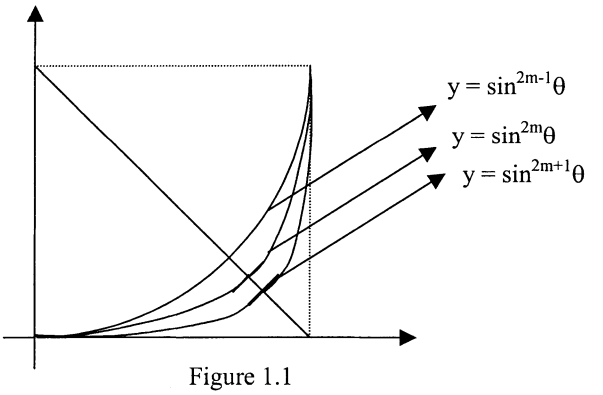


Figure 1.1

Dividing by I_{2m+1} gives

$$1 < I_{2m}/I_{2m+1} < I_{2m-1}/I_{2m+1} \qquad (*)$$

Here the extreme right hand term is equal to $(2m+1)/2m$ (see (1) and (2)), which tends to 1 as $m \to \infty$. So letting $m \to \infty$ in (*), we get $1 \le \lim_{m\to\infty} I_{2m}/I_{2m+1} \le 1$. However using (1) and (2), we get

$$\frac{I_{2m}}{I_{2m+1}} = \left(\frac{\pi}{2}\right)\left(\frac{1}{2}\right)\left(\frac{3}{2}\right)\left(\frac{3}{4}\right)\cdots\left(\frac{2m-1}{2m-2}\right)\left(\frac{2m-1}{2m}\right)\left(\frac{2m+1}{2m}\right),$$

and letting $m \to \infty$, we get the result.

Corollary 1.3.6.

$$\lim_{m\to\infty} \frac{(2^m\, m!)^4}{((2m)!)^2 \cdot (2m+1)} = \frac{\pi}{2}.$$

Proof. The right side of Wallis' product is

$$\begin{aligned}
\frac{2\cdot 2\cdot 4\cdot 4\cdots 2m\cdot 2m}{1\cdot 3\cdot 5\cdots(2m-1)(2m+1)} &= \frac{(2\cdot 4\cdots 2m)^2}{(1\cdot 3\cdot 5\cdots(2m-1))^2(2m+1)} \\
&= \frac{(2^m\cdot m!)^2}{(1\cdot 3\cdot 5\cdots(2m-1))^2(2m+1)} \cdot \frac{(2\cdot 4\cdots 2m)^2}{(2\cdot 4\cdots 2m)^2} \\
&= \frac{(2^m\cdot m!)^2\cdot(2^m\cdot m!)^2}{(1\cdot 2\cdot 3\cdot 4\cdots(2m-1)\cdot 2m)^2(2m+1)} \\
&= \frac{(2^m\cdot m!)^4}{((2m)!)^2(2m+1)}.
\end{aligned}$$

We shall now describe some methods to compute the value of π.

1. By inscribed and circumscribed polygons. Euclid, in the fourth century B.C., had proved that $3 < \pi < 4$, but it was not until the third century B.C. that Archimedes attacked the problem of the determination of π systematically. Using polygons inscribed in and circumscribed to the circle, whose number of sides are successively doubled, he obtained for π the bounds $3\frac{10}{71} < \pi < 3\frac{1}{7}$. The bound $3\frac{1}{7} = \frac{22}{7}$ is often referred to, erroneously, as the Archimedian value. Archimedes meant this as an upper bound on the value of π.

Archimedes' method remained essentially unchanged except for better approximations to π obtained by taking larger and larger doublings, until the advent of Calculus.

2. An analytic expression for π . Consider the identity

$$\begin{aligned}
(\sin\theta)/\theta &= \cos(\theta/2)\cdot\sin(\theta/2)/(\theta/2) \\
&= \cos(\theta/2)\cdot\cos(\theta/4)\cdot\sin(\theta/4)/(\theta/4) \\
&\quad \ldots \qquad \ldots \qquad \ldots \\
&= \cos(\theta/2)\cdot\cos(\theta/4)\cdot\ \ldots\ \cdot\cos(\theta/2^n)\cdot\sin(\theta/2^n)/(\theta/2^n).
\end{aligned}$$

As $n \to \infty$, $\sin\theta/2^n/(\theta/2^n) \to 1$ and we obtain Euler's formula

$$(\sin\theta)/\theta = \lim_{n\to\infty}(\cos(\theta/2)\cdots\cos(\theta/2^n)) = \cos(\theta/2)\cdot\cos(\theta/4)\cdot\cos(\theta/8)\ldots$$

Putting $\theta = \pi/2$, this gives, on use of the formula $\cos(\theta/2) = \sqrt{(1+\cos\theta)/2}$, the following result, giving $2/\pi$ as a limit of a sequence, which was first given by Fancois Vieta in 1593:

$$2/\pi = \sqrt{(1/2+1/2\sqrt{(1/2)})}\cdot\sqrt{(1/2+1/2\sqrt{(1/2+1/2\sqrt{(1/2)})})}\cdot\cdots\ .$$

The convergence of this expression was proved by F.Rudio in 1891 (see [98]). Vieta's formula is the first analytical expression ever obtained for π.

Now, taking logarithms in the Euler's formula above (noting that the logarithm is a continuous function):

$$\log \sin\theta - \log\theta = \log\cos\theta/2 + \log\cos\theta/4 + \log\cos\theta/8 + \cdots$$

On differentiating with respect to θ, this gives

$$1/\theta = \cot\theta + (1/2)\tan(\theta/2) + (1/4)\tan(\theta/4) + (1/8)\tan(\theta/8) + \cdots$$

Putting $\theta = \pi/4$, we obtain

$$4/\pi = 1 + (1/2)\tan(\pi/8) + (1/4)\tan(\pi/16) + (1/8)\tan(\pi/32) + \cdots .$$

The calculation of the number π by this formula is equivalent to a geometrical calculation. It was undertaken by Rene Descartes in the seventeenth century .

There have since been many formulae giving π as a rapidly converging series. These make use of various methods, as for example Euler's summation formula, Γ-function and the Taylor's series for $\tan^{-1} x$. We give here an arctan series formula for the calculation of π.

3. The arctan method. The formula

$$\pi/4 \;=\; 4\,\tan^{-1}(1/5) \;-\; \tan^{-1}(1/239) \qquad \text{(i)}$$

can be derived as follows:

Let α be the angle given by $\tan\alpha = 1/5$; then $\tan 2\alpha = 2\tan\alpha/(1 - \tan^2\alpha) = 5/12$ and $\tan 4\alpha = 2\tan 2\alpha/(1 - \tan^2 2\alpha) = 120/119 = \tan(\pi/4) + 1/119$. Thus $4\alpha > \pi/4$, say $4\alpha - \pi/4 = \beta$, so that $\tan\beta = (\tan 4\alpha - \tan\pi/4)/(1 + \tan 4\alpha\tan\pi/4) = 1/239$, which finally gives $4\alpha - \pi/4 = \beta = \tan^{-1}(1/239)$, i.e., $\pi/4 = 4\tan^{-1}(1/5) - \tan^{-1}(1/239)$, as required. This formula was discovered by John Machin in 1706.

We also have the expression $1/(1 + x^2) = 1 - x^2 + x^4 - x^6 + \cdots$, $|\,x\,| < 1$. As the above series is uniformly convergent, we may integrate term by term to get

$$\tan^{-1} x = \sum_{n=0}^{\infty} (-1)^n x^{2n+1}/(2n+1) \qquad (|\,x\,| < 1) \qquad \text{(ii)}$$

This is the series discovered by James Gregory in 1671, which we shall obtain in Chapter 7. For $x = 1$, it gives the Leibnitz's celebrated series $\pi/4 = 1 - 1/3 + 1/5 - \cdots$ which requires 2000 terms to give three decimal figures of π. Using (i) and (ii), we get

$$\begin{aligned}\pi = {} & (16/5)(1 - 1/(3\cdot 25) + 1/(5\cdot 25^2) - 1/(7\cdot 25^3) + \cdots) \\ & -(4/(239))(1 - 1/(3\cdot 57121) + 1/(5\cdot 57121^2) - 1/(7\cdot 57121^3) + \cdots)\end{aligned}$$

which is well suited for the calculation of π. In 1706, using this series, Machin did the computation to 100 decimal digits. Similar series for arctan have been used by many to compute π accurately. The reader is referred to pages 90–96 of [22].

In 1882, Lindemann proved that π is transcendental (see [74]). Here we give Niven's ingenious proof (see [75]) that π is irrational.

Theorem 1.3.7. *The real number π is irrational.*

Proof. Suppose $\pi = a/b$ $(a, b \in \mathbb{N})$. Define polynomials $f(x) = x^n(a-bx)^n/n!$ and $g(x) = f(x) - f''(x) + f^{(4)}(x) - \cdots + (-1)^n f^{(2n)}(x)$, where the positive integer n will be chosen later. Now,

$$\begin{aligned} n!\,f(x) &= x^n\left(a^n + \binom{n}{1}a^{n-1}(-bx) + \binom{n}{2}a^{n-2}(-bx)^2 + \cdots + (-bx)^n\right) \\ &= a^n x^n - \binom{n}{1}a^{n-1}bx^{n+1} + \binom{n}{2}a^{n-2}b^2x^{n+2} - \cdots + (-1)^n b^n x^{2n}\,. \end{aligned}$$

Differentiating successively we get

$$\begin{aligned} n!f'(x) =& na^n x^{n-1} - \binom{n}{1}(n+1)a^{n-1}bx^n + \binom{n}{2}(n+2)a^{n-2}b^2x^{n+1} - \cdots + \\ & (-1)^n b^n 2n x^{2n-1}\,, \\ n!f''(x) =& n(n-1)a^n x^{n-2} - \binom{n}{1}(n+1)na^{n-1}bx^{n-1} + \\ & + \binom{n}{2}(n+2)(n+1)a^{n-2}b^2x^n - \cdots + (-1)^n b^n 2n(2n-1)x^{2n-2}\,, \end{aligned}$$

$$\cdots \quad \cdots \quad \cdots \quad \cdots$$

$$n!f^{(2n)}(x) = (-1)^n b^n.2n(2n-1).\ldots.2.1\,.$$

The last equation is the only one which is free of x. It follows that $f(0)=f'(0)=\ldots f^{(2n-1)}(0) = 0$, while $n!f^{(2n)}(0) = (2n)!\,(-1)^n\,b^n$, so that $f^{(2n)}(0)$ is an integer. In any case:

$$f(0)\,,\, f'(0)\,,\, \ldots\,,\, f^{(2n)}(0) \text{ are all integers.} \tag{1}$$

Further, it can be easily checked that $f(x) = f(a/b - x)$ for all x and so differentiating successively, we get $f'(x) = -f'(a/b - x)$, $f''(x) = +f''(a/b - x)$, ... Hence $f(\pi) = f(a/b - \pi) = f(0) = 0$, $f'(\pi) = f'(0) = 0$, ..., $f^{(2n-1)}(\pi) = f^{(2n)}(0) = ((2n)!/n!)(-1)^n b^n \in \mathbb{Z}$. In any case

$$f(\pi),\, f'(\pi),\, f''(\pi), \ldots\,,\, f^{(2n-1)}(\pi),\, f^{(2n)}(\pi) \text{ are all integers.} \tag{2}$$

Note that in (1) and (2), only $f^{(2n)}(0)$ and $f^{(2n)}(\pi)$ are nonzero, the rest are in fact zero. Now,

$$\frac{d(g'(x)\sin x - g(x)\cos x)}{dx} = g'(x)\cos x + \sin x g''(x) + g(x)\sin x - \cos x g'(x)$$
$$= (g(x) + g''(x))\sin x\,.$$

But $g = f - f^{(2)} + f^{(4)} - \cdots + (-1)^n f^{(2n)}$, and so $g'' = f^{(2)} - f^{(4)} + f^{(6)} - \cdots + (-1)^n f^{(2n+2)}$. Adding, we get $g + g'' = f + (-1)^n f^{(2n+2)} = f$, as $f^{(2n)} = (-1)^n b^n (2n)!/n!$, and so $f^{(2n+1)} = f^{(2n+2)} = 0$. Thus $\frac{d(g'(x)\sin x - g(x)\cos x)}{dx} = f(x)\sin x$. Hence

$$\begin{aligned}\int_0^\pi f(x)\sin x\,dx &= \Big[g'(x)\sin x - g(x)\cos x\Big]_0^\pi \\ &= g(\pi) + g(0) \\ &= f(\pi) - f^{(2)}(\pi) + \cdots + (-1)^n f^{(2n)}(\pi) + f(0) - f^{(2)}(0) + \cdots \\ &\quad + (-1)^n f^{(2n)}(0) \\ &= (-1)^n (f^{(2n)}(\pi) + f^{(2n)}(0)) \quad \text{(the rest being all zero)}\,, \\ &= 2(-1)^n (2n)!\, b^n / n!\,,\end{aligned}$$

which is an integer, since $(2n)!/n!$ is an integer. Thus

$$\int_0^\pi f(x)\sin x\,dx \quad \text{is an integer.} \tag{3}$$

However, for $0 < x < \pi$, we have $0 < f(x)\sin x < \pi^n a^n / n! 2^{2n}$, because, $f(x) = x^n(a - bx)^n/n!$; hence $f'(x) = x^n.n(a - bx)^{n-1}(-b) + (a - bx)^n.nx^{n-1}/n! = 0, i.e., x^{n-1}(a - bx)^{n-1}(-bx + (a - bx)) = 0$, giving $x = 0$, or $a/b = \pi$, or $a/2b = \pi/2$. But as $0 < x < \pi$, the only relevant solution is $x = a/2b = \pi/2$. The value at 0 and π is 0, so that $\pi/2$ is a maximum and this maximum value is $f(a/2b)\sin \pi/2 = (a/2b)^n(a - b.a/2b)^n/n! = (a/2b)^n(a^n)/2^{2n}n! = \pi^n a^n / 2^{2n} n!$. Now, $\pi^n a^n / 2^{2n} n! \to 0$ as $n \to \infty$, so if we choose n large enough, we can ensure that $0 < f(x)\sin x < 1/5$, and so

$$0 < \int_0^\pi f(x)\sin x\,dx < \pi/5 < 1\,,$$

which is a contradiction to (3). This proves that π is irrational.

We now turn to the transcendence of e and π, which is considerably more difficult to establish than their irrationality.

A complex number is said to be algebraic if, as in the case of real numbers, it is a root of some non-constant polynomial with rational coefficients. The transcendence of e and π can be deduced from:

Theorem 1.3.8 (Lindemann). *If $A_1, \ldots, A_n$; $a_1, \ldots, a_n$ are algebraic numbers (possibly complex), $a_i \neq a_j$ for $i \neq j$, $A_i \neq 0$ for all i, then $\sum_{i=1}^n A_i e^{a_i} \neq 0$.*

For a proof of this result, the reader is referred to [101].

Corollary 1.3.9. *The number e is transcendental.*

Proof. Take arbitrary $A_1, \ldots, A_n \in \mathbb{Q}$, then $A_1 e + A_2 e^2 + \cdots + A_n e^n \neq 0$.

Corollary 1.3.10. *The number π is transcendental .*

Proof. If π is algebraic, then, as observed in Remark 1.2.13, $\pi\iota$ is algebraic (as ι obviously is) and hence by Theorem 1.3.8, $e^{\pi\iota} + e^0 \neq 0$, but $e^{\pi\iota} = -1$ and e^0=1; which gives a contradiction.

It is not known whether the numbers πe and $\pi + e$ are transcendental or algebraic. However , we have the following interesting:

Theorem 1.3.11 ([17]). *At least one of the numbers πe and $\pi + e$ is transcendental.*

Proof. Suppose to the contrary, that both πe and $\pi + e$ are algebraic. By Remark 1.2.13, the number $A = (\pi + e)^2 - 4\pi e = (\pi - e)^2$ is algebraic. It is easy to see that if α^2 is an algebraic number, then so is α. Thus $\pi - e$ and hence $(\pi + e) + (\pi - e) = 2\pi$ would be algebraic, which is not true.

§1.4. The Cantor ternary set

In this section we describe and establish some properties of the Cantor ternary set, which is a source of several counter examples in real function theory. As a preparation, we first discuss some results in point set topology.

Theorem 1.4.1. *Let O be an open and bounded subset of $\mathbb{R}$. Then O can be uniquely expressed as a disjoint union of (at most) countably many open intervals.*

Proof. Let $a \in O$. Since O is open, there exist an open interval $(a-\epsilon, a+\epsilon) \subseteq O$. Now consider the closed set $S = (-\infty, a] \cap O^c$ (here O^c denotes the complement of O), i.e., the points not in O, but lying to the left of a. Let $\alpha = \sup S$, which we know belongs to S, since S is closed and bounded above. It follows that $(\alpha, a] \subseteq O$.

Similarly, let $\beta = \inf([a, \infty) \cap O^c)$, so that β is the smallest element of $[a, \infty) \cap O^c$. Then again $[a, \beta) \subseteq O$. Hence $(\alpha, \beta) \subseteq O, \alpha, \beta \notin O$, i.e., (α, β) is the 'largest' open interval contained in O which contains the point a. It is called a *component interval* of O and is clearly uniquely determined by a (or indeed by any other point of (α, β) as the starting point instead of the point a).

Thus, O is a disjoint union of open intervals, the component intervals. To see that they are at most countable, we select a rational number in each of them. Being disjoint intervals, these rationals are distinct so that there is a 1–1 correspondence between these component intervals of O and a subset of the rationals.

Now let F be a closed, bounded nonempty set and let $\alpha = \inf F$, $\beta = \sup F$, so that $\alpha, \beta \in F$, since F is closed. The interval $[\alpha, \beta]$ is called *the smallest*

closed interval containing F. Let $O = [\alpha, \beta] \setminus F$. It is easy to see that O is open. Indeed, $O = F^c \cap [\alpha, \beta] = F^c \cap (\alpha, \beta)$ (since $\alpha, \beta \notin F^c$), which is an open set, since F^c and (α, β) are both open. It follows, by Theorem 1.4.1 that O is a disjoint union of at most countably many open intervals. These are called *the complementary intervals of* F (five of them are shown in Figure 1.2).

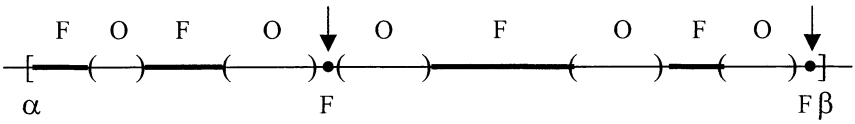


Figure 1.2

Since F is closed, F', the set of all limit points of F, is contained in F. We ask: When is $F \subseteq F'$, i.e., when is each point of F a limit point of F? Such closed subsets of $\mathbb{R}$ are called perfect sets. Now a point of F (indeed of any set) is either an *isolated point (i.e., a point such that there exists some open interval centred at the point which contains no point of* F *other than itself*) of F or a *limit point* of F. Thus, if F has no isolated point, then all points of F are limit points of F, i.e., $F \subseteq F'$ (i.e., F is dense in itself). Since $F' \subseteq F$ (F being closed), we get $F = F'$, i.e., F is perfect. We characterize all the isolated points of F in the following:

Theorem 1.4.2 *Let* F *be a closed, bounded, nonempty subset of* $\mathbb{R}$. *A point* γ *of* F *is an isolated point of* F *if and only if*
(i) γ *is a common end point of two complementary intervals of* F,
or *(ii)* $\gamma = \alpha$, *where* $(\alpha, \alpha + \delta)$ *is a complementary interval of* F,
or *(iii)* $\gamma = \beta$, *where* $(\beta - \delta', \beta)$ *is a complementary interval of* F.

Remark 1.4.3. The theorem says that either points of the type y shown in Figure 1.3, i.e.,

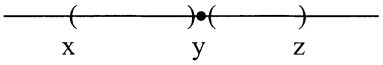


Figure 1.3

the common end points of the two complementary intervals (x, y) and (y, z) of F are isolated points of F or α is an isolated point of F if a complementary interval of F begins at α, i.e., is $(\alpha, \alpha + \delta)$

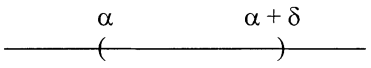


Figure 1.4

or β is an isolated point of F if a complementary interval of F ends at β, i.e., is $(\beta - \delta', \beta)$ and that there are no other isolated points of F.

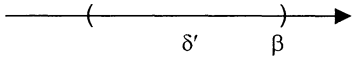


Figure 1.5

Proof of Theorem 1.4.2. The points mentioned in the theorem are clearly isolated points of F. Conversely, let γ be an isolated point of F. Then there exists an $\epsilon > 0$ such that $(\gamma - \epsilon, \gamma + \epsilon)$ has no point of F except γ, i.e., $(\gamma - \epsilon, \gamma) \subseteq O$, where O is $[\alpha, \beta] \setminus F$, as defined above (proof to be modified for $\gamma = \alpha$) and $(\gamma, \gamma + \epsilon) \subseteq O$ (proof to be modified for $\gamma = \beta$), i.e., $(\gamma - \epsilon, \gamma)$ and $(\gamma, \gamma + \epsilon)$ are both subsets of two complementary intervals of F meeting at γ, as required.

For $\gamma = \alpha$ or $\gamma = \beta$, a similar argument yields the result.

Theorem 1.4.4. *The set S of all sequences of the type $(a_1, a_2, a_3, \dots)$ where $a_i = 0$ or 1, is uncountable.*

Proof. Indeed there is a one-to-one correspondence between S and all the real numbers in $[0, 1]$, except for at most countably many exceptions. In fact, writing each real number $r \in [0, 1]$ in its binary expansion , i.e., $r = .a_1 a_2 a_3 \dots$ $(a_i = 0, 1)$, we see that the mapping $r \leftrightarrow (a_1, a_2, a_3, \dots)$ is a one to one, onto mapping between S and $[0, 1]$, the exceptions mentioned being those reals which have two expansions, viz. certain rational numbers as for example $1/2 = .011111\dots = .100000\dots$, and being a subset of rational numbers, there are at most countably many of these.

We are now in a position to define and establish some properties of the curious Cantor ternary set $\mathcal{C}$. Let $F = [0, 1]$. Now remove the middle third open interval $(1/3, 2/3)$. From the two remaining intervals, again remove the middle third open intervals, i.e., remove $(1/9, 2/9)$ and $(7/9, 8/9)$, and so on. What remains, is the Cantor ternary set $\mathcal{C}$. Thus, for example, the points

$$0,\ 1\,;\ 1/3\,,\ 2/3\,;\ 1/9,\ 2/9,\ 7/9,\ 8/9\,;\dots$$

will never be removed and therefore belong to $\mathcal{C}$ (see Figure 1.6):

Figure 1.6

As $\mathcal{C}$ is a countable intersection of closed sets $F_1 = [0, 1]$, $F_2 = [0, \frac{1}{3}] \cup [\frac{2}{3}, 1]$, $F_3 = [0, \frac{1}{9}] \cup [\frac{2}{9}, \frac{1}{3}] \cup [\frac{2}{3}, \frac{7}{9}] \cup [\frac{8}{9}, 1]$, $\dots$, it follows that $\mathcal{C}$ is a closed set. Since no two removed intervals are of the type (x, y), (y, z), we see that $\mathcal{C}$ has no isolated point (see Theorem 1.4.2); in fact, we get:

Theorem 1.4.5. *The subset $\mathcal{C}$ of $\mathbb{R}$ is perfect, i.e., $\mathcal{C}' = \mathcal{C}$.*

To understand $\mathcal{C}$ more clearly, it is essential to look at the *ternary expansion*

of real numbers. Let us therefore start first with a digression on the decimal expansion of real numbers $r \in [0,1]$.

Each r has a unique decimal expansion: $r = .r_1r_2r_3\ldots$ ($r_i = 0,1,2,\ldots,9$), except those r for which $r_i = 0$ for all $i \geq i_0$, for some i_0 or for which $r_i = 9$ for all $i \geq i_0'$ for some i_0'. For example:

$$\begin{aligned} 1/2 = .500000\ldots &= .499999\ldots\ , \quad \text{i.e. } r_i = 0 \text{ or } 9 \text{ for } i \geq 2\ ; \\ 1/4 = .250000\ldots &= .249999\ldots\ , \quad \text{i.e. } r_i = 0 \text{ or } 9 \text{ for } i \geq 3\ ; \\ 1/125 = .008000\ldots &= .007999\ldots\ , \quad \text{i.e. } r_i = 0 \text{ or } 9 \text{ for } i \geq 4\ ; \end{aligned}$$

etc. Such numbers are a subset of the rational numbers and therefore countable. Note further that

- the numbers in the interval $[0, 1/10]$ will have an expansion of the type $r = .0r_2r_3\ldots$, beginning with a 0. For example, $1/10 = .09999\ldots (= .10000\ldots)$, $1/11 = .09090\ldots$, $1/12 = .08333\ldots$, etc. ,
- the numbers in the interval $[1/10, 2/10]$ have an expansion of the type $r = .1r_2r_3\ldots$, beginning with a 1. For example $1/10 = .1000\ldots$ (this is in $[0, 1/10]$ as well as in $[1/10, 2/10]$), $1/5 = .1999\ldots$ $(= .2000\ldots)$ etc. ,

and so on. Finally ,

- the numbers in the interval $[9/10, 1]$ have an expansion of the type $r = .9r_2r_3\ldots$, beginning with a 9.

Next we divide each of $[0, 1/10]$, $[1/10, 2/10]$, $\ldots$, $[9/10, 1]$ into ten equal parts again. Take the interval $[0, 1/10]$, which gets divided into $[0, 1/100]$, $[1/100, 2/100]$, $\ldots$, $[9/100, 10/100]$. Here the numbers in the first part have a 0 in the second place of their decimal expansion, those in the second part have a 1 in the second place and so on; and similarly for the third place, ...

Keeping this in mind, we now work in ternary expansions, so that we have just three digits $0, 1, 2$. Here each number in $[0, 1/3]$ will begin its ternary expansion with a 0, while numbers in $[1/3, 2/3]$ begin with a 1, and numbers in $[2/3, 1]$ with a 2. Note that $1/3$ and $2/3$ both fall into two categories.

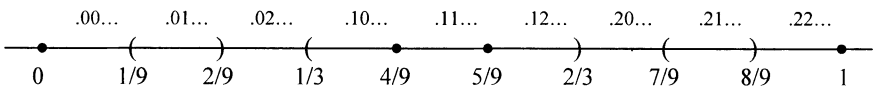


Figure 1.7

Next when we divide $[0, 1/3]$ into three equal parts, the numbers in $[0, 1/9]$ will have a 0 in the second place, those in $[1/9, 2/9]$ a 1 in the second place and those in $[2/9, 1/3]$ a 2 in the second place and so on; and so on for the third, fourth, ..., place. The full chart, up to nine parts, is as in Figure 1.7. Looking at the above chart, we easily get the following:

Theorem 1.4.6. *In the construction of the Cantor set $\mathcal{C}$, the removed intervals consist of those numbers of $[0,1]$ that have a 1 somewhere in their ternary expansion, i.e., $\mathcal{C}$ is the set of those real numbers in $[0,1]$ that can be written with only 0's and 2's in their ternary expansion.*

We now have the following bijective mappings:

$$\begin{aligned}
\mathcal{C} &\leftrightarrow \{r \in [0,1] \mid r = .r_1 r_2 \ldots \ r_i = 0 \text{ or } 2\} \\
&\leftrightarrow \{s \in [0,1] \mid s = .s_1 s_2 \ldots \ s_i = 0 \text{ or } 1\} && \text{(replacing 2's by 1's)} \\
&\leftrightarrow \{(s_1, s_2, \ldots) \mid s_i = 0 \text{ or } 1\} = S && \text{(in the notation of Theorem 1.4.4)} \\
&\leftrightarrow \text{all real numbers in } [0,1] \text{ except countably many.} && \text{(by Theorem 1.4.4)}
\end{aligned}$$

Hence we get the following:

Theorem 1.4.7. *The Cantor ternary set $\mathcal{C}$ is uncountable.*

This would seem even more surprising when we note that the total of the lengths of the intervals removed in the construction of $\mathcal{C}$ is equal to 1. Indeed, the removed lengths equal $1/3 + 2 \cdot 1/9 + 4 \cdot 1/27 + \cdots = (1/3) \cdot (1 + (2/3) + (2/3)^2 + \cdots) = (1/3)(1/(1 - 2/3)) = 1/(3 - 2) = 1$. So it would seem that the set of leftover points (i.e., the set $\mathcal{C}$) is rather small (as far as set inclusions go or in the language of measure theory, not discussed here, $\mathcal{C}$ is of "measure zero") compared to the set of all real numbers in $[0,1]$, but is not as small as it appears to be, at first sight, as Theorem 1.4.7 shows.

Remark 1.4.8. Since the end points of the removed open intervals form a countable set, it follows that $\mathcal{C}$ contains points other than these end points. In ternary expansion, these end points of the removed open intervals are $.0222\ldots$, $.2000\ldots$, $.00222\ldots$, $.02000\ldots$, $.2022\ldots$, $.22000\ldots$, $\ldots$, i.e., with either all 0's or all 2's after a stage. An example of a point of $\mathcal{C}$, which is not amongst the countably many listed above is $.20202020\ldots$.

Although $\mathcal{C}$ is very thinly scattered over $[0,1]$, the following even more surprising result holds:

Theorem 1.4.9. *The set of distances between points of $\mathcal{C}$ fills the unit interval, i.e., given $c \in [0,1]$, there exist $x, y \in \mathcal{C}$ such that $y - x = c$.*

This theorem was first proved using geometric methods and published in the Polish journal *Wektor* in 1917, by H.Steinhauss. In 1940, Randolph gave a proof that appeared in [90]. There is plenty of literature on this subject. Here we give a very neat proof of a slightly more general result, which is due to W.R.Utz [112].

Theorem 1.4.10. *Let $c \in [0,1]$ and let $1/3 \leq\mid m \mid\leq 3$. Then there exist $x, y \in \mathcal{C}$ such that $y - mx = c$.*

Taking $m = 1$, we get Theorem 1.4.9.

Proof. Consider the Cantor set $\mathcal{C}$ on the x-axis as well as the y-axis. Let Σ consists of all points (x, y) of the unit square with corners $(0,0)$, $(0,1)$, $(1,1)$,

$(1,0)$ such that x belongs to the Cantor subset $\mathcal{C}$ of $[0,1]$ on the x-axis and y belongs to the Cantor subset $\mathcal{C}$ of $[0,1]$ on the y-axis. The result will follow if we can show that for c, m, as above, the line $y - mx = c$ meets Σ in at least one point (x, y).

We divide the unit square $[0,1] \times [0,1] = S$ into three equal vertical strips, by inserting two vertical line segments in S. Delete from S the interior of the middle strip as well as its top and bottom open segments (see Figure 1.8).

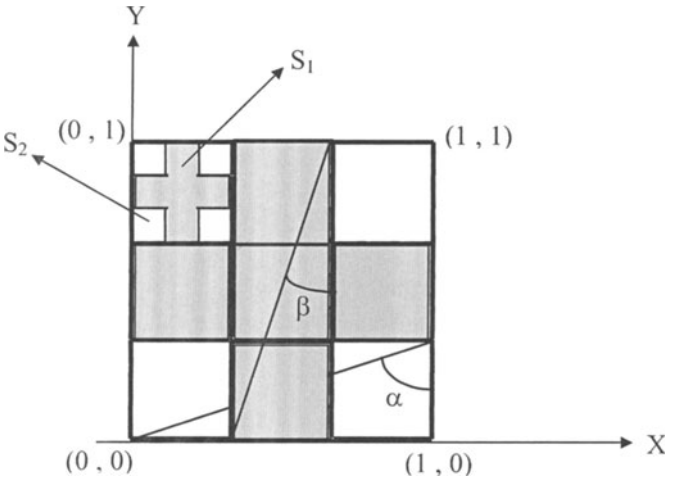


Figure 1.8

Next delete the interior of a similar horizontal middle third strip so that there remains four closed corner squares, each of area $1/9$ and separated by a region shaped like a cross. It is easily observed that:

- In case the general line $\lambda : y = mx + c$ meets S and if $1/3 \leq| m |\leq 3$, then λ meets at least one of the four smaller squares.

- • Indeed, the above observation is true of any square besides S, whose sides are parallel to the two axes.

Now start with the unit square S and let $1/3 \leq| m |\leq 3$. If $\lambda : y = mx + c$ meets S, then by the first observation above, λ meets at least one of the four corner squares that remain when the cross is removed from S. Call that square S_1. Since λ meets S_1, we may repeat the above process (using the second observation) to get S_2 (see Figure 1.8) such that λ meets S_2, and so on. This leads to a shrinking sequence of *compact* squares: $S = S_0 \supset S_1 \supset S_2 \supset \cdots$; their diameters tending to zero, such that $\lambda \cap S_i \neq \emptyset$. By Cantor intersection theorem (see [103]), the intersection of all S_i's is nonempty. But the intersection of all S_i's is contained in Σ, which completes the proof of the theorem.

Taking $m = -1$ in the above result yields:

Corollary 1.4.11. *For $c \in [0,1]$, there exist $x, y \in \mathcal{C}$ such that $x + y = c$.*

Taking $m = -1$ and replacing c by $2c$ (so that the line λ becomes $y = -x + 2c$, and it still meets S for all $c \in (0,1)$), we get:

Corollary 1.4.12. *If $c \in (0,1)$, then c lies midway between two points x, y of $\mathcal{C}$.*

Example 1.4.13. *(A perfect nowhere dense set).* A nowhere dense set is a set whose closure has no interior point. The above discussion shows that the Cantor ternary set $\mathcal{C}$, constructed above is an example of a perfect nowhere dense subset of $\mathbb{R}$.

Exercises

1.1. Let $E \subseteq \mathbb{R}$ and $\delta > 0$. For $x \in E$, define

$$E(x, \delta) = E \cap (x - \delta, x + \delta).$$

(i) If for some δ, $E(x, \delta) = \{x\}$, then x is said to be an isolated point of E.
(ii) If for some δ, $E(x, \delta)$ is an infinite set, but countable, then x is said to be a sparse point of E.
(iii) If for all δ, $E(x, \delta)$ is uncountable, then x is said to be a dense point of E.
By considering intervals with rational end points, prove that the set of isolated points of E and the set of sparse points of E are both countable.

1.2. Let E be any subset of $\mathbb{R}$. Prove that the following statements are equivalent:
(i) Any bounded monotone sequence of points in E converge to a point of E.
(ii) For any $A \subseteq E$ ($A \neq \emptyset$, A bounded), $\sup A$ and $\inf A$ both belongs to E.
(iii) Every sequence $\{x_n\}$ of points of E satisfying

$$\lim_{N \to \infty} \{\sup\{|\, x_n - x_m \,|\, | \, n, m \geq N\}\} = 0$$

converge to a point in E.

1.3. Find a sequence $\{a_n\}$ of real numbers such that $a_n \to 0$ as $n \to \infty$, but na_n converge to a transcendental number.

Chapter 2

Functions: pathological, peculiar and extraordinary

§2.1. Some pathological functions

One of the first standard examples of a real-valued function that is discontinuous everywhere is

$$f(x) = \begin{cases} 1\,, & \text{if } x \in \mathbb{Q} \\ -1\,, & \text{if } x \notin \mathbb{Q} \end{cases}$$

Example 2.1.1. The function defined above is discontinuous everywhere but its absolute value is continuous everywhere.

A slightly more difficult job is to construct an example of a function that is continuous at only one point (e.g. at $x = 0$) but discontinuous everywhere else. The following function is one such example:

$$f(x) = \begin{cases} x\,, & \text{if } x \in \mathbb{Q} \\ -x\,, & \text{if } x \notin \mathbb{Q}\,. \end{cases}$$

Now if one is asked to construct a function, defined in say $[0,1]$, that is *continuous at each irrational but discontinuous at each rational*, it would require quite some thought. The function

$$f(x) = \begin{cases} 1/q\,, & \text{if } x = p/q \in \mathbb{Q}\,,\ x \neq 0,\ (p,q) = 1\,, \\ 1\,, & \text{if } x = 0 \\ 0\,, & \text{if } x \notin \mathbb{Q}\ , \end{cases}$$

where (p,q) denotes the g.c.d. of p and q, serves the purpose. The proof goes as follows:

It is immediately clear that f is discontinuous at each rational, for whatever a rational $x_0 = p/q$, $(p,q) = 1$, may be, there exist irrational numbers α as close to $x_0$as we like and then given $\epsilon > 0$, $\mid f(x_0) - f(\alpha) \mid = \mid 1/q - 0 \mid = 1/q$, which is not less than any given ϵ for $\mid x_0 - \alpha \mid$ less than any δ if $\epsilon \leq 1/q$. Now to prove that f is continuous at each irrational $\alpha \in [0,1]$, we proceed as

follows: Let $\epsilon > 0$ be given. Define a set S of rational numbers by putting the rational p/q in S if and only if $q \leq 1/\epsilon$ and $p/q \in (\alpha - 1, \alpha + 1)$, i.e.,

$$S = \{p/q \in (\alpha - 1, \alpha + 1) \mid q = 1, 2, \ldots, [1/\epsilon]\}.$$

Then S is obviously a finite set, as q takes finitely many values and so does p. Consequently, we may choose a δ-neighbourhood of α which is free of elements of S (see Figure 2.1), i.e., there exists a δ such that $(\alpha - \delta, \alpha + \delta) \cap S = \emptyset$. Thus, given $\epsilon > 0$ we arrived at a δ and we now show that $|f(x) - f(\alpha)| < \epsilon$ if $|x - \alpha| < \delta$.

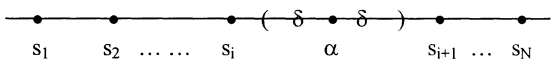


Figure 2.1

So let $x \in (\alpha - \delta, \alpha + \delta)$. If x is irrational, then $|f(x) - f(\alpha)| = 0 < \epsilon$. If x is rational, then x, not being in S, has denominator larger than $1/\epsilon$, i.e., $x = p/q$, $(p, q) = 1$, $q > 1/\epsilon$, so $|f(x) - f(\alpha)| = |1/q - 0| = 1/q < \epsilon$. Thus, for all $x \in (\alpha - \delta, \alpha + \delta)$, rational or irrational, we have $|f(x) - f(\alpha)| < \epsilon$, which means that $f(x)$ is continuous at α.

Remark 2.1.2. The above proof, as well as a proof using sequences (rather than neighbourhoods, as in the above) are both given by J.H. Staib, in [105].

Remark 2.1.3. It is instructive to draw a rough graph of this function. First note that the function can be defined on whole of $\mathbb{R}$ by the same definition and that the function is periodic with period 1 in $(0, \infty)$ and also in $(-\infty, 0)$, for if x is irrational, so is $x + 1$ and so $f(x) = f(x + 1) = 0$, while if $x = p/q$, gcd $(p, q) = 1$, then $x + 1 = (p + q)/q$, gcd $(p + q, q) = 1$, and again $f(x + 1) = 1/q = f(x)$ (here p may have any sign) and of course $f(n) = 1$ if $n \in \mathbb{N}$. Further, $f(x) = f(-x)$, i.e., f is symmetrical about the y-axis, so it is enough to draw the graph of f in $[0, 1]$. A rough sketch is given in Figure 2.2.

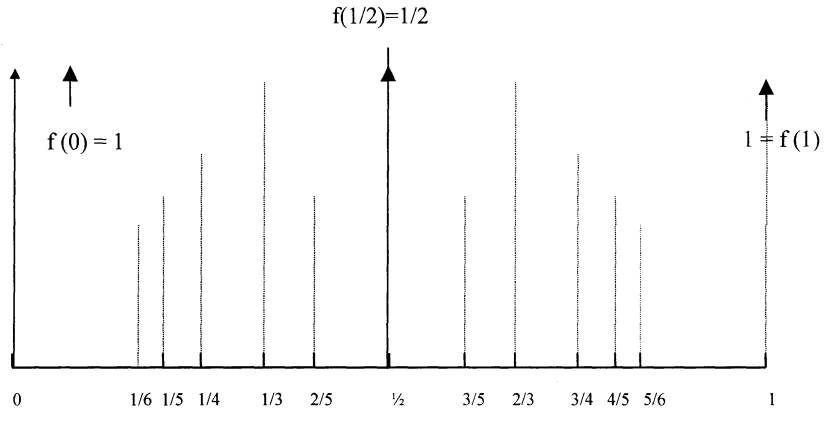


Figure 2.2

Remark 2.1.4. Classically, the function f given above has been defined to be 0 at $x = 0$. This was because if we take a sequence p_n/q_n of rationals tending to 0, then q_n must tend to infinity and as such $f(p_n/q_n) = 1/q_n \to 0$ hence it is natural to define $f(0) = 0$ so that f becomes continuous at $x = 0$. However, 0 is a rational number and we want f to be discontinuous at $x = 0$, so it is even more natural to put $f(0) = 1$, say.

Example 2.1.5 The function defined above also serves as an example of a function which is continuous at uncountably many points (i.e., all irrational numbers) but is nowhere differentiable. To this end, it is clearly enough to look for points of differentiability in $0 \leq x \leq 1$. Since f is not even continuous at nonzero rationals, it is enough to look at the irrationals in $[0, 1]$ and the point 0. We show that f is not differentiable at any of these points. First let $x = 0$. Let $\{h_i\}$ be a sequence of irrationals with limit equal to 0, (e.g.$\sqrt{2} - 1.4, \sqrt{2} - 1.41, \sqrt{2} - 1.414, \ldots$). Then $\lim(f(0 + h_i) - f(0)/h_i) = \lim(-1/h_i)$, which does not exist, and so $f(x)$ is not differentiable at $x = 0$.

Let now x be an irrational in $[0, 1]$, so that $0 < x < 1$. First choose a sequence $\{h_i\}$ of reals having limit 0 and such that $x + h_i$ is irrational for each i (e.g. $h_i = 1/i$). Then $\lim(f(x + h_i) - f(x))/h_i = \lim(0/h_i) = 0$.

Next suppose that x has decimal representation $.a_1a_2 \ldots a_n \ldots$ and choose $h_i = .a_1a_2 \ldots a_i - x$ (which is an irrational). Since $x \neq 0$ we have $a_i \neq 0$ for some i and indeed for i as large as we like, for otherwise, x would be a rational. Let N be the least (first) integer such that $a_N \neq 0$, i.e.,

$$x = .\overbrace{000\ldots00}^{N-1 \ zeros} a_N a_{N+1} \ldots .$$

Then $f(x + h_i) = f(.a_1a_2 \ldots a_i) \geq 1/10^i$ if $i \geq N$, for $.a_1a_2 \ldots a_i = a_1/10 + a_2/10^2 + \ldots + a_i/10^i = (a_1 10^{i-1} + \cdots + a_i)/10^i$ and so $f(.a_1a_2 \ldots a_i) \geq 1/10^i$ (note that any cancellation in this fraction (only 2 or 5 can be cancelled) will reduce the denominator and so increase the value of f), while $\mid h_i \mid \leq 1/10^i$ for $\mid h_i \mid = .00 \ldots 0a_{i+1} \ldots = 0/10 + 0/10^2 + \cdots + 0/10^i + a_{i+1}/10^{i+1} + \cdots \leq 1/10^i$. It follows that $(f(x + h_i) - f(x))/h_i \geq 1$ if $i \geq N$ and so f is not differentiable at x, by the observation in the last paragraph .

This example qualifies as pathological, at least to a beginner and clearly reveals the power of careful analytic reasoning.

Example 2.1.6. The above example thus gives us a function f defined on the entire real line $\mathbb{R}$ which is continuous at each irrational point of $\mathbb{R}$ but discontinuous at each rational point. Note that the set of discontinuities for f is a dense set and that each of the points of discontinuities is removable. Because if the function is redefined at a to have the value 0, (also for $a = 0$) then since $\lim_{x \to a} f(x) = 0 = f(a)$, as redefined, f becomes continuous at a. Another observation about the function f is that it gives the example of a function which is everywhere finite but everywhere locally unbounded. For, if f were bounded in an ϵ-neighbourhood N of a, then for all m/n in N, the denominator n would be bounded, and hence the numerator m would be too.

But this would permit only finitely many rational numbers in the interval N, which is not true.

Since a function, continuous at all irrational numbers but discontinuous at all rational numbers exist, it is tempting to believe that a function that is continuous at each rational but discontinuous at each irrational also exists. However the surprise here is that *no such function can exist !* A beautiful elementary proof of this curious result was given in 1881 by Vito Volterra. We have:

Theorem 2.1.7 *Let (a,b) be any open interval (possibly $(-\infty,\infty)$). There exists no function $g\colon (a,b) \to \mathbb{R}$ that is continuous at each rational number but discontinuous at each irrational number.*

Proof (Volterra). Suppose to the contrary that such a function g exists, defined on an open interval (a,b). Let f be the function constructed above, that is continuous at the irrational numbers but discontinuous at the rational numbers. Let x_0 be a rational in (a,b) and take $\alpha = 1$. By the continuity of g at x_0, there exists a $\delta > 0$ such that

$$(x_0 - \delta, x_0 + \delta) \subset (a,b)\,, \mid g(x) - g(x_0) \mid < 1/2 \quad \text{for all } x \in (x_0 - \delta, x_0 + \delta)\,.$$

Choose $c < d$ so that $[c,d]$ is a closed subinterval of $(x_0 - \delta, x_0 + \delta)$. Then for any $x, y \in [c,d]$,

$$\mid g(x) - g(y) \mid \leq \mid g(x) - g(x_0) \mid + \mid g(x_0) - g(y) \mid < 1/2 + 1/2 = 1 = \alpha\,.$$

Now, take any irrational ξ in (c,d), so that f is continuous at ξ and by the preceding argument, there exist $a_1 < b_1$, with $[a_1,b_1] \subset (c,d)$ such that $\mid f(x) - f(y) \mid < 1$ for all $x, y \in [a_1,b_1]$. To summarize then, for all $x, y \in [a_1,b_1]$, (note the inclusions $[a_1,b_1] \subset (c,d) \subset (x_0 - \delta, x_0 + \delta) \subset (a,b)$), we have $\mid g(x) - g(y) \mid < 1$ and $\mid f(x) - f(y) \mid < 1$ (remember $\alpha = 1$).

We next repeat this argument, starting with the open interval (a_1,b_1) and taking $\alpha = 1/2$, then again $\alpha = 1/4$, ... , and generally $\alpha = 1/2^n$. This generates a strictly decreasing sequence of closed intervals (except for (a,b)):

$$(a,b) \supset [a_1,b_1] \supset [a_2,b_2] \supset \ldots \supset [a_n,b_n] \supset \ldots$$

such that for all $x, y \in [a_n,b_n]$, we have $\mid g(x) - g(y) \mid < 1/2^n$ and $\mid f(x) - f(y) \mid < 1/2^n$. The denseness property of the rationals and irrationals plays a central role at each step, providing at least one point of continuity for g and f alternatively in any open subinterval. By the nested interval property, there exists at least one point p contained in all the $[a_i,b_i]$ and thus both g and f are continuous at p, giving a contradiction, since g is continuous where f is not or vice versa.

An immediate corollary to this is the following:

Theorem 2.1.8. *There can exist no continuous function $\varphi : \mathbb{R} \to \mathbb{R}$, mapping the rational numbers to irrationals and the irrational numbers to rationals.*

Proof. Consider the function f which is continuous at each irrational but discontinuous at each rational. Suppose, to the contrary, that a continuous function $\varphi\colon \mathbb{R} \to \mathbb{R}$ exists that maps rationals to the irrationals and the irrationals to the rationals. Let $h(x) = f(\varphi(x))$. Then for $x \in \mathbb{Q}$, $\varphi(x)$ is irrational and so f is continuous at $\varphi(x)$ and φ is continuous everywhere by hypothesis; hence we observe that h is continuous at each $x \in \mathbb{Q}$.

For an irrational number y, let $\{x_n\}$ be a sequence of rationals that tends to y. Then $\lim_{n\to\infty} h(x_n) = \lim_{n\to\infty} f(\varphi(x_n)) = 0$, since $x_n \in \mathbb{Q}$ implies that $\varphi(x_n)$ is irrational and hence $f(\varphi(x_n)) = 0$. However, $h(y) = f(\varphi(y)) \neq 0$, since y is irrational and so $\varphi(y)$ is rational. Hence h is discontinuous at each irrational number. These observations contradict Theorem 2.1.7, thus completing the proof.

For some historical remarks on the difficulties Volterra encountered see [31].

Remark 2.1.9. A direct proof of Theorem 2.1.8, using cardinality arguments goes as follows: Suppose to the contrary that an everywhere continuous function $\varphi\colon \mathbb{R} \to \mathbb{R}$ exists with $\varphi(\mathbb{Q}) \subseteq \mathbb{I}$ and $\varphi(\mathbb{I}) \subseteq \mathbb{Q}$. It follows that φ takes at least two values, one rational and one irrational, say ρ and σ. By the intermediate value theorem, since φ is continuous, φ assumes all values in $[\rho, \sigma]$, where without loss of generality, $\rho < \sigma$, and since $\rho \neq \sigma$, we get that the set $[\rho, \sigma]$ is uncountable. On the other hand, the set $\{\varphi(\mathbb{Q})\}$ is countable, since $\mathbb{Q}$ is countable. Further, since $\varphi(\mathbb{I}) \subset \mathbb{Q}$, $\varphi(\mathbb{I})$ is countable and consequently $\varphi(\mathbb{R})$ is countable. This contradiction shows that φ can not exist.

Remark 2.1.10. *This paragraph requires some knowledge of measure theory.* We have constructed a function f that is continuous at the irrationals and discontinuous at the rationals, i.e., points of continuity of f constitute a set of measure 1 while points of discontinuity of f form a set of measure 0. After Volterra's theorem, it may be rather surprising to note that there do exist functions $h(x)$ with these characteristics reversed, i.e., points of continuity of h constitute a set of measure 0 while points of discontinuity of h form a set of measure 1. The interested reader may look up [110].

§2.2. The Cantor function

Recall that in §1.4 we constructed the Cantor ternary set $\mathcal{C}$. We now give certain labels to the intervals left out after each stage, as follows:

First step : On deleting the open interval $(1/3, 2/3)$ from $I = [0, 1]$, we are left with the two intervals $I_1 = [0, 1/3]$ and $I_2 = [2/3, 1]$, each of length $1/3$.

Second step : On deleting the open middle thirds $(1/9, 2/9)$ from I_1 and $(7/9, 8/9)$ from I_2, we are left with $2^2 = 4$ intervals $I_{11} = [0, 1/9]$, $I_{12} = [2/9, 3/9]$, $I_{21} = [6/9, 7/9]$, $I_{22} = [8/9, 9/9]$, each of length $(1/3)^2$,

and so on, i.e., by induction, we complete the construction as follows:

Induction step : Suppose the k^{th} step is complete ($k \in \mathbb{N}$), i.e., on removal of the open middle thirds of 2^{k-1} intervals, there remain 2^k closed intervals, each of length $(1/3)^k$. Then the $(k+1)^{\text{th}}$ step consists in removing the open middle thirds of these 2^k intervals to be left with 2^{k+1} closed intervals, each of length $(1/3)^{k+1}$.

We continue this process indefinitely and define $\mathcal{C}$ to be the points of I that do not get removed. Example of such points are the end points of each removed interval viz:

$$\{1/3, 2/3\,;\, 1/9, 2/9, 7/9, 8/9\,;\, 1/27, 2/27, \ldots\,;\, \ldots\} \qquad (*)$$

which form only a countable subset of $\mathcal{C}$. More precisely, if $F_1 = [0,1]$, $F_2 = [0, \frac{1}{3}] \cup [\frac{2}{3}, 1]$, ..., $F_n = [0, \frac{1}{3^{n-1}}] \cup [\frac{2}{3^{n-1}}, \frac{3}{3^{n-1}}] \cup \ldots \cup [\frac{3^{n-1}-1}{3^{n-1}}, 1]$, then $\mathcal{C} = \cap_{n=1}^{\infty} F_n$.

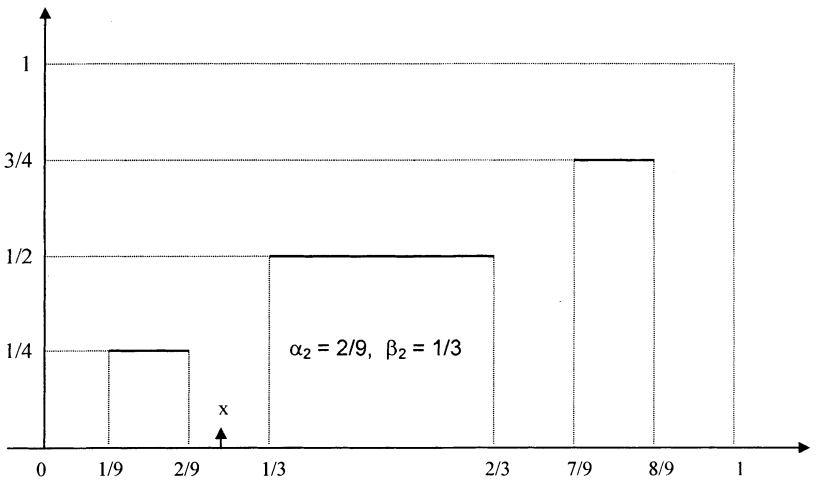


Figure 2.3

We are now in a position to define the *Cantor function* $\Phi : [0,1] \to [0,1]$. Define $\Phi(0) = 0, \Phi(1) = 1$, and $\Phi(x) = 1/2$ if $1/3 \leq x \leq 2/3$. This defines Φ at the end points of I_1, I_2 and on the deleted open interval $(1/3, 2/3)$. Next, define $\Phi(x) = 1/4$ if $1/9 \leq x \leq 2/9$ and $\Phi(x) = 3/4$ if $7/9 \leq x \leq 8/9$ (see Figure 2.3). We repeat the process indefinitely, thereby defining Φ on each deleted interval and its end points and at $0, 1$, i.e., Φ is defined on each deleted open interval and at countable set (*) defined above and at 0 and 1.

Let now x be a point at which Φ is not defined (i.e., let $x \in \mathcal{C}$). Then for each $k \in \mathbb{N}$, x belongs to the interior of exactly one of the 2^k left over intervals of length $(1/3)^k$ after the k^{th} step of our construction of $\mathcal{C}$. Call this interval $[\alpha_k, \beta_k]$. Being of length $(1/3)^k$, we have $\beta_k = \alpha_k + (1/3)^k$. Further, Φ has been defined at the end points of these left over intervals, viz. , $\Phi(\beta_k) =$

$\Phi(\alpha_k) + (1/2^k)$. This should best become clear by an example (see Figure 2.3): $\Phi(\alpha_2) = 1/4$, $\Phi(\beta_2) = 1/2$, and $\Phi(\beta_2) = \Phi(\alpha_2) + (1/2)^2$, or $1/2 = 1/4 + 1/4$, as required. Now when the $(k+1)^{th}$ deletion takes place, $[\alpha_k, \beta_k]$ is subdivided into three equal parts and the middle third is removed. Then, x belongs to one of the two leftovers $[\alpha_{k+1}, \beta_{k+1}]$, say.

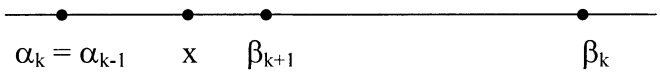


Figure 2.4

Hence:

$$\Phi(\alpha_k) \leq \Phi(\alpha_{k+1}) < \Phi(\beta_{k+1}) \leq \Phi(\beta_k) \qquad (\dagger)$$

We now define Φ at x by :

$$\Phi(x) = \lim_{k\to\infty} \Phi(\alpha_k) = \lim_{k\to\infty} \Phi(\beta_k) \qquad (\ddagger)$$

which is natural. Indeed, by (†), $\Phi(\alpha_k)$ is increasing and is bounded above by 1, $\Phi(\beta_k)$ is decreasing and is bounded below by 0, so both sequences $\{\Phi(\alpha_k)\}$ and $\{\Phi(\beta_k)\}$ tend to limits and since $\Phi(\beta_k) = \Phi(\alpha_k) + (1/2^k)$, these limits are equal and this common value is defined to be $\Phi(x)$. This defines Φ on $[0,1]$ as a continuous, nondecreasing function with values in $[0,1]$. Moreover Φ is continuous in $[0,1]$, as it is clearly continuous at each point of the deleted intervals and at there end points too, as a little thought will show. For continuity at points of $\mathcal{C}$, the defining equation (‡) is equivalent to the continuity of Φ at x! We have thus proved (i) and (ii) of the following

Theorem 2.2.1 *The Cantor function* $\Phi\colon [0,1] \to [0,1]$ *with* $\Phi(0) = 0$, $\Phi(1) = 1$, *has the following properties:*

(i) Φ *is continuous,*

(ii) Φ *is monotone increasing (not strictly increasing),*

(iii) Φ *is not differentiable at any point* x *of* $\mathcal{C}$,

(iv) Φ *is differentiable at each point* x *which is not in* $\mathcal{C}$ *and for such* x, $\Phi'(x) = 0$.

Proof. (i) and (ii) have been proved, while (iv) is trivial since $x \notin \mathcal{C}$ implies that x is in a removed interval and hence is constant in a neighbourhood of x. Thus $\Phi'(x)$ exists and is 0. We now proceed to prove (iii).

At each step of deletion of open intervals, a part of the Cantor function gets defined. Let us join the end points of this partially defined function Φ, at each step, by straight lines. Thus, for $k = 1$, we have the graph as in Figure 2.5 and for $k = 2$, the graph is shown in Figure 2.6.

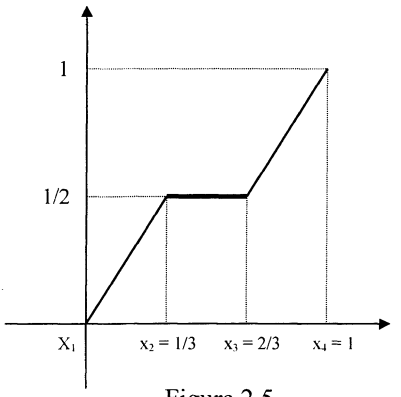


Figure 2.5

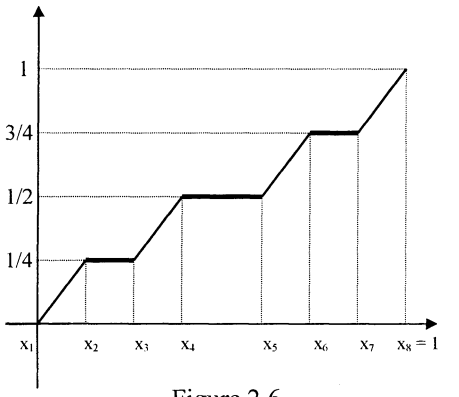


Figure 2.6

Here note that the horizontal graph (i.e., the actual Φ) continues undisturbed, once it appears, whereas the other parts consist of 2^k congruent portions with positive slope. Indeed, for $k = 1$, the slope equals $(1/2)/(1/3) = 3/2$ while for $k = 2$, the slope is $(1/4)/(1/9) = 9/4 = (3/2)^2$, and so on.

Now to prove (iii). Since $x \in \mathcal{C}$, so x is in a closed left over interval $[x_{2i-1}, x_{2i}]$ at each step of deletion. It follows that $\Phi(x_{2i}) - \Phi(x_{2i-1}) = 1/2^k$; but $x_{2i} - x_{2i-1} = 1/3^k$. These give: $\Phi(x_{2i}) - \Phi(x_{2i-1}) = (3/2)^k(x_{2i} - x_{2i-1})$. Now recall the inequality (easy to check): $\max(\alpha/\beta, \gamma/\delta) \geq (\alpha + \gamma)/(\beta + \delta)$, for $\alpha, \gamma \geq 0$; $\beta, \delta > 0$. Using this, we get

$$\max[(\Phi(x_{2i}) - \Phi(x))/(x_{2i} - x), (\Phi(x) - \Phi(x_{2i-1}))/(x - x_{2i-1})] \geq (3/2)^k ,$$

where $x \in (x_{2i-1}, x_{2i})$. So either the left or the right derivative of Φ at x can not exist, i.e., Φ is not differentiable at x.

Example 2.2.2. The above discussion also yields an example of a nowhere dense set $\mathcal{A}$ of real numbers and a continuous mapping of $\mathcal{A}$ onto the closed unit interval $[0, 1]$. Let $\mathcal{A}$ be the Cantor set $\mathcal{C}$ defined previously. Let $x \in \mathcal{C}$, and let $0.c_1c_2c_3\ldots$ be its ternary expansion, where $c_n = 0$ or 2 for $n = 1, 2, \ldots$ and let

$$\phi(x) = 0 . \frac{c_1}{2}\frac{c_2}{2}\frac{c_3}{2}\cdots ,$$

where the expansion on the right is to be regarded as a binary expansion in terms of the digits 0 and 1. Clearly $\phi(\mathcal{C})$ is a subset of $[0, 1]$. To see that $[0, 1] \subseteq \phi(\mathcal{C})$, choose an arbitrary $y \in [0, 1]$ and a binary expansion of y:

$$y = 0.b_1b_2b_3\ldots$$

Then

$$x = 0.(2b_1)(2b_2)(2b_3)\ldots ,$$

evaluated in the ternary system, is a point of $\mathcal{C}$ such that $\phi(x) = y$. Continuity of ϕ is not difficult to establish, using geometric considerations given in the above description of the Cantor function.

§2.3. The length of the Cantor function

The Cantor function Φ has another very astonishing property, viz., the length (from 0 to 1) of its graph is 2. Before we prove this, we need to define what we mean by the length of a curve that is continuous and nondecreasing (i.e., increasing but not strictly increasing). Denote by G_f the graph of the function $f\colon [a,b] \to \mathbb{R}$. The range of f is then $[f(a), f(b)]$. We define the length $\lambda(G_f)$ of G_f by the equation

$$\lambda(G_f) = \sup_{\wp} \left[\sum \left((x_i - x_{i-1})^2 + (f(x_i) - f(x_{i-1})^2\right)^{1/2}\right] ,$$

where $\wp$ is any partition: $a = x_1 < x_2 < \ldots < x_{n-1} < x_n = b$ of $[a,b]$. Since for $\alpha, \beta \in \mathbb{R}$, $(\alpha^2 + \beta^2)^{1/2} \leq |\alpha| + |\beta|$ (with strict inequality when α, β are both non-zero), we have $((x_i - x_{i-1})^2 + (f(x_i) - f(x_{i-1}))^2)^{1/2} < (x_i - x_{i-1}) + (f(x_i) - f(x_{i-1}))$ and so $\sum((x_i - x_{i-1})^2 + (f(x_i) - f(x_{i-1}))^2)^{1/2} \leq b - a + f(b) - f(a)$.

This proves the following

Lemma 2.3.1. *If $\lambda(G_f)$ denotes the length of the graph G_f, then $\lambda(G_f) \leq b - a + f(b) - f(a)$. Moreover, if f is a continuous nondecreasing function defined from $[0,1]$ onto itself, with $f(0) = 0$, $f(1) = 1$, then $\lambda(G_f) \leq 2$.*

Suppose now that f is continuously differentiable (i.e., that the derivative f' is continuous in $[0,1]$). Using the Mean Value Theorem, we get:

Lemma 2.3.2. *Let f be a continuous nondecreasing function defined from $[0,1]$ onto itself, with $f(0) = 0$, $f(1) = 1$. If $f'(x)$ exists in $[0,1]$ and is continuous throughout, then $\lambda(G_f) < 2$.*

Proof. Since f is nondecreasing, $f' \geq 0$. If f' were equal to 0 throughout $[0,1]$, then $f(x) = k$ and $\lambda(G_f) = 1$. Next, $f'(x)$ can not be larger than 1 throughout $[0,1]$, nor it can be less than 1 throughout $[0,1]$ (see Figures 2.7 and 2.8).

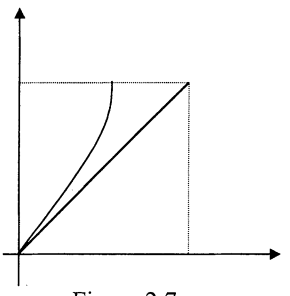


Figure 2.7

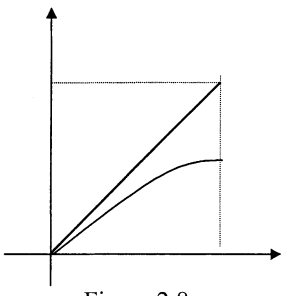


Figure 2.8

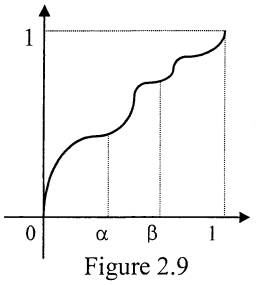


Figure 2.9

So, there exists $\gamma, \delta > 0$ and an interval $[\alpha, \beta] \subset (0,1)$ such that (see Figure 2.9)

$$0 < \gamma \leq f'(t) \leq \delta < 1 \quad (\text{for all } t \in [\alpha, \beta]) \tag{1}$$

Denote by λ_1, λ_2, λ_3, the lengths of the parts of G_f, lying respectively above $[0,\alpha]$, $[\alpha,\beta]$, $[\beta,1]$. Then $\lambda(G_f) = \lambda_1+\lambda_2+\lambda_3$ and the obvious estimates (Lemma 2.3.1) yields:

$$\begin{aligned}
\lambda_1 &\le \alpha - 0 + f(\alpha) - f(\beta) = \alpha + f(\alpha)\,,\\
\lambda_2 &\le \beta - \alpha + f(\beta) - f(\alpha)\,,\\
\lambda_3 &\le 1 - \beta + f(1) - f(\beta) = 2 - \beta - f(\beta)\,,
\end{aligned}$$

which only imply that $\lambda(G_f) \le 2$. However, we shall show that $\lambda_2 \le A((\beta-\alpha)+(f(\beta)-f(\alpha)))$, where $0<A<1$, and that would prove the Lemma.

Now, $\xi \in [\gamma,\delta]$ gives $1+\xi = (1+\xi)^{1/2}(1+\xi)^{1/2} \ge (1+\gamma)^{1/2}(1+\xi^2)^{1/2}$, since $\xi \ge \gamma$, $\xi \ge \xi^2$, i.e.,

$$(1+\xi^2)^{1/2} \le (1+\gamma)^{-1/2}(1+\xi) = A(1+\xi) \tag{2}$$

with $A = (1+\gamma)^{-1/2} < 1$. Hence, when $\alpha \le u < v \le \beta$, there exists a point $\eta \in (u,v)$, such that

$$f(v) - f(u) = f'(\eta)(v-u) \tag{3}$$

and of course $f'(\eta) \in [\gamma,\delta]$ (see (1)). Hence

$$\begin{aligned}
((v-u)^2+(f(v)-f(u))^2)^{1/2} &= ((v-u)^2 + f'^2(\eta)(v-u)^2)^{1/2} && \text{(by (3))}\\
&= (v-u)\{1+f'^2(\eta)\}^{1/2}\\
&\le (v-u)A(1+f'(\eta)) && \text{(by (2))}\\
&= A(v-u)(1+(f(v)-f(u))/(v-u)) && \text{(by (3))}\\
&= A.(v-u+f(v)-f(u))\,, && (4)
\end{aligned}$$

with $A<1$. Now let $\alpha = t_0 < t_1 < \ldots < t_{n-1} < t_n = \beta$ be any partition of $[\alpha,\beta]$. Then

$$\begin{aligned}
&\sum_{i=1}^{n} \left((t_i - t_{i-1})^2 + (f(t_i) - f(t_{i-1}))^2\right)^{1/2}\\
&\le \sum_{i=1}^{n} A\,(t_i - t_{i-1} + f(t_i) - f(t_{i-1})) && \text{(with } A<1\text{, by (4))}\\
&= A((\beta-\alpha)+(f(\beta)-f(\alpha))) && \text{(with } A<1\text{)}\\
&\le A(\beta-\alpha)+(f(\beta)-f(\alpha)) && \text{(as } A<1\text{).}
\end{aligned}$$

Taking supremum, it follows that $\lambda_2 \le A(\beta-\alpha)+(f(\beta)-f(\alpha))$ and hence $\lambda_2 < (\beta-\alpha)+(f(\beta)-f(\alpha))$, as $0<A<1$. It now follows that $\lambda_1+\lambda_2+\lambda_3 < 2$, as required.

Remark 2.3.3. It is not straightforward to produce an example of an increasing or even a nondecreasing continuous function f on $[0,1]$ (or indeed any

increasing function f) with $\lambda(G_f) = 2$. By Lemma 2.3.2, such a function, if constructed, can not be continuously differentiable. The following example, of a continuously differentiable function, goes part of the way towards this. Let $f_n(x) = x^n$, $0 \leq x \leq 1$. The graphs of these f_n's are shown in Figure 2.10. As n increases, the graph flattens out to nearing a horizontal line, almost up to 1.

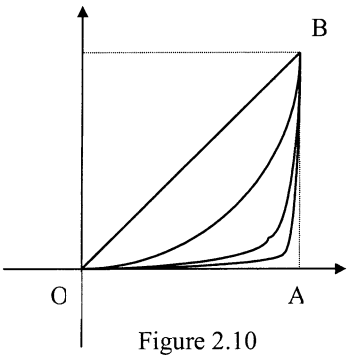


Figure 2.10

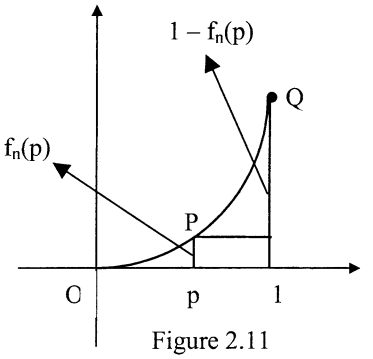


Figure 2.11

It is clear from the figure that $\lambda(G(f_n))$ increases with n and gradually approaches the length $OA + OB = 2$, but it really never makes it to $OA + OB$ as $f_n \to f$, where $f(x) = 0$ if $0 \leq x < 1$ and $f(1) = 1$, and so the best we can say is that given $\epsilon > 0$, there exists a natural number N such that $\lambda(G_{f_n}) > 2 - \epsilon$ if $n \geq N$. As $\epsilon \to 0$, $N \to \infty$ and it follows that

$$\lim_{n\to\infty} \lambda(G_{f_n}) \geq 2 \tag{6}$$

However, $\lim \lambda(G_{f_n}) \neq \lambda(\lim G_{f_n})$, for the left hand side is greater than or equal to 2 (by (6)), whereas, the limit of G_{f_n} is the graph of f, which is discontinuous at 1 and so $\lambda(\lim G_{f_n}) = 1$. To improve on (6), we proceed as follows (see Figure 2.11).

Let $0 < p < 1$. Then clearly $\lambda(G_{f_n}) = \text{Arc}\, OP + \text{Arc}\, PQ > p + 1 - f_n(p)$, and letting $n \to \infty$, this gives that $\lim \lambda(G_{f_n}) \geq 1 + p$, provided $p < 1$. Thus

$$L = \lim_{n\to\infty} \lambda(G_{f_n}) \geq p + 1 \text{ for all } 0 < p < 1 \tag{7}$$

We may now give the following:

Corollary 2.3.4. *Let* $f_n(x) = x^n$ *in* $[0,1]$. *Then* $\lim_{n\to\infty} \lambda(G_{f_n}) = 2$.

Proof. Let $L = \lim \lambda(G_{f_N})$. If $L < 2$, say $L = 2 - \delta$ ($\delta > 0$), it would follow, by (7) above that $2 - \delta \geq p + 1$ for all $0 < p < 1$, i.e., that $p \leq 1 - \delta$ for all $0 < p < 1$, which is false if p is chosen to be larger than $1 - \delta$.

Example 2.3.5. A more artificial example of a continuous, nondecreasing function $f_\epsilon(x)$ on $[0,1]$ with $\lambda(G_{f_\epsilon}) > 2 - \epsilon$ is the one shown in Figure 2.12; or indeed, a continuous strictly increasing function, with this property, is shown in Figure 2.13 below.

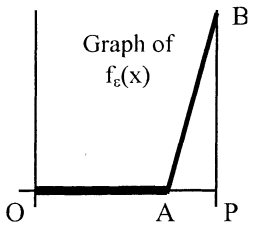


Figure 2.12

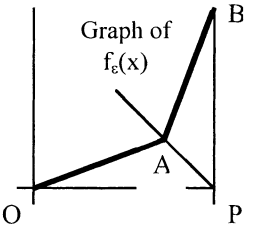


Figure 2.13

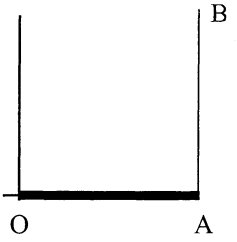


Figure 2.14

In Figure 2.12, $\lambda(G_{f_\epsilon}) = OA + AB = 1 - \epsilon + AB > 1 - \epsilon + BP = 2 - \epsilon$, as required.

However, as $\epsilon \to 0$, $f_\epsilon(x) \to f(x) = 0$ in $[0,1)$; 1 at $x = 1$ (see Figure 2.14), and the length of this function is 1.

Similar conditions hold in Figure 2.13, and indeed, if we want a continuous and differentiable, strictly increasing function $f_\epsilon(x)$, we may round off the corners in Figure 2.13 and get what is required.

We are now in a position to prove the following:

Theorem 2.3.6. *For the Cantor function $\Phi(x)$ defined on the closed interval $[0,1]$, the length $\lambda(G_\Phi)$ of its graph G_Φ is equal to 2.*

Proof. In view of Lemma 2.3.1, it is enough to prove that $\lambda(G_\Phi) \geq 2$. To this end, choose a positive integer k and compute the approximation of the length for the partition

$$\wp_k \,:\, 0 = x_1 < x_2 < \ldots < x_{2^{k+1}} = 1\,,$$

consisting of end points x_i of the 2^k intervals that remain after the k^{th} step of our construction of $\mathcal{C}$. Thus, for example, for $k = 1$, $\wp_1$ is given by

$$\wp_1 \,:\, 0 = x_1 < x_2 = \frac{1}{3} < x_3 = \frac{2}{3} < x_4 = 1$$

and for $k = 2$, $\wp_2$ is given by (see Figure 2.6)

$$\wp_2 : 0 = x_1 < x_2 = \frac{1}{9} < x_3 = \frac{2}{9} < x_4 = \frac{1}{3} < x_5 = \frac{2}{3} < x_6 = \frac{7}{9} < x_7 = \frac{8}{9} < x_8 = 1.$$

Call the corresponding sum approximating $\lambda(G_\Phi)$ to be L_k. Then

$$L_k = \sum_{i=2}^{2^{k+1}} \left((x_i - x_{i-1})^2 + (\Phi(x_i) - \Phi(x_{i-1}))^2\right)^{1/2}.$$

Here L_k is the length of the approximating polygon. We now have to calculate the value of $x_i - x_{i-1}$ and of $\Phi(x_i) - \Phi(x_{i-1})$ for all i and each k. Looking at

the cases $k = 1, 2$ (Figures 2.5 and 2.6), we see that the contribution of the horizontal portions in L_k equals

$$((2/3 - 1/3)^2 + 0)^{1/2} + ((2/9 - 1/9)^2 + 0)^{1/2} + ((8/9 - 7/9)^2 + 0)^{1/2} + \cdots$$

Observe that the first term is equal to L_1, sum of the first three terms is equal to L_2, and so on. Thus, the contribution is

$$\begin{aligned} 1/3 + 1/9 + 1/9 + \cdots &= 1/3 + 2/9 + 4/27 + \cdots + 2^{k-1}/3^k \\ &= (1/3)(1 + (2/3) + (2/3)^2 + \cdots + (2/3)^{k-1}) \\ &= 1 - (2/3)^k. \end{aligned}$$

The other portion of the graph consists of 2^k congruent portions with slope $3/2$ for $k = 1$, $(3/2)^2$ for $k = 2$, $(3/2)^3$ for $k = 3$ and so on. Then the contribution to L_k from this portion equals $2^k((1/3^k)^2 + (1/2^k)^2)^{1/2} = (2^{2k}(1/3^{2k} + 1/2^{2k}))^{1/2} = ((2/3)^{2k} + 1)^{1/2}$. So now $L_k = 1 - (2/3)^k + (1 + (2/3)^{2k})^{1/2}$, and since $\lambda(G_\Phi) = \sup L_\wp$, we see that this contribution is less than or equal to $\lambda(G_\Phi)$. Letting $k \to \infty$, we get $\lim L_k = 1 + 1 = 2 \leq \lambda(G_\Phi)$.

§2.4. Some more curious functions

It is not surprising that the Cantor Function $\Phi(x)$ has its derivative equal to 0 almost everywhere, since on all the removed intervals, it is, by definition, horizontal. As such, it fails to be a strictly increasing function. For a strictly increasing function, one would think it to be impossible that the derivative is 0 almost everywhere and the existence of such a function therefore, would definitely come as a surprise to most. We now describe such a function (see [108]), i.e., we construct:

Example 2.4.1. A function $\Theta(x)$ on the closed interval $[0, 1]$ which is
(i) continuous therein ,
(ii) strictly increasing, with $\Theta(0) = 0$, $\Theta(1) = 1$,
(iii) $\Theta'(x) = 0$ wherever $\Theta'(x)$ exists and is finite.

Let $x \in (0, 1]$ and write x in binary form as:

$$x = 1/2^{a_0} + 1/2^{a_1} + 1/2^{a_2} + \cdots = \sum_{r=0}^{\infty} 1/2^{a_r} \qquad (a_i \in \mathbb{N},\ a_0 < a_1 < a_2 < \ldots)$$

Now define $\Theta(0) = 0$, and $\Theta(x) = \sum_{r=0}^{\infty} \rho^r/(1+\rho)^{a_r}$, where ρ is a positive real number. For $x = 1$, we have $1 = 1/2 + 1/2^2 + 1/2^3 + \cdots$, so that $a_r = r + 1$ for each $r \geq 0$. It follows that $\Theta(1) = 1/(1+\rho) + \rho/(1+\rho)^2 + \rho^2/(1+\rho)^3 + \cdots = 1$, on summing a geometric series. Thus $\Theta(0) = 0$, $\Theta(1) = 1$, as required in (ii) above.

Now let $0 < x \leq 1$. Then by the definition of Θ, $0 = \Theta(0) < \Theta(x)$. Let

$0 < x < y \leq 1$ and write

$$x = 1/2^{a_0} + 1/2^{a_1} + 1/2^{a_2} + \cdots, \qquad (a_0 < a_1 < a_2 \ldots),$$
$$y = 1/2^{b_0} + 1/2^{b_1} + 1/2^{b_2} + \cdots, \qquad (b_0 < b_1 < b_2 \ldots).$$

Let s be the smallest integer such that $a_s \neq b_s$ (i.e., $a_0 = b_0$, $a_1 = b_1$, $\ldots, a_{s-1} = b_{s-1}$, $a_s \neq b_s$). Since $x < y$, we see that $a_s > b_s$ and we have

$$\Theta(x) = 1/(1+\rho)^{a_0} + \rho/(1+\rho)^{a_1} + \cdots + \rho^{s-1}/(1+\rho)^{a_{s-1}} + \rho^s/(1+\rho)^{a_s} + \cdots$$
$$\Theta(y) = 1/(1+\rho)^{b_0} + \rho/(1+\rho)^{b_1} + \cdots + \rho^{s-1}/(1+\rho)^{b_{s-1}} + \rho^s/(1+\rho)^{b_s} + \cdots$$

Here the first s terms are equal and since $b_s < a_s$, we have $1/(1+\rho)^{b_s} > 1/(1+\rho)^{a_s}$. It follows that $\Theta(y) > \Theta(x)$ (even if $\rho = 1$), i.e., we have thus proved (ii) above, i.e., that $\Theta(x)$ *is strictly increasing in* $[0, 1]$.

Next, for $x = 1/2^{a_0} + 1/2^{a_1} + \cdots$ and $n \geq 0$, let $x_n = \sum_{a_r \leq n} 1/2^{a_r}$ and let $y_n = x_n + 1/2^n$. Then $x_n \leq x \leq y_n$, because $x_n \leq x$ is trivial, while $x = x_n + \sum 1/2^{a_r}$ (summation over $a_r > n$) $\leq x_n + 1/2^{n+1} + 1/2^{n+2} + \cdots = x_n + 1/2^n = y_n$. Let k_n be the number of those suffixes for which $a_i \leq n$. Then

$$\Theta(y_n) - \Theta(x_n) = \frac{\rho^{k_n}}{(1+\rho)^n} \qquad (\dagger)$$

where clearly $0 \leq k_n \leq n$ (note that even if we allow a_r's to be 0, $0 \leq k_n \leq n+1$). This is best seen by the following:

(i) Take, for example,

$$\begin{aligned} x &= 1/2^2 + 1/2^3 + 1/2^5 + 1/2^6 + \cdots \\ &= 0/2^1 + 1/2^2 + 1/2^3 + 0/2^4 + 1/2^5 + 1/2^6 + \cdots \\ &= .011011\ldots, \end{aligned}$$

and so $a_0 = 2, a_1 = 3, a_2 = 5, a_3 = 6, \ldots$. Take $n = 5$, say; then $x_5 = \sum_{a_r \leq 5} 1/2^{a_r} = 1/2^2 + 1/2^3 + 1/2^5$, and $y_5 = 1/2^2 + 1/2^3 + 1/2^5 + 1/2^5 = 1/2^2 + 1/2^3 + 2/2^5 = 1/2^2 + 1/2^3 + 1/2^4$. So now $\Theta(y_5) = 1/(1+\rho)^2 + \rho/(1+\rho)^3 + \rho^2/(1+\rho)^4$ and $\Theta(x_5) = 1/(1+\rho)^2 + \rho/(1+\rho)^3 + \rho^2/(1+\rho)^5$. Hence $\Theta(y_5) - \Theta(x_5) = \rho^2/(1+\rho)^4 - \rho^2/(1+\rho)^5 = \rho^3/(1+\rho)^5 = \rho^{k_5}/(1+\rho)^5$, where $k_5 = 3$ is equal to the number of r, $0 \leq r \leq 5$, with $a_r \leq 5$.

(ii) If all the terms in x are present, say, $x = 1/2 + 1/2^2 + 1/2^3 + 1/2^4 + \cdots = .1111111\ldots$, then

$$\begin{aligned} x_5 &= 1/2 + 1/2^2 + 1/2^3 + 1/2^4 + 1/2^5, \\ y_5 &= 1/2 + 1/2^2 + 1/2^3 + 1/2^4 + 1/2^5 + 1/2^5 \\ &= 1/2 + 1/2^2 + 1/2^3 + 1/2^4 + 2/2^5 \\ &= 1/2 + 1/2^2 + 1/2^3 + (1/2^4 + 1/2^4) \\ &= 1/2 + 1/2^2 + 1/2^3 + 1/2^3 \\ &= 1. \end{aligned}$$

Then $\Theta(y_5) = 1$, $\Theta(x_5) = 1/(1+\rho) + \rho/(1+\rho)^2 + \cdots + \rho^4/(1+\rho)^5$, and so, after simplifications, $\Theta(y_5) - \Theta(x_5) = \rho^5/(1+\rho)^5 = \rho^{k_n}/(1+\rho)^5$, where k_n is the number of r's such that $a_r \leq 5$, which is 5 in this case. It now follows that

$$\lim_{n\to\infty} (\Theta(y_n) - \Theta(x_n)) = \lim_{n\to\infty} \frac{\rho^{k_n}}{(1+\rho)^n} = 0\,,$$

since for $\rho < 1$, we have $1/(1+\rho)^n \to 0$ and ρ^{k_n} is bounded (even if k_n does not tend to 0), while if $\rho > 1$, we have $\rho^{k_n}/(1+\rho)^n \leq \rho^n/(1+\rho)^n = 1/(1+1/\rho)^n \to 0$. So Θ is continuous in $(0,1]$. As for continuity at $x = 0$, note that $\lim_{x\to 0} \Theta(x) = 0$ (since as $x \to 0$, more and more of the a_r's in the representation $x = \sum 1/2^{a_r}$ are zero) and as $\Theta(0) = 0$, Θ is continuous at $x = 0$ too. Finally if $\rho = 1$, we have $\Theta(x) = \sum_{r=0}^{\infty} 1^r/(1+1)^{a_r} = \sum_{r=0}^{\infty} 1/2^{a_r} = x$ and hence Θ is continuous at 1. We have thus proved (i) above, i.e., that Θ is continuous in $[0,1]$.

Let now $\rho \neq 1$. We prove the curious result that if, for a point $a \in [0,1]$, $\Theta'(a)$ exists, then $\Theta'(a) = 0$.

Using (†) above, we get $(\Theta(y_n) - \Theta(x_n))/(y_n - x_n) = (\rho^{k_n}/(1+\rho)^n).2^n$, where $x_n < a < y_n$ and where $\{x_n\}$, $\{y_n\}$ are sequences constructed above for a point a. Since $\Theta'(a) = \lim_{n\to\infty}(\Theta(y_n) - \Theta(x_n))/(y_n - x_n)$, we see that $\lim_{n\to\infty} \rho^{k_n}.2^n/(1+\rho)^n$ exists and is finite and we have to show that this limit is 0 (under the assumption that $\rho \neq 1$). Suppose that the limit in question is not 0. We get a contradiction as follows: We know that if $\lim a_n$ is finite and nonzero, then $\lim(a_n/a_{n-1}) = 1$. It follows, then, that $\lim(\rho^{k_n}((2/(1+\rho))^n)/\rho^{k_{n-1}}((2/(1+\rho))^{n-1})) = 1$, i.e., that

$$\lim_{n\to\infty} (\rho^{k_n - k_{n-1}}) = (1+\rho)/2 \qquad (\ddagger)$$

Since $\rho \neq 1$, this means $\lim(k_n - k_{n-1})$ exists and equals k, say. But now $\alpha_n \to A$ implies that $(\alpha_1 + \alpha_2 + \cdots + \alpha_n)/n \to A$. This gives that $(k_2 - k_1 + k_3 - k_2 + \cdots + k_n - k_{n-1})/n \to k$, i.e., $(k_n - k_1)/n \to k$, i.e. $k_n/n \to k$. But $0 \leq k_n \leq n$, so $k_n/n \to 0$ or 1, depending on how k_n behaves as $n \to \infty$, i.e., $k = 0$ or 1. Now use (‡) to give $\rho^0 = (1+\rho)/2$ or $1 = (1+\rho)/2$, i.e., $\rho = 1$, giving a contradiction.

Remark 2.4.3. *This remark requires some knowledge of measure theory.* The property (iii) above says that either $\Theta'(x)$ does not exist or if it does, it is 0. By Lebesgue's theorem [97], $\Theta'(x)$ does exist and is finite almost everywhere, since Θ is an increasing function. It follows that $\Theta'(x) = 0$, almost everywhere.

It is worthwhile to compare this function with the Cantor function. In the latter, $\Phi(x)$ is a horizontal line in $(1/3, 2/3)$; $(1/9, 2/9)$, $(7/9, 8/9)$; ... ; i.e., almost on the entire line and as such, the fact that $\Phi'(x) = 0$ almost everywhere, is obvious. Further, Φ is increasing, but not strictly increasing. In the present case, the function Θ is strictly increasing and still $\Theta'(x) = 0$ almost everywhere. The reader is advised to make a rough sketch of the function Θ. He should plot points with $x = 1/2, 1/4, 3/4, \ldots$ and take, say, $\rho = 1/2$. For

$x = 1/2$, $a_0 = 1, a_1 = a_2 = a_3 = \ldots = 0$, but this does not satisfy the requirements given in the beginning, i.e., $a_0 < a_1 < a_2 < \ldots$. The way to do this is to write $1/2 = 1/4 + 1/8 + 1/16 + \cdots = 1/2^2 + 1/2^3 + 1/2^4 + \cdots$, giving $a_0 = 2, a_1 = 3, a_2 = 4, \ldots$. Hence $\Theta(1/2) = \sum \rho^r/(1+\rho)^{a_r} = 1/(1+\rho) = 2/3$ (on summing the geometric progression, since we are taking $\rho = 1/2$). Similarly for points $x = 3/4$ or $x = 1/8$.

A much shorter, but slightly more difficult example, that does all the above, is given by Freilich in [36].

There is no dearth of functions that are peculiar in one way or the other. There is one more example that we would like to take up in detail, which we think, is quite curious. The function Ω in question has the following astounding properties:

- $\Omega(x)$ is differentiable (and therefore continuous) for all real x.
- $\Omega(x)$ is monotone in no interval (however small).

We now give a series of lemmas that gradually build up to our required function Ω (also see [54]).

Lemma 2.4.4. *Let $r, s \in \mathbb{R}$.*
(i) $r > s > 0 \Rightarrow (r-s)/(r^2 - s^2) < 2/r$,
(ii) $r, s > 1 \Rightarrow (r+s-2)/(r^2+s^2-2) < 2/s$.

Proof. Easy.

Lemma 2.4.5. *Let $\varphi(x) = 1/\sqrt{(1+\mid x \mid)}$, $x \in \mathbb{R}$. Then, for two real numbers a, b, $a \neq b$,*

$$\frac{1}{(b-a)} \int_a^b \varphi(x)\, dx < 4 \min\{\varphi(a), \varphi(b)\}.$$

Proof. Without loss of generality we may suppose $a < b$ (otherwise consider the interval $[b, a]$ and the integral from b to a equals the negative of the integral from a to b).

Case 1: $0 \leq a < b$. Then by Lemma 2.4.4

$$\begin{aligned}
\frac{1}{(b-a)} \int_a^b \varphi(x)\, dx &= \frac{1}{(b-a)} \int_a^b \frac{dx}{\sqrt{(1+x)}} && \text{(since } \mid x \mid = x \text{ as } 0 \leq a < b\text{)} \\
&= 2\left(\frac{\sqrt{(1+b)} - \sqrt{(1+a)}}{(b-a)} \right) && \text{(putting } x = \tan^2\theta\text{)} \\
&= 2\left(\frac{\sqrt{(1+b)} - \sqrt{(1+a)}}{(1+b)-(1+a)} \right) \\
&< \frac{4}{\sqrt{(1+b)}} && \text{(by Lemma 2.4.4)} \\
&= 4 \min\{\varphi(a), \varphi(b)\},
\end{aligned}$$

since φ is decreasing, so the minimum is at b.
Case 2: $a < b \leq 0$. Now $\varphi(-x) = \varphi(x)$, so changing x to $-x$, we land up in Case 1, with φ unchanged and so Case 2 follows.

Case 3: $a < 0 < b$. In this case $\int_a^b dx/\sqrt{(1+\mid x \mid)} = \int_a^0 dx/\sqrt{(1-x)} + \int_0^b dx/\sqrt{(1+x)}$. Evaluating by the substitution $x = \sin^2\theta$, $x = \tan^2\theta$ respectively, this equals $2(\sqrt{(1-a)} - 1) + 2(\sqrt{(1+b)} - 1)$. Hence

$$I = \frac{1}{b-a}\int_a^b \frac{dx}{\sqrt{1+\mid x \mid}} = 2\left(\frac{\sqrt{1+b}+\sqrt{1-a}-2}{(1+b)+(1-a)-2}\right)$$
$$< \frac{4}{\sqrt{1-a}},$$

again by Lemma 2.4.4. But I is also equal to $2(\sqrt{1-a}+\sqrt{1+b}-2)/((1-a)+(1+b)-2)$. It follows that $I < 4\min\{1/\sqrt{1-a}\,,\, 1/\sqrt{1+b}\} = 4\min\{\varphi(a), \varphi(b)\}$. This completes the proof of the Lemma.

The graph Γ of φ is drawn in Figure 2.15. At the point $(0,1)$, it is easy to check that the right hand derivative equals -1, while the left hand derivative equals $+1$. Thus φ is not differentiable in $(0,1)$ and the slope of Γ (on the right and the left) is shown in the Figure 2.15. As $x \to \pm\infty$, $y \to 0$.

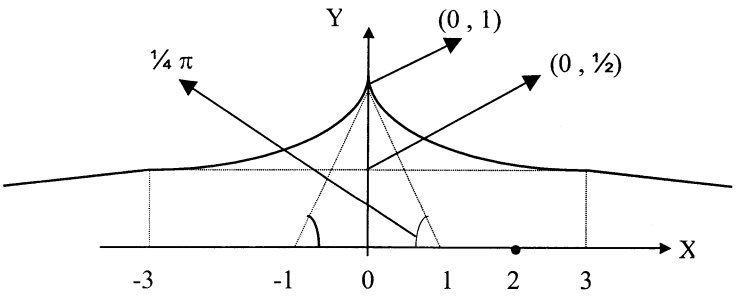


Figure 2.15

Lemma 2.4.6. *Let φ be as in Lemma 2.4.5 and let*

$$\psi(x) = \sum_{i=1}^{n} c_i \varphi(\lambda_i(x - \alpha_i))$$

where $c_1, c_2, \ldots, c_n$; $\lambda_1, \lambda_2, \ldots, \lambda_n$ are positive real numbers; $\alpha_1, \alpha_2, \ldots, \alpha_n$ are any real numbers. Then

$$\frac{1}{(b-a)}\int_a^b \psi(x)\,dx < 4\min\{\psi(a), \psi(b)\},$$

where a, b are distinct real numbers.

Proof. On putting $t = \lambda(x - a)$, we see that

$$\frac{1}{b-a}\int_a^b \varphi(\lambda(x-\alpha))\,dx = \frac{1}{b-a}\int_{\lambda(a-\alpha)}^{\lambda(b-\alpha)} \frac{\varphi(t)\,dt}{\lambda}$$
$$= \frac{1}{\lambda(b-\alpha)-\lambda(a-\alpha)}\int_{\lambda(a-\alpha)}^{\lambda(b-\alpha)} \varphi(t)\,dt\,.$$

Now, Lemma 2.4.5 immediately gives the result.

Figure 2.16

Remark 2.4.7. To understand the function ψ, it is helpful to build it up in stages:

1. The function $\varphi(x - \alpha)$ is just the translate of $\varphi(x)$ by an amount of α.

2. For the function $\varphi(\lambda x)$, taking $\lambda = 2, 3, 4, \ldots$, the graph looks like the following:
   The right hand gradient of $\varphi(\lambda x)$, at $x = 0$, is equal to $-\lambda$, which tends to $-\infty$ as $\lambda \to \infty$, as is clear from Figure 2.16. Similarly, the left hand gradient is equal to $+\infty$.

3. The function $c\varphi(x)$ has graph with each ordinate multiplied by c.

Then ψ is just a finite sum of such functions $c\varphi(\lambda(x - \alpha))$.

Lemma 2.4.8. *Let $\psi_1, \psi_2, \ldots$ be any sequence of functions as in Lemma 2.4.6. For any real number x and any positive integer n, define*

$$\Psi_n(x) = \int_0^x \psi_n(t)\,dt\,. \qquad (*)$$

Suppose that $\sum_{n=1}^{\infty} \psi_n(a)$ is convergent to, say, $s < \infty$, for some real number a. Then
(i) the series $\sum_{n=1}^{\infty} \Psi_n(x) = \sum_{n=1}^{\infty}(\int_0^x \psi_n(t)dt)$ is uniformly convergent on every bounded subset of $\mathbb{R}$, with limit function $F(x)$, say;
(ii) $F(x)$ is differentiable at $x = a$ and $F'(a) = s$(i.e., $F'(a) = \sum_{n=1}^{\infty} \Psi'(a) = \sum_{n=1}^{\infty} \psi_n(a) = s$, by hypothesis (*)). In particular, if $\sum_{n=1}^{\infty} \psi_n(t) = f(t) < \infty$, for all $t \in \mathbb{R}$, then F is differentiable everywhere on $\mathbb{R}$ and $F' = f$.

Proof. Let b be a real number with $b \geq |\, a \,|$. For $-b \leq x \leq b$, we have

$$|\, \Psi_n(x) \,| = \left| \int_0^x \psi_n(t)dt \right| \leq \int_0^x |\psi_n(t)|dt = \int_0^a |\psi_n(t)|dt + \int_a^x |\psi_n(t)|dt\,,$$

which, by Lemma 2.4.6, is less than or equal to $4|a| \min\{\psi_n(a), \psi_n(0)\} + 4|x - a|.\min\{\psi_n(x), \psi_n(a)\} = 4|a|\psi_n(a) + 4|x - a|.\psi_n(a) \leq 12b.\psi_n(a)$, by looking at the graph of ψ_n and since $\min\{\psi_n(x), \psi_n(a)\} \leq \psi_n(a)$. By Weierstrass' M-test, uniform convergence on $[-b, b]$ now follows.

To prove $F'(a) = s$, let $\epsilon > 0$ be given. Choose N such that

$$10 \sum_{n=N+1}^{\infty} \psi_n(a) < \epsilon/2\,. \tag{1}$$

Since each $\psi_n(x)$ is continuous at $x = a$, there exists a $\delta > 0$, such that for all $0 < |h| < \delta$, we have

$$\left| \frac{1}{h} \int_a^{a+h} \psi_n(t)dt \; - \; \psi_n(a) \right| < \epsilon/2N\,, \tag{2}$$

for all $n = 1, 2, \ldots, N$. Using Lemma 2.4.6 again, we have, for $0 < |h| < \delta$,

$$\begin{aligned}
\left| \frac{F(a+h) - F(a)}{h} - s \right| &= \left| \sum_{n=1}^{\infty} \left\{ \left(\frac{1}{h} \int_a^{a+h} \psi_n(t)dt \right) - \psi_n(a) \right\} \right| \\
&\leq \sum_{n=1}^{N} \left| \left(\frac{1}{h} \int_a^{a+h} \psi_n(t)dt \right) - \psi_n(a) \right| \\
&\quad + \sum_{n=N+1}^{\infty} \left\{ \left(\frac{1}{h} \int_a^{a+h} \psi_n(t)dt \right) + \psi_n(a) \right\} \\
&< \epsilon/2 + \sum_{n=N+1}^{\infty} 5\,\psi_n(a), \quad \text{(by (2) and as by Lemma 2.4.6,} \\
&\qquad\qquad \tfrac{1}{h} \textstyle\int_a^{a+h} \psi_n(t)dt \leq 4\psi_n(a)). \\
&< \epsilon/2 + \epsilon/2 = \epsilon \qquad \text{(by (1)).}
\end{aligned}$$

This completes the proof of Lemma 2.4.8.

Lemma 2.4.9. *Let $I_1, I_2, \ldots, I_n$ be disjoint open intervals and let α_i be the mid points of the I_i. Let ϵ; $d_1, d_2, \ldots, d_n$ be positive real numbers. Then there exists a function $\psi(x)$, as in Lemma 2.4.6, such that for each i, we have*

(i) $\psi(\alpha_i) > d_i$,

(ii) $\psi(x) < d_i + \epsilon$, *if* $x \in I_i$,

(iii) $\psi(x) < \epsilon$, *if* $x \notin I_1 \cup I_2 \cup \ldots \cup I_n$.

Proof. Choose $c_i = d_i + \epsilon/2$ and write $\varphi_i(x) = c_i\varphi(\lambda_i(x - \alpha_i))$, where the function φ is as defined in Lemma 2.4.5 and λ_i are chosen so large that

$$\varphi_i(x) < \epsilon/2n \text{ if } x \notin I_i \tag{3}$$

(one only needs to check this inequality at an end point of I_i, since here, the quantity $x - \alpha_i$ is least). Now take $\psi(x) = \varphi_1(x) + \varphi_2(x) + \cdots + \varphi_n(x)$. Since the I_i are disjoint and since $\varphi_i(x)$ takes its maximum value c_i at α_i, properties (i), (ii), (iii) are easy to verify. Indeed, take $i = 1$, say. Then

$$\begin{aligned} \psi(\alpha_1) &= \varphi_1(\alpha_1) + \varphi_2(\alpha_1) + \cdots \\ &= c_1\varphi(0) + \varphi_2(\alpha_1) + \cdots \\ &= (d_1 + \epsilon/2)\varphi(0) + \varphi_2(\alpha_1) + \cdots \\ &> d_1 , \end{aligned} \tag{4}$$

since $\varphi(0) = 1$ and since each φ is greater than 0 everywhere. This yields (i).

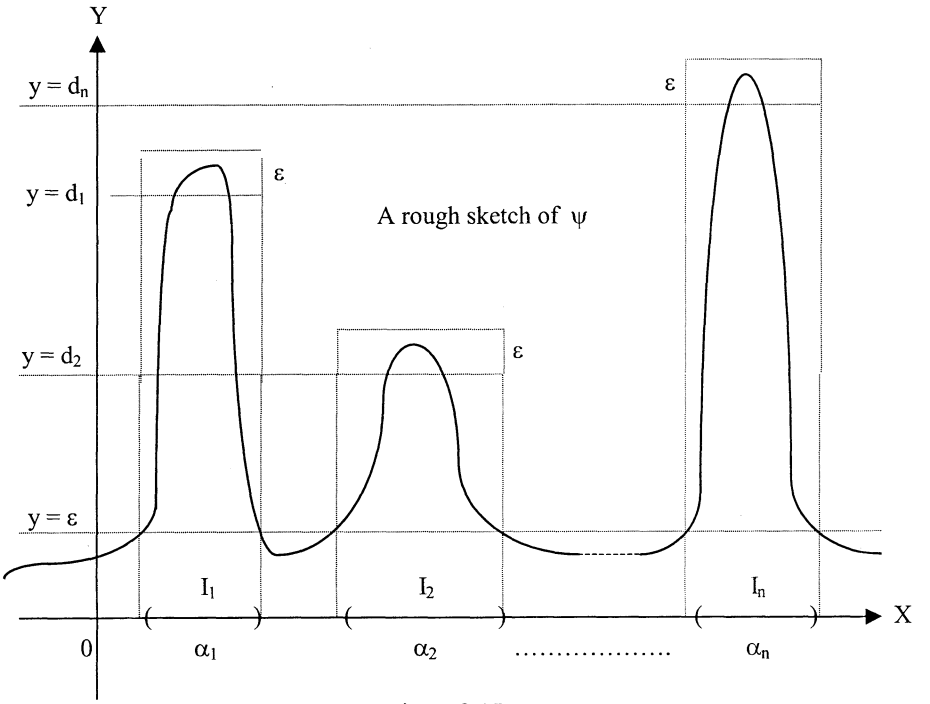


Figure 2.17

Also,

$$\psi(\alpha_1) \leq (d_1 + \epsilon/2).1 + \underbrace{\epsilon/2n + \epsilon/2n + \cdots + \epsilon/2n}_{n-1 \text{ terms}} < d_1 + \epsilon \text{ (by (3))},$$

which proves (ii).

For (iii), observe that if $x \notin \cup_{i=1}^{n} I_i$, then as φ_1 takes its maximum value c_1 at α_1 and the rest of φ's are, in any case, less than $\epsilon/2n$, we have

$$\psi(x) \leq \epsilon/2n + \epsilon/2n + \cdots + \epsilon/2n < \epsilon\,.$$

Theorem 2.4.10. *Let $\{\alpha_i\}$ and $\{\beta_i\}$, $i = 1, 2, 3, \ldots$, be disjoint countable subsets of $\mathbb{R}$. Then there exists a real valued, everywhere differentiable function $F(x)$ on $\mathbb{R}$, satisfying the following properties:*
(i) $F'(\alpha_i) = 1$,
(ii) $F'(\beta_i) < 1$ for all i ,
(iii) $0 < F'(x) \leq 1$, for all x.

Proof. We construct the partial sums $f_n(x) = \sum_{k=1}^{n} \psi_k(x)$ with the following three properties, by induction on n:

A_n : $f_n(\alpha_i) > 1 - 1/n$ for all $i = 1, 2, \ldots, n$;

B_n : $f_n(x) < 1 - 1/(n+1)$ for all $x \in \mathbb{R}$;

C_n : $\psi_n(\beta_i) < 1/2n.2^n$, for all $i = 1, 2, \ldots, n$.

First, let $n = 1$. Choose an open interval I_1, with midpoint α_1 such that $\beta_1 \notin I_1$. By Lemma 2.4.9, with $\epsilon = d_1 = 1/4$ so that $c_1 = d_1 + \epsilon/2 = 3/8$, we obtain $f_1(x) = \psi_1(x)$, which satisfies: (i) $\psi_1(\alpha_1) > 1/4$, (ii) $\psi_1(x) < 1/4 + 1/4 = 1/2$ if $x \in I_1$, (iii) $\psi_1(x) < 1/4$ if $x \notin I_1$. Here (i) implies that $\psi_1(\alpha_1) > 0$, which gives A_1, while (ii) and (iii) imply that $\psi_1(x) < 1/2$ for all x, which is B_1. Finally (iii) is C_1 and now our induction starts.

We next have the induction step. Let $n > 1$ and suppose $f_{n-1}(x) = \sum_{k=1}^{n-1} \psi_k(x)$ has been constructed to satisfy $A_{n-1}, B_{n-1}, C_{n-1}$. Select disjoint open intervals $I_1, I_2, \ldots, I_n$ such that for each $i \in \{1, 2, \ldots, n\}$, α_i is the mid-point of I_i and $I_i \cap \{\beta_1, \beta_2, \ldots, \beta_n\} = \emptyset$ and

$$f_{n-1}(x) < f_{n-1}(\alpha_i) + \delta\,, \tag{5}$$

for all $x \in I_i$ and where $\delta = 1/(n(n+1)) - 1/2n.2^n > 0$. Now apply Lemma 2.4.9, with

$$\epsilon = 1/2n.2^n\,, d_i = 1 - 1/n - f_{n-1}(\alpha_i)\,, i = 1, 2, \ldots, n \tag{6}$$

to obtain $\psi_n(x)$, and take $f_n(x) = \sum_{k=1}^{n} \psi_k(x)$.

We now verify A_n, B_n, C_n for these functions $\psi_n(x)$ and $f_n(x)$.

C_n follows trivially; for, according to (iii) of Lemma 2.4.9, $\psi_n(x) < \epsilon$ if $x \notin I_1 \cup \ldots \cup I_n$ and since $\beta_i \notin I_1 \cup \ldots \cup I_n$ for all $i = 1, 2, 3, \ldots, n$, we get $\psi_n(\beta_i) < \epsilon = 1/2n.2^n$ for all $i = 1, 2, \ldots n$, giving C_n, as required.

Next, to check A_n, observe that for $i = 1, 2, \ldots, n$,

$$\begin{aligned} f_n(\alpha_i) &= f_{n-1}(\alpha_i) + \psi_n(\alpha_i) \\ &> f_{n-1}(\alpha_i) + d_i \quad \text{(by (i) of Lemma 2.4.9)} \\ &= 1 - 1/n \qquad\quad \text{(by (6))}. \end{aligned}$$

Finally, to check Property B_n, if $x \in I_i$ (for some i), then we have

$$\begin{aligned} f_n(x) &= f_{n-1}(x) + \psi_n(x) \\ &< (f_{n-1}(\alpha_i) + \delta) + (d_i + \epsilon) && \text{(by (5) and by (ii) of Lemma 2.4.9)} \\ &= (f_{n-1}(\alpha_i) + d_i) + \delta + \epsilon && \text{(on rewriting)} \\ &= 1 - 1/n + 1/(n(n+1)) && \text{(by (6) and since } \delta = \frac{1}{n(n+1)} \\ & && \quad - \frac{1}{2n.2^n}\ ,\ \epsilon = \frac{1}{2n.2^n}) \\ &= 1 - 1/(n+1); \end{aligned}$$

while if $x \notin I_1 \cup I_2 \cup \ldots I_n$, then

$$\begin{aligned} f_n(x) &= f_{n-1}(x) + \psi_n(x) \\ &< 1 - 1/n + \epsilon && \text{(by } B_{n-1} \text{ and by (ii) of Lemma 2.4.9)} \\ &< 1 - 1/(n+1) && \text{(since } \epsilon = 1/2n.2^n). \end{aligned}$$

This verifies B_n.

Construction of the function $F(x)$ on the real line, as required in the theorem, can now be achieved. By Lemma 2.4.8, $\lim_{n\to\infty} \sum_{k=1}^{n} \psi_k(x) = \lim_{n\to\infty} f_n(x)$ exists (call it $F'(x)$) and is the term-by-term derivative of $F(x) = \sum_{n=1}^{\infty} \Psi_n(x)$, i.e., $F'(x) = \lim_{n\to\infty} f_n(x)$. Hence:

(i) $F'(\alpha_i) = \lim_{n\to\infty} f_n(\alpha_i) \geq 1$, by A_n, and also ≤ 1 by B_n; giving $F'(\alpha_i) = 1$, as required.

(ii) $0 < F'(x)$ for all x, since $F'(x) = \sum_{n=1}^{\infty} \Psi'_n(x) = \sum_{n=1}^{\infty} \psi_n(x) > 0$, as each $\psi_n(x) > 0$; also $F'(x) = \lim_{n\to\infty} f_n(x) \leq 1$, by B_n.

(iii) Choose $n > i$. We have

$$\begin{aligned} F'(\beta_i) &= \sum_{k=1}^{\infty} \psi_k(\beta_i) \\ &= \sum_{k=1}^{n-1} \psi_k(\beta_i) + \sum_{k=n}^{\infty} \psi_k(\beta_i) \\ &= f_{n-1}(\beta_i) + \sum_{k=n}^{\infty} \psi_k(\beta_i) \\ &< 1 - 1/n + \sum_{k=n}^{\infty} 1/2k.2^k && \text{(by } B_n \text{ and by } C_n) \\ &< 1 - 1/n + (1/2n).1 && \text{(estimating the geometric progression)} \\ &< 1 - 1/2n < 1, \end{aligned}$$

which finally completes the proof of Theorem 2.4.10.

Theorem 2.4.10 gives an elegant construction of a function $\Omega(x)$, satisfying the most unexpected properties mentioned earlier.

Example 2.4.11. There exists a real valued function Ω such that

(i) Ω is differentiable for all real x and Ω' is bounded for all real x,

(ii) Ω is not monotone in any subinterval, however small, of the real line.

Indeed, let $\{\alpha_i\}$ and $\{\beta_i\}$, $i = 1, 2, 3, \ldots$, be disjoint dense subsets of the real line (e.g., the $\{\alpha_i\}$ could be the set of all rationals and the $\{\beta_i\}$ the set of rationals + a fixed irrational such as $\sqrt{2}$, say). Using Theorem 2.4.10 construct a function F on $\mathbb{R}$ such that

(i) $F'(\alpha_i) = 1$ for all i ,

(ii) $F'(\beta_i) < 1$ for all i ,

(iii) $0 < F'(x) \leq 1$ for all real x ,

and a function G on $\mathbb{R}$ such that

$(i)'$ $G'(\alpha_i) < 1$ for all i ,

$(ii)'$ $G'(\beta_i) = 1$ for all i ,

$(iii)'$ $0 < G'(x) \leq 1$ for all real x ,

and write $\Omega(x) = F(x) - G(x)$. Then

$$\Omega'(\alpha_i) = F'(\alpha_i) - G'(\alpha_i) > 0\,, \quad \text{(using } (i) \text{ and } (i)') \tag{7}$$

$$\Omega'(\beta_i) = F'(\beta_i) - G'(\beta_i) < 0\,, \quad \text{(using } (ii) \text{ and } (ii)') \tag{8}$$

$$-1 < \Omega'(x) = F'(x) - G'(x) < 1\,, \quad \text{(using } (iii) \text{ and } (iii)') \tag{9}$$

In particilar, (9) shows that $\Omega'(x)$ is bounded. Finally, since the sets $\{\alpha_i\}$ and $\{\beta_i\}$ are both dense, Ω can not be monotone in any interval.

This function Ω has many more remarkable properties; we give a few here:

Corollary 2.4.12. *$\Omega(x)$ has a local maximum and a local minimum in every interval of $\mathbb{R}$.*

Proof. Let $[a, b]$ be any interval, $a < b$. Choose $\alpha, \beta \in [a, b]$ from our dense subsets, $\alpha < \beta$; then $\Omega'(\alpha) > 0$, $\Omega'(\beta) < 0$. It follows that $\Omega(x)$ takes an absolute maximum value in $[\alpha, \beta]$ at some $\gamma \in (\alpha, \beta)$. Hence $\Omega(\gamma)$ is a local maximum value for $\Omega(x)$.

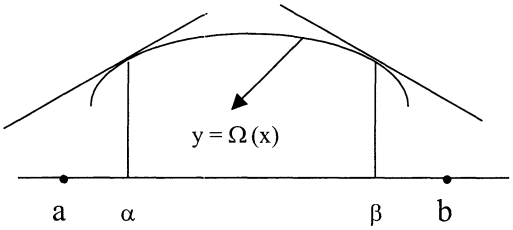


Figure 2.18

Similarly there exists a local minimum.

Corollary 2.4.13. *(For readers familiar with Lebesgue integration). The function $\Omega'(x)$ is Lebesgue integrable but not Riemann integrable.*

Proof. (i) We first note that the function $\Omega(x)$ is absolutely continuous in $[a,b]$, for, let $a = x_0 < x_1 < \ldots < x_{n-1} < x_n$ be any set of points in $[a,b]$ such that

$$\sum_{i=1}^{n}(x_i - x_{i-1}) < \delta\,.$$

By the first mean value theorem, $\Omega(x_i) - \Omega(x_{i-1}) = (x_i - x_{i-1})\Omega'(\xi_i)$, $x_{i-1} < \xi_i < x_i$. It follows that

$$\sum_{i=1}^{n}|\Omega(x_i) - \Omega(x_{i-1})| \leq \sum_{i=1}^{n}(x_i - x_{i-1})|\Omega'(\xi_i)| \leq \sum_{i=1}^{n}(x_i - x_{i-1}) < \delta\,,$$

since $|\Omega'(x)| < 1$. Thus $\Omega(x)$ is absolutely continuous on $[a,b]$, as claimed. Hence by results from the theory of Lebesgue integration (see [97]), it follows that $\Omega'(x)$ is Lebesgue integrable, as required; and indeed,

$$\Omega(x) = \Omega(a) + \int_a^x \Omega'(t)dt \tag{10}$$

Next, we claim that $\Omega'(x)$ is not Riemann integrable in $[a,b]$. Suppose, if possible, that $\Omega'(x)$ is Riemann integrable in $[a,b]$. Then by Lebesgue's well-known theorem (see [97]), $\Omega'(x)$ is continuous almost everywhere in $[a,b]$. However, if $\Omega'(t)$ is continuous at t, then it is easy to see that $\Omega'(t) = 0$ (for, say $\Omega'(t) > 0$; then there exists a neighbourhood $(t-\delta, t+\delta)$ of t, throughout which, $\Omega'(x) > 0$. However, there exists a β of our dense set in $(t-\delta, t+\delta)$ at which $\Omega'(\beta) < 0$, a contradiction).

Thus $\Omega'(t) = 0$ almost everywhere on $[a,b]$. Hence by (10), $\Omega(x) = \Omega(a)$ for all $x \in [a,b]$, i.e., $\Omega(x)$ is a constant, which gives a contradiction, since Ω is not monotone anywhere.

Next, we would like to mention an example of a function $f\colon \mathbb{R} \to \mathbb{R}$ with a *proper local maximum* at each point of a set S that is everywhere dense in $\mathbb{R}$ (for example S could be the set of rational numbers $\mathbb{Q}$).

Definition 2.4.14. We say that a function $f : \mathbb{R} \to \mathbb{R}$ has a proper local maximum at a point $x = a$, if there exists an open neighbourhood $N(a)$ of a, such that $f(x) < f(a)$ for all $x \in N(a) \setminus \{a\}$.

It is easy to see (A.Schoenflies, *Die Entwickelung der Lehre von den punktmannigfaltigkeiten*, Bericht, Erstattet der Dentschen Mathematiker - Vereinigung, 1900) that for any function f, the set S of points at which f has a proper local maximum, is at most countable. Indeed, let $\mathcal{B}$ be the set of all open intervals in $\mathbb{R}$ with rational end points. For each $a \in S$, pick a neighbourhood $N(a) \in \mathcal{B}$ such that $f(x) < f(a)$ for all $x \in N(a) \setminus \{a\}$. As $\mathcal{B}$ is countable, S is at most countable.

Example 2.4.15. It is trivial to sketch the graph of a continuous function $f\colon \mathbb{R} \to \mathbb{R}$ which has a proper local maximum at infinitely many points in $[0, 1]$, say at each of the points $a = 1, 1/2, 1/3, \ldots$; see Figure 2.19.

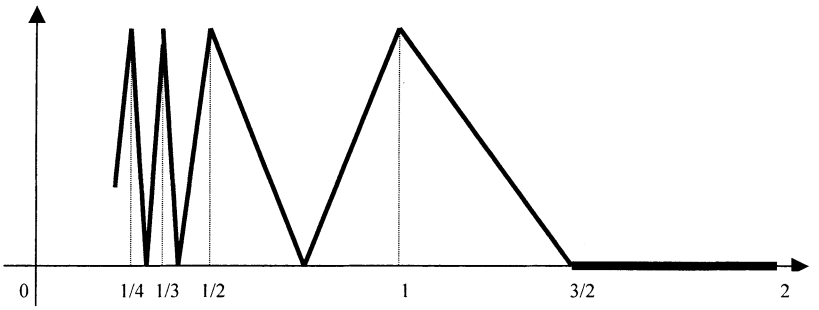


Figure 2.19

This brings up the question:

- Does there exist a continuous function $f : \mathbb{R} \to \mathbb{R}$, which has a proper local maximum at each point of a countable dense set?

It is not easy to visualize the graph of such a continuous function, but the answer to the question is known to be yes; see [81], page 63, Theorem 3. For a straightforward and simple example, also see [82].

Finally, we record some more examples/counterexamples in the spirit of the above discussion.

Example 2.4.16. Let f be a function defined on the compact set $[0, 1]$ by

$$f(x) = \begin{cases} \frac{(-1)^n n}{n+1}, & \text{if } x \text{ is a rational, } x = \frac{m}{n} \text{ in lowest terms, } n > 0 \\ 0, & \text{if } x \text{ is irrational.} \end{cases}$$

Then in any neighbourhood of every point of $[0, 1]$, the values of f come arbitrary close to the number 1 and -1 while always lying between them. This

gives an example of a bounded function having no relative extreme value on a compact domain.

Example 2.4.17 [37]. We now give an example of a monotonic function whose points of discontinuity form an arbitrary countable set.

Let A be any arbitrary nonempty countable set of real numbers $a_1, a_2, \ldots$ and let $\sum p_n$ be a finite or convergent infinite series of positive numbers with sum p. If A is bounded below and x is less than every point of A, let $f(x) = 0$. Otherwise, define $f(x)$ to be the sum of all terms p_m of $\sum p_n$ such that $a_m \leq x$. The function f is increasing on $\mathbb{R}$, continuous at every point not in A, and discontinuous with a jump equal to p_n at each point a_n, i.e.,

$$\lim_{x \to a_n +} f(x) - \lim_{x \to a_n -} f(x) = p_n \,.$$

Example 2.4.18 *(R.E.Megginson; Math. Mag., 57(1984), 112).* This is an example of a one-to-one onto function $f : \mathbb{R} \to \mathbb{R}$ which is continuous at some $a \in \mathbb{R}$, but f^{-1} is not continuous at $f(a)$. Let

$$f(x) = \begin{cases} \frac{1}{2n}\,, & \text{if } x = \frac{1}{n},\ n \in \mathbb{N} \\ \frac{1}{x}\,, & \text{if } x = 3, 5, 7, \ldots \\ \frac{x}{2}\,, & \text{if } x = 2, 4, 6, \ldots \\ x\,, & \text{otherwise} \end{cases}$$

i.e., $f(x) = x$, except when $x = 1, 2, 3, \ldots$; $1/2, 1/3, 1/4, \ldots$; but for $1, 1/2$, $1/3, 1/4, \ldots$, the image under f is $1/2, 1/4, 1/6, 1/8, \ldots$, respectively; and for $3, 5, 7, \ldots$, the image under f is $1/3, 1/5, 1/7, \ldots$ respectively; and for $2, 4, 6, \ldots$, the image is $1, 2, 3, \ldots$ respectively. So the set of values $1, 2, 3, \ldots$; $1/2, 1/3, 1/4, \ldots$ are just reshuffled by f and so, since everywhere else $f(x) = x$, we see that f is onto. So f^{-1} exists and f is continuous at 0 (check !). However, f^{-1} is not continuous at $f(0) = 0$, since $f^{-1}(1/n) = n$ for $n = 3, 5, 7, \ldots$.

§2.5. Algebraic functions

Definition 2.5.1. A function f defined on a subset E of $\mathbb{R}$ is called an algebraic function if for some positive integer n, there exists a sequence $P_0(x), P_1(x), \ldots,$ $P_n(x)$ of polynomials such that for each $x \in E$, the following relation holds:

$$P_0(x) + P_1(x).f(x) + P_2(x).f^2(x) + \cdots + P_n(x).f^n(x) = 0 \qquad (*)$$

For such a function f, there exists a least positive integer m such that the degrees of each of the polynomials $P_0(x), P_1(x), \ldots, P_n(x)$ is less than or equal to m. *We call* $(*)$*, with this minimal pair* n, m *the minimum* $(*)$. Our object is to prove the following

Theorem 2.5.2. *There is no algebraic function* f *whose derivative exists and is* $1/x$ *for each* x *in the domain of definition of* f.

Proof (H.R.Rouse [96]). Suppose that such a function f exists. Since $f'(x) = 1/x$, 0 does not belongs to the domain of definition of f. Now differentiate *minimum (*)* with respect to x:

$$\{P_0'(x) + P_1'(x)f(x) + P_2'(x)f^2(x) + \cdots + P_n'(x)f^n(x)\}$$
$$+ \{P_1(x)f'(x) + 2P_2(x)f(x)f'(x) + \cdots + nP_n(x)f^{n-1}(x)f'(x)\} = 0.$$

Since $f'(x) = 1/x$, this gives

$$\{P_0'(x) + P_1'(x)f(x) + P_2'(x)f^2(x) + \cdots + P_n'(x)f^n(x)\}$$
$$+\{P_1(x) + 2P_2(x)f(x) + \cdots + nP_n(x)f^{n-1}(x)\}.(1/x) = 0.$$

Multiply this by x and regroup to get

$$(xP_0'(x) + P_1(x)) + (xP_1'(x) + 2P_2(x))f(x)$$
$$+ (xP_2'(x) + 3P_3(x))f^2(x) + \cdots$$
$$+ (xP_{n-1}'(x) + nP_n(x))f^{n-1}(x) + xP_n'(x)f^n(x) = 0 \qquad (\sharp)$$

Now $(\sharp)$ shows that there exists an equation $(*)$ with the same n, m but such that the last polynomial has a factor x. So without loss of generality, let $(*)$ be such that $x \mid P_n(x)$. But then $(\sharp)$ implies that the last but one term, viz. $xP_{n-1}'(x) + nP_n(x)$ also has a factor x, since $P_n(x)$ has factor x, and so on. Repeating this argument n times, we see that we may assume that each of $P_0(x), \ldots, P_n(x)$ in $(*)$ has a factor x. But 0 does not belong to the domain of f, so $(*)$ can be divided by x, which reduces the degree of each $P_i(x)$, i.e., m can be reduced, contradicting the assumption that m is minimal.

Definition 2.5.3. If $n = 1$, this minimum $(*)$ becomes $P_0(x) + P_1(x)f(x) = 0$. Such an f is called a rational function .

By an argument, similar to the one above, it can be proved that there is no rational function ϕ such that $\phi'(x) = 1/x$.

An immediate application of Theorem 2.5.2 is the following

Proposition 2.5.4. *The function* $\log x$ *is not algebraic.*

Proof. The function $\log x$ has derivative equal to $1/x$ in its domain of definition, namely the set $x > 0$.

Remark 2.5.4. It can also be shown that there is no nonzero algebraic function f for which $f'(x) = f(x)$ for all x belonging to the domain of definition of f.

Exercises

2.1. Let f be a continuous function on $[a, b]$. Prove that if s, t are distinct values taken by f in $[a, b]$, then

$$\delta(s, t) = \inf\{|x - y| \mid f(x) = s, f(y) = t\}$$

is positive. Prove that if r, s, t are values taken by f in $[a, b]$, where $r < s < t$, then $\delta(r, s) \leq \delta(r, t)$. Prove further that $\delta(r, s) < \delta(r, t)$.

2.2. Let f be defined on the entire real line and for $\delta > 0$, define

$$\varphi(x, \delta) = \sup\{|f(x)-f(y)| \mid x-\delta < y < x+\delta\} \text{ and } \psi(\delta) = \sup\{\varphi(x, \delta) | x \in \mathbb{R}\}.$$

Prove that

(i) ψ is nondecreasing,

(ii) ψ is bounded,

(iii) $\psi(\delta) \to 0$ as $\delta \to 0$ if and only if f is uniformly continuous on $\mathbb{R}$.

2.3. Let f be defined on $\mathbb{R}$. Prove that the following statements are equivalent:

(i) For each sequence $\{x_n\}$, convergent to a, the sequence $\{f(x_n)\}$ is convergent,

(ii) For some A, $f(x) \to A$ as $x \to a$,

(iii) For each $\epsilon > 0$, there exists a $\delta > 0$ such that $|f(x) - f(y)| < \epsilon$ whenever $0 < |x - a| < \delta$ and $0 < |y - a| < \delta$.

2.4. Let $r_1, r_2, \ldots,$ be an enumeration of the rationals in the interval $[0, 1]$. For each $x \in [0, 1]$, define the function ϕ_x on $[0, 1]$ by

$$\phi_x(y) = \begin{cases} 1 & \text{if } y \leq x, \\ 0 & \text{if } y > x. \end{cases}$$

Let f be defined on $[0, 1]$ by

$$f(x) = \sum_{n=1}^{\infty} \phi_x(r_n)/2^n .$$

Prove that

(i) f is strictly increasing,

(ii) f is continuous at $x \in [0, 1]$ if and only if $x \notin \mathbb{Q}$.

2.5. Let f be a continuous function on $[0, 1]$ with $g(0) = 0 = g(1)$. Suppose, for every $0 < x < 1$, there exists a $k > 0$ such that (i) $0 \leq x-k < x < x+k \leq 1$ (ii) $g(x) = \frac{1}{2}(g(x + k) + g(x - k))$. Show that g is identically zero in $[0, 1]$.

2.6. Let f be a function such that for all real numbers x, y,

$$|f(x) - f(y)| \leq k\,|x - y|$$

where $0 \leq k \leq 1$ is a constant.

By considering the sequence $x, f(x), f(f(x)), \ldots$, for a real x, prove that there exists a unique real number x_0 such that $f(x_0) = x_0$.

By considering the function $(1+x^2)^{1/2}$, or otherwise, show that this result is false if the above condition is replaced by the corresponding condition without the k.

2.7. Show that a continuous, rational-valued function must be a constant.

2.8. Prove that if f is a continuous real-valued function on $[0, \infty)$ and $f(x)$ tends to a finite limit as $x \to \infty$, then f is uniformly continuous on $[0, \infty)$. Is the converse true? Justify your answer.

Prove that if $g(x)$ is uniformly continuous on $(-\infty, \infty)$, then $g(\sqrt{x})$ is uniformly continuous on $[0, \infty)$.

2.9. Let $f_1, f_2, \ldots$, be continuous functions on $[0,1]$ with values in $[0,1]$. Define functions $g_1, g_2, \ldots$, from $[0,1]$ to $\mathbb{R}$ by

$$g_{k+1}(x) = \begin{cases} f_1(x) & \text{if } k = 0, \\ g_k(x) & \text{if } g_k(x) < 1 - 2^{-k}, \\ g_k(x) + 2^{-k} f_{k+1}(x) & \text{if } g_k(x) \geq 1 - 2^{-k}. \end{cases}$$

For $x \in [0,1]$, set $g(x) = \lim_{k \to \infty} g_k(x)$. Show that for $x \in [0,1]$ and $\epsilon > 0$, there exists a $\delta > 0$ such that

$$|x - y| < \delta\,,\ y \in [0,1] \Rightarrow g(y) < g(x) + \epsilon\,.$$

Deduce that $g(x_0) = \sup_{0 \leq x \leq 1} g(x)$ for some $x_0 \in [0,1]$.

2.10. For $n \geq 0$, let $P_n(x) = 1 + x + x^2/2! + \cdots + x^n/n!$. Show that P_n has no real zero if n is even and exactly one real zero if n is odd (use induction on n).

Using uniform convergence, justify that $\exp(x) = \sum_{r=0}^{\infty} x^r/r!$ is a differentiable function on $\mathbb{R}$, and that $(\exp(x))' = \exp(x)$. Deduce that $\exp(x)$ is a nondecreasing function of x.

2.11. Suppose that $f\colon (a,b) \to \mathbb{R}$ be continuous. Prove that a necessary and sufficient condition that f is uniformly continuous in (a,b) is that $\lim_{x \to b-} f(x)$ and $\lim_{x \to a+} f(x)$ both exist.

2.12. Which one of the following statements is sufficient to determine the value $f(0)$ of $f(x)$? Give the values when determined:

1. $f(x)$ is continuous at $x = 0$ and takes both positive and negative values in any neighbourhood of $x = 0$.
2. Given $\epsilon > 0$, there exists a $\delta > 0$ such that $|f(x)| < \epsilon$ for $0 < |x| < \delta$.
3. $(f(h) + f(-h) - 2f(0))/h \to 1$ and $f(h) \to a$ as $h \to 0$.

2.13. f is defined in $[-1,1]$ by: $f(0) = 0$, $f(x) = \operatorname{sign}(\sin(\pi/x))$, if $x \neq 0$, where $\operatorname{sign} 0 = 0$, $\operatorname{sign} y = y/|y|$ $(y \neq 0)$. Prove that $f(x)$ is integrable in

$[-1, 1]$.

2.14. Determine whether the given conditions are sufficient to ensure that $f(x)$ tends to a finite limit as $x \to \infty$

1. $f(x) > 0$, $g(x) < 0$, $f(x) - g(x) \to 0$ as $x \to \infty$.
2. $(f(x))^2 \to \lambda$ as $x \to \infty$ $(\lambda < \infty)$.

Distinguish between the cases $\lambda = 0$, $\lambda \neq 0$.

2.15. Let $f(x + y) = f(x).f(y)$. Prove that if f is not identically equal to 0, then $f(x) > 0$. Prove also that if f is bounded in some interval $[-X, X]$, then $\inf_{x \in [-X,X]} f(x) > 0$.

2.16. Let f be defined in $[0, 1]$, $f(0) = 0$. For $x \in [0, 1]$, there exists an $h(x) > 0$ such that $f(x) - f(x') \in \mathbb{Q}$ if $x' \in (x - h(x), x + h(x))$, for all $0 \leq x' \leq 1$. Prove that $f(x) \in \mathbb{Q}$ for all $x \in [0, 1]$.

2.17. The functions $f(x) = 0$ and $f(x) = x^n$ have the property that for any integer m, $(f(x))^m = f(x^m)$. Prove that there exists a function with this property which is discontinuous at each $x \in (-\infty, \infty)$.

2.18. Does there exist a real nonconstant function f such that $f(x+p) = f(x)$ for all $p > 0$?

2.19. Let f be integrable in $[a, b]$ and suppose that $f(x) \geq 0$ and $\int_a^b f = 0$. Given $\epsilon > 0$, show that there exists a closed interval $[c, d] \subset [a, b]$ such that $f(x) < 1/r$ for all $x \in [c, d]$. Deduce that there exists a nested sequence of intervals

$$[a, b] \supset [a, b_1] \supset [a_2, b_2] \supset \cdots$$

such that $f(x) < 1/r$ for all $x \in [a_r, b_r]$. Hence show that $f(\xi) = 0$ for some point $\xi \in [a, b]$.

2.20. Give two examples to show that a function f can satisfy one of the following properties without satisfying the other :

- Given $\epsilon > 0$, there exists a $\delta > 0$, such that $|x - 1| < \delta$ implies that $|f(x) - 1| < \epsilon$.
- Given $\delta > 0$, there exists a $\epsilon > 0$, such that $|x - 1| < \delta$ implies that $|f(x) - 1| < \epsilon$.

2.21. A continuous function f is such that $f(x + \pi) = f(x)$ for all $x \in \mathbb{R}$. Find necessary and sufficient conditions on f under which one can find another function g satisfying $g(x + \pi) = g(x)$ for all $x \in \mathbb{R}$, such that $\frac{dg}{dx} = f(x)$.

2.22. Give an example of a real-valued continuous function f for which $f(0) = 0$, and for which the sequences

$$1.f(1), 3.f(1/3), 5.f(1/5), \ldots, (2n + 1).f(1/(2n + 1)), \ldots$$

and

$$2.f(1/2), 4.f(1/4), 6.f(1/6), \ldots, 2n.f(1/2n), \ldots$$

converge respectively to 1 and -1. Can such an f be differentiable at 0? Justify your answer.

2.23. For $n = 2, 3, 4, \ldots$, let $f_n : [0,1] \to [0,1]$ be the function whose graph is shown below;

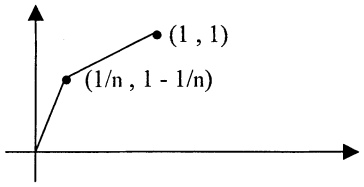


Figure 2.20

Calculate:

1. $\lim_{n\to\infty}(f_n^{-1}(x))$, for all $x \in [0,1]$.
2. $\lim_{n\to\infty} \underbrace{g \circ g \circ \cdots \circ g}_{n \text{ times}}$, where g is the function f_3 above.

2.24. Given $f(x) = p\sin x + q\cos x + x^2$, $f(2) = 3$, is $f(-2)$ uniquely determined or not?

2.25. Prove that the function $f(x) = \frac{1}{\sin x} - \frac{1}{x}$, $0 < x \leq \pi/2$, is positive and increasing.

2.26. Prove that for $0 < x < \pi/4$, $(\sin x)^{\sin x} < (\cos x)^{\cos x}$.

2.27. Prove that:

1. $\cos(\sin x) > \sin(\cos x)$, for all real x.
2. $\cos(\sin^{-1} x) < \sin^{-1}(\cos x)$, $0 \leq x \leq 1$.
3. $|\sin rx| < r|\sin x|$, for integer $r \geq 2$, $\sin x \neq 0$.

2.28. Suppose $f : \mathbb{R} \to \mathbb{R}$ is continuous and for every rational number q, there exists an integer n such that $\overbrace{f(f(\ldots f}^{n}(q))\ldots)$, i.e., the n^{th} iterate of f evaluated at q is 0. Prove or disprove: for every real number t, there exists an integer n, such that $\overbrace{f(f(\ldots f}^{n}(t))\ldots) = 0$.

Chapter 3

The famous everywhere continuous, nowhere differentiable functions : van der Waerden's and others

§3.1. van der Waerden's function.

One of the most weird functions is that which is defined for all real x, is continuous at each x but is differentiable at no x. Geometrically, it would appear to be some kind of limit of the saw tooth function,

Figure 3.1

provided the limit does not degenerate into a straight line. The first example of a continuous nowhere differentiable function was given by Weirstrass (1815-1897), namely $f(x) = \sum_{n=0}^{\infty} b^n \cos(a^n \pi x)$, where b is an odd integer and a is such that $0 < a < 1$ and $ab > 1 + \frac{3}{2}\pi$. Now there are many elegant constructions of such functions (sometimes called the Weierstrass functions or the blancmange functions, see [109]); some very geometric, others very analytical (i.e., pictures very difficult to visualize), yet others a compromise. We begin with a delightfully simple example given by van der Waerden. The function is simply

$$\Phi(x) = \sum_{k=0}^{\infty} \phi_k(x) = \sum_{k=0}^{\infty} \frac{1}{2^k} \phi_0(2^k x)\,,$$

where $\phi_0(x)$ is the distance of x to the nearest integer .

To elucidate with diagrams and explanations: $\phi_0(x)$ is simply equal to x, if $0 < x \leq 1/2$, and to $-x + 1$ if $1/2 \leq x \leq 1$, in $[0, 1]$ and extended periodically on the entire real line. Its graph is shown in Figure 3.2:

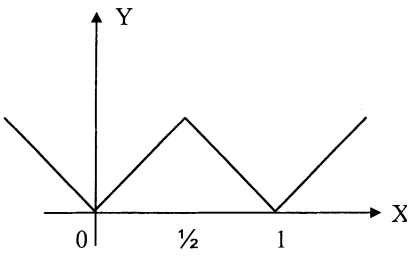


Figure 3.2

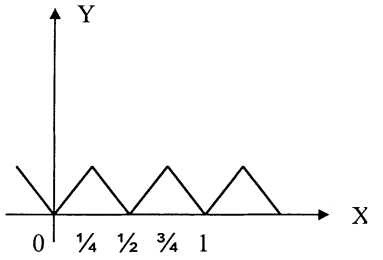


Figure 3.3

The next function $\phi_1(x)$ is equal to $(1/2).\phi_0(2x)$ and so has two peaks in $[0,1]$, each half as high as in ϕ_0; thus

$$\phi_1(x) = \begin{cases} x & \text{if } 0 \le x \le 1/4\ , \\ -x + 1/2 & \text{if } 1/4 \le x \le 1/2\ , \\ x - 1/2 & \text{if } 1/2 \le x \le 3/4\ , \\ x + 1 & \text{if } 3/4 \le x \le 1\ , \end{cases}$$

in $[0,1]$ and extended periodically on the entire real line. Its graph is shown in Figure 3.3; and so it goes, the number of peaks doubling at each stage but each peak half as high as the previous one. Then one adds them " all " up; " all " meaning add up the first n to get

$$\Phi_n(x) = \phi_0(x) + (1/2)\phi_0(2x) + \cdots + (1/2^n)\phi_0(2^n x)$$

and take the limit as $n \to \infty$ to get $\Phi(x) = \lim_{n\to\infty} \Phi_n(x)$, which exists, since for each x, $\Phi_n(x)$ increases and is bounded above by $(1/2)+(1/2)^2+\cdots+(1/2)^n$, i.e., $0 \le \Phi_{n-1} \le \Phi_n \le 1/2 + (1/2)^2 + \cdots + (1/2)^n \le 1$.

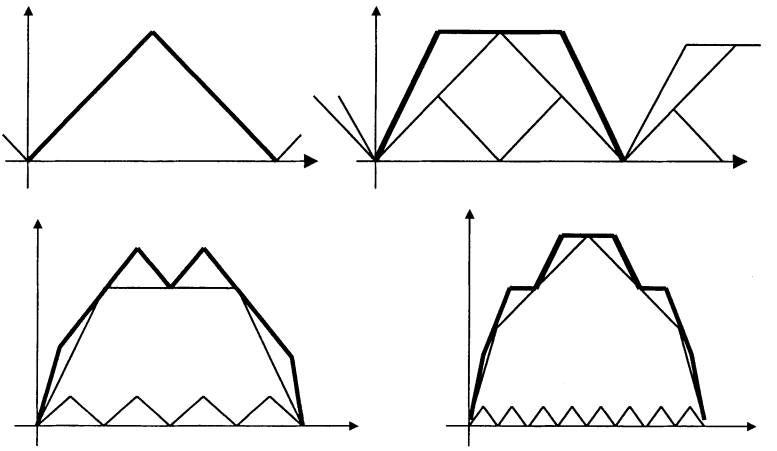


Figure 3.4

We note that as n increases, the contribution to Φ_n from the n^{th} term, viz. $(1/2^n)\phi_0(2^n x)$, gets more and more insignificant; indeed, even for $n = 5, 6, \ldots$, the visual difference between Φ_5 and Φ_6, is negligible. The graphs of $\Phi_1, \Phi_2, \ldots$ are drawn in Figure 3.4.

Since $|\phi_0(x)| \leq 1/2$ for all x, we have $|\phi_n(x)| = |1/2^n.\phi_0(2^n x)| \leq 1/2^{n+1} = M_{n+1}$, say. Since $\sum M_n$ is convergent, it follows, by Weierstrass' M-test, that $\Phi(x)$ is continuous for all real x, since each $\phi_n(x)$ is continuous for all real x. We shall now show that $\Phi'(x)$ does not exist at any point. The following proof is given in [12].

Lemma 3.1.1. *Suppose that the function $\Phi(x)$ is differentiable at some $\xi \in \mathbb{R}$. Let $u_n \leq \xi \leq v_n$, $u_n \leq v_n$, $v_n - u_n \to 0$ as $n \to \infty$. Then*

$$\frac{\Phi(v_n) - \Phi(u_n)}{v_n - u_n} \longrightarrow \Phi'(\xi) ,$$

as $n \to \infty$, or in other words,

$$\lim_{(h\,,\,k)\to(0\,,\,0)} \frac{\Phi(\xi + h) - \Phi(\xi - k)}{h + k} = \Phi'(\xi)$$

.

Proof. From the following diagram (Figure 3.5) it is clear that if $\theta_1 \leq \theta \leq \theta_2$ $(\theta_1, \theta, \theta_2 \in (0, \pi/2))$, then $\tan\theta_1 \leq \tan\theta \leq \tan\theta_2$, i.e.,

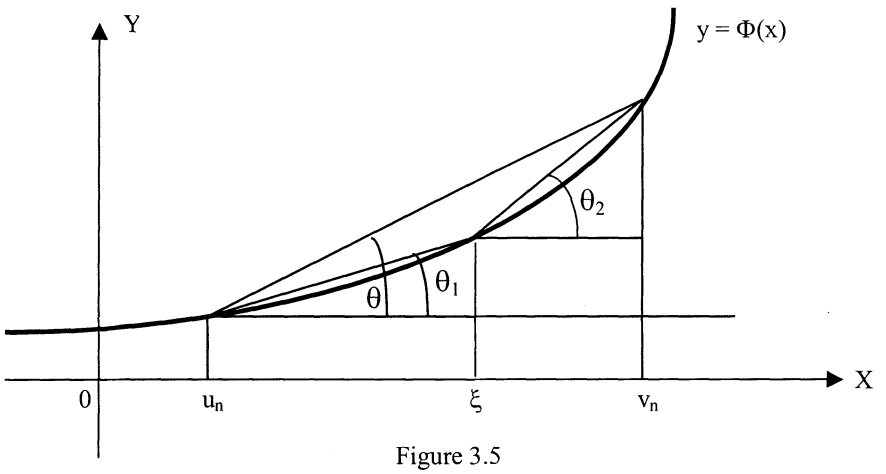


Figure 3.5

$$\frac{\Phi(\xi) - \Phi(u_n)}{\xi - u_n} \leq \frac{\Phi(v_n) - \Phi(u_n)}{v_n - u_n} \leq \frac{\Phi(v_n) - \Phi(\xi)}{v_n - \xi} ,$$

and letting $n \to \infty$ in this, the lemma follows, since the extreme members tend to $\Phi'(\xi)$, which is given to exist.

We construct suitable $u_n \leq \xi \leq v_n$ (in the notation of Lemma 3.1.1) as follows:

For $n = 1, 2, \ldots$, find $i_n \in \mathbb{Z}$ such that $i_n/2^n \leq \xi < (i_n + 1)/2^n$ (indeed, given n, $2^n\xi$ lies between some consecutive integers: $i_n \leq 2^n\xi < i_n + 1$). Take $u_n = i_n/2^n$ and $v_n = (i_n + 1)/2^n$. We now calculate the value of $\Phi(u_n)$ and $\Phi(v_n)$. We have

$$\begin{aligned}\phi_k(u_n) &= (1/2^k)\phi_0(2^k u_n)\\ &= (1/2^k)\phi_0(2^k.i_n/2)^n\\ &= 0\,;\end{aligned}$$

if $k \geq n$, because if $k \geq n$, then $z = 2^k.i_n/2^n$ is an integer. Hence $\Phi(u_n) = \sum_{k=0}^{\infty} \phi_k(u_n) = \sum_{k=0}^{n-1} \phi_k(u_n)$ and similarly $\Phi(v_n) = \sum_{k=0}^{n-1} \phi_k(v_n)$. So now

$$\frac{\Phi(v_n) - \Phi(u_n)}{v_n - u_n} = \sum_{k=0}^{n-1} \frac{\phi_k(v_n) - \phi_k(u_n)}{v_n - u_n} \qquad (*)$$

We next show that $\phi_k(x)$ is linear in $[u_n, v_n]$ if $0 \leq k < n$, i.e., that neither a peak nor a trough of any of $\phi_0, \phi_1, \ldots, \phi_{n-1}$ can occur in $[u_n, v_n]$. Indeed, peaks and troughs of $\phi_0, \phi_1, \ldots, \phi_{n-1}$ occur at the following points:

$$1; 1/2; 1/4, 3/4; 1/8, 3/8, 5/8, 7/8; \ldots; 1/2^{n-1}, 3/2^{n-1}, \ldots, (2^{n-1} - 1)/2^{n-1}\,;$$

and none of these can lie strictly between u_n and v_n, for $u_n = i_n/2^n < t_1/2^j < (i_n + 1)/2^n = v_n$, for some odd integer t_1 and $0 \leq j \leq n - 1$, which in turn implies that $i_n < t_2.2^{n-j} < i_n + 1$, for some odd integer t_2, which is impossible, as required.

It follows that the quotient $(\phi_k(v_n) - \phi_k(u_n))/(v_n - u_n) = \pm 1$ (see the diagrams (Figure 3.6)).

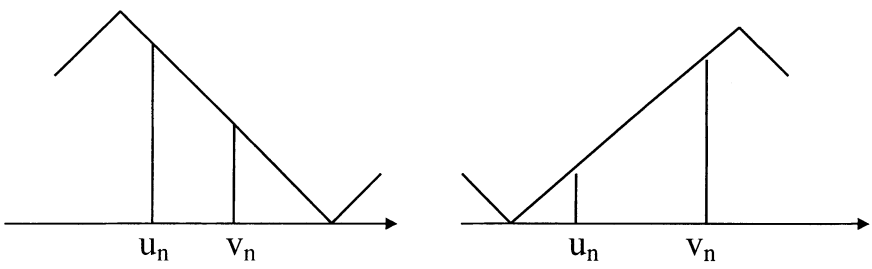


Figure 3.6

So now the (*) above becomes $(\Phi(v_n) - \Phi(u_n))/(v_n - u_n) = \sum_{k=0}^{n-1}(\pm 1)$. Letting $n \to \infty$, this gives (using Lemma 3.1.1) that $\Phi'(\xi) = \sum_{k=0}^{\infty}(\pm 1)$, which is absurd, since the right hand series, being convergent to $\Phi'(\xi)$, must have n^{th} term tending to zero, which is not the case. That completes the proof of the

following striking

Theorem 3.1.2. *The van der Waerden function*

$$\Phi(x) = \sum_{k=0}^{\infty} \phi_k(x)\,,$$

where $\phi_k(x) = (1/2^k).\phi_0(2^k x)$, $\phi_0(x)$ *being the distance of* x *from the nearest integer, is continuous for all real* x *but differentiable for no real* x.

By making slight modifications in the proof of Theorem 3.1.2, we can prove the following

Theorem 3.1.3 ([21]). *The function* $\Phi(x)$ *has no finite one-sided derivative at any point.*

Proof. Assume, to the contrary, that $\Phi'_+(\xi)$ exists at some point ξ and equals R, say (R for right). As before, for an integer $n \geq 1$, let i_n be the integer such that

$$(i_n - 1)/2^n \leq \xi < i_n/2^n\,.$$

Let $u_n = i_n/2^n$, $v_n = (i_n + 1)/2^n$ (we want both of u_n, v_n to the right of ξ), so that as $n \to \infty$, both u_n and v_n tend to ξ.

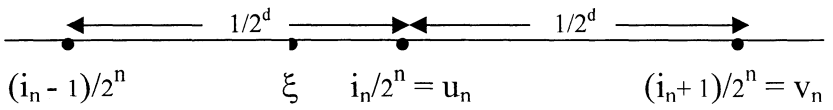


Figure 3.7

Define r_n, s_n by

$$\Phi(u_n) - \Phi(\xi) = (R + r_n)(u_n - \xi) \qquad (i)$$
$$\Phi(v_n) - \Phi(\xi) = (R + s_n)(v_n - \xi)\,. \qquad (i)'$$

Then we have

$$0 < u_n - \xi < 1/2^n \qquad (ii)$$
$$0 < v_n - \xi < 2/2^n\,. \qquad (ii)'$$

Since $(\Phi(u_n) - \Phi(\xi))/(u_n - \xi) \to R$ and $(\Phi(v_n) - \Phi(\xi))/(v_n - \xi) \to R$ as $v_n, u_n \to \xi$ (i.e., as $n \to \infty$), it follows, by (i) and $(i)'$, that $r_n\,,\ s_n \to 0$ as $n \to \infty$. Now

$$\begin{aligned}\Phi(v_n) - \Phi(u_n) &= (\Phi(v_n) - \Phi(\xi)) - (\Phi(u_n) - \Phi(\xi)) \\ &= (R + s_n)(v_n - \xi) - (R + r_n)(u_n - \xi) \quad \text{(by } (i) \text{ and } (i)'\text{)} \\ &= R(v_n - u_n) + s_n(v_n - \xi) - r_n(u_n - \xi) \\ &= R.(1/2^n) + s_n(v_n - \xi) - r_n(u_n - \xi)\,.\end{aligned}$$

Hence, $2^n(\Phi(v_n)-\Phi(u_n)) = R+2^n s_n(v_n-\xi)-2^n r_n(u_n-\xi)$, or $0 \leq |2^n\{\Phi(v_n)-\Phi(u_n)\}-R| \leq 2^n|s_n|.2/2^n-2^n|r_n|.1/2^n = 2|s_n|-|r_n| \to 0$, by (ii) and $(ii)'$ and because r_n , $s_n \to 0$ as $n \to \infty$, i.e., $(\Phi(v_n)-\Phi(u_n))/(v_n-u_n) \to R$ as $n \to \infty$ (since $1/(v-n-u_n) = 2^n$). However, as before, $(\Phi(v_n)-\Phi(u_n))/(v_n-u_n) = \sum_{k=0}^{n-1} \phi_k^{'+}(\xi)$, where $\phi_k^{'+}(\xi)$ is the right derivative of $\phi_k(x)$ at $x = \xi$, and again, as before, $\phi_k(x)$ is linear for each k with $0 \leq k < n$ and so $\phi_k^{'+} = \pm 1$, giving $(\Phi(v_n)-\Phi(u_n))/(v_n-u_n) = \sum_{k=0}^{n-1}(\pm 1)$. Letting $n \to \infty$, this gives $R = \sum_{k=0}^{\infty}(\pm 1)$, which is impossible.

Similarly, the left hand derivative L at ξ can not exist.

For yet another variant and proof of Theorem 3.1.3, the reader is referred to [67].

Another interesting variation of the function Φ above is the function

$$\Psi(x) = \sum_{k=0}^{\infty} \psi_k(x)\,,$$

where $\psi_k(x) = (1/2^k)\phi_0(2^{3k}x)$ and $\phi_0(x)$, as before, being the distance of x from the nearest integer. This function may be shown to be continuous everywhere, with an infinite left derivative and an infinite right derivative at each point.

§3.2. Some more properties of van der Waerden's function

There are further questions, answers to which would throw more light on this beautiful function $\Phi(x)$.

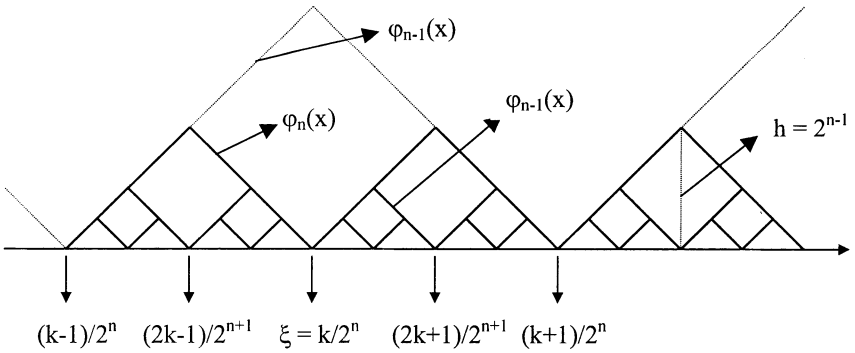


Figure 3.8

A few pertinent properties of the functions Φ_n and Φ are the following; the proofs are vividly clear from the graphs:

(i) If $\phi_n(\alpha) = 0$, at a point $x = \alpha$, then all subsequent ϕ's are 0 at α, i.e., $\phi_{n+1}(\alpha) = \phi_{n+2}(\alpha) = \ldots = 0$.

(ii) $\Phi(k/2^n) = \phi_0(k/2^n)+\phi_1(k/2^n)+\cdots+\phi_{n-1}(k/2^n)$, for $\phi_n(k/2^n) = \phi_n(u_n) =$

0 (the u_n's are same as in Theorem 3.1.2), and so (ii) follows from (i). Let $\xi = k/2^n$ $(0 \leq k \leq 2^n)$, k odd, $n = 0, 1, 2, \ldots$. Then the graph of ϕ_n, in a neighbourhood of $\xi = k/2^n$ is drawn in Figure 3.8.

For $x \in (k/2^n, k/2^n + \varepsilon)$, a right neighbourhood of $\xi = k/2^n$,

$$\text{the slope of } \phi_i = \begin{cases} 1 & \text{if } i \geq n\ , \\ -1 & \text{if } i = n-1\ , \\ \pm 1 & \text{if } 0 \leq i \leq n-2\ ; \end{cases}$$

while for $x \in ((k/2^n) - \varepsilon, k/2^n)$, a left neighbourhood of ξ,

$$\text{the slope of } \phi_i = \begin{cases} -1 & \text{if } i \geq n\ , \\ 1 & \text{if } i = n-1\ , \\ \pm 1 & \text{if } 0 \leq i \leq n-2\ . \end{cases}$$

We thus have :
(iii) At the point $x = \xi = k/2^n$, ϕ_i has

$$\text{the right derivative} = \begin{cases} 1 & \text{if } i \geq n\ , \\ -1 & \text{if } i = n-1\ , \\ \pm 1 & \text{if } 0 \leq i \leq n-2\ ; \end{cases}$$

$$\text{the left derivative} = \begin{cases} -1 & \text{if } i \geq n\ , \\ 1 & \text{if } i = n-1\ , \\ \pm 1 & \text{if } 0 \leq i \leq n-2\ . \end{cases}$$

All this is clear from the graphs of ϕ_n and of the other ϕ_i.

Now let $m > n$ (n fixed); then $\Phi_m(x) = \Phi_{n-1}(x) + \phi_n(x) + \cdots + \phi_m(x)$. Hence for $x \in (k/2^n, k/2^n + \varepsilon)$, $\Phi'_m(x) = \Phi'_{n-1}(x) - (m - (n-1)) \to -\infty$ as $m \to \infty$, by (iii). Similarly for $x \in (k/2^n - \varepsilon, k/2^n)$, $\Phi'_m(x) \to \infty$ as $m \to \infty$. This gives
(iv) It follows from (iii) that for $\xi = k/2^n$ (n fixed, k odd),

$$\Phi'_m(x) \longrightarrow \begin{cases} \infty & \text{if } \xi < x < \xi + \varepsilon\ , \\ -\infty & \text{if } \xi - \varepsilon < x < \xi\ , \end{cases}$$

i.e., the right derivative $\Phi'_+(\xi)$ of $\Phi(x)$ at $x = \xi$ equals $+\infty$, while the left derivative $\Phi'_-(\xi)$ equals $-\infty$.

In particular, we get:

(i) $\Phi(x)$ is not differentiable at points x of the form $k/2^n$ $(n = 0, 1, 2, \ldots)$.

(ii) The graphical considerations given below also provide another proof of

the nondifferentiability of the function Φ at points not of the form $k/2^n$ (see T.H.Hildebrandt, [43]).

Let ξ be a real number, not of the form $k/2^n$. Suppose, to the contrary, that $\Phi'(\xi)$ exists and is finite. Let $(\xi-\delta, \xi+\delta)$ be any open neighbourhood of ξ. We shall construct three suitable points P, Q, R on the graph of Φ, with abscissa in $(\xi-\delta, \xi+\delta)$, as shown in Figure 3.9, such that

$$\theta = \text{slope } PQ - \text{slope } PR \ (= \tan\beta - \tan\gamma) = 1\,.$$

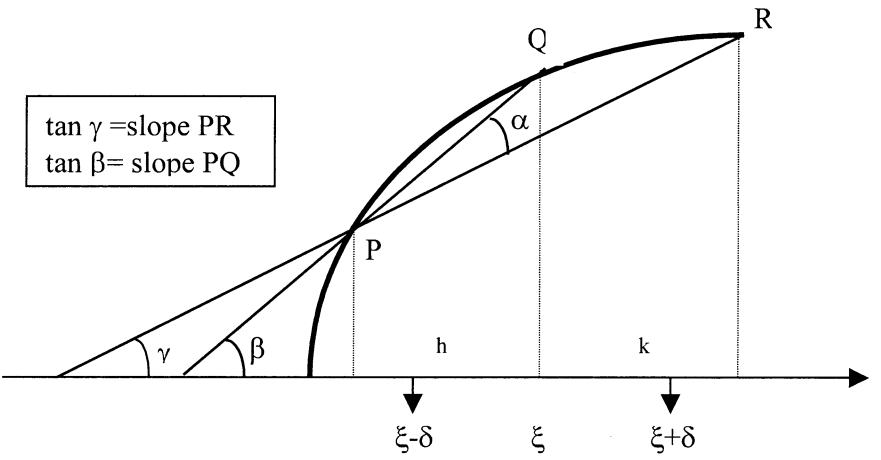


Figure 3.9

In view of Lemma 3.1.1, slope $PR \to \Phi'(\xi)$ as $\delta \to 0$ and so PQ can not possibly make a nonzero angle with PR, since, as $\delta \to 0$, PQ and PR will tend to coincide, making a zero angle with each other, giving the required contradiction.

To construct P, Q, R, find, for each $n = 0, 1, 2, \ldots$, an integer k_n such that $k_n/2^n \le \xi < (k_n+1)/2^n$; choose n so large that $[k_n/2^n, (k_n+1)/2^n] \subset (\xi-\delta, \xi+\delta)$. Let

$$\begin{aligned} P &\equiv (k_n/2^n, \Phi(k_n/2^n))\,, \\ R &\equiv ((k_n+1)/2^n, \Phi((k_n+1)/2^n))\,, \\ Q &\equiv ((2k_n+1)/2^{n+1}, \Phi((2k_n+1)/2^{n+1}))\,, \end{aligned}$$

where Q is the mid-point of PR.

Remembering that some of the powers of 2 in the denominator of the coordinates of P, Q, R are superfluous (e.g. $2k/2^{n+1} = k/2^n$, where, if k is even n

will further get reduced), we get

$$\begin{aligned}
\text{slope of } PQ &= \frac{\Phi((2k+1)/2^{n+1}) - \Phi(2k/2^{n+1})}{(2k+1)/2^{n+1} - 2k/2^{n+1}} \\
&= \frac{(\phi_0((2k+1)/2^{n+1}) + \cdots + \phi_n(2k/2^{n+1}))}{(1/2^{n+1})} \\
&\quad - \frac{(\phi_0(2k/2^{n+1}) + \cdots + \phi_n(2k/2^{n+1}))}{(1/2^{n+1})} \\
&= \frac{\phi_0((2k+1)/2^{n+1}) - \phi_0(2k/2^{n+1})}{1/2^{n+1}} + \cdots \\
&\quad + \frac{\phi_n((2k+1)2^{n+1}) - \phi_n(2k/2^{n+1})}{1/2^{n+1}} \\
&= \varepsilon_0 + \varepsilon_1 + \cdots + \varepsilon_{n-1} + \varepsilon_n \text{ , say } , \qquad (\dagger)
\end{aligned}$$

where $\varepsilon_i = \pm 1$, $\varepsilon_n = 1$ (this will be clear by looking at the Figure 3.10).

In exactly the same way, cancelling the 2 in either of the coordinates, we find that

$$\begin{aligned}
\text{slope of } PR &= \frac{\Phi((k+1)/2^n) - \Phi(k/2^n)}{(1/2^n)} \\
&= \frac{(\phi_0((k+1)/2^n) - \phi_0(k/2^n)}{(1/2^n)} + \cdots \\
&\quad + \frac{\phi_{n-1}((k+1)/2^n) - \phi_{n-1}(k/2^n)}{(1/2^n)} \\
&= \varepsilon_0 + \varepsilon_1 + \cdots + \varepsilon_{n-1} + \varepsilon_{n-1} \text{ ;}
\end{aligned}$$

where the same ε_i feature as in (†) above (this too may be seen immediately by looking at the Figure 3.10). It follows that

$$\text{slope } PQ - \text{slope } PR = \varepsilon_n = 1 \text{ ,}$$

as required.

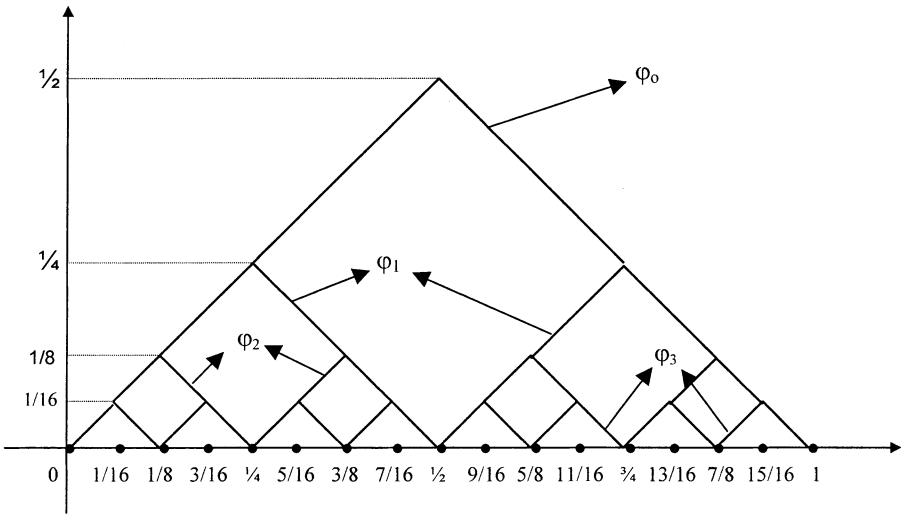


Figure 3.10

In addition to Figure 3.10, the following particular cases should clarify further:

(i) Take $n = 3, k = 0$; then $P=(0,0)$, $R=(1/8,\Phi(1/8))$, $Q=(1/16,\Phi(1/16))$;

$$\begin{aligned} \text{slope } PQ &= \varepsilon_0+\varepsilon_1+\varepsilon_2+\varepsilon_3 = 1+1+1+1, \\ \text{slope } PR &= \varepsilon_0+\varepsilon_1+\varepsilon_2 \qquad\quad = 1+1+1. \end{aligned}$$

(ii) Take $n = 3, k = 3$; then $P = (3/8, \Phi(3/8))$, $R = (1/2, \Phi(1/2))$, $Q = (7/16, \Phi(7/16))$;

$$\begin{aligned} \text{slope } PQ &= \varepsilon_0+\varepsilon_1+\varepsilon_2+\varepsilon_3 = 1-1-1+1, \\ \text{slope } PR &= \varepsilon_0+\varepsilon_1+\varepsilon_2 \qquad\quad = 1-1-1. \end{aligned}$$

At the end of Hildebrandt's paper, referred to above, the editors ask the following question: For what values of x (if any), does $\Phi(x)$ has a definite infinite derivative? i.e., for what x (if any) is

$$\lim_{(h,k)\to(0,0)} \frac{\Phi(x+h)-\Phi(x-k)}{h+k} = \infty \text{ or } -\infty \text{ ?}$$

This question is answered by Begle and Ayres in [9]. Their result is given below.

Let $\xi \in \mathbb{R}$, ξ not of the form $k/2^n$, where, without loss of generality, $\xi \in (0,1)$. Let I_n and O_n be the number of 1's and 0's respectively, in the first n terms of the dyadic (binary) expansion for ξ and let $D_n = O_n - I_n$. Then we have:

Theorem 3.2.1. *Let $\xi \in (0,1)$ be a real number which is not of the form $k/2^n$. Then*

$$\Phi'(\xi) = \begin{cases} \infty & \text{if } D_n \to \infty \text{ as } n \to \infty, \\ -\infty & \text{if } D_n \to -\infty \text{ as } n \to \infty; \end{cases}$$

and if $\lim_{n\to\infty} D_n$ does not exist (as $\pm\infty$), then $\Phi'(\xi)$ does not exist (even as $\pm\infty$).

Remark 3.2.2. One of the following

(i) $D_n \to \infty$,

(ii) $D_n \to -\infty$,

(iii) D_n does not tend to any limit,

must always hold since $D_n = O_n - I_n$ is a sequence of integers with $D_1 = 1$ or -1; $D_2 = D_1 \pm 1 = 2, -2, 0; \ldots$, $D_n = D_{n-1} \pm 1 = \pm 1 \pm 1 \pm 1 \cdots$ does not tend to any finite limit, since the n^{th} term of the series $\pm 1 \pm 1 \pm 1 \cdots$ equals ± 1, which does not tend to 0.

For numbers ξ of the form $k/2^n$, the theorem is not valid, but for such numbers, we already know that $\Phi'(\xi)$ does not exist, even as $\pm\infty$.

The numbers $\xi = 0, 1$ are both of the form $k/2^n$, so without loss of generality, we may take ξ to be in $(0,1)$.

Proof of Theorem 3.2.1 For every power 2^n of 2, there exists an integer k_n such that

$$k_n/2^n < \xi < (k_n + 1)/2^n$$

(indeed, $2^n\xi$ lies strictly between two consecutive integers k_n and k_{n+1}; strictly, since $\xi \neq k/2^n$). Now refer to Lemma 3.1.1, which says that $\Phi'(\xi)$ exists if and only if

$$L = \lim_{\substack{(h,k)\to(0,0) \\ h,\, k \geq 0}} \frac{\Phi(\xi + h) - \Phi(\xi - k)}{h + k} \qquad (\ddagger)$$

exists and then $\Phi'(\xi) = L$ (the implication $\Leftarrow$ is trivial).

Let $\xi + h = (k_n + 1)/2^n$, $\xi - k = k_n/2^n$ $(n = 0, 1, 2, \ldots)$. If the limit in $(\ddagger)$ exists, then with these special values of h and k also it will exist. For the converse, we shall need the following results:

Lemma 3.2.3. *For the function $\Phi(x)$, if the limit in $(\ddagger)$, using the special values $\xi + h = (k_n + 1)/2^n$, $\xi - k = k_n/2^n$ exists, then the limit in $(\ddagger)$, for general h and k also exists and equals $\Phi'(\xi)$.*

Proof. For these special values of h and k, since $\xi + h$, $\xi - k$ have the form $m/2^n$, we have $\Phi(\xi + h) = \Phi_{n-1}(\xi + h)$, $\Phi(\xi - k) = \Phi_{n-1}(\xi - k)$ (see (ii)). So in the above limit $(\ddagger)$, the value of the numerator equals

$$\frac{\Phi_{n-1}((k_n + 1)/2^n) - \Phi_{n-1}(k_n/2^n)}{1/2^n}.$$

Now, each of $\phi_0, \phi_1, \ldots, \phi_{n-1}$ is linear in the interval $(k_n/2^n, (k_n+1)/2^n)$; indeed $\phi_i(x) = \pm x + c_i$ ($c_i = \pm d_i/2^m$ $(i = 0, 1, \ldots, n-1)$). The function $\phi_n(x)$ is the first to have a peak at the mid-point $(2k_n+1)/2^{n+1}$ of $k_n/2^n$ and $(k_n+1)/2^n$. It follows that (see Figure 3.11):

$$\Phi(x) = \Phi_{n-1}(x) = \phi_0(x) + \cdots + \phi_{n-1}(x) = \lambda_n x + \mu_n\,,$$

with $\lambda_n = \sum_{i=0}^{n-1} \pm 1$, $\mu_n = \sum_{i=0}^{n-1} c_i$, i.e., a linear function of x and so the numerator in (‡) is the slope of $\Phi_{n-1}(x)$ in $(k_n/2^n, (k_n+1)/2^n)$, which is equal to λ_n.

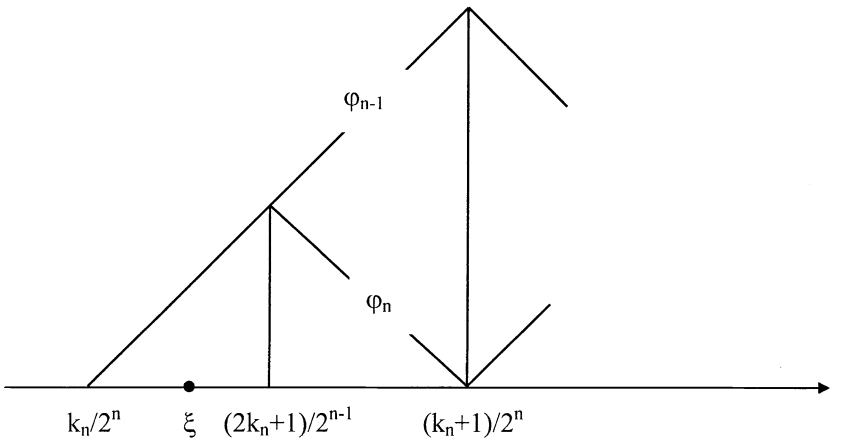


Figure 3.11

Now

$$\text{the slope of } \Phi_n(x) = \begin{cases} \lambda_n + 1 & \text{if } x \in (k_n/2^n, \xi)\ , \\ \\ \lambda_n - 1 & \text{if } x \in (\xi, (k_n+1)/2^n)\,. \end{cases} \qquad \text{(i)}$$

As n increases to N say, slope $\Phi_N = \lambda_n \pm 1 \pm 1 \ldots$ $(N - (n-1)$ ± 1's) and so if the limit in (‡) exists, it has to be $+\infty$ or $-\infty$. We consider these two cases separately and being similar, it is enough to treat one of them, say the $+\infty$ case.

Let the limit in (‡) (for the special h, k) equal ∞ and we have to show that the general limit in (‡) is equal to ∞, i.e. we must show that given M, however large, there exists a $\delta > 0$ such that $((\Phi(\xi+\eta) - \Phi(\xi))/\eta) > M$ whenever $|\eta| < \delta$. Here again, we take $\eta > 0$, as the case $\eta < 0$ is similar.

Since, for the special set of values of $\xi + h$, $\xi - k$, limit in (‡) exists and is ∞, so there exists an integer N such that for $n > N$, the slope

$$(\Phi_n(\xi + h) - \Phi_n(\xi - k))/(h + k) > M + 10/3\,. \qquad \text{(ii)}$$

Choose s with $s/2^N < \xi < (s+1)/2^N$, and let $\delta = \min(\xi - s/2^N, (s+1)/2^N - \xi)$. Find n such that

$$r/2^n < \xi < (r+1)/2^n < \xi + \eta < (r+2)/2^n \,. \tag{iii}$$

Then either ξ lies in the first half of the interval or $\xi + \eta$ in the last half of the interval.

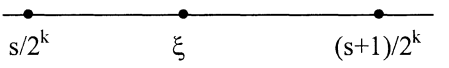


Figure 3.12

Since $\eta < \delta$, we have $n > N$. For later use, note also (using (iii)) that

$$\xi - r/2^n < 1/2^n \,,\ \eta > 1/2^{n+1} \tag{iv}$$

By (ii), we see (using (i)) that

$$\text{the slope of } \Phi_n(x), \text{ between } r/2^n \text{ and } (r+1)/2^n, \text{ is } > M + 10/3 \tag{v}$$

and

$$\text{the slope of } \Phi_n(x), \text{ between } (r+1)/2^n \text{ and } (r+2)/2^n, \text{is} > M + 4/3 \tag{vi}$$

In particular, with $x = \xi$, (i) gives, since Φ_n is linear, $(\Phi_n(\xi) - \Phi_n(r/2^n))/(\xi - r/2^n) > M + 10/3$. However, $\xi - r/2^n < 0$, so this gives $(\Phi_n(\xi) - \Phi_n(r/2^n)) < (M + 10/3)(\xi - r/2^n)$, i.e.,

$$(\Phi_n(\xi) < \Phi_n(r/2^n)) + (M + 10/3)(\xi - r/2^n) \tag{vii}$$

Also, for every x, $\Phi(x) - \Phi_n(x) < 2/3.2^n$, because the left hand side is equal to

$$\begin{aligned}
\phi_{n+1}(x) + \phi_{n+2}(x) + \cdots &= \frac{1}{2^{n+1}}\phi_0(2^{n+1}x) + \frac{1}{2^{n+2}}\phi_0(2^{n+2}x) + \cdots \\
&\le \frac{1}{2}\left(\frac{1}{2^{n+1}} + \frac{1}{2^{n+2}} + \cdots\right) \qquad \left(\text{since } \phi_0(x) \le \frac{1}{2}\right) \\
&= \frac{1}{2}\frac{1}{2^n} \\
&= \frac{1}{4}\frac{1}{2^{n-1}} < \frac{1}{3}\frac{1}{2^{n-1}} \,,
\end{aligned}$$

as claimed. Taking $x = \xi$, this together with (vii) gives:

$$\Phi(\xi) < \Phi(r/2^n) + (M + 10/3)(\xi - r/2^n) + 2/3.2^n \tag{viii}$$

In exactly the same way we get, (using (vi)):

$$\Phi(\xi + \eta) > \Phi(r/2^n) + (M + 10/3)(1/2^n) + (M + 4/3)(\xi + \eta - (r+1)/2^n) \tag{ix}$$

Subtracting (viii) from (ix) and dividing by η gives, after a straightforward simplification,

$$(\Phi(\xi+\eta)-\Phi(\xi))/\eta > (M+4/3)-(2\xi-2r/2^n-(4/3).(1/2^n))/\eta \qquad \text{(x)}$$

But $2\xi-2r/2^n-(4/3).(1/2^n) < 4/3$, for, by (iv), the left hand side is less than $(1/2^{n+1}-(4/3).(1/2^n))/\eta < 2^{n+1}((1/2^{n-1})-(4/3).(1/2^n)) = 4-8/3 = 4/3$. Hence, finally, (x) gives $(\Phi(\xi+\eta)-\Phi(\xi))/\eta > M$. This completes the proof of the lemma.

Now let $\xi = .\varepsilon_1\varepsilon_2\varepsilon_3\ldots$ ($\varepsilon_i = 0$ or 1) and let us see what the inequalities

$$k_n/2^n < \xi < (k_n+1)/2^n \qquad (*)$$

say about the ε_i and the $D_n = O_n - I_n$.

For $n = 0$, $k_1 = 0$ and (*) becomes $0 < \xi < 1$.

For $n = 1$,

$$k_1 = \begin{cases} 0 & \text{if } 0 < \xi < 1/2\,, \\ 1 & \text{if } 1/2 < \xi < 1\,, \end{cases}$$

For $n = 2$,

$$k_2 = \begin{cases} 0 & \text{if } 0 < \xi < 1/4\,, \\ 1 & \text{if } 1/4 < \xi < 1/2\,, \\ 2 & \text{if } 1/2 < \xi < 3/4\,, \\ 3 & \text{if } 3/4 < \xi < 1\,, \end{cases}$$

Lemma 3.2.4. *The slope of $\Phi_{n-1}(\xi) = D_n$ (for our ξ), $n \geq 1$.*

Proof. Use induction. For $n = 1$, we have to show that slope $\phi_0(\xi) = D_1$. Here, the left hand side is equal to 1 if $0 < \xi < 1/2$ and is -1 if $1/2 < \xi < 1$; while the right hand side is equal to $\varepsilon_1 = 1$ if ξ belongs to the left half of $(0,1)$ and is equal to -1 if ξ belongs to the right half of $(0,1)$ (see how the binary expansions go in $(0,1)$). That starts our induction.

Suppose the result is true for n, i.e., suppose that slope $\Phi_{n-1} = D_n$ (for ξ); then we have to show that slope $\Phi_n = D_{n+1}$ (for ξ). Here, the left hand side is equal to

$$\text{slope } \Phi_{n-1}(\xi) + \begin{cases} 1 & \text{if } \xi \in (k_n/2^n, (2k_n+1)/2^{n+1})\,, \\ -1 & \text{if } \xi \in ((2k_n+1)/2^{n+1}, (k_n+1)/2^n)\,, \end{cases}$$

which is equal to $D_n \pm 1 = D_{n+1}$, using induction hypothesis.

Lemma 3.2.5.

$$\Phi'(\xi) = \begin{cases} \infty\,, \\ -\infty\,, \\ \textit{nonexistent (always, by Theorem 3.1.2), even as } \pm\infty\,, \end{cases}$$

according as

$$\lim_{n\to\infty} D_n = \begin{cases} \infty\,, \\ -\infty\,, \\ \textit{nonexistent}\ . \end{cases}$$

Proof.

$$\begin{aligned} \Phi'(\xi) &= \lim_{n\to\infty}[(\Phi((k_n+1)/2^n) - \Phi(k_n/2^n))/(1/2^n)] \\ &= \lim_{n\to\infty}[(\Phi_{n-1}((k_n+1)/2^n) - \Phi_{n-1}(k_n/2^n))/(1/2^n)] \\ &= \lim_{n\to\infty}(\text{slope of } \Phi_{n-1} \text{ in } (k_n/2^n, (k_n+1)/2^n)) \\ &= \lim_{n\to\infty} D_n \qquad \text{(by Lemma 3.2.4)}\ . \end{aligned}$$

The proof of Theorem 3.2.1 now follows easily from the above lemmas.

Corollary 3.2.6. *Let ξ be a rational number in $(0,1)$, which is not of the form $k/2^n$. If, in a complete period of the dyadic (binary) expansion of ξ,*

(i) the number of 1's exceed the number of 0's, then $\Phi'(\xi) = -\infty$,

(ii) the number of 0's exceed the number of 1's, then $\Phi'(\xi) = \infty$,

(iii) the number of 1's and 0's are equal, then $\Phi'(\xi)$ does not exist, even as $\pm\infty$.

Proof. This is quite obvious once one works out one or two typical examples.

The following examples will clarify the above statements.

Example 3.2.7. Let $\xi = 1/3 = .010101\ldots$ in the binary representation; the period is 01. We have the following table:

n	1	2	3	4	5	$\cdots$ $\cdots$
I_n	0	1	1	2	2	$\cdots$ $\cdots$
O_n	1	1	2	2	3	$\cdots$ $\cdots$
D_n	1	0	1	0	1	$\cdots$ $\cdots$

We see that D_n does not tend to any limit.

Example 3.2.8. Let $\xi = .01100|01100|01100\ldots$, then we have the following chart:

n	1 2 3 4 5	6 7 8 9 10	11 12 ...
I_n	0 1 2 2 2	2 3 4 4 4	4 ...
O_n	1 1 1 2 3	4 4 4 5 6	7 ...
D_n	1 0 -1 0 1	2 1 0 1 2	3 ...

We see that $D_n \to \infty$

Example 3.2.9 Let $\xi = 101|01100|01100|\ldots$, i.e., ξ starts off with say, three off period terms initially. A table giving the values of I_n, O_n, D_n will immediately make it clear that the initial 1, 0, 1 do not affect $\lim D_n$.

So far we have basically only considered van der Waerden's function $\Phi(x)$ or its variants. This function is, geometrically, quite interesting and analytically reasonably straight forward in the demonstration of continuity and nondifferentiability. There are lots of other ways such continuous, nowhere differentiable functions can be constructed; some of them give a clear geometric picture of the functions while others lend themselves to a simple analytic proof of continuity and nondifferentiability. We close this chapter with two extreme examples: one where the geometric picture is very revealing; the other, where the analytic verification of continuity and nondifferentiability is very simple.

§3.3. A geometric example

We first look at the purely geometric construction, where it will be seen that verification of its continuity and nondifferentiability will be found to be no more difficult than in any of the previous cases. We shall see that the idea behind this geometric construction is extremely simple and elegant (see A.A. Blank, [13]). Given a line joining the two points $(x_1, y_1), (x_2, y_2)$, $x_1 < x_2$, $y_1 \neq y_2$ (Figure 3.13), we construct a zigzag of three lines from it (Figure 3.14) by the following procedure:

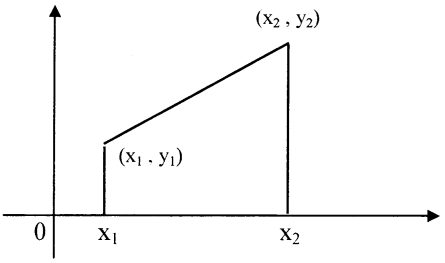


Figure 3.13

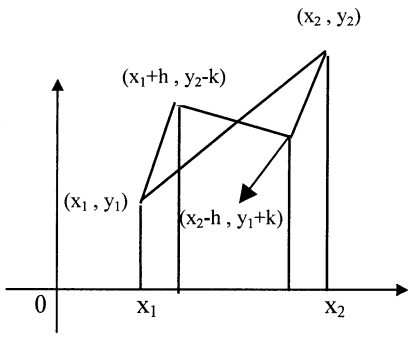


Figure 3.14

Let $h = (x_2 - x_1)/3$, $k = \lambda(y_2 - y_1)$, where λ is a fixed positive number, to be chosen later. Then Figure 3.14 is got simply by joining the vertices (x_1, y_1),

$(x_1 + h, y_2 - k)$, $(x_2 - h, y_1 + k)$, (x_2, y_2). The differences in the ordinates of successive vertices are, respectively $(y_1 - k) - y_1$, $(y_1 + k) - (y_2 - k)$, $y_2 - (y_1 + k)$, i.e., $(1 - \lambda)(y_2 - y_1)$, $(2\lambda - 1)(y_2 - y_1)$, $(1 - \lambda)(y_2 - y_1)$.

First condition on λ: Choose λ (if possible) such that

1. the maximum of the absolute values of these three differences equals $(1 - \lambda)|y_2 - y_1| = \mu|y_2 - y_1|$, say; and

2. $\mu\ (= 1 - \lambda) < 1$.

Next, let the slope of the line joining (x_1, y_1) to (x_2, y_2) be m, so that $m = \frac{(y_2 - y_1)}{(x_2 - x_1)}$ (m may have any sign). The slope of the zigzag lines are then (check)

$$3(1 - \lambda)m,\ 3(2\lambda - 1)m,\ 3(1 - \lambda)m.$$

Second condition on λ: Choose λ (if possible) such that

1. the minimum of the absolute value of these three slopes equals $3(1 - 2\lambda)|m| = \nu|m|$, say; and

2. $\nu\ (= 3(1 - 2\lambda)) > 1$.

Both of these sets of conditions are satisfied if $0 < \lambda < 1/3$. Indeed, the first condition on λ means that (i) $\max(|1 - \lambda|, |2\lambda - 1|) = 1 - \lambda$ and (ii) $1 - \lambda < 1$. Here (ii) implies that $0 < \lambda$, while (i) implies that $|2\lambda - 1| \leq |1 - \lambda|$ and $|1 - \lambda| = 1 - \lambda$. The latter condition means $\lambda < 1$, while the former says $-|1 - \lambda| \leq 2\lambda - 1 \leq |1 - \lambda|$ or since $\lambda < 1$, $-1 + \lambda \leq 2\lambda - 1 \leq 1 - \lambda$, i.e., $0 \leq \lambda \leq 2/3$. The conditions above then yields

$$0 < \lambda \leq 2/3 \qquad \text{(A)}$$

The second condition on λ means that (i) $\min(3|2\lambda - 1|, 3|1 - \lambda|) = 3(1 - 2\lambda)$ and (ii) $3(1 - 2\lambda) > 1$. Here (ii) implies that $\lambda < 1/3$, while (i) implies that $|2\lambda - 1| \leq |1 - \lambda|$ and that $|2\lambda - 1| = 1 - 2\lambda$. This latter condition means $2\lambda - 1 \leq 0$ or $\lambda \leq 1/2$, while the former says $-(1 - \lambda) \leq 2\lambda - 1 \leq 1 - \lambda$ (remember that $|1 - \lambda| = 1 - \lambda$, since $1 - \lambda > 0$), i.e., $0 \leq \lambda \leq 2/3$. The conditions above then give

$$0 \leq \lambda < 1/3 \qquad \text{(B)}$$

Both (A) and (B) are satisfied if and only if $0 < \lambda < 1/3$, as required.

These conditions on λ imply that the entire zigzag of three lines on the open interval (x_1, x_2) remains between the lines $y = y_1$ and $y = y_2$.

We take $\lambda = 3/10$ for drawing a graph. Any λ satisfying $0 < \lambda < 1/3$, will, of course, do.

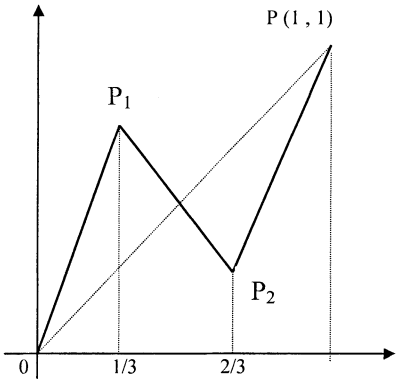


Figure 3.15

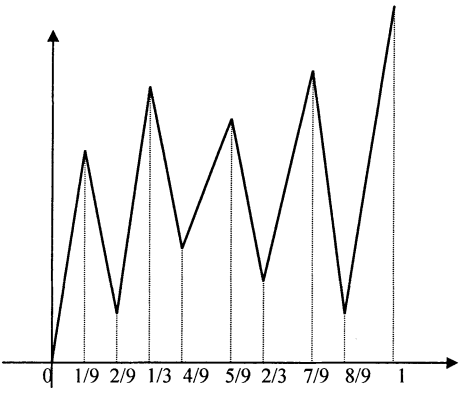


Figure 3.16

For the construction of our function, we fix a λ (with $0 < \lambda < 1/3$). Start with the line segment joining $(0,0)$, $(1,1)$. Then construct the first zigzag of the three lines as described above to get Figure 3.15. Next construct zigzags on each of OP_1, P_1P_2 and P_2P by the same procedure to get Figure 3.16 comprising a zigzag of nine lines and so on. Figure 3.17 shows the fourth step of this zigzag procedure, i.e., a zigzag of eighty one lines, drawn with λ taken as .3.

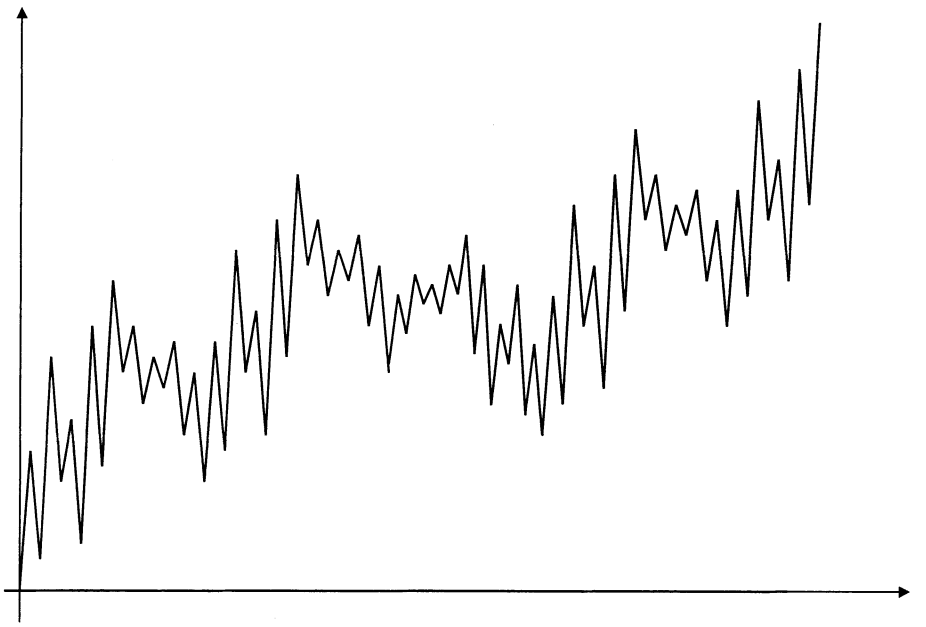


Figure 3.17

The abscissa of successive vertices of the n^{th} step zigzag (n^{th} approximating polygon) are the ternary points

$$x_{n,i} = \frac{i}{3^n}\ ,\ \ i = 0, 1, \ldots, 3^n$$

and each of these vertices continue to be vertices of all succeeding zigzags. They will be points on the graph of Weierstrass' function $\Omega(x)$.

This sequential construction yields a function f defined on all ternary points $x_{n,i}$ and $\Omega(x)$ is defined as the continuous extension of f to each point of the interval $[0, 1]$. We give below the details:

Let $x_{n,i}$, $x_{n,i-1}$ be successive ternary points of order n. Then by the first condition on λ, we have

$$|f(x_{n,i}) - f(x_{n,i-1})| \leq \mu^n\ ,\ \ i = 0, 1, \ldots, 3^n\ . \qquad \text{(I)}$$

where $\mu = 1 - \lambda$. Moreover, the slope of the the segment joining the corresponding vertices satisfies, by the second condition on λ,

$$\frac{f(x_{n,i}) - f(x_{n,i-1})}{x_{n,i} - x_{n,i-1}} \geq \nu^n\ , i = 1, 2, \ldots, 3^n \qquad \text{(II)}$$

where $\nu = 3(1 - 2\lambda)$.

Let now $r \in (0, 1)$ be any real number. Then there exists a unique integer i_n (for a given n), such that $i_n - 1 \leq 3^n.r < i_n$, i.e., $(i_n - 1)/3^n \leq r < i_n/3^n$, i.e., $x_{n,i_n-1} \leq r < x_{n,i_n}$, or say $a_n \leq r \leq b_n$. It is easy to verify that $a_n \leq a_{n+1}$, $b_{n+1} \leq b_n$. Indeed, we have to prove that $(i_n - 1)/3^n \leq (i_{n+1} - 1)/3^{n+1}$ or that $3(i_n - 1) \leq i_{n+1} - 1$. Now, by the definition of i_n, i_{n+1}, we have that (i) $i_n - 1 \leq 3^n.r < i_n$ (ii) $i_{n+1} - 1 \leq 3^{n+1}.r < i_{n+1}$. Multiply (i) by 3 to get $3.i_n - 3 \leq 3^{n+1}.r < 3.i_n$, i.e., $3^{n+1}.r$ can be anywhere in the interval $[3\,i_n - 3, 3\,i_n)$. In any case the integer on its left is greater than or equal to $3\,i_n - 3$, i.e., $i_{n+1} - 1 \geq 3\,i_n - 3$, as required.

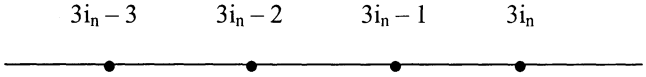


Figure 3.18

Also, $f(a_{n+1})$ and $f(b_{n+1})$ both lie in the interval $[f(a_n), f(b_n)]$ (or $[f(b_n), f(a_n)]$). This gives us the sequence of nested intervals

$$[f(a_1), f(b_1)] \supset [f(a_2), f(b_2)] \supset \cdots \supset [f(a_n), f(b_n)] \supset \cdots$$

(in case $f(a_i) > f(b_i)$, consider $[f(b_i), f(a_i)]$ etc.). The length of the n^{th} interval is equal to $|f(b_n) - f(a_n)| \leq \mu^n$ (by (I)), which tends to 0 as $n \to \infty$,

since $\mu < 1$. Hence by the nested interval property, there exists a real number ξ such that $\xi \in [f(a_n), f(b_n)]$ (for all n) and we define

$$\Omega(r) = \xi \qquad \text{(III)}$$

Proof of continuity of $\Omega(x)$: Let $\epsilon > 0$ be given and choose n so large that $\mu^n < \epsilon/2$ and take $\delta = 1/4^n$. We have to show that $|\Omega(x) - \Omega(r)| < \epsilon$ if $|x - r| < \delta$. First, let $x < r$, and of course, $|x - r| < \delta$. Then (see Figure 3.19)

$$|\Omega(x) - \Omega(r)| \leq |\Omega(x) - \Omega(a_n)| + |\Omega(a_n) - \Omega(r)| \,.$$

But

$$\begin{aligned} |\Omega(a_n) - \Omega(r)| &= |\Omega(a_n) - \xi| \\ &\leq |\Omega(a_n) - \Omega(b_n)| \quad (\text{since } \xi \in [\Omega(a_n), \Omega(b_n)]) \\ &= \Omega(a_n) - \Omega(b_n) \quad (\text{since } a_n \text{ and } b_n \text{ are ternary points}) \\ &\leq \mu^n < \epsilon/2 \,, \end{aligned}$$

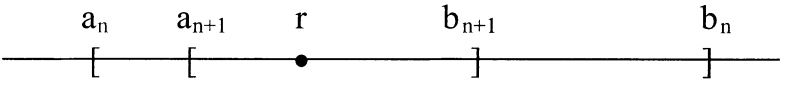


Figure 3.19

while $|\Omega(x) - \Omega(a_n)| \leq \mu^n < \epsilon/2$. That completes the proof of continuity.

Analytic proof of nondifferentiability. We have

$$\frac{|\Omega(b_n) - \Omega(r)|}{b_n - r} + \frac{|\Omega(r) - \Omega(a_n)|}{r - a_n} > \frac{|\Omega(b_n) - \Omega(r)|}{b_n - a_n} + \frac{|\Omega(r) - \Omega(a_n)|}{b_n - a_n}$$

(in fact either term on the left side is more than the corresponding term on the right side). Hence the above expression is more than or equal to $|\Omega(b_n) - \Omega(a_n)|/(b_n - a_n) \geq \nu^n$. Thus, at least one of the slopes $\frac{\Omega(b_n)-\Omega(r)}{b_n-r}$, $\frac{\Omega(r)-\Omega(a_n)}{r-a_n}$ has absolute value more than $\nu^n/2$, which means that we can not place a bound on the steepness of the chords with end point at $(r, \Omega(r))$ in any neighbourhood of r (as any neighbourhood contains $[a_n, b_n]$, if n is large), i.e., Ω is not differentiable at r.

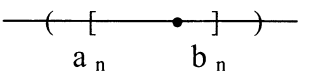


Figure 3.20

Geometric proof of nondifferentiability. We consider two cases.

Case 1: Suppose that r is a ternary point of order n, say $r = i/3^n$, then r is a ternary point of every higher order: $r = 3^k.i/3^{n+k}$ and it follows from (II) that no bound can be placed on the steepness of the chords of the graph of Ω with one point at $(r, \Omega(r))$, so Ω is not differentiable at r.

Case 2: Suppose that r is not a ternary point. Then $a_n < r < b_n$, with strict inequality for all n. Then the steepness of one of the two chords joining $(r, \Omega(r))$ to $(a_n, \Omega(a_n))$ and $(b_n, \Omega(b_n))$ is at least as great as the steepness of the chord C joining $(a_n, \Omega(a_n))$ to $(b_n, \Omega(b_n))$. For example, if $\Omega(a_n) < \Omega(b_n)$, then in case $(r, \Omega(r))$ lies above C, the chord joining $(r, \Omega(r))$ to $(a_n, \Omega(a_n))$ is steeper (Figure 3.21 (left)), while if $(r, \Omega(r))$ lies below C, the chord joining $(r, \Omega(r))$ to $(b_n, \Omega(b_n))$ is steeper (Figure 3.21 (right)).

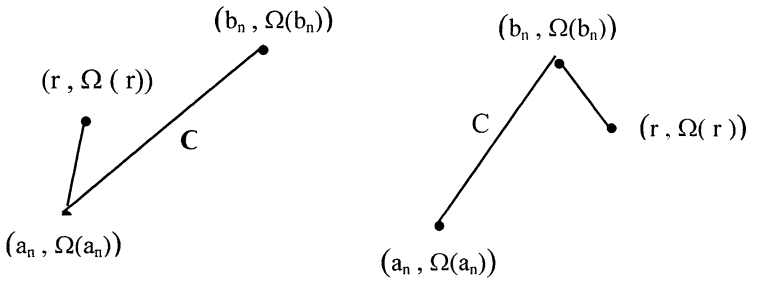


Figure 3.21

A similar argument holds if $\Omega(a_n) > \Omega(b_n)$, and again by (II), nondifferentiability follows.

§3.4. An analytic example

We end this discussion on everywhere continuous, nowhere differentiable function with the following example, which is not only short in its definition but also in the proof of its being continuous and nondifferentiable (see William C. Swift, [107]), but it is difficult to visualize its graph.

It is, as usual, enough to define our function Φ on $[0, 1]$. So let $x \in [0, 1]$ and write x in its ternary expansion

$$x = .x_1 x_2 \cdots x_n \cdots \text{(base 3)}$$

so that each $x_n = 0, 1$ or 2. Define $\Phi(x)$, written out in the its binary expansion as

$$\Phi(x) = .y_1 y_2 \cdots y_n \cdots \text{(base 2)}$$

so that each $y_n = 0$ or 1, where

$y_1 = 1$ if and only if $x_1 = 1$ and $y_{n+1} = y_n$ if and only if $x_{n+1} = x_n$. (*)

Example 3.4.1. Let $x = .2101120001122\ldots = .x_1x_2x_3x_4x_5x_6\ldots$, then $\Phi(x) = .0101101110011\ldots = .y_1y_2y_3y_4y_5y_6\ldots$.

It is easy to verify that if x is expressible in two distinct ternary expansions, the two binary expansions for $\Phi(x)$ define the same real number, for example: If $x = .21122222\ldots = .21200000\ldots$, then according to the first expression for x, we have $\Phi(x) = .01100000\ldots$, while according to the second expression for x, we have $\Phi(x) = .01011111\ldots$ and these two are equal since $.100000\ldots = .011111\ldots$.

Proof of nondifferentiability of $\Phi(x)$. Let $x = .x_1x_2\ldots x_nx_{n+1}x_{n+2}\ldots$ (base 3), and $\Phi(x) = .y_1y_2\ldots y_ny_{n+1}y_{n+2}\ldots$ (base 2). We construct a sequence of real numbers $\{\xi_n\}$ tending to x, such that the differential quotient $(\Phi(\xi_n) - \Phi(x))/(\xi_n - x)$ does not tend to any finite limit as the sequence $\{\xi_n\}$ approaches x. For each given n, let $\xi_n = .x_1x_2\ldots x_nx'_{n+1}x'_{n+2}\ldots$ (base 3), where we select $x'_{n+1}, x'_{n+2},\ldots$ such that $\Phi(\xi_n) = .y_1y_2\ldots y_ny'_{n+1}y'_{n+2}$ (base 2), and where $y'_{n+1} \neq y_{n+1}$, but otherwise the corresponding digits of $\Phi(x)$ and $\Phi(\xi_n)$ are identical. This can be done, using (*), as a moment's thought reveals. This gives our required sequence $\{\xi_n\}$, for which we have

$$0 < |\xi_n - x| \leq 1/3^n \quad \text{and} \quad |\Phi(\xi_n) - \Phi(x)| = 1/2^{n+1},$$

and therefore, we have $|(\Phi(\xi_n) - \Phi(x))/(\xi_n - x)| \geq (1/2)(3/2)^n$, which shows that $\Phi(\xi_n) - \Phi(x))/(\xi_n - x)$ can not tend to a finite limit as ξ_n tends to x, i.e., that $\Phi'(x)$ does not exist.

Remark 3.4.2. The function $\Phi(x)$ can be extended periodically on the entire real line. The only point to observe, then, is that $\Phi(x)$ is continuous at $x = 0$ and $x = 1$. For this, it is enough to check that $\Phi(0) = \Phi(1)$. Since $0 = .000\ldots$ (base 3), we have $\Phi(0) = .000\ldots$ (base 2), (by (*)), and since $1 = .222\ldots$ (base 3), we have $\Phi(1) = .000\ldots$ (base 2) (again by (*)).

There have been many new constructions of everywhere continuous, nowhere differentiable functions, treated by many authors. Since we have had a good many examples of such functions already, we end this chapter here.

Chapter 4

Functions : continuous, periodic, locally recurrent and others

In this chapter we shall study some important properties of functions. The stress will be on various examples, counter examples and nonconventional proofs. We begin with:

§4.1. The intermediate value property

We say that a real-valued function f, defined on an interval $\mathcal{I}$ has *intermediate value property* if for $a, b \in \mathcal{I}$ and for any point c between $f(a)$ and $f(b)$, there exists a point ξ between a and b such that $f(\xi) = c$. We call this property *IVP* for short.

One of the few results that one proves about continuous functions is that every continuous function has IVP. The continuity of the function is only a sufficient condition for it to have IVP.

Example 4.1.1. A function may satisfy IVP, and yet fail to be continuous. For example, the function f defined on $[-1, 1]$ by

$$f(x) = \begin{cases} \sin \frac{1}{x} , & \text{if } x \neq 0 \\ 0 , & \text{if } x = 0 \ , \end{cases}$$

has IVP in $[-1, 1]$, but fails to be continuous at the point $x = 0$.

Indeed, it is possible to construct a function that has IVP in every interval, however small, but fails to be continuous.

Example 4.1.2. Let us take our domain to be $(0, 1)$. For $x \in (0, 1)$, write x in its decimal expansion: $x = .a_1 a_2 a_3 \ldots$ and let $z = .a_1 a_3 a_5 \ldots$. Define the function f as follows:

(1) If z is not periodic, put $f(x) = 0$.

(2) If z is periodic with its first period beginning with a_{2n-1}, put $f(x) = .a_{2n} a_{2n+2} a_{2n+4} \ldots$.

(Note that we can always regard the period as beginning with an odd suffix, e.g., $1/6 = .1\overset{\uparrow}{6}66\ldots$, with period starting at $\uparrow$, i.e., at a_2 but also $1/6 = .16\overset{\uparrow}{6}66\ldots$, with period beginning at $\uparrow$, i.e., at a_3).

To see that f has the required property, let $\mathcal{I} \subset (0,1)$ be an open interval. Find n so large that $\mathcal{I}$ contains a terminating decimal $.a_1a_2\ldots a_{2n-1}000\ldots$ and with it all the numbers $.a_1a_2\ldots a_{2n-1}a_{2n}a_{2n+1}\ldots$ that begin with the same first $2n-1$ digits (the latter number is very near the former and $\mathcal{I}$ is open).

Now, let $y = .b_1b_2\ldots$ be any number in $(0,1)$. Choose a_{2n+1}, $a_{2n+3}, \ldots$, so that $.a_1a_3\ldots a_{2n-1}a_{2n+1}a_{2n+3}\ldots$ is periodic with its first period starting at a_{2n-1}, and hence, by our construction (2), the number

$$x = .a_1a_2\ldots a_{2n-1}b_1a_{2n+1}b_2a_{2n+3}b_3\ldots$$

is such that $f(x) = y$. Using (1), it is not difficult to see that f is not continuous.

We now make the formal :

Definition 4.1.3. We say that the function f defined on an interval $\mathcal{I}$ has intermediate value property (also called the Darboux property), if for each $x, y \in \mathcal{I}$, $x \leq y$, the closed interval $[f(x), f(y)](or[f(y), f(x)]) \subseteq \{f(z) \mid x \leq z \leq y\}$.

A striking result (proved in the next chapter) is the following:

Theorem 4.1.4 (Darboux). *If $f'(x)$ (the derivative of f at x) exists for every x in $[a,b]$, then f' has IVP in $[a,b]$.*

Since f' may exist, but fail to be continuous (for example, the function $f(x) = x^2 \sin \frac{1}{x}$, if $x \neq 0$, $f(0) = 0$), we get a class of noncontinuous functions satisfying IVP.

If f and g are continuous in $[a,b]$, then $f+g$ has IVP, as it is continuous. However, we have:

Example 2.1.5. If the functions f and g have the IVP, it does not follow that $f+g$ has the IVP. Let

$$\varphi(t) = t^2 \sin \frac{1}{t}, \text{ if } t \neq 0, \varphi(0) = 0 \quad \text{and} \quad \psi(t) = t^2 \cos \frac{1}{t}, \text{ if } t \neq 0, \psi(0) = 0.$$

Then

$$\varphi'(t) = 2t \sin \frac{1}{t} - \cos \frac{1}{t} \quad \text{if } t \neq 0,\ \varphi'(0) = 0$$
$$\psi'(t) = 2t \cos \frac{1}{t} + \sin \frac{1}{t} \quad \text{if } t \neq 0,\ \psi'(0) = 0.$$

Let $f(t) = (\varphi'(t))^2$, $g(t) = (\psi'(t))^2$. Then f, g have IVP, since φ', ψ' do have it, by Theorem 4.1.4 (Note that if F has IVP, then F^2 has IVP, for: $F^2(x) \leq c \leq F^2(y)$ implies that either $F(x) \leq \sqrt{c} \leq F(y)$ or $F(y) \leq -\sqrt{c} \leq F(x)$. Thus, there exists $z \in [x,y]$ such that $F(z) = \sqrt{c}$ (or $F(z) = -\sqrt{c}$), so that $F^2(z) = c$, as required). But $(f+g)(t) = 4t^2 + 1$ if $t \neq 0$, $(f+g)(0) = 0$, so $f+g$ does not have the IVP on any interval containing 0.

Remark 4.1.6. Discontinuous functions with IVP can be such that their sum

has IVP, for example, let f, g be such that f', g' exists in $[a, b]$, then f', g' have IVP and so has $f' + g' = (f + g)'$ (by Theorem 4.1.4). Now consider the question: Suppose f is continuous on $[a, b]$, g has IVP on $[a, b]$, must $f + g$ have IVP in $[a, b]$? It is surprising that the answer is *no*. A counterexample is given in [73].

There is a partial converse to the IVP of continuous functions, namely:

Theorem 4.1.7. *Let the function f, defined in $[a, b]$, have the property that*

(i) for any $c, d \in [a, b]$, f assumes every value between $f(c)$ and $f(d)$,

(ii) no rational value is assumed infinitely often by f,

then f is continuous in $[a, b]$.

Proof. Suppose f is discontinuous at $x = \xi$, say, on the left; then the statement

"given $\epsilon > 0$, $\exists$ a $\delta > 0$ such that $|f(x) - f(\xi)| < \epsilon$ for all $x \in (\xi - \delta, \xi)$"

is false, i.e., for some $\epsilon > 0$, $f(x) > f(\xi) + \epsilon$ for some x in $(\xi - \delta, \xi)$ for each δ, however small. So, there exists an increasing sequence $x_1 < x_2 < \ldots < x_n < \ldots$, approaching ξ, and with $f(x_n) > f(\xi) + \epsilon$ (for each n).

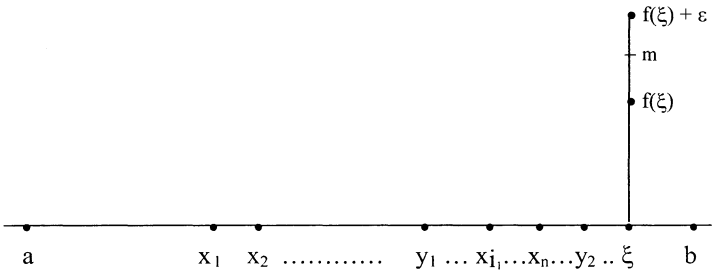


Figure 4.1

Let m be any rational in $[f(\xi), f(\xi) + \epsilon] \subset [f(\xi), f(x_1)]$ (by above). So by (i), there exists a $y_1 \in (x_1, \xi)$ with $f(y_1) = m$. Select an x_{i_1} of the sequence $\{x_n\}$ to the right of y_1: $y_1 < x_{i_1}$.

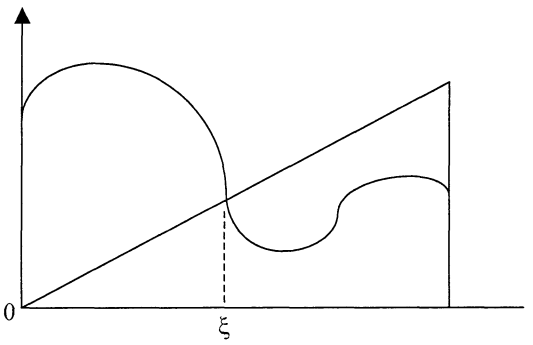


Figure 4.2

Again, as above, $m \in [f(\xi), f(\xi)+\epsilon] \subset [f(\xi), f(x_{i_1})]$. So by (i), there exists a $y_2 \in (x_{i_1}, \xi)$ with $f(y_2) = m$, and so on with $y_3, y_4, \ldots$. Thus f assumes the rational value m infinitely often in $[x_1, \xi]$, contradicting (ii).

The IVP also yields the following very simple fixed-point theorem (see Figure 4.2):

Theorem 4.1.8. *Let $f : [a,b] \to [a,b]$ be a continuous functions. Then there exists a $\xi \in [a,b]$ such that $f(\xi) = \xi$.*

Proof. The function $g(x) = f(x) - x$ has the property that $g(a) = f(a) - a \geq 0$ and $g(b) = f(b) - b \leq 0$ (since $f(a), f(b)$ both lie in $[a,b]$) and so by IVP, $g(\xi) = 0$ for some $\xi \in [a,b]$.

Another way of saying this is that if $f : [0,1] \to [0,1]$ is continuous, then the graph of f must cross the line $y = x$.

§4.2. Boundedness and attainment of bounds

We know that a continuous function defined on a closed interval is bounded and attains its bounds. We now give two proofs of this important result, both different from the ones given in the usual Calculus courses.

Theorem 4.2.1. *Let the function f be continuous on $[a,b]$. Then there exists a $\xi \in [a,b]$ such that $f(x) \leq f(\xi)$ for all $x \in [a,b]$.*

First Proof (W.B.Pennington, [79]). Let $a_0 = a$, $b_0 = b$. Define a_{n+1}, b_{n+1} inductively from a_n, b_n as follows. Let $c_n = (a_n + b_n)/2$. If there exists a number $\alpha \in [a_n, c_n]$ such that $f(x) \leq f(\alpha)$ for all $x \in [c_n, b_n]$, we let $a_{n+1} = a_n$, $b_{n+1} = c_n$. If not (i.e., if for each $\alpha \in [a_n, c_n]$, there exists a number $x \in [c_n, b_n]$, with $f(\alpha) < f(x)$), we let $a_{n+1} = c_n, b_{n+1} = b_n$.

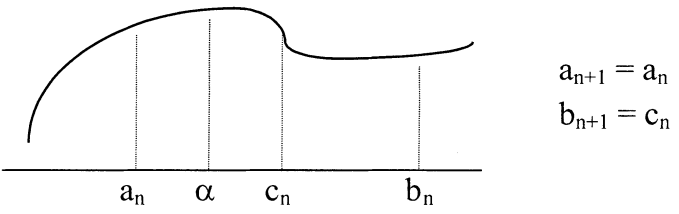


Figure 4.3

In either case, we have

(*i*) $a_n \leq a_{n+1} \leq b_{n+1} \leq b_n$,

(*ii*) $b_{n+1} - a_{n+1} = (b_n - a_n)/2$,

(*iii*) if $x \in [a_n, b_n]$, there exists a $y \in [a_{n+1}, b_{n+1}]$ such that $f(x) \leq f(y)$ (easily seen).

This inductive process defines a sequence of intervals $[a_n, b_n]$, $n = 0, 1, 2, \ldots$, such that

$(i)'$ $a_m \leq a_n \leq b_m \leq b_0$, if $n \geq m \geq 0$,

$(ii)'$ $b_n - a_n \to 0$ as $n \to \infty$,

$(iii)'$ if $x \in [a, b]$ and $N \in \mathbb{N} \cup \{0\}$, there exists a $y \in [a_N, b_N]$ such that $f(x) \leq f(y)$.

By $(i)'$, $a_n \to \xi$ as $n \to \infty$ and $\xi \in [a_m, b_m] \subset [a, b]$ (for each $m = 0, 1, 2, \ldots$).

We shall prove that $f(x) \leq f(\xi)$ for all $x \in [a, b]$. So, let $\alpha \in [a, b]$ be any point. We shall show that $f(\alpha) \leq f(\xi)$. Choose $t > f(\xi)$. Since f is continuous in $[a, b]$, $t > f(y)$ for all y in a neighbourhood of ξ, i.e., there exists a δ such that

$$x \in (\xi - \delta, \xi + \delta) \cap [a, b] \Longrightarrow f(x) < t \,. \tag{1}$$

By $(ii)'$, choose N large enough to ensure that $b_N - a_N < \delta$ and then by $(iii)'$, choose $\beta \in [a_N, b_N]$ such that

$$f(\alpha) \leq f(\beta) \,. \tag{2}$$

Since $\xi \in [a_N, b_N]$, we have $|\beta - \xi| \leq b_N - a_N < \delta$ (and also $\beta \in [a, b]$). Hence, by (1)

$$f(\beta) < t \,. \tag{3}$$

Thus, (2) and (3) imply that

$$f(\alpha) < t \,. \tag{4}$$

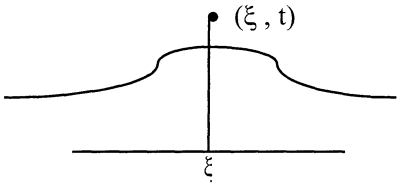


Figure 4.4

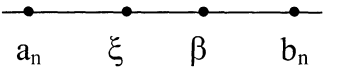


Figure 4.5

It follows that $f(\alpha) < f(\xi)$ (for otherwise $f(\alpha) > f(\xi)$ and taking $t = f(\alpha)$, (4) would give $f(\alpha) < f(\beta)$, which is a contradiction).

Second proof (Gerald Jungck [53]). If one of $f(a)$ or $f(b)$ is a maximum, we are through. So, suppose that there exists a point $c \in (a, b)$ such that $f(c) > f(a)$ and $f(b)$. By the continuity of f, $f(c) > f(x)$ throughout small neighbourhoods of a, b :

$$f(c) > f(x) \text{ for all } x \in [a, a + \epsilon) \cup (b - \epsilon, b] \tag{1}$$

Thus $c \in [a + \epsilon, b - \epsilon]$.

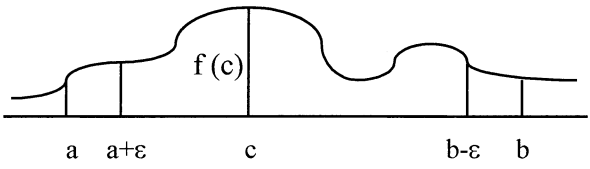


Figure 4.6

Now let

$$S = \{\gamma \in (a, b) \mid \exists \text{ a point } d \in [\gamma, b] \text{ with } f(d) > f(x) \ , \forall\, x \in [a, \gamma)\}.$$

We see that $S \neq \emptyset$, since, by (1), $a + \epsilon \in S$, with $d = c$. Moreover, b is an upper bound for S, i.e., S is bounded. Let $\xi = \sup S$. Then as $a + \epsilon \in S$, we see that

$$a + \epsilon \leq \xi \ . \tag{2}$$

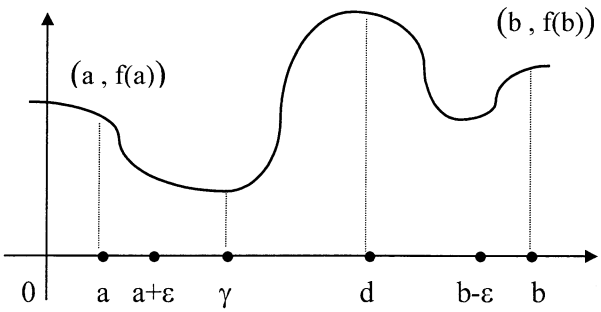


Figure 4.7

We prove $\xi \leq b - \epsilon$. Otherwise, there exists a point $\sigma \in (b - \epsilon, \xi]$, which is in S and this means that there exists $d \in [\sigma, b]$, such that $f(d) > f(x)$ for all $x \in [a, \sigma)$, in particular for $x = c$, i.e.,

$$f(d) > f(c) \ . \tag{3}$$

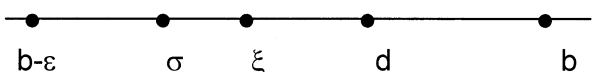


Figure 4.8

But by (1), $f(c) > f(x)$ for all $x \in [a, a + \epsilon) \cup (b - \epsilon, b]$, in particular, for $x = d$; i.e.,

$$f(c) > f(d) \ . \tag{4}$$

As (3) and (4) contradict each other, $\xi \leq b - \epsilon$, as claimed, i.e., (see (2)), $\xi \in [a + \epsilon, b - \epsilon]$.

We now prove that $f(\xi)$ is maximum, i.e., that $f(\xi) \geq f(x)$ for all $x \in [a, b]$. Suppose not; then $f(\zeta) > f(\xi)$ for some $\zeta \in [a, b]$, and so by the continuity of f, there exists δ $(< \epsilon)$ such that

$$f(\zeta) > f(x) \text{ for all } x \in (\xi - \delta, \xi + \delta) \ . \tag{5}$$

But by the definition of ξ as $\sup S$, there exists a point $d \in [c, d]$ such that

$$f(d) > f(x) \text{ for all } x \in [a, c) \text{ and } [a, c) \supset [a, \xi - \delta] \tag{6}$$

Let $f(\eta)$ be the greater of $f(\zeta), f(d)$. Then

$$f(\eta) \geq f(\zeta) > f(x) \text{ (by(5)) for all } x \in (\xi - \delta, \xi + \delta) \tag{7}$$

and

$$f(\eta) \geq f(d) > f(x) \text{ (by(6)) for all } x \in [a, \xi - \delta] \ . \tag{8}$$

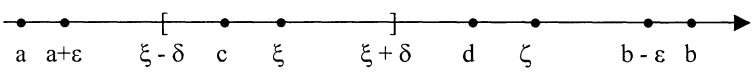


Figure 4.9

Equations (7) and (8) means $f(\eta) > f(x)$ for all $x \in [a, \xi + \delta)$ and so clearly that $\eta > \xi + \delta$. Hence $\xi + \delta \in S$, which contradicts the definition of ξ as $\sup S$.

Remark 4.2.2. This point ξ is the (unique) infimum of all points in $[a, b]$ that yields the maximum value of $f(x)$, i.e., $\xi = \inf\{z | f(z) \geq f(x) \text{ for all } x \in [a, b]\}$.

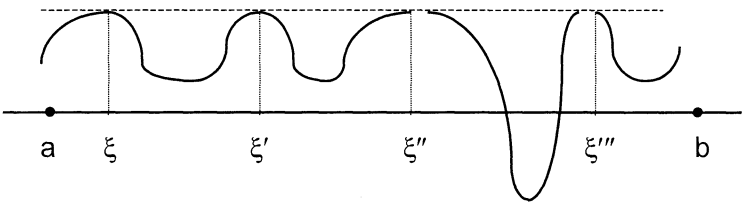


Figure 4.10

§4.3. Locally recurrent functions

This small class of functions, still in its developing stages, is defined as follows:

Definition 4.3.1. A function f, defined in the interval $[a, b]$, is said to be

locally recurrent at the point $c \in (a, b)$ if every deleted neighbourhood $N_D(c) = (c - \epsilon, c + \epsilon) \setminus \{c\}$ of c contains a point x such that $f(x) = f(c)$.

A function f is said to be *locally recurrent everywhere* in (a, b) if it is locally recurrent at each point c of (a, b).

Example 4.3.2. (i) The function $f(x) = x \sin \frac{1}{x}$, if $x \neq 0$, $f(0) = 0$, is locally recurrent at $x = 0$.
(ii) The function $f(x) = 1$ if x is rational, 0 otherwise, is locally recurrent everywhere.

Definition 4.3.3. The function f defined in $[a, b]$ is said to be *locally recurrent over right neighbourhoods at the point* $c \in (a, b)$, if every right neighbourhood $(c, c + \epsilon)$ of c contains a point x such that $f(x) = f(c)$.

Similarly we define *locally recurrent over left neighbourhoods at* c and *locally recurrent over right (or left) neighbourhoods everywhere.*

The following questions arise immediately:

Question 1. Does there exist an everywhere locally recurrent function which is nonconstant and continuous?
Question 2. Does there exist a function which is locally recurrent over right neighbourhoods everywhere and which is nonconstant and continuous?

Remark 4.3.4 At first it may seem that if f is locally recurrent everywhere, then it is locally recurrent over right neighbourhoods everywhere. In fact this is not the case. Locally recurrent everywhere means that for each point c, there exists either a left or a right neighbourhood of c where the value $f(c)$ occurs again. So for some point c, it may be that $f(c)$ occurs in the left neighbourhood but not the right neighbourhood (or vice versa), so f need not be locally recurrent over right neighbourhoods (or left neighbourhoods). This remark may therefore make the answers to the two questions above less surprising, viz. *yes* to question 1 and *no* to question 2.

In [20], K.A.Bush gives an example of a continuous nonconstant function on $[0, 1]$ which is locally recurrent everywhere. Interested reader may look up this example.

Our object here is to show that the answer to question 2 is *no*, i.e., we have the following:

Theorem 4.3.5. *There is no nonconstant continuous function, which is locally recurrent over right neighbourhoods everywhere.*

Proof. Suppose, to the contrary, that f is a nonconstant, continuous function that is locally recurrent over right neighbourhoods everywhere in $[0, 1]$, say. Fix $c \in (0, 1)$. Let $S = \{x \in (0, 1) | f(x) = f(c)\}$. Then $S^c = (0, 1) \setminus S$ is an open set. To see this, let $\alpha \in S^c$; then $f(\alpha) \neq f(c)$, say $f(\alpha) > f(c)$. By the continuity of f, $f(x) > f(c)$ in some open neighbourhood of α and so this open neighbourhood of α is contained in S^c, showing that S^c is open.

Write this open set S^c as a countable disjoint union of open intervals (we

can suppose that these are contained in $(0,1)$), i.e.,

$$S^c = \bigcup_{n=1}^{\infty} (a_n, b_n) .$$

Here a_n, $b_n \notin S^c$ and so they are in S, i.e., $f(a_n) = f(c)$. But f is locally recurrent over right neighbourhoods everywhere and so in particular at $x = a_n$, i.e., there exists a point β in a right neighbourhood of a_n at which $f(\beta) = f(a_n)$, and so $f(\beta) = f(c)$, showing that $\beta \in S$. But $\beta \in (a_n, b_n) \subset S^c$, which gives the required contradiction (see Figure 4.11).

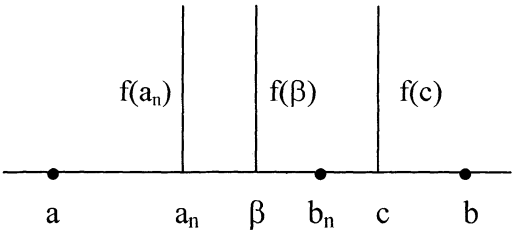


Figure 4.11

§4.4. Periodic functions

We begin by defining periodic functions.

Definition 4.4.1. Let f be a real-valued function on $\mathbb{R}$. We say that f is periodic if there exists a real number $p \neq 0$, such that $f(x+p) = f(x)$ for all real numbers x.

Definition 4.4.2. The real number p in Definition 4.4.1 is called a period of f.

Remark 4.4.3. If p is a period of f, clearly so is $-p$ and hence also $|p|$.

Definition 4.4.4. If there is a positive period ω such that $\omega \leq p$ for all positive periods p of f, then ω is called the fundamental period (abbreviated as FP) of f (ω is clearly unique, since if ω' is another one, then by the definition of the FP, $\omega \leq \omega'$, and $\omega' \leq \omega$).

Theorem 4.4.5. *Any period p is an integral multiple of ω.*

Proof. Since ω is the FP, $0 < \omega \leq |p|$. If $\omega = |p|$, then $p = \pm\omega$, as required; so suppose $0 < \omega < |p|$. Subtracting a suitable positive multiple $n\omega$ of ω from $|p|$, we see that there exists a unique positive integer n and a unique real number r such that $|p| = n\omega + r$ $(n > 0, 0 \leq r < \omega)$ and then $f(x) = f(x + |p|) = f(x + n\omega + r) = f(x + r)$ for all real numbers x. Thus, if $r \neq 0$, f would have a positive period r smaller than ω. Hence $r = 0$, giving $|p| = n\omega$.

Theorem 4.4.6. *Let f be periodic with a nonzero period and let*

$$\delta^* = \inf\{\delta > 0 \mid f(x+\delta) = f(x) \text{ for all } x \in \mathbb{R}\}.$$

Then either $\delta^* = 0$ or every period of f is an integral multiple of δ^*.

Proof. Suppose that $\delta^* \neq 0$ and let p be any nonzero period of f. If $p = 0$, then $p = 0.\delta^*$ is an integral multiple of δ^*. If δ^* is a period of f, then by Theorem 4.4.5, p is a multiple of δ^*. If δ^* is not a period of f, there exists arbitrarily small numbers ε such that

$$\delta^* + \varepsilon \text{ is a period of } f \qquad (*)$$

(this is one of the two characterizing properties of the infimum, in particular, of δ^*). Write (uniquely) $|p| = n\delta^* + r$, $n > 0$ an integer and $0 \leq r < \delta^*$. We wish to show that $r = 0$. Suppose, to the contrary, that $r > 0$. Let $\varepsilon > 0$ be a positive real number such that $\delta^* + \varepsilon$ is a period of f and $r - n\varepsilon > 0$ (possible by (*)). Then, for all real x, we have

$$\begin{aligned} f(x) &= f(x + |p|) \\ &= f(x + n\delta^* + r) \\ &= f(x + n(\delta^* + \varepsilon) + r - n\varepsilon) \\ &= f(x + (r - n\varepsilon)) \qquad \text{(by (*)).} \end{aligned}$$

Thus, $r - n\varepsilon$ is a period of f, with $0 < r - n\varepsilon < r < \delta^*$, which contradicts the definition of δ^*. It follows that $r = 0$ and so $|p| = n\delta^*$ is an integral multiple of δ^*.

It is still not clear that this number δ^*, defined above, is actually a period of f. This is the content of the following:

Theorem 4.4.7. *If the infimum δ^* of the positive periods of the periodic function f is nonzero, then it is a period of f.*

Proof. Suppose that $\delta^* > 0$ and is not a period. By a property of the infimum, there exists a small positive number, say ε, $0 < \varepsilon < \delta^*$, such that $\delta^* + \varepsilon$ is a period of f. But by Theorem 4.4.6, $\delta^* + \varepsilon = n\delta^*$ (for some positive integer n) and this is impossible as then $\delta^* < \delta^* + \varepsilon < 2\delta^*$.

The following matter, up to Theorem 4.4.15, calls for a stronger background on the part of the reader than what is required for the rest of the matter in the book.

Theorem 4.4.6 has a group theoretic counterpart which is worth noting (see [29]). Let f be a real valued function. Let

$$\begin{aligned} P_f &= \{p \in \mathbb{R} \mid f(x+p) = f(x) \text{ for all } x \in \mathbb{R}\} \\ &= \text{the set of all periods of } f \text{ together with } 0 \end{aligned}$$

Theorem 4.4.8. *P_f is an additive subgroup of $\mathbb{R}$.*

Proof. Clearly, $0 \in P_f$. Let $p, q \in P_f$. Then

$$\begin{aligned} f(x + (p - q)) &= f(x - q) \\ &= f((x - q) + q) = f(x) , \end{aligned}$$

shows that $p - q \in P_f$. Hence, P_f is a subgroup of $\mathbb{R}$.

Definition 4.4.9. P_f is called the *group of periods of* f and f is said to be periodic if $|P_f| > 1$ (note that 0 is always in P_f).

Also, if G is any additive subgroup of $\mathbb{R}$, then there exists a function g on $\mathbb{R}$ whose periods are precisely the elements of G, namely, the characteristic function χ_G of G (χ_G =1 if $x \in G$, and 0 otherwise).

With this relationship between periodic functions on $\mathbb{R}$ and additive subgroups of $\mathbb{R}$, it is not difficult to show that Theorem 4.4.6 is equivalent to:

Theorem 4.4.10. *Every proper closed additive subgroup of $\mathbb{R}$ is a discrete subgroup of the form $a\mathbb{Z}$, where $a > 0$.*

The above theorem can be seen to be equivalent to

Theorem 4.4.11. *Every nontrivial additive subgroup G of $\mathbb{R}$ is one of the following two types:*
(a) If there exists a least positive element $\omega \in G$, then $G =< \omega >$, the cyclic group generated by ω,
(b) G is dense in $\mathbb{R}$ (in the usual topology).

Let f be a function with $P_f =< \omega >$, where ω is the least positive element of P_f. We call ω the fundamental period of f. We now have the following:

Theorem 4.4.12. *Let f be periodic and suppose that f is continuous at some point ξ. Then either f has a fundamental period, or f is a constant function.*

Proof. Suppose f has no fundamental period. Then P_f is dense in $\mathbb{R}$. For every $\epsilon > 0$ there exists a $\delta > 0$ such that $|f(x) - f(\xi)| < \epsilon$ if $|x - \xi| < \delta$. Let x' be any point in $\mathbb{R}$. Then $x' - \xi \in \mathbb{R}$ and P_f is dense in $\mathbb{R}$, so there exists a $\gamma \in P_f$, at a distance less than δ from $x' - \xi$, i.e., $x' - \xi - \gamma = t$, say, is such that $|t| < \delta$, i.e., $x' = \xi + \gamma + t$, where $\gamma \in P_f$, $|t| < \delta$. Hence

$$\begin{aligned} |f(x') - f(\xi)| &= |f(\xi + \gamma + t) - f(\xi)| \\ &= |f(\xi + t) - f(\xi)| \\ &< \epsilon , \end{aligned}$$

since $|t| < \delta$. As ϵ is arbitrary, $f(x') = f(\xi)$. Here ξ is a fixed point while x' is any point. It follows that f is a constant.

Example 4.4.12. The converse of the above theorem is not true, for example,

let

$$f(x) = \begin{cases} 0\ , & \text{if } x \text{ is rational} \\ 2 + \cos 2\pi x\ , & \text{otherwise.} \end{cases}$$

This function has fundamental period equal to 1, but is discontinuous everywhere.

Now consider the quotient group $\mathbb{R}/\mathbb{Q}$, whose elements are cosets $\mathbb{Q} + r$ ($r \in \mathbb{R}$). Let $f^* : \mathbb{R}/\mathbb{Q} \to \mathbb{R}$ be a function on $\mathbb{R}/\mathbb{Q}$. This induces a periodic function $f : \mathbb{R} \to \mathbb{R}$, with $P_f \supset \mathbb{Q}$, by defining $f(x) = f^*(\mathbb{Q}+x)$. For this function f, clearly $P_f \supset \mathbb{Q}$, as let $\rho \in \mathbb{Q}$, then $f(x+\rho) = f^*(\mathbb{Q}+x+\rho) = f^*(\mathbb{Q}+x) = f(x)$ (since $\mathbb{Q} + \rho = \mathbb{Q}$), i.e., ρ is a period and so $\mathbb{Q} \subset P_f$, as required.

Further, if f^* is one to one, then $\mathbb{Q} \supset P_f$, for let $p \in P_f$ be a period; then $f(x + p) = f(x)$ for all x, i.e., $f^*(\mathbb{Q} + x + p) = f^*(\mathbb{Q} + x)$ for all x. But f^* is one to one, so $\mathbb{Q} + p + x = \mathbb{Q} + x$, i.e., $p \in \mathbb{Q}$, as required. Hence we have proved the following

Theorem 4.4.14. *If $f^* : \mathbb{R}/\mathbb{Q} \to \mathbb{R}$ is a one to one mapping, then the induced function $f : \mathbb{R} \to \mathbb{R}$, given by $f(x) = f^*(\mathbb{Q} + x)$, is periodic and its group of periods P_f equals $\mathbb{Q}$.*

Since $\text{card}(\mathbb{R}/\mathbb{Q}) = \text{card } \mathbb{R}$, it is possible to choose a map $f^* : \mathbb{R}/\mathbb{Q} \to \mathbb{R}$ that is one to one. The resulting function $f : \mathbb{R} \to \mathbb{R}$ has $P_f = \mathbb{Q}$, which is dense in $\mathbb{R}$ and the range of f is uncountable. This construction will easily yield a periodic function whose graph is dense in the plane. Note also that there exist uncountable proper subgroups of $\mathbb{R}$ and hence we get:

Theorem 4.4.15. *There exist a nonconstant periodic function with uncountably many distinct periods.*

Let now f and g be two periodic functions, with fundamental periods α, β respectively. We ask the question: *When is the sum $f + g$ periodic ?* If we admit only continuous functions, the following result holds :

Theorem 4.4.16. *Suppose f, g are continuous periodic functions with fundamental periods α, β respectively. Then $f + g$ is periodic if and only if $\alpha/\beta \in \mathbb{Q}$*

Proof. Let f and g be continuous periodic functions. If a function f is periodic, then so is $-f$, therefore it is enough to prove that *if f, g and $h = f+g$ are periodic, continuous functions, with periods α, β, γ respectively and $\alpha/\gamma, \beta/\gamma \notin \mathbb{Q}$, then h is a constant function.* Because this shows that at least one of $\alpha/\gamma, \beta/\gamma$ is rational and since the relation $f + g = h$ may be written as $f + (-h) = -g$ or $g + (-h) = -f$, this also shows that at least one of $\alpha/\beta, \gamma/\beta$ is rational and that at least one of $\beta/\alpha, \gamma/\alpha$ is rational. Thus at least two of $\alpha/\beta, \beta/\gamma, \gamma/\alpha$ are rational; but then so is the third, i.e., all three are rational. In particular, if $f, g, f + g$ $(= h)$ are continuous and periodic, then α/β is rational as required. So now it remains to prove the above assertion.

Let x, y be any given real numbers. Since α/γ is irrational, the set $\{p\alpha+q\gamma \mid$

$p, q \in \mathbb{Z}\}$ is dense in $\mathbb{R}$ (see Theorem 1.1.6) and so there exists sequences $\{p_n\}, \{q_n\}$ of integers such that $\{p_n\alpha + q_n\gamma\} \to x$ and similarly there exists sequences $\{s_n\}$, $\{t_n\}$ of integers such that $\{s_n\alpha + t_n\gamma\} \to y$. Now we have

$$h(p_n\alpha + q_n\gamma) = h(p_n\alpha) = f(p_n\alpha) + g(p_n\alpha) = f(0) + g(p_n\alpha + s_n\beta)\,,$$

and

$$h(s_n\beta + t_n\gamma) = h(s_n\beta) = f(s_n\beta) + g(s_n\beta) = f(p_n\alpha + s_n\beta) + g(0).$$

Adding, we get

$$h(p_n\alpha+q_n\gamma)+h(s_n\beta+t_n\gamma) = h(0)+h(p_n\alpha+s_n\beta) = h(0)+h(p_n\alpha+q_n\gamma+s_n\beta+t_n\gamma).$$

Letting $n \to \infty$, we get (by the continuity of f,g,h) that $h(x) + h(y) = h(0) + h(x + y)$, or $(h(y) - h(0)) + (h(y) - h(0)) = h(x + y) - h(0)$, which means that the function $H(x) = h(x) - h(0)$ is additive, and so, being continuous, $H(x) = cx$ (the equation $\varphi(x + y) = \varphi(x) + \varphi(y)$ for all $x, y \in \mathbb{R}$ is known as Cauchy's equation. It will be proved in Appendix I, that if φ is also continuous, then $\varphi(x) = cx$, for some constant c). But H is periodic too and so $c = 0$, i.e., $h(x) = h(y)$ for all $x, y \in \mathbb{R}$, i.e., that $h(x)$ is a constant, as required.

The converse is trivial: If α, β are periods of f, g and $\alpha/\beta = m/n$ $(m, n \in \mathbb{Z})$, then, for example $\alpha n+\beta m$ is a period of $f+g$, for, we have $(f+g)(x+\alpha n+\beta m) = f(x + \alpha n + \beta m) + g(x + \alpha n + \beta m) = f(x + 2\alpha n) + g(x + 2\beta m) = f(x) + g(x)$ (since $\alpha n = \beta m$), as required.

For general f and g with fundamental periods α and β respectively, we need to consider the two cases, namely $\alpha/\beta \in \mathbb{Q}$ and $\alpha/\beta \notin \mathbb{Q}$.

Definition 4.4.17. We say that two non-zero real numbers α, β are commensurable if their quotient is a rational number, i.e., if there exists integers i, j such that $i\alpha + j\beta = 0$. Also, α, β are called incommensurable if α/β is not rational, i.e., if $i\alpha + j\beta = 0$, $i, j \in \mathbb{Z}$ implies that $i = j = 0$.

Definition 4.4.18. Suppose that α/β is a rational number, so that there exist integers i, j such that $i\alpha + j\beta = 0$; then $\gamma = i\alpha = -j\beta$. Such a γ is called a common multiple of α and β. The smallest positive such γ is called the least common multiple (lcm) of α and β.

Going back to the question posed just before the statement of Theorem 4.4.16, we note that it is easy to deal with the case $\alpha/\beta \in \mathbb{Q}$. For, in this case $i\alpha = -j\beta$ (for some $i, j \in \mathbb{Z}) = \gamma$, say. Then the sum $h = f + g$ satisfies

$$\begin{aligned} h(x + \gamma) &= f(x + \gamma) + g(x + \gamma) \\ &= f(x + i\alpha) + g(x - j\beta) \\ &= f(x) + g(x) \qquad \text{(since } \alpha, \beta \text{ are FP's of } f, g) \\ &= h(x). \end{aligned}$$

We have thus proved the following:

Theorem 4.4.19. *Let f, g be periodic, with FP's α, β respectively. Suppose that α/β is a rational number. Then $h = f + g$ is periodic and any common multiple $\gamma = i\alpha = -j\beta$ is a period of $f + g$.*

Remark 4.4.20. In general, the lcm γ of α, β need not be the FP of $f + g$. However, in the special case when $f(x) = a \sin \alpha x$ and $g(x) = b \cos \beta x$ (so that the FPs of f and g are $2\pi/\alpha$ and $2\pi/\beta$ respectively), where a, b, α, β are real numbers and α/β is rational, the lcm of $2\pi/\alpha, 2\pi/\beta$ (which is same as that of α and β) is the FP of $f + g$, i.e., we have the following

Theorem 4.4.21. *Let $f(x) = a \sin \alpha x$, $g(x) = b \cos \beta x$, where a, b, α, β are any real numbers and α, β are commensurable (i.e., α/β is rational). Then the function $h = f + g$ is periodic and the lcm γ of the FPs of f and g is the FP of $f + g$.*

Proof. Without loss of generality, we may suppose that $\alpha > 0, \beta > 0$ (for if, say $\alpha < 0$, we consider the function $-a \sin(-\alpha x)$, which equals $a \sin \alpha x$, with $-\alpha > 0$). Now as we have seen, the lcm γ of the FPs of f and g is certainly a period of $h = f + g$ (Theorem 4.4.19). It remains to show that γ is the FP.

If $\alpha = \beta$, then $h(x) = a \sin \alpha x + b \cos \alpha x = \sqrt{a^2 + b^2} \sin(\theta + \alpha x)$, where $\cos \theta = \frac{a}{\sqrt{a^2+b^2}}, \sin \theta = \frac{b}{\sqrt{a^2+b^2}}$, and hence h is periodic, as required. So assume that $\alpha \neq \beta$.

Let T be the FP of $h = f + g$. Then

$$h(T + x) = h(x) \text{ (for all real } x) \tag{*}$$

Taking $x = 0$, (*) gives

$$a \sin \alpha T + b \cos \beta T = b \tag{1}$$

Differentiating (*) gives $h'(T + x) = h'(x)$ and putting $x = 0$ in this gives

$$a\alpha \cos \alpha T - b\beta \sin \beta T = a\alpha \tag{2}$$

As above, differentiating again and putting $x = 0$, gives

$$-a\alpha^2 \sin \alpha T - b\beta^2 \cos \beta T = -b\beta^2 \tag{3}$$

and differentiating yet again and putting $x = 0$, finally gives

$$-a\alpha^3 \cos \alpha T + b\beta^3 \sin \beta T = -a\alpha^3 \tag{4}$$

Solving (1)–(4) in pairs, viz. (1), (3) and (2), (4), we get

$$\sin \alpha T = 0 \, , \; \cos \alpha T = 1 \tag{5}$$

$$\sin \beta T = 0 \, , \; \cos \beta T = 1 \tag{6}$$

Now, (5) implies that $\alpha T = n\pi$, $\alpha T = 2n\pi$ (n integer), i.e., $\alpha T = 2n\pi$ (as this covers both), i.e., $T = n.(2\pi)/\alpha =$ a multiple of the FP of $\sin \alpha x$ (viz. $2\pi/\alpha$). Similarly, (6) implies that T is a multiple of the FP of $\cos \beta x$ (viz. $2\pi/\beta$).

Since T is the FP of $h(x)$, it is the least common multiple of the two FPs viz. $2\pi/\alpha, 2\pi/\beta$, as required. This completes the proof of the theorem.

For a purely trigonometric proof of the above theorem and for the related material, the reader is referred to [10].

Let now f and g have FPs α, β respectively, where α/β is not rational. Theorem 4.4.16 tells us that if f and g are continuous then $f+g$ cannot be periodic. However, in general, $f+g$ can be periodic. Here is a beautiful

Example 4.4.22. We now give an example of bounded functions f, g with FPs α, β respectively, α/β not rational, such that $f+g$ is periodic.

Define a set

$$S = \{x \in \mathbb{R} \mid x = i\alpha + j\beta\,, i, j \in \mathbb{Z}\}.$$

Select a real number γ such that for no $k \in \mathbb{Z}$, we have $k\gamma \in S$, i.e., none of $0, \pm\gamma, \pm 2\gamma, \ldots \in \mathbb{Z}$. This is possible since the set $\{i\alpha/k + j\beta/k \mid i, j, k \in \mathbb{Z}, k \neq 0\}$ is countable, so we may select any γ in $\mathbb{R}$, outside of this set. Let

$$T = \{y \in \mathbb{R} \mid y = i\alpha + j\beta + k\gamma\,,\ i, j, k \in \mathbb{Z}\}.$$

For the sets S, T the following properties are easily verified:

1. If $x \in S$, the integers i, j, for which $x = i\alpha + j\beta$, are unique,
2. $S + S = \{s_1 + s_2 \mid s_1, s_2 \in S\} \subset S$,
3. $S^c + S \subset S^c$, where S^c is the complement of S in the reals, i.e., $S^c = \mathbb{R} \setminus S$,

and

(1)′ if $y \in T$, the integers i, j, k, for which $y = i\alpha + j\beta + k\gamma$ are unique,

(2)′ $T + T \subset T$,

(3)′ $T^c + T \subset T^c$.

Now, finally define

$$f(x) = \begin{cases} 2^{-|j|} + 2^{-|k|}\,, & \text{if } x \in T : x = i\alpha + j\beta + k\gamma, \\ 0\ , \text{ if } x \in T^c\,, \end{cases}$$

and

$$g(x) = \begin{cases} 2^{-|i|} - 2^{-|k|}\,, & \text{if } x \in T : x = i\alpha + j\beta + k\gamma, \\ 0\ , \text{ if } x \in T^c\,. \end{cases}$$

Then

$$h(x) = f(x) + g(x) = \begin{cases} 2^{-|i|} + 2^{-|j|}\,, & \text{if } x \in T : x = i\alpha + j\beta + k\gamma, \\ 0\ , \text{ if } x \in T^c\,. \end{cases}$$

We shall now show that
(a) f is periodic with FP equal to α,
(b) g is periodic with FP equal to β,
(c) h is periodic with FP equal to γ.
Because of the similar form of f, g, h, we shall only verify (a). To see that α is a period of f, note that

(i) if $x \in T^c$, then $f(x) = 0$ and $x + \alpha \in T^c + T \subset T^c$ and so $f(x + \alpha) = 0$, giving $f(x) = f(x + \alpha)$,

(ii) if $x \in T$, say $x = i\alpha + j\beta + k\gamma$, then

$$\begin{aligned} f(x+\alpha) &= f((i+1)\alpha + j\beta + k\gamma) \\ &= 2^{-|j|} + 2^{-|k|} \qquad \text{(since } x+\alpha \in T\text{)} \\ &= f(i\alpha + j\beta + k\gamma) \\ &= f(x)\ , \end{aligned}$$

as required. To prove that α is the FP of f, we need to show that every period of f is an integral multiple of α. So let ρ be any period of f. If $\rho \in T^c$, then for $x \in T$, $x + \rho \in T^c$ and so $f(x+\rho) = 0$, but $f(x) \neq 0$ because $x \in T$. So $\rho \notin T^c$, i.e., any period ρ must belong to T and so $\rho = l\alpha + m\beta + n\gamma$ $(l, m, n \in \mathbb{Z})$ and since ρ is a period, $f(0 + \rho) = f(0)$, i.e., $2^{-|m|} + 2^{-|n|} = 2^0 + 2^0 = 2$. This is possible if and only if $m = n = 0$ and then $\rho = l\alpha$, an integral multiple of α, as required.

In the above example, we chose γ such that none of $k\gamma$ $(k = 0, \pm 1, \pm 2, \ldots)$ belongs to S. That this is essential follows from the following:

Theorem 4.4.23. *Let f, g have FPs α, β respectively, where $\alpha/\beta \notin \mathbb{Q}$ and suppose that not both of f, g are unbounded (i.e., at least one of f, g is bounded). If γ is a real number such that $k\gamma \in S$ for some integer $k \neq 0$, then the sum $f + g$ can not have γ as a period .*

Proof. Without loss of generality, let f be bounded and suppose to the contrary, that γ is a period of $f + g$. then $k\gamma$ too is a period and so for all real y and $x = i\alpha + j\beta \in S$, we have (since α is a period of f),

$$f(y + x + k\gamma) = f(y + j\beta + k\gamma) - f(y + j\beta) \tag{1}$$

Now, since $k\gamma$ is a period of $f + g$, we have (as $j\beta$ is a period of g)

$$f(y + j\beta + k\gamma) + g(y + j\beta + k\gamma) = f(y + j\beta) + g(y + j\beta)$$

i.e.,

$$f(y + j\beta + k\gamma) - f(y + j\beta) = g(y + j\beta) - g(y + j\beta + k\gamma) = g(y) - g(y + k\gamma) \tag{2}$$

So, using (1) and (2), we get $f(y+x+k\gamma)-f(y+x)=g(y)-g(y+k\gamma)$. In this, take $x=\mu.(k\gamma)\in S$, $\mu=0,1,2,\ldots n-1$, and we get respectively, the following equations:

$$\begin{aligned} f(y+k\gamma)-f(y)&=g(y)-g(y+k\gamma)\\ f(y+2k\gamma)-f(y+k\gamma)&=g(y)-g(y+k\gamma)\\ &\ldots\quad\ldots\quad\ldots\quad\ldots\\ f(y+nk\gamma)-f(y+(n-1)k\gamma)&=g(y)-g(y+k\gamma). \end{aligned}$$

Adding, we get $f(y+nk\gamma)=f(y)+n(g(y)-g(y+k\gamma))$. Since f is bounded, this gives

$$g(y)-g(y+k\gamma)=0\ ,\quad \forall y\in\mathbb{R}. \tag{3}$$

Again, $f(y+k\gamma)+g(y+k\gamma)=f(y)+g(y)$, since $k\gamma$ is a period of $f+g$, so $g(y)-g(y+k\gamma)=f(y+k\gamma)-f(y)$, and it follows, by (3), that $f(y+k\gamma)=f(y)$ for all real numbers y, and hence that $k\gamma$ is a period of both f and g and so $k\gamma$ is an integral multiple of both α and β. Thus, $k\gamma=p\alpha=q\beta$, i.e., $\alpha/\beta=q/p$ is a rational number, which gives the required contradiction.

By allowing f and g both to be unbounded, it is possible to have a sum $f+g$ with a period γ in the set S. We have:

Example 4.4.24. Let

$$f(x)=\begin{cases} j & \text{when } x=i\alpha+j\beta\in S,\\ 0 & \text{when } x\in S^c, \end{cases}$$

and let

$$g(x)=\begin{cases} -i & \text{when } x=i\alpha+j\beta\in S,\\ 0 & \text{when } x\in S^c, \end{cases}$$

then

$$f(x)+g(x)=\begin{cases} j-i & \text{when } x=i\alpha+j\beta\in S,\\ 0 & \text{when } x\in S^c. \end{cases}$$

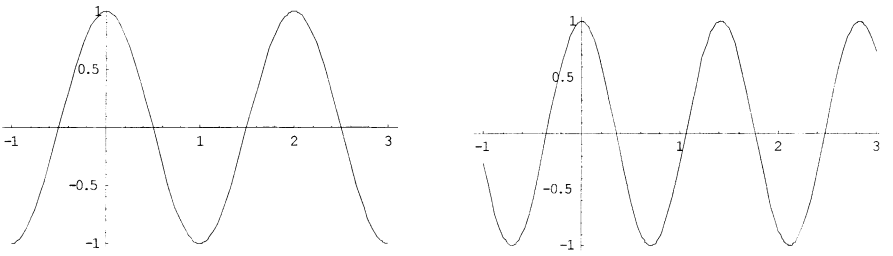
The unbounded functions f and g have FPs α and β respectively (as in example 4.2.22), while their sum $f+g$ has FP equal to $\gamma=\alpha+\beta\in S$ (check).

Remark 4.4.25. Note that the set $S=\{x\in\mathbb{R}\mid x=i\alpha+j\beta, i,j\in\mathbb{Z}\}$ is dense in $\mathbb{R}$ if $\alpha/\beta\notin\mathbb{Q}$. This fact can be generalized in the context of complex function theory and then used to show that *the only triply periodic analytic functions are constants* – a result due to Jacobi (see E.Hille, *Analytic Function Theory*, Vol. II, Ginn, 1962).

In Theorem 4.4.16, the continuity conditions may be replaced by others that guarantee $\alpha/\beta\in\mathbb{Q}$; for example, we have the following

Theorem 4.4.26. *Let f, g be periodic and have (without loss of generality) positive periods α, β respectively and suppose that f, g both have strict absolute maxima at the origin, i.e., $f(0) > f(x)$ for all $x \in (0, \alpha)$ (note that $f(\alpha) = f(0)$) and $g(0) > g(x)$ for all $x \in (0, \beta)$ (note again that $g(\beta) = g(0)$). If $f + g$ is periodic, then $\alpha/\beta \in \mathbb{Q}$.*

Proof. Let γ be a positive period of $f + g$. Then $(f + g)(0) = (f + g)(\gamma)$ i.e., $f(0) + g(0) = f(\gamma) + g(\gamma)$. But $f(0) \geq f(\gamma)$, $g(0) \geq g(\gamma)$. So this is impossible unless equality holds in both places: $f(\gamma) = f(0)$, $g(\gamma) = g(0)$. Hence γ is a multiple of α and a multiple of β: $\gamma = m\alpha = n\beta$, giving $\alpha/\beta = n/m$.



Graph of y = cos π x **Graph of y = cos √2π x**

Figure 4.12

Example 4.4.27. The above mentioned simple condition suffices to show, for example, that $\cos \pi x + \cos \sqrt{2}\pi x$ is not periodic, for if $f(x) = \cos \pi x$ and $g(x) = \cos \sqrt{2}\pi x$, then the FPs of f, g are given by $\alpha = 2$, $\beta = 2/\sqrt{2}$; further, both f, g have absolute maxima at the origin (Figure 4.12). So, if $f + g$ were periodic, $\alpha/\beta = 2/(2/\sqrt{2}) = \sqrt{2}$, would be rational, which is not the case, hence $f + g$ can not be periodic.

There are other similar conditions, which, together with the periodicity of f, g and $f + g$ imply that $\alpha/\beta \in \mathbb{Q}$. the interested reader may look up [23].

Example 4.4.28. Let

$$f(x) = \begin{cases} 0\ , \text{ if } x = (2n+1)/2 \text{ for some } n \in \mathbb{Z}, \\ \tan \pi x\ , \text{ otherwise}\ ; \end{cases}$$

and

$$g(x) = \begin{cases} 0\ , \text{ if } x = (2n+1)/2\sqrt{2} \text{ for some } n \in \mathbb{Z}, \\ \tan \sqrt{2}\pi x\ , \text{ otherwise}\ . \end{cases}$$

Here, f is periodic with FP equal to 1, g is periodic with FP equal to $1/\sqrt{2}$, but $f + g$ is not periodic, $\alpha/\beta \notin \mathbb{Q}$.

§4.5. Horizontal chords

Let $f : [a, b] \to \mathbb{R}$ be a continuous function.

Definition 4.5.1. We say that f has a horizontal chord of length λ if there is a point x such that both of $x, x + \lambda \in [a, b]$ and $f(x) = f(x + \lambda)$.

Example 4.5.2. (1) The function in Figure 4.13 has two horizontal chords of length 2 and one of length 4, viz. OA, OB, AB of length 2, OB of length 4.

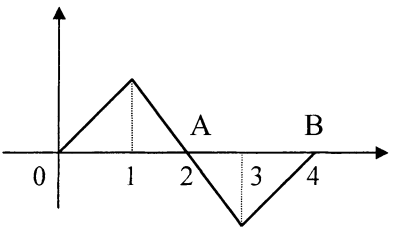


Figure 4.13

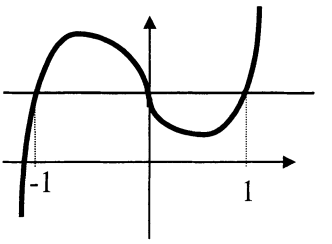


Figure 4.14

(2) The function $f(x) = 1$ defined on the entire real line, has horizontal chords of all lengths.
(3) The function $f(x) = x^3 - x + 1$ has a horizontal chord of length 2, viz. $(-1, 1)$ to $(1, 1)$ and two horizontal chords of length 1, viz. $(-1, 1)$ to $(0, 1)$ and $(0, 1)$ to $(1, 1)$ (see Figure 4.14).
(4) The function $f(x) = x^3$ has no horizontal chords.

Remark 4.5.3. Suppose that f is a continuous periodic function of period p. Then it is easy to see that (see Figure 4.15) the integral $\int_a^{a+p} f(x)\,dx$ has value independent of a.

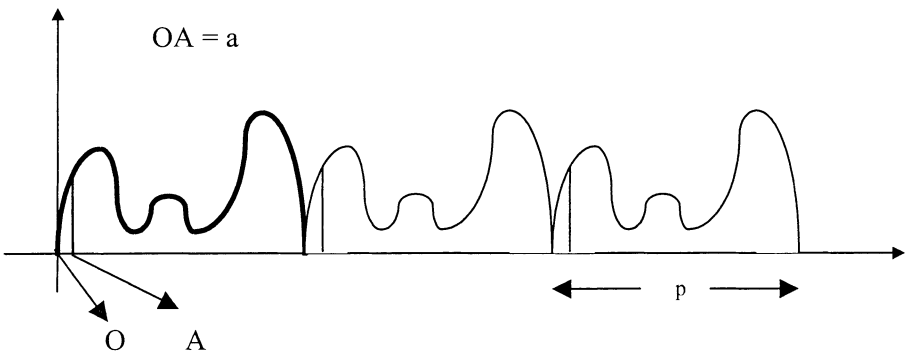


Figure 4.15

Using this, we can prove the following result:

Proposition 4.5.4. *Let f be a continuous and periodic function. Then f has horizontal chords of all lengths.*

Proof. Let p be the period. We have to show that for any real number λ, there exists a number ξ such that $f(\xi + \lambda) = f(\xi)$. To see this, consider the integral $\int_0^p (f(x + \lambda) - f(x))\, dx$, which is zero, by Remark 4.5.3. Hence $g(x) = f(x + \lambda) - f(x)$ can not have a fixed sign on the interval $(0, p)$ and so by the IVP (which holds since the function g is continuous), $g(\xi) = 0$ for some $\xi \in (0, p)$, as required.

Indeed, we have the following result:

Proposition 4.5.5. *Let f be continuous and periodic with period p. Then f has two horizontal chords of any given length λ, with their left hand end points at different points of $[0, p)$.*

Proof. Consider the graph of the function $g(x) = f(x + \lambda) - f(x)$.

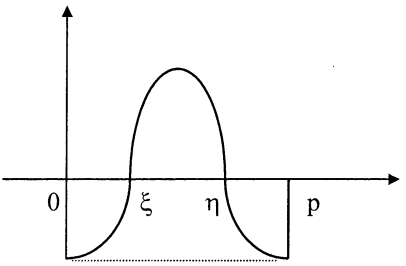


Figure 4.16

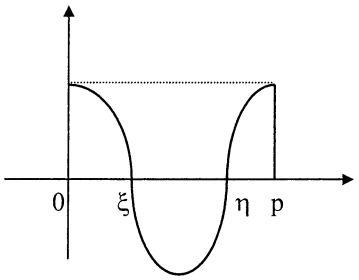


Figure 4.17

We have $g(p) = g(0)$; but g changes sign between 0 and p, by a reasoning similar to the one used above. So the graph of g must have two possibilities as shown in the Figures 4.16 and 4.17, and so g vanishes at least twice between $[0, p)$; say at ξ and η; i.e., $f(\xi + \lambda) = f(\xi)$, $f(\eta + \lambda) = f(\eta)$ $(\xi \neq \eta)$. This completes the proof.

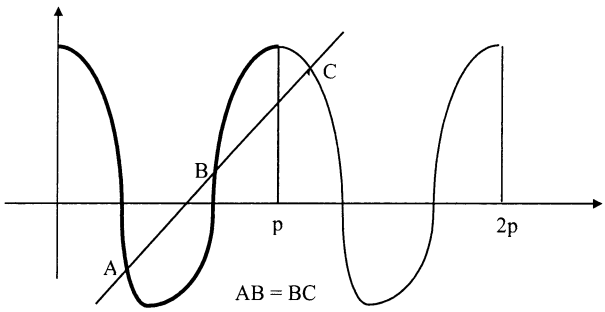


Figure 4.18

By considering the function $g(x) = \varphi(x) - \varphi(x - \lambda)$, where $\varphi(x) = f(x + \lambda) - f(x)$, we see, as before, that $\int_0^p g(x)\, dx = 0$ and so g changes sign in $[0, p)$. It follows that g vanishes at some $\xi \in [0, p)$ (by the IVP, since g is continuous),

i.e., $\varphi(\xi) = \varphi(\xi - \lambda)$, giving $f(\xi+\lambda) - f(\xi) = f(\xi) - f(\xi-\lambda)$. This last equation may be interpreted as the following interesting :

Proposition 4.5.6. *Let $f\colon [a,b] \to \mathbb{R}$ be periodic and continuous. Then it has a chord of any prescribed span (length), not necessarily horizontal, with its mid-point on the graph (see Figure 4.18).*

For functions that are continuous but not periodic, the situation can be quite different. A continuous function may have no horizontal chord, as for example, a strictly increasing function. However, the following striking result holds:

Theorem 4.5.7 (The Horizontal Chord Theorem). *Suppose that $f : [0,1] \to \mathbb{R}$ has a horizontal chord of length λ; say $f(0) = f(1)$, i.e., $\lambda = 1$. Then there are horizontal chords of lengths $1/2, 1/3, 1/4, \ldots$, but not necessarily a horizontal chord of any other length.*

The positive half of this theorem may be proved as follows: Let $k = 2, 3, 4, \ldots$ and let $g(x) = f(x + 1/k) - f(x)$, so that g has domain $[0, 1 - 1/k]$. We shall show that 0 belongs to the range of g, i.e., that there exists ξ such that $g(\xi) = 0$, giving $f(\xi + 1/k) = f(\xi)$, showing that f has a horizontal chord of length $1/k$ as required.

Suppose not; then g is either positive for all $x \in [0, 1 - 1/k]$ or negative for all $x \in [0, 1 - 1/k]$, for otherwise, by IVP, it would vanish at some ξ, as required. Say $g(x) > 0$ throughout $[0, 1 - 1/k]$, i.e., $f(x) < f(x + 1/k)$ for all x in $[0, 1 - 1/k]$. Taking $x = 0, 1/k, 2/k, \ldots, (k-1)/k$ (all in the interval $[0, 1 - 1/k]$), we get

$$f(0) < f(1/k) < f(2/k) < \ldots < f((k-1)/k) < f(k/k) = f(1) = f(0) ,$$

giving $f(0) < f(0)$, a contradiction.

Now consider the function shown in the Figure 4.19. It has a horizontal chord of length 1 but none of length λ with $1/2 < \lambda < 1$.

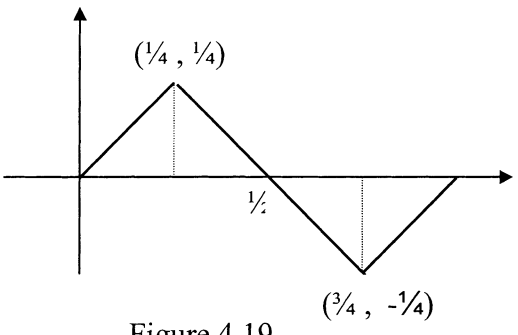


Figure 4.19

However, we require to show that for each $\rho \neq 1/k$ ($k \in \mathbb{N}$), there exists a function, that is continuous, has a horizontal chord of length 1, but none of

length ρ for all $\rho \in [0, 1/2]$; so other examples have to be furnished. The following example is due to P.Levy (see [62]).

Let $a \in (0,1)$ be fixed with $a \neq 1/2, 1/3, 1/4, \ldots$. Consider the function

$$f(x) = \sin^2(\pi x/a) - x \sin^2(\pi/a).$$

Since $f(0) = 0, f(1) = 0$, we see that f has a horizontal chord of length 1; see Figure 4.20.

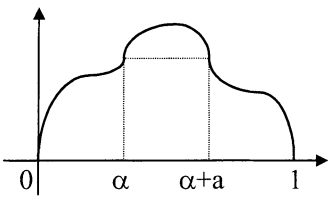


Figure 4.20

Now, if f has a horizontal chord of length a, then, there is an α such that $f(\alpha) = f(\alpha + a)$, i.e., $\sin^2(\pi\alpha/a) - \alpha \sin^2 \pi/a = \sin^2(\pi(\alpha + a)/a) - (\alpha + a)\sin^2(\pi/a)$, i.e., $a \sin^2 \pi/a = 0$. i.e., $\pi/a = n\pi$, with $n \in \mathbb{N}$, since $a > 0$ (being the length of a horizontal chord), i.e., $a = 1/n$ $(n \in \mathbb{N})$. This contradicts the choice of a.

An interesting complement to the Horizontal Chord Theorem is the following result due to P.Levy:

Theorem 4.5.8. *Suppose $f\colon [0,1] \to \mathbb{R}$ is continuous with $f(0) = f(1)$ and suppose f has no horizontal chord of length a. Then f has two horizontal chords of length $1 - a$ $(0 < a < 1)$.*

Proof. Let g be defined on $\mathbb{R}$ as the periodic extension of f with period 1. As a periodic continuous function, g has two horizontal chords of length a, with left hand end points γ_1, γ_2 in $[0,1)$ (see the result above Theorem 4.5.6). A horizontal chord of length a for g, starting at $\gamma \in [0,1)$, is a horizontal chord for f, unless $\gamma + a > 1$, but then

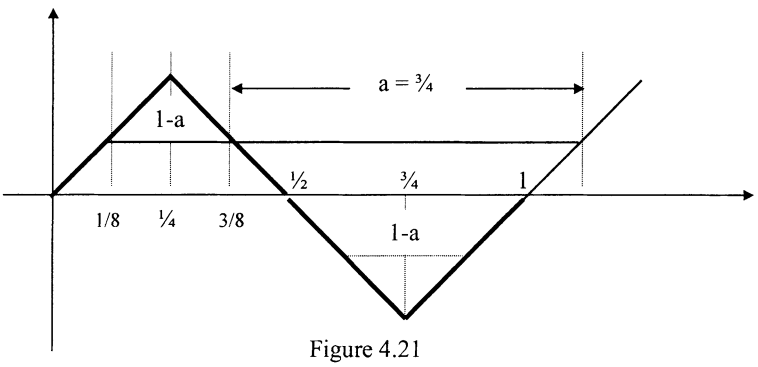


Figure 4.21

$0 < \gamma + a - 1 < 1$ and so a horizontal chord of length $1 - a$ for g (and also for f) starts at $\gamma + a - 1$ and ends at γ.

As an immediate corollary of this, we prove the following result due to Rosenbaum (see [94]):

Theorem 4.5.9. *Let $f : [0,1] \to \mathbb{R}$ be continuous, with $f(0) = f(1)$ and let $n \in \mathbb{N}$. Then the graph of f has at least n horizontal chords whose lengths are integral multiples of $1/n$.*

Proof. For each of $k = 1, 2, \ldots, n-1$, if f has no horizontal chord of length $a = k/n$, then it has two horizontal chords of length $1 - a = (n-k)/n$ and vice versa. Thus, there always are 2 horizontal chords, either of length k/n or of length $(n-k)/n$ or one each of length $k/n, (n-k)/n$ and this is the case for each of $k = 1, 2, \ldots, n-1$.

For n odd, this gives $2.(n-1)/2 = n-1$ horizontal chords of the required type and for $k = n$, there always is a horizontal chord of length $1 = n.(1/n)$ (an integral multiple of n). This gives the full set of n horizontal chords, as required.

For n even, $n-1$ is odd and the counting is different, viz. either there are 2 horizontal chords of length k/n or two of length $(n-k)/n$ or one for each of $k = 1, 2, \ldots, (n-2)/2$; in all $2.(n-2)/2 = n-2$ plus one of length $1 = n.(1/n)$, giving the full set of n again.

§4.6. Continuous functions and integration

The usual proof of the result that a continuous function f defined in $[a,b]$ is Riemann integrable uses the more sophisticated result that a continuous function in $[a,b]$ is uniformly continuous. We here give a proof that blends well with the level of sophistication of the statement of the theorem.

With the usual notation, let f be a bounded function in $[a,b]$ and let

$$\wp : a = x_0 < x_1 < \ldots < x_n = b$$

be a partition of $[a,b]$. Let $m_i = \inf_{x \in [x_{i-1}, x_i]} f(x)$ and $M_i = \sup_{x \in [x_{i-1}, x_i]} f(x)$ and let $s_\wp$ and $S_\wp$ denote respectively the lower and upper (Darboux) sums, i.e.,

$$s_\wp = \sum_{i=1}^{n} m_i(x_i - x_{i-1}) \qquad \text{and} \qquad S_\wp = \sum_{i=1}^{n} M_i(x_i - x_{i-1}).$$

Then

(i) $s_\wp \leq S_\wp$,

(ii) If $\wp_2$ is a refinement of $\wp_1$, then $s_{\wp_1} \leq s_{\wp_2}$, $S_{\wp_2} \leq S_{\wp_1}$,

(iii) For any two partitions $\wp_1$, $\wp_2$, $s_{\wp_1} \leq S_{\wp_2}$.

Letting

$$\underline{\int_a^b} f = \sup_{\wp} s_\wp \qquad \text{and} \qquad \overline{\int_a^b} f = \inf_{\wp} S_\wp \,,$$

we get

(iv) $m(b-a) \le \underline{\int_a^b} f \le \overline{\int_a^b} f \le M(b-a)$

(v) If $a < c < b$, then $\underline{\int_a^c} f + \underline{\int_c^b} f = \underline{\int_a^b} f$ and $\overline{\int_a^c} f + \overline{\int_c^b} f = \overline{\int_a^b} f$.

Recall that we say that f is Riemann integrable in $[a,b]$ if $\underline{\int_a^b} f = \overline{\int_a^b} f$. We have

Theorem 4.6.1. *Let f be continuous in $[a,b]$, then f is Riemann integrable in $[a,b]$.*

Proof. (M.J.Norris, [76]). The function f is bounded in $[a,b]$. Let

$$\varphi(x) = \overline{\int_a^x} f(t)\,dt - \underline{\int_a^x} f(t)\,dt \qquad (a \le x \le b).$$

For $x \le b-h$, $h > 0$, we have

$$\varphi(x+h) - \varphi(h) = \Big\{\overline{\int_a^{x+h}} f - \underline{\int_a^{x+h}} f\Big\} - \Big\{\overline{\int_a^{x}} f - \underline{\int_a^{x}} f\Big\} = \overline{\int_a^{x+h}} f - \underline{\int_a^{x+h}} f.$$

Let $m_h \le f \le M_h$ in $[x, x+h]$, then $0 \le (\varphi(x+h) - \varphi(x)) \le M_h.h - m_h.h$, i.e.,

$$0 \le \frac{\varphi(x+h) - \varphi(x)}{h} \le M_h - m_h \qquad (*)$$

Now $\lim_{h\to 0} M_h$ and $\lim_{h\to 0} m_h$ both exist and equal $f(x)$. Hence, by (*), $\lim_{h\to 0} \frac{\varphi(x+h)-\varphi(x)}{h}$ exists and is equal to 0. Here we are looking at the interval $[x, x+h]$, $h > 0$. A similar argument is available if $a - h < x$, $h < 0$. It follows that $\varphi'(x)$ exists and is 0 in (a,b). Further, $\varphi(x)$ is continuous in $[a,b]$. Thus, $\varphi(x) = K$, (a constant) in $[a,b]$ and since $\lim_{x\to a} \varphi(x) = \overline{\int_a^a} f - \underline{\int_a^a} f = 0$, we see that $\varphi(x) \equiv 0$ in $[a,b]$, i.e.,

$$\overline{\int_a^b} f = \int_a^a f \,.$$

Let f be a bounded function in $[a,b]$. Let $\varphi_1(x) = \overline{\int_a^x} f$, $\varphi_2(x) = \underline{\int_a^x} f$, $x \in [a,b]$. The argument, similar to the above, easily yields that $\varphi_1'(x) = f(x) = \varphi_2'(x)$, at any point x at which f is continuous. Hence, if f is Riemann integrable, then $\frac{d}{dx}(\int_a^x f) = f(x)$; and the fundamental theorem of calculus also falls out at almost no extra cost, namely :

Theorem 4.6.2. *Let f be a bounded and Riemann integrable function in $[a,b]$ and let $F(x) = \int_a^x f$, $a \le x \le b$. If f is continuous at a point $c \in [a,b]$, then F is differentiable at c and $F'(c) = f(c)$.*

Remark 4.6.3. Using the argument similar to the above, it is easy to show that if f is continuous in (a,b) and bounded in $[a,b]$, then f is Riemann integrable in $[a,b]$. Indeed, with φ_1, φ_2 as above, $(\varphi_1 - \varphi_2)'(x) \equiv 0$ in (a,b) and $\varphi_1 - \varphi_2$ is continuous in $[a,b]$ and $(\varphi_1 - \varphi_2)(a) = 0$.

Another interesting result can be proved using the nested interval property.

Theorem 4.6.4. *Let f be Riemann integrable in $[a,b]$. Then the set of points of $[a,b]$, at which f is continuous, is dense in $[a,b]$.*

Proof. (K.E.Hirst, [45]). All we need to show is that given any interval $[c,d] \subset [a,b]$, however small, it has a point of continuity of f, so it is enough to prove that $[a,b]$ has a point of continuity of f (for then, similarly so has $[c,d]$).

Let ϵ be given, then there exists a δ such that for any partition $\wp$ of $[a,b]$, with norm $\wp < \delta$, we have $S_\wp(f) - s_\wp(f) < \epsilon$. Choose $\wp$ with m equal parts so that each part is of length $(b-a)/m$ and that $(b-a)/m < \delta$ and choose $\epsilon = (b-a)/2$. Then

$$S_\wp(f) - s_\wp(f) = \sum_{i=1}^{m}(M_i - m_i)(x_i - x_{i-1}) = \frac{b-a}{m}\sum_{i=1}^{m}(M_i - m_i) < \epsilon = \frac{b-a}{2}\ ,$$

i.e.,

$$\sum_{i=1}^{m}(M_i - m_i) < m/2 \le m - 2\,, \quad \text{if } m \ge 4\ , \tag{1}$$

which is the case if we choose m large enough to ensure the following:
(i) $m \ge 4$ (ii) Norm $\wp = (b-a)/m < \delta$ (δ corresponding to $\epsilon = (b-a)/2$). Also

$$\text{(i)} \Rightarrow \exists \text{ at least three integers } i,\ 1 \le i \le m, \text{ for which } M_i - m_i < 1. \tag{2}$$

We may thus choose $[x_{i-1}, x_i] = [a_1, b_1]$ say, such that (i) $a \ne x_{i-1}, b \ne x_i$ (ii) $M_i - m_i < 1$ ($M_i = \sup f$ in $[x_{i-1,x_i}]$, $m_i = \inf f$ in $[x_{i-1,x_i}]$) (iii) $x_i - x_{i-1} = (b-a)/m \le (b-a)/4$ (since $m \ge 4$).

Now work with $[a_1, b_1]$ in exactly the same way as we did for $[a,b]$, the only difference being that we wish to make $M_i - m_i < 1/2$ in (ii) and in general

$M_i - m_i < 1/n$ in the interval $[a_n, b_n]$. Here the only difference is in the selection of ϵ. We take $\epsilon = (b-a)/(n+1)$ (in $[a,b] = [a_0, b_0]$, we took it to be $(b-a)/2$). Then for a suitable δ, for all partitions $\wp$ of $[a_{n-1}, b_{n-1}]$, with Norm $\wp < \delta$ and length of each subinterval being $x_i - x_{i-1} = (b_{n-1} - a_{n-1})/m$ (note that $b_{n-1} - a_{n-1})/m < \delta$), we have

$$S_\wp - s_\wp = \sum_{i=1}^{m} (M_i - m_i)(x_i - x_{i-1}) = \frac{b_{n-1} - a_{n-1}}{m} \sum_{i=1}^{m} (M_i - m_i) < \epsilon = \frac{b_{n-1} - a_{n-1}}{(n+1)} .$$

This implies that

$$\sum_{i=1}^{m} (M_i - m_i) < m/(n+1), \quad \text{if Norm } \wp < \delta \ . \tag{1}'$$

It follows that for at least three integers i, $1 \le i \le m$, we have $M_i - m_i < 1/n$, for, if not, then $M_i - m_i \ge 1/n$ for $m-2$ values of i and so

$$\sum_{i=1}^{m} (M_i - m_i) \ge (m-2)/n \ . \tag{2}'$$

Now (1)′ and (2)′ imply that $m - 2/n < m/n + 1$ and hence $mn + m - 2n - 2 < mn \Rightarrow m < 2(n+1)$, which is false if m is chosen large enough to ensure
(i) $m > 2(n+1)$ (for $n = 1$, i.e., for $[a_0, b_0] = [a, b]$, this condition was $m \ge 4$),
(ii) Norm $\wp = (b_{n-1} - a_{n-1})/m < \delta$ (δ corresponding to $\epsilon = (b_{n-1} - a_{n-1})/(n+1)$), as required.

We thus get a nested sequence $[a_n, b_n]$ of intervals. Special care has been taken to ensure that at each stage, the subinterval does not share an end point with the "parent" interval. This guarantees that the sequence $< a_n >$ is strictly increasing and $< b_n >$ is strictly decreasing and hence we get a point $\xi \in \cap\, [a_n, b_n]$ which is actually in $\cap\, (a_n, b_n)$. It is easy to see that f is continuous at $x = \xi$. For, let $\epsilon > 0$ be given. Then by (ii), we have $\sup f - \inf f < 1/(n+1) < \epsilon$ (the sup and inf are taken over $[a_n, b_n]$), if n is large enough. Hence $|f(x) - f(\xi)| < \sup f - \inf f < \epsilon$, if $x, \xi \in (a_n, b_n)$; for, $x, \xi \in (a_n, b_n)$ is equivalent to $\xi \in (a_n, b_n)$ and $|x - \xi| < \min\{b_n - \xi, \xi - a_n\} = \delta$.

We shall close this section by recording some interesting examples/counterexamples regarding Riemann integration.

Example 4.6.5. It is well known that the function f on $[0,1]$ defined by $f(x) = 0$, if x is rational and $f(x) = 1$, if x is irrational, is not Riemann integrable. It is discontinuous everywhere. Similarly, the function h on $[0,1]$ defined by $h(x) = 1$, if x is rational and $f(x) = -1$, if x is irrational, is not Riemann integrable, whereas the function $|h|$ is. However, it can be shown that the function g defined on $[0,1]$ by

$$g(x) = \begin{cases} 0\,, \text{when } x \text{ is irrational or zero} \\ \frac{1}{q}\,, \text{when } 0 \ne x \text{ is any rational number } \frac{p}{q}, \text{ in lowest terms and } q > 0 \end{cases}$$

is Riemann integrable on $[0,1]$ and the value of the integral is zero. It can also be shown that g is discontinuous at each rational and is continuous at each irrational.

Example 4.6.6. Let f be defined on $[-1,1]$ by

$$f(x) = \begin{cases} 1\ , & \text{if } x > 0 \\ 0\ , & \text{if } x = 0 \\ -1\ , & \text{if } x < 0\ . \end{cases}$$

Then f is Riemann integrable on $[-1,1]$, but there is no function F on $[-1,1]$ such that $f(x) = F'(x)$ for all $x \in [-1,1]$, i.e., f fails to have a primitive.

Example 4.6.7. The function defined by

$$f(x) = \begin{cases} x^2 \sin \frac{1}{x^2}\ , & \text{if } x \neq 0 \\ 0\ , & \text{if } x = 0 \end{cases}$$

gives an example of a function having a derivative $g(x)$ at each point x of the closed interval $[-1,1]$. The function g, therefore, has a primitive, but since g is unbounded, it is not Riemann integrable on $[-1,1]$.

Example 4.6.8. Let $f(x) \equiv 1$, if $0 < x \leq 1$ and $f(0) = 0$, and if g is the function defined on $[0,1]$ by

$$g(x) = \begin{cases} 0\ , & \text{if } x \text{ is irrational or zero} \\ \frac{1}{n}\ , & \text{if } x \text{ is a rational number } \frac{m}{n} \text{ in lowest terms and } n > 0\ , \end{cases}$$

then $f(g(x))$ is the function which is 1 for rationals and 0 for irrationals. Thus, $f \circ g$ is not Riemann integrable on $[0,1]$, whereas both f and g are.

§4.7. Continuous functions and their graphs

Let $f : \mathbb{R} \to \mathbb{R}$ be a function and denote its graph by G_f. Many would tend to think that *f is continuous* **if and only if** *G_f (as a subset of $\mathbb{R}^2$) is connected.* This two sided equivalence is false, the following being a standard counter example:

Example 4.7.1. Let $f(x) = \sin 1/x$, if $x \neq 0$, $f(0) = 0$. Then the graph of f is connected, as it is easy to see. However, f fails to be continuous at $x = 0$.

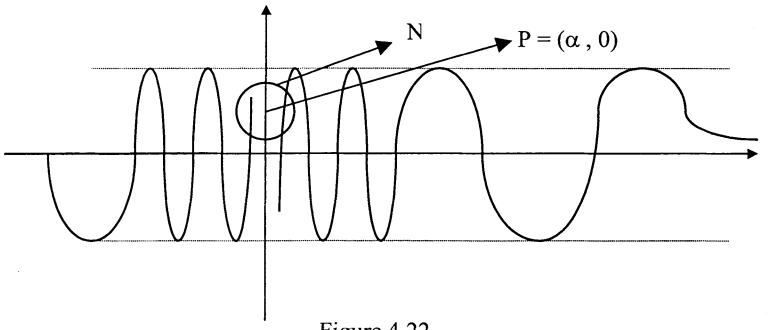


Figure 4.22

The converse however is true and is contained in the following:

Theorem 4.7.2. *The function $f : \mathbb{R} \to \mathbb{R}$ is continuous if and only if its graph G_f is a closed and connected subset of $\mathbb{R}^2$.*

Remark 4.7.3. In Example 4.7.1 above, the graph G_f of the given function, is not closed. Indeed, all the points $(0, \alpha)$, $0 < \alpha \leq 1$ are limit points of G_f but do not belong to G_f. To see this, let $P = (0, \alpha)$, with $0 < \alpha \leq 1$. Then any neighbourhood N of P will have G_f passing through it infinitely often (see Figure 4.22), and so P is a limit point of G_f.

Proof of Theorem 4.7.2. First, let G_f be closed and connected. Since G_f is connected, it is easy to see that f will have IVP. So, $f : \mathbb{R} \to \mathbb{R}$ has the IVP and G_f is closed (and connected) and we are to show that f is continuous. Suppose f is not continuous at $x = c$. Then for some $\epsilon > 0$, there exists a sequence $\{b_n\}$ of real numbers tending to c, such that $|f(b_n) - f(c)| > \epsilon$ for each n, i.e., $-\epsilon > f(b_n) - f(c) > \epsilon$. Let us consider the right hand inequality $f(b_n) > f(c) + \epsilon$. By the IVP, there exists c_n between b_n and c such that $f(c_n) = f(c) + \epsilon$. Thus, the sequence of points $\{(c_n, f(c_n))\} \to (c, f(c) + \epsilon) \notin G_f$, a contradiction, since G_f is closed.

Conversely, suppose f is continuous and we have to show that (i) G_f is closed and (ii) G_f is connected.

Proof of (i). If not, then there exists a sequence of points $(\alpha_n, f(\alpha_n)) \in G_f$, $\alpha_n \to \alpha$, with limit not in G_f. However, this limit is $(\alpha, f(\alpha))$ as $\alpha_n \to \alpha$ and $f(\alpha_n) \to f(\alpha)$ (since f is continuous), so $(\alpha, f(\alpha)) \notin G_f$, which is absurd since for every $x \in \mathbb{R}$, $(x, f(x)) \in G_f$, by the definition of G_f.

Proof of (ii). If not, let $G_f \subset H \cup K$, where $G_f = \{(\alpha, f(\alpha)) \mid \alpha \in \mathbb{R}\}$, $H, K \neq \emptyset$, $H \cap K = \emptyset$ and H, K are open. Now put $S_1 = \{\alpha \mid (\alpha, f(\alpha)) \in H\}$, $S_2 = \{\alpha \mid (\alpha, f(\alpha)) \in K\}$. Then S_1, S_2 are non-empty, open and disjoint subsets of $\mathbb{R}$ such that $S_1 \cup S_2 = \mathbb{R}$, giving a contradiction, since $\mathbb{R}$ is connected.

Definition 4.7.4. We say that a function $f : \mathbb{R} \to \mathbb{R}$ is locally bounded on a set S if, for each $x \in S$, there exists a neighbourhood $(x - \epsilon_x, x + \epsilon_x)$ of x, in which f is bounded.

Clearly, f bounded implies f locally bounded. The converse, however, is false, for example:

Example 4.7.5. $f(x) = x$ is locally bounded on $\mathbb{R}$ but not bounded. Another example would be $f(x) = 1/x$, if $x \neq 0$, $f(0) = 0$. This function is locally bounded in $(0, \infty)$ but not bounded in $(0, \infty)$. The following interesting result holds (parts of which have already been proved):

Theorem 4.7.6. *Suppose that $f\colon \mathbb{R} \to \mathbb{R}$ has a closed graph G_f (closed as a subset of $\mathbb{R}^2$). Then each of the following conditions implies that f is continuous:*
(i) f is locally bounded,
(ii) f has IVP,
(iii) G_f is connected.

Proof. (i) $\Rightarrow$ f is continuous: Suppose f is not continuous at the point $x = \xi$. Then there exists a sequence $\{\alpha_n\} \to \xi$ such that $f(\alpha_n) \to \eta \neq f(\xi)$ with $\eta \in \mathbb{R}$, as f is locally bounded. Thus, the sequence of points $\{(\alpha_n, f(\alpha_n))\} \to (\xi, \eta)$ in $\mathbb{R}^2$, contradicting the fact that G_f is closed, since the point $(\xi, \eta) \notin G_f$ and is a limit point of G_f.
(ii) $\Rightarrow$ f is continuous: Already done in the proof of Theorem 4.7.2.
(iii) $\Rightarrow$ f is continuous: This is the implication $\Leftarrow$ of Theorem 4.7.2.

In looking at the converse of Theorem 4.7.6, the hypothesis on G_f being closed (in $\mathbb{R}^2$) is redundant, as it follows from the continuity of f (see the converse in Theorem 4.7.2). So the converse to Theorem 4.7.6 would read:

Theorem 4.7.7. *Suppose that $f : \mathbb{R} \to \mathbb{R}$ is continuous, then*
(i) f is locally bounded ,
(ii) f has the IVP ,
(iii) G_f is connected.

Proof. Here (i) is trivially true; (ii) is a well-known result, while (iii) has already been proved in Theorem 4.7.2 (the converse, (ii)).

However, it would be interesting to see which of the following implications are true or false:

$$f \text{ locally bounded} \Leftrightarrow f \text{ has the IVP} \Leftrightarrow G_f \text{ is connected},$$

with no assumption about G_f being closed.

Example 4.7.8. Consider the function $f(x) = (1/x)\sin(1/x)$, if $x \neq 0$, $f(0) = 0$. A rough sketch of the graph of this function may be drawn by the reader. To arrive at it, we note the following properties of this function:
(1) $f(-x) = f(x)$, so we need only to look at the region $x \geq 0$.
(2) If x is large, $1/x$ is small and positive and so is $\sin(1/x)$, hence $(1/x)\sin(1/x)$ is small and positive. The last point where the function is zero, is where $1/x = \pi$, i.e., $x = 1/\pi \cong .32...$ and when $x > 1/\pi$, $0 < 1/x < \pi$ and so $\sin(1/x)$ never vanishes.

(3) For $0 < x < 1/\pi$, $\sin(1/x)$ vanishes infinitely often, viz. at each of the points $x = 1/(n\pi)$.
(4) If need be, we may look at the points at which $f'(x)$ vanishes. We have $f'(x) = -(1/x^3)\cos(1/x) - (1/x^2)\sin(1/x)$ and this is equal to 0 if $\tan(1/x) = -(1/x)$. Solutions of this are awkward. We may draw graphs of $y = \tan x$ and of $y = -1/x$ and check the intersections. All this if need be, but for a rough sketch, it is not necessary.

This function has the IVP but it is not locally bounded at the point $x = 0$; hence *f has IVP does not imply f is locally bounded.* Further, for this very function f, its graph is connected, showing *G_f is connected does not imply f is locally bounded.*

Example 4.7.9. Now consider the function

$$g(x) = \begin{cases} 1\ , \text{ if } x > 0 \\ 0\ , \text{ if } x \leq 0 \end{cases}$$

This immediately shows that *f is locally bounded does not imply f has the IVP* and also that *f is locally bounded does not imply its graph G_f is connected.*

Remark 4.7.10. It has already been proved (see step 1 in the proof of Theorem 4.7.2) that *G_f is connected implies f has the IVP.* It finally remains to analyze the implication or otherwise: *f has the IVP implies its graph G_f is connected,* and it turns out that the answer is negative, i.e., *f has the IVP does not imply that its graph G_f is connected.* The counterexample is not easy; it is a slight modification of a function described by F.B.Jones in [51]. This modified function, and other relevant material is given in [19].

§4.8. Convex functions

The function $f\colon [a,b] \to \mathbb{R}$ is said to be convex (downwards) if between any two points $x_1, x_2 \in [a,b]$, the curve $y = f(x)$ lies below or on the chord joining the points $A_1 \equiv (x_1, f(x_1))$ and $A_2 \equiv (x_2, f(x_2))$ of the graph G_f of f (see Figure 4.23).

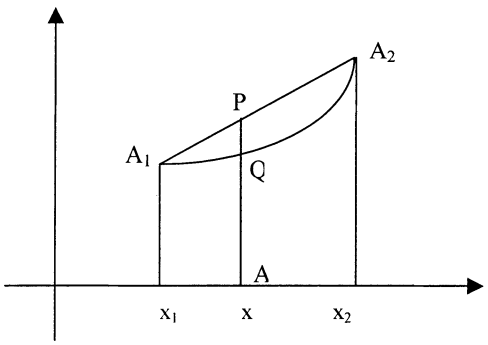


Figure 4.23

The condition simply means that $AQ \leq AP$, i.e., $f(x)$ is less than or equal to the y-coordinate of P, i.e., $f(x) \leq (f(x_2)-f(x_1))((x-x_1)/(x_2-x_1))+f(x_1)$, which on simplification becomes

$$f(x) \leq ((x_2 - x)/(x_2 - x_1)).f(x_1) + ((x - x_1)/(x_2 - x_1)).f(x_2)\,. \qquad (*)$$

Since $x = \lambda x_1 + (1-\lambda)x_2$, $0 \leq \lambda \leq 1$, this condition $(*)$ becomes the equivalent condition

$$f(\lambda x_1 + (1 - \lambda)x_2) \leq \lambda f(x_1) + (1 - \lambda)f(x_2). \qquad (*)_1$$

Another equivalent condition is the symmetrical form (easy to check):

$$\frac{(x_3 - x_2).f(x_1) + (x_1 - x_3).f(x_2) + (x_2 - x_1).f(x_3)}{(x_1 - x_2)(x_2 - x_3)(x_3 - x_1)} \geq 0\,, \qquad (*)_2$$

for all distinct $x_1, x_2, x_3 \in [a, b]$, $x_1 < x_2 < x_3$, which may be written in the form of a determinant:

$$\begin{vmatrix} 1 & x_1 & f(x_1) \\ 1 & x_2 & f(x_2) \\ 1 & x_3 & f(x_3) \end{vmatrix} \geq 0\,.$$

Theorem 4.8.1. *Suppose f is convex in $[a, b]$. Then f is continuous in $[a, b]$.*

Proof. Let x_1, x_2, x_3 be any three points of $[a, b]$. By $(*)$, we have

$$f(x_2) \leq ((x_3 - x_2)/(x_3 - x_1))f(x_1) + ((x_2 - x_1)/(x_3 - x_1)).f(x_3)\,. \qquad (\dagger)$$

Take $x \in (a, b)$ and $h > 0$, sufficiently small, and $x_1 = x-h, x_2 = x, x_3 = x+h$; we get

$$f(x) \leq \frac{h}{2h}f(x) + \frac{h}{2h}f(x + h)\,,$$

which implies that (taking $h \to 0$) $f(x) \leq \lim_{t\to x+} f(t)$. Also, taking $x_1 = x, x_2 = x + h, x_3 = x + 2h$, we get

$$f(x + h) \leq \frac{1}{2}f(x) + \frac{1}{2}f(x + 2h)\,,$$

so that $\lim_{t\to x+} f(t) \leq f(x)$. Thus, $f(x) = \lim_{t\to x+} f(t)$. Similar considerations yields that $f(x) = \lim_{t\to x-} f(t)$. So that $f(x) = \lim_{t\to x} f(t)$ and f is continuous in $[a, b]$

Remark 4.8.2. The function f may fail to be continuous at a and/or at b; for example, if $f(x) \equiv 0$ in $(0, 1)$, while at $x = 0, 1$, $f(x) = 1$.

Example 4.8.3. A convex function, in an open or a half-open interval, may be unbounded, for example, let $f(x) = \tan x$ in $[0, \pi/2)$.

Remark 4.8.4. If we put $x = (x_1 + x_2)/2$ in $(*)$, we obtain

$$f((x_1 + x_2)/2) \leq (f(x_1) + f(x_2))/2\,, \qquad (**)$$

which is sometimes taken as the definition of convexity (rather than (∗)). We call it the *mid-convexity condition* and any function f satisfying it, is called *mid-convex*. It is less restrictive than (∗), since it assumes (∗) for the mid-point $(x_1 + x_2)/2$ of x_1 and x_2. Indeed, this condition does not imply continuity of the function. To achieve continuity, we have to put further restrictions on f. One such restriction is that f be bounded above. Indeed, the following stronger result is also known to hold, namely: *Suppose that f is mid-convex on $[a, b]$, and that f be bounded above in a neighbourhood of a single point ξ of $[a, b]$, then f is continuous at all points of (a, b)*. The condition f be bounded above, is essential. Indeed, there exists mid-convex functions that are discontinuous everywhere but not bounded in any interval. Here is an example:

Example 4.8.5. Let $\mathcal{B} = \{b_i\}_{i \in I}$ be a basis of the vector space $\mathbb{R}$ over $\mathbb{Q}$. Define f on $\mathcal{B}$ by putting $f(b_i)$ to be equal to an arbitrary $r_i \in \mathbb{Q}$. Then if $x = \sum \alpha_i b_i$ ($\alpha_i \in \mathbb{Q}, \alpha_i = 0$ for all except finitely many i), we let $f(x) = \sum \alpha_i f(b_i) = \sum \alpha_i r_i \in \mathbb{Q}$. It is now trivial to check that if $y = \sum \beta_i b_i$ ($\beta_i \in \mathbb{Q}, \beta_i = 0$ for all but finitely many i), then (∗∗) holds, with equality sign, i.e., $f((x + y)/2) \leq (f(x) + f(y))/2$; so f is mid-convex. However, f can not be continuous, as a continuous rational valued function would be a constant, which our f is not.

For all this, and much more, see the book by Wayne Roberts and Varberg [116].

Exercises

4.1. Show that the function $F(x) = \int_0^x \sin(1/t)dt$ is differentiable at $x = 0$ (note that $\sin(1/t)$ is bounded and continuous everywhere, except at $x = 0$ and so is Riemann integrable over any bounded interval; thus $F(x)$ is defined).

4.2. Let A be the area under the curve $y = e^x/x$, between the x-axis and the two ordinates $x = \alpha$, $x = \alpha + 1$. Locate α for which A has the least value.

4.3. Prove the geometric, logarithmic, arithmetic mean inequalities:

$$\sqrt{ab} < \frac{b - a}{\log b - \log a} < \frac{a + b}{2}\ ,\ 0 < a < b\,,$$

by considering the area under the curve $y = e^x$, between the ordinates $x = \log a$, $x = \log b$ and certain trapezoia involving the point $C = (\frac{\log a + \log b}{2}, 0)$.

4.4. Suppose f is continuous in $[0, n]$ and $f(0) = f(n) = 0$. Then $f(x') = f(x'')$ has at least n different solutions with $x'' - x' \in \mathbb{N}$.

4.5. Let $f : \mathbb{R} \to \mathbb{R}$ be a continuous function and denote the n^{th} iterate of f by f^n, i.e., $f^1(x) = f(x), f^2(x) = f(f(x)), \ldots, f^n(x) = f(f^{n-1}(x))$. A number x is said to be a point of period k for f if $f^k(x) = x$, but $f^i(x) \neq x$ for $0 < i < k$. Prove that if f has points of period $k > 1$, then f has a fixed point, i.e., a point

of period 1.

4.6. R. Kannan proved (in Bull. Calcutta Math. Soc., 60 (1968),71-76) that if f is a real-valued function such that $|f(x) - f(y)| = c(|x - f(x)| + |y - f(y)|)$ for all $x, y \in \mathbb{R}$, $0 \leq c < 1/2$, then f has a unique fixed point, i.e., there exists a unique ξ such that $f(\xi) = \xi$.

Prove that if f is a real-valued function such that $|f(x) - f(y)| = c(|x - f(y)| + |y - f(x)|)$, for all $x, y \in \mathbb{R}$, where $0 \leq c < 1/2$, then f has a unique fixed point, i.e., there exists a real number ξ such that $f(\xi) = \xi$ (B.Fisher, Math. Mag., **48** (1975), 223–225).

Chapter 5

The Derivative and Higher Derivatives

§5.1. Continuity of the first derivative

The derivative of a function, even if it exist everywhere, can be discontinuous. For example, if $f(x) = x^2 \sin \frac{1}{x}$, for $x \neq 0$, $f(0) = 0$, then for $x \neq 0$, $f'(x) = -\cos \frac{1}{x} + 2x \sin \frac{1}{x}$, which does not tend to any limit as x tends to 0, although $f'(0)$ exists and equals 0, as may be easily checked. Thus, $f'(x)$ exists everywhere, but is discontinuous at $x = 0$.

Recall that a function f is said to have a removable discontinuity at $x = c$ if $\lim_{x \to c} f(x)$ exists but is not equal to $f(c)$ (which may or may not exist). The function f is said to have a discontinuity of the first kind (or a jump discontinuity) at a point c, if both $\lim_{x \to c+} f(x)$ and $\lim_{x \to c-} f(x)$ exist but are not equal and f is said to have a discontinuity of the second kind at a point c if either one or both $\lim_{x \to c+} f(x)$ and $\lim_{x \to c-} f(x)$ do not exist. A remarkable result is:

Theorem 5.1.1. *The derivative of a function can have a discontinuity only of the second kind.*

Proof. We must rule out the two possibilities:

- f' has a removable discontinuity at a point $x = a$, say;
- f' has a jump discontinuity at $x = a$.

Now

$$f'(a) = \lim_{x \to a} \frac{f(x) - f(a)}{x - a}$$

and here, since the numerator and the denominator both vanish at $x = a$, L'Hospital's rule (see §5.7) may be applied to evaluate this limit and yields

$$f'(a) = \lim_{x \to a} \frac{\frac{d}{dx}(f(x) - f(a))}{\frac{d}{dx}(x - a)} = \lim_{x \to a} f'(x)\,.$$

Thus, f' can not have a removable discontinuity or singularity.

To rule out the second possibility, it is enough to prove IVP for f', i.e., if $f'(a) < c < f'(b)$, then there exists a number $\xi \in (a, b)$ such that $f'(\xi) = c$

To this end, let $\phi(x) = f(x) - cx$. Then $\phi'(a) < 0, \phi'(b) > 0$ and we have to find a number ξ in (a, b) such that $\phi'(\xi) = 0$.

Since ϕ' exists in $[a,b]$, ϕ is continuous and so bounded below and attains its infimum in $[a,b]$, call it m, i.e., $\phi(\xi) = m = \inf \phi(x)$. Then $\phi'(a) < 0$ implies that $\{\phi(x) - \phi(a)\}/(x-a) < 0$ for all $x \in (a, a+\delta)$. Thus, $\phi(x) < \phi(a)$ for all $x \in (a, a+\delta)$, and hence $\xi \neq a$; similarly $\xi \neq b$, i.e., $\xi \in (a,b)$.

Now, $(\phi(x) - \phi(\xi))/(x-\xi) \geq 0$ as $x \to \xi+$, and is less than or equal to 0 as $x \to \xi-$, i.e., $\phi'_+(\xi) \geq 0$, $\phi'_-(\xi) \leq 0$, but $\phi'_+(\xi) = \phi'_-(\xi) = \phi'(\xi)$, since $\phi'(\xi)$ exists, hence $\phi'(\xi) = 0$, as required.

Remark 5.1.2. The following argument is offered as a challenge to students by Albert Wilansky (Math. Mag., **38** (1965), 108).
(i) Give an example of a function, differentiable everywhere, but with a discontinuous derivative at $x = 0$,
(ii) pinpoint precisely where the fallacy in the argument below lies:

Let f be differentiable in an open interval I and let $[a,b] \subset I$. Let $\xi \in [a,b]$. For $k \neq 0$, let $\epsilon = (f(\xi+k) - f(\xi))/k - f'(\xi)$. Then

$$f(\xi+k) = f(\xi) + kf'(\xi) + k\epsilon\,, \qquad (*)$$

where $\epsilon \to 0$ as $k \to 0$. In $(*)$, taking successively (a) $\xi = 0$, $k = 2h$ (b) $\xi = 0$, $k = h$ (c) $\xi = k = h$, we get

$$f(2h) = f(0) + 2hf'(0) + 2h\epsilon_1 \qquad (1)$$
$$f(h) = f(0) + hf'(0) + h\epsilon_2 \qquad (2)$$
$$f(2h) = f(h) + hf'(h) + h\epsilon_3 \qquad (3)$$

Now (3) implies $f'(h) = (f(2h) - f(h))/h - \epsilon_3$. Substitute for $f(2h)$ and $f(h)$ from (1) and (2) and we get $f'(h) = f'(0) + 2\epsilon_1 - \epsilon_2 - \epsilon_3 \to 0$ as $h \to 0$, showing that $f'(x)$ is continuous at $x = 0$!

The following necessary and sufficient condition for the derivative f' of a function f to be continuous is due to Weinstock [117]:

Theorem 5.1.3. *Let I be an open interval of which $[a,b]$ is a closed subinterval and suppose f is differentiable in I. Then a necessary and sufficient condition for f' to be continuous in $[a,b]$ is that f be uniformly differentiable in $[a,b]$, i.e., given $\epsilon > 0$, there exists a $\delta = \delta(\epsilon)$, δ independent of x, such that*

$$|(f(x+h) - f(x))/h - f'(x)| < \epsilon$$

for all $0 < |h| < \delta$ and for all $a \leq x \leq b$, $x \pm h \in I$.

Proof. By hypothesis, if $\{h_i\}$ is any sequence of non-zero numbers, with $x \pm h_i \in I$ for all $x \in [a,b]$, which tends to 0, then the sequence of continuous functions $\{f(x+h_i) - f(x)/h_i\}$ is convergent uniformly to $f'(x)$ in $[a,b]$. Hence the limit $f'(x)$ of this sequence is continuous in $[a,b]$, as required.

Conversely, by the Mean Value Theorem (see Theorem 5.4.2)

$$\frac{f(x+h) - f(x)}{h} - f'(x) = f'(x+\theta h) - f'(x)\ ,\ 0 < \theta < 1\,, \qquad (i)$$

for each $x \in [a, b]$, $x + h \in I$.

Since, by hypothesis, f' is continuous and so uniformly continuous in $[a, b]$, there exists, for each $\epsilon > 0$, a $\delta = \delta(\epsilon) > 0$ (δ independent of x) such that $|f'(x + \theta h) - f'(x)| < \epsilon$, whenever $0 < |\theta h| < \delta$, i.e., whenever $0 < |h| < \delta$. Therefore, f is uniformly differentiable by (i) and the above observation.

However, the derivative of a function can be unbounded:

Example 5.1.4. The function

$$f(x) = \begin{cases} x^2 \sin \frac{1}{x^2} \text{ , if } x \neq 0 \\ 0 \text{ , if } x = 0 \end{cases}$$

has the derivative

$$f'(x) = \begin{cases} 2x \sin \frac{1}{x^2} - \frac{2}{x} \cos \frac{1}{x^2} \text{ , if } x \neq 0 \\ 0 \text{ , if } x = 0 \end{cases}$$

which is unbounded on the closed interval $[-1, 1]$.

After introducing the derivative, the following results are usually discussed in a first course in Calculus:
(i) The relationship between the derivative at a point P of $y = f(x)$ and the tangent to the graph of $y = f(x)$ at the point P,
(ii) Rolle's Theorem,
(iii) The First Mean Value Theorem,
(iv) $f'(x) = 0$ implies that f is a constant function,
(v) Taylor's Theorem etc.

We shall look at these results and ideas one by one, starting with the first.

§5.2. The tangent and the first derivative

Intuitively, aided by geometry, one may tend to infer that *the tangent to the graph G_f of $y = f(x)$ exists at the point $(c, f(c))$ of G_f if and only if f is differentiable at $x = c$.* Intuition is certainly a correct approach to take. However, intuitive results must always be coupled with logical reasoning. We have:

Example 5.2.1. Let $f(x) = \sqrt{1 - x^2}$ if $|x| \leq 1$, x rational and $f(x) = -\sqrt{1 - x^2}$ if $|x| \leq 1$, x irrational.

The graph of f is a "ghost circle" (see Figure 5.1) and has a tangent at the point $(0, 1)$ (see Definition 5.2.3); but f is not differentiable at $x = 0$, since it is not even continuous (see [111])!

The correct result in this direction is the following:

Theorem 5.2.2. (H.Thurston [111]). *If the derivative of a function f exists at a point $x = c$, then the tangent to the graph G_f of $y = f(x)$ exists at the point $P \equiv (c, f(c))$. If the tangent to G_f exists at a point $P \equiv (c, f(c))$ and f*

is continuous at $x = c$, then the derivative of f exists at $x = c$.

Before we give a proof of this result, we would like to give the precise definition of what we mean by a *tangent to a curve X at a point P on the curve* (without going into the tricky question of what a curve is).

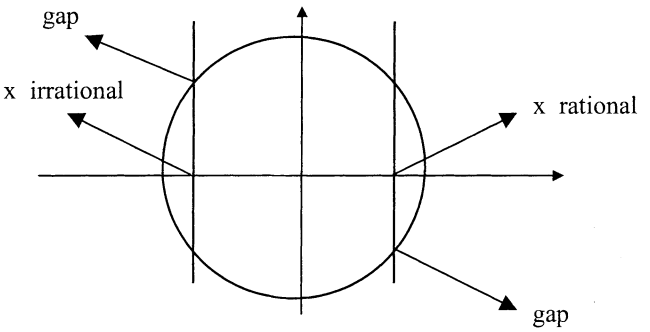


Figure 5.1

Definition 5.2.3. Let $\mathcal{F}$ be any set of points in the plane $\mathbb{R}^2$ and let P_0 be a nonisolated point of $\mathcal{F}$. A line L through P_0 is a tangent to $\mathcal{F}$ at P_0 if for given $\epsilon > 0$, there exists a $\delta > 0$ such that the angle θ between P_0P and L is less than ϵ for every point P of $\mathcal{F}$ within a distance δ of P_0 (see Figure 5.2).

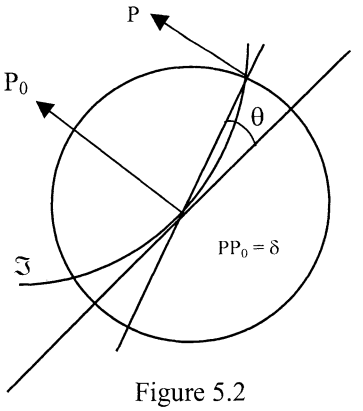


Figure 5.2

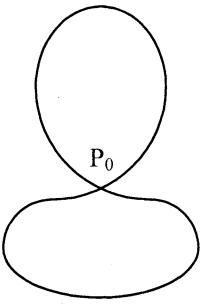


Figure 5.3

Thus, according to this definition, the 'figure of **8**' (Figure 5.3) has no tangent at P_0 where it crosses itself.

Proof of Theorem 5.2.2. We first show that if f is differentiable at $x = c$, the line L through P_0 with slope $f'(c)$ is a tangent to the graph of f at P_0. P_0 is not an isolated point of the graph because f is continuous at $x = c$ and c is a limit point of the domain of f.

Now, for each $\epsilon > 0$, there exists an $\eta > 0$ such that the lines with slopes between $f'(c) - \eta$ and $f'(c) + \eta$ make angles less than ϵ with L (continuity of

trigonometric functions). For this η, there exists a $\delta > 0$ such that

$$|\tfrac{f(c+h)-f(c)}{h}| < \eta, \text{ whenever } 0 < |h| < \delta \ \ (c+h \text{ in the domain of } f), \quad (*)$$

because f is differentiable at $x = c$.

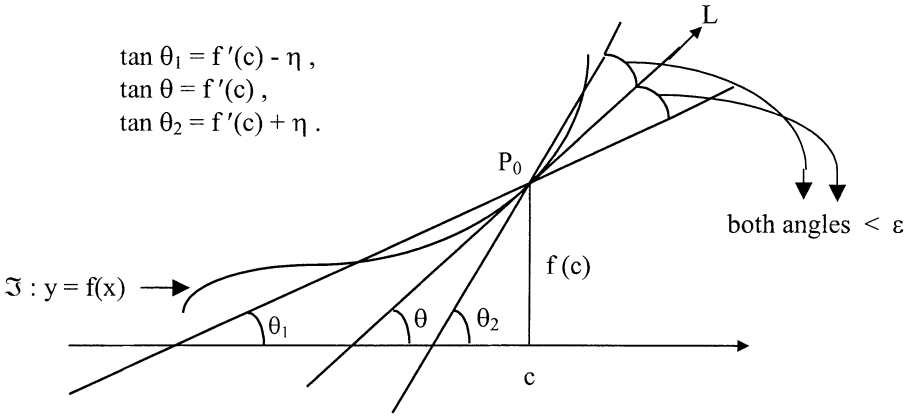


Figure 5.4

Thus, whenever $0 < |P_0P_h| < \delta$, $c+h$ in the domain of f and $0 < |h| < \delta$ (see Figure 5.5), (*) holds, i.e., $f'(c) - \eta < (f(c+h) - f(c))/h < f'(c) + \eta$, i.e.,

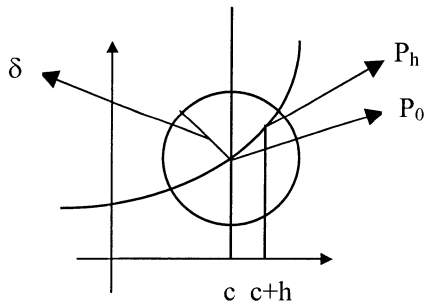


Figure 5.5

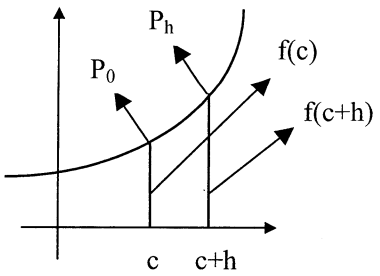


Figure 5.6

the slope of P_0P_h lies between $f'(c) \pm \eta$ and so P_0P_h makes an angle less than ϵ with L, i.e., by definition, L is the tangent at the point $(c, f(c))$ to G_f, as required.

We now come to the proof of the second part of the theorem. Suppose that f is continuous at $x = c$ and the graph of f has a nonvertical tangent τ at P_0. We then show that f is differentiable at $x = c$.

Let $\epsilon > 0$ be given and let L be the tangent at P_0 making an angle α with the x–axis, so that $\tan\alpha = m$ (m = the slope of L). Let P_h be a point on the graph close to P_0 (see Figure 5.6) (note that in general, h close to 0 does not

mean that P_h is close to P_0; for example, in our ghost circle, if we choose h small but irrational, P_h is in the lower semi circle and the chord P_0P_h is not small !). Then by the definition of "tangent", the angle θ between P_0P_h and L is small and so $\tan\theta$ is small, i.e., $\tan(\beta-\alpha)$ is small, i.e., |slope of P_0P_h − slope of L| is small, i.e.,

$$|\tfrac{f(c+h)-f(c)}{h} - m| \text{ is small, i.e., small enough to be } < \epsilon\ , \qquad (**)$$

whenever $|P_0P_h| < \eta$ (see Figure 5.7).

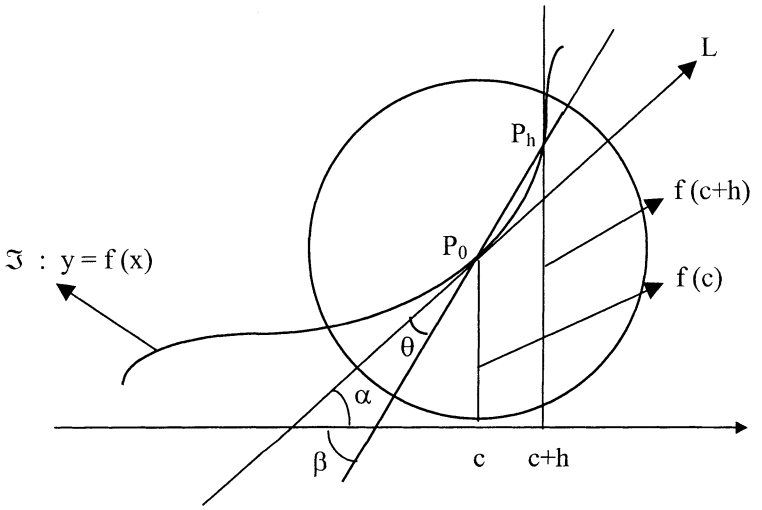


Figure 5.7

As f is continuous at $x = c$, $|P_0P_h| < \eta$ if $|h| < \delta$ (h small implies that $|f(c+h) - f(c)|$ is small and hence $|P_0P_h|$ is small). Hence $|h| < \delta \Rightarrow |P_0P_h| < \eta \Rightarrow$ $(**)$, i.e., $\lim_{h\to 0}(f(c+h)-f(c))/h$ exists and equals m, i.e., f is differentiable at c and $f'(c) = m$.

§5.3. Implications of first derivative being zero

In this section we analyse the situation when the first derivative of a function is zero. The usual proofs of the result that *if $f'(x) = 0$ for all $x \in [a,b]$, for a continuous function defined on $[a,b]$, then f is a constant function*, uses the First Mean Value Theorem, which uses properties of continuous functions such as boundedness, attainment of the bounds etc. We now give two direct proofs of this result. We first prove:

Theorem 5.3.1. *Let f be a continuous function defined on $[a,b]$. If $f'(x) > 0$ for $a < x < b$, then f is strictly increasing in $[a,b]$. Also, if $f'(x) \geq 0$ for $a < x < b$, then f is nondecreasing in $[a,b]$.*

Proof ([11]). Let $p \in (a, b)$ be any point. We wish to prove that

$$a < q < p \Rightarrow f(q) < f(p) \qquad (*)$$

Let $S = \{x \mid a < x < p, f(x) \geq f(p)\}$. Then it is enough to prove that $S = \emptyset$. Suppose not. Let $q = \sup S$. Now $f'(p) > 0$, so $(f(p+h) - f(p))/h > 0$ if $|h|$ is small. Taking $h < 0$, this shows $f(p+h) < f(p)$ if h is sufficiently small and negative. It follows that $q < p$, i.e., that $a < q < p$.

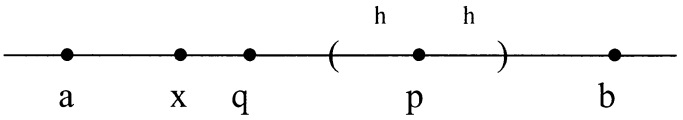


Figure 5.8

Case 1: $f(q) \geq f(p)$. Since $f'(q) > 0$, $(f(q+h) - f(q))/h > 0$, if h is small. Take h small and positive. Then $f(q+h) > f(q) \geq f(p)$, i.e., $f(q+h) > f(p)$, so $q + h \in S$, which is a contradiction since $q = \sup S$.
Case 2: $f(q) < f(p)$. Then $q \notin S$ and by the continuity of f (at q), no point of a small neighbourhood of q is in S ($f(q) < f(p) \Rightarrow f(x) < f(p)$ if x belongs to a small neighbourhood of q). In particular $[q - h, q] \cap S = \emptyset$, which is again a contradiction since $q = \sup S$.

For the second part, let $g(x) = f(x) + \epsilon x$, where $\epsilon > 0$ is chosen suitably small. Then $g'(x) = f'(x) + \epsilon > 0$ for $a \leq x \leq b$, so, a fortiori, for $a < x < b$ we can apply the above result to g to conclude that g is strictly increasing, i.e.,

$$\begin{aligned} x_1 > x_2 &\Rightarrow g(x_1) > g(x_2) \\ &\Rightarrow f(x_1) + \epsilon x_1 > f(x_2) + \epsilon x_2 \\ &\Rightarrow f(x_1) > f(x_2) - \epsilon(x_1 - x_2) \\ &\Rightarrow f(x_1) \geq f(x_2)\,, \end{aligned}$$

since ϵ is arbitrary small.

Theorem 5.3.2. *Let f be a continuous function in $[a, b]$ such that $f'(x) = 0$ for all $x \in (a, b)$. Then f is a constant function.*

Proof. Observe that $f'(x) = 0 \Rightarrow f'(x) \geq 0$ and $f'(x) \leq 0$, which together imply that f is nondecreasing and that f is nonincreasing. Thus f is a constant function.

Another direct proof of Theorem 5.3.2 ([92]). Let $f'(x) \equiv 0$ on $[a, b]$. If f is not a constant, as claimed, then, for some $a \leq u < v \leq b$, $f(u) \neq f(v)$. Hence the chord joining $(u, f(u))$ and $(v, f(v))$ has a nonzero slope, i.e.,

$$f(v) - f(u) = c(v - u) \quad (c \neq 0)\,. \qquad (*)$$

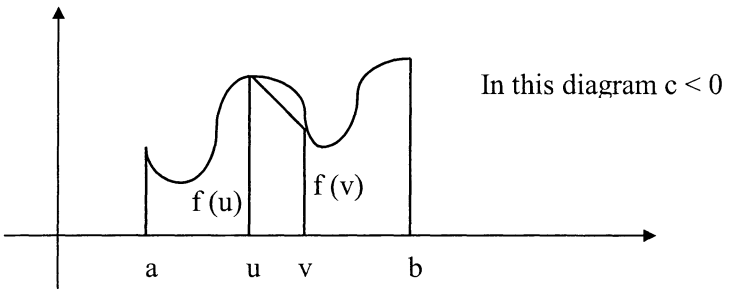


Figure 5.9

First, let $c > 0$. Bisect $[u, v]$ at w. If

$$f(v) - f(w) < c(v - w) \text{ and } f(w) - f(u) < c(w - u) \qquad (\dagger)$$

then $f(v) - f(u) < c(v - u)$, which contradicts (*). Hence, for at least one of the intervals $[u, w], [w, v]$, (†) fails. Call this interval $[u_1, v_1]$. By repeated bisection, we obtain a sequence of nested intervals $[u, v] = [u_0, v_0] \supset [u_1, v_1] \supset \ldots$ for each of which $f(v_j) - f(u_j) \geq c(v_j - u_j)$. Let $\xi = \cap [u_j, v_j] \subseteq [u, v]$. Hence, in the case $c > 0$, $(f(v_j) - f(u_j))/(v_j - u_j) \to f'(\xi) \geq c > 0$, which is a contradiction, since $f'(x) \equiv 0$ on $[a, b]$.

If $c < 0$, the same argument as above, with inequalities reversed, yields a contradiction.

It follows that $c = 0$, which is again a contradiction to (*). Hence $f(u) = f(v)$.

Remark 5.3.3. The same argument shows that if $f'(x) \geq 0$ on $[a, b]$, then $f(v) \geq f(u)$, while if $f'(x) \leq 0$ on $[a, b]$, then $f(v) \leq f(u)$. Indeed, the argument may be generalized to show that if $m \leq f'(x) \leq M$ in $[a, b]$, then $m(v - u) \leq f(v) - f(u) \leq M(v - u)$, namely, it is sufficient to replace $c > 0$ by $c > M$ and $c < 0$ by $c < m$.

Some more direct proofs of Theorem 5.3.2 are given in [26] and [83].

A rather specialized result is the following:

Theorem 5.3.4. *Let f be a continuous function defined in an interval I. If the right derivative of f at each interior point of I exists and is zero, then f is constant in I.*

The following lemmas are needed for the proof.

Lemma 5.3.5. *Let f be continuous in $[a, b]$, $f(a) = f(b) = 0$. If the right derivative $f'_+(x)$ of f exists at each point $x \in (a, b)$, then there exists a point $\xi \in (a, b)$ at which $f'_+(\xi) \leq 0$.*

Proof. If f is a constant, the lemma is trivially true; so suppose otherwise. Since f is continuous, it assumes either a positive maximum or a negative minimum. We consider these cases separately.

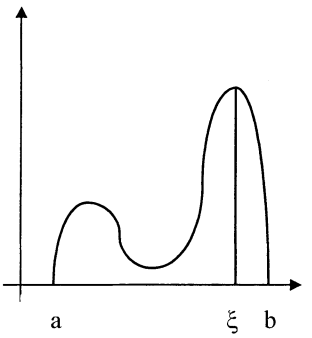


Figure 5.10

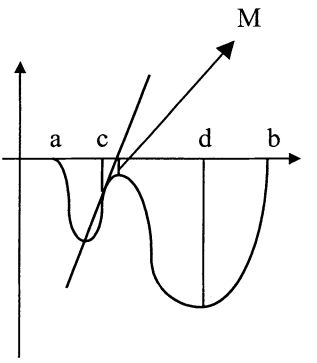


Figure 5.11

Case 1: *f has a positive maximum*, say at $\xi \in (a,b)$, see Figure 5.10 ($\xi \neq a, b$ as $f(a) = f(b) = 0$); so that $f(x) \leq f(\xi)$ for all $x \in [a,b]$. Then $(f(x) - f(\xi))/(x - \xi) \leq 0$ for all $\xi < x \leq b$ and letting $x \to \xi+$ gives $f'_+(\xi) \leq 0$, as required.
Case 2: *f has a negative minimum*, say at $d \in (a,b)$, see Figure 5.11. Now look at $f'_+(x)$ for $x \in (a,d)$. If $f'_+(x) \leq 0$ for all $x \in (a,d)$, then the result is true. If not, choose a point $c \in (a,d)$ such that $f'_+(c) > 0$. Then for all x sufficiently near c, on the right of c, we have $(f(x) - f(c))/(x - c) > 0$ for $x \in (c, c+\delta)$. Hence $f(x) > f(c)$ for $x \in (c, c+\delta)$. Now, let M be the maximum value of f on $[c,d]$. Since $f(x) > f(c)$ for all $x \in (c, c+\delta)$ and $f(d)$ is the negative minimum, we have $M > f(c) \geq f(d)$, so f assumes the value M at some point $\xi \in (c,d)$. This leads us to case 1, with $[c,d]$ as the initial interval I and M the maximum of f in $[c,d]$, attained at $x = \xi \in (c,d)$, *i.e.*, $f(\xi) = M$. Hence by case 1, $f'_+(\xi) \leq 0$.

Lemma 5.3.6. *Let f be continuous in* $[a,b]$, $f(a) = f(b) = 0$. *If f has a right derivative at each point in* (a,b), *then there exists a point* $\eta \in (a,b)$ *at which* $f'_+(\eta) \geq 0$.

Proof. Apply Lemma 5.3.5 to $-f$.

Lemma 5.3.7. *Let f be continuous in* $[a,b]$. *If f has a right derivative at each point of* (a,b), *then there exist points* ξ, η *in* (a,b) *such that*

$$f'_+(\xi) \leq \frac{f(b) - f(a)}{b - a} \leq f'_+(\eta).$$

Proof. Let $g(x) = f(x) - f(a) - [(f(b) - f(a))/(b - a)].(x - a)$, so that g is continuous in the interval $[a,b]$, $g(a) = g(b) = 0$ and $g'_+(x)$ exists for all $x \in (a,b)$. Hence Lemmas 5.3.5 and 5.3.6 hold for g, giving the required result.

Proof of Theorem 5.3.4. Choose points $a, b \in I$, $a < b$, and apply Lemma 5.3.7 to f in $[a,b]$. Since $f'_+(x) \equiv 0$ in (a,b), by hypothesis, it follows from

above that $f(a) = f(b)$. We have thus proved that for any two points $a, b \in I$, $f(a) = f(b)$ and hence f is a constant function.

Remark 5.3.8. We would like to remind the reader that the first derivative being equal to zero implies that the function is constant, is true only for intervals. Otherwise, if $f : [1,2] \cup [3,4] \to \mathbb{R}$ is defined by $f(x) = 1$ for $x \in [1,2]$ and $f(x) = 2$ for $x \in [3,4]$, then $f'(x) = 0$ for all $x \in (1,2) \cup (3,4)$.

Some relevant counterexamples regarding monotonicity and extreme values are:

Example 5.3.9. The function

$$f(x) = \begin{cases} x^4(2 + \sin \frac{1}{x}) \ , \text{ if } x \neq 0 \\ 0 \ , \text{ if } x = 0 \ , \end{cases}$$

has an absolute minimum value at $x = 0$. Its derivative is

$$f'(x) = \begin{cases} x^2(4x(2 + \sin \frac{1}{x}) - \cos \frac{1}{x}) \ , \text{ if } x \neq 0 \\ 0 \ , \text{ if } x = 0 \ , \end{cases}$$

which has both positive and negative values in every neighbourhood of 0. So f is not monotone in any interval of the form $(a, 0)$ or $(0, b)$. Thus f has an extreme value at a point where the derivative does not make a simple change in sign.

Example 5.3.10. The function

$$f(x) = \begin{cases} x + 2x^2 \sin \frac{1}{x} \ , \text{ if } x \neq 0 \\ 0 \ , \text{ if } x = 0 \ , \end{cases}$$

has the derivative

$$f'(x) = \begin{cases} 1 + 4x \sin \frac{1}{x} - 2 \cos \frac{1}{x} \ , \text{ if } x \neq 0 \\ 0 \ , \text{ if } x = 0. \end{cases}$$

In every neighbourhood of 0, the function $f'(x)$ has both positive and negative values, however, $f'(0) > 0$, yet f is not monotone in any neighbourhood of 0.

Example 5.3.11 [37]. The function

$$f(x) = \begin{cases} x^4 e^{-\frac{1}{4}x^2} \sin \frac{8}{x^3} \ , \text{ if } x \neq 0 \\ 0 \ , \text{ if } x = 0 \end{cases}$$

has the derivative

$$f'(x) = \begin{cases} e^{-\frac{1}{4}x^2}[(4x^3 - \frac{1}{2}x^5) \sin \frac{8}{x^3} - 24 \cos \frac{8}{x^3}] \ , \text{ if } x \neq 0 \\ 0 \ , \text{ if } x = 0. \end{cases}$$

In every neighbourhood of the origin, this derivative has values arbitrarily near both 24 and -24. Also, for $0 < h \leq 1$ and for $x = h$,

$$0 < e^{-\frac{1}{4}x^2} < 1 - \frac{1}{4}h^2 e^{-\frac{1}{4}h^2} < 1 - \frac{3}{16}h^2 ,$$

and

$$\left| \left(4x^3 - \frac{1}{2}x^5\right) \sin\frac{8}{x^3} - 24\cos\frac{8}{x^3} \right| \leq 24 + \frac{9}{2}h^3 .$$

Therefore, $0 < h \leq 1$ implies that

$$|f'(x)| < \left(1 - \frac{3}{16}h^2\right)\left(24 + \frac{9}{2}h^3\right) < 24 - \frac{9}{2}h^2(1-h) \leq 24 .$$

Thus, on the closed interval $[-1, 1]$, the range of f' has supremum equal to 24 and infimum equal to -24 and neither of these numbers is assumed as a value for f'.

§5.4. Rolle's Theorem and the first Mean Value Theorem

The usual proof of Rolle's Theorem uses the result that the function f is bounded, f attains its maximum at $x = \xi$, say, and then show that $f'(\xi) = 0$. We give here a new proof using only the intermediate value theorem.

Theorem 5.4.1 (Rolle's Theorem). *Suppose f' exists in (a, b), f is continuous in $[a, b]$, and $f(a) = f(b)$; then there exists $\xi \in (a, b)$ such that $f'(\xi) = 0$.*

Proof (H.Samelson, [99]). Write $[a, b] = [a_0, b_0]$. We first prove that:

$$\begin{array}{c} \textit{there exists an interval } [a_1, b_1] \subset [a_0, b_0], \textit{ with } b_1 - a_1 = (b_0 - a_0)/2, \\ \textit{such that } f(a_1) = f(b_1). \end{array} \qquad (*)$$

Let $g(x) = f(x + (b-a)/2) - f(x)$. Then

$$\begin{aligned} g((a+b)/2) &= f((a+b)/2 + (b-a)/2) - f((a+b)/2) \\ &= f(b) - f((a+b)/2) \\ &= -f((a+b)/2) + f(b) \\ &= -f(a + (b-a)/2) + f(a) \qquad \text{(since } f(a) = f(b)\text{)} \\ &= -(f(a + (b-a)/2) - f(a)) \\ &= -g(a). \end{aligned}$$

Hence, by the intermediate value theorem, $g(\xi) = 0$ for some $\xi \in (a, (a+b)/2)$, i.e., $f(\xi + (b-a)/2) = f(\xi)$ and letting $[a_1, b_1] = [\xi, \xi + (b-a)/2]$, (*) follows.

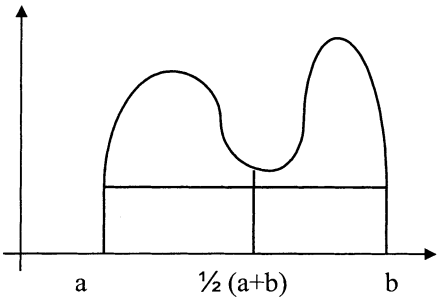


Figure 5.12

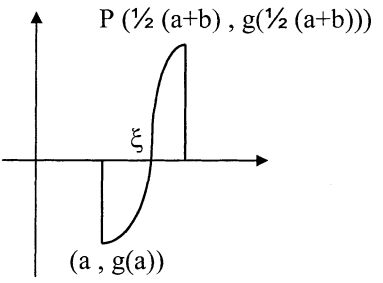


Figure 5.13

Note that $b_1 - a_1 = \xi + (b-a)/2 - \xi = (b-a)/2$, as required and also that since $\xi < (a+b)/2$, we have $\xi + (b-a)/2 < (a+b)/2 + (b-a)/2 = b$ and so $[\xi, \xi + (b-a)/2] = [a_1, b_1] \subset [a, b]$.

Now apply (*) repeatedly to get a sequence $[a_n, b_n]$ of nested intervals such that

(i) $b_n - a_n = (b-a)/2^n$,

(ii) $[a_n, b_n] \subset [a_{n-1}, b_{n-1}]$, $n = 1, 2, \ldots$,

(iii) $f(a_n) = f(b_n)$.

Then a_n increases, b_n decreases, and both are bounded and so they both tend to limits, say ξ, η where $\xi = \eta$ (by (i)) and $a_n \leq \xi \leq b_n$.

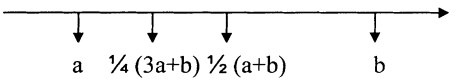


Figure 5.14

It follows that $(f(b_n) - f(a_n))/(b_n - a_n) \to f'(\xi)$; but by (iii), the quantities $(f(b_n) - f(a_n))(b_n - a_n)$ are all equal to 0, giving $f'(\xi) = 0$.

Finally, note that this $\xi \neq a$ or b for, suppose for example that $\xi = a$. Then $[a_1, b_1] = [\xi, \xi + (b-a)/2] = [a, (a+b)/2]$, and $[a_2, b_2] = [\xi, \xi + (b_1 - a_1)/2] = [a, (3a+b)/4]$ (check this last one), so that $f(a) = f((a+b)/2) = f((3a+b)/4)$ (observe that $f(a_n) = f(b_n)$). But then we may choose $[a_2, b_2]$ to be equal to $[(3a+b)/4, (a+b)/2]$ rather than equal to $[a, (3a+b)/4]$ and then subsequently there is no danger of the limit ξ of a_n (or b_n) being equal to a. Similarly, $\xi \neq b$.

Observe that this really is a different proof. The point ξ is not necessarily an absolute maximum or a minimum, nor even necessarily a local maximum or minimum. For yet another proof of Rolle's Theorem see [1].

Following Chorlton [25], we now discuss mean value theorems. The first

mean value theorem (MVT) is:

Theorem 5.4.2. *Suppose f is continuous in $[a, a+h]$ and differentiable in $(a, a+h)$; then there exists a real number θ, $0 < \theta < 1$, such that*

$$\frac{f(a+h) - f(a)}{h} = f'(a + \theta h) .$$

In general, this number θ depends on a and h. However, if f is a quadratic polynomial: $f(x) = A + Bx + Cx^2$, then in any interval $[a, a+h]$, it turns out that $\theta = 1/2$. For, we have

$$\begin{aligned} \frac{f(a+h) - f(a)}{h} &= \frac{(A + B(a+h) + C(a+h)^2) - (A + Ba + Ca^2)}{h} \\ &= B + 2C(a + h/2) \\ &= f'(a + h/2) , \end{aligned}$$

as required.

Geometrically, this simply means the following: Consider the parabola

$$\mathcal{P} : y = A + Bx + Cx^2$$

represented by a general polynomial of degree 2. Let $P_1 \equiv (x_1, y_1)$, $P_2 \equiv (x_2, y_2)$ be any two points on $\mathcal{P}$. Then the abscissa x^* of the point $P^* = (x^*, y^*)$ at which the tangent to $\mathcal{P}$ is parallel to the chord P_1P_2 is the arithmetic mean of the abscissae of the end points of the chord P_1P_2, i.e., $x^* = (x_1 + x_2)/2$. This may be directly verified using co-ordinate geometry. Analytically it translates into the following:

Theorem 5.4.3. *The chord P_1P_2 is parallel to the tangent at P^*, where, if $P_1 \equiv (x_1, y_1)$, $P_2 \equiv (x_2, y_2)$, then $P^* \equiv ((x_1 + x_2)/2, y^*)$.*

Proof. We are required to prove that the slope of $P_1P_2 = (dy/dx)_{P^*}$. Here the left hand side equals (see Figure 5.15)

$$\begin{aligned} \frac{y_2 - y_1}{x_2 - x_1} &= \frac{f(x_2) - f(x_1)}{x_2 - x_1} \\ &= \frac{A + Bx_2 + Cx_2^2 - A - Bx_1 - Cx_1^2}{x_2 - x_1} \\ &= B + C(x_1 + x_2) , \end{aligned}$$

while the right hand side equals $(B + 2C\,x)_{P^*} = B + 2C(x_1 + x_2)/2$, as required.

Incidentally, this affords a means of constructing the tangent at any point $P \equiv (x, y)$ of $\mathcal{P}$. For, let P_1, P_2 be any points with abscissa $x - h, x + h$ (where x is the abscissa of P). Then the line through P, parallel to P_1P_2 is the tangent to $\mathcal{P}$ at P.

Our object now is to investigate the converse problem, i.e., *to determine which real valued functions f have the property that there exists a fixed θ such that for all a, h, $f(a+h) - f(a) = hf'(a+\theta h)$.*

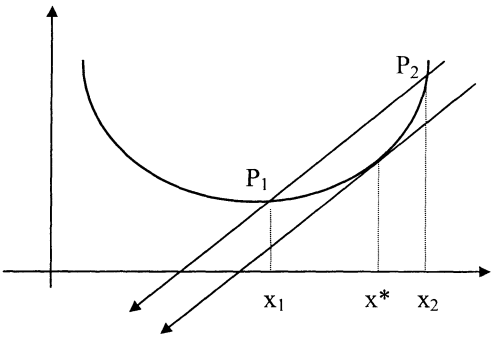


Figure 5.15

We have the following remarkable

Theorem 5.4.4. *Let f be a real valued differentiable function and suppose that the equation (i.e., the first MVT)*

$$f(a+h) - f(a) = hf'(a+\theta h) \qquad (*)$$

holds for all values of a and h, with θ independent of a and h. Then

$$f(x) = C + Bx + Ax^2 \, .$$

Proof (T.W. Chaundy, [24]). In (*), first write $-h$ for h and then replace a by $a+h$ to get

$$f(a+h) - f(a) = hf'(a+(1-\theta)h) \, ; \qquad (1)$$

then (*) and (1) imply, that for all a, h

$$f'(a+\theta h) = f'(a+(1-\theta)h) \, . \qquad (2)$$

But $x = a+\theta h$ and $y = a+(1-\theta)h$ can take all real values x and y, for suitable a and h, provided $\theta \neq 1/2$. Indeed, given x, y, subtracting the second from the first, we get $h = (x-y)/(2\theta-1)$ and then $a = x - \theta((x-y)/(2\theta-1))$, so for these values of h and a, any given x and y can be got. So now (2) can be interpreted as saying

$$f'(x) = f'(y) \quad \text{for all } x, y \, ,$$

i.e., that f' is a constant, say B, which then gives $f(x) = C + Bx$.

Now let $\theta = 1/2$. Then (*) reads:

$$f(a+h) - f(a) = hf'(a+h/2)\ ; \tag{3}$$

In this, let (i) $a \to a$, $h \to -h$ (ii) $a \to a-h$, $h \to 2h$, to get the equations

$$f(a-h) - f(a) = -hf'(a-h/2)\,; \tag{4}$$

$$f(a+h) - f(a-h) = 2hf'(a)\,. \tag{5}$$

Now (4)+(5)−(3) implies that $f'(a+h/2) + f'(a-h/2) = 2f'(a)$. In this equation substituting (i) $a = (x+t)/2$, $h = x+t$ and (ii) $a = (x+t)/2$, $h = x-t$ yields respectively $f'(x+t)+f'(0) = 2f'((x+t)/2)$, $f'(x)+f'(t) = 2f'((x+t)/2)$ and these give

$$f'(x+t) - f'(x) = f'(t) - f'(0)\,. \tag{6}$$

Now let $\phi(t) = f'(t) - f'(0) = f'(x+t) - f'(x)$ (for all x, by (6)). Then

$$\begin{aligned}(f(x+y) - f(x)) - (f(y) - f(0)) &= yf'(x+y/2) - yf'(y/2) && \text{(by (3))}\\ &= y(f'(x) - f'(0)) && \text{(by (6))}\\ &= y\phi(x) && \text{(by definition of } \phi)\\ &= x\phi(y)\ ,\end{aligned}$$

by symmetry in x, y, since the extreme left hand side of the above is symmetric in x and y.

Hence $\phi(x)/x = \phi(y)/y$ (for all x, y) and so is a constant, say K, i.e., $\phi(x) = Kx$ and so by the definition of ϕ, we get $f'(x+t) - f'(x) = Kt = K(x+t) - Kx$, i.e., $f'(x+t) - K(x+t) = f'(x) - Kx$ (for all t). Now writing y for $x+t$, this gives $f'(y) - Ky = f'(x) - Kx$ (for all x, y), i.e., that $f'(x) - Kx$ is a constant, say B, or $f'(x) = B + Kx$, or $f(x) = Kx^2/2 + Bx + C = Ax^2 + Bx + C$.

In this connection, see also [89].

Remark 5.4.5. If we assume f to be 3 times differentiable, we can get our result much more easily as follows:

Differentiate the equation (*), i.e., $f(a+h) - f(a) = hf'(a+\theta h)$, with respect to h, partially, r times to get (using Leibnitz theorem)

$$\begin{aligned}f^{(r)}(a+h) &= hf^{(r+1)}(a+\theta h).\theta^r + \binom{r}{1}\frac{d}{dh}f^{(r)}(a+\theta h).\theta^{r-1} + 0 + 0 + \dots\\ &= h\theta^r f^{(r+1)}(a+\theta h) + r\theta^{r-1}f^{(r)}(a+\theta h)\,.\end{aligned}$$

Now put $h = 0$ in this to get $f^{(r)}(a) = r\theta^{r-1}f^{(r)}(a)$. Here θ is fixed, a is any real number and r is any positive integer less than or equal to 3.

$$\begin{aligned}&r = 1 \Rightarrow f'(a) = f'(a) \ \text{(nothing new)}\ ,\\ &r = 2 \Rightarrow \text{either } f''(a) = 0 \text{ for all } a \text{ or } 2\theta = 1\ ,\\ &r = 3 \Rightarrow \text{either } f'''(a) = 0 \text{ for all } a \text{ or } 3\theta^2 = 1\ .\end{aligned}$$

If $f'''(a) = 0$ for all a, well and good, while if $3\theta^2 = 1$, then $2\theta \neq 1$ and so $f''(a) = 0$ for all a and hence $f'''(a) = 0$ for all a. Thus $f'''(a) = 0$ for all a always, which then implies that $f(x) = A + Bx + Cx^2$, as required.

The number θ appearing in MVT depends on h, i.e., $\theta = \theta_h$. Another interesting property which this θ has, is the following:

Theorem 5.4.6. *Suppose $f''(a)$ exists and is nonzero and f'' exists in a neighbourhood of a. Then $\lim_{h\to 0} \theta_h$ exists and equals $1/2$.*

Proof. We have, by L'Hospital's rule (Theorem 5.7.12),

$$\lim_{h\to 0} \frac{f(a+h) - f(a) - hf'(a)}{h^2} = \lim_{h\to 0} \left(\frac{d(f(a+h) - f(a) - hf'(a))}{dh} \Big/ \frac{dh^2}{dh} \right)$$
$$= \lim_{h\to 0} \frac{f'(a+h) - f'(a)}{2h} = \frac{f''(a)}{2}$$

On the other hand, using $f(a+h) - f(a) = hf'(a+\theta h)$, the extreme left side of the above equation is $(hf'(a+\theta h) - hf'(a))/h^2 = ((f'(a+\theta_h h) - f'(a))/\theta_h h).\theta_h$, and letting $h \to 0$, this tends to $f''(a).\lim_{h\to 0} \theta_h$, since $\theta_h h \to 0$ as $h \to 0$, because $0 < \theta < 1$. The result now follows from above.

Example 5.4.7. The condition $f''(a) \neq 0$ is essential, e.g., for the functions $f(x) = x^3$ or $f(x) = x^4$, at the point $a = 0$, $f(a+h) - f(a) = hf'(a+\theta h)$ implies, for $f(x) = x^3$:

$$f(h) - f(0) = hf'(\theta h) \Rightarrow h^3 = h.3(\theta h)^2 \Rightarrow \theta = \frac{1}{\sqrt{3}}\,, \text{which does not tend to } \tfrac{1}{2}$$

and for $f(x) = x^4$:

$$f(h) - f(0) = hf'(\theta h) \Rightarrow h^4 = h.4(\theta h)^3 \Rightarrow \theta = \frac{1}{\sqrt[3]{4}}\,, \text{which does not tend to } \tfrac{1}{2}.$$

§5.5. Points of inflection

The derivative f' of a given function f is often used to determine extreme values (the maxima, the minima and the flexes) and the following result is often *thought* to be true:

$f'(c) = 0$ implies that c is a local maximum, a local minimum or a point of inflection (for short, a flex) according as $f''(c) < 0$, $f''(c) > 0$, or $f''(c) = 0$ respectively.

This result, as it stands, is incorrect. Let us look at the following examples (A.A.Ball, [7]):

Example 5.5.1. Let (see Figure 5.16)

$$y = f(x) = \begin{cases} 3x^2/8 - 4x + 29/2\,, & \text{if } 2 \leq x < 6\,, \\ -5x^2/16 + 17x/4 - 41/4\,, & \text{if } 6 \leq x \leq 10\,. \end{cases}$$

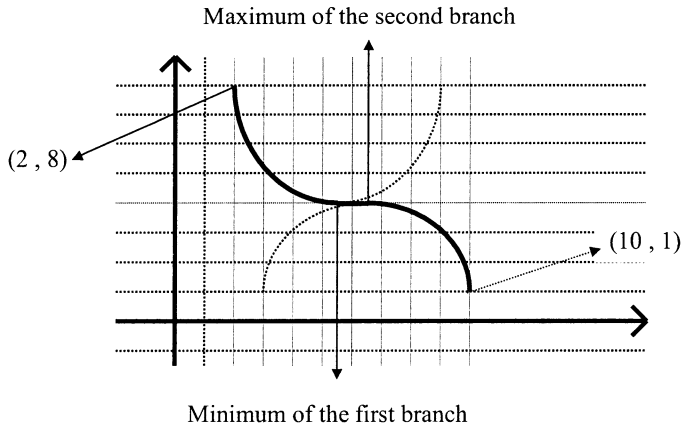


Figure 5.16

On differentiating we obtain:

$$\frac{dy}{dx} = \begin{cases} 3x/4 - 4\ , & \text{if } 2 \le x < 6\ , \\ -5x/8 + 17/4\ , & \text{if } 6 \le x \le 10\ . \end{cases}$$

At $x = 6$, $dy/dx = 1/2$ and is continuous. Now differentiating again, we get

$$\frac{d^2y}{dx^2} = \begin{cases} 3/4\ , & \text{if } 2 \le x < 6\ , \\ -5/8\ , & \text{if } 6 \le x \le 10\ . \end{cases}$$

In the figure, the point $(6, 4)$ belongs to the first branch. The minimum of the first branch is at $x = 15/3$, while the maximum of the second branch is at $x = 34/5$. The point $(6, 4)$ is a point of transition between the two branches where the respective second derivatives are of opposite signs (check). It is, therefore a flex, but this flex is not defined by the condition $d^2y/dx^2 = 0$. Indeed, $d^2y/dx^2 \neq 0$ for any value of x and at $x = 6$, d^2y/dx^2 is actually undefined, as it has a jump discontinuity at the flex.

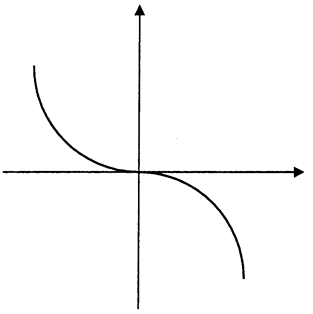


Figure 5.17

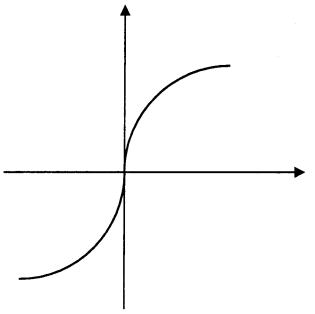


Figure 5.18

Example 5.5.2.(a). Let $y = -x^3$, then $dy/dx = -3x^2$, $d^2y/dx^2 = -6x$. This is defined for all x and $y'' = 0$ holds for $x = 0$ and gives the flex at $x = 0$, i.e., at the point $(0,0)$ (see Figure 5.17).
(b) Now consider the function $y = x^{1/3}$. Its graph is shown in Figure 5.18. Turning it clockwise by an angle $\pi/2$, it becomes exactly the same as the graph of the function $y = -x^3$ (i.e., Figure 5.17).

However, in the case $y = x^{1/3}$, $dy/dx = x^{-2/3}/3$ and $d^2y/dx^2 = -2x^{-5/3}/9$, which is defined for all $x \neq 0$. Thus $y'' > 0$ when $x < 0$ and $y'' < 0$ when $x > 0$. Consequently, the point of transition, i.e., the point $(0,0)$, is a flex but y'' is not defined at $(0,0)$.

Why this failure? Because the inflectional tangent happens to be parallel to the y-axis. In this circumstance y'' can never be defined at the flex, since approaching from one side it needs to be very large and positive $(+\infty)$ and from the other side, it needs to be very large and negative $(-\infty)$.

Thus y'' may be undefined at a flex. However, the curvature plots (see § 5.6 for curvature) of the two curves (i) the parabola described in example 5.5.1 above and (ii) $y = x^{1/3}$ of Example 5.5.2 (b) above are respectively given in Figures 5.19 and 5.20 and it is clear that the two examples are intrinsically different.

Indeed, they illustrate the only two instances in which y'' may be undefined at a flex, viz.

1. where there is a discontinuity in $1/\rho$, the curvature,
2. where the inflectional tangent is parallel to the y-axis.

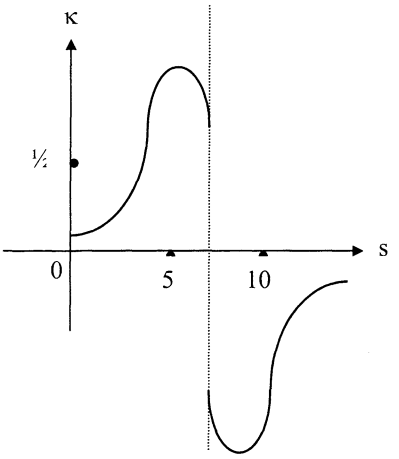


Figure 5.19

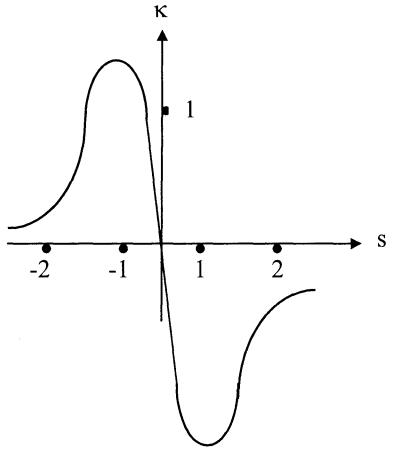


Figure 5.20

So then, how does one get all the flexes? Suppose a smooth curve is defined by $y = f(x)$. All the flexes may be found by the following procedure:

(a) Identify the points where $y'' = 0$ or is undefined.

(b) Determine the signs of y'' over the intervals separating the points determined by (a).

(c) Identify a flex if the signs of y'' are different over the intervals on either side of the point.

So, things can go wrong if one is not careful enough. Our first expedient, therefore, is to give various definitions of a flex, geometric as well as analytic, that are found in various texts and try to understand their equivalence or otherwise.

Definition 5.5.3 (Geometric). A point of inflection (flex, for short) is a point of a smooth curve C at which the tangent not only touches C but also crosses it.

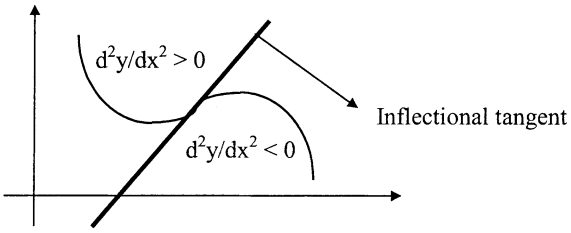


Figure 5.21

The tangent at a flex is called the inflectional tangent at the flex (see Figure 5.21). We see, from the figure, that a flex occurs when the curve stops bending to the left and starts bending to the right or vice versa.

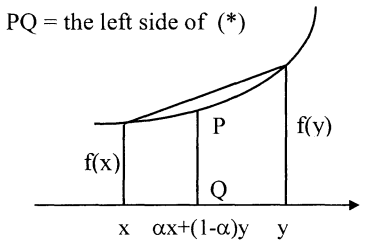
Let now C be defined by the equation $y = f(x)$. Then the direction of bending can be related to the value of d^2y/dx^2 as follows: If $d^2y/dx^2 > 0$, then the gradient dy/dx increases with x and the curve bends to the right, i.e., we have the following definition:

Definition 5.5.4 (Analytic). A flex is a point of transition between two curve segments, one where the second derivative d^2y/dx^2 is positive and one where it is negative.

A function f is said to be concave up on the interval (a, b) if for any two points $x, y \in (a, b)$, and for all $\alpha \in [0, 1]$, we have (see Figure 5.22):

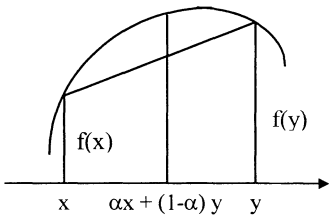
$$f(\alpha x + (1 - \alpha)y) \leq \alpha f(x) + (1 - \alpha)f(y) \qquad (*)$$

If f is differentiable, (*) is equivalent to saying that f' is increasing in (a, b). Figure 5.23 shows a curve that is convex up and hence the inequality in (*) is reversed.



f(x) **concave** up

Figure 5.22



f(x) **convex** up

Figure 5.23

We may rewrite Definition 5.5.3 in terms of concavity up (or down) as follows:

Definition 5.5.5 (Geometric-Analytic). The point c is a flex of the function f, if sufficiently close to c, the function is concave up on one side of c and concave down on the other side of c.

Let us now analyse the following four definitions, which are in current use, under the assumption that $f'(x)$ exists (but $f''(x)$ need not exist) in a neighbourhood I of the point $c \in (a, b)$ (see [18]).

Definitions 5.5.6. We say that a point c is a flex of a function f defined on an interval (a, b) if:
(I) There exists an open interval $I \subset (a, b)$, with $c \in I$ such that in I, f' is increasing on one side of c and decreasing on the other side of c.
(II) There exists an open interval $I \subset (a, b)$, with $c \in I$ such that in I, f' attains either a maximum or a minimum at c.
(III) There exists an open interval $I \subset (a, b)$, with $c \in I$ such that in I, the tangent T at c, to the graph of f, is greater than or equal to $f(x)$ on one side of c, and is less than or equal to $f(x)$ on the other side of c.
(IV) Assuming that $f''(x)$ exists in a neighbourhood I of c; the function $f''(x) \leq 0$ on one side of c, but $f''(x) \geq 0$ on the other side of c.

We now look at the relationship between these definitions. Since $f''(x) \leq 0$ implies that f' is decreasing and $f''(x) \geq 0$ implies that f' is decreasing, it follows that:
(a) c is a flex according to Definition 5.5.6 (IV) then c is a flex of f according to Definition 5.5.6(I).
(b) It is clear, from the graphs of functions that are convex up that c is a flex of f according to Definition 5.5.6 (III) if and only if c is a flex of f according to Definition 5.5.5.
(c) Let f be differentiable on (a, b) and let $c \in (a, b)$. Then c is a flex of f according to Definition 5.5.6 (I) implies that c is a flex of f according to definition 5.5.6 (II) and this implies that c is a flex of f according to definition 5.5.6 (III).

The most interesting aspect of these definitions are the following two neg-

ative results:
(d) There exist functions, differentiable infinitely many times, in a neighbourhood of a point c, which satisfy Definition 5.5.6 (III) at c but not Definition 5.5.6 (II) at c.
(e) There exist functions, differentiable infinitely many times, in a neighbourhood of a point c, which satisfy Definition 5.5.6 (II) at c but not Definition 5.5.6 (I) at c.

Example 5.5.7. To substantiate (d) above, we take

$$f(x) = \begin{cases} (e^{-x^{-2}} . \sin \frac{1}{x})^2 \ , \ \text{if } x > 0 \ , \\ 0 \ , \ \text{if } x = 0 \ , \\ -(e^{-x^{-2}} . \sin \frac{1}{x})^2 \ , \ \text{if } x < 0 \ , \end{cases}$$

and $c = 0$. A rough sketch of this function is shown in Figure 5.24.

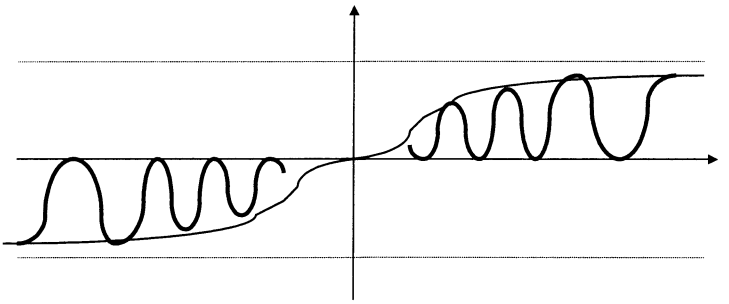


Figure 5.24

The following may now be checked easily:
(1) f is differentiable infinitely many times for all x.
(2) $f^{(n)}(0) = 0$, for all $n = 1, 2, \ldots$.
(3) Requirements of 5.5.6 (III) are satisfied by f around the origin, since the x-axis is the tangent to the curve at the origin.
(4) Requirements of 5.5.6 (II) are not satisfied by f with $c = 0$, for $f'(x)$ takes positive and negative values in any interval having the origin as an end point, so that $f'(x)$ does not attain a relative maximum or a minimum at the origin.

Example 5.5.8. To illustrate (e) above, we take $f(x) = \int_0^x h(t)dt$, where

$$h(x) = \begin{cases} (e^{-x^{-2}} . \sin \frac{1}{x})^2 \ , \ \text{if } x \neq 0 \ , \\ 0 \ , \ \text{if } x = 0 \ . \end{cases}$$

A rough sketch of the function $h(x)$ is given in Figure 5.25.

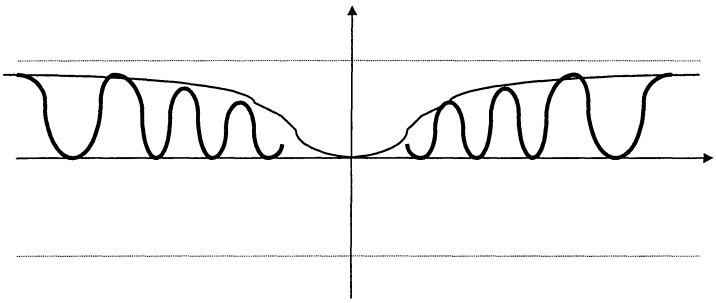


Figure 5.25

The following may now be easily verified:
(1) f is differentiable infinitely many times in a neighbourhood of the origin.
(2) $f'(x) = h(x)$, by the fundamental theorem of calculus, which is non-negative for all x.
(3) $f'(0) = 0$.
(4) $f''(x)$ takes on both positive and negative values in any interval having the origin as an end point.

Results (d) and (e) show that no amount of differentiability makes the three Definitions 5.5.6 (I), (II), (III) equivalent. We have, however, the following result:

Proposition 5.5.9. *Let f be analytic on an interval (a, b) (i.e., expandable in Taylor series around any point of the interval) and let $c \in (a, b)$. The conditions given in Definitions (I), (II) and (III) are equivalent.*

Proof. First note that the hypothesis implies that f'' can not vanish on a set of points having c as an accumulation point (unless f is linear). For, f'' is also analytic and hence determined by its values on such a set.

Thus, for $c \in (a, b)$, there exists an open interval $I \subset (a, b)$, $c \in I$, such that on I, f' is monotone on each side of c. If now the conditions of Definition 5.5.6 (III) are satisfied by f at $x = c$, then this monotonicity of f' must be of opposite kinds (i.e., increasing on one side, decreasing on the other), i.e., the conditions of Definition 5.5.6 (I) are satisfied. Thus conditions of Definitions 5.5.6 (I) $\Rightarrow$(II) $\Rightarrow$ (III) $\Rightarrow$ (I), as required.

For more details, the reader is referred to [113] and [32].

Two more examples of functions differentiable infinitely many times with special properties are:

Example 5.5.10. The function

$$f(x) = \begin{cases} e^{-1/x^2}\ , \text{ if } x > 0 \\ 0\ , \text{ if } x \leq 0 \end{cases}$$

is differentiable infinitely many times, all of its derivatives at $x = 0$ being equal to 0. This function is positive for positive x and vanishes for negative x.

Example 5.5.11. The function

$$f(x) = \begin{cases} e^{-1/(x^2(1-x)^2)} \text{ , if } 0 < x < 1 \\ 0 \text{ , otherwise} \end{cases}$$

is differentiable infinitely many times, is positive in $(0, 1)$ and vanishes outside.

§5.6. Geometric interpretation of the second and the third derivatives

For a point P on a curve $\boldsymbol{C} : y = f(x)$, the tangent of the angle θ, which the tangent line τ to $\boldsymbol{C}$ at P makes with the axis of x, is related to the derivative $(dy/dx)_P$ of f, evaluated at the point P, the relation being:

$$\left(\frac{dy}{dx}\right)_P = \tan\theta \tag{1}$$

The first derivative, therefore, has a geometric significance.

A natural question that comes to mind is whether any of the higher order derivatives, in particular, the second and the third derivative, have some kind of geometric interpretation related to the graph of $\boldsymbol{C}$.

The tangent τ at P to $\boldsymbol{C}$ may be thought of as the limit, as $P' \to P$, of a secant line joining P and a neighbouring point P' of $\boldsymbol{C}$. We say τ makes a *2-point contact* with $\boldsymbol{C}$ at P. We also note that τ and $\boldsymbol{C}$ as well as their derivatives, take same values at P. So, instead of saying 2-point contact, we sometimes say that τ has a *first order contact* with $\boldsymbol{C}$ at P.

Our object now is to relate the second derivative $(d^2y/dx^2)_P$ at P to the so called curvature $\kappa = (1/\rho)$ of $\boldsymbol{C}$ at the point P which is defined by the equation

$$\kappa = \textit{the rate of change of the direction with respect to the arc length, i.e., } d\theta/ds \textit{ at } P \tag{2}$$

Intuitively, we may think of the curvature of one curve $\boldsymbol{C}_1$ being greater than that of another curve $\boldsymbol{C}_2$, if $\boldsymbol{C}_1$ recedes more rapidly from its tangent than $\boldsymbol{C}_2$ does. To make this idea more precise, we first consider the case of a circle. Its curvature increases as its radius decreases; it is therefore natural to select as a measure of its curvature κ, the simplest function $f(R)$ of its radius R, such that $f(R)$ increases as R decreases, viz. $\kappa = f(R) = 1/R$. Let us look at Figure 5.26: Here the arc $AB = s = R\omega$. Hence the measure of the curvature κ of the circle is

$$\kappa = 1/R = \omega/s \tag{*}$$

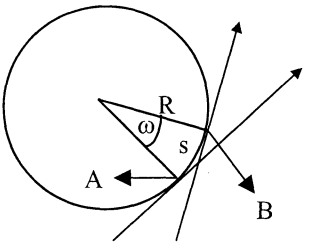


Figure 5.26

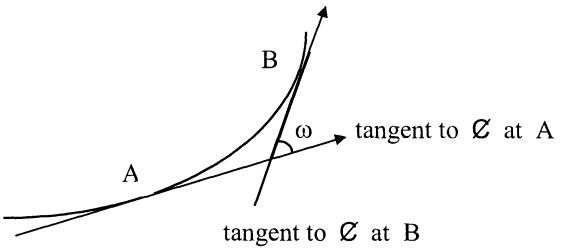


Figure 5.27

We extend (*) to any curve. So let AB be an arc of a curve $\boldsymbol{C}$, which has no flex. Consider the tangents to $\boldsymbol{C}$ at A and B and take the directions of these tangents in the same sense as the arc AB (see Figure 5.27). We call $\omega/\text{Arc}\,AB$ the *average curvature* of the arc AB (cf. (*)). As $B \to A$, this quotient $\omega/\text{Arc}\,AB$ tends to a limit $\kappa_{\boldsymbol{C}}(A)$, which is called the curvature of $\boldsymbol{C}$ at the point A. The radius of curvature of $\boldsymbol{C}$ at A is defined to be the radius of the circle which would have the same curvature as $\boldsymbol{C}$ has at A, i.e., $\kappa_{\boldsymbol{C}}(A) = \lim(\text{Arc}\,AB/\omega)$. To calculate this limit, we proceed as follows (see Figure 5.28).

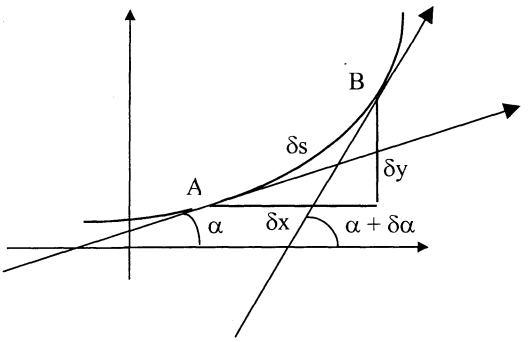


Figure 5.28

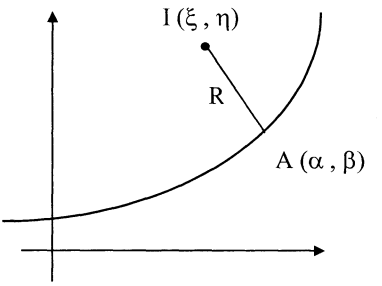


Figure 5.29

Let $y = f(x)$ be the equation of $\boldsymbol{C}$ and let α be the angle which the tangent to $\boldsymbol{C}$ at A makes with the axis of x. Let B be a point near A, say $B = (x+\delta x, y+\delta y)$, where $A = (x, y)$, and let $\text{Arc}\,AB = \delta s$. Let the tangent at B to $\boldsymbol{C}$ makes an angle $\alpha + \delta\alpha$ with the x-axis. Then

$$\kappa_{\boldsymbol{C}}(A) = \pm \lim_{B \to A} \frac{\delta s}{\delta \alpha} = \pm \frac{d\,s}{d\,\alpha} \quad \text{(sign is chosen to makes } \kappa > 0).$$

However, $\tan\alpha = dy/dx$, and so $\sec^2\alpha\,(d\alpha/dx) = d^2y/dx^2$ and $\delta s^2 = \delta x^2 +$

δy^2, whence $(ds/dx)^2 = 1 + (dy/dx)^2$. Hence,

$$\begin{aligned}\frac{ds}{d\alpha} &= \frac{ds}{dx}.\frac{dx}{d\alpha} \\ &= \sqrt{1 + \left(\frac{dy}{dx}\right)^2}\cdot\frac{\sec^2\alpha}{d^2y/dx^2} \\ &= \frac{\left(1 + (\frac{dy}{dx})^2\right)^{3/2}}{d^2y/dx^2},\end{aligned}$$

and so $\kappa = |(1 + (y')^2)^{3/2}/y''|$.

If a length equal to R be marked off from A on the normal to $\boldsymbol{C}$, on the side towards which $\boldsymbol{C}$ is concave, the extremity E is called the centre of curvature (see Figure 5.29). The circle with center E and radius R is called the circle of curvature. The coordinates (ξ, η) of E satisfy the two conditions:

(i) (ξ, η) belongs to the normal to $\boldsymbol{C}$ at A,

(ii) $EA = R$.

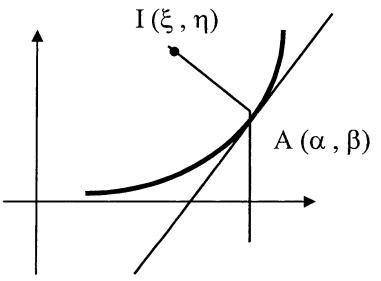


Figure 5.30

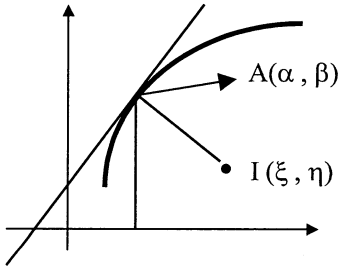


Figure 5.31

Now, the normal to $\boldsymbol{C}$ at $A \equiv (\alpha, \beta)$ is the line given by $(x - \alpha) + (y - \beta).dy/dx = 0$, so (i) becomes $(\xi - \alpha) + (\eta - \beta).dy/dx = 0$, while (ii) becomes $(\xi - \alpha)^2 + (\eta - \beta)^2 = (1 + (dy/dx)^2)^3/(d^2y/dx^2)^2$. On eliminating $\xi - \alpha$ we get

$$\eta - \beta = \pm(1 + (y')^2)/y'' , \tag{3}$$

which then gives

$$-\alpha + \xi = \pm y'(1 + (y')^2)/y'' . \tag{4}$$

We now show that in (3) (and so also in (4)), the + sign holds. To see this note that if y'' is positive, then $\eta > \beta$ (see Figure 5.30) and so the + sign holds in (3) and so also in (4). If, however, y'' is negative, then $\eta < \beta$ (see Figure 5.31)

and so $\eta - \beta < 0$, but so is y'', hence again the + sign holds in (3) and so also in (4). Hence, finally the coordinates of the centre of curvature $E \equiv (\xi, \eta)$ are given by

$$\xi = \alpha - y'(1 + (y')^2)/y'' \ , \ \eta = \beta + (1 + (y')^2)/y'' \ .$$

We see how the second derivative y'' enters naturally into a geometric configuration.

We may avoid explaining the motivation behind the intuitive approach and define κ by the equation $\kappa = d\theta/ds$, without any further ado. Then

$$\kappa = \frac{d\theta}{ds} = \frac{d\theta}{dx} \cdot \frac{dx}{ds} \ . \qquad (i)$$

But $ds^2 = dx^2 + dy^2$, i.e.,

$$\frac{ds}{dx} = \left(1 + \left(\frac{dy}{dx}\right)^2\right)^{3/2} \ . \qquad (ii)$$

The equation $dy/dx = \tan\theta$, on differentiating, gives

$$\sec^2\theta \frac{d\theta}{dx} = \frac{d^2y}{dx^2} \ . \qquad (iii)$$

Also, (i),(ii) and (iii) immediately give

$$\frac{d\theta}{ds} = \frac{d^2y/dx^2}{(1 + (dy/dx)^2)^{3/2}} \ . \qquad (5)$$

We define ρ, the radius of curvature, by the equation $\rho = 1/\kappa$. Thus ρ is the radius of the circle σ (called the circle of curvature at P) through P and two neighbouring points P_1, P_2 on $\boldsymbol{C}$ as $P_1, P_2 \to P$. This σ is called the osculating circle of $\boldsymbol{C}$ at P. Equivalently, σ has y, y', y'' equal to those of $\boldsymbol{C}$ at P and so, rather than talking about a 3-point contact, we sometimes say σ has a 2^{nd} order contact with $\boldsymbol{C}$ at P (for an interesting problem/solution, see [80]).

Let us now go over, briefly, to the third derivative d^3y/dx^3 and see how we may relate it to a geometrical quantity, the way dy/dx was related to the slope and d^2y/dx^2 to the curvature. Merely saying that d^3y/dx^3 is the rate of change of d^2y/dx^2 with respect to x, is not good enough.

The required geometrical entity is the so called aberrancy of $\boldsymbol{C}$ at P, viz. the tangent of the angle δ between the normal to $\boldsymbol{C}$ at P and the limiting position of the line joining P to the mid point M of chords, parallel to the tangent to $\boldsymbol{C}$ at P, as the chord approaches P, assuming that there is a limiting position (see Figure 5.32, where, for convenience, we have taken the tangent to $\boldsymbol{C}$ at P to be the x-axis and P to be the origin).

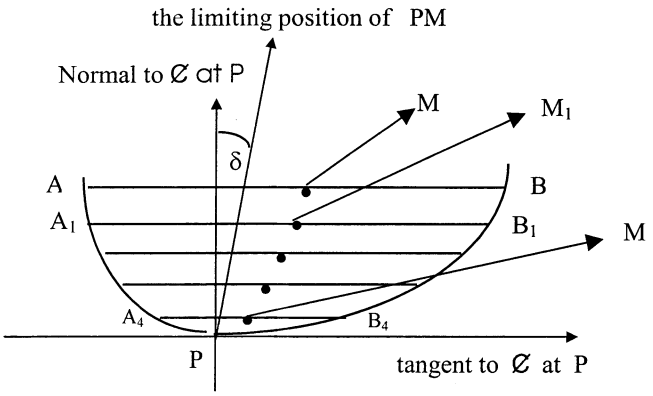


Figure 5.32

By carrying out this limiting process, it is not difficult to see that the aberrancy is given by:

Theorem 5.6.1.

$$\tan\delta = \frac{(d\rho/d\theta)}{3\rho} = \frac{dy}{dx} - \frac{\left(1+(dy/dx)^2\right)}{3(d^2y/dx^2)^2}\cdot\frac{d^3y}{dx^3}\,.$$

Aberrancy is thus a quantity which does indeed depend on the third derivative y''' at P. The angle δ is easily constructed and visualized. The aberrancy is, roughly speaking, the measure of the *asymmetry* of $\boldsymbol{C}$ with respect to the normal to $\boldsymbol{C}$ at P.

As in the case of the tangent and of the curvature, it is possible to connect this concept with 4-point contact or equivalently the third order contact and osculants to $\boldsymbol{C}$ as follows: In this case, the appropriate osculant is the osculating parabola, i.e., the parabola $\mathcal{P}$, which is the limit of the parabolas through P and three neighbouring points P_1, P_2, P_3 of $\boldsymbol{C}$ to P, as $P_1, P_2, P_3 \to P$ (hence 4-point contact). This osculating parabola is also the parabola which is such that its value as well as the values of its 1st, 2nd and 3rd derivatives at P are equal to the corresponding values of $\boldsymbol{C}$ and its derivatives at P (hence third-order contact).

We may now define aberrancy in another manner as the tangent of the angle between the axis of the osculating parabola $\mathcal{P}$ and the normal to $\boldsymbol{C}$ at P.

Physically, the third derivative y''' (and so also aberrancy) has an instructive interpretation, viz. *the jerk*. Indeed, the rate of change of distance with respect to time, i.e., ds/dt, is the velocity; the rate of change of velocity with respect to time, i.e., d^2s/dt^2, is the acceleration; and, finally, the rate of change of acceleration with respect to time, i.e., d^3s/dt^3 is the *jerk*.

In calculus, one defines the centre of curvature O of a point P on $\boldsymbol{C}$ to be the intersection of neighbouring normals at P, P' and then the radius of curvature ρ at P works out to be the line segment PO. Analogously, the limiting position A of the point of intersection of neighbouring axis of aberrancy at P, P' is called

the center of aberrancy and the length PA is termed the radius of aberrancy of C at P and is denoted by R. Complicated Cartesian expression for R and the coordinates of A can be derived but let us leave this interesting topic at that. The interested reader may look up the paper [102] by Steven H. Schot.

§5.7. Taylor's theorem and L'Hospital's rule

In its usual form, Taylor's theorem, with Lagrange form of remainder, is stated as:

Theorem 5.7.1. *Suppose that $f^{(n-1)}(x)$ is continuous in $[a,b]$ and that $f^{(n)}(x)$ exists in (a,b), then*

$$f(b) = f(a) + \frac{b-a}{1!}f'(a) + \frac{(b-a)^2}{2!}f''(a) + \cdots + \frac{(b-a)^{n-1}}{(n-1)!}f^{(n-1)} + R_n\,,$$

where $R_n = \frac{(b-a)^n}{n!}f^{(n)}(\xi)$ (Lagrange form of remainder), for some $a < \xi < b$.

Proof. The proof uses Cauchy's mean value theorem. Let

$$F(x) = f(b) - f(x) - \frac{b-x}{1!}f'(x) - \cdots - \frac{(b-x)^{n-1}}{(n-1)!}f^{(n-1)}(x)\ ,\ G(x) = (b-x)^n.$$

Here, $F(x)$ and $G(x)$ are both continuous in $[a,b]$ and $F'(x) = -((b-x)^{n-1}/(n-1)!)f^{(n)}(x)$, $G'(x) = -x(b-x)^{n-1} \neq 0$ in (a,b). Hence by the Cauchy mean value theorem,

$$\frac{F(b)-F(a)}{G(b)-G(a)} = \frac{F'(\xi)}{G'(\xi)}\ ,$$

for some $a < \xi < b$, which on noting that $G(b) = F(b) = 0$, gives the theorem.

Remark 5.7.2. If we take $G(x) = (b-x)^p$ $(p > 0)$, we get

$$\frac{F(a)}{G(a)} = \frac{F'(\eta)}{G'(\eta)} = \frac{(b-\eta)^{n-1}f^{(n)}(\eta)}{(n-1)!p(b-\eta)^{p-1}}\ ,\ (a < \eta < b),$$

which is Theorem 5.7.1 with the remainder R_n in Schlomilch's form as $((b-a)^p/(n-1)!p).(b-\eta)^{n-p}.f^{(n)}(\eta)$. Writing $\eta = a + \theta_1(b-a)$, $0 < \theta_1 < 1$, we get $b - \eta = (b-a)(1-\theta_1)$ and the remainder takes the form:

$$R_n = \frac{(b-a)^n}{(n-1)!p}(1-\theta_1)^{n-p}f^{(n)}(a+\theta_1(b-a)) \quad \text{(Schlomilch's form of remainder)}.$$

Putting $p = n$ in the above, we get the Lagrange's form of the remainder, while putting $p = 1$, we get Cauchy's form of remainder, viz.

$$R_n = \frac{(b-a)^n}{(n-1)!}(1-\theta_2)^{n-1}f^{(n)}(a+\theta_2(b-a)) \quad \text{(Cauchy's form of remainder)}.$$

Combining these results together, we see that Taylor's theorem shows, on writing $b = a + h$, that if $|h| < k$, then

$$f(a+h) = f(a) + \frac{h}{1!}f'(a) + \frac{h^2}{2!}f''(a) + \cdots + \frac{h^{n-1}}{(n-1)!} + R_n(h) \tag{4}$$

where

$$R_n = \begin{cases} \dfrac{h^n}{n!}f^{(n)}(a+\theta h)\ ,\ 0<\theta<1 \text{ (Lagrange's)} \\ \dfrac{h^n}{(n-1)!p}(1-\theta_1)^{n-p}f^{(n)}(a+\theta_1 h),\ 0<\theta_1<1 \text{ (Schlomilch's)} \\ \dfrac{h^n}{(n-1)!}(1-\theta_2)^{n-1}f^{(n)}(a+\theta_2 h),\ 0<\theta_2<1 \text{ (Cauchy's).} \end{cases}$$

Definition 5.7.3. Suppose that f has derivatives of all orders in $(a-k, a+k)$, for some $k > 0$. Then the series

$$\sum_{n=0}^{\infty} \frac{h^n}{n!}f^{(n)}(a) = f(a) + \frac{h}{1!}f'(a) + \frac{h^2}{2!}f''(a) + \cdots \tag{5}$$

is called the Taylor series of the function $f(x)$ at $x = a$.

Equation (4) immediately gives the following:

Theorem 5.7.4. *The series (5) converges to $f(a+h)$ if and only if $R_n(h) \to 0$, as $n \to \infty$.*

Note that R_n involves $f^{(n)}(a + \theta_n h)$, where θ_n is an unknown quantity and depends on n.

Example 5.7.5. It may happen that the Taylor series (5), although convergent, for a certain a, is convergent to the "wrong" sum, i.e., not to $f(a + h)$. For example, let

$$f(x) = \begin{cases} e^{-1/x^2}\ , \text{ if } x \neq 0, \\ 0\ , \text{ if } x = 0. \end{cases}$$

Then for $x \neq 0$,

$$\begin{aligned} f'(x) &= (1/x^3).2e^{-1/x^2}\ , \\ f''(x) &= (1/x^4).2(-3+2/x^2)e^{-1/x^2}\ , \\ f'''(x) &= (1/x^5).2(12-18/x^2+4/x^4)e^{-1/x^2}\ , \\ \ldots &= \ldots \quad \ldots \quad \ldots \\ f^{(n)}(x) &= (1/x^{n+2})(a_0 + a_2/x^2 + a_4/x^4 + \cdots + a_{2n-2}/x^{2n-2})e^{-1/x^2}\ , \end{aligned}$$

by induction. For $x = 0$, $|(f(h) - f(0))/h| = |e^{-1/h^2}/h| = \sqrt{(1/h^2)/e^{1/h^2}} = \sqrt{y}/e^y$ (where $y = 1/h^2$), which tends to 0 as $y \to \infty$, i.e., as $h \to 0$. So $f'(0)$ exists and equals 0.

Next, $f''(0) = \lim_{h\to 0}((f'(h) - f'(0))/h) = \lim_{h\to 0}((2/h^3).e^{-1/h^2} - 0)/h = \lim_{h\to 0}(2/h^4)/e^{1/h^2} = \lim_{y\to\infty} 2y^2/e^y = 0$, where $y = h^2$. So $f''(0)$ exists and equals 0.

Similarly, by induction, since $y^\alpha/e^y \to 0$ as $y \to \infty$, for any α, we see that $f^{(n)}(0)$ exists and equals 0.

Now taking the Taylor expansion at the point $a = 0$, we find that the Taylor series is $f(0)+(h/1!)f'(0)+(h^2/2!)f''(0)+\cdots = 0+0+0+\cdots$, which converges to 0, and not to $f(0+h) = f(h)$, as required.

Also, it is easy to check that in this case $R_n(h)$ does not tend to 0 as $n \to \infty$ (of course, if the contrary were the case, then by Theorem 5.7.4, the Taylor series would converge to $f(a+h)$). Indeed,

$$\begin{aligned} R_n(h) &= \frac{h^n}{n!} f^{(n)}(0+\theta h) \text{ (in Lagrange's form)} \\ &= \frac{h^n}{n!} \times (\text{a polynomial in } 1/\theta h \text{ of degree } 3n)(1/e^{1/h^2}) \\ &= (\frac{1}{n!} y^{n/2} e^y) \times (\text{a polynomial in } \frac{\sqrt{y}}{\theta} \text{ of degree } 3n)(\text{where } y = 1/h^2) \\ &= (a_0 + a_1\sqrt{y}/\theta + a_2(\sqrt{y})^2/\theta^2 + \cdots + a_{3n}(\sqrt{y})^{3n}/\theta^{3n})/n!(\sqrt{y})^n.e^y \end{aligned}$$

and as $n \to \infty$ ($y = 1/h^2$ fixed), this does not tend to 0 as required.

As an interesting and useful example, we prove the following:

Theorem 5.7.6. $\log(1+x)$ *(defined for* $1+x > 0$, *i.e., for* $x > -1$*), with* $f(0) = \log 1 = 0$, *has the Taylor expansion*

$$\log(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \cdots \quad , \text{ if } -1 < x \leq 1.$$

Proof. Let $f(x) = \log(1+x)$. Then $f'(x) = 1/(1+x)$ and indeed, $f^{(n)}(x) = (-1)^{n+1}(n-1)!/(1+x)^n$. Hence, by Taylor's theorem

$$\log(1+h) = h - \frac{h^2}{2} + \cdots + (-1)^N \frac{h^{N-1}}{N-1} + R_N(h).$$

Further, we have already shown that the Taylor series converges to $\log(1+h)$ if and only if $R_N(h) \to 0$ as $N \to \infty$. We shall establish the convergence for $-1 < h \leq 1$. It is interesting to note that

(i) for $0 < h \leq 1$, we use Lagrange form of the remainder, viz. $R_N(h) = (-1)^{N+1}h^N/N(1+\theta_N h)^N$, $0 < \theta_N < 1$,

(ii) for $-1 < h < 1$, we use Cauchy form of the remainder, viz. $R_N(h) = (-1)^{N+1}h^N(1-\theta'_N)^{N-1}/(1+\theta'_N h)^N$, $0 < \theta'_N < 1$.

So, first let $0 < h \leq 1$. The Lagrange form gives

$$|R_N| \leq h^N/N \to 0 \quad \text{as} \quad N \to \infty \qquad (\text{since } 1+\theta_N h \geq 1).$$

Next, let $-1 < h < 1$. Then Cauchy form gives

$$|R_N| \leq |h|^N/(1+\theta'_N h)\ ,\ \text{(since } |(1-\theta'_N)/(1+\theta'_N h)| < 1 \text{ if } |h|<1),$$
$$\leq |h|^N/(1-|h|) \to 0 \text{ as } N \to \infty \text{ if } |h| < 1.$$

This completes the proof of the theorem.

Let us now look at the point ξ, sometimes written as $\xi = a + \theta(b-a)$, $0 < \theta < 1$, appearing in Theorem 5.7.3. Let f be continuous in $[0,x]$ and suppose that $f^{(n-1)}(0)$ exists and $f^{(n)}(t)$ exists in $(0,x)$. By Taylor's theorem, there exists $\xi(=\xi_x) \in (0,x)$ such that

$$f(x) = f(0) + \frac{f'(0)}{1!}x + \frac{f''(0)}{2!}x^2 + \cdots + \frac{f^{(n-1)}(0)}{(n-1)!}x^{n-1} + \frac{f^{(n)}(\xi_x)}{n!}x^n \qquad (7)$$
$$= P_{n-1}(x) + \frac{f^{(n)}(\xi_x)}{n!}x^n\ ,$$

say. Then the following result holds:

Theorem 5.7.7. *If $f^{(n+1)}(t)$ exists in $[0,x]$, and is continuous from the right at $t=0$, and if $f^{(n+1)}(0) \neq 0$, then $\lim_{x\to 0+}(\xi_x/x) = 1/(n+1)$.*
(For $n=2$, we get Theorem 5.4.6.)

Proof [80]. First of all note that the assumptions of the theorem imply that $\xi = \xi_x$ is uniquely determined if x is small enough.

Apply the mean value theorem to $f^{(n)}(\xi)$ in (7). So, there exists $\tau \in (0,\xi)$ such that

$$f(x) = P_{n-1}(x) + (f^{(n)}(\xi)/n!)x^n$$
$$= P_{n-1}(x) + (x^n/n!)(f^{(n)}(0) + \xi f^{(n+1)}(\tau))$$
$$= P_n(x) + \xi f^{(n+1)}(\tau).(x^n/n!)\ . \qquad (8)$$

On the other hand, (7) gives, with $n+1$ instead of n,

$$f(x) = P_n(x) + (f^{(n+1)}(\sigma)/(n+1)!)x^{n+1} \quad (\text{for some } \sigma \in (0,x)). \qquad (8)'$$

Comparing (8) and (8)', we get $\xi f^{(n+1)}(\tau)(x^n/n!) = f^{(n+1)}(\sigma)(x^{n+1}/(n+1)!)$, i.e.,

$$(\xi/x).f^{(n+1)}(\tau) = f^{(n+1)}(\sigma)/(n+1) \qquad (9)$$

Now, let $x \to 0$; since $\xi, \tau, \sigma \in (0,x)$, we see that ξ, τ, σ all tend to 0 and since $f^{(n+1)}(t)$ is right continuous and nonzero at $t=0$, we may cancel $\lim f^{(n+1)}(t)$ in (9) to get the required result.

We note that under the usual conditions

$$f(\xi) = f(a) + \frac{(\xi-a)}{1!}f'(a) + \frac{(\xi-a)^2}{2!}f''(a) + \cdots +$$
$$+ \frac{(\xi-a)^{n-1}}{(n-1)!}f^{(n-1)}(a) + \frac{(\xi-a)^n}{n!}f^{(n)}(\gamma) \qquad (\alpha)$$

Our aim is to obtain $\lim_{\xi\to a}\frac{\gamma-a}{\xi-a}$, under certain conditions.

Theorem 5.7.8 ([4]). *Suppose a, ξ belong to an open interval I, and suppose that*

(i) $f^{(n+p)}(x)$ exists for all $x \in I$ and is continuous at a $(n \geq 1,\ p \geq 1)$,

(ii) $f^{(n+1)}(a) = f^{(n+2)}(a) = \ldots = f^{(n+p-1)}(a) = 0$,

(iii) $f^{(n+p)}(a) \neq 0$;

then

$$\lim_{\xi\to a}\frac{\gamma-a}{\xi-a} = \frac{1}{\left[\binom{n+p}{n}\right]^{1/p}},$$

where, $\gamma \in (a, \xi)$ is as in the equation (α) above.

Proof. Under the assumption of the theorem, we may expand $f^{(n)}(\gamma)$ in (α) up to the p^{th} power, to give

$$f^{(n)}(\gamma) = f^{(n)}(a) + \frac{(\gamma-a)}{1!}f^{(n+1)}(a) + \frac{(\gamma-a)^2}{2!}f^{(n+2)}(a) + \cdots + \frac{(\gamma-a)^{p-1}}{(p-1)!}f^{(n+p-1)}(a) + \frac{(\gamma-a)^p}{p!}f^{(n+p)}(\gamma_1),$$

which on using (ii), equals $f^{(n)}(a) + ((\gamma-a)^p/p!)f^{(n+p)}(\gamma_1)$. Substituting this in (α) gives:

$$f(\xi) = f(a) + \frac{(\xi-a)}{1!}f'(a) + \frac{(\xi-a)^2}{2!}f''(a) + \cdots + \frac{(\xi-a)^{n-1}}{(n-1)!}f^{(n-1)}(a) + \frac{(\xi-a)^n}{n!}\left(f^{(n)}(a) + \frac{(\gamma-a)^p}{p!}f^{(n+p)}(\gamma_1)\right), \quad a < \gamma_1 < \gamma. \qquad (\dagger)$$

Now expand $f(\xi)$ once again, but up to $(n+p)^{\text{th}}$ power to give

$$f(\xi) = f(a) + \frac{(\xi-a)}{1!}f'(a) + \cdots + \frac{(\xi-a)^n}{n!}f^{(n)}(a) + \frac{(\xi-a)^{n+1}}{(n+1)!}f^{(n+1)}(a) + \cdots + \frac{(\xi-a)^{n+p-1}}{(n+p-1)!}f^{(n+p-1)}(a) + \frac{(\xi-a)^{n+p}}{(n+p)!}f^{(n+p)}(\gamma_2), \quad a < \gamma_2 < \xi,$$

and again on using hypothesis (ii), this gives

$$f(\xi) = f(a) + \frac{(\xi-a)}{1!}f'(a) + \cdots + \frac{(\xi-a)^n}{n!}f^{(n)}(a) + \frac{(\xi-a)^{n+p}}{(n+p)!}f^{(n+p)}(\gamma_2). \qquad (\ddagger)$$

From $(\dagger)$ and $(\ddagger)$ it follows that

$$\frac{(\xi-a)^n}{n!}\frac{(\gamma-a)^p}{p!}f^{(n+p)}(\gamma_1) = \frac{(\xi-a)^{n+p}}{(n+p)!}f^{(n+p)}(\gamma_2),$$

i.e.,

$$\frac{(\gamma-a)^p}{n!p\,!}f^{(n+p)}(\gamma_1)=\frac{(\xi-a)^p}{(n+p)!}f^{(n+p)}(\gamma_2)\,,$$

giving

$$\left(\frac{\gamma-a}{\xi-a}\right)^p=\frac{n!p\,!}{(n+p)!}\frac{f^{(n+p)}(\gamma_2)}{f^{(n+p)}(\gamma_1)}\,.$$

Letting $\xi\to a$, we see that γ,γ_1,γ_2 all tend to a and the theorem follows.

Remark 5.7.9 (a). Take $n=p=1$ and let $f''(a)\neq 0$, $f''(x)$ exists and is continuous, then $f(\xi)=f(a)+(\xi-a)f'(\gamma)$. By the theorem, $(\gamma-a)/(\xi-a)\to 1/2$ as $\xi\to a$, i.e., as $\xi\to a$, γ approaches the midpoint of a and ξ, for $(\gamma-a)/(\xi-a)\to 1/2\Rightarrow(\gamma-a)/(\xi-a)=1/2+\epsilon(\xi)$, where $\epsilon(\xi)\to 0$. Hence, $\gamma=a+(\xi-a)/2+(\xi-a)\epsilon(\xi)=(\xi+a)/2+(\xi-a)\epsilon(\xi)\to(\xi+a)/2$, as required.
(b). Condition (ii) in the theorem is necessary. For take $n=p=1$, $f(x)=cx+d$. Then γ can be chosen anywhere in $[a,\xi]$, for example γ can be such $(\gamma-a)/(\xi-a)=1/3$, so that the conclusion of the theorem, viz. $(\gamma-a)/(\xi-a)\to 1/2$, does not hold.
(c). Let $f(x)=x^N+x^{N+p}$, $a=0, n=N, p+P$; then $(\gamma-a)/(\xi-a)=(n!p\,!/(n+p)!)^{1/p}$.

The first mean value theorem for integrals states that if f is continuous in $[a,x]$, then there exists a point c, $a<c<x$, such that

$$\int_a^x f(t)dt=f(c)(x-a).$$

About this number c, we have the following:

Theorem 5.7.10 ([49]). *Suppose f is differentiable at $x=a$, $f'(a)\neq 0$, and that f is continuous in $[a,x]$ (otherwise such a number c would not exist). Then $\lim_{x\to a}(c-a)/(x-a)$ exists and equals $1/2$.*

Proof. Consider

$$L=\lim_{x\to a}\left(\left(\int_a^x f(t)dt-xf(a)+af(a)\right)\Big/(x-a)^2\right)\,.$$

By the mean value theorem for integrals, L is equal to

$$\begin{aligned}\lim_{x\to a}\frac{f(c)(x-a)-(x-a)f(a)}{(x-a)^2}&=\lim_{x\to a}\frac{f(c)-f(a)}{x-a}\\&=\lim_{x\to a}\frac{f(c)-f(a)}{c-a}\frac{c-a}{x-a}\\&=f'(a).\lim_{x\to a}\frac{c-a}{x-a}\,.\end{aligned}\qquad(*)$$

But on applying L'Hospital's rule to evaluate L, we get

$$L=\lim_{x\to a}\frac{f(x)-f(a)}{2(x-a)}=\frac{f'(a)}{2}\qquad(**)$$

Equating (*) and (**) gives the theorem.

We have looked extensively at the Taylor expansion of a function f, about a point $x = a$. What is it supposed to tell us about the function f? It tells us that, under certain conditions, f can be represented, in a neighbourhood of the point $x = a$, as a power series (or a polynomial with a remainder). There are several ways of proving Taylor's theorem, all of which use auxiliary functions, whose choice can not be motivated. Proofs using integration by parts (see, for example, [65]) are better motivated but get the reader lost in technical details.

What really matters in Taylor's theorem, is not the form of the remainder but that the function can be approximated by a Taylor polynomial, with an error term (the remainder), which can be estimated. With this in mind, a proof is given below, which depends only on the representation of a function as the integral of its derivative and uses the trivial formula

$$\left| \int_a^b f \right| \leq M(b-a) \ , \quad \text{where } M = \sup_{x \in [a\,,\,b]} f(x) \ .$$

Theorem 5.7.11 ([16]). *Suppose that for some $n \in \mathbb{N} \cup \{0\}$, the functions f, f', f'', $\ldots$, $f^{(n)}$ are continuous in $[a,b]$ and that $f^{(n+1)}$ exists and is bounded in (a,b), i.e., $|f^{(n+1)}(x)| \leq M$ for $x \in (a,b)$; then for $x \in [a,b]$,*

$$f(x) = f(a) + \frac{f'(a)}{1!}(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \cdots + \frac{f^{(n)}}{n!}(x-a)^n + R_n(x) \ , \quad (*)$$

where $|R_n(x)| \leq (M/(n+1)!).|x-a|^{n+1}$.

Proof. The polynomial

$$P_n(x) = f(a) + \frac{f'(a)}{1!}(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \cdots + \frac{f^{(n)}}{n!}(x-a)^n$$

(called the Taylor polynomial) is chosen to have the properties:

$$f(a) = P_n(a) \ , f'(a) = P_n'(a) \ , \cdots \ , f^{(n)}(a) = P_n^{(n)}(a) \qquad (1)$$

Since P_n is a polynomial of degree n,

$$P_n^{(n+1)}(x) \equiv 0 \ , \forall \ x \in [a,b] \ . \qquad (2)$$

The remainder R_n is defined by (*), so that $R_n(x) = f(x) - P_n(x)$, and by (1)

$$R_n'(a) = R_n''(a) = \ldots = R_n^{(n)}(a) = 0 \qquad (3)$$

and by (2),

$$f^{(n+1)}(t) = R_n^{(n+1)}(t) \ , \quad a < t < x \qquad (4)$$

Integrating (4) from a to x, and using (3), gives

$$R_n^{(n)}(x) = R_n^{(n)}(a) + \int_a^x R_n^{(n+1)}(t)dt = \int_a^x f_n^{(n+1)}(t)dt$$

and the bound $|f^{(n+1)}(t)| \leq M$ gives $|R_n^{(n)}(x)| \leq \int_a^x |f^{(n+1)}(t)|dt| \leq M(x-a)$. A second integration gives $R_n^{(n-1)}(x) = R_n^{(n-1)}(a) + \int_a^x R_n^{(n)}(t)dt$. So $|R_n^{(n-1)}(x)| \leq \int_a^x |R_n^{(n)}(t)|dt \leq \int_a^x M(t-a)dt = M(x-a)^2/2$. Continuing (or using induction), we see that for $j = 0, 1, 2, \ldots, n$,

$$|R_n^{(n-j)}(x)| \leq (M/(j+1)!).(x-a)^{j+1} \quad , \quad a \leq x \leq b.$$

For the desired estimate, we simply take $j = n$.

Marquis De L' Hospital's rule (to give him, for once, as R.P.Boas Jr. puts it, the spellings that he himself used), was actually discovered by John Bernoulli. Whatever form it appears in and whatever hypothesis are assumed to prove it, it is exclusively used for evaluating indeterminate forms such as $\frac{0}{0}$, 0^0, $\frac{\infty}{\infty}$, $0.\infty$ etc. The simplest form it takes (usually to tackle expressions like $\frac{0}{0}$) is the following:

Theorem 5.7.12 (L'Hospital's rule). *Suppose that $f(a) = g(a) = 0$, and that $A = \lim_{x\to a} f'(x)/g'(x)$ exists. Then $\lim_{x\to a} f(x)/g(x)$ exists and is equal to A.*

Proof. Observe that

$$\frac{f(x)}{g(x)} = \frac{f(x)-0}{g(x)-0} = \frac{f(x)-f(a)}{g(x)-g(a)}$$
$$= \frac{(f(x)-f(a))/(x-a)}{(g(x)-g(a))/(x-a)} \quad . \tag{1}$$

But by the mean value theorem, applied to f and g separately, in the interval between a and x (i.e., $[a,x]$ if $a < x$, $[x,a]$ if $x < a$), there exist ξ, η between a and x, such that $(f(x)-f(a))/(x-a) = f'(\xi)$ and $(g(x)-g(a))/(x-a) = g'(\eta)$. Substituting in (1), we get $f(x)/g(x) = f'(\xi)/g'(\eta) \to A$, by the hypothesis, since $\xi, \eta \to a$ as $x \to a$.

This proof uses the mean value theorem. With a little stronger hypothesis, that suffices for all applications, the rule can be proved without using the mean value theorem. Indeed, we have the following:

Theorem 5.7.13. *Suppose that*

(i) the derivatives f', g' of the functions f and g are continuous,

(ii) f, g both tend to 0 or both tend to ∞ as $x \to a$,

(iii) $g'(x) \neq 0$,

(iv) $f'(x)/g'(x) \to L$ as $x \to a$;

then $f(x)/g(x) \to L$ as $x \to a$.

Proof ([14]). The theorem holds true if we replace, throughout, the limit

$x \to a$ by right hand or left hand limits $x \to a+$ or $x \to a-$; and so it is enough to prove the theorem for $a = \infty$, since the same proof works for $a = -\infty$ or for any finite value of a.

Given $\epsilon > 0$, (iv) implies that $-\epsilon < f'(t)/g'(t) - L < \epsilon$, if t is large enough, say $t > \Delta$. Since g' is continuous and never 0, it has a fixed sign, say, without loss of generality, that $g'(t) > 0$. Then the above gives

$$-\epsilon g'(t) < f'(t) - Lg'(t) < \epsilon g'(t) \text{ , if } t > \Delta \,. \tag{1}$$

The right hand inequality implies that $f'(t) - (\epsilon + L)g'(t) < 0$, if $t > \Delta$, which implies that $\int_y^x (f'(t) - (\epsilon + L)g'(t))dt < 0$, if $x > y$ and hence

$$f(x) - f(y) - (\epsilon + L)(g(x) - g(y)) < 0 \ . \tag{2}$$

Case 1 : f, g both tend to 0. In (2), fix y and let $x \to \infty$, to get

$$-f(y) + (\epsilon + L)g(y) \leq 0 \,. \tag{3}$$

But $g'(y) > 0$ and $g(y) \to 0$, so $g(y) < 0$ for large y. Hence, by (3), $-f(y)/g(y) + L \geq -\epsilon$ if $y > \Delta_2$, say. Similarly, the left hand inequality of (1) implies $-f(y)/g(y) + L \leq \epsilon$, if $y > \Delta_2$, say. These imply that $f(y)/g(y) \to L$ as $y \to \infty$.

Case 2 : f, g both tend to ∞. Then by (2), (noting that $g(x) > 0$ if $x > \Delta$) $f(x)/g(x) - (\epsilon + L) < (f(y) - (\epsilon + L)g(y))/g(x)$. In this, fix y. Then, for sufficiently large x, the right hand side is less than ϵ, since $g(x) \to \infty$; so $f(x)/g(x) - L < 2\epsilon$, for $x > \Delta_3$, say. Similarly, the left hand inequality of (1) implies that $-2\epsilon < f(x)/g(x) - L$, for $x > \Delta_4$, say. These imply $f(x)/g(x) \to L$ as $x \to \infty$.

Remark 5.7.14. The hypothesis that $g'(x)$ is continuous is redundant; all we really use is that if $g' \neq 0$ in an interval, then g' has a fixed sign (because, a derivative can not have a jump discontinuity).

In handling indeterminates of the form 0^0, the usual examples that one meets are

$$\lim_{x \to 0}(x^x) \quad \text{and} \quad \lim_{x \to 0}((\sin x)^{\tan x}) \,,$$

and both of these limits, and many others, that generally come to mind, equal 1. That this is not always the case, is shown by the following example.

Example 5.7.15. Consider $\lim_{x \to 0+}(x^{a/\log x})$, where a is some constant. This leads to the indeterminate form under consideration. Letting $y = x^{a/\log x}$, we find $\log y = a$, giving $\log_{x \to 0+}(\log y) = a$. Hence $e^a = e^{\lim_{x \to 0+} \log y} = \log_{x \to 0+} e^{\log y} = \lim_{x \to 0+} y = \lim_{x \to 0+}(x^{a/\log x})$. Thus, we may vary $\lim_{x \to 0+}(x^{a/\log x})$ by varying a.

Note that the equation $\lim e^{\varphi(x)} = e^{\lim \varphi(x)}$ is valid, since exponentiation is continuous.

On the same lines, we prove the following useful:

Theorem 5.7.16 ([115]). *Suppose f, g, h are functions, defined in a neighbourhood of the origin, that satisfy*

(i) g *is continuous at* $x = 0$,

(ii) $\lim_{x\to 0} f(x)/x = b$ $(0 < b < \infty)$,

(iii) $\lim_{x\to 0} h(x)/\log x = c$ $(0 < c < \infty)$,

(iv) these functions and their derivatives have the necessary properties for applying L'Hospital's rule near the origin ;

then

$$\lim_{x\to 0} f(x)^{g(x)/h(x)} = e^{g(0)/c}\,.$$

Proof. We have that $b = \lim_{x\to 0+} f(x)/x = \lim_{x\to 0+} f'(x)/1$ (by (iv)) and similarly $c = \lim_{x\to 0+} h(x)/\log x = \lim_{x\to 0+} h'(x)/(1/x)$ (by (iv)) ($f(x)/x$ is of type $0/0$ while $h(x)/\log x$ is of type ∞/∞).

Now, let $y = f(x)^{g(x)/h(x)}$; then $\log y = (g(x)/h(x)).\log f(x)$. Hence

$$\begin{aligned}
\lim_{x\to 0+} \log y &= \lim_{x\to 0+} (g(x)/h(x)).\log f(x) \\
&= \lim_{x\to 0+} g(x).\ \lim_{x\to 0+} (\log f(x))/h(x) \\
&= g(0).\ \lim_{x\to 0+} ((1/f(x)).f'(x)/h'(x)) \quad \text{(by (i) and L'Hospital's rule)} \\
&= g(0)\ \lim_{x\to 0+} f'(x).((x/f(x))/x.h'(x)) \\
&= g(0)b.(1/b)/c \qquad\qquad \text{(by above)} \\
&= g(0)/c\,.
\end{aligned}$$

Hence, as before, $e^{\lim_{x\to 0+} \log y} = e^{g(0)/c}$, i.e., $\lim_{x\to 0+} e^{\log y} = e^{g(0)/c}$, giving

$$\lim_{x\to 0+} y = e^{g(0)/c}\,,$$

as required.

In attempting to construct specific examples of the indeterminate form 0^0, we immediately think of $x^{f(x)}$ where $f(0) = 0$ and where $f'(x)$ exists in a neighbourhood $(0-\epsilon, 0+\epsilon)$ of the origin. But under these hypotheses, we have the following surprising result:

Theorem 5.7.17. *If the limit, as x tends to 0, of $x^{f(x)}$ exists, where $f(0) = 0$, $f'(x)$ exists in $(-\epsilon, \epsilon)$ for sufficiently small ϵ, then it must be equal to 1.*

Proof ([77]). This limit is equal to $\lim_{x\to 0} e^{\log(x^{f(x)})} = e^{\lim_{x\to 0} \log(x^{f(x)})} = e^{\lambda}$, say, where, $\lambda = \lim_{x\to 0}(f(x).\log(x)) = \lim_{x\to 0}(\log x/(1/f(x)))$ (leading to the indeterminate form ∞/∞). Applying L'Hospital's rule, we find

$$\lambda = \lim_{x\to 0}((1/x)/(-f'(x)/f^2(x))) = -\lim_{x\to 0}(f(x)/x).(f(x)/f'(x)).$$

We now claim that *under the hypothesis of Theorem 5.7.17, if*

$$\lim_{x\to 0}(f(x)/f'(x))$$

exists, it must be 0. Assuming this, Theorem 5.7.17 follows, for then

$$\lim_{x\to 0} f(x)/x = \lim_{x\to 0}((f(x)-f(0))/(x-0)) = f'(0)\ , \quad (\text{since } f(0)=0)$$

(which exists and is finite by the hypothesis of Theorem 5.7.17) and so

$$\lim_{x\to 0}(f(x)/x).(f(x)/f'(x)) = f'(0).0 = 0 \quad (\text{by } (4))$$

(since $f'(0)$ exists and is finite). Thus $\lambda = 0$ and so $\lim_{x\to 0} x^{f(x)} = e^{\lambda} = 1$, as required.

It remains to prove the claim. So suppose that $\lim_{x\to 0} f(x)/f'(x)$ exists. Then 0 can not be a limit point of zeros of $f'(x)$, for otherwise, 0 becomes a limit point of points where $f(x)/f'(x)$ is either infinite or undefined, so $\lim_{x\to 0} f(x)/f'(x)$ can not exist. Hence we conclude that

$$f'(x) \neq 0 \quad \text{for } 0 < |x| < \delta \qquad (*)$$

and so $f(x) \neq 0$ in $(0,\delta)$, for otherwise, if for example, $f(x_1) = 0$ $(0 < x_1 < \delta)$, then $(f(x_1) - f(0))/(x_1 - 0) = f'(\xi)$, for some $0 < \xi < x_1$ (by the mean value theorem), i.e., $f'(\xi) = 0$ (since $f(x_1) = 0 = f(0)$), which is not possible. So, without loss of generality, let

$$f(x) > 0 \quad \text{in} \quad (0,\delta) \qquad (4)$$

and we must show that $\lim f(x)/f'(x) = L = 0$. Suppose, to the contrary, that $L > 0$ (say). Then there exists a number δ' such that $|f(x)/f'(x) - L| < \epsilon$ if $0 < x < \delta'$. Take $\epsilon = L/2$ and the inequality reads $L/2 < f(x)/f'(x) < 3L/2$ if $0 < x < \delta'$, i.e.,

$$2/L > f'(x)/f(x) > 2/3L \quad \text{if } 0 < x < \delta' \qquad (5)$$

Now, by the mean value theorem $(f(x)-f(0))/(x-0) = f'(\xi)$ for some $0 < \xi < x < \delta'$, i.e., $f(x)/f'(x) = xf'(\xi)/f'(x) = x(f(\xi)/f(x))(f(x)/f'(x))(f'(\xi)/f(\xi))$. But by (5), $f'(x)/f(x)$ is bounded by $3L/2$, $f'(\xi)/f(\xi)$ is bounded by $2/L$ and $f(\xi)/f(x)$ is bounded by 1, since $f'(x) > 0$ (because $f(x)/f'(x) \to L > 0$ and $f(x) > 0$, by (4)). Therefore $f(x)/f'(x) < x.1.(3L/2).(2/L) = 3x$ in $(0,\delta'')$, which implies that the limit of $f(x)/f'(x)$, as x tends to 0+, is 0, which is a contradiction, since $L \neq 0$. Hence $L = 0$.

Remark 5.7.18. In Example 5.7.15, i.e., $\lim_{x\to 0+} x^{a/\log x}$, the hypothesis of Theorem 5.7.17 is not satisfied: $f'(x) = -a/x(\log x)^2$ does not exist at 0, so the conclusion fails too.

Under a different kind of hypothesis, this limit, viz., $\lim_{x\to 0+}(f^g)$ equals 1. More precisely, we have the following:

Theorem 5.7.19 ([95]). *Let f, g be nonzero real analytic functions at $x = 0$ (i.e., representable, in a neighbourhood of $x = 0$, by a Taylor series) for which $f(x) \geq 0$ for all possible x close to 0. If $\lim_{x\to 0+} f(x) = \lim_{x\to 0+} g(x) = 0$, then $\lim_{x\to 0+} f(x)^{g(x)} = 1$.*

Proof. We have

$$\begin{aligned}\lim_{x\to 0+} f(x)^{g(x)} &= \lim_{x\to 0+} (\exp(\log(f(x)^{g(x)}))) \\ &= \exp(\lim_{x\to 0+} (\log(f(x)/(1/g(x)))) \\ &= \exp(\lim_{x\to 0+} ((1/f(x)).f'(x)/(-g^{-2}(x).g'(x)))) \text{ (L'Hospital's rule)} \\ &= \exp(\lim_{x\to 0+} (-f'(x)g^2(x)/f(x)g'(x))) \,.\end{aligned}$$

Now f, g are analytic and tend to 0 as $x \to 0$, so, by continuity, $f(0) = g(0) = 0$. It follows that in some neighbourhood of 0, f and g have the form $f = x^m.F(x)$, $F(0) \neq 0$, $g(x) = x^n.G(x)$, $G(0) \neq 0$; $m, n \in \mathbb{N}$; F, G analytic at $x = 0$. Substituting these in the above, gives

$$\begin{aligned}\lim_{x\to 0+} f(x)^{g(x)} &= \exp\left(\lim_{x\to 0+} \frac{x^n(xF'(x) + mF(x))G^2(x)}{-F(x)(xG'(x) + nG(x))}\right) \\ &= \exp\left(\lim_{x\to 0+} \frac{N(x)}{D(x)}\right) \,,\end{aligned}$$

say, where, $N(x)$ and $D(x)$ are analytic at $x = 0$. Since $D(0) \neq 0$, by the continuity at $x = 0$, $\lim_{x\to 0+} D(x) \neq 0$ and so $\lim_{x\to 0+} N(x)/D(x) = 0$.

Remark 5.7.20. The case $\lim(f(x)^{g(x)})$, as $x \to a$, is dealt with by a simple linear transformation of x.

The case when the limit is taken as $x \to 0-$ is similar to the above theorem. The only requirement needed is $f(x) \geq 0$ for x negative, close to 0.

Example 5.7.21. If " f, g analytic " is replaced by " f, g infinitely many times differentiable ", say at the point $x = 0$, the conclusion of the theorem is false, for example, take

$$g(x) = x \qquad \text{and} \qquad f(x) = \begin{cases} e^{-1/x^2} & \text{if } x \neq 0 \,, \\ 0 & \text{if } x = 0 \,; \end{cases}$$

then $\lim_{x\to 0} f(x)^{g(x)} = 0$ and $\lim_{x\to 0-} f(x)^{g(x)} = \infty$.

In this context, see also [77] and [115].

The converse of L'Hospital's rule is rarely looked into and it is generally thought that if f and g have continuous derivatives and $f(a) = g(a) = 0$, then

$$\lim_{x\to a} f(x)/g(x) = \lim_{x\to a} f'(x)/g'(x) \,,$$

in the sense that if either limit exists, so does the other and the two are equal. The following is a counterexample to show that this is not correct.

Example 5.7.22. Following [93], let

$$f(x) = \begin{cases} x\sin(1/x^4).e^{-1/x^2}\ , \text{ if } x \neq 0 \\ 0\ , \text{ if } x = 0 \end{cases}$$

and

$$g(x) = \begin{cases} e^{-1/x^2}\ , \text{ if } x \neq 0 \\ 0\ , \text{ if } x = 0. \end{cases}$$

Then f, g are continuous at $x = 0$, since both $f(x) \to 0$ and $g(x) \to 0$ as $x \to 0$. Further, for $x \neq 0$, $f(x)/g(x) = x\sin(1/x^4) \to 0$ as $x \to 0$ and for $x \neq 0$, it is easy to check that $f'(x)/g'(x) = -(2/x)\cos(1/x^4) + x\sin(1/x^4).(1 + x^2/2)$, and for small x, this is $-(2/x).\cos(1/x^4) + \epsilon(x)$, where $\epsilon(x) \to 0$ as $x \to 0$, and this does not tend to any limit. Indeed, f and g are differentiable infinitely many times, as is easy to check.

There is, however, an occasion when the converse does hold; in fact we have the following:

Theorem 5.7.23 (The North-East Theorem, [47]). *Let f and g be differentiable near $x = a$. Suppose that*

$$(i)\ \lim_{x\to a} f(x) = \lim_{x\to a} g(x) = \infty \qquad \text{and} \qquad (ii)\ \lim_{x\to a} f'(x)/g'(x) = \lambda\ ,$$

then $\lim_{x\to a} f(x)/g(x)$ exists and is equal to λ.

Proof. Without loss of generality, we may suppose $\lambda > 0$, for otherwise, use $f_1(x) = f(x) + cg(x)$, instead of $f(x)$, for a large enough c.

Let $\epsilon > 0$ be given. Then there exists $\eta > 0$ such that $|\log f'(x)/g'(x) - \log\lambda| < \epsilon/2$ if $0 < |x - a| < \eta$. Select x_1 such that $0 < |x_1 - a| < \eta$ and fix it. Then there exists $\eta' < \eta$ such that if $0 < |x - a| < \eta'$ then $f(x) > f(x_1)$ and $g(x) > g(x_1)$. This is because $f(x)$ and $g(x)$ tend to ∞ as $x \to a$, so $f(x), g(x)$ become large (larger than $f(x_1), g(x_1)$ respectively) near a. Then we have

$$\begin{aligned} \left|\log\left(\frac{f(x)}{g(x)}\right) - \log\lambda\right| &= \left|\log\left(\frac{f(x) - f(x_1)}{g(x) - g(x_1)}\right) - \log\lambda - \log\left(\frac{1 - f(x_1)/f(x)}{1 - g(x_1)/g(x)}\right)\right| \\ &\leq \left|\log\left(\frac{f'(\xi)}{g'(\xi)}\right) - \log\lambda\right| + \left|\log\left(\frac{1 - f(x_1)/f(x)}{1 - g(x_1)/g(x)}\right)\right|, \end{aligned}$$

where ξ lies between x and x_1. Here the first term is less than $\epsilon/2$ by hypothesis ($x \in (a-\eta', a+\eta')$ so $\xi \in (a-\eta, a+\eta)$). Further, the function $\log x$ is continuous and $\log 1 = 0$, so we may choose $\delta < \eta'$ such that $|f(x_1)/f(x)|$ and $|g(x_1)/g(x)|$ are small enough to ensure that $(1 - f(x_1)/f(x))/(1 - g(x_1)/g(x))$ is near 1 whenever $x \in (a - \delta, a + \delta)$; in fact so near 1, that the absolute value of the second term is less than $\epsilon/2$ if $|x-a| < \delta$ (this is possible, since $f(x), g(x) \to \infty$ as $x \to a$). Hence $|\log f(x)/g(x) - \log\lambda| < \epsilon$ if $|x - a| < \delta$, i.e.,

$$\lim_{x\to a}(\log f(x)/g(x)) = \log\lambda$$

and the theorem follows.

Exercises

5.1. Prove that
(i) If $f'(x) \to 0$ as $x \to \infty$, then $f(x)/x \to 0$ as $x \to \infty$.
(ii) If $f(x) \to 0$ as $x \to \infty$, and $f'(x)$ exists, does it follow that $f'(x) \to 0$ as $x \to \infty$?
(iii) If $g'(x)$ exists for all $x > 1$ and if $g(x)$ tends to a limit as $x \to \infty$, then there exists a sequence of points c_n, $2^n < c_n < 2^{n+1}$, such that $c_n.g'(c_n) \to 0$ as $n \to \infty$. Give an example to show that $x.g'(x)$ need not tend to 0 as $x \to \infty$.

5.2. Let $f(x)$ be a real-valued function, defined on (a, b) such that the second derivative $f''(x)$ is continuous in (a, b). Show that for any $x \in (a, b)$

$$f''(x) = \lim_{h \to 0} [(f(x+h) + f(x-h) - 2f(x))/h^2] .$$

5.3. Let $\alpha > 0$ and suppose that the differentiable function $f : (1, \infty) \to \mathbb{R}$ is such that $f'(t) + \alpha f(t) \to 0$ as $t \to \infty$. By considering the function $xf(\log x/\alpha)$, prove that if $\epsilon > 0$, then

$$|e^{\alpha t_1}.f(t_1) - e^{\alpha t_2}.f(t_2)| \leq \epsilon |e^{\alpha t_1} - e^{\alpha t_2}|$$

provided t_1, t_2 are sufficiently large. Deduce that $f(t) \to 0$ as $t \to \infty$.

5.4. Using the above exercise twice, prove that if $f : (1, \infty) \to \mathbb{R}$ is twice differentiable and $f''(t) + 3f'(t) + 2f(t) \to 2$ as $t \to \infty$, then $f(t) \to 1$ as $t \to \infty$.

5.5. Let $\mathcal{F}$ be the class of all functions $f\colon [0, \infty) \to \mathbb{R}$ that are twice differentiable, such that

$$f(0) = 0,\ f'(0) = 1,\ f'(t) \leq 2,\ |f''(t)| \leq 4 \quad \text{(for all } t > 0).$$

Let $m(x) = \inf_{f \in \mathcal{F}} |f(x)|$. Find $m(x)$, considering separately the cases $x \leq 1/2$, $x > 1/2$. Find a function $f_{x_0} \in \mathcal{F}$ such that $f_{x_0}(x_0) = m(x_0)$, in each of the two cases $x_0 \leq 1/2$, $x_0 > 1/2$.

5.6. Let f have a finite derivative $f'(x_0)$ at x_0. Prove that

$$(f(x_0 + h) - f(x_0 - k))/(h + k) \to f'(x_0)$$

as $h, k \to 0$ through positive values. By considering the function $x^2 \sin(\pi/x)$, show that the result need not hold if k is allowed to take those negative values for which $h + k \neq 0$.

5.7. The function f is continuous for $x > X$ and $f'(x)$ exists and is finite for all $x > X$. Prove that:
(i) $f'(x) \to \lambda$ as $x \to \infty$ implies that $f(x)/x \to \lambda$ (λ finite or infinite).
(i) If $f(x)$ tends to a finite limit as x tends to infinity, then $f'(x)$ can not tend

to an infinite limit nor to a nonzero finite limit.

5.8. A function f is such that $f(0) = 0$ and $f''(x)$ exists and is finite for $x \geq 0$. Prove that for some ξ, $0 < \xi < x$, $f'(x) - f(x)/x = (1/2)xf''(\xi)$. Hence show that if $f(0) = 0$, $f''(x) > 0$ for $x > 0$, then $f(x)/x$ is an increasing function of x for $x > 0$. Explain the geometric significance of this result.

5.9. Let f be differentiable twice in $[0,1]$ and suppose $f(0) = 0$. Let $a > 0$. Prove that there exists a $c \in (0,1)$ such that $f'(c) = ac^{a-1}.f(c)/(1 - c^a)$.

5.10. Let f, g be differentiable functions on $[a,b]$, and suppose $g'(x) \neq 0$ for all $x \in [a,b]$. Prove that there exists $c \in (a,b)$ such that

$$f'(c)/g'(c) = (f(c) - f(a))/(g(b) - g(c)).$$

5.11. Let f be differentiable on $[0,1]$, $f(0) = 0$, $f(1) = 1$. For each $n \in \mathbb{N}$, show that there exist distinct $x_1, x_2, \ldots, x_n$ such that

$$\sum_{i=1}^{n} 1/f'(x_i) = n\,.$$

5.12. Let $f : (0,\infty) \to \mathbb{R}$ be a differentiable function and suppose that $f(x) + f'(x) \to 0$ as $x \to \infty$. Then $f(x) \to 0$ as $x \to \infty$.

5.13. If f is a function such that (i) $f(x) \to c$ as $x \to \infty$ (where c is a fixed real number), and (ii) $f'''(x) \to 0$ as $x \to \infty$. Prove that $f'(x)$ and $f''(x)$ both tend to 0 as $x \to \infty$.

5.14. Let f be defined for $x \geq 1$ and suppose that $f(1) = f(2) = 4$ and that $f''(x)$ exists and is positive in $[1,3]$. What is the sign of $f'(3)$?

5.15. Let $f_1, f_2, \ldots, f_n$ be differentiable functions in $[a,b]$, and suppose $f_1(a) = f_n(b) = 0, f_i(x) \neq 0$ for $x \in (a,b)$, $i = 1,2,\ldots,n$. Show that there exists a point $c \in (a,b)$, such that

$$f_1'(c)/f_1(c) + f_2'(c)/f_2(c) + \cdots + f_n'(c)/f_n(c) = 0\,.$$

5.16. Functions defined by $f(x) = 0$ or $f(x) = x^n$ satisfy $(f(x))^m = f(x^m)$ for all $m \in \mathbb{Z}$. Find a real valued function with this property, which is discontinuous at each $x \in \mathbb{R}$.

5.17. Let f, g be nonnegative and continuous functions in $[1,\infty)$. Suppose f is decreasing, g' is continuous in $[1,\infty)$ and suppose $g(x) \to \infty$ as $x \to \infty$. Then

$$\lim_{n\to\infty} (f(1) + f(2) + \cdots + f(n))/g(n) = \lim_{n\to\infty} (f(n)/g'(n))\,,$$

provided the right hand side limit exists.

5.18. (Theorem of proportional parts). Prove the following result: *Let $a < b < c$ and let f be a real-valued function, defined in $[a,c]$, such that*

(i) f'' exists in (a, c),
(ii) f is continuous in $[a, c]$;
then there exists a $\zeta \in (a, c)$ such that

$$f(b) - f(a) = ((b-a)/(c-a)).(f(c) - f(a)) + (b-a)(b-c).f''(\zeta)/2$$

5.19. Let f and g be differentiable functions in $(0, \infty)$ and for $x > 0$, suppose that $f'(x) = -g(x)/x$ and $g'(x) = -f(x)/x$. Characterize all such functions.

5.20. A function f is defined on $[a, b]$, and f', f'', $\ldots$, $f^{(n+1)}$ all exist in $[a, b]$. Suppose $f^{(i)}(a) = f^{(i)}(b) = 0$ for $i = 0, 1, \ldots, n$. Prove that there exists a $\xi \in (a, b)$ such that $f^{(n+1)}(\xi) = f(\xi)$.

5.21. Let f be continuous in (a, b) and suppose $f'(x)$ exists in (a, b), except perhaps at c, but the limit of $f'(x)$, as x tends to c exists and equals A. Prove that $f'(c)$ exists and equals A.

5.22. Let f be defined and be continuously differentiable in $[a, b]$ (i.e., f' exists in $[a, b]$ and is continuous therein). Suppose there exists a point $c \in (a, b]$ such that $f'(c) = 0$. Then there exists a point $\xi \in (a, b)$ such that $f'(\xi) = (f(\xi) - f(a))/(b - a)$.

5.23. Suppose that f is differentiable twice in $[0, 2]$ and satisfies (i) $|f(x)| \leq 1$, (ii) $|f''(x)| \leq 1$. Then $|f'(x)| \leq 2$ for all $x \in [0, 2]$.

5.24. Show that there exists a unique $c \in \mathbb{R}$ such that for each differentiable function f on $[0, 1]$, with $f(0) = 0$, $f(1) = 1$, the equation $f'(x) = cx$ has a solution in $(0, 1)$.

5.25. We know that $dy/dx = 1/(dx/dy)$ always. However this is false for d^2y/dx^2 and d^2x/dy^2. Determine all functions $y = f(x)$ for which $d^2y/dx^2 = 1/(d^2x/dy^2)$.

5.26. (i) Sketch the graph of the function $y = \cos x . \cos(x + 2) - \cos^2(x + 1)$. (ii) Make a picture of the planar set $A = \{(x, y) : |x - 1| + |y| = 1\}$. Find all points of A at which $(x - 1)^2 + y^2$ is maximum or minimum.

5.27. Let f be continuous in (a, b) and suppose $f'(c)$ exists in a neighbourhood of the point $x = c \in (a, b)$. Suppose that there exists a continuous function $\theta(h)$, with $0 < \theta(h) < 1$, satisfying the equation $f(c+h) - f(c) = hf'(c + h.\theta(h))$. Show that $f'(x)$ is continuous at $x = c$.

5.28. A continuous function f is such that $(f(1 + h) - f(1 - h))/h$ tends to a finite limit as $h \to 0$. Does it follow that $f'(1)$ exists and is finite? Justify your answer.

5.29. The functions f and g are nonconstant, differentiable, real valued functions such that for each pair x, y of real numbers

$$f(x + y) = f(x)f(y) - g(x)g(y) \tag{1}$$

$$g(x+y) = f(x)g(y) + g(x)f(y) \tag{2}$$

If $f'(0) = 0$, prove that $f^2(x) + g^2(x) = 1$ for all x.

5.30. Show that for $x \in (1, e)$, there exists a unique $y \in (e, \infty)$ such that $(\log y)/y = (\log x)/x$. Show that for such x, y, we have $x + y > x \log y + y \log x$.

5.31. Find points, one on each graph, where the functions $y = e^x$ and its inverse function $y = \log_e x$ are closest.

5.32. Find a point a, for which the function $y = \log_a x$ $(a > 1)$ intersects its inverse $y = a^x$ at only one (real) point.

5.33. Prove that if $k \geq 3$, the equation $(\log x)^k = x$ $(x \geq 1)$ has just 2 solutions r_k, s_k and $r_k \to e$, $s_k \to \infty$ as $k \to \infty$.

5.34. Prove that for $x > 1$, the function

$$f(x) = \log((x+2)/(x+1))/(\log((x+1)/x))$$

is an increasing function of x.

5.35. Prove that the function $f(x) = x^n.e^{-x}$ $(x \geq 0, n \geq 2)$ is increasing in $(0, n)$, decreasing in (n, ∞) and has a flex at $x = n \pm \sqrt{n}$.

5.36. Find

(i) all continuous functions$f : \mathbb{R} \to \mathbb{R}$, that satisfy

$$f(f(x)) = f(x) \quad \text{for all } x \in \mathbb{R}\,, \tag{$*$}$$

(ii) all differentiable functions $f : \mathbb{R} \to \mathbb{R}$, that satisfy (*).

5.37. Construct functions, f, that satisfy the following properties (one each) or show that no such function can exist:

I. (i) f' exists everywhere, (ii) $f(\mathbb{Q}) \subset \mathbb{Q}$, (iii) $f'(\mathbb{Q}) \subset \mathbb{R} \setminus \mathbb{Q}$.

II. (i) $f : [0, 1] \to \mathbb{R}$, (ii) f is discontinuous on an everywhere dense set, (iii) f is differentiable on an everywhere dense set.

III. (i) $f : [-1, 1] \to \mathbb{R}$, (ii) $\sum_{n=1}^{\infty} f(1/n)$ is convergent, (iii) $\sum_{n=1}^{\infty} |f(1/n)|$ is divergent.

Chapter 6

Sequences, Harmonic Series, Alternating Series and Related Topics

§6.1. Sequences and series of real numbers

Recall that a function whose domain is the set $\mathbb{N}$ and range a set of real numbers is called a sequence of real numbers. We shall denote a sequence by $\{s_n\}$. A sequence is said to converge to a real number l, if for each $\epsilon > 0$, there exists a positive number m (depending on ϵ) such that $|s_n - l| < \epsilon$, for all $n \geq m$; and we write $s_n \to l$; otherwise it is said to be divergent. Every convergent sequence can easily be seen to be bounded (i.e., its range is bounded), however, the converse is not true:

Example 6.1.1. The sequence $s_n = (-1)^n$ is bounded, but not convergent. The sequence for which $s_n = 1$ for n odd and $s_n = 2$ for n even also has the same property and has positive terms.

A real number ξ is said to be a limit point of a sequence $\{s_n\}$ if every neighbourhood of ξ contains an infinite number of terms of the sequence. For a sequence of real numbers $\{s_n\}$, limit inferior and limit superior are defined respectively as

$$\underline{\lim}_{n\to\infty} s_n = \liminf_{n\to\infty} s_n = \sup_n \{\inf\{a_n, a_{n+1}, \dots\}\}$$

and

$$\overline{\lim}_{n\to\infty} s_n = \limsup_{n\to\infty} s_n = \inf_n \{\sup\{a_n, a_{n+1}, \dots\}\}.$$

It follows that $\inf s_n \leq \underline{\lim}\, s_n \leq \overline{\lim}\, s_n \leq \sup s_n$ and that if $\{s_n\}$ is bounded, then $-\infty < \underline{\lim}\, s_n \leq \overline{\lim}\, s_n < \infty$.

Example 6.1.2. Let $\{a_n\}$ and $\{b_n\}$ be sequences given by

$$\{a_n\} : 0, 1, 2, 1, 0, 1, 2, 1, 0, 1, 2, 1, 0, 1, 2, 1, \dots$$

$$\{b_n\} : 2, 1, 1, 0, 2, 1, 1, 0, 2, 1, 1, 0, 2, 1, 1, 0, \dots .$$

Then

$$\underline{\lim}\, a_n + \underline{\lim}\, b_n < \underline{\lim}\,(a_n + b_n) < \underline{\lim}\, a_n + \overline{\lim}\, b_n < \overline{\lim}\,(a_n + b_n) < \overline{\lim}\, a_n + \overline{\lim}\, b_n ,$$

because the five numbers in question can easily be checked to be $0 < 1 < 2 < 3 < 4$.

By the completeness property of real numbers, it follows that every bounded sequence with a unique limit point is convergent and conversely. Thus, a bounded sequence $\{s_n\}$ converges if and only if $\underline{\lim}\, s_n = \overline{\lim}\, s_n = s$. We also have (the proof can be found in any text book of Analysis):

Theorem 6.1.3 (Cauchy's general principle of convergence). *A necessary and sufficient condition for the convergence of a sequence $\{s_n\}$ is that for each $\epsilon > 0$, there exist a positive integer m such that*

$$|s_{n+p} - s_n| < \epsilon ,$$

for all $n \geq m$ and for all $p \geq 1$.

A sequence $\{s_n\}$ is said to be monotonically increasing if $s_{n+1} \geq s_n$ for all n and monotonically decreasing if $s_{n+1} \leq s_n$ for all n. A necessary and sufficient condition for the convergence of a monotonic sequence is that it is bounded.

A series is the sum of the terms of a sequence. Thus, if $u_1, u_2, u_3, \ldots$ is a sequence, then the sum $u_1 + u_2 + u_3 + \cdots$ of all terms is called an infinite series and is denoted by $\sum_{n=1}^{\infty} u_n$ or simply by $\sum u_n$. Given the series $\sum u_n$, we form, for each $n \geq 1$, $s_n = \sum_{k=1}^{n} u_k$, called the partial sums of the series. If the sequence $\{s_n\}$ of partial sums converges, then the series is regarded as convergent and $\lim_{n\to\infty} s_n$ is said to be the sum of the series. If $\{s_n\}$ does not converge, we say that the series $\sum u_n$ does not converge or that it diverges. A necessary condition (easily provable) for the convergence of an infinite series $\sum u_n$ is that $\lim_{n\to\infty} u_n = 0$. The converse, however, is not true. We have the harmonic series:

Example 6.1.4. The series $\sum_{n=1}^{\infty} \frac{1}{n}$ is not convergent, i.e., it is divergent.

If $\sum \frac{1}{n}$ was convergent to s, say, then the series $(1 + 1/2) + (1/3 + 1/4) + (1/5 + 1/6) + \cdots$ would also converge to s. But the bracketed series is strictly greater than $(1/2+1/2)+(1/4+1/4)+(1/6+1/6)+\cdots = 1+1/2+1/3+\cdots = s$, giving $s > s$, which is a contradiction.

Example 6.1.5. Let $a_n = 1 + 1/2 + \cdots + 1/n$. Then the sequence $\{a_n\}$ is divergent, but for $p > 0$, $a_{n+p} - a_n = 1/(n+1)+\cdots+1/(n+p) \leq p/(n+1) \to 0$. This gives an example of a divergent sequence $\{a_n\}$ for which $\lim_{n\to\infty}(a_{n+p} - a_n) = 0$.

As in the case of sequences, for series also, we have:

Theorem 6.1.6 (Cauchy's general principle of convergence for series). *A series $\sum u_n$ converges if and only if for each $\epsilon > 0$, there exists a positive integer m such that*

$$|u_{n+1} + u_{n+2} + \cdots + u_{n+p}| < \epsilon ,$$

for all $n \geq m$ *and for all* $p \geq 1$, *i.e., if* $\{s_n\}$ *denotes the partial sums of the series, then* $|s_{n+p} - s_n| < \epsilon$, *for all* $n \geq m$ *and* $p \geq 1$.

Series with nonnegative terms are the simplest and the most well behaved. It can be easily proved that a series with positive terms converges if and only if the sequence of its partial terms is bounded above.

We now recall some common tests of convergence for series of positive terms:

(I) *Comparison Test.* If $\sum u_n$ and $\sum v_n$ are two series of positive terms and $k \neq 0$, a fixed real number (independent of n) and if there exists a positive integer m such that $u_n \leq kv_n$, for all $n \geq m$, then
(a) $\sum u_n$ is convergent, if $\sum v_n$ is convergent; and
(b) $\sum v_n$ is divergent, if $\sum u_n$ is divergent.

(II) *Limit Form of Comparison Test.* If $\sum u_n$ and $\sum v_n$ are two series with positive terms such that $\lim_{n\to\infty}(u_n/v_n) = l$, where l is a nonzero finite real number, then the two series converge or diverge together.

(III) *Cauchy's Root Test.* If $\sum u_n$ is a series with positive terms such that $\lim_{n\to\infty}(u_n)^{1/n} = l$, then the series $\sum u_n$
(a) converges, if $l < 1$,
(b) diverges, if $l > 1$; and
(c) the test fails to give any definite information, if $l = 1$.

(IV) *D'Alembert's Ratio Test.* If $\sum u_n$ is a series with positive terms, such that $\lim_{n\to\infty}(u_{n+1}/u_n) = l$, then the series
(a) converges, if $l < 1$,
(b) diverges, if $l > 1$; and
(c) the test fails, if $l = 1$.

(V) *Raabe's Test.* If $\sum u_n$ is a series with positive term, such that $\lim_{n\to\infty} n(\frac{u_n}{u_{n+1}} - 1) = l$, then the series
(a) converges, if $l > 1$,
(b) diverges, if $l < 1$,
(c) the test fails, if $l = 1$.

(VI) *Cauchy's Integral Test.* If $u(x)$ is a nonnegative monotonic decreasing integrable function such that $u(n) = u_n$, for all positive integral values of n, then the series $\sum u_n$ and $\int_1^\infty u(x)dx$ converge or diverge together (the integral $\int_a^\infty u(x)dx$ is said to converge if $t(x) = \int_a^x u(s)ds$ tends to a finite limit as $x \to \infty$, otherwise, the integral is said to diverge).

Example 6.1.7. A series for which the ratio test fails: The series $\sum_{n=1}^{\infty} \frac{1}{n^2}$ and $\sum_{n=1}^{\infty} \frac{1}{n}$ are such that $\lim_{n\to\infty} \frac{(n+1)^2}{n^2} = 1 = \lim_{n\to\infty} \frac{n+1}{n}$, but $\sum_{n=1}^{\infty} \frac{1}{n}$ diverges, whereas $\sum_{n=1}^{\infty} \frac{1}{n^2}$ converges. Indeed, $\frac{1}{1^2} = 1$, $\frac{1}{2^2} + \frac{1}{3^2} < \frac{2}{2^2} = \frac{1}{2}$, $\frac{1}{4^2} + \frac{1}{5^2} + \frac{1}{6^2} + \frac{1}{7^2} < \frac{4}{4^2} = \frac{1}{4} = (\frac{1}{2})^2, \ldots$, shows that $\sum_{n=1}^{\infty} \frac{1}{n^2} < \sum_{n=1}^{\infty} (\frac{1}{2})^n < \infty$, being a convergent geometric series.

Example 6.1.8. The ratio test may also fail by virtue of the nonexistence of the limit l of $\frac{a_{n+1}}{a_n}$. Examples for convergent and divergent series are $\sum_{n=1}^{\infty} 2^{(-1)^n - n}$ and $\sum_{n=1}^{\infty} 2^{n-(-1)^n}$. In the first case $\overline{\lim_{n\to\infty}} \frac{a_{n+1}}{a_n} = 2$ and $\underline{\lim_{n\to\infty}} \frac{a_{n+1}}{a_n} = \frac{1}{8}$, whereas in the second case $\overline{\lim_{n\to\infty}} \frac{a_{n+1}}{a_n} = 8$ and $\underline{\lim_{n\to\infty}} \frac{a_{n+1}}{a_n} = \frac{1}{2}$.

Example 6.1.9. A series for which the root test fails: The root test fails for the convergent series $\sum_{n=1}^{\infty} \left(\frac{5+(-1)^n}{2}\right)^{-n} = \frac{1}{2} + \frac{1}{3^2} + \frac{1}{2^3} + \frac{1}{3^4} + \frac{1}{2^5} + \frac{1}{3^6} + \cdots$, since $\overline{\lim_{n\to\infty}} \sqrt[n]{a_n} = \frac{1}{2}$ and $\underline{\lim_{n\to\infty}} \sqrt[n]{a_n} = \frac{1}{3}$. It also fails for the divergent series $\sum_{n=1}^{\infty} \left(\frac{5+(-1)^n}{2}\right)^{n}$, since $\overline{\lim_{n\to\infty}} \sqrt[n]{a_n} = 3$ and $\underline{\lim_{n\to\infty}} \sqrt[n]{a_n} = 2$.

Example 6.1.10. A series for which the root test succeeds but the ratio test fails:

The convergent series $\sum_{n=1}^{\infty} 2^{(-1)^n - n}$ of Example 6.1.8 is the one for which the root test succeeds. Indeed, $\sqrt[n]{a_n} = 2^{\frac{(-1)^n - n}{n}} \to 2^{-1} = \frac{1}{2} < 1$. Similarly, for the divergent series $\sum_{n=1}^{\infty} 2^{n-(-1)^n}$, of Example 6.1.8, $\sqrt[n]{a_n} = 2^{\frac{n-(-1)^n}{n}} \to 2^1 = 2 > 1$.

Example 6.1.11. A series for which the ratio test fails, but Raabe's test succeeds.

For example take the series

$$\frac{\alpha}{\beta} + \frac{1+\alpha}{1+\beta} + \frac{(1+\alpha)(2+\alpha)}{(1+\beta)(2+\beta)} + \cdots$$

Here

$$u_n = \frac{(1+\alpha)(2+\alpha)\ldots(n-1+\alpha)}{(1+\beta)(2+\beta)\ldots(n-1+\beta)} .$$

Thus $\lim_{n\to\infty} \frac{u_{n+1}}{u_n} = \lim_{n\to\infty} \frac{n+\alpha}{n+\beta} = 1$, and hence the ratio test fails. Also,

$$\lim_{n\to\infty} n\left(\frac{u_n}{u_{n+1}} - 1\right) = \lim_{n\to\infty} n\left(\frac{n+\beta}{n+\alpha} - 1\right) = \lim_{n\to\infty} \frac{\beta-\alpha}{1+\frac{\alpha}{n}} = \beta - \alpha .$$

Thus, by Raabe's test, the series converges if $\beta - \alpha > 1$ or $\beta > \alpha + 1$ and diverges if $\beta < \alpha + 1$. The test fails for $\beta = \alpha + 1$.

We now turn to series with arbitrary terms.

Definition 6.1.12. A series whose terms are alternately positive and negative is called an alternating series.

For such series, we have:

Theorem 6.1.13 (Leibnitz Test). *If the alternating series*

$$u_1 - u_2 + u_3 - u_4 + \cdots \quad (u_n > 0, \text{ for all } n),$$

is such that
(i) $u_{n+1} \leq u_n$, *for all* n,
(ii) $\lim_{n\to\infty} u_n = 0$,
then the series converges.

Definition 6.1.14. A series $\sum u_n$ is said to be absolutely convergent if the series $\sum |u_n|$ is convergent.

Definition 6.1.15. A series which is convergent but not absolutely convergent is called a conditionally convergent series.

Alternating and conditionally convergent series are dealt with in more detail in §6.5. We end this introductory section with:

Proposition 6.1.16. *If* $\sum u_n$ *is absolutely convergent, then* $\sum u_n$ *is convergent.*

Proof. The series $\sum |u_n|$ convergent implies that $|u_{n+1}| + \cdots + |u_m| < \epsilon$, if $m \geq n \geq N$ and hence $|u_{n+1} + \cdots + u_m| < \epsilon$, if $m \geq n \geq N$, so that $\sum u_n$ is convergent, by Cauchy's general principle of convergence.

§6.2. The series $\sum \frac{1}{n}$ and Euler's constant γ

We have seen in Example 6.1.4 that the series $\sum_{n=1}^{\infty} 1/n$ is divergent. It is equally well known that $\sum 1/n$ diverges extremely slowly. While attempting to explore how large the sum $1 + 1/2 + 1/3 + \cdots + 1/n$ is, a beautiful constant γ emerges from the picture. We turn our attention to this famous Euler's constant γ. Letting $S_n = 1 + 1/2 + \cdots + 1/n$ to be the n^{th} partial sum of the series, so that $S_n \to \infty$, we ask: *How fast does* S_n *go to* ∞ *with* n? The answer is the following:

Theorem 6.2.1. $S_n / \log n \to 1$ *as* $n \to \infty$.

Proof. Let $\wp$ be the partition: $1 < 2 < 3 < \ldots < n-1 < n$ of the interval $[1, n]$. Then $\int_1^n \frac{dx}{x}$ lies between the upper and lower Riemann sums (see Figure 6.1), i.e., $s_\wp < \int_1^n \frac{dx}{x} < S_\wp$, i.e.,

$$1/2 + 1/3 + \cdots + 1/(n-1) + 1/n < \log n < 1 + 1/2 + \cdots + 1/(n-1) \quad (1)$$

i.e., $S_n < 1 + \log n$ and $\log n < S_n - 1/n$, i.e., $\log n + 1/n < S_n < \log n + 1$, i.e.,

$$1 + 1/(n \log n) < S_n / \log n < 1 + (1/\log n)$$

and letting $n \to \infty$, Theorem 6.2.1 follows.

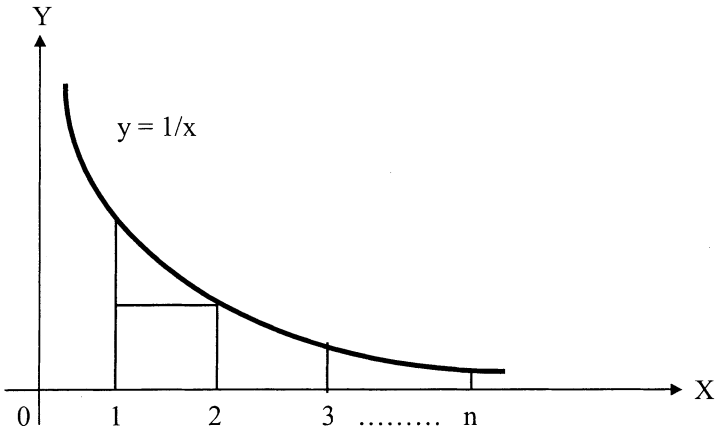


Figure 6.1

However, a much stronger result holds, viz. :

Theorem 6.2.2. *Let* $T_n = S_n - \log n = 1 + 1/2 + \cdots + 1/n - \log n$, *then* T_n *converges to a limit* γ *(called Euler constant* γ*) as* $n \to \infty$, *where* $0 \leq \gamma \leq 1$.

Proof. Rewrite (1) above as

$$1/n < 1 + 1/2 + \cdots + 1/(n-1) + 1/n - \log n < 1 \tag{2}$$

If we can show that $S_n - \log n$ tends to a limit, say, γ, as $n \to \infty$, it will follow from (2) that $0 \leq \gamma \leq 1$, as required. But by (1)

$$0 < \{\log n - (1/2 + \cdots + 1/n)\} < 1 - 1/n < 1 \tag{3}$$

and the quantity τ_n in curly brackets, being the sum of the shaded regions in Figure 6.2, is monotone increasing and bounded above by 1 (by (3)) and hence tends to a limit.

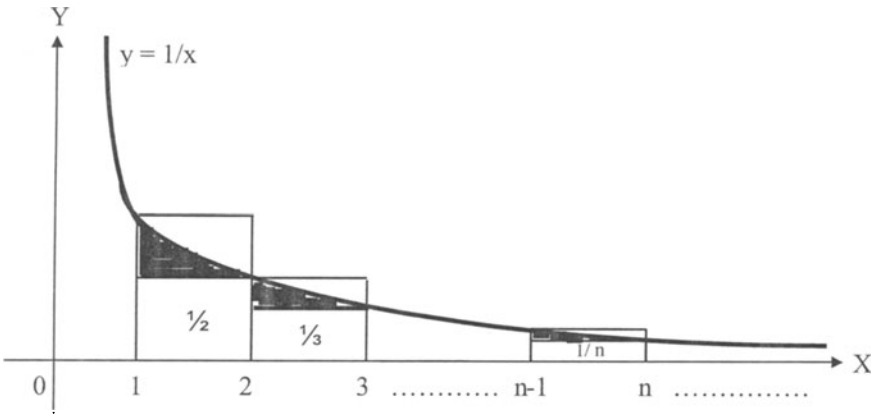


Figure 6.2

It follows that $T_n = 1 - \tau_n$ also tends to a limit (call it γ) as n tends to infinity, completing the proof of the theorem.

Using a slightly more refined argument, we can easily prove the following

Theorem 6.2.3. $1/2 \leq \gamma \leq 1$.

Proof. In the proof of Theorem 6.2.2, we considered rectangles and the area under the curve $y = 1/x$. The refinement is to consider trapezia as follows: We have

$$\frac{1}{2}+\frac{1}{3}+\cdots+\frac{1}{n} < \int_1^n \frac{dx}{x} < \frac{1}{2}\left(1+\frac{1}{2}\right)+\frac{1}{2}\left(\frac{1}{2}+\frac{1}{3}\right)+\cdots+\frac{1}{2}\left(\frac{1}{n-1}+\frac{1}{n}\right) \quad (4)$$

Here the first inequality is got by using rectangles (as in (1) above), while the second one comes from using trapezia, for, the area of the $(r-1)^{\text{th}}$ trapezium is equal to (see Figure 6.3)

$$\frac{1}{2}h(f(r-1)+f(r)) = \frac{1}{2}.1.\left(\frac{1}{r-1}+\frac{1}{r}\right),$$

so starting with $r = 2$ and going up to $r = n$, (4) follows.

The extreme right side in (4) equals $(1/2)(-1 + 2S_n - 1/n)$, while the extreme left side is $S_n - 1$. Hence

$$1/2 + 1/2n < S_n - \log n < 1 \quad (5)$$

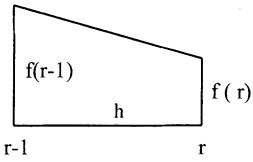


Figure 6.3

and letting $n \to \infty$, Theorem 6.2.3 follows, provided that the limit in question exists; but this has already been established in the proof of Theorem 6.2.2.

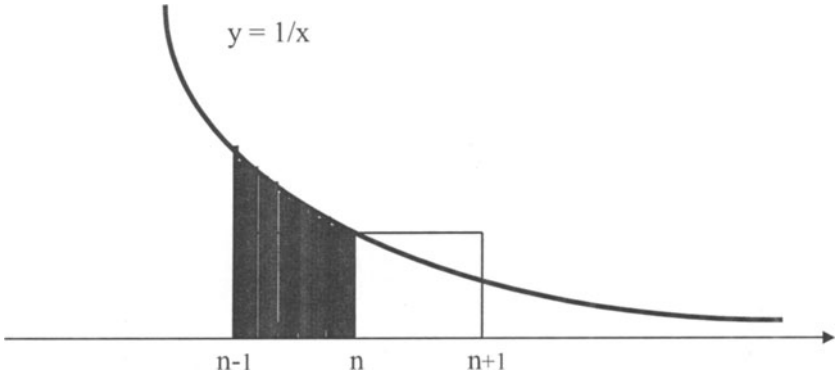


Figure 6.4

Remark 6.2.4. Actually, if we let $X_n = \{S_{n-1} - \log(n-1)\} - \{S_n - \log n\}$, it is easy to see, using a diagram, that $X_n > 0$ for all n.

Indeed

$$X_n = \{\log n - \log(n-1)\} - (S_n - S_{n-1})\,.$$

Here the quantities $\log n - \log(n-1)$ and $S_n - S_{n-1}$ are just the shaded areas shown in Figures 6.4 and 6.5 respectively. The difference is the shaded area shown in Figure 6.6, which is clearly greater than 0. Thus $S_n - \log n$ decreases as n increases and so, by (5), γ exists, as required.

Using the *Mercator expansion* (see Theorem 6.2.5 below)

$$\log(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \cdots \quad (-1 < x \leq 1)\,,$$

so that $X_n = (S_{n-1} - S_n) - \log(1 - 1/n) = (-1/n) + (1/2).1/n^2 + (1/3).1/n^3 + \cdots > 0$, as required. However, the proof using a diagram is far more instructive.

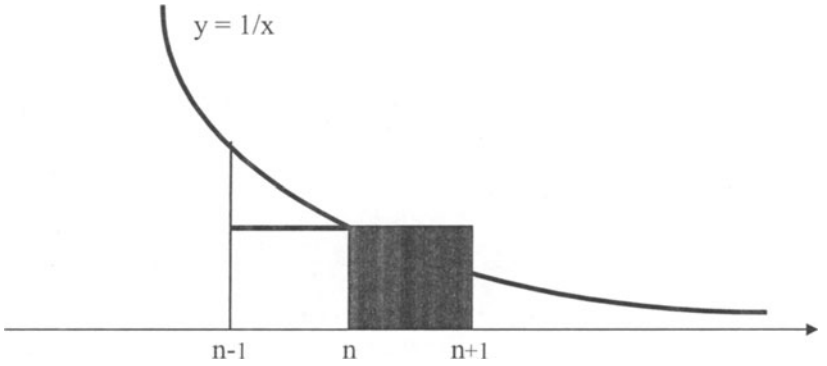


Figure 6.5

We now take a digression and give elementary proofs of the following two well known expansions. Proof of the second of these has already been given using Taylor series in Chapter 5.

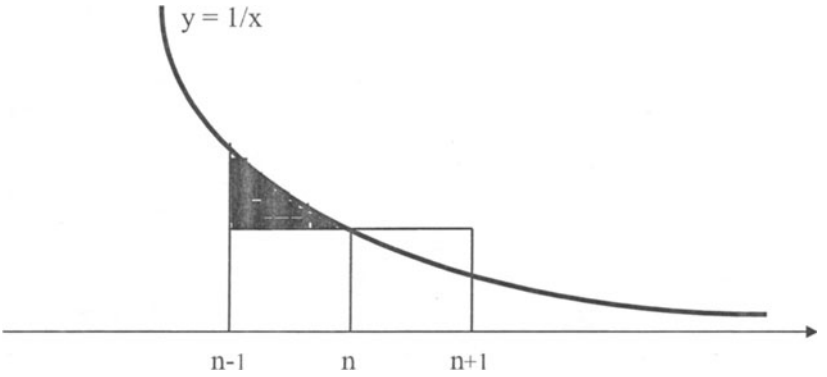


Figure 6.6

Theorem 6.2.5. I. (James Gregory, 1668).

$$\tan^{-1} x = x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \cdots \text{ , } \textit{if } |x| \leq 1 \text{ .}$$

II. (Nicolaus Mercator, 1669).

$$\log(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots \text{ , } \textit{if } -1 < x \leq 1.$$

Proof. Sum the geometric progression $S = 1 - x + x^2 - \cdots + (-x)^{n-1}$ to get $S(1+x) = 1 - (-x)^n$, or $1/(1+x) = (1 - x + x^2 - x^3 + \cdots + (-x)^{n-1}) + (-x)^n/(1+x)$. Now replace x by t^2 and integrate from 0 to x to get

$$\int_0^x \frac{dt}{1+t^2} = x - \frac{x^3}{3} + \frac{x^5}{5} - \cdots + (-1)^n \frac{x^{2n-1}}{2n-1} + \int_0^x (-1)^n \frac{t^{2n} dt}{1+t^2},$$

i.e.,

$$\tan^{-1} x = x - \frac{x^3}{3} + \cdots + (-1)^n \frac{x^{2n-1}}{2n-1} + R_n(x),$$

where

$$R_n(x) = \int_0^x (-1)^n \frac{t^{2n} dt}{1+t^2} \text{ .} \tag{1}$$

In the range $[0, x]$ (or $[x, 0]$ if $x < 0$), $1 \leq 1 + t^2$, i.e., $1/(1+t^2) \leq 1$ and so

$$\frac{t^{2n}}{1+t^2} \leq t^{2n} \text{ .} \tag{2}$$

Hence

$$\begin{aligned} |R_n(x)| &\leq \left| \int_0^x (-1)^n \frac{t^{2n} dt}{1+t^2} \right| \\ &\leq \left| \int_0^x t^{2n} dt \right| \quad \text{(by (2))} \\ &= \left| \frac{x^{2n+1}}{2n+1} \right| \\ &\leq \frac{1}{2n+1}, \end{aligned}$$

which tends to 0 as $n \to \infty$, if $|x| \leq 1$. However, if $|x| > 1$, say $|x| = 1 + h$ $(h > 0)$, then $|x^n|/n = |x|^n/n = (1 + nh + \cdots + h^n)/n > nh/n$ which does not tend to 0 as $n \to \infty$, i.e., (1) is not convergent. That completes the proof of the first result.

Also, $1/(1+t) = S_n + (-t)^n/(1+t)$, where $S_n = 1 - t + t^2 - \cdots (-t)^{n-1}$. Integrating from 0 to x, we get

$$\log(1+x) = \int_0^x \{(S_n + ((-t)^n)/(1+t))\} dt$$
$$= x - x^2/2 + x^3/3 - \cdots + (-1)^{n-1} x^n/n + \int_0^x ((-t)^n/(1+t))\, dt\,.$$

Let $E_n(x) = \int_0^x ((-t)^n/(1+t))\, dt$. We prove that $E_n(x) \to 0$ as $n \to \infty$, if $-1 < x \leq 1$.

First, let $0 \leq x \leq 1$. If $t > 0$, then $1 + t > 1$ and so $1/(1+t) < 1$ or $t^n/(1+t) < t^n$. Hence $|E_n(x)| = \int_0^x (t^n/(1+t))\, dt < \int_0^x t^n dt = x^{n+1}/(n+1) \leq 1/(n+1) \to 0$ as $n \to \infty$.

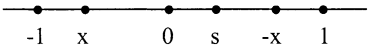


Figure 6.7

Next, let $-1 < x < 0$. Put $s = -t$, then $E_n(x) = (-1)^n \int_0^{-x} (-s)^n ds/(1-s)$, and since $1 - s > 1 + x$, $s^n/(1-s) < s^n/(1+x)$, we get $|E_n(x)| < \int_0^x s^n ds/(1+x) = (-x)^{n+1}/(1+x)(1+n) < 1/(1+x)(n+1)$ (since $0 < -x < 1$), which tends to 0 as $n \to \infty$, i.e., $E_n(x) \to 0$ as $n \to \infty$.

Observe that if $x > 1$, the series does not converge as x^n/n does not tend to 0, while if $x \leq -1$, $\log(1+x)$ is not defined, so the question of convergence to $\log(1+x)$ does not arise.

Putting $x = 1$ in the above expansion, we get

Corollary 6.2.6.

$$\frac{\pi}{4} = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \cdots$$

and

$$\log 2 = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots\,.$$

The second result of this corollary may also be derived in a neat way as follows (see [35]):

We know that

$$\int_1^2 \frac{dx}{x} = \log 2\,.$$

Subdivide the interval $[1, 2]$ into n equal parts and consider lower approximating rectangles of height $1/(1+1/n)$, $1/(1+2/n)$, ... As $n \to \infty$, the sum of the areas of these n rectangles tend to (see Figure 6.8) the integral $\int_1^2 dx/x$, i.e.,

$$\lim_{n\to\infty} \left[\frac{1}{1+1/n}\frac{1}{n} + \frac{1}{1+2/n}\frac{1}{n} + \cdots + \frac{1}{1+n/n}\frac{1}{n}\right] = \log 2$$

i.e.,

$$\lim_{n\to\infty}\left(\frac{1}{n+1}+\frac{1}{n+2}+\cdots+\frac{1}{n+n}\right)=\log 2 \qquad (*)$$

Now, let

$$\begin{aligned} S_{2n} &= 1-1/2+1/3-1/4+\cdots+1/(2n-1)-1/2n \\ &= 1+1/2+1/3+\cdots+1/2n-2(1/2+1/4+\cdots+1/2n) \\ &= 1+1/2+\cdots 1/2n-(1+1/2+\cdots+1/n) \\ &= 1/(n+1)+1/(n+2)+\cdots+1/(n+n)\ , \end{aligned}$$

which tends to $\log 2$ (by $(*)$). Also $S_{2n+1}=S_{2n}+1/(2n+1)$ tends to $\log 2$ too.

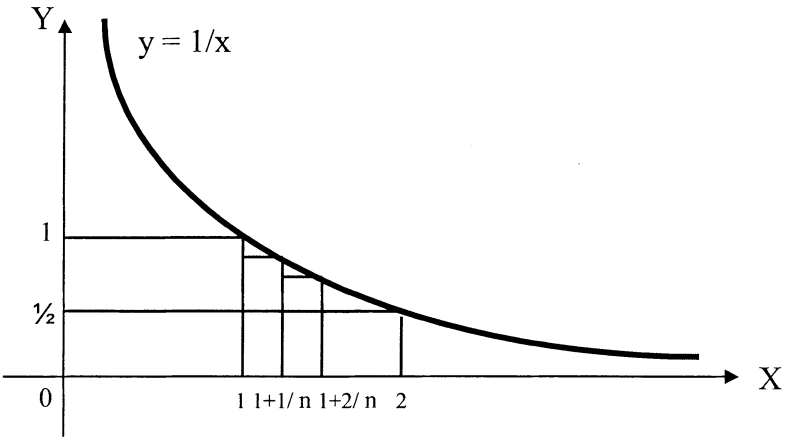


Figure 6.8

Before we go on to some more properties of the Euler constant γ, we give another simple derivation of Leibnitz-Gregory series for $\pi/4$ and Mercator series for $\log 2$ (see [55]) i.e.,

$$\pi/4=1-1/3+1/5-1/7+\cdots+(-1)^k/(2k+1)+\cdots=\sum_{k=1}^{\infty}(-1)^k/(2k+1) \qquad (i)$$

$$\log 2=1-1/2+1/3-1/4+\cdots+(-1)^k/k+\cdots=\sum_{k=1}^{\infty}(-1)^{k-1}/k \qquad (ii)$$

Consider the integral $I_n=\int_0^{\pi/4}\tan^n x\,dx$ $(n\geq 2)$. Since, for $0\leq x\leq \pi/4$, we have $0\leq \tan x\leq 1$, we see that $0\leq \tan^{n+1} x\leq \tan^n x$, for all $x\in[0,\pi/4]$ and for all $n=0,1,2,\ldots$. So, being the area under $\tan^n x$,

$$I_n \text{ decreases as } n \text{ increases .} \qquad (1)$$

Next, it is easy to verify the formulae

$$I_n + I_{n-2} = 1/(n-1) \tag{2}$$

$$I_n + I_{n+2} = 1/(n+1) \tag{2}'$$

Using (1), these give

$$1/(2(n+1)) < I_n < 1/(2(n-1)) \tag{3}$$

The reduction formula (2) gives

$$I_{2n} = 1/(2n-1) - 1/(2n-3) + \cdots \pm 1 - (\pm\pi/4) \tag{4}$$

$$I_{2n+1} = 1/(2n) - 1/(2n-2) + \cdots \pm 1/2 - (\pm \log 2)/2 \tag{5}$$

Let us analyse (4) for signs. Using (2), we get the following :

$$I_2 = 1 - I_0 = 1 - \pi/4 ,$$
$$I_4 = 1/3 - I_2 = 1/3 - 1 + \pi/4 = (-1)(-1/3 + 1 - \pi/4) ,$$
$$I_6 = 1/5 - 1/3 + 1 - \pi/4 ,$$
$$I_8 = 1/7 - 1/5 + 1/3 - 1 + \pi/4 = (-1)(-1/7 + 1/5 - 1/3 + 1 - \pi/4) ,$$
$$\ldots \quad \ldots \quad \ldots \quad \ldots .$$

Taking absolute values, we get, for all even suffixes

$$I_{2n} = \left| \sum_{k=0}^{n} \frac{(-1)^k}{2k+1} - \frac{\pi}{4} \right| , \quad n = 1, 2, \ldots .$$

Now using (3), this gives

$$\frac{1}{2(2n+1)} < \left| \sum_{k=0}^{n} \frac{(-1)^k}{2k+1} - \frac{\pi}{4} \right| < \frac{1}{2(2n-1)} , \quad n = 1, 2, \ldots$$

Letting $n \to \infty$, we obtain (i).

Next, analyzing (5) for signs, we get, using (2):

$$I_1 = \int_0^{\pi/4} \tan x \, dx = [-\log \cos x]_0^{\pi/4} = (\log 2)/2 ,$$
$$I_3 = 1/2 - I_1 = 1/2 - (\log 2)/2 ,$$
$$I_5 = 1/4 - I_3 = 1/4 - 1/2 + (\log 2)/2 = (-1)(-1/4 + 1/2 - (\log 2)/2) ,$$
$$I_7 = 1/6 - I_5 = 1/6 - 1/4 + 1/2 - (\log 2)/2 ,$$
$$\ldots \quad \ldots \quad \ldots \quad \ldots .$$

Taking absolute values, we get, for all odd suffixes

$$I_{2n+1} = \left| \sum_{k=1}^{n} \frac{(-1)^{k-1}}{2k} - \frac{\log 2}{2} \right| , \quad n = 1, 2, \ldots .$$

Now using (3), this gives

$$\frac{1}{4(n+1)} < \left| \sum_{k=1}^{n} \frac{(-1)^{k-1}}{2k} - \frac{\log 2}{2} \right| < \frac{1}{4n} \ , \ n = 1, 2, \ldots .$$

Letting $n \to \infty$, we obtain (ii).

For geometrical ways of looking at the expansion of $\log 2$, see [59] and [34].

Now, some more properties of γ. As already said before (in Chapter I), it is not known if γ is rational, irrational or transcendental. It has been calculated to many significant figures; the value is approximately given by $\gamma = .57721566490\ldots$. The following mnemonic for remembering the value of γ has been composed by Morgan Ward [114]:

These numbers proceed to a limit Euler's subtle mind discerned.

Here the number of letters in each word agrees with the decimal digits of the value of γ. Since the next significant figure in the decimal expansion of γ is 0, the sentence comes to a natural stopping place!

Our first aim is to calculate γ to say, 5 significant figures. We write

$$S_n = \log n + \gamma + \epsilon_n \ , \ \text{where } \epsilon_n \to 0 \text{ as } n \to \infty \ . \tag{6}$$

Since $\log(n+a) - \log n \to 0$ as $n \to \infty$, it follows that $S_n - \log(n+a) \to \gamma$ as $n \to \infty$ and so we may write (6) as

$$S_n = \log(n+a) + \gamma + \epsilon_n \ , \tag{7}$$

where $\epsilon_n \to 0$ as $n \to \infty$ and the number a is arbitrary (but fixed), to be chosen suitably. Equation (7), with $n-1$ in place of n, gives

$$S_{n-1} = \log(n-1+a) + \gamma + \epsilon_{n-1} \tag{8}$$

Subtracting (8) from (7) gives $1/n = \log((n+a)/(n-1+a)) + \epsilon_n - \epsilon_{n-1}$.

Now take $a = 1/2$, as then the terms $1/n$ and $1/n^2$ will not occur. Indeed, we have $1/n = \log((n+1/2)/(n-1/2)) + \epsilon_n - \epsilon_{n-1} = \log(1+1/2n) - \log(1-1/2n) + \epsilon_n - \epsilon_{n-1}$. Now use the expansions $\log(1+x) = x - x^2/2 + x^3/3 - \cdots$ and $\log(1-x) = -x - x^2/2 - x^3/3 - \cdots$, so the above becomes $1/n = 1/n + 2((1/2n)^3/3 + (1/2n)^5/5 + \cdots) + \epsilon_{n-1} - \epsilon_n$, giving $\epsilon_n - \epsilon_{n-1} = 2((1/2n)^3/3 + (1/2n)^5/5 + \cdots)$, which is greater than the first tem $1/(12n^3)$, but

$$\begin{aligned} \epsilon_{n-1} - \epsilon_n &< (1 + 3(1/2n)^2/5 + 3(1/2n)^4/7 + \cdots)/(12n^3) \\ &< (1 + 1/n^2 + 1/n^4 + \cdots)/(12n^3) \quad \text{(to estimate crudely)} \\ &= 1/(12n^3(1 - 1/n^2)) \\ &= 1/(12n(n^2-1)) \ . \end{aligned}$$

Thus, $1/12n^3 < \epsilon_{n-1} - \epsilon_n < 1/(12n(n-1)(n+1))$. Now sum these inequalities from n to m $(n < m)$ to obtain

$$\sum_{r=n}^{m} 1/(12r^3) < \epsilon_{n-1} - \epsilon_m < \sum_{r=n}^{m} 1/(12(r-1)r(r+1))\,.$$

Here the last sum above, using partial fractions, works out to be equal to $1/(12.2n(n-1))$ and letting $m \to \infty$, we get, since $\epsilon_m \to 0$,

$$\sum_{r=n}^{\infty} 1/(12r^3) < \epsilon_{n-1} < 1/(24n(n-1)) \tag{9}$$

Now, when $\alpha < x < \beta$, we say that "*x is less than β by less than $\beta - \alpha$*" meaning x falls short of β by a quantity less than $\beta - \alpha$. Thus (9) can be read as: ϵ_{n-1} is less than $1/24n(n-1)$ by less than $A = (1/24n(n-1) - \sum_{r=n}^{\infty} 1/12r^3)$.

We now wish to estimate A. We have $A = (1/12)\sum_{r=n}^{\infty} 1/(r-1)r(r+1) - (1/12)\sum_{r=n}^{\infty} 1/r^3$ (going back to the summation, so that earlier, we need not have used partial fractions and summed !), i.e., $A = (1/12)\sum_{r=n}^{\infty} 1/r^3(r-1)(r+1) < (1/12)\sum_{r=n}^{\infty} 1/(r-2)(r-1)r(r+1)(r+2) = X/12$, say, since $1/r^3(r-1)(r+1) < 1/(r^2-4)r(r-1)(r+1)$. But now using partial fractions to sum this series, we get $X = 1/4(n-2)(n-1)n(n+1)$. Indeed,

$$\begin{aligned} 24X &= \sum_{r=n}^{\infty}(1/(r-2) - 4/(r-1) + 6/r - 4/(r+1) + 1/(r+2)) \\ &= 1/(n-2) - 3/(n-1) + 3/n - 1/(n+1) \\ &= 6/((n-2)(n-1)n(n+1))\,, \end{aligned}$$

as required. So now finally we have:

$$\epsilon_{n-1} \text{ is less than } 1/24n(n-1) \text{ by less than } 1/48(n-2)(n-1)n(n+1). \tag{10}$$

Choose a numerical value of n, say $n = 12$. Then $S_{12} = \log 12\tfrac{1}{2} + \gamma + \epsilon_{12}$ (by (7) rather than (6)). But $S_{12} = 3.103210678\ldots$ and $\log 12\tfrac{1}{2} = \log 100/8 = 2\log 10 - 3\log 2 = 2.525728654\ldots$. Hence

$$\gamma + \epsilon_{12} = .577482024\ldots\,, \tag{11}$$

where ϵ_{12} is less than $.0002671\ldots$ by less than $.0000009\ldots$ (a check: ϵ_{12} is less than $1/24.13.12$ by less than $1/48.11.12.13.14$, i.e., the given figures), i.e., $.0002662 < \epsilon_{12} < .0002671$. This together with (11) gives

$$.5772149 < \gamma < .5772158\,,$$

the actual value of γ being $.57721566\ldots$. So we have calculated γ to within an error of 5 in the sixth decimal place and have found that:

$$S_n = \log(n+1/2) + \gamma + 1/24(n(n+1)) - \delta\,, \tag{12}$$

where $\delta < 1/(48(n-1)n(n+1)(n+2))$ (by (7) and (10)).

Our aim now is to prove the following striking result

Theorem 6.2.7. $\lim_{n\to\infty} n.((1+1/2+1/3+\cdots+1/n-\log n)-\gamma)$ *exists and equals* $1/2$.

To prove this result, we shall find an interesting *integral representation of* γ. Indeed, we have the following:

Theorem 6.2.8. $\lim_{n\to\infty}(1+1/2+1/3+\cdots+1/n-\log n)$ *exists and equals*

$$\int_0^\infty (e^{-x}/(1-e^{-x})-e^{-x}/x)dx.$$

Proof (Lakshman Rao [60]). Verify that

$$1/r = \int_0^\infty e^{-rx}dx \ , \ r>0 \tag{1}$$

Now integrate (1) from 1 to n to get

$$\log n = \int_1^n dr/r = \int_1^n \left(\int_0^\infty e^{-rx}dx\right)dr = \int_0^\infty \left(\int_1^n e^{-rx}dr\right)dx$$
$$= \int_0^\infty (e^{-x}-e^{-nx})dx/x \tag{2}$$

(it is easy to check that interchange of the two integrals is valid–see Lemma 6.2.9; (2) is called the Frullani integral). Hence

$$a_n = 1+\frac{1}{2}+\cdots+\frac{1}{n}-\log n$$
$$= \int_0^\infty \left(e^{-x}+e^{-2x}+\cdots+e^{-nx}\right)dx - \int_0^\infty \frac{e^{-x}-e^{-nx}}{x}dx \text{ (using (1) and (2))}$$
$$= \int_0^\infty \left(\frac{e^{-x}-e^{-(n+1)x}}{1-e^{-x}} - \frac{e^{-x}-e^{-nx}}{x}\right)dx \qquad \text{(on summing the G.P.)}$$
$$= \int_0^\infty \left(\frac{e^{-x}}{1-e^{-x}} - \frac{e^{-x}}{x}\right)dx + \int_0^\infty e^{-nx}\left(\frac{1}{x}-\frac{1}{e^x-1}\right)dx \text{ (on regrouping)} \tag{3}$$

We now verify the inequalities:

$$1/2 - x/8 < 1/x - 1/(e^x-1) < 1/2 \ , \qquad (\text{for } x>0) \tag{4}$$

First note that $e^x - 1 > 0$ if $x > 0$. and so we may multiply throughout by $x(e^x-1)$ and clear denominators. Thus, it is enough to prove

(i) $(x^2-4x+8).e^x > x^2+4x+8$,

(ii) $(x-2).e^x + x + 2 > 0$.

Checking (ii). Expanding $e^x = 1+x+x^2/2!+x^3/3!+\cdots$, terms upto x^2 cancel out. For $n \geq 3$, it is enough to prove that $1/(n-1)! > 1/n!$, i.e., that $n > 2$, which is true.

Checking (i). Here too, terms up to x^2 cancel. For $n \geq 3$, coefficient of x^n works out to be $n^2 - 5n + 8$ and so it is enough to prove that this coefficient is greater than 0, which is clear since it is equal to $(n-5/2)^2 + 7/4$.

The second integral in (3) can be estimated as $\int_0^\infty e^{-nx}(1/x - 1/(e^x - 1))dx < \int_0^\infty e^{-nx}dx/2 = 1/2n$ (using the second inequality of (4)). Hence, (3) gives

$$0 \leq \left| a_n - \int_0^\infty \left(\frac{e^{-x}}{1-e^{-x}} - \frac{e^{-x}}{x} \right) dx \right| = \left| \int_0^\infty e^{-nx} \left(\frac{1}{x} - - \frac{1}{e^x - 1} \right) dx \right| < \frac{1}{2n} .$$

Letting $n \to \infty$, we get Theorem 6.2.8. The result used in the proof is the following:

Lemma 6.2.9.

$$\int_1^n \left(\int_0^\infty e^{-rx} dx \right) dr = \int_0^\infty \left(\int_1^n e^{-rx} dr \right) dx .$$

Proof. Write $\varphi(r) = \int_0^\infty e^{-rx}dx$ and let $\epsilon > 0$ be given. Then there exists a Δ such that $|\int_0^X e^{-rx}dx - \int_0^\infty e^{-rx}dx| < \epsilon$ whenever $X \geq \Delta$, for the left side is equal to $|\int_X^\infty e^{-rx}dx| = [e^{-rx}/-r]_X^\infty = 1/re^{rx}$. Here, $re^{rx} > 1.e^x$ (since $r \in [1,n]$ (see the second integral)), which tends to ∞ as $X \to \infty$, so $1/re^{rX} < \epsilon$ if $X \geq \Delta$, as required.

Now choose Δ such that $|\int_X^\infty e^{-rx}dx| < \epsilon/n$, whenever $X \geq \Delta$. Then

$$\int_1^n \left(\left| \int_X^\infty e^{-rx} dx \right| \right) dr < \int_1^n \epsilon\, dr/n = \epsilon(n-1)/n < \epsilon \qquad (*)$$

and so $\int_1^n (\int_0^\infty e^{-rx}dx)dr = \int_1^n (\int_0^X e^{-rx}dx)dr + \int_1^n (\int_X^\infty e^{-rx}dx)dr$. In the first integral on the right, the order can be interchanged, since e^{-rx} is a continuous function of r and x. Here, bringing that first integral to the left side, we get

$$\left| \int_1^n \left(\int_0^\infty e^{-rx} dx \right) dr - \int_0^X \left(\int_1^n e^{-rx} dr \right) dx \right| = \left| \int_1^n \left(\int_X^\infty e^{-rx} dx \right) dr \right| < \epsilon ,$$

by (*) above, if $X > \Delta$, and the lemma follows.

Proof of Theorem 6.2.7. Write equation (3) as

$$a_n - \gamma = \int_0^\infty e^{-nx}(1/x - 1/(e^x - 1))dx\,,$$

and so by (4) we get

$$\int_0^\infty e^{-nx}(1/2 - x/8)dx < a_n - \gamma < \int_0^\infty e^{-nx}dx/2\,,$$

or $1/2n - 1/8n^2 < a_n - \gamma < 1/2n$. Multiplying by n and letting $n \to \infty$, Theorem 6.2.7 follows.

Example 6.2.10. Writing $S_n = 1 + 1/2 + \cdots + 1/n$, it follows that $\gamma < S_r + S_s - S_{rs} \leq 1$, for all integers $r, s \geq 1$. Indeed, we have $S_r + S_s - S_{rs} = (S_r + S_s - S_r) - (S_{2r} - S_r) - (S_{3r} - S_{2r}) - \cdots - (S_{sr} - S_{(s-1)r}) \leq S_s - r(1/2r) - r(1/3r) - \cdots - r(1/sr) = 1$, as required. Further, we have already seen that

$$S_n - \log n \text{ decreases and tends to } \gamma \text{ as } n \text{ tends to } \infty. \qquad (**)$$

So, $S_{rs} - \log rs \leq S_r - \log r$ (by (**)), i.e.,

$$0 \leq S_r - S_{rs} - \log r + \log rs \quad \text{and} \quad \gamma < S_s - \log s \ \text{(by(**))}\,.$$

Adding these two inequalities gives that $\gamma < S_r + S_s - S_{rs}$.

Another way of proving the above is as follows:
Put $f(r,s) = S_r + S_s - S_{rs}$. Then

$$\begin{aligned} f(r,s) - f(r-1,s) &= (S_r + S_s - S_{rs}) - (S_{r-1} + S_s - S_{(r-1)s}) \\ &= 1/r - (1/(rs - s + 1) + 1/(rs - s + 2) + \cdots + 1/rs) \\ &< 1/r - s/rs = 0 \end{aligned}$$

Hence, $f(r,s)$ is a decreasing function of r and so also of s, by symmetry.

(i) Let $r = s = 1$. Then, f being decreasing, $f(1,1)$ is maximum, and so $f(r,s) \leq f(1,1) = 1$, as required.

(ii) Let $r, s \to \infty$. Then

$$\begin{aligned} f(r,s) &= S_r + S_s - S_{rs} \\ &= (\log r + \gamma + \epsilon_r) + (\log s + \gamma + \epsilon_s) - (\log rs + \gamma + \epsilon_{rs}) \\ &= \gamma + \epsilon_r + \epsilon_s - \epsilon_{rs} \\ &\to \gamma\,, \end{aligned}$$

as $r, s \to \infty$, since all the ϵ's tend to 0. Further, since $f(r,s)$ decreases, it is greater than γ.

Example 6.2.11. We can use the above notions to find the sum of the series

$$S = (\log 1)/1 - (\log 2)/2 + (\log 3)/3 - \cdots = \sum_{r=1}^{\infty} (-1)^{r+1} (\log r)/r \ .$$

The series is convergent since the terms decrease and tend to 0. Let $f(n) = \sum_{r=1}^{n} (\log r)/r - (\log^2 n)/2$. By Cauchy's integral test, $\sum_{r=1}^{n} (\log r)/r - (\log^2 n)/2$ is convergent if and only if the integral $\int_1^n \log x dx/x - (\log^2 n)/2$ is convergent, which is the case (since the expression works out to be 0). Thus, $f(n)$ converges to a limit λ, say, as $n \to \infty$ and so $f(2n) - f(n) \to \lambda - \lambda = 0$, as $n \to \infty$. But now

$$\begin{aligned}
f(2n) - f(n) &= \sum_{r=1}^{2n} \frac{\log r}{r} - \frac{\log^2 2n}{2} - \sum_{r=1}^{n} \frac{\log r}{r} + \frac{\log^2 n}{2} \\
&= \left\{ \sum_{r=1}^{2n} \frac{(-1)^{r+1} \log r}{r} + 2 \sum_{r=1}^{n} \frac{\log 2r}{2r} \right\} - \frac{\log^2 2n}{2} - \sum_{r=1}^{n} \frac{\log r}{r} + \frac{\log^2 n}{2} \\
&= \sum_{r=1}^{2n} \frac{(-1)^{r+1} \log r}{r} + \left\{ 2 \sum_{r=1}^{n} \frac{\log 2r}{2r} - \sum_{r=1}^{n} \frac{\log r}{r} \right\} - \frac{(\log 2 + \log n)^2}{2} + \\
&\qquad + \frac{\log^2 n}{2} \\
&= \sum_{r=1}^{2n} \frac{(-1)^{r+1} \log r}{r} + \left\{ \log 2 . \sum_{r=1}^{n} 1/r \right\} - \frac{\log^2 2}{2} - \log 2 . \log n \\
&= \sum_{r=1}^{2n} \frac{(-1)^{r+1} \log r}{r} + \log 2 \left\{ \sum_{r=1}^{n} \frac{1}{r} - \log n \right\} - \frac{\log^2 2}{2} \quad \text{(on shuffling)},
\end{aligned}$$

and letting $n \to \infty$, we get that $0 = S + \gamma(\log 2) - \frac{1}{2} . \log 2 . \log 2$, which gives

$$S = \log 2 \, (\log \sqrt{2} - \gamma) \ .$$

§6.3. Some number-theoretic aspects of the harmonic series

Since $\sum 1/n$ is divergent, the partial sums $S_N = \sum_{n=1}^{N} 1/n$ are larger than every positive integer as N increases. It is a remarkable fact that S_N is never actually equal to an integer if $N > 1$, i.e., we have the following:

Theorem 6.3.1 *The partial sums $S_N = 1 + 1/2 + 1/3 + \cdots + 1/N$ of the harmonic series are never an integer, for any value of $N \geq 2$.*

Proof. Let 2^t be the highest power of 2 which is less than or equal to N, so

that $1 < 2^t \leq N$, but $2^t > N$. Then

$$S_N = \frac{1}{1} + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{2^t} + \frac{1}{2^t+1} + \cdots + \frac{1}{N}$$
$$= \frac{2^t u + 2^{t-1}u + 2^t u/3 + \cdots + u + \cdots + 2^t u/N}{2^t u} ,$$

where u is odd. Here each term in the numerator, except u, is even; so the numerator is odd and hence the 2^t in the denominator can not cancel out, showing that S_N is not an integer.

Remark 6.3.2. Since $\sum 1/n$ diverges very slowly, it will take quite a number of terms to add up to even a small number like say 10. The problem is to look for the integer n_A such that $S_{n_A} \geq A$ but $S_{n_A-1} < A$. It may amuse the reader to note, for example, that the sum of a quarter of a million terms of the series is still less than 20, whereas in order to surpass 100, it is necessary to add more than 15 million trillion trillion trillion terms! In [15], R.P.Boas 'almost' solves the problem. Indeed, he says that "it is 'almost' true that $n_A = [e^{A-\gamma} + \frac{1}{2}]$". Heuristically, since we know that $S_n = \log n + \gamma - \epsilon_n$, where $\epsilon_n \to 0$ as $n \to \infty$, so $S_n > A$ if and only if $\log n + \gamma - \epsilon_n > A$, i.e., if and only if $n > e^{A-\gamma}.e^{\epsilon_n}$. Since e^{ϵ_n} is close to 1, we see that n_A is close to $[e^{A-\gamma}]$, as required (more or less!).

Boas refines the equation $S_n = \log n + \gamma - \epsilon_n$, using the Euler-Maclaurin formula, to the equation

$$S_n = \log n + \gamma + 1/2n + \int_n^\infty (t - [t] - 1/2).t^{-2} dt .$$

Consequently, S_n is larger than A, if $\log n > A - \gamma - 1/2n + 1/8n^2$ and is less than A if $\log n < A - \gamma - 1/2n$. This leads to Boas' result. For details see the paper of Boas cited above.

Example 6.3.3. Let $A = 3$. We have, approximately, $1 = 1.00000\ldots$, $1/2 = 0.50000\ldots$, $1/3 = 0.33333\ldots$, $1/4 = 0.25000\ldots$, $1/5 = 0.20000\ldots$, $1/6 = 0.16666\ldots$, $1/7 = 0.142857\ldots$, $1/8 = 0.12500\ldots$, $1/9 = 0.11111\ldots$, $1/10 = 0.10000\ldots$; adding, we get $S_{10} = 2.9288\ldots$, which is less than 3, but $S_{11} = S_{10} + 1/11 = 2.9288\ldots + 0.0909\ldots = 3.0197$, which is greater than 3. Thus $S_{10} < 3$, $S_{11} > 3$. It follows that $n_3 = 11$.

Let now $A = 10$ and let us try to determine n_{10}. Clearly, the crude method used above for $A = 3$, will be most impractical. By equation (12) (immediately above Theorem 6.2.7), we have $S_n = \log(n + 1/2) + \gamma + (1/24n(n+1) - \delta)$, where $\delta < 1/48(n-1)n(n+1)(n+2)$. We try to solve $10 = \log(\nu + 1/2) + \gamma$, i.e., $\log(\nu + 1/2) = 10 - .57722 = 9.42278$, i.e., $\nu + 1/2 = e^{9.42278}$ and using tables this gives $\nu + 1/2$ as just under 12367 and so ν lies between 12366 and 12367 (it is very near $12366 + 1/2$, to the left of it (see Figure 6.9)).

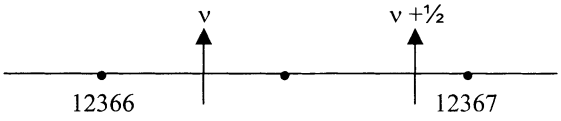


Figure 6.9

The value of S_{12367} is thus equal to $\log(12367 + 1/2) + \gamma + \delta'$, which is *just* larger than $\log(\nu + 1/2) + \gamma + \delta' = 10 + \delta' > 10$ (*just* because δ' is very small) and further $S_{12366} = S_{12367} - 1/12367 = 10 + \delta'' - 1/12367 < 10$, since δ'' is very very small. Thus $n_{10} = 12367$ (for a more detailed account of this, see R.G.Lyness, Note on γ, *Math. Mag.*, **24** (1940), 206–209).

Let us now consider the following similar question. We know that as n increases, the quantity $S(m,n) = 1/(m+1) + 1/(m+2) + \cdots + 1/(m+n)$ tends to ∞ for any fixed value of m. We have the following:

Theorem 6.3.4. *The quantity $S(m,n)$ crosses 1 for some value of n between the two numbers $[me] + 1$ and $[(m+1)e]$, i.e.,*

$$S(m,[me]) < 1 < S(m,[(m+1)e]) .$$

Proof. From Figure 6.10, we see that

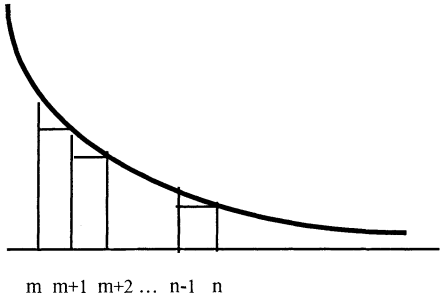


Figure 6.10

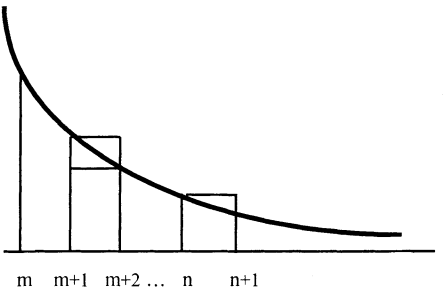


Figure 6.11

$$S(m,n) < \int_m^n \frac{dx}{x} ,$$

while from the Figure 6.11, we see that

$$\int_{m+1}^{n+1} \frac{dx}{x} < S(m,n) .$$

These two give $\log_e((n+1)/(m+1)) < S(m,n) < \log_e(n/m)$. Since we want $1 < S(m,n)$, we must ensure that $1 < \log_e((n+1)/(m+1))$, i.e., that $e < (n+1)/(m+1)$, or $(m+1)e - 1 < n$, while $S(m,n)$ will definitely be less than

1 if $\log_e n/m < 1$, i.e., if $n < m.e$. So, as soon as n exceeds $m.e$, i.e., as soon as $n \geq [m.e] + 1$, $S(m,n)$ may cross 1 and it will definitely cross 1 as soon as n exceeds $m.e + e - 1$, and so being an integer, for some value of n between $[m.e] + 1$ and $[(m+1)e]$ (both inclusive), $S(m,n)$ will cross 1.

We now prove the following striking result

Theorem 6.3.5 (See [81]). *Every positive rational number is the sum of a finite number of distinct terms of the harmonic series (without the condition "distinct", the result is trivial: any positive rational A/B equals*

$$\underbrace{1/B + 1/B + \cdots + 1/B}_{(A \text{ terms})} \ )$$

Proof. Let a/b be any given positive rational and first let $A/B > 1$. Since $\sum 1/n$ is divergent, there exists a unique positive integer n_0 such that

$$S_{n_0} = \sum_{i=1}^{n_0} 1/i < A/B \leq \sum_{i=1}^{n_0+1} 1/i = S_{n_0} + 1/(n_0+1)\,. \qquad (I)$$

If the second inequality in (I) is an equality, we are through; so suppose that strict inequality holds in (I), at the second place too, i.e.,

$$A/B - S_{n_0}\ (= A_1/B_1\,,\ \text{say})\ < 1/(n_0+1)\,. \qquad (II)_1$$

Let n_1 be the unique positive integer such that

$$1/(n_1+1) \leq A_1/B_1 < 1/n_1\,, \qquad (III)_1$$

i.e., n_1 is determined by $n_1 < B_1/A_1 \leq n_1 + 1$. If equality holds here, we are through, for $A/B = S_{n_0} + A_1/B_1$ (by $(II)_1$) $= S_{n_0} + 1/(n_1+1)$ (by the supposed equality in $(III)_1$). So suppose that strict inequality holds in $(III)_1$. Then

$$0 < (A_1(n_1+1) - B_1)/B_1(n_1+1) = A_2/B_2\,(\text{say}) \ \text{ with } (A_2, B_2) = 1\,, \qquad (II)_2$$

i.e., $0 < (A_1(n_1+1) - B_1)/B_1(n_1+1) = A_2/B_2$ (A_2, B_2 in their lowest terms), so $A_2 \leq A_1(n_1+1) - B_1$, while the right hand inequality of $(III)_1$ implies that $A_1(n_1+1) - B_1 < A_1$ and these imply

$$A_2 < A_1\,. \qquad (\dagger)_1$$

Now we check that

$$A_2/B_2 < 1/(n_1(n_1+1))\,, \qquad (IV)$$

i.e., $(A_1(n_1+1) - B_1)/B_1(n_1+1) < 1/n_1(n_1+1)$, or $A_1/B_1 - 1/(n_1+1) < 1/n_1(n_1+1)$ or $A_1/B_1 < 1/n_1$, which is true by $(III)_1$. Let n_2 be the unique integer such that

$$1/(n_2+1) \leq A_2/B_2 < 1/n_2\,, \qquad (III)_2$$

i.e., $n_2 < B_2/A_2 \leq n_2 + 1$, while (IV) implies that $N_1(n_1 + 1) < B_2/A_2$; hence $n_1(n_1 + 1) < n_2 + 1$, i.e., $n_1 < n_1(n_1 + 1)$, so

$$n_1 < n_2 \ . \qquad (\ddagger)_1$$

If equality holds in the left inequality of $(III)_2$, we are again through, for then

$$\begin{aligned} A/B &= S_{n_0} + A_1/B_1 && \text{(by } (II)_1\text{)} \\ &= S_{n_0} + 1/(n_1 + 1) + A_2/B_2 && \text{(by } (II)_2\text{)} \\ &= S_{n_0} + 1/(n_1 + 1) + 1/(n_2 + 1) && \text{(by the assumed equality of } (III)_2\text{).} \end{aligned}$$

So suppose strict inequality holds in $(III)_2$. Now proceed as from $(II)_2$ to get

$$0 < A_2/B_2 - 1/(n_2 + 1) = A_3/B_3 \text{ (say)} \ , \text{ with } (A_3, B_3) = 1 \ , \qquad (II)_3$$

i.e., $0 < (A_2(n_2 + 1) - B_2)/B_2(n_2 + 1) = A_3/B_3$, $((A_3, B_3) = 1)$, so $A_3 \leq A_2(n_2+1)-B_2$, while the right hand side of $(III)_2$ implies $A_2(n_2+1)-B_2 < A_2$. These imply

$$A_3 < A_2 \ . \qquad (\dagger)_2$$

Now check that

$$A_3/B_3 < 1/(n_2(n_2 + 1)) \ . \qquad (IV)_2$$

Let n_3 be the unique integer such that

$$1/(n_3+1) \leq A_3/B_3 < 1/n_3 \ . \qquad (III)_3$$

Prove, as above (see after $(III)_2$), that

$$n_2 < n_3 \ , \qquad (\ddagger)_2$$

and so on. The process must terminate after finitely many steps, since at each step, the numerator strictly decreases (see $(\dagger)_1$, $(\dagger)_2$, ...) and so a stage comes when A_m/B_m becomes $1/B_m$, which is a term of the series $\sum 1/n$, giving

$$A/B = S_{n_0} + 1/(n_1 + 1) + 1/(n_2 + 1) + \cdots + 1/(n_{m-1} + 1) + 1/B_m \ .$$

Further, by $(\ddagger)_1$, $(\ddagger)_2$, ..., the n_i are all distinct, so the condition that the terms be distinct is satisfied too. This completes the case $A/B > 1$.

So, now let $A/B < 1$. Let n_0 be uniquely determined by $1/n_0 < A/B < 1/(n_0 - 1)$, where clearly we may suppose strict inequality is valid on the right above, otherwise we are through. Let $0 < A/B - 1/n_0 = A_1/B_1$. Let n_1 be determined (uniquely) by $1/(n_1 + 1) \leq A_1/B_1 < 1/n_1$. This is exactly the equation $(III)_1$ and so we may proceed as in the case $A/B > 1$. This completes the proof of the theorem.

Remark 6.3.6. For actual computations, we summarize the procedure below:

- Let $A/B > 1$.
- Let n_0 be determined by

$$S_{n_0} = 1 + 1/2 + \cdots + 1/n_0 < A/B < 1 + 1/2 + \cdots + 1/n_0 + 1/(n_0 + 1)\,.$$

- Let $A_1/B_1 = A/B - S_{n_0}$.
- Let n_1 be determined by $n_1 < B_1/A_1 \leq n_1 + 1$. Let $A_2/B_2 = A_1/B_1 - 1/(n_1 + 1)$.
- Let n_2 be determined by $n_2 < B_2/A_2 \leq n_2 + 1$. Let $A_3/B_3 = A_2/B_2 - 1/(n_2 + 1)$

and so on. Since $\ldots < A_3 < A_2 < A_1$, in finitely many steps, we get $A_m = 1$, i.e., $1/B_m = A_m/B_m = A_{m-1}/B_{m-1} - 1/(n_{m-1} + 1)$, and then

$$A/B = S_{n_0} + 1/(n_1 + 1) + 1/(n_2 + 1) + \cdots + 1/(n_{m-1} + 1) + 1/B_m\,.$$

Let us take some actual examples to illustrate this.

Example 6.3.7. $2 = 1 + 1/2 + 1/3 + 1/6$.

The number 3 requires hard work. We proceed as follows:
We first exhibit the representation, then explain the working.

$$3 = 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5} + \frac{1}{6} + \frac{1}{7} + \frac{1}{8} + \frac{1}{9} + \frac{1}{10} + \frac{1}{29} + \frac{1}{1239} + \frac{1}{4311720}\,,$$

i.e., $3 = S_{10} + 1/29 + 1/1239 + 1/4311720$. Indeed, $S_{10} = 2 + 2431/2520 < 3$, but $S_{11} = 2 + 2431/2520 + 1/11 = 2 + 29261/27720 > 3$. Thus

$$0 < A_1/B_1 = 3 - S_{10} = 89/2520 < 1/11\,. \tag{1}$$

Define n_1 by $n_1 < B_1/A_1 \leq n_1 + 1$, i.e., $28 < 2520/89 = 28 + 28/89 < 29$, giving $1/29 < 89/2520 < 1/28$. So,

$$0 < A_1/B_1 - 1/(n_1+1) = 89/2520 - 1/29 = 59/73080 = A_2/B_2\,. \tag{2}$$

Define n_2 by $n_2 < 73080/59 \leq n_2 + 1$, i.e., $1238 < 73080/59 \leq 1239$. So

$$\frac{A_3}{B_3} = \frac{A_2}{B_2} - \frac{1}{n_2 + 1} = \frac{59}{73080} - \frac{1}{1239} = \frac{59}{8.5.7.9.29} - \frac{1}{3.7.59} = \frac{1}{4311720}\,. \tag{3}$$

Hence

$$\begin{aligned} 3 &= S_{10} + 89/2520 && \text{(by (1))} \\ &= S_{10} + 1/29 + 59/73080 && \text{(by (2))} \\ &= S_{10} + 1/29 + 1/1239 + 1/4311720 && \text{(by (3))} \end{aligned}$$

Proposition 6.3.8. *For any positive integer r,*

$$r(1-1/(n+1)^{1/r}) < 1+1/2+\cdots+1/n \leq r(n^{1/r}-1)+1\,.$$

Proof. Consider the curve $y = 1/(x+1)^{1+1/r}$ and the area A between its graph, the x-axis and the ordinates $x = 0$, $x = n$ (see Figure 6.12). We have

$$A < 1+1/2+\cdots+1/n\,.$$

However, $A = \int_0^n (x+1)^{-1-1/r}dx = \left[-r/(x+1)^{1/r}\right]_0^n = r(1-1/(n+1)^{1/r})$, which proves the first inequality.

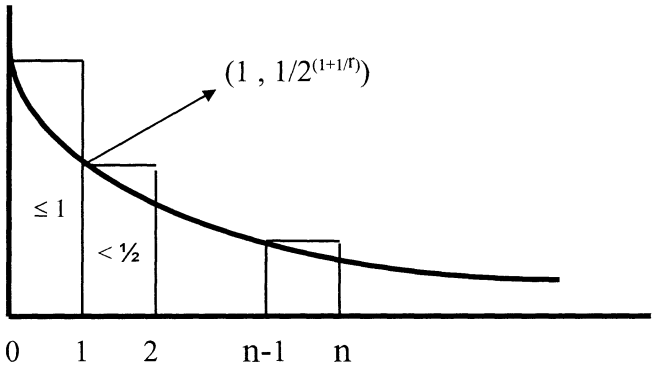


Figure 6.12

For the second inequality, consider the curve $y = 1/x^{1-\frac{1}{r}}$ and the area B between its graph, the x-axis and the ordinates $x = 1$, $x = n$ (see Figure 6.13). We have $B > 1/2+1/3+\cdots+1/n$. However, $B = \int_0^n x^{-1+\frac{1}{r}}dx = r\left[x^{1/r}\right]_1^n = r(n^{1/r}-1)$. Hence $1+1/2+1/3+\cdots+1/n \leq r(n^{1/r}-1)+1$, as required. Observe that in Figure 6.13, the rectangles are respectively of area greater than $1/2, 1/3, \ldots 1/n$.

For an alternative method of proving this result, see [87].

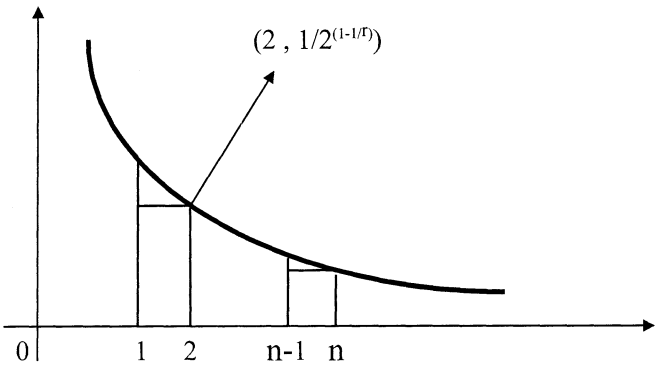


Figure 6.13

§6.4. The restricted harmonic series

Let S_9 denote the series obtained from the harmonic series upon the deletion of all terms which contain the digit 9 (it could be any other digit; for example, if it is 8, the series would be called S_8) i.e.,

$$S_9 = \sum_{9 \notin n} \frac{1}{n} = 1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{8} + \frac{1}{10} + \frac{1}{11} + \cdots + \frac{1}{18} + \frac{1}{20} + \cdots + \\ + \frac{1}{88} + \frac{1}{100} + \frac{1}{101} + \cdots + \frac{1}{108} + \frac{1}{110} + \cdots$$

Since such a reduction seems to leave a richer series than that with prime denominators, which is known to be divergent (see [33]), one would expect this series to be very much divergent. However, a big surprise is that this series is convergent. This was first proved in 1914 by A.J.Kempner (see [56], and also [48]). It is easy to prove that all these series S_i, with sum σ_i, say, are convergent (see [46]).

Let us first consider $S_0 = \sum_{0 \notin n} 1/n$, i.e., the sum is taken over all those n's which do not contain the digit 0. Let σ_0 denote its sum. We have the following:

Theorem 6.4.1. $20.18 < \sigma_0 < 28.29$.

Proof.(i) We leave as it is, the sum from all the one digit terms: $1 + 1/2 + \cdots + 1/9$, since no zero appears in any of the terms.
(ii) Next, the sum from the two digit terms without a zero is $(1/11 + 1/12 + \cdots + 1/19) + (1/21 + 1/22 + \cdots + 1/29) + \cdots + (1/91 + 1/92 + \cdots + 1/99) < 9 \cdot 1/10 + 9 \cdot 1/20 + \cdots + 9 \cdot 1/90$.
(iii) The sum from the three digit terms without a zero equals $(1/111 + 1/112 + \cdots + 1/119) + (1/121 + 1/122 + \cdots + 1/129) + \cdots + (1/191 + \cdots + 1/199) + (1/211 + \cdots + 1/219) + \cdots + (1/291 + \cdots + 1/299) + \cdots + \cdots < 9^2 \cdot 1/100 + 9^2 \cdot 1/200 + \cdots + 9^2 \cdot 1/900$; and so on.

Hence

$$\begin{aligned} S_0 &< (1 + 1/2 + \cdots + 1/9) + (9/10)(1 + 1/2 + \cdots + 1/9) + \\ &\qquad + (9/10)^2(1 + 1/2 + \cdots + 1/9) + \cdots \\ &= (1 + 1/2 + \cdots + 1/9)(1 + 9/10 + (9/10)^2 + \cdots) \\ &= (1 + 1/2 + \cdots + 1/9)10 \cong 28.29\,. \end{aligned}$$

The convergence of S_0 now follows.

To get a lower bound for S_0, we proceed exactly as above:
$((i)')$ Leave the one-digit terms as they are, since they have no zero in them, to give the sum $1 + 1/2 + \cdots + 1/9$.
$((ii)')$ The sum of the two-digit terms without a zero is greatèr than $9/20 + 9/30 + \cdots + 9/100$.
$((iii)')$ The sum of the three-digit terms without a zero is greater than $9 \cdot$

$1/200 + 9^2 \cdot 1/300 + \cdots + 9^2 \cdot 1/1000$; and so on, hence

$$\begin{aligned}
\sigma_0 &> \left(1 + \frac{1}{2} + \cdots + \frac{1}{9}\right) + \frac{9}{10}\left(\frac{1}{2} + \cdots + \frac{1}{10}\right) + \left(\frac{9}{10}\right)^2 \left(\frac{1}{2} + \cdots \frac{1}{10}\right) + \cdots \\
&= \left(1 + \frac{1}{2} + \cdots + \frac{1}{9}\right) + \frac{9}{10}\left(\frac{1}{2} + \cdots + \frac{1}{10}\right)\left(1 + \frac{9}{10} + (\frac{9}{10})^2 + \cdots\right) \\
&= \left(1 + \frac{1}{2} + \cdots + \frac{1}{9}\right) + \frac{9}{10}10\left(1 + \frac{1}{2} + \cdots + \frac{1}{10} - 1\right) \\
&= \left(1 + \frac{1}{2} + \cdots + \frac{1}{9}\right)(1 + 9) - \frac{9 \cdot 9}{10} \\
&\cong 20.18
\end{aligned}$$

The approximate calculations of the other σ's is of about the same order of difficulty. For example, we prove, with equal ease, the following:

Theorem 6.4.2 (See [38]). *Let* $S_1 = \sum_{1 \notin n} 1/n$. *Then* S_1 *is convergent with sum* σ_1, *where* $14.68 < \sigma_1 < 18.29$.

Proof. Exactly as before, the one-digit numbers, without a 1 contribute $1/2 + 1/3 + \cdots + 1/9$; the two-digit numbers, without a 1 contribute $(1/20 + 1/22 + \cdots + 1/29) + (1/30 + 1/32 + \cdots + 1/39) + \cdots + (1/90 + 1/92 + \cdots + 1/99) < 9.1/20 + 9.1/30 + \cdots + 9.1/90 = (9/10)(1/2 + 1/3 + \cdots + 1/9)$. The three-digit numbers, without a 1, similarly contribute $(9/10)^2(1/2 + 1/3 + \cdots + 1/9)$; and so on. Hence, we get

$$\begin{aligned}
\sigma_1 &< (1/2 + 1/3 + \cdots + 1/9)(1 + (9/10) + (9/10)^2 + \cdots) \\
&= (1/2 + 1/3 + \cdots + 1/9).10 \\
&\cong 18.29\,.
\end{aligned}$$

A lower estimate can be worked out as follows: The contribution from the one-digit numbers, without a 1, equals $(1/2 + 1/3 + \cdots + 1/9)$; the contribution from the two-digit numbers, without a 1, by above, is more than $9.1/30 + 9.1/40 + \cdots + 9.1/100 = (9/10)(1/3 + 1/4 + \cdots + 1/10)$; the contribution from the three-digit numbers, without a 1, is similarly more than $(9/10)^2(1/3 + 1/4 + \cdots + 1/10)$; and so on. Hence, we get

$$\begin{aligned}
\sigma_1 &> \left(\frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{9}\right) + \left(\frac{1}{3} + \frac{1}{4} + \cdots + \frac{1}{10}\right)\left(\frac{9}{10} + \left(\frac{9}{10}\right)^2 + \cdots\right) \\
&= \left(\frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{9}\right) + \left(\frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{10}\right)9 \quad \text{(summing the G.P.)} \\
&= \left(\frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{9}\right)10 - 9\left(\frac{1}{2} - \frac{1}{10}\right) \\
&= \left(\frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{9}\right)10 - 3.6 \\
&\cong 14.68
\end{aligned}$$

Remark 6.4.3. This argument can be refined and better bounds can be obtained. In [5], Robert Baillie gave a method of finding the value of σ_i, $i = 1, 2, \ldots, 9$, to a high degree of accuracy and also gave a table giving these values to 20 decimal places. Baillie remarks that " it seems plausible that the σ_i are all irrational ".

Remark 6.4.4. We would like to give an explanation as to why the convergence of these series S_i is to be expected. Let us fix an i, say $i = 1$. When we look at the number of removed terms, we see that, in the beginning at least, they are very few and so we tend to think that the left over series would diverge. However, as we look at large n, we see that if for example, n contains 50 digits, in all probability it would contain $i = 1$ as a digit and so would get removed. Thus, as n increases, most n get removed, which is what leads to the convergence.

More precisely, it is easy to check that
(i) the proportion removed from the one-digit numbers is 1/10 (1 from $1, 1/2, \ldots, 1/10$);
(ii) the proportion removed from the two-digit numbers is 18/90 ($1/10, 1/11, \ldots, 1/19, 1/21, 1/31, 1/41, \ldots, 1/91$ from $1/10, 1/11, \ldots, 1/99$);
(iii) the proportion removed from the three-digit numbers is 252/900 (more cumbersome to calculate);
and so on and we see that $1/10 < 1/5 < 252/900 < \ldots$.

§6.5. Rearrangements of the alternating series $\sum(-1)^n/n$

We now look at the alternating series $1 - 1/2 + 1/3 - \cdots$ (convergent to $\log 2 = S$, say) which is so very closely related to the harmonic series. This is a typical example of a *conditionally convergent series.* Unexpected things can happen with such a series, for example, if we rearrange its terms as

$$1 - 1/2 - 1/4 + 1/3 - 1/6 - 1/8 + 1/5 - \cdots ,$$

where each positive term is followed by two negative ones, it is easy to see that this rearranged series converges to $\frac{1}{2}S$. Indeed,

$$\begin{aligned} S_{3n} &= (1 - 1/2 - 1/4) + (1/3 - 1/6 - 1/8) + \cdots + \\ &\qquad + (1/(2n-1) - 1/2(2n-1) - 1/4n) \\ &= (1/2 - 1/4) + (1/6 - 1/8) + \cdots + (1/2(2n-1) - 1/2(2n)) \\ &= \frac{1}{2}(1 - 1/2 + 1/3 - 1/4 + \cdots + 1/(2n-1) - 1/2n)) , \end{aligned}$$

converges to $\frac{1}{2}S$, as $n \to \infty$. Further, $S_{3n+1} = S_{3n} + 1/(2n+1) \to \frac{1}{2}S$ and $S_{3n+2} = S_{3n} + 1/(2n+1) - 1/2(2n+1) \to \frac{1}{2}S$ as $n \to \infty$. It follows that $S_n \to \frac{1}{2}S$, as required.

Such a phenomena is not possible with, for example, the alternating series

$$1 - 1/2 + 1/2^2 - 1/2^3 + 1/2^4 - \cdots ,$$

a geometric progression with sum 2/3. By making a similar rearrangement, we get the series $(1-1/2-1/2^3)+(1/2^2-1/2^5-1/2^7)+\cdots$, which also converges to the same sum, viz. 2/3, for

$$\begin{aligned} S_{3n} &= \left(1-\frac{1}{2}-\frac{1}{2^3}\right)+\left(\frac{1}{2^2}-\frac{1}{2^5}-\frac{1}{2^7}\right)+\cdots+\left(\frac{1}{2^{2n-2}}-\frac{1}{2^{4n-3}}-\frac{1}{2^{4n-1}}\right) \\ &= (1-5/2^3)+(1/2^2-5/2^7)+\cdots+(1/2^{2n-2}-5/2^{4n-1}) \\ &= \left(1+\frac{1}{2^2}+\frac{1}{2^4}+\cdots+\frac{1}{2^{2n-3}}\right)-\left(\frac{5}{2^3}\right)\left(1+\frac{1}{2^4}+\frac{1}{2^8}+\cdots+\frac{1}{2^{4n-4}}\right) \\ &\to 4/3-2/3=2/3\,, \end{aligned}$$

as required.

The reason for *why this happens for* $1-1/2+1/3-\cdots$ *but not for* $1-1/2+1/2^2-1/2^3+\cdots$? is due to the fact that this latter series is not only *convergent* but also *absolutely convergent.*

We now develop the theory of *unconditionally convergent series* and prove the beautiful theorem of Riemann in the sequel. This will enable us to understand the reason behind the two interesting phenomena described above.

Definition 6.5.1. The series $\sum a'_n$ is called a rearrangement of the series $\sum a_n$ if each a_n is an a'_n and each a'_n is an a_n, once and once only.

Definition 6.5.2. We say that $\sum a_n$ is unconditionally convergent if every rearrangement of $\sum a_n$ is convergent.

Our object now is to prove the following three results:

Theorem 6.5.3. *If all rearrangements of* $\sum a_n$ *converge (i.e., if* $\sum a_n$ *is unconditionally convergent, according to Definition 6.5.2), then they all converge to the same sum.*

In view of our opening remarks about the alternating harmonic series, Theorem 6.5.3 shows that all rearrangements of $1-1/2+1/3-1/4+\cdots$ do not converge.

Theorem 6.5.4. $\sum a_n$ *is unconditionally convergent if and only if* $\sum a_n$ *is absolutely convergent.*

Theorem 6.5.5 (Riemann). *Let* $\alpha \le \beta$ *be any two real numbers and let* $\sum a_n$ *be a series of real numbers, with partial sums* S_n*, which is convergent but not absolutely convergent. Then there exists a rearrangement* $\sum a'_n$ *of* $\sum a_n$ *with partial sums* S'_n*, such that*

$$\limsup_{n\to\infty} S'_n = \beta \;,\quad \liminf_{n\to\infty} S'_n = \alpha\,.$$

Proof. Let $p_n = (|a_n| + a_n)/2$ and $q_n = (|a_n| - a_n)/2$, i.e.,

$$p_n = \begin{cases} a_n \text{ , if } a_n \geq 0 \\ 0 \text{ , if } a_n \leq 0 , \end{cases}$$

and

$$q_n = \begin{cases} 0 \text{ , if } a_n \geq 0 \\ -a_n \text{ , if } a_n \leq 0 . \end{cases}$$

The series $\sum p_n$ and $\sum q_n$ both diverge to $+\infty$, for if both were convergent, then $\sum |a_n| = \sum(p_n + q_n) = \sum p_n + \sum q_n$ would be convergent, which is not, while if $\sum p_n$ were convergent and $\sum q_n$ divergent or vice versa, then $\sum a_n = \sum(p_n - q_n) = \sum p_n - \sum q_n$ would be divergent, which it is not.

Now let $P_1, P_2, \ldots$ be the nonnegative terms of $\sum a_n$, in the order in which they occur and let $Q_1, Q_2, \ldots$ be the absolute values of the negative terms of $\sum a_n$, also in the order in which they occur. Thus, $\sum P_n$ and $\sum Q_n$ are same as $\sum p_n$ and $\sum q_n$ and are therefore divergent, i.e. :

$$\sum P_n \text{ and } \sum Q_n \text{ are divergent.} \qquad (*)$$

Now start adding terms of $\sum P_n$ one by one till $P_1 + P_2 + \cdots + P_{n_1 - 1} < \beta$ and $P_1 + P_2 + \cdots + P_{n_1 - 1} + P_{n_1} \geq \beta$. This is possible since $\sum P_n$ is divergent (if $\beta < 0$, take $n_1 = 1$).

Next, subtract sufficiently many Q's from $P_1 + P_2 + \cdots + P_{n_1}$, one by one till $P_1 + P_2 + \cdots + P_{n_1} - Q_1 - Q_2 - \cdots Q_{m_1 - 1} > \alpha$ and $P_1 + P_2 + \cdots + P_{n_1} - Q_1 - Q_2 - \cdots Q_{m_1 - 1} - Q_{m_1} \leq \alpha$. Again this is possible since $\sum Q_n$ is divergent.

Now again add the P's from $n_1 + 1$ onwards, one by one, to the last sum till $P_1 + P_2 + \cdots + P_{n_1} - Q_1 - Q_2 - \cdots Q_{m_1 - 1} - Q_{m_1} + P_{n_1 + 1} + \cdots + P_{n_2 - 1} < \beta$ and $P_1 + P_2 + \cdots + P_{n_1} - Q_1 - Q_2 - \cdots Q_{m_1 - 1} - Q_{m_1} + P_{n_1 + 1} + \cdots + P_{n_2 - 1} + P_{n_2} \geq \beta$. Then again subtract from this sum the Q's from $m_1 + 1$ onwards, one by on till $P_1 + P_2 + \cdots + P_{n_1} - Q_1 - Q_2 - \cdots Q_{m_1 - 1} - Q_{m_1} + P_{n_1 + 1} + \cdots + P_{n_2 - 1} + P_{n_2} - Q_{m_1 + 1} - \cdots Q_{m_2 - 1} > \alpha$ and $P_1 + P_2 + \cdots + P_{n_1} - Q_1 - Q_2 - \cdots Q_{m_1 - 1} - Q_{m_1} + P_{n_1 + 1} + \cdots + P_{n_2 - 1} + P_{n_2} - Q_{m_1 + 1} - \cdots Q_{m_2 - 1} - Q_{m_2} \leq \alpha$, and so on, alternate "just" crossing β to the right and then "just" crossing α to the left.

This gives us a rearrangement of $\sum a_n$, say $\sum a'_n$, viz. n_1 positive terms of $\sum a_n$ followed by m_1 negative terms, then n_2 positive terms of $\sum a_n$ followed by m_2 negative terms; and so on.

Let S'_n be the partial sums of $\sum a'_n$. The subsequence $\{S'_k\}$ of $\{S'_n\}$, whose last terms are $P_{n_1}, P_{n_2}, \ldots$ is convergent to β, as for such a S'_k,

$$\begin{aligned} |S'_k - \beta| &= |(P_1 + \cdots + P_{n_1} - Q_1 - \cdots - Q_{m_1}) + \cdots + \\ &\quad + (P_{n_{k-1}+1} + \cdots + P_{n_k - 1} + P_{n_k}) - \beta| \\ &\leq P_{n_k} \\ &\to 0 , \end{aligned}$$

as $k \to \infty$ (see Figure 6.14).

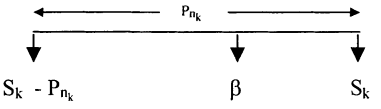


Figure 6.14

Similarly, the subsequence, whose last terms are $Q_{m_1}, Q_{m_2}, \ldots$ is convergent to α.

Remark 6.5.6. If β, say, is ∞, we may choose an increasing sequence of real numbers, for example $1, 2, 3, \ldots$ and then construct our rearrangement as follows:

$$\begin{aligned}
&P_1 + \cdots P_{n_1 - 1} < 1 \,, \\
&P_1 + \cdots P_{n_1 - 1} + P_{n_1} \geq 1 \,; \\
&P_1 + \cdots P_{n_1} - Q_1 - \cdots Q_{m_1} + P_{n_1 + 1} + \cdots + P_{n_2 - 1} < 2 \,, \\
&P_1 + \cdots P_{n_1} - Q_1 - \cdots Q_{m_1} + P_{n_1 + 1} + \cdots + P_{n_2 - 1} + P_{n_2} \geq 2 \,;
\end{aligned}$$

and so on. Then as above, the subsequence whose last terms are $P_{n_1}, P_{n_2}, \ldots$ is divergent to ∞.

Similarly, if $\alpha = -\infty$, β finite, or if $\alpha = -\infty$ and $\beta = \infty$, we may suitably modify the proof.

Proofs of Theorems 6.5.3 and 6.5.4. First let $\sum a_n$ be absolutely convergent. Then by the general principle of convergence, given $\epsilon > 0$, there exists a $N = N(\epsilon)$ such that

$$|a_{N+1}| + |a_{N+2}| + \cdots < \epsilon \,. \tag{†}$$

Let $\sum a'_n$ be an arbitrary rearrangement of $\sum a_n$ and let the partial sums of $\sum a'_n$ and $\sum a_n$ be respectively S'_n and S_n. Since $\sum a_n$ is convergent, $S_n \to S$, say. Being a rearrangement of $\sum a_n$, we see that

$$\begin{aligned}
&\textit{The numbers } a_1, a_2, \ldots, a_N \textit{ appear in the sum } a'_1 + a'_2 + \cdots + a'_M, \\
&\qquad\qquad \textit{if } M \textit{ is large enough.}
\end{aligned} \tag{‡}$$

Then, since $a_N, a_{N-1}, \ldots, a_2, a_1$ cancel out (perhaps more of them cancel out), we have

$$\begin{aligned}
|S'_M - S_M| &\leq |a_{N+1} + a_{N+2} + \cdots| \\
&\leq |a_{N+1}| + |a_{N+2}| + \cdots \\
&< \epsilon \,,
\end{aligned}$$

if M is sufficiently large (to ensure (†) and (‡)). Since $S_M \to S$, we see that $S'_M \to S$, showing that the arbitrary rearrangement $\sum a'_n$ is convergent and converges to the same sum as $\sum a_n$ is convergent to. This proves Theorem 6.5.3 and one implication of Theorem 6.5.4.

Conversely, let $\sum a_n$ be not absolutely convergent (but of course $\sum a_n$ is convergent). Then by Theorem 6.5.5, there is a rearrangement of $\sum a_n$ which is not convergent (chose $\beta \neq \alpha$ in Theorem 6.5.5), i.e., $\sum a_n$ is not unconditionally convergent, which proves the other implication of Theorem 6.5.4.

As a very instructive example to illustrate all this, we look at the alternating harmonic series, which is convergent but not absolutely convergent. We have the following beautiful:

Theorem 6.5.7. *The rearrangement of* $\sum(-1)^n/n$ *given by*

$$\{(1+1/3+1/5+\cdots+1/(2p-1))-(1/2+1/4+\cdots+1/2q)\}+$$
$$+\{(1/(2p+1)+\cdots+1/(4p-1))-(1/2(q+1)+\cdots+1/4q)\}+\cdots$$

in which, at each stage, p positive terms are followed by q negative ones, has sum equal to $\frac{1}{2}\log(4p/q)$.

Proof. Let T_N be the sum of the first N terms of the new series, each term being a sum of terms inside the (curly) brackets above. These N brackets contain $(p+q)N$ terms of the original series. We add and subtract $1/2+1/4+\cdots+1/2p$ to the first bracket and $1/2(p+1)+1/2(p+2)+\cdots+1/4p$ to the second bracket, and so on, to get

$$\begin{aligned}
T_N &= S_{(p+q)N} \\
&= \left\{\left(1+\frac{1}{3}+\cdots+\frac{1}{2p-1}\right)-\left(\frac{1}{2}+\frac{1}{4}+\cdots+\frac{1}{2q}\right)+\right. \\
&\quad \left.+\left(\frac{1}{2}+\frac{1}{4}+\cdots+\frac{1}{2p}\right)-\left(\frac{1}{2}+\frac{1}{4}+\cdots+\frac{1}{2p}\right)\right\}+ \\
&\quad +\left\{\left(\frac{1}{2p+1}+\cdots+\frac{1}{4p-1}\right)-\left(\frac{1}{2(q+1)}+\cdots+\frac{1}{4q}\right)+\right. \\
&\quad \left.+\left(\frac{1}{2(p+1)}+\cdots+\frac{1}{4p}\right)-\left(\frac{1}{2(p+1)}+\cdots+\frac{1}{4p}\right)\right\}+\text{up to } N \text{ brackets} \\
&= \left\{\left(1+\frac{1}{2}+\cdots+\frac{1}{2p}\right)-\frac{1}{2}\left(1+\frac{1}{2}+\cdots+\frac{1}{q}\right)-\frac{1}{2}\left(1+\frac{1}{2}+\cdots+\frac{1}{p}\right)\right\} \\
&\quad +\left\{\left(\frac{1}{2p+1}+\cdots+\frac{1}{4p}\right)-\frac{1}{2}\left(\frac{1}{q+1}+\cdots+\frac{1}{2q}\right)-\right. \\
&\quad \left.-\frac{1}{2}\left(\frac{1}{p+1}+\cdots+\frac{1}{2p}\right)\right\}+\cdots+ \\
&\quad +\left\{\left(\frac{1}{2(N-1)p+1}+\cdots+\frac{1}{2Np}\right)-\frac{1}{2}\left(\frac{1}{(N-1)q}+\cdots+\frac{1}{Nq}\right)\right. \\
&\quad \left.-\frac{1}{2}\left(\frac{1}{(N-1)p}+\cdots+\frac{1}{Np}\right)\right\}.
\end{aligned}$$

But by (6) in §6.2, $1+1/2+\cdots+1/n=\log n+\gamma+\epsilon_n$, where $\epsilon_n \to 0$ as $n \to \infty$.

It follows that

$$T_N = \log 2Np + \gamma + \epsilon_{2Np} - \frac{1}{2}(\log Nq + \gamma + \epsilon_{Nq}) - \frac{1}{2}(\log Np + \gamma + \epsilon_{Np})$$
$$= \frac{1}{2}\log 4N^2p^2/Nq.Np + \epsilon_{2Np} - \frac{1}{2}\epsilon_{Nq} - \frac{1}{2}\epsilon_{Np}\ ,$$

which converges to $\frac{1}{2}\log 4p/q$, as $n \to \infty$.

We may now drop the (curly) brackets and then the partial sums of the rearranged series will not just be groups of $p+q$ terms but will have $T_{(p+q)N+1}$, $T_{(p+q)N+2}, \ldots, T_{(p+q)N+p+q} = T_{(p+q+1)N}$ as additional partial sums, but they all converge to the same limit clearly, for example

$$T_{(p+q)N+2} = T_{(p+q)N} + (\pm 1/((N-1)p+1)) + (\pm 1/((N-1)p+2))$$

and the second and the third terms tend to 0 as $N \to \infty$, giving what is required. This completes the proof of the theorem.

With conditionally convergent series, other peculiar things can happen, for example with the *Cauchy product*. We know that:

Theorem 6.5.8. *If $\sum a_n$ and $\sum b_n$ are absolutely convergent, then their Cauchy product $\sum c_n$, where $c_n = a_0b_n + a_1b_{n-1} + \cdots + a_nb_0$, is absolutely convergent and $\sum c_n = (\sum a_n)(\sum b_n)$*

This fails with conditionally convergent series, for example:

Example 6.5.9. Let

$$S = 1/\sqrt{1} - 1/\sqrt{2} + 1/\sqrt{3} - 1/\sqrt{4} + \cdots$$

which is conditionally convergent. The Cauchy product of S with itself is the series

$$1/\sqrt{1}.\sqrt{1} - (1/\sqrt{1}\sqrt{2} + 1/\sqrt{2}\sqrt{1}) + (1/\sqrt{1}.\sqrt{3} + 1/\sqrt{2}.\sqrt{2} + 1/\sqrt{3}.\sqrt{1}) - \cdots\ .$$

Hence the absolute value of the n^{th} term is equal to

$$1/\sqrt{1}.\sqrt{n} + 1/\sqrt{2}.\sqrt{n-1} + \cdots + 1/\sqrt{n}.\sqrt{1}\ ,$$

and since $\sqrt{r}.\sqrt{n-r+1} \leq (n+1)/2$ (as the geometric mean is less than or equal to the arithmetic mean), the above term is larger than or equal to $(2/(n+1)).n$, which does not tend to 0 as $n \to \infty$, so the series is not convergent.

Example 6.5.10. The Cauchy product of the two series

$$2 + 2 + 2^2 + 2^3 + \cdots + 2^n + \cdots$$

$$-1 + 1 + 1 + 1 + \cdots + 1^n + \cdots$$

is

$$-2 + 0 + 0 + 0 + \cdots + 0^n + \cdots\ .$$

This gives the example of two divergent series whose Cauchy product converges absolutely.

We have only considered what are called simple rearrangements of the alternating series and we shall continue doing so, i.e., we do not consider, for example, the rearrangement $1+1/7-1/4+1/3-1/2+\cdots$ of the alternating harmonic series. We may put this down as the following:

Definition 6.5.11. We say that a rearrangement of $\sum(-1)^{n-1}/n$ is a simple rearrangement if the subsequences of positive terms and negative terms occur in their original order.

In any convergent series, we know that the convergence is unaffected if we tamper with any number of initial terms (however, the convergence may be to a different limit (sum)). In Theorem 6.5.5, in particular, all we require is that eventually p positive terms should be followed by q negative ones. With this in mind, for a rearrangement $\sum a_n$ of the alternating harmonic series, we let $\alpha = \lim_{n\to\infty} p_n/n$ (if this limit exists), where p_n = the number of positive terms in $\{a_1, a_2, \ldots, a_n\}$. We have the following (essentially a restatement of theorem 6.5.5).

Theorem 6.5.12 (see [28]). *A simple rearrangement $\sum a_n$ of $\sum(-1)^{n-1}/n$ converges (in $\mathbb{R}\cup\{\pm\infty\}$) if and only if α, as defined above, exists. Moreover, the sum of the rearranged series is then equal to $\log 2 + \frac{1}{2}\log(\alpha/(1-\alpha))$, if $\alpha \neq 1$ and $\log 2$, if $\alpha = 1$.*

Proof. Letting q_n = the number of negative terms in $\{a_1, a_2, \ldots, a_n\}$, we see that $p_n + q_n = n$. Then

$$\sum_{k=1}^{n} a_k = \sum_{i=1}^{p_n} 1/(2i-1) - \sum_{i=1}^{q_n} 1/2i\,.$$

Let $\Gamma_n = \sum_{k=1}^{n} 1/k - \log n$, so that Γ_n is positive and decreasing and tends to γ, the Euler constant, as n tends to infinity. Now we have

$$\sum_{i=1}^{q_n} 1/2i = \frac{1}{2}\sum_{i=1}^{q_n} 1/i = \frac{1}{2}(\log q_n + \Gamma_{q_n})$$

and

$$\begin{aligned}\sum_{i=1}^{p_n} 1/(2i-1) &= \sum_{i=1}^{2p_n} 1/i - \sum_{i=1}^{p_n} 1/2i \\ &= (\log 2p_n + \Gamma_{2p_n}) - \frac{1}{2}(\log p_n + \Gamma_{p_n})\,.\end{aligned}$$

Hence

$$\begin{aligned}\lim_{n\to\infty}\sum_{k=1}^{n} a_k &= \lim_{n\to\infty}\left((\log 2p_n + \Gamma_{2p_n}) - \frac{1}{2}(\log p_n + \Gamma_{p_n}) - \frac{1}{2}(\log q_n + \Gamma_{q_n})\right)\\ &= \lim_{n\to\infty}\left(\log 2 + \frac{1}{2}\log(p_n/q_n) + \Gamma_{2p_n} - \frac{1}{2}\Gamma_{p_n} - \frac{1}{2}\Gamma_{q_n}\right)\\ &= \log 2 + \frac{1}{2}\lim_{n\to\infty}(\log(p_n/q_n)) + \gamma - \frac{1}{2}\gamma - \frac{1}{2}\gamma \quad \text{(if the limit exists)}\\ &= \log 2 + \frac{1}{2}\lim_{n\to\infty}(\log(p_n/q_n)), \quad \text{(if the limit exists)} \qquad (*)\end{aligned}$$

i.e., $\sum a_n$ converges if and only if $\lim p_n/q_n$ exists. Further, (*) is exactly Theorem 6.5.12. To this end, we have $1/\alpha = \lim_{n\to\infty} n/p_n = \lim_{n\to\infty}(p_n + q_n)/p_n = 1 + \lim_{n\to\infty} q_n/p_n$, and so $\lim_{n\to\infty} q_n/p_n = (1/\alpha) - 1 = (1-\alpha)/\alpha$, as required. This completes the proof.

For integers p, q, define

$$\begin{aligned}S_{p,q} = &\frac{1}{2} + \frac{1}{4} + \cdots + \frac{1}{2p} - \frac{1}{3} - \frac{1}{5} - \cdots - \frac{1}{2q+1} + \frac{1}{2p+2} +\\ &+ \frac{1}{2p+4} + \cdots + \frac{1}{4p} - \frac{1}{2q+3} - \frac{1}{2q+5} - \cdots - \frac{1}{4q+1} + \cdots,\end{aligned}$$

as the series with blocks of p positive terms followed by blocks of q negative terms alternatively, (a rearrangement of the alternating harmonic series), with sum $\log 2 + \frac{1}{2}\log(p/q)$. Also put

$$\begin{aligned}P_{p,q} = &\left(1+\frac{1}{2}\right)\left(1+\frac{1}{4}\right)\cdots\left(1+\frac{1}{2p}\right)\left(1-\frac{1}{3}\right)\cdots\left(1-\frac{1}{2q+1}\right)\left(1+\frac{1}{2p+2}\right)\cdot\\ &\left(1+\frac{1}{2p+4}\right)\cdots\left(1+\frac{1}{4p}\right)\left(1-\frac{1}{2q+3}\right)\left(1-\frac{1}{2q+5}\right)\cdots\left(1-\frac{1}{4q+1}\right)\cdots,\end{aligned}$$

as the product obtained by rearranging terms of the product

$$\left(1+\frac{1}{2}\right)\left(1-\frac{1}{3}\right)\left(1+\frac{1}{4}\right)\left(1-\frac{1}{5}\right)\cdots$$

correspondingly. We have the following striking

Theorem 6.5.13 (See [88]). $P_{p,q} = \sqrt{p/q}$.

Proof. Let

$$\begin{aligned}\varphi_n &= \left(1+\frac{1}{2}\right)\left(1+\frac{1}{4}\right)\left(1+\frac{1}{6}\right)\cdots\left(1+\frac{1}{2n}\right)\\ &= \frac{3}{2}\frac{5}{4}\frac{7}{6}\cdots\frac{2n+1}{2n}\\ &= \frac{(2n+1)!}{(2.4.6.\ldots.(2n))^2}\\ &= \frac{(2n+1).(2n)!}{2^{2n}.(n!)^2}\\ &= \binom{2n}{n}.\frac{2n+1}{2^{2n}}\,.\end{aligned}$$

But by Wallis' product (see §1.3), we have

$$\frac{2}{1}\frac{2}{3}\frac{4}{3}\frac{4}{5}\cdots\frac{2n}{2n-1}\frac{2n}{2n+1}\cdots \;\to\; \frac{\pi}{2}\,,$$

as $n\to\infty$, i.e.,

$$\left(\frac{2}{3}\frac{4}{5}\cdots\frac{2n}{2n+1}\right)^2(2n+1) \;\to\; \frac{\pi}{2}\,,$$

which, on taking reciprocals, gives that

$$\varphi_n = (3/2)(5/4)\cdots((2n+1)/(2n)) \simeq 2\sqrt{n}.\sqrt{(1+1/2n)}/\sqrt{\pi} \simeq 2\sqrt{n}/\sqrt{\pi}$$

and we have

$$(1-1/3)(1-1/5)\cdots(1-1/(2n+1)) = (2/3)(4/5)\cdots(2n/(2n+1)) = 1/\varphi_n\,.$$

Hence $\varphi_{pn}/\varphi_{qn} = (1+1/2)(1+1/4)\cdots(1+1/(2pn))(1-1/3)(1-1/5)\cdots(1-1/(2qn+1))$ is equal to the partial product $P_{pn,\,qn}$ (i.e., product with p_n positive terms followed by q_n negative ones). Letting $n\to\infty$, we get

$$P_{p,\,q} = \lim_{n\to\infty}\frac{\varphi_{pn}}{\varphi_{qn}} = \lim_{n\to\infty}\frac{2\sqrt{p_n}/\sqrt{\pi}}{2\sqrt{q_n}/\sqrt{\pi}} = \sqrt{p/q}\,.$$

Corollary 6.5.14. $S_{p,\,q} - S_{1,1} = \log\sqrt{(p/q)}$.

Proof. We have

$$\begin{aligned}\log P_{1,1} - S_{1,1} &= \log((1+1/2)(1-1/3)(1+1/4)(1-1/5)\cdots) -\\ &\quad -(1/2-1/3+1/4-1/5+\cdots)\\ &= (\log(1+1/2)-1/2)+(\log(1-1/3)+1/3)+\\ &\quad +(\log(1+1/4)-1/4)+\cdots\end{aligned}$$

and this series is absolutely convergent since $|\log(1-x)+x| = |x^2/2+x^3/3+\cdots| \le (|x|^2/2)(1+2|x|/3+2|x|^2/4+\cdots) < \frac{1}{2}|x|^2(1+|x|+|x|^2+\cdots) =$

$|x|^2/(2(1-|x|)) \leq |x|^2$, if $1 \leq 2(1-|x|)$, i.e., if $|x| \leq \frac{1}{2}$, which is the case for the above series. Hence the value of the series is unaltered by rearranging it and so is equal to $\log P_{p,q} - S_{p,q}$. Thus $\log P_{1,1} - S_{1,1} = \log P_{p,q} - S_{p,q}$ or $S_{p,q} - S_{1,1} = \log P_{p,q} - \log P_{1,1} = \log \sqrt{(p/q)} - \log \sqrt{1/1}$ (by Theorem 6.5.13) $= \log \sqrt{(p/q)}$, as required.

Example 6.5.15. What convergent rearrangements of the alternating harmonic series will have sum which is a rational number?

Indeed, let p_n be equal to the number of positive terms in the first n terms and let q_n be equal to the number of negative terms in the first n terms and let $p_n/q_n \to k$ as $n \to \infty$. Then by Theorem 6.5.12, the sum of this rearranged series is $\log 2 + \frac{1}{2}\log k$. If $S(= \log 2)$ is the sum of the alternating harmonic series, then the change Λ in the sum is $\frac{1}{2}\log k$. Write $k = c^2/4$ so that the sum of the rearranged series equals $\log 2 + \frac{1}{2}(2\log c - 2\log 2) = \log c$ and we want this to be a rational number m/n, say. Thus $c = e^{m/n}$ giving $k = \frac{1}{4}e^{2m/n}$.

So now take any sequence $\{a_n/b_n\}$ of rationals having limit $\frac{1}{4}e^{2m/n}$ (for example convergents of its continued fraction expansion) and take $p_n = a_n$ and $q_n = b_n$.

For example, take $k = 1/4$. Then the rearrangement is

$$1 - \frac{1}{2} - \frac{1}{4} - \frac{1}{6} - \frac{1}{8} + \frac{1}{3} - \frac{1}{10} - \frac{1}{12} - \frac{1}{14} - \frac{1}{16} + \frac{1}{5} - \cdots$$

and the sum is $\log 2 + \frac{1}{2}\log k = 0$, since $k = 1/4$.

As an example of the conditional convergence of series, we show:

Proposition 6.5.16. *The series*

$$S = 1 - \left(\frac{1}{2} + \frac{1}{3}\right) + \left(\frac{1}{4} + \frac{1}{5} + \frac{1}{6}\right) - \left(\frac{1}{7} + \frac{1}{8} + \frac{1}{9} + \frac{1}{10}\right) + \left(\frac{1}{11} + \cdots\right) - \cdots$$

is convergent.

Proof. Let u_n be the sum of blocks of consecutive terms of the same sign. Thus

$$\begin{aligned} u_1 &= 1\,, \\ u_2 &= 1/2 + 1/3\,, \\ u_3 &= 1/4 + 1/5 + 1/6\,, \\ u_4 &= 1/7 + 1/8 + 1/9 + 1/10\,, \\ &\cdots \quad \cdots \qquad \cdots\,, \end{aligned}$$

To get a formula for the n^{th} term u_n, we note that $1+2+\cdots+(n-1)$ terms of the harmonic series have preceded it, so it begins with $1/(\frac{1}{2}(n-1)n+1)$ and goes up to $1/(\frac{1}{2}(n-1)n+n)\,(= 2/(n^2+n))$, i.e.,

$$u_n = 2/(n^2-n+2) + 2/(n^2-n+3) + \cdots + 2/(n^2+n)$$

and putting $n+1$ for n we get

$$u_{n+1} = 2/(n^2+n+2) + \cdots + 2/(n^2+3n) + 2/(n^2+3n+2)\,.$$

So $0 < u_n < n(2/(n^2-n+2))$ (number of terms times the largest term $= n$ times the first term). Letting $n \to \infty$, we see that $u_n \to 0$.

Next, we estimate $u_n - u_{n+1}$. We have $u_n > n(2/(n^2+n)) = 2/(n+1)$ (number of terms times the smallest term $= n$ times the last term) and $u_{n+1} < 2(n+1)/(n^2+n+2)$ (by the above inequality, with $n+1$ for n), so $-u_{n+1} > -2(n+1)/(n^2+n+2)$. It follows that $u_n - u_{n+1} > 2/(n+1) - 2(n+1)/(n^2+n+2) = 2(-n+1)$ which is not greater than 0, so we need a better estimate. The inequality $u_n > 2/(n+1)$ is, as it is, good enough. As for u_{n+1}, we have $u_{n+1} <$ (smallest of the first n terms times the number of terms + the last term) $= n(2/n(n+3)) + 2/(n^2+3n+2)$. Hence $u_n - u_{n+1} > 2/(n+1) - 2/(n+3) - 2/(n+1)(n+2) = 2(n+1)/(n+1)(n+2)(n+3) = 2/(n+3)(n+2) > 0$. It follows that u_n is a decreasing sequence and we have seen above that $u_n \to 0$. Hence the series $\sum(-1)^{n+1}.u_n$ is convergent. But then S is also convergent, for let S_N be the partial sum of the first N terms of S, and U_n the partial sums of $u_1 - u_2 + u_3 - u_4 + \cdots$. Then $S_N = U_n + r_n$ for a suitable n and r_n is the sum of the first m terms of u_{n+1} (sign $\pm$ included), for a suitable m (for example if $N = 18$, then $S_{18} = U_5 - 1/16 - 1/17 - 1/18 = U_5 + r_5$, where r_5 is the sum of the first 3 terms of u_6). As $N \to \infty$, so does n and U_n tends to a limit U, say. Also $|r_m| < u_{m+1} \to 0$, hence $S_N \to U$.

Following [36], we shall now give estimates for the sum S of the alternating series $1 - 1/2 + 1/3 - 1/4 + \cdots$.

Grouping two terms at a time and collecting them into a fraction we get

$$\begin{aligned}
S &= (1-1/2) + (1/3-1/4) + \cdots \\
&= 1/1.2 + 1/3.4 + 1/5.6 + 1/7.8 + \cdots \\
&> 1/1.2 + 1/3.5 + 1/5.7 + 1/7.9 + \cdots \\
&= 1/1.2 + \frac{1}{2}(1/3-1/5) + \frac{1}{2}(1/5-1/7) + \frac{1}{2}(1/7-1/9) + \cdots \\
&= \frac{1}{2}(1 + 1/3 - 1/5 + 1/5 - 1/7 + 1/7 - 1/9 + \cdots) \\
&= \frac{1}{2}(1+1/3) = 2/3\,.
\end{aligned}$$

Moreover, $S = 1 - (1/2 - 1/3 + 1/4 - 1/5 + 1/6 - 1/7 + \cdots)$ and again making

a common denominator of two terms at a time, we find that

$$\begin{aligned} S &= 1 - (1/2.3 + 1/4.5 + 1/6.7 + \cdots) \\ &< 1 - (1/2.4 + 1/4.6 + 1/6.8 + \cdots) \\ &= 1 - \frac{1}{4}(1/1.2 + 1/2.3 + 1/3.4 + \cdots) \\ &= 1 - \frac{1}{4}(1 - 1/2 + 1/2 - 1/3 + 1/3 - 1/4 + \cdots) \\ &= 1 - 1/4 = 3/4\,. \end{aligned}$$

Hence

$$2/3 < S < 3/4\,.$$

Other estimates for the partial sums of this alternating harmonic series are also available.

Let $\sigma_n = 1 - 1/2 + 1/3 - 1/4 + \cdots + (-1)^{n+1}/n$. Since $\sigma_n \to \sigma = \log 2$, as $n \to \infty$, we have

$$|\sigma - \sigma_n| < \epsilon \text{ if } n \geq N\,. \qquad (*)$$

Now, given $\epsilon = 1/k$ (k a positive integer), we can calculate at what stage (*) begins to hold, i.e., the least value on N such that (*) holds. Indeed, we have

$$\begin{aligned} |\sigma - \sigma_n| &= 1/(n+1) - 1/(n+2) + 1/(n+3) - \cdots \\ &= 1/(n+1) - (1/(n+2) - 1/(n+3)) - \cdots \\ &< 1/(n+1) \end{aligned}$$

and this is less than $1/k$ as soon as $n + 1 > k$, so

$$|\sigma - \sigma_n| < 1/k \text{ if } n \geq k. \text{ (i.e., } N = k \text{ will do)} \qquad (\sharp)$$

But of course we have used very crude estimates here and by no means will k be the smallest value of N. The following result improves this value.

Theorem 6.5.17. $|\sigma - \sigma_n| < 1/2k$, if $n \geq k$.

Proof. Let $R_n = \sigma - \sigma_n$. Then

$$\begin{aligned} |R_{n-1}| &= 1/n - 1/(n+1) + 1/(n+2) - 1/(n+3) + \cdots \\ &= 1/(n(n+1)) + 1/(n+2)(n+3) + \cdots\,, \end{aligned}$$

and

$$\begin{aligned} |R_n| &= 1/(n+1) - 1/(n+2) + 1/(n+3) - 1/(n+4) + \cdots \\ &= 1/((n+1)(n+2)) + 1/((n+3)(n+4)) + \cdots\,. \end{aligned}$$

Now compare these with the series

$$\begin{aligned} 1/2n &= \frac{1}{2}(1/n - 1/(n+2) + 1/(n+2) - 1/(n+4) + 1/(n+4) - \cdots) \\ &= 1/n(n+2) + 1/(n+2)(n+4) + \cdots , \end{aligned}$$

and we at once see that $|R_{n-1}| > 1/2n > |R_n|$ (since $1/n(n+1) > 1/n(n+2) > 1/(n+1)(n+2)$ and so on for each term) and this is what is required.

There is another very simple, even closer, approximation to σ. Let

$$\begin{aligned} \sigma_n^* &= 1 - 1/2 + 1/3 - 1/4 + \cdots + (-1)^{n+1}/n + (-1)^{n+2}\frac{1}{2}(1/(n+1)) \\ &= \sigma_n + (-1)^n\frac{1}{2}(1/(n+1)) . \end{aligned}$$

We then have:

Theorem 6.5.18. $|\sigma - \sigma_n^*| < 1/2n(n+1)$.

Proof. The preceding inequality gives

$$1/2n > |R_n| > 1/2(n+1)$$

and on subtracting $1/2(n+1)$ throughout, we get

$$1/2n(n+1) > |R_n| - 1/2(n+1) > 0 .$$

Then

$$\begin{aligned} \sigma - \sigma_n^* &= \sigma - \sigma_n + \sigma_n - \sigma_n^* \\ &= R_n - (-1)^n\frac{1}{2}(1/(n+1)) \\ &= (-1)^n(|R_n| - 1/2(n+1)) . \end{aligned}$$

Hence $|\sigma - \sigma_n^*| < 1/2n(n+1)$.

Thus, for example, if $2k = 1000$, then by (♯), $|\sigma - \sigma_n| < 1/1000$, if $n \geq 1000$, whereas, according to Theorem 6.5.17, $|\sigma - \sigma_n| < 1/1000$, if $n \geq 500$. Also, $|\sigma - \sigma_n^*| < 1/1000$, when $n \geq 22$, by Theorem 6.5.18.

Exercises

6.1. True or false? Justify

(i) $\sum a_n^4$ convergent $\Rightarrow \sum a_n^5$ convergent.

(ii) $a_n \geq 0$, $\sum a_n$ convergent $\Rightarrow \sum a_n^5$ convergent.

(iii) $\sum a_n$ convergent $\Rightarrow \sum |a_n|/n$ convergent.

(iv) The sequence $\{a_n\}$ converges if for each $r \geq 2$, $\{a_{r^n}\}$ converges.

(v) The sequence $\{a_n\}$ converges if there exists a k, $0 \leq k < 1$, such that $|a_{n+1}/a_n| \leq k$ for all sufficiently large n.

(vi) The sequence $\{a_n\}$ converges if a_n's are bounded and $a_{n+1} - a_n \to 0$ as $n \to \infty$.

(vii) The series $\sum a_n$ is convergent, $a_n \neq 0, 1$, then $\sum a_n^2$ is convergent.

(viii) The series $\sum a_n$ is convergent, $a_n \neq 0, 1$, then $\sum a_n/1 - a_n$ is convergent.

(ix) The series $\sum a_n$ is convergent, $a_n \neq 0, 1$. If $a_n > 0$ for all n, then $\sum a_n^2$ is convergent.

(x) The series $\sum a_n$ is convergent, $a_n \neq 0, 1$. If $a_n > 0$ for all n, then $\sum a_n/(1 - a_n)$ is convergent.

6.2. Let $\{a_n\}$, $\{b_n\}$ be sequences such that $a_n \neq 0$, $\sum a_n$ is convergent and $b_n/a_n \to 1$ as $n \to \infty$. Prove that if $a_n > 0$ for all n, then $\sum b_n$ is convergent. Give an example to show that $\sum b_n$ need not converge if the restriction $a_n > 0$ is removed.

6.3. Let $\{a_n\}$ be a decreasing sequence of positive numbers. Using the general principle of convergence, or otherwise, prove that if there is a positive number k such that $a_n \geq k/n$ for infinitely many n, then $\sum a_n$ diverges.

6.4. $\sum(c_n + id_n)$ is convergent and all the c_n's , d_n's are greater than or equal to 0. Let S_m be the point in the complex plane representing $\sum(c_n + id_n)$. Show that the length of the polygonal path $S_1S_2 \ldots S_N$ remains bounded as $N \to \infty$. Does the result remain true if the condition $c_n, d_n \geq 0$ is replaced by $c_n + d_n \geq 0$? Justify your answer.

6.5. If $d_n \geq 1$ for all n and the sequence $\{d_n + d_n^{-1}\}$ converges, prove that the sequence $\{d_n\}$ converges.

Produce an example to show that this result is no longer true if the condition $d_n \geq 1$ is replaced by $d_n \geq k$, where $k < 1$.

6.6. $\sum a_n$ is convergent and $a_n > 0$. The numbers P_n are defined by

$$P_n = (1 + a_1)(1 + a_2) \cdots (1 + a_n) .$$

By using the inequality $1 + x < e^x$ $(x > 0)$, or otherwise, prove that P_n tends to a limit as $n \to \infty$. Hence or otherwise prove that if $R_n = \sum_{i=n+1}^{\infty} a_i$, then $\sum_{n=1}^{\infty}(a_n/R_n)$ is divergent.

Chapter 7

The infinite exponential $x^{x^{x^{\cdot^{\cdot^{\cdot}}}}}$ and Related Topics

The object of this chapter is to answer the following attractive and pertinent questions:

Q1. When is $x^y = y^x$ (x, y positive real numbers)?

Q2. More generally, when is $x^y \leq y^x$ (x, y positive real numbers)?

Q3. Is there an explicit formula for y as a function of x, that gives the solutions of the equation $x^y = y^x$ (this is where the title of the chapter comes in)?

§7.1. The equation $x^y = y^x$ and the parameterization

The first question is most interesting and has attracted attention for some 250 years; from the time of Bernoulli (1728), who was looking for integral solutions of $x^y = y^x$, to the work of Maher and Breuch (1963) and of Sato (1972), who characterized the algebraic solutions of $x^y = y^x$. For an exhaustive set of references and an excellent account of a part of the material of this chapter, see [58].

Let us begin by looking at the graph of the function

$$f(x, y) = x^y - y^x = 0 \,,$$

so that the solutions of $f = 0$ are precisely the ones to satisfy $x^y = y^x$. Moreover, by plotting $f = 0$, we get the various regions in which the sign of f is maintained, thus answering the second question above. Taking xy^{th} roots, we need only consider the equation

$$x^{1/x} = y^{1/y} \,.$$

To get the solutions of this equation, we first plot the graph of the auxiliary function $\phi(x) = x^{1/x}$. One easily checks the following features of this function:

1. ϕ has a maximum equal to $e^{1/e}$ at $x = e$,

2. $\phi(x) \to 0$ as $x \to 0$, $\phi(x) \to 1$ as $x \to \infty$,

3. $\phi(x)$ increases in $(0, e]$ and decreases in $[e, \infty)$ ($d\phi/dx > 0$ in $(0, e)$ but is less than 0 in (e, ∞)).

We may thus plot the graph of $\phi(x) = x^{1/x}$ (see Figure 7.1).

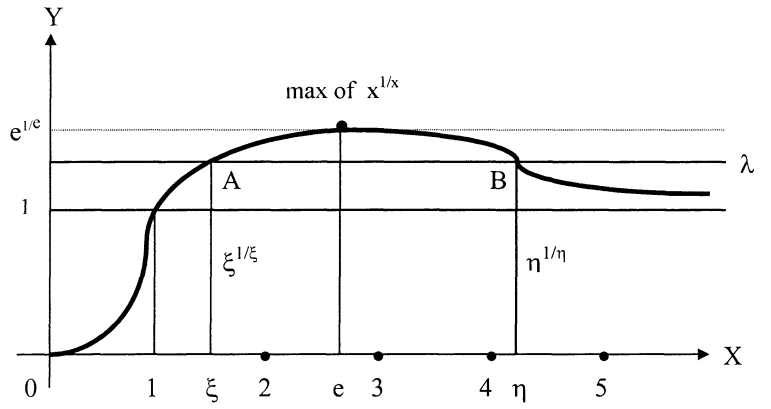


Figure 7.1

Now to determine a pair (x, y), $x \neq y$ such that $x^y = y^x$, i.e., $x^{1/x} = y^{1/y}$, all we need to do is to take a line λ parallel to the x-axis to meet the graph of ϕ in points A, B, say, with $A = (\xi, \xi^{1/\xi})$, $B = (\eta, \eta^{1/\eta})$. But $\xi^{1/\xi} = \eta^{1/\eta}$, since λ is parallel to the x-axis. So we have determined ξ, η such that $\xi^{1/\xi} = \eta^{1/\eta}$, as required. From the graph of ϕ, it is now clear that if

(i) λ is above or at $e^{1/e}$ (the maximum of ϕ), no such pair (ξ, η), $\xi \neq \eta$ exists.

(ii) λ is below or at 1, again no such pair (ξ, η), $\xi \neq \eta$ exists.

(iii) λ is strictly between 1 and $e^{1/e}$, then for each $\xi \in (1, e)$, there exists a unique $\eta \in (e, \infty)$ such that $\xi^{1/\xi} = \eta^{1/\eta}$ (i.e. $\xi^\eta = \eta^\xi$).

(iv) As $\xi \to 1$, $\eta \to \infty$ and vice versa and as $\xi \to e$, $\eta \to e$.

It follows that the graph of $x^y = y^x$, i.e., the set $\{(x, y) \mid x^y = y^x, x, y \in \mathbb{R}\}$ consists of the following:

(1) The line $x = y$, the branch of equal points called the *trivial branch.*

(2) The points (ξ, η) determined above.

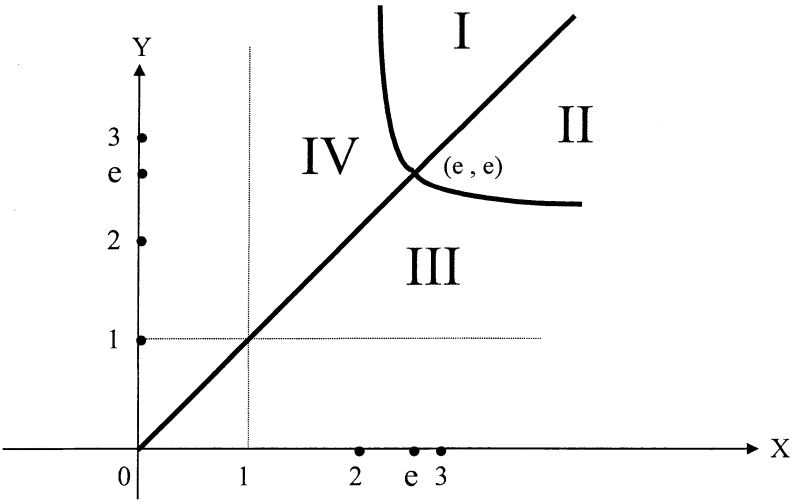


Figure 7.2

By (iv) above, this graph is as shown in Figure 7.2. These two branches divide the first quadrant into 4 parts denoted by I, II, III, IV, shown in the figure. In IV and II, $f(x,y) = x^y - y^x < 0$, while in I and III, $f(x,y) > 0$. To see this we merely have to check on any one (suitable) point in each part, for example in part IV, we may take the point $(1,2)$ and get $f(1,2) = 1^2 - 2^1 = -1 < 0$ and so $f(x,y) < 0$ for all points of IV. Similarly in I, we take the point $(3,4)$ and get $f(3,4) = 17 > 0$, and so $f(x,y) > 0$ for all points of I, etc.

Example 7.1.1. Which of π^e and e^π is bigger?

Indeed, the line $y = e$ manages to always remain in parts IV and II, only crossing the point (e,e) on the boundary. It follows that for any point (x,e) on this line $(x \neq e)$, $x^y - y^x < 0$, i.e., $x^e - e^x < 0$ for all $x \neq e$, $0 < x < \infty$, which itself is an interesting inequality. Taking $x = \pi$, we get $\pi^e < e^\pi$.

Although, technically all the required pairs (x,y) for which $x^y = y^x$ (and even those for which $x^y < y^x$ or $x^y > y^x$) are now available, taking any value of x $(1 < x)$ the unique corresponding y for which $x^y = y^x$ is still not explicitly available; all we have is that given $x = \xi$, the corresponding $y = \eta$ is given by $\xi^y = y^\xi$. So how does one get y as a function of ξ?

The nontrivial branch (which is what matters) can be *parameterized* as

$$x = s^{1/(s-1)} \ , \ y = s^{s/(s-1)} \ , \ \text{where } s \in \mathbb{R}^+ \ .$$

Observe that $s = y/x$ and $x^s = y$. So $y^x = x^{sx} = x^y$. Thus for each unequal pair (ξ,η), there exists a unique $\sigma \in \mathbb{R}^+$, such that $\xi = \sigma^{1/(\sigma-1)}$, $\eta = \sigma^{\sigma/(\sigma-1)}$. In fact we have the following :

Theorem 7.1.2. *Let ξ, η, σ be as above, i.e., $\xi = \sigma^{1/(\sigma-1)}$, $\eta = \sigma^{\sigma/(\sigma-1)}$; then*

(i) $\xi^\eta = \eta^\xi$,

(ii) $\sigma = \eta/\xi$,

(iii) (Carlini-1889) $\xi.\sigma = \xi^{\sigma}$.

Proof. (i) We have to prove that

$$(\sigma^{1/(\sigma-1)})^{\sigma^{\sigma/(\sigma-1)}} = (\sigma^{\sigma/(\sigma-1)})^{\sigma^{1/(\sigma-1)}}$$

or, taking logarithms, that $(\sigma^{\sigma/(\sigma-1)})(1/(\sigma-1))\log\sigma = \sigma^{1/(\sigma-1)}(\sigma/(\sigma-1))\log\sigma$. Cancelling $(\log\sigma)/(\sigma-1)$, it remains to show that $\sigma^{\sigma/(\sigma-1)} = \sigma.\sigma^{1/(\sigma-1)}$, which is easily checked.
(ii) $\eta/\xi = \sigma^{\sigma/(\sigma-1)}/\sigma^{1/(\sigma-1)} = \sigma^{(\sigma/(\sigma-1)-1/(\sigma-1))} = \sigma^{(\sigma-1)/(\sigma-1)} = \sigma$.
(iii) We must show that $(\sigma^{1/(\sigma-1)}).\sigma = (\sigma^{1/(\sigma-1)})^{\sigma}$. Here, the left hand side is equal to $\sigma^{(1/(\sigma-1)+1)} = \sigma^{\sigma/(\sigma-1)}$, while the right hand side is equal to $\sigma^{(1/(\sigma-1)).\sigma} = \sigma^{\sigma/(\sigma-1)}$, too.

Observe that (iii) gives us the solution of when the product and power functions are equal.

With the above parameterization, we can produce lots of simple solutions (ξ,η) of $\xi^{\eta} = \eta^{\xi}$: Take $s = 1/2, 2/3, 3/2, 2, 3, \ldots$, which give respectively

$$\xi = 4, 27/8, 9/4, 2, \sqrt{3}, \ldots \text{ and } \eta = 2, 9/4, 27/8, 4, 3\sqrt{3}, \ldots .$$

Thus, for example $(\sqrt{3})^{3\sqrt{3}} = (3\sqrt{3})^{\sqrt{3}}$. Note that if s gives the pair (ξ,η), then $1/s$ gives the pair (η,ξ).

For $s = 1$, there is an apparent singularity, but it is only apparent, for if we take the limit as $s \to 1$ and substitute $s = t + 1$, we get the usual expression for e, viz.

$$\lim_{t\to 0}(1+t)^{1/t} = x = e = y = \lim_{t\to 0}(1+t)^{(1+1/t)} .$$

This parameterization enables us to characterize some other solutions of significance, for example we shall prove the following results:

1. The only integer solutions of $x^y = y^x$ are $(\xi,\eta) = (2,4)$ and $(4,2)$.
2. Solutions (ξ,η) of $x^y = y^x$, with $\xi,\eta \in \mathbb{Q}$ are given by $s = m/(m+1)$ or $(n+1)/n$ $(m,n \in \mathbb{N})$.
3. Solutions (ξ,η) of $x^y = y^x$, with ξ,η algebraic numbers are given by $s = m/n \in \mathbb{Q}^+$.
4. Solutions (ξ,η) of $x^y = y^x$, with ξ,η algebraic integers (*a real number α is called an algebraic integer if α satisfies a monic polynomial with rational integral coefficients*) are given by $s = m$ or $1/n$ $(m,n \in \mathbb{N})$.

Theorem 7.1.3. *The only pair (m,n) that satisfies $x^y = y^x$, with $m,n \in \mathbb{N}$, $m > n$, is* $(4,2)$.

First Proof. Write $m = n + a$, $a \in \mathbb{N}$. Substituting in the equation gives

$$(n+a)^n = n^{(n+a)} \quad \text{or} \quad (1+a/n)^n = n^a . \qquad (*)$$

Here, the left hand side is equal to

$$
\begin{aligned}
&1 + \frac{n.a}{n} + \frac{n(n-1)}{2!}\frac{a^2}{n^2} + \cdots + \frac{n(n-1)\ldots(n-r+1)}{r!}\frac{a^r}{n^r} + \cdots + \\
&\qquad + \frac{n(n-1)\ldots(n-(n-1))}{n!}\frac{a^n}{n^n} \\
&= 1 + a + \frac{a^2}{2!}\left(1-\frac{1}{n}\right) + \frac{a^3}{3!}\left(1-\frac{1}{n}\right)\left(1-\frac{2}{n}\right) + \cdots + \\
&+ \frac{a^r}{r!}\left(1-\frac{1}{n}\right)\left(1-\frac{2}{n}\right)\ldots\left(1-\frac{r-1}{n}\right) + \cdots + \frac{a^n}{n!}\left(1-\frac{1}{n}\right)\ldots\left(1-\frac{n-1}{n}\right) \\
&< 1 + a + \frac{a^2}{2!} + \cdots + \frac{a^r}{r!} + \cdots \frac{a^n}{n!} \\
&< e^a .
\end{aligned}
$$

Now (*) gives $n^a < e^a$ or $n < e$. It follows that $n = 1, 2$. Here $n = 1$ gives $m^1 = 1^m$, i.e., $m = 1$, giving the trivial solution $(1, 1)$; while $n = 2$ gives $m^2 = 2^m$, i.e., $m = 4$ (since for $m > 4$ we get $2^m > m^2$).

Second Proof. Write (x, y) for a solution, so that $x > y > 0$. Write $x = y + a$ $(a \in \mathbb{N})$. Then $x^y = y^x$ implies that $(y + a)^y = y^{(y+a)}$.

Factor $y = p_1^{\alpha_1} p_2^{\alpha_2} \ldots p_s^{\alpha_s}$ as a product of prime powers. Then

$$(y + a)^y = y^a . (p_1^{\alpha_1} p_2^{\alpha_2} \ldots p_s^{\alpha_s})^y$$

and so $p_i^{\alpha_i y} | (y + a)^y$, i.e., $p_i^{\alpha_i} | y + a$ and so $y | y + a$, giving $y | a$, say $a = ky$. Then $x = y + a = y + ky$ $(k \geq 1) = y(1 + k) = yK$, say, $K \geq 2$, and so $x^y = y^x$ implies $(Ky)^y = y^{Ky}$ implies that $(Ky)^y = (y^K)^y$ and hence $Ky = y^K$, so that

$$K = y^{K-1} \quad (K \geq 2) . \tag{♯}$$

$K = 2$ gives $2 = y$ and so $x = 2y = 4$.

Now, let $K > 2$. Write $y = 1 + t$, where $t \geq 1$ (as $t = 0$ implies $y = 1$, giving the trivial solution). Then by (♯),

$$
\begin{aligned}
K = (1 + t)^{K-1} &= 1 + (K-1)t + ((K-1)(K-2)/2!).t^2 + \cdots + t^{K-1} \\
&> 1 + (K-1)t \\
&\geq 1 + (K-1) \qquad (\text{since } t \geq 1) \\
&= K ,
\end{aligned}
$$

i.e., $K > K$, which is a contradiction. So $K > 2$ does not give any solution.

We now consider the rational solutions of the equation $x^y = y^x$, which we write as (see [104])

$$(\log x)/x = (\log y)/y . \tag{*}'$$

If $y = \log x$, $dy/dx = 1/x \to 0$ as $x \to \infty$.

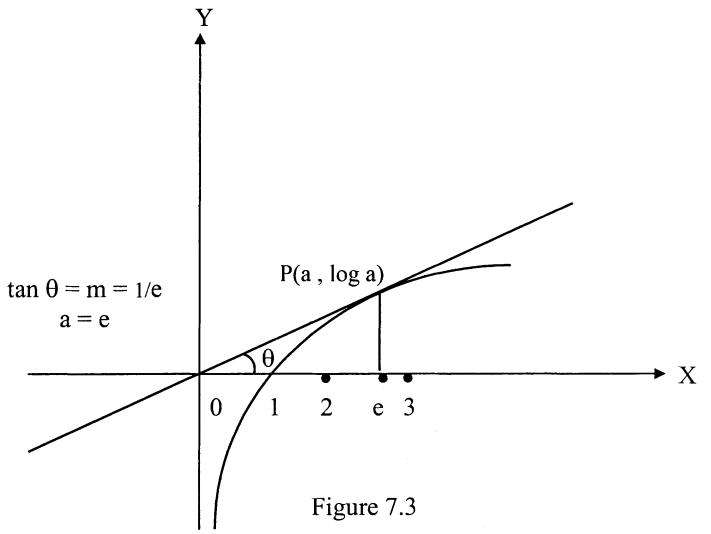


Figure 7.3

So the graph of $y = \log x$ becomes parallel to the x-axis as $x \to \infty$. First we determine m for which the line $y = mx$ is tangent to the curve $y = \log x$. Suppose the tangent at $P \equiv (a, \log a)$ to $y = \log x$ goes through $(0,0)$. Then $m = (dy/dx)_P = 1/a$. So the tangent is $y = x/a$ and (see Figure 7.3) since P lies on it, we have $\log a = (1/a).a = 1$, giving $a = e$, i.e., $m = 1/e$. Hence, we have essentially proved the following:

Lemma 7.1.4. *The line $y = mx$ meets the curve $y = \log x$ in no real point, one real point, two real points, according as m satisfies*

$$1/e < m < \infty \ ; \ m = 1/e \ \text{ or } \ m < 0 \ , \ 0 < m < 1/e \, .$$

Now take $0 < m < 1/e$ and let $\{y = \log x\} \cap \{y = mx\} = \{x_1, x_2\}$. Then

$$m = (\log x_1)/x_1 = (\log x_2)/x_2 \qquad (*)''$$

and we simply have to determine all the rational solutions of $(*)''$.

Let x_1, x_2 be rational solutions and let $x_2/x_1 = r \in \mathbb{Q}$. We let $x_1 < x_2$ as in Figure 7.4, so that $r > 1$ and we write $r = s + 1$ (s rational and positive). Substituting in $(*)''$ gives $(\log x_1)/x_1 = (\log rx_1)/rx_1$, which on simplification, gives $x_1^{(r-1)/r} = r^{1/r}$, i.e., $x_1 = r^{1/(r-1)}$ and so $x_2 = r^{r/(r-1)}$, i.e., $x_1 = (1+s)^{1/s}$, $x_2 = (1+s)^{(1+s)/s}$. Write $s = a/b$ ($a, b \in \mathbb{N}$, (a,b)=1). Then

$$x_1 = ((a+b)/b)^{b/a} \ , \ x_2 = ((a+b)/b)^{(a+b)/a}$$

and it is easy to see, since $(a, b) = 1$, that both $(a+b)^{1/a}$ and $b^{1/a}$ must be integers: say $(a+b)^{1/a} = n_1$, $b^{1/a} = n_2$, so that $n_1 > n_2$, say $n_1 = n_2 + t$ (t positive integer).

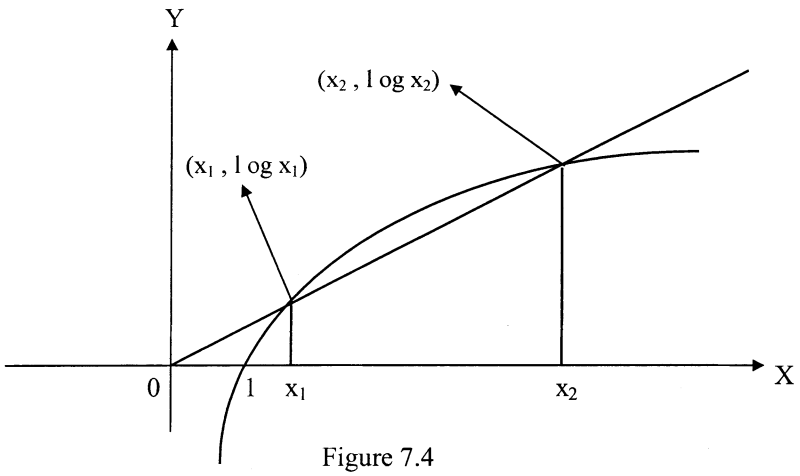


Figure 7.4

Then

$$x_1 = (n_1/n_2)^b \ , \ x_2 = (n_1/n_2)^{(a+b)}$$

so that $(n_2+t)^a = n_1^a = a+b$ and $n_2^a = b$. Subtracting, we get $(n_2+t)^a - n_2^a = a$, i.e.,

$$a.n_2^{a-1}t + (a(a-1)/2!).n_2^{a-2}t^2 + \cdots + t^a = a \ .$$

Here, n_2, t are positive integers and so, unless $a = 1$ (and then $n_1 = 1+b, n_2 = b, t = 1$), the left hand side is clearly greater than a, i.e., $a > a$, which is impossible. It follows that $a = 1$ and hence $s = 1/b$ $(b = 1, 2, \ldots)$. Taking $b = 1, 2, \ldots$ successively, we get respectively

$$x_1 = \left(\frac{1+b}{b}\right)^b = 2, \left(\frac{3}{2}\right)^2, \left(\frac{4}{3}\right)^3, \ldots \text{which is an increasing sequence,} \quad (**)$$

and

$$x_2 = \left(\frac{1+b}{b}\right)^{1+b} = 4, \left(\frac{3}{2}\right)^3, \left(\frac{4}{3}\right)^4, \ldots \text{which is a decreasing sequence,} \quad (**)'$$

and as m increases to $1/e$, x_1 increases to e, while x_2 decreases to e. However, if $m < (\log 2)/2$, the corresponding x_1 is less than 2. But we have seen that 2 is the smallest value of x_1 for which x_1, x_2 are both rational. It follows that if $0 < m < (\log 2)/2$, at least one of the corresponding x_1, x_2 is not rational. This gives the following surprise (see Figure 7.5):

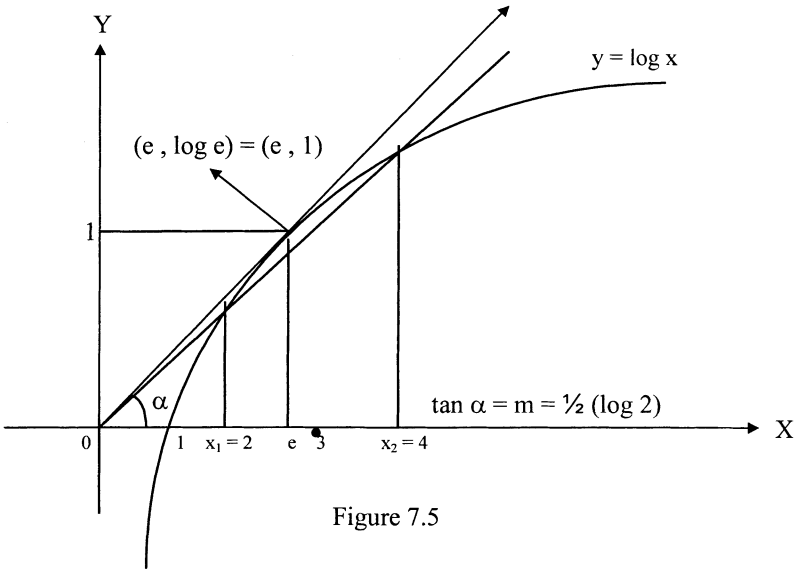


Figure 7.5

Theorem 7.1.5 *If* $(\log 2)/2 \leq m < 1/e$, *the line* $y = mx$ *meets the graph of* $y = \log x$ *in* x_1, x_2 $(x_1 < x_2)$ *and all possible such rational pairs* (x_1, x_2) *are given by* $(**)$ *and* $(**)'$*above. If, however,* $0 < m < (\log 2)/2$, *at least one of the corresponding* x_1, x_2 *is not rational.*

Remark 7.1.6. Observe that $(**)$ and $(**)'$ give the result (2) of the four results stated above Theorem 7.1.3, as may be easily checked. Remember that result (2) gives the full set (ξ, η) and (η, ξ) of solutions, whereas $(**)$ and $(**)'$ gives only one pair.

Theorem 7.1.7. *Solutions of* $x^y = y^x$ $(0 < x < y)$ *with* x, y *real algebraic numbers are parameterized by*

$$x = s^{s/(s-1)} \quad , \quad y = s^{1/(s-1)} \qquad (*)'$$

where $s \in \mathbb{Q} \cap (0, 1)$. *The solutions with* x, y *real algebraic integers are parameterized by*

$$x = (k+1)^{1/k} \quad , \quad y = (k+1)^{1+1/k} \quad , \quad k = 1, 2, \ldots \; . \qquad (*)''$$

Proof. It is easy to check that the x, y given in $(*)'$ are algebraic numbers where as those in $(*)''$ are algebraic integers and that either set satisfies the equation $x^y = y^x$. In fact this has already been done for the set $(*)'$. For the set $(*)''$, we have

$$x^y = \left((k+1)^{1/k}\right)^{(k+1)^{(k+1)/k}} = (k+1)^{(((k+1)^{(k+1)/k})/k)}$$

and

$$y^x = \left((k+1)^{(k+1)/k}\right)^{(k+1)^{1/k}} = (k+1)^{(((k+1)^{1+1/k})/k)} \; ,$$

which is as required. Note also that the condition $0 < x < y$ entails $s \in (0,1)$, for $x/y = s^{s/(s-1)-1/(s-1)} = s$ and so $s < 1$ if and only if $x < y$; and $0 < x$ implies $0 < s$. Conversely, let $0 < x < y$, where x, y are real algebraic numbers and satisfy $x^y = y^x$. Write $s = x/y$ and let $u = s/(s-1) = x/(x-y)$ (check), $v = 1/(s-1) = y/(x-y)$ (check), so that s, u, v are all real algebraic numbers. We are to show that s is rational (note that with this value of s (viz. x/y), $s^{s/(s-1)} = x$ and $s^{1/(s-1)} = y$, as before).

Suppose s is irrational. Then u, v would be irrational too (if, for example, u is rational, i.e., $s/(s-1) = a/b$, then $sb = sa - a$, i.e., $s = a/(a-b) \in \mathbb{Q}$). But $s^u = x, s^v = y$ (check: $s^v = (x/y)^{y/(x-y)} = y$ since $(x/y)^y = y^{x-y}$ as $x^y = y^x$) and so by the Gelfond-Schneider Theorem (Chapter 1), x, y would be transcendental, which is a contradiction as we are supposing (x, y) to be an algebraic solution. So s is rational, say $s = m/n$, $0 < m < n$ (since $0 < s < 1$), as required.

Next, let x be an algebraic integer; then so is x^{n-m}. But $x^{n-m} = (s^{\frac{s}{s-1}})^{n-m}$ $= (m/n)^{(m/n).(n-m)/((m-n)/n)} = (m/n)^{-m} = n^m/m^m$, with $(n^m, m^m) = 1$. It follows that $m = 1$ and since $0 < m < n$, we see that $2 \leq n$. We then get $x = s^{s/(s-1)} = (1/n)^{(1/n)/(1/n-1)} = n^{(1/(n-1))}$ (since $s = m/n = 1/n$), and similarly $y = n^{n/(n-1)}$ $(n = 2, 3, \ldots)$. Letting $k = n - 1$ (so that $k = 1, 2, \ldots$) this gives

$$x = (k+1)^{1/k} \ , \ y = (k+1)^{(k+1)/k} \ , \ k = 1, 2, 3, \ldots \ .$$

That completes the proof of statements (3) and (4) stated above Theorem 7.1.3.

The above proof is due to D.Sato [100].

Remark 7.1.8. Questions (3) and (4) were parts of a problem posed by K.Mahler in 1963 (see [64]) and solved by many (see [60]). The result may also be written as follows:

Suppose m, n are algebraic numbers satisfying $m^n = n^m$ with $mn(n-1)(m-1)(m-n) \neq 0$ (i.e., $m, n \neq 0, 1, m \neq n$). Then there exist integers h, k with $h, k \neq 0, h \neq k$, $(h, k) = 1$, such that $m = (h/k)^{k/(h-k)}$, $n = (k/h)^{h/(k-h)}$. If further, m, n are algebraic integers, then h or k equals 1.

Example 7.1.9. We show that $m^{(n^m)} = n^{(m^n)}$ has no solution in positive integers m, n with $m \neq n$.

Indeed, without loss of generality, let $n > m \geq 1$.

Case 1: $m = 1$. Then our equation becomes $1 = n^n$, so $n = 1 = m$, as required.

Case 2: $m = 2$, $n = 3$ or 4. Our equation becomes $2^{(n^2)} = n^{(2^n)}$.

For $n = 3$, the left hand side equals 2^9 while the right hand side equals 3^8, so the two can not be equal. For $n = 4$, the left hand side equals 2^{16} while the right hand side equals 4^{16}, and again the two sides are different.

Case 3: $m = 2$, $n \geq 5$. Now the function $f(x) = x^{1/x}$ is decreasing for $x > e$ and so $4^{1/4} > n^{1/n}$ (since $n \geq 5 > e$), i.e., $4^n > n^4$, or taking square roots, $2^n > n^2$ and so, a fortiori, $2^n(\log n) > n^2(\log 2)$, or $n^{(2^n)} > 2^{(n^2)}$ and since $m = 2$, this gives $n^{(m^n)} > m^{(n^m)}$.

Case 4: $n > m \geq 3$. Again, since $x^{1/x}$ decreases, $m^{1/m} > n^{1/n}$ or $m^n > n^m$ or $m^n(\log n) > n^m(\log m)$ (a fortiori) or $n^{(m^n)} > m^{(n^m)}$.

§7.2. The infinite exponential $x^{x^{x^{\cdot^{\cdot^{\cdot}}}}}$

We have already mentioned that given ξ, the corresponding η satisfying the equation $\xi^\eta = \eta^\xi$, is not readily available, even after the parameterization $x = s^{1/(s-1)}$, $y = s^{s/(s-1)}$, where $s \in \mathbb{R}^+$. For example, if say $\xi = \sqrt{2}$ (so that $1 < \xi < 2$), trying to get s from the equation $\sqrt{2} = s^{1/(s-1)}$ is not possible and without this s, the corresponding $\eta = s^{s/(s-1)}$ is not available. Thus we really have to have something even better than the above parameterization. This is exactly where the beautiful "function"

$$y = h(x) = x^{x^{x^{\cdot^{\cdot^{\cdot}}}}}, \text{ i.e., } y \text{ equal to the limit of the sequence } x, x^x, x^{x^x}, \ldots, \quad (*)$$

comes into the picture. Before we consider the connection of this function with that of $f(x,y) = x^y - y^x$, we must determine the values of x for which $h(x)$ is meaningful. Thus, for example, if $x = 2$, the sequence $2, 2^2, 2^{2^2}, \ldots$ is obviously divergent and it would seem that for $x > 1$, the sequence would diverge. However, it surprises most to learn that there are values of x greater than 1, for which $h(x)$ converges. As it turns out:

Theorem 7.2.1. *The infinite exponential, as defined above, converges for all values of x in the range $[e^{-e}, e^{1/e}]$, i.e., for all x satisfying*

$$.06598803585\ldots = e^{-e} \leq x \leq e^{1/e} = 1.444667861\ldots .$$

We first give a graphical motivation of this result (see D.F.Barrow [8] or M.C.Mitchelmore [68]).

Graphical proof (motivation) of Theorem 7.2.1. Let $\zeta_1 = a$, $\zeta_2 = a^a$, $\zeta_3 = a^{a^a} = a^{\zeta_2}, \ldots, \zeta_n = a^{\zeta_{n-1}}$ and ask for what values of a, the limit, as n tends to infinity, of ζ_n exists. The trick is to consider the graph Γ of the function $y = a^x$, which, in spite of having a more or less uniform shape, varies a great deal, for different values of a, in its orientation and its relative position to the graph G of the line $y = x$, and it is this fact that will be seen to account for convergence or divergence of the sequence $\{\zeta_n\}$ for various values of a. We need to split different cases:
Case 1: $1 < a < e^{1/e}$. To fix our ideas, let us take $a = \sqrt{2} = 1.4142\ldots$. The graph Γ and G of $y = (\sqrt{2})^x$ and $y = x$ respectively are shown in Figure 7.6. We see that the set $\Gamma \cap G = \{(2,2),(4,4)\}$ and the points $Q_1 = (\sqrt{2}, \sqrt{2}^{\sqrt{2}})$, $Q_2 = (\sqrt{2}^{\sqrt{2}}, \sqrt{2}^{\sqrt{2}^{\sqrt{2}}}), \ldots$ can be successively got by chasing the zigzag shown in Figure 7.6, starting with $a = \sqrt{2}$ on the x-axis and going up to Q_1 on Γ,

then to P_1 on G, onto Q_2 on Γ, and so on. It is clear from the graph that $Q_i \to (2,2)$, showing that $\sqrt{2}^{\sqrt{2}^{\sqrt{2}^{\cdot^{\cdot^{\cdot}}}}} \to 2$.

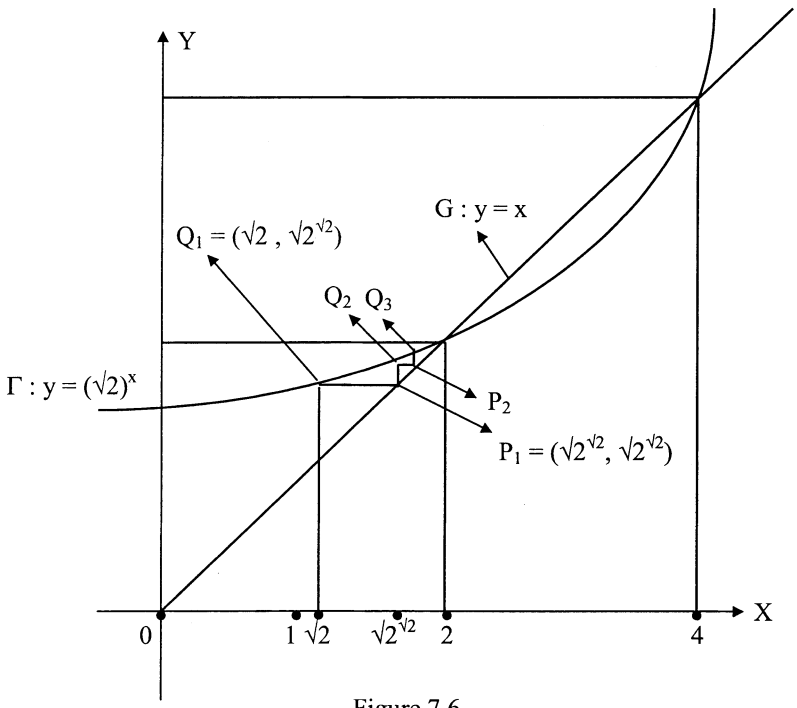


Figure 7.6

However, there is a second point of intersection of Γ and G, viz. $(4,4)$, but clearly the process stops at the point with smaller coordinates (we shall prove the existence of this limit analytically as well; however, the graphical "proof" is far more revealing, illuminating and instructive).

Case 2: $a = e^{1/e}$. This zigzag process described above fails if Γ and G do not intersect in real points. As a increases, a time will come when G is tangent to Γ. The critical value, when this happens, is $a = e^{1/e}$. So let us draw the graph Γ of $y = (e^{1/e})^x$ (see Figure 7.7).

The point $(e,e) \in \Gamma$. Let us determine the tangent to Γ at (e,e). We have $y = e^{x/e}$ and so $\log y = x/e$, giving $dy/dx = y/e = e^{x/e}/e$. At the point (e,e), the value of dy/dx is equal to $e/e = 1$. The equation of the required tangent is therefore $y = x + c$. Since this goes through (e,e), $c = 1$ and so the tangent is the line $y = x$, as required.

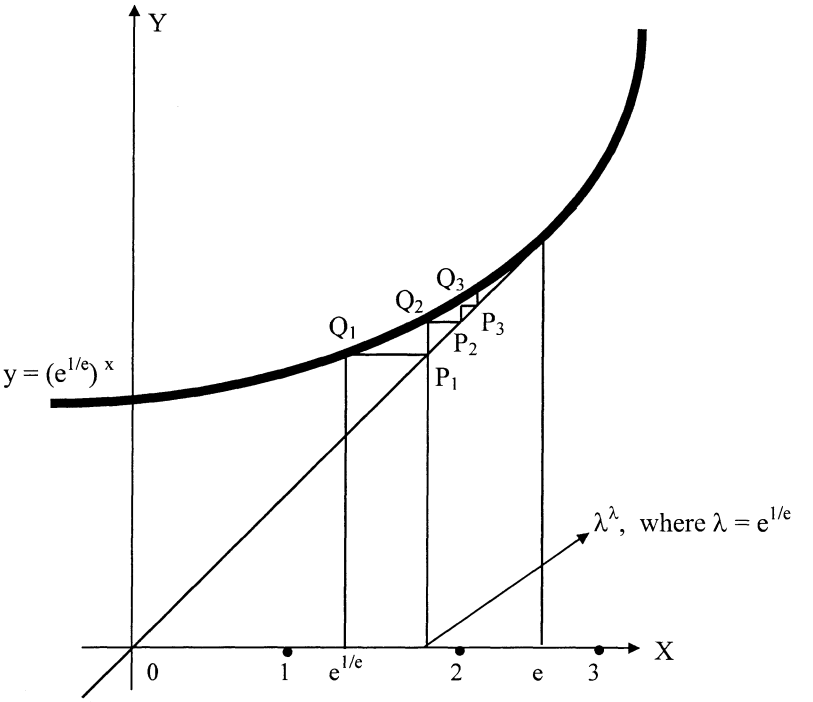


Figure 7.7

Now, taking $x = e^{1/e}$ on the x-axis, we see that the zigzag of points $Q_1, Q_2, \ldots \to (e, e)$, in particular,

$$(e^{1/e})(e^{1/e})^{\cdot^{\cdot}} \to e\,.$$

Case 3: $a > e^{1/e}$. The curve $\Gamma : y = a^x$ has graph shown in Figure 7.8, and it is obvious that for such values of a, the infinite exponential $a^{a^{\cdot^{\cdot}}}$ diverges. *Case 4:* $1 < a < e^{1/e}$. Here, exactly as for the case $a = \sqrt{2}$ in case 1, the sequence

$$a^{a^{\cdot^{\cdot}}}$$

converges. There will, of course, be 2 points of intersection of Γ and G, since $a^x \to \infty$ faster than x (since $a > 1$) and at $x = 0, y = 1$, so Γ must cross G twice. The graph of various a, $1 \leq a \leq e^{1/e}$ are shown in Figure 7.9 and $a^{a^{\cdot^{\cdot}}}$ converges to the smaller of the x-coordinates of the two points of intersection.

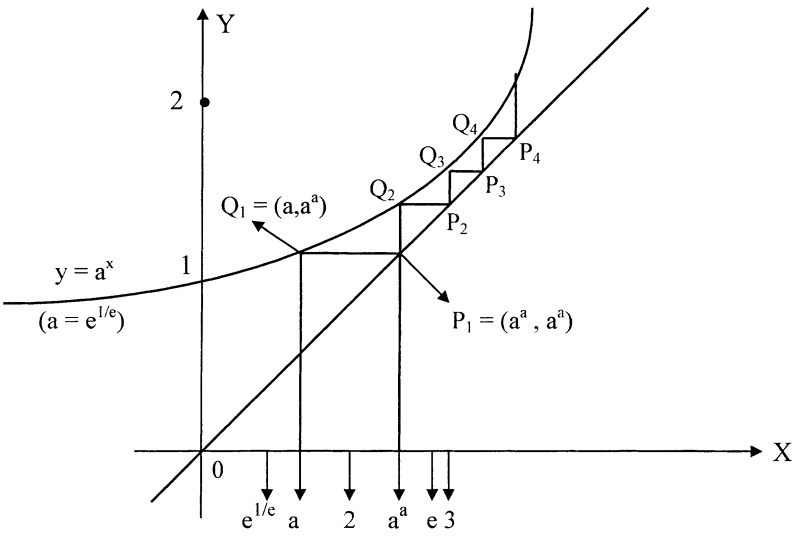


Figure 7.8

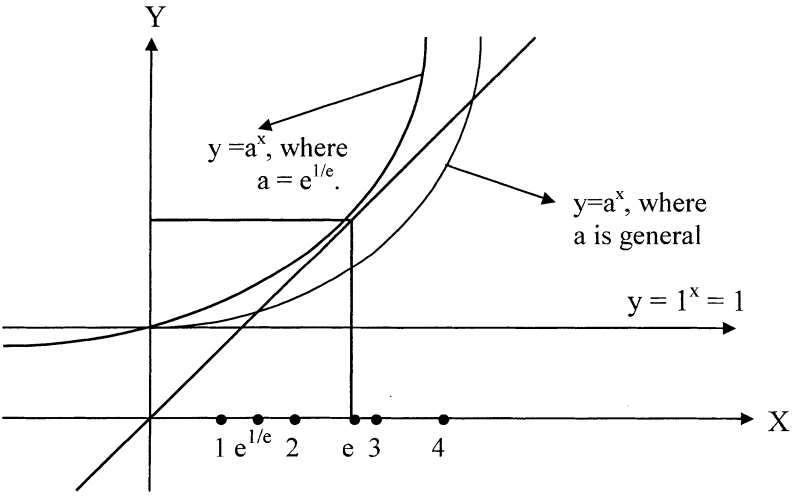


Figure 7.9

For various $a < 1$, the graphs of $y = a^x$ behave differently, since, as $x \to \infty$, $a^x \to 0$ and as $x \to -\infty$, $a^x \to \infty$ and so the graphs Γ have shape shown in Figure 7.10.

As it turns out, we need to consider the graph

$$\Gamma^* : x = a^y$$

as well, which is the mirror image of $\Gamma : y = a^x$ in the line $G : y = x$ and we further need to determine the points $\Gamma \cap \Gamma^*$ in this case (i.e., when $0 < a < 1$),

i.e., all the real solutions of the pair of equations

$$(i)\ y = a^x \qquad \text{and} \qquad (ii)\ x = a^y \qquad (*)$$

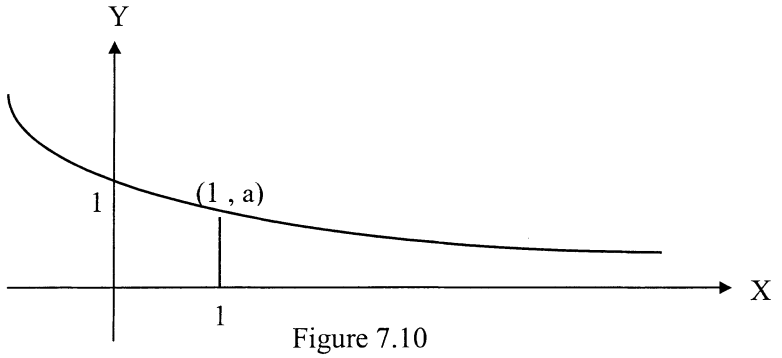


Figure 7.10

Now one solution of (*) is $x = y = \zeta$, say. Let P be the point (ζ, ζ). If the absolute value of the gradient of (i) at P is greater than 1, then clearly the graph of (i) will drop below the graph of (ii) as x moves to the right of P and further since it stays above the x-axis, it follows that the two graphs must cross each other once more, at Q, say. By symmetry about G, there is then a third point R of intersection of Γ and Γ^* (see Figure 7.11).

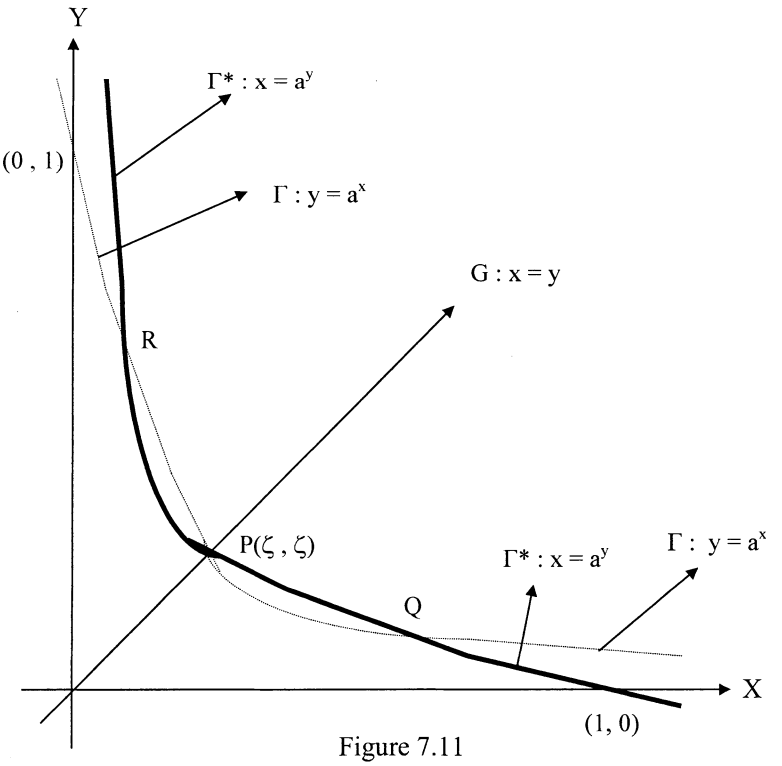


Figure 7.11

The problem now is to find when (i.e., for what values of a) the graph Γ has absolute value of the gradient greater than 1 at P. We shall see that the critical value is at $a = e^{-e}$. For then, our Γ is $y = (e^{-e})^x = e^{-ex}$ and so $dy/dx = e^{-ex}.\log(e^{-e}) = -e.e^{-ex}$. To get P, we solve $y = e^{-ex}$ with $y = x$, giving $e^{-ex} = x$. Since there is only one real solution (because Γ is decreasing) and $x = 1/e$ is a solution, it is the solution, i.e., $P = (1/e, 1/e)$. Hence $(dy/dx)_P = (-e)e^{-e(1/e)} = -1$.

So now the gradient of Γ, at the point $P \in \Gamma \cap G$ changes from -1 to 1 as a varies from e^{-e} to $e^{1/e}$. For $a > e^{1/e}$, $(dy/dx)_P > 1$ while for $a < e^{-e}$, $(dy/dx)_P < -1$ and as $a \to 0$, $(dy/dx)_P \to -\infty$ and so in the range $0 < a < e^{-e}$, $|(dy/dx)_P| > 1$ and $\Gamma \cap \Gamma^*$ consists of 3 points, as shown in Figure 7.11. In case this graphical consideration does not satisfy the reader, we may prove the above analytically as follows: Since $y = a^x$, $dy/dx = a^x \log a$. Then $(dy/dx)_P = x \log a = \log a^x = \log x$ (since $a^x = y = x$ at P) and this is less than -1 if and only if

$$
\begin{aligned}
\log x < -1 &\Leftrightarrow x < e^{-1} && (*)\\
&\Leftrightarrow a^x < e^{-1} && (\text{as } x = a^x)\\
&\Leftrightarrow x \log a < -1\\
&\Leftrightarrow \log a < -1/x\\
&\Leftrightarrow \log a < -e && (\text{since } -1/x < -e \text{ by } (*))\\
&\Leftrightarrow a < e^{-e}\,,
\end{aligned}
$$

as required. Now let us collect all the information that we have observed so far that is relevant to the behaviour of $\Gamma : y = a^x$.

Let $a > 0$. Mark off, on the real line, certain points as shown below:

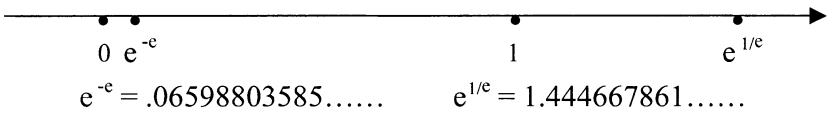


Figure 7.12

Let

$$(i)\ \Gamma : y = a^x \qquad (ii)\ \Gamma^* : x = a^y \qquad (iii)\ G : y = x\,,$$

so that Γ, Γ^* are mirror images of each other in G. Then the following results hold:

1. If $e^{-e} \leq a \leq 1$, $\Gamma \cap \Gamma^*$ consists of only one point, say $P = (\alpha, \alpha) \in G$. The gradient of (i) at P is greater than or equal to -1 (it is equal to -1 when $a = e^{-e}$ and is 0 when $a = 1$).

2. If $0 < a < e^{-e}$, $\Gamma \cap \Gamma^*$ consists of 3 points, say, $R = (\alpha, \beta)$, $P = (\gamma, \gamma)$, $Q = (\beta, \alpha)$, $\alpha < \beta$, and the gradient of (i) at P is less than or equal to -1 and as $a \to 0$, this gradient tends to $-\infty$.

3. If $1 < a < e^{1/e}$, $\Gamma \cap \Gamma^*$ consists of 2 points, say, $P = (\alpha, \alpha)$, $Q = (\beta, \beta)$, $\alpha < \beta$ and the gradient of (i) at P lies between 0 and 1 (it is 0 if $a = 1$ and is 1 if $a = e^{1/e}$; for $a = 1$, Q does not exist, while for $a = e^{1/e}$, Q and P coincide).

4. If $a = e^{1/e}$, $\Gamma \cap \Gamma^*$ consists of one point, say, $P = (\alpha, \alpha) \in G$, in fact G is tangent to both of Γ and Γ^*, and the gradient of (i) at P is 1.

5. If $e^{1/e} < a$, $\Gamma \cap \Gamma^* = \emptyset$.

For the convergence of $\zeta_n = a^{\zeta_{n-1}}$, $\zeta_1 = a$, for $0 < a < 1$, two cases arise:

(i) $0 < a < e^{-e}$,

(ii) $e^{-e} \leq a < 1$.

It will be found that in case (ii), ζ_n is convergent to the unique point $P = \Gamma \cap \Gamma^*$ while in case (i), ζ_n has no limit but two limit points (accumulation points) R and Q, as described above and indeed $\zeta_{2n} \to R$, $\zeta_{2n-1} \to Q$, as $n \to \infty$. In fact we have $R \to (0,1)$, $Q \to (1,0)$, as $a \to 0$.
Case 5: $e^{-e} \leq a < 1$. Let us take a typical value of a, say $a = 1/2$ ($e^{-e} \leq 1/2 < 1$). The graph of $y = (1/2)^x$ is shown in Figure 7.13.

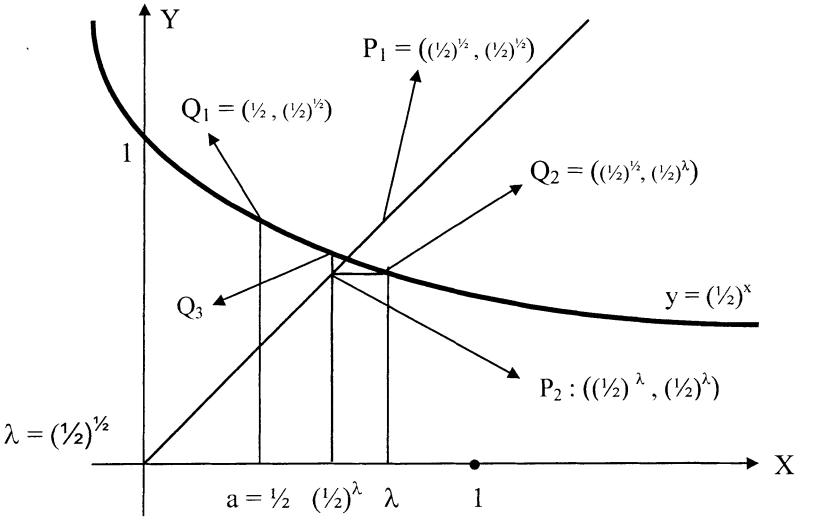


Figure 7.13

The sequence ζ_n, $n = 1, 2, \ldots$, is the x-coordinate of Q_n and these Q_n's are given by a spiral instead of a zigzag as was the case for $1 < a \leq e^{1/e}$. It is now

clear, but perhaps not as clear as in the zigzag case, that the spiral of points Q_i, P_i tends to the unique point of intersection P of Γ and G.

Case 6: $0 < a < e^{-e}$.The relevant graph for a typical a with $0 < a < e^{-e}$, now looks like Figure 7.14. Here, the spirals alternatively tend to the points R and Q, as required.

Analytic proof of Theorem 7.2.1. We now give an analytic proof of all these cases discussed above. More specifically, we prove:

Let $a > 0$ be a real number and let $\zeta_1 = a, \zeta_2 = a^{\zeta_1} = a^a, \ldots, \zeta_n = a^{\zeta_{n-1}}$.

- The sequence $\{\zeta_n\}$ is convergent to a limit Λ if and only if $e^{-e} \leq a \leq e^{1/e}$.
- If $1 < a < e^{1/e}$, the equation $x = a^x$ has two roots α, β ($\alpha < \beta$, say) and Λ equals the smaller root α.
- If $a = e^{1/e}$, the equation $x = a^x$ has a unique root α and $\Lambda = \alpha$, while if $a = 1$, trivially $\Lambda = 1$.
- If $e^{1/e} < a$, ζ_n diverges to ∞.
- If $0 < a < 1$, the ζ_i satisfy

$$0 < \zeta_1 < \zeta_3 < \zeta_5 < \ldots < \zeta_6 < \zeta_4 < \zeta_2 < 1.$$

Writing $A = \lim \zeta_{2n-1}$, $B = \lim \zeta_{2n}$, we have that if $e^{-e} \leq a \leq 1$, $A = B = \Lambda$. If $0 < a < e^{-e}$, then $A < B$ and ζ_n oscillates with A, B as two limit points.

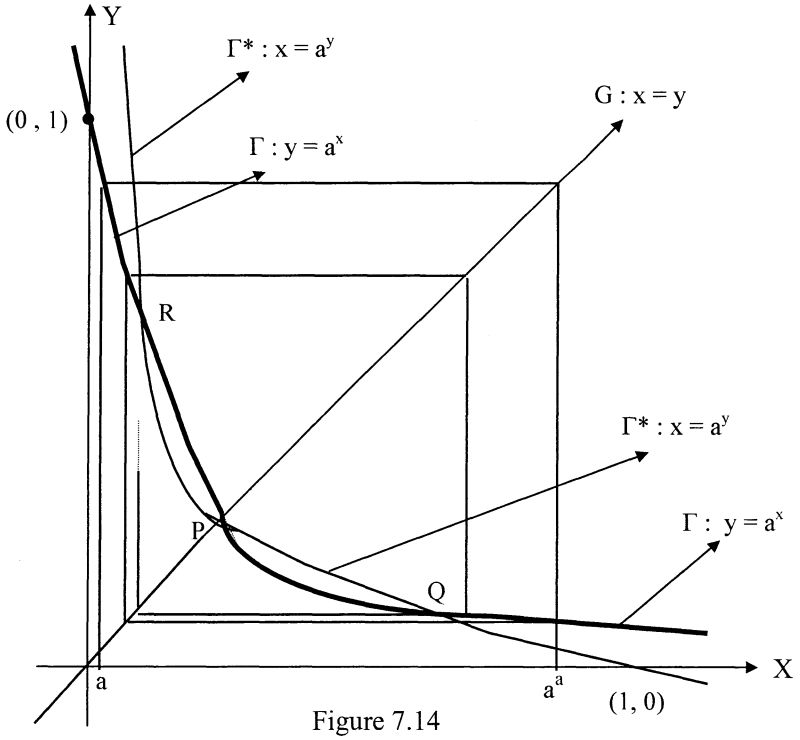


Figure 7.14

First, let $1 < a \leq e^{1/e}$. We have $\zeta_n/\zeta_{n-1} = a^{\zeta_{n-1}}/a^{\zeta_{n-2}} = a^{(\zeta_{n-1}-\zeta_{n-2})}$. Hence $\zeta_n > \zeta_{n-1}$ if and only if $\zeta_{n-1} > \zeta_{n-2}$. But $\zeta_2 = a^a > a = \zeta_1$, so by induction $\zeta_n > \zeta_{n-1}$ for all n, i.e. the sequence $\{\zeta_n\}$ is increasing.

Next we show that the sequence $\{\zeta_n\}$ is bounded above by e. First notice that, since $a \leq e^{1/e} < e$, so $a^a \leq (e^{1/e})^a < (e^{1/e})^e = e$ (since $a < e$). Now use induction. If $\zeta_{n-1} < e$, then $\zeta_n = a^{\zeta_{n-1}} \leq (e^{1/e})^{\zeta_{n-1}} < (e^{1/e})^e = e$ (by induction hypothesis). It follows that ζ_n tends to a limit, Λ say, which is less than or equal to e. Letting $n \to \infty$ in the defining equation $\zeta_n = a^{\zeta_{n-1}}$, we get $\Lambda = a^\Lambda$, where $\Lambda \leq e$, i.e.,

$$\Lambda \text{ is a root of the equation } x = a^x \text{ and } \Lambda \leq e. \qquad (\sharp)$$

To get this root, we draw the graphs $\Gamma : y = a^x$ and $G : y = x$. In the range $1 < a \leq e^{1/e}$, the graphs Γ and G, for $a = 1$, $a = e^{1/e}$ and a typical a are shown in Figure 7.15.

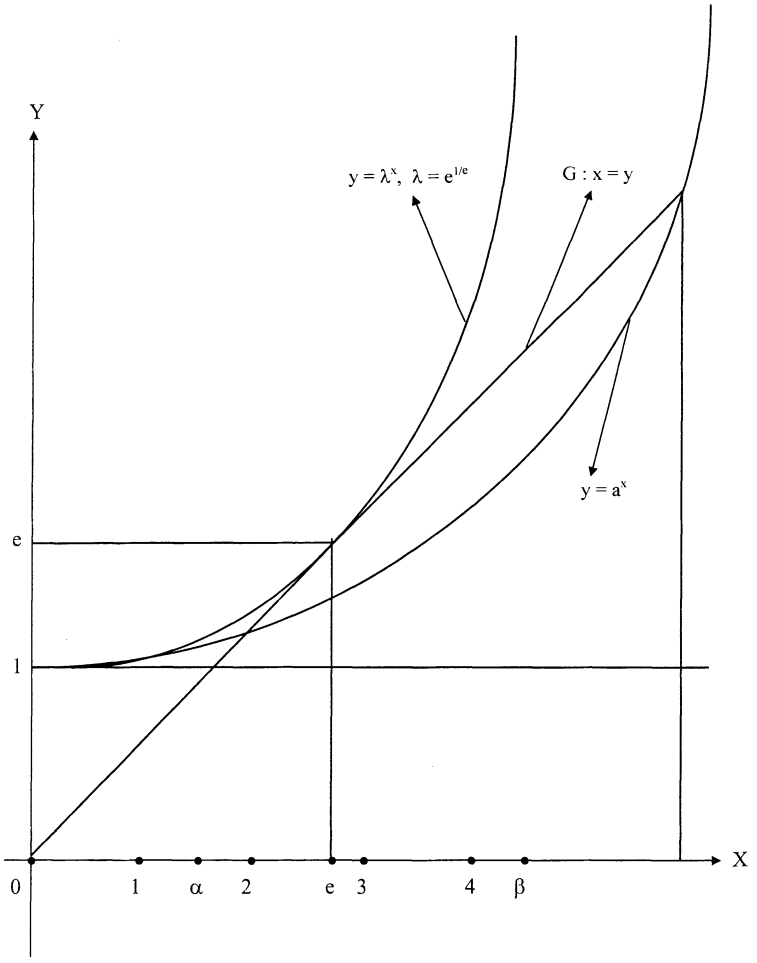


Figure 7.15

For $a \neq e^{1/e}$, the equation $x = a^x$ has two roots α, β ($\alpha < \beta$, say) and indeed $\alpha < e < \beta$. Also, since $\Lambda \leq e$, it follows that $\Lambda = \alpha$, the smaller of the two roots.

Next, let $e^{1/e} < a$. We have seen that in this case, the two curves Γ and G have no real intersection (see case 3 and Figure 7.8 of the graphical motivation). Suppose $\lim \zeta_n$ exists and equals Λ. Then

$$0 = a^\Lambda - a^\Lambda = \lim(a^{\zeta_n} - a^\Lambda) = \lim(\zeta_{n+1} - a^\Lambda) = \Lambda - a^\Lambda \,,$$

i.e., that Λ is a root of the equation $x = a^x$, showing that $\Gamma \cap G \neq \emptyset$.

Let now $0 < a < 1$. We show, by induction, that

$$0 < \zeta_1 < \zeta_3 < \zeta_5 < \ldots < \zeta_6 < \zeta_4 < \zeta_2 < 1 \,.$$

We need one pertinent inequality throughout, viz., if $0 < a < 1$ (as is the case) and if $0 < p < q$, then

$$0 < a^q < a^p < 1 \,. \qquad (*)$$

The proof of this is immediate from the graph Γ of $y = a^x$ ($0 < a < 1$), which has the shape shown in Figure 7.16.

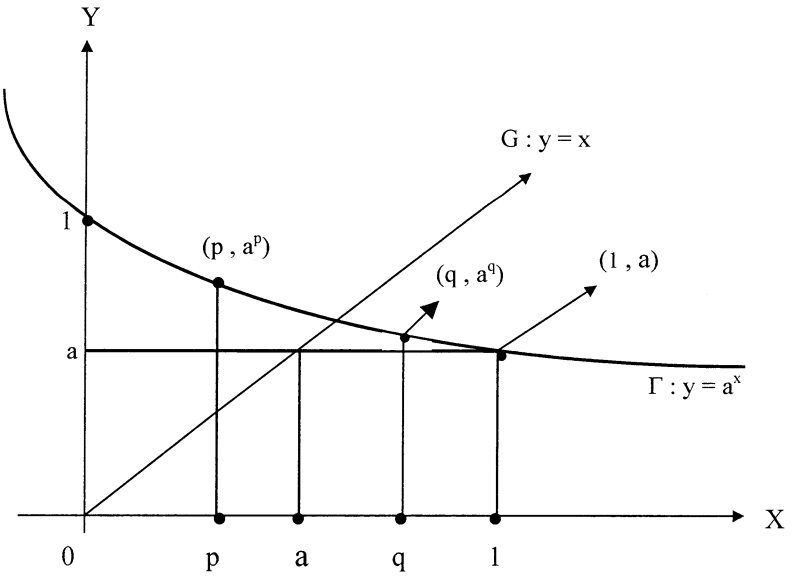


Figure 7.16

Since, as x increases, Γ decreases (i.e., comes down onto the x-axis), it follows that $a^q < a^p$ and both these numbers are less than 1 (even if q is to the right of 1).

Now using this inequality repeatedly, we have

$$\begin{aligned} 0 < \zeta_1 = a^1 < a^a = \zeta_2 < 1\,, & \\ \zeta_3 = a^{(a^a)} < a^a = \zeta_2\,, & \qquad (\text{since } a < a^a) \\ \zeta_1 = a^1 < a^{(a^a)} = \zeta_3\,, & \qquad (\text{since } a^a < 1) \\ \zeta_2 = a^a > a^{(a^{(a^a)})} = \zeta_4\,, & \end{aligned}$$

since $a < a^{(a^a)}$ and since $\zeta_3 < \zeta_2$, so $a^{\zeta_2} < a^{\zeta_3}$, i.e., $\zeta_3 < \zeta_4$. These inequalities imply $\zeta_1 < \zeta_3 < \zeta_4 < \zeta_2$, which starts off our induction.

Now the induction step. Suppose (induction hypothesis)

$$\zeta_1 < \zeta_3 < \ldots < \zeta_{2r-1} < \zeta_{2r} < \ldots < \zeta_4 < \zeta_2\,.$$

Then

$$\begin{aligned} \zeta_{2r+1} = a^{\zeta_{2r}} < a^{\zeta_{2r-1}} = \zeta_{2r}\,, & \qquad (\text{since } \zeta_{2r-1} < \zeta_{2r}) \\ \zeta_{2r-1} = a^{\zeta_{2r-2}} < a^{\zeta_{2r}} = \zeta_{2r+1}\,, & \qquad (\text{since } \zeta_{2r} < \zeta_{2r-2}) \\ \zeta_{2r+2} = a^{\zeta_{2r+1}} < a^{\zeta_{2r-1}} = \zeta_{2r}\,, & \qquad (\text{since } \zeta_{2r-1} < \zeta_{2r+1}) \\ \zeta_{2r+1} = a^{\zeta_{2r}} < a^{\zeta_{2r+1}} = \zeta_{2r+2}\,, & \qquad (\text{since } \zeta_{2r+1} < \zeta_{2r}) \end{aligned}$$

i.e.,

$$\zeta_1 < \zeta_3 < \ldots < \zeta_{2r-1} < \zeta_{2r+1} < \zeta_{2r+2} < \zeta_{2r} < \zeta_{2r-2} \cdots < \zeta_4 < \zeta_2\,,$$

which gives the induction step.

Finally, to see that $0 < \zeta_n < 1$ for all n, we proceed as follows:

Since $0 < a < 1$, so $0 < a^a < 1$, i.e., $0 < \zeta_2 < 1$. If $0 < \zeta_{n-1} < 1$, then $0 < a^{\zeta_{n-1}} < 1$, i.e., $0 < \zeta_n < 1$, so by induction, the result follows.

Thus ζ_{2n-1} increases while ζ_{2n} decreases and both are bounded by 0 and 1. Hence they tend to limits, say ζ_{2n-1} tends to A, ζ_{2n} tends to B, where

$$0 < A \leq B \leq 1\,.$$

Now, in the relation $\zeta_n = a^{\zeta_{n-1}}$, let n tend to ∞, first through odd values and then through even values and we get $A = a^B$, $B = a^A$, i.e., (A, B) is a point common to the two curves $\Gamma : y = a^x$ and $\Gamma^* : x = a^y$. But we have seen that if $e^{-e} \leq a \leq 1$, then $\Gamma \cap \Gamma^*$ is a single point P, which belongs to $G : y = x$. It follows that $A = B = \lim \zeta_n$ exists (in case $e^{-e} \leq a \leq 1$) and is the unique solution of the equation $x = a^x$ (i.e., the unique intersection of Γ, Γ^* and G), as required.

Let $0 < a < e^{-e}$. In this case $\Gamma \cap \Gamma^*$ consists of three distinct points $R = (\alpha, \beta)$, $P = (\gamma, \gamma)$, $Q = (\beta, \alpha)$, $(\alpha < \beta)$. Since $(A, B) \in \Gamma \cap \Gamma^*$, we see that if $A \neq B$, then $A = \alpha$, $B = \beta$ and although $\{\zeta_n\}$ has two limit points, it has no limit. It remains therefore to show that $A \neq B$.

The equations $A = a^B$, $B = a^A$ (established above) imply $A = a^B = a^{(a^A)}$ and $B = a^A = a^{(a^B)}$, i.e., that A, B are roots of the equation $x = a^{(a^x)}$. Let

$$\phi(x) = x - a^{(a^x)} . \tag{1}$$

If Λ is a root of ϕ, i.e., $\Lambda = a^{(a^\Lambda)}$, then $\log\Lambda = a^\Lambda \log a$ or $\log(1/\Lambda) = a^\Lambda \log(1/a)$ and so $\log\log(1/\Lambda) = \Lambda \log a + \log\log(1/a)$. Let

$$\psi(x) = \log\log(1/a) + x\log a - \log\log(1/x) . \tag{2}$$

Then it is easy to check that $\phi(x) = 0$ if and only if $\psi(x) = 0$, and indeed

$$\phi(x) \text{ is } \begin{matrix} > \\ = \\ < \end{matrix} 0 \Leftrightarrow \psi(x) \text{ is } \begin{matrix} > \\ = \\ < \end{matrix} 0 \tag{3}$$

(for example, $\phi(x_0) > 0 \Leftrightarrow x_0 > a^{(a^{x_0})} \Leftrightarrow \log x_0 > a^{x_0}\log a \Leftrightarrow \log(1/x_0) < -a^{x_0}.\log a \Leftrightarrow \log\log(1/x_0) < x_0 \log a + \log(-\log a) \Leftrightarrow \log\log(1/x_0) < x_0 \log a + \log\log(1/a) \Leftrightarrow \psi(x_0) > 0$). Also

$$\psi'(x) = \log a - (1/\log(1/x)).(1/(1/x)).(-1/x^2) = \log a - 1/(x\log x) . \tag{4}$$

So $\psi'(x) = 0$ when $\log x^x = 1/\log a$, i.e., when $x^x = e^{1/\log a}$.

We now consider the equation $x^x = c$ $(0 < c < 1)$ and record:

Lemma 7.2.2. *The equation $x^x = c$ $(0 < c < 1)$ has no real root if $0 < c < 1/e^{1/e}$; one real root if $c = 1/e^{1/e}$; two real roots if $1/e^{1/e} < c < 1$.*

Proof. Consider the graph T of $y = x^x$ (sketch shown in Figure 7.17). Taking logarithms and differentiating gives $(1/y).dy/dx = 1 + \log x$; so there is an extreme at $x = 1/e$ and indeed it is a minimum, equal to $e^{-1/e}$. As x increases from 0 to 1, $y = x^x$ decreases from 1 to the value $1/e^{1/e}$ at $x = 1/e$ and then increases from $1/e^{1/e}$ to 1.

To continue with the analytic proof, as $\psi'(x) = 0$ when $x^x = e^{1/\log a}$ $(= c$, say), we get that $\psi'(x) = 0$ has no root, one root and two roots according as $0 < e^{1/\log a} < e^{-1/e}$, $e^{1/\log a} = e^{-1/e}$ and $e^{-1/e} < e^{1/\log a} < 1$ respectively, i.e., according as $a > e^{-e}$, $a = e^{-e}$ and $a < e^{-e}$, respectively (check that $1/\log a$ is less than, equal to or greater than $-1/e \Leftrightarrow \log a$ is less than, equal to or greater than $-e \Leftrightarrow a$ is less than, equal to or greater than e^{-e} respectively).

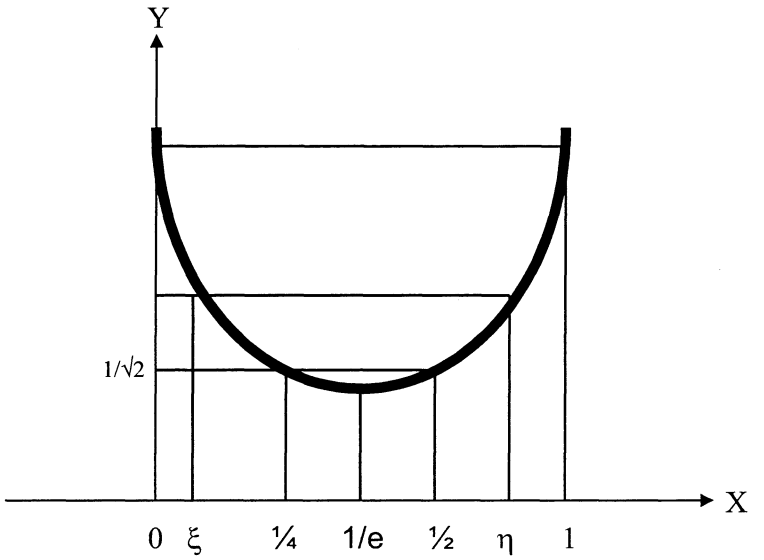


Figure 7.17

Now consider the graph of $y = \psi(x)$ and note that as $x \to 0$, $\psi(x) \to -\infty$ and as $x \to 1$, $\psi(x) \to \infty$. It follows that $\psi(x)$ crosses the $x-$axis an odd number of times (see Figures 7.18–7.21), i.e., that $\psi(x) = 0$ has an odd number of roots. We now split two cases.

Case (i): $e^{-e} \leq a < 1$. Then by above $\psi'(x) = 0$ has either no root or one root and so only the two possibilities (i), $(i)'$ in the diagrams (Figures 7.18–7.21) hold, since in the other possibilities, the equation $\psi'(x) = 0$ has $2, 4, \ldots$ roots respectively.

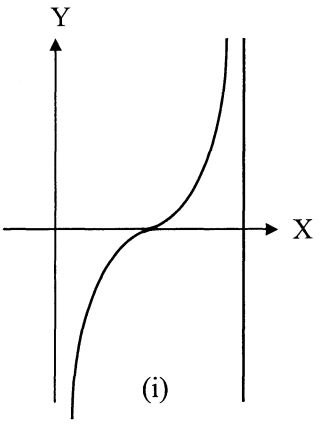


Figure 7.18

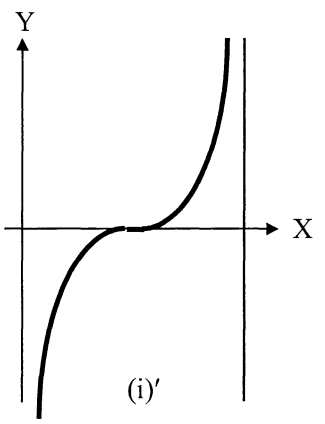


Figure 7.19

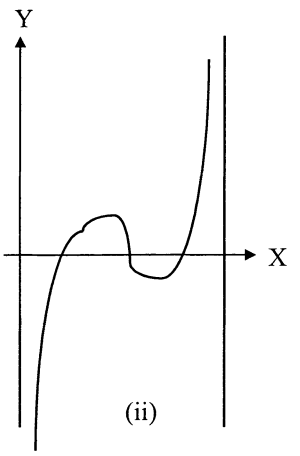


Figure 7.20

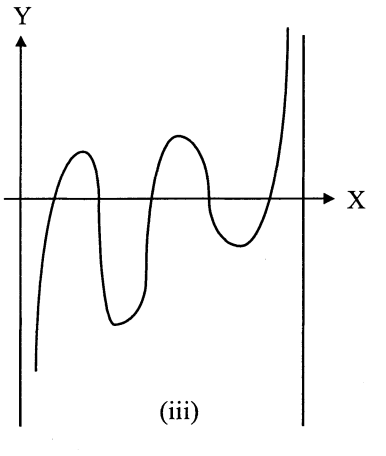


Figure 7.21

Thus $\psi(x) = 0$ has only one root and hence $\phi(x) = 0$ also has only one root (see (3)). Since A, B are roots of $\phi(x) = 0$, it follows that $A = B$; so in this case the sequence ζ_n converge to $A = B$.

Case (ii): $0 < a < e^{-e}$. We have seen in this case that the equation $x^x = e^{1/\log a}$ $(= c)$ has two roots x_1, x_2, say, in the interval $(0, 1)$, i.e.,

$$x_2^{x_2} = x_1^{x_1} = e^{1/\log a} \tag{5}$$

(for we must check that $c = e^{1/\log a} > 1/e^{1/e}$, i.e., that $1/\log a > -1/e$, i.e., that $\log a < -e$, i.e., that $a < e^{-e}$, which is true in this case).

We have also seen in Lemma 7.2.2 that these two roots lie on different sides of $1/e$, i.e.,

$$x_1 < 1/e < x_2 \,. \tag{6}$$

Then $x < x_1$ implies that $x^x > x_1^{x_1} = e^{1/\log a} \Rightarrow \psi'(x) > 0$ (see Figure 7.22).

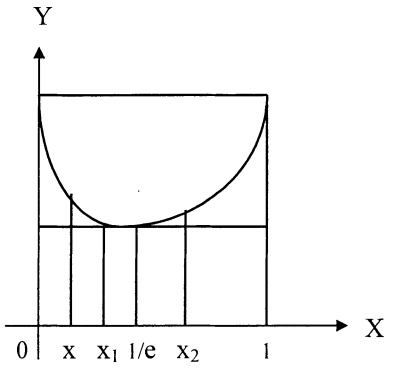


Figure 7.22

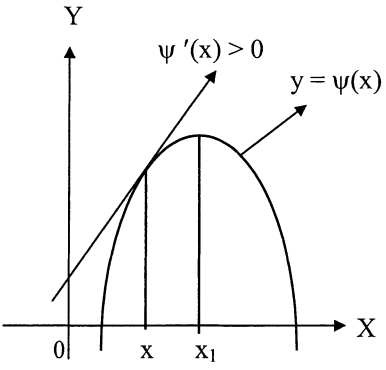


Figure 7.23

To check the above implications; the first one follows by looking at the graph of the function $y = x^x$ in $[0, 1]$, since $x < x_1$. To see the second implication, we have $\psi'(x) > 0 \Leftrightarrow \log a > 1/\log(x^x)$ (see (4)) $\Leftrightarrow 1/\log a < \log(x^x) \Leftrightarrow x^x > e^{1/\log a}$, as required.

Similarly, $x > x_1 \Rightarrow x^x < e^{1/\log a} \Rightarrow \psi'(x) < 0$. So (see Figure 7.23)

$$x = x_1 \textit{ is a maximum of } \psi(x) \textit{ \& similarly } x = x_2 \textit{ is a minimum of } \psi(x). \quad (7)$$

In particular,

$$\psi'(x_1) = \psi'(x_2) = 0\,. \quad (8)$$

So now, by (2), we have $\psi(x_1) = \log\log(1/a) + x_1 \log a - \log\log(1/x_1)$. However, by (8), $\psi'(x_1) = 0$ and so by (4), $\log a = (1/x_1)\log x_1$, so $x_1 \log a = 1/\log x_1 = -1/\log(1/x_1)$. Therefore, we have

$$\begin{aligned}\log\log(1/a) &= \log 1 - \log(-x_1 \log x_1)\\ &= \log(1/(-x_1 \log x_1))\\ &= \log(1/x_1 \log(1/x_1))\\ &= \log((1/x_1)/\log(1/x_1))\\ &= \log(1/x_1) - \log\log(1/x_1).\end{aligned}$$

So

$$\begin{aligned}\psi(x_1) &= (\log(1/x_1) - \log\log(1/x_1)) - 1/(\log(1/x_1)) - \log\log(1/x_1)\\ &= r - 1/r - 2\log r = \lambda(r)\ ,\ \text{say}\ ,\end{aligned} \quad (9)$$

where

$$r = \log(1/x_1) > \log e = 1 \quad (\text{since } x_1 < 1/e). \quad (10)$$

Then $\lambda'(r) = 1 + 1/r^2 - 2/r = (r-1)^2/r^2 \geq 0$ and in fact it is greater than 0, since $r > 1$, by (10), i.e., $\lambda'(r) > 0$, showing that $\lambda(r)$ is increasing and so $\lambda(r) > \lambda(1) = 0$, or by (9) we have that

$$\psi(x_1) > 0\,. \quad (11)$$

Further, by (2)

$$\psi(1/e) = \log\log(1/a) + (1/e)\log a = \log\log(1/a) - (1/e)\log(1/a). \quad (12)$$

But

$$(\log z)/z < 1/e\ ,\ \text{if } z > e \quad (13)$$

(check: We have to show that $h(z) = z - e\log z > 0$. We have $dh/dz = 1 - e/z > 0$ if $z > e$, and so $h(x)$ is increasing and $h(e) = 0$, so $h(z) > 0$).

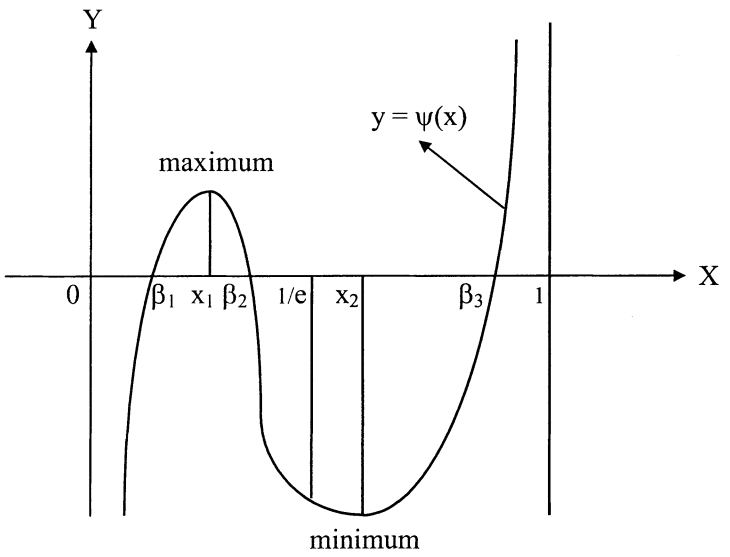


Figure 7.24

Taking $z = \log(1/a)$, this shows that $(\log\log(1/a))/\log(1/a) < 1/e$ (for, $z = \log(1/a)$ is greater than e since $1/a > e^e$ as $a < e^{-e}$), i.e., $\log\log(1/a) < (1/e)\log(1/a)$ (since $\log(1/a) > 0$ as $a < 1$), and so, by (12),

$$\psi(1/e) < 0\,. \tag{14}$$

It follows that the graph of $\psi(x) = 0$ is as shown in Figure 7.24 (see (6), (7) and (11)), showing that $\psi(x) = 0$ has three roots in $(0,1)$; call them $\beta_1, \beta_2, \beta_3$, where

$$0 < \beta_1 < x_1 < \beta_2 < 1/e < x_2 < \beta_3 < 1.$$

Since, by (3), $\phi(x)$ is larger than, equal to or less than 0 if and only if $\psi(x)$ is larger than, equal to or less than 0, respectively, it follows that $\phi(x)$ has the same three roots $\beta_1, \beta_2, \beta_3$. Remembering again that $0 < a < e^{-e}$, we now show that $B = \beta_3$, $A = \beta_1$. Indeed, we shall show, by induction, that $\zeta_{2n} > 1/e$ for all n. We have $\zeta_0 = 1 > 1/e$. If $\zeta_{2n-2} > 1/e$, then $\zeta_{2n} = a^{a^{\zeta_{2n-2}}} > a^{a^{1/e}} > 1/e$ (this last inequality, since $\psi(1/e) < 0$ by (14) and so $\phi(1/e) < 0$). Thus, $\zeta_{2n} > 1/e$ and so $\lim \zeta_{2n} = B \geq 1/e$. But B is a zero of ϕ and the only zero of ϕ larger than $1/e$ is β_3. So $B = \beta_3$.

Next we show, again by induction, that $\zeta_{2n+1} < x_1$ for all n. The induction step is simple. Suppose $\zeta_{2n-1} < x_1$, then $\zeta_{2n+1} = a^{a^{\zeta_{2n-1}}} < a^{a^{x_1}} < x_1$ (the last inequality comes from (11) because $\psi(x_1) > 0$ and so $\phi(x_1) > 0$), completing the induction step.

To start the induction, it remains to show that $a < x_1$. Now $z \geq e\log z$ for all $z > 0$ (check: It is enough to prove that $g(z) = z - e\log z \leq 0$. Now

$g'(z) = 1 - e/z = 0$ if $z = e$ and $g''(z) = e/z^2 > 0$ at $z = e$, and so $z = e$ is a minimum and $g(e) = 0$, so $g(z) \leq 0$). Hence $1/\sqrt{x_1} \geq e \log 1/\sqrt{x_1} > 2 \log 1/\sqrt{x_1} = \log 1/x_1$, i.e., $1/x_1 > (\log 1/x_1)^2$, i.e., $1/x_1 \log(1/x_1) > \log 1/x_1$ or $1/(-x_1 \log 1/x_1) > -\log x_1$. But $\log a = 1/(x_1 \log x_1) < \log x_1$ (taking logarithms in (5)), by above, we get that $a < x_1$, as required.

So now, $\zeta_{2n+1} < x_1$ for all n and ζ_{2n+1} is increasing and tends to A, so $A \leq x_1$. But A is a zero of ϕ and the only zero of ϕ less than or equal to x_1 is β_1. Hence $A = \beta_1$. Thus $A \neq B$ and so ζ_n oscillates with A, B as limits.

This at last completes the analytic proof of Theorem 7.2.1.

§7.3. Applications and examples

In the last section we have given graphical and analytic proofs of the convergence or otherwise of the infinite exponential of the title of this chapter. Our object now is to establish the connection of this infinite exponential with the function $f(x, y) = x^y - y^x$ in the range of convergence of the exponential. We have the following striking result:

Theorem 7.3.1. *In its interval of convergence, i.e., for* $z \in [e^{-e}, e^{1/e}]$, *the functions*

$$x = h(z) = z^{z^{z^{\cdot^{\cdot^{\cdot}}}}} \quad \text{and} \quad z = g(x) = x^{1/x}$$

are partial inverses of each other. More specifically

(i) $g(h(z)) = z$, *for all* $z \in [e^{-e}, e^{1/e}]$ *(= the full range of convergence of* $h(z)$*),*

(ii) $h(g(x))=x$, *for all* $x \in [1/e, e]$.

Proof. Since exponentiation is continuous (and increasing), it follows that

$$x = h(z) \Rightarrow z^x = z^{h(z)} \Rightarrow z^x = z^{(z^{z^{\cdot^{\cdot^{\cdot}}}})} = h(z) = x \Rightarrow z = x^{1/x} = g(x).$$

Hence, we have

(i) $g(h(z)) = g(z^{z^{z^{\cdot^{\cdot^{\cdot}}}}}) = g(x) = z$; and

(ii) $h(g(x)) = h(x^{1/x}) = h(z) = x$.

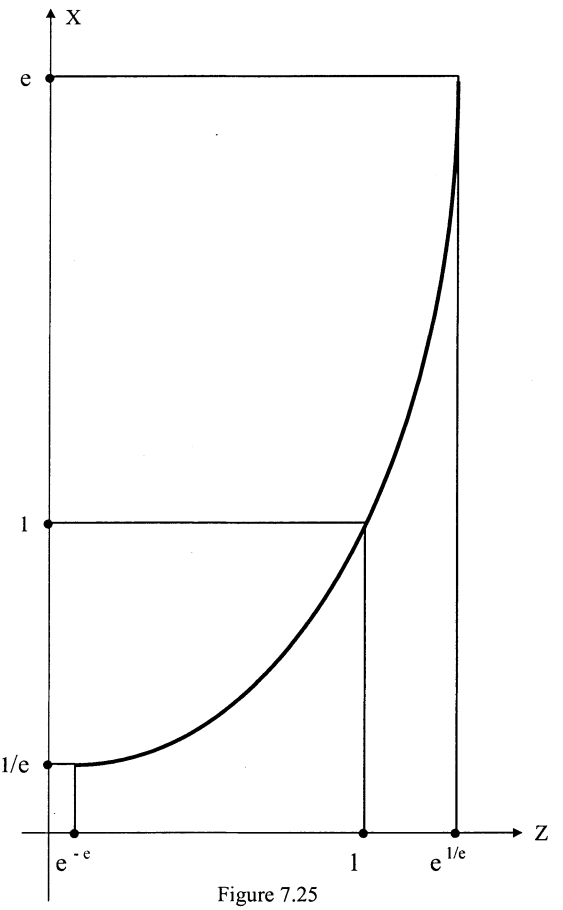


Figure 7.25

We now determine the range where these functions are meaningful. We take them up by turn.

In (i), $h(z)$ is meaningful only in the range of its convergence, i.e., when $z \in [e^{-e}, e^{1/e}]$, as required. Note that g is defined for all $x > 0$.

For (ii) to hold, $g(x)$ must lie in the range of convergence of h, i.e.,

$$e^{-e} \leq x^{1/x} \leq e^{1/e} .$$

But this is the same thing as saying that $1/e \leq x \leq e$, since at $x = 1/e$, we have $x^{1/x} = e^{-e}$ and $x^{1/x}$ increases from $x = 1/e$ right up to $x = e$ (where $e^{1/e}$ is the maximum of $x^{1/x}$), so $1/e^e \leq x^{1/x}$. Similarly, at $x = e$, we have $x^{1/x} = e^{1/e}$. For a clear understanding of this, we sketch, in Figure 7.25, the graph of the function

$$x = h(z) = z^{z^{z^{\cdot^{\cdot^{\cdot}}}}} .$$

Example 7.3.2. Evaluate

$$\sqrt{\left(1+2\sqrt{\left(1+3\sqrt{(1+4\sqrt{(1+5\sqrt{(1+\cdots)})})}\right)}\right)}\,.$$

Indeed, for $x>0$, let

$$f_n(x)=\sqrt{\left(1+x\sqrt{(1+(x+1)\sqrt{(1+\cdots+(x+n-1)\sqrt{(1+(x+n))})})}\right)}. \quad (1)$$

Clearly, $f_n(x)$ increases as n increases, but

$$\begin{aligned} x+1 &= \sqrt{(1+x(x+2))} \\ &= \sqrt{(1+x\sqrt{(1+(x+1)(x+3))})} \\ &= \ldots \quad \ldots \quad \ldots \\ &= \sqrt{(1+x\sqrt{((1+(x+1)\sqrt{(1+\cdots+(x+n-1)(x+n+1)})))})} \\ &> f_n(x)\,, \end{aligned}$$

since $\sqrt{(1+(x+n))}<x+n+1$, i.e., $f_n(x)$ is bounded above by $x+1$ and so tends to a limit, say $f(x)$. Thus $f(x)\le x+1$. Also

$$f(x)>\sqrt{(1+x\sqrt{(1+x\sqrt{(1+x\sqrt{(\cdots}}}}} = R \text{ ,say.}$$

For this, compare $f_n(x)$ with the right hand side; i.e., $f_n(x)>R$, for all n and so $f(x)\ge f_n(x)>R$. Then

$$R^2=1+x\sqrt{(1+x\sqrt{(1+x\sqrt{(1+x\sqrt{(\cdots}}}}} = 1+xR,$$

giving $R=(x\pm\sqrt{(x^2+4)})/2$ and since $R>0$, the $+$ sign holds, so $R=(x+\sqrt{(x^2+4)})/2>x$. Thus

$$x<f(x)\le x+1 \quad (2)$$

Let $x+1-f(x)=\varphi(x)$, so that (2) becomes

$$0\le\varphi(x)<1\,. \quad (2)'$$

Now squaring (1), we get

$$\begin{aligned} f_n^2(x) &= 1+x\sqrt{(1+(x+1)\sqrt{(1+\cdots+(x+n-1)\sqrt{(1+(x+n))})})} \\ &= 1+xf_{n-1}(x+1) \end{aligned}$$

and letting $n \to \infty$, this gives $f^2(x) = 1 + xf(x+1)$, i.e.,

$$\begin{aligned}
&(1+f(x))(1-f(x)) + xf(x+1) = 0\\
\text{or}\quad &(1+f(x))(\varphi(x)-x) + x(x+2-\varphi(x+1)) = 0\\
\text{or}\quad &(1+f(x))\varphi(x) + x(x+2) - x(1+f(x)) = x\varphi(x+1)\\
\text{or}\quad &(1+f(x))\varphi(x) + x((x+1-f(x))) = x\varphi(x+1)\\
\text{or}\quad &(1+x+f(x))\varphi(x) = x\varphi(x+1)\\
\text{or}\quad &\varphi(x)/x = \varphi(x+1)/(x+1+f(x)) \le \varphi(x+1)/(x+1)\,.
\end{aligned}$$

Hence,

$$0 \le \varphi(x)/x \le \varphi(x+1)/(x+1) \le \ldots \le \varphi(x+n)/(x+n) < 1/(x+n)$$

(this last inequality by (2)′) and this tends to 0 as $n \to \infty$, i.e., $\varphi(x) = 0$, giving $f(x) = x+1$.

So now, the given expression of the example is equal to $f(2) = 2+1 = 3$.

Example 7.3.3. For the equation $y^x = x^y$,

(i) determine those positive values of x, for which the equation has only trivial solution $y = x$,

(ii) for those $x > 0$ for which nontrivial positive solutions y exist, determine all such solutions,

(iii) if x is selected from the range $(0, e)$, determine the probability that a nontrivial solution for y, satisfying $y^x = x^y$, lies in the same interval.

Indeed, since $x^y = \exp(y \log x)$, it is enough to compare $y \log x$ and $x \log y$. Further, the function exp is increasing, so

$$x^y \text{ is } \begin{matrix}>\\=\\<\end{matrix} y^x \Leftrightarrow (\log x)/x \text{ is } \begin{matrix}>\\=\\<\end{matrix} (\log y)/y\,;$$

thus it is enough to consider the function $g(x) = (\log x)/x$, defined for $x > 0$.

Now $g'(x) = (1 - \log x)/x^2$, $g''(x) = (-3 + 2\log x)/x^3$. Therefore g is negative for $0 < x < 1$, increasing for $0 < x \le e$, attains a maximum of $1/e$ at $x = e$ and is decreasing for $x \ge e$. Further, $g(1) = 0$ and $g(x)$ tends to the x-axis asymptotically as $x \to \infty$. Hence

(i) the equation $y^x = x^y$ has only the trivial solution $y = x$, only when either $x = e$ or $0 < x \le 1$.

(ii) If $1 < x$, $x \ne e$, there exists exactly one nontrivial solution of $y^x = x^y$. Let y be this solution. Then, if $1 < x < e$, we have $y > e$, while if $x > e$, then $1 < y < e$.

(iii) The probability is 0, because nontrivial solutions of $y^x = x^y$ exist, in this case, only if $1 < x < e$ and then the corresponding y is greater than e.

We shall now discuss the convergence or otherwise of the infinite continued radical

$$\sqrt[r]{(a + \sqrt[r]{(a + \sqrt[r]{a + \cdots}}}$$

where $\sqrt[r]{\ }$ stands for the r^{th} root and $r \in \mathbb{N}$, $a \in \mathbb{R}^+$, the set of positive real numbers. Writing $u_1 = \sqrt[r]{a}$, $u_2 = \sqrt[r]{(a + u_1)}$, $\ldots$, $u_n = \sqrt[r]{(a + u_{n-1})}$, $\ldots$, we have the following:

Lemma 7.3.4. *The sequence $\{u_n\}$ is convergent.*

Proof. $u_{n+1}^r = a + u_n$, $u_n^r = a + u_{n-1} \Rightarrow u_{n+1}^r - u_n^r = u_n - u_{n-1}$. Hence, if $u_n > u_{n-1}$ then $u_{n+1}^r > u_n^r$ or $u_{n+1} > u_n$. But $u_2 > u_1$ (for it is enough to prove that $u_2^r > u_1^r$, i.e., that $a + u_1 > a$, which is true since $u_1 = \sqrt[r]{a} > 0$ as $a > 0$). So, by induction, u_n is an increasing sequence.

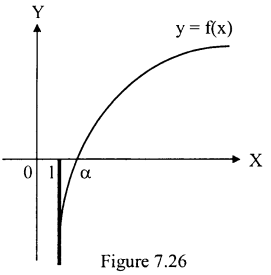


Figure 7.26

Now we prove that the equation $f(X) = X^r - X - a = 0$ has a unique positive root; call it α, for, its derivative is $rX^{r-1} - 1$ which is greater than 0 (i.e., $X^{r-1} > 1/r$) if $X \geq 1$, so $X^r - X - a$ increases after $X \geq 1$. At $X = 1$, $X^r - X - a = 1^r - 1 - a = -a < 0$, so $X^r - X - a$ vanishes just once after $X = 1$, say at $X = \alpha$. Further, in $[0, 1)$, i.e., if $X \in [0, 1)$, $X^r < X$, i.e., $X^r - X < 0$. So $X^r - X - a < 0$.

Thus, for all $X \in [0, 1)$, $f(X) < 0$. It follows that $f(X) = 0$, only at $X = \alpha$ in the range $[0, \infty)$. Now $u_n^r < u_{n+1}^r = a + u_n$ (since $\{u_n\}$ is increasing), i.e., $u_n^r - u_n - a < 0$ so $u_n < \alpha$, i.e., $\{u_n\}$ is bounded above by α. Hence, $\{u_n\}$ is a convergent sequence, which converges to u, say. Letting $n \to \infty$ in $u_n^r = a + u_n$, we get $u^r = a + u$.

From the above it follows that u is the positive real root α of $X^r - X - a = 0$, i.e., we have proved:

Theorem 7.3.5. *Let $r \geq 2$ be an integer and let $a \in \mathbb{R}^+$. Then the expression*

$$\sqrt[r]{(a + \sqrt[r]{(a + \sqrt[r]{a + \cdots}}}$$

is meaningful, i.e., convergent, with limit α, which is the unique positive real root of the equation $X^r - X - a = 0$.

Example 7.3.6. Taking $r = 2$ in the above, we get

$$\sqrt{(a + \sqrt{(a + \sqrt{a + \cdots}} = \alpha ,$$

where $\alpha^2 - \alpha - a = 0$, $\alpha > 0$, i.e., $\alpha = (1 + \sqrt{(1 + 4a)})/2$.

Taking $a = 6$, for example, we get

$$\sqrt{(6 + \sqrt{(6 + \sqrt{6 + \cdots}} = (1 + \sqrt{(1 + 24)})/2 = 3 .$$

In general, if $a = k(k - 1)$,

$$\sqrt{(a + \sqrt{(a + \sqrt{a + \cdots}} = k .$$

Taking $r = 2$, $a = 1$, we get

$$\sqrt{(1 + \sqrt{(1 + \sqrt{1 + \cdots}} = (1 + \sqrt{5})/2 ,$$

the golden ratio, which incidentally, is also equal to the continued fraction

$$1 + \cfrac{1}{1 + \cfrac{1}{1 + \cfrac{1}{1 + \cdots}}} .$$

Finally, taking $r = a = 2$, we get

$$\sqrt{(2 + \sqrt{(2 + \sqrt{2 + \cdots}} = 2 .$$

There is a similar infinite expression involving logarithms and fits in the above pattern ideally. The interested reader may look up [40].

Appendix I

§A.1. Stirling's formula and the trapezoidal rule

The object of Stirling's formula is to find an asymptotic value for $n!$. We have:

Theorem A.1.1 (Stirling's formula). *$n!$ is asymptotically equal to*

$$\sqrt{n}.n^n.\sqrt{2\pi/e^n}\,.$$

Proof. Consider the curve $y = \log x$ (see Figure A.1.1). The gradient $dy/dx = 1/x$, which is greater than 0 and decreasing as x increases; so the curve is convex downwards.

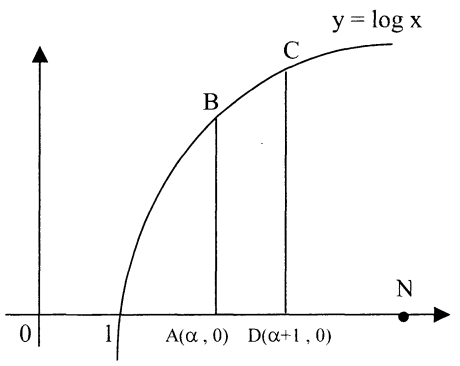


Figure A.1.1

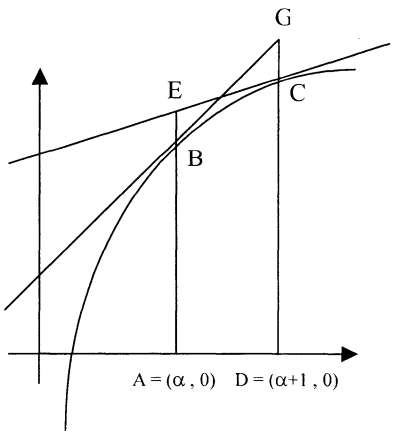


Figure A.1.2

Hence, area of the trapezium $ABCD$ is less than $\int_\alpha^{\alpha+1} \log t\,dt$, i.e., $\frac{1}{2}(\log\alpha + \log(\alpha+1)) < \int_\alpha^{\alpha+1} \log t\,dt$. Summing from $\alpha = 1$ to $\alpha = N-1$ gives

$$\frac{1}{2}((\log 1 + \log 2) + (\log 2 + \log 3) + \cdots + (\log(N-1) + \log N)) < \int_1^N \log t\,dt\,.$$

Here the left hand side is equal to $\log N! - \frac{1}{2}\log N$, while the right hand side is equal to $N\log N - N + 1$. Hence

$$C_N = \left(N + \frac{1}{2}\right)\log N - N + 1 - \log N! > 0\,. \tag{1}$$

However, C_N is equal to the difference of area under $y = \log x$ from 1 to N and the trapezoidal approximation to this area. Hence

$$C_N \text{ increases as } N \text{ increases .} \tag{2}$$

We now show that C_N is bounded above; indeed that

$$C_N < 1/4 \ . \tag{3}$$

For this we look at the Figure A.1.2. Let the tangents to $y = \log x$ at B, C meet DC, AB in G, E respectively. The coordinates of G are $(\alpha + 1, (\log \alpha) + 1/\alpha)$ and the coordinates of E are $(\alpha, \log(\alpha + 1) - 1/(\alpha + 1))$. This is because $B = (\alpha, \log \alpha)$, the equation of the tangent at B is $y = x/\alpha + (\log \alpha - 1)$, the slope of this tangent is $(dy/dx)_B = 1/\alpha$ and this tangent passes through B. This gives G.

Similarly, since $C = (\alpha + 1, \log(\alpha + 1))$, the equation of the tangent at C is $y = x/(\alpha + 1) + \log(\alpha + 1) - 1$, which then gives E. So now

$$\text{Area of trapezium } ABGD = \frac{1}{2}(\log \alpha + (1/\alpha + \log \alpha))$$

and

$$\text{Area of trapezium } AECD = \frac{1}{2}(\log(\alpha + 1) + \log(\alpha + 1) - 1/(\alpha + 1)).$$

Since the area of each trapezium is greater than the area under $y = \log x$, it follows that the average area of these trapezia is greater than the area under the curve, i.e.

$$\frac{1}{2}\left(\frac{1}{2}(2\log \alpha + 1/\alpha\right) + \frac{1}{2}\left(2\log(\alpha + 1) - 1/(\alpha + 1)\right) > \int_{\alpha}^{\alpha+1} \log t \, dt \ .$$

Again summing this from $\alpha = 1$ to $\alpha = N - 1$, we get (see (3))

$$\int_1^N \log t \, dt < \log N! - \frac{1}{2}\log N + \frac{1}{4} - \frac{1}{4N} \ ,$$

which gives $C_N < 1/4 - 1/4N < 1/4$. Also (2) and (3) imply that $C_N \to C$ as $N \to \infty$, where $0 < C \leq 1/4$. But now $C_N = \log(N^N\sqrt{N}) - \log e^N + \log e - \log N! = \log(N^N\sqrt{N}.e^{-N}.e/N!)$. Since $C_N \to C$, we get $e^{C_N} \to e^C$, i.e., $N^N\sqrt{N}/N!.e^N \to e^{C-1}$, or

$$\lim_{N\to\infty} N!.e^N/N^N\sqrt{N} \text{ exists and equals } e^{1-C} \ . \tag{4}$$

The existence of this limit is the essential part of Stirling's formula. We have, however, still to find the value of the limit. This is done by using the expression for π in Corollary 1.3.6 as follows: Let the left side of (4) be denoted by f_n

(replacing N by n), so that $n! = \sqrt{n}n^n e^{-n} f_n$. Then the left side of Corollary 1.3.6 equals

$$\frac{2^{4n}n^2n^{4n}e^{-4n}f_n^4}{(\sqrt{2n}(2n)^{2n}e^{-2n}f_{2n})^2(2n+1)} = \frac{f_n^4}{2f_{2n}^2((2n+1)/n)} ,$$

(on cancellation) and letting $n \to \infty$, we see that this tends to $e^{4(1-C)}/4e^{2(1-C)} = e^{2(1-C)}/4$. This, coupled with Corollary 1.3.6, gives $e^{2(1-C)}/4 = \pi/2$, giving $e^{1-C} = \sqrt{(2\pi)}$, which completes the proof of Stirling's formula.

The above proof is due to A.J.Coleman [27].

Theorem A.1.2 (The trapezoidal rule). *Let f, f' be continuous in $[a,b]$ and suppose f'' exists and is continuous in (a,b); then, there exists a $\xi \in (a,b)$, such that*

$$E = \int_a^b f(x)dx - (b-a)(f(a)+f(b))/2 = -f''(\xi)(b-a)^3/12 \qquad (*)$$

Proof. Let $y = \phi(x)$ be the equation of the line joining the point A with B, so that (see Figure A.1.3):

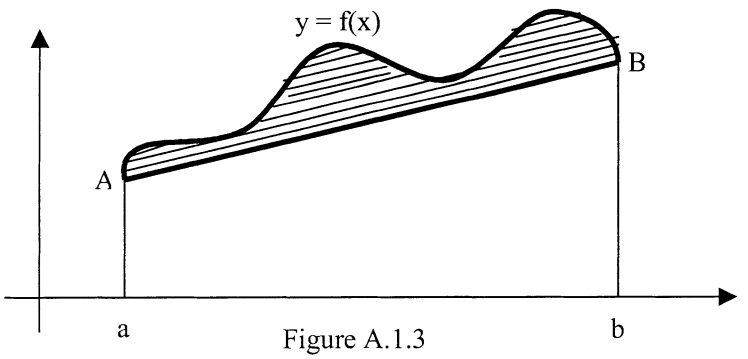


Figure A.1.3

$$E = \int_a^x f(t)\,dt - \int_a^x \phi(t)\,dt .$$

Put $g(x) = \int_a^x f(t)\,dt - \int_a^x \phi(t)\,dt + (2((x-a)/(b-a))^3 - 3((x-a)/(b-a))^2)E$. Then $g(a) = g(b) = 0$ and so by Rolle's theorem, there exists u, $a < u < b$, with $g'(u) = 0$.

Now, $g'(x) = f(x) - \phi(x) + (6.((x-a)/(b-a))^2.(1/(b-a)) - 6.((x-a)/(b-a)).(1/(b-a)))E$, and so $g'(a) = g'(b) = 0$ and so again by Rolle's theorem, there exist points c, d such that $a < c < u < d < b$, and with $g''(c) = g''(d) = 0$, and we conclude, once more, by Rolle's theorem, that there exists a point ξ, $c < \xi < d$, with $g'''(\xi) = 0$, i.e.,

$$0 = g'''(\xi) = f''(\xi) + 12.E/(b-a)^3 ,$$

since $\phi''(x) \equiv 0$, ϕ being linear; and that proves the theorem.

As a corollary to the trapezoidal rule we prove the existence part of the limit in Stirling's formula:

Corollary A.1.3. $\lim_{n\to\infty} K_n = \lim_{n\to\infty} n^n\sqrt{n}/n!e^n$ *exists (and equals* K*, say).*

Proof ([50]). Let $f(t) = \log t$, $a = k-1$, $b = k$ in the trapezoidal rule, to obtain (see Figure A.1.4)

$$\int_{k-1}^{k} \log t\, dt - (\log(k-1)+\log k)/2 = (-1/12).f''(\xi_k) = 1/12\xi_k^2 \; k-1 < \xi_k < k.$$

Summing this from $k = 2$ to $k = n$ gives

$$\begin{aligned}\int_1^n \log t\, dt - \frac{1}{2}((\log 1 + \log 2) + (\log 2 + \log 3) + \cdots + (\log(n-1) + \log n)) \\ = (n\log n - n + 1) - (\log 2 + \log 3 + \cdots + \log n) + \frac{1}{2}\log n \\ = (\log n^n - \log e^n + 1) - \log n! + \log\sqrt{n} \\ = (\log n^n\sqrt{n}/n!e^n) + 1 \\ = \log K_n + 1.\end{aligned}$$

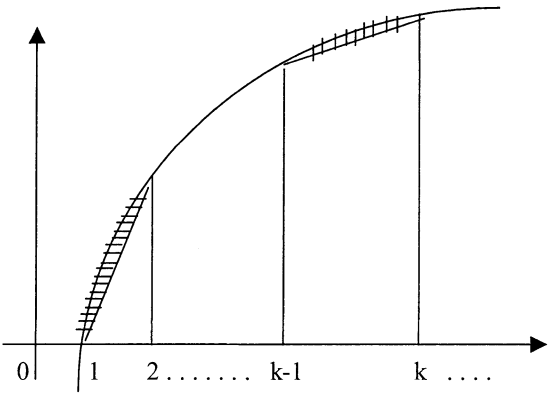


Figure A.1.4

It follows that $\log K_n = \sum_{k=2}^{n} 1/12\xi_k^2 - 1$. But now, since $k-1 < \xi_k < k$, we have $1/\xi_k < 1/(k-1)$, and hence $\sum_{k=2}^{\infty} 1/\xi_k^2 < \sum_{k=2}^{\infty} 1/(k-1)^2 = \sum_{k=1}^{\infty} 1/k^2 = \pi^2/6$, i.e., $\log K_n$ is bounded above. Moreover, $\log K_n$ increases as n increases, because positive terms get added to it, when n increases. It follows that $\log K_n$ tends to a limit and therefore so does K_n, as required.

The value of this limit can not be determined by this method; however, we

can give an estimate for it as follows: We have

$$\sum_{k=n+1}^{\infty} 1/(12\xi_k^2) < (1/12) \sum_{k=n+1}^{\infty} 1/(k-1)^2 < (1/12) \int_{n-1}^{\infty} (1/x^2)\, dx = 1/(12(n-1))$$

(here the inequality between summation and integral follows from Figure A.1.5, where the shaded area is less than the area under the curve $y = 1/x^2$ from $n-1$ to ∞).

Similarly, since $1/\xi_k > 1/k$, so $1/\xi_k^2 > 1/k^2$, we have

$$\sum_{k=n+1}^{\infty} 1/(12\xi_k^2) > (1/12) \sum_{k=n+1}^{\infty} 1/k^2 > (1/12) \int_{n+1}^{\infty} (1/x^2)\, dx = 1/(12(n+1))$$

(again the inequality between the summation and integral follows from Figure A.1.6 where the shaded area is greater than the area under the curve $y = 1/x^2$ from $n+1$ to ∞).

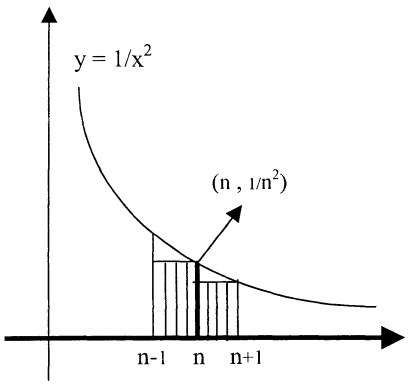


Figure A.1.5

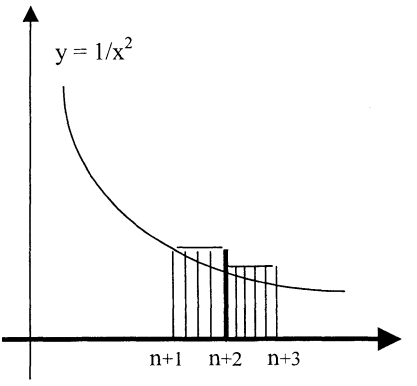


Figure A.1.6

Thus

$$1/12(n+1) < \sum_{k=n+1}^{\infty} 1/12\xi_k^2 < 1/12(n-1)\,,$$

as required. Taking $n = 2$, this gives

$$1/36 < \sum_{k=3}^{\infty} 1/12\xi_k^2 < 1/12\,.$$

Adding $1/12\xi_2^2 - 1$ throughout, gives

$$1/12\xi_2^2 + 1/36 - 1 < \log K < 1/12\xi_2^2 + 1/12 - 1.$$

Putting in the estimate $1 < \xi_2 < 2$, i.e., $1/4 < 1/\xi_2^2 < 1$, this gives $1/48 + 1/36 - 1 < \log K < 1/12 + 1/12 - 1$, i.e., $-137/144 < \log K < -5/6$, which

gives the following

Corollary A.1.4. *$e^{-137/144} < K < e^{-5/6}$, where $K = \lim_{n\to\infty} n^n\sqrt{n}/n!e^n$ (which we know, from Stirling's formula, to be equal to $1/\sqrt{(2\pi)}$).*

We now estimate the error in trapezoidal rule for a concave curve.

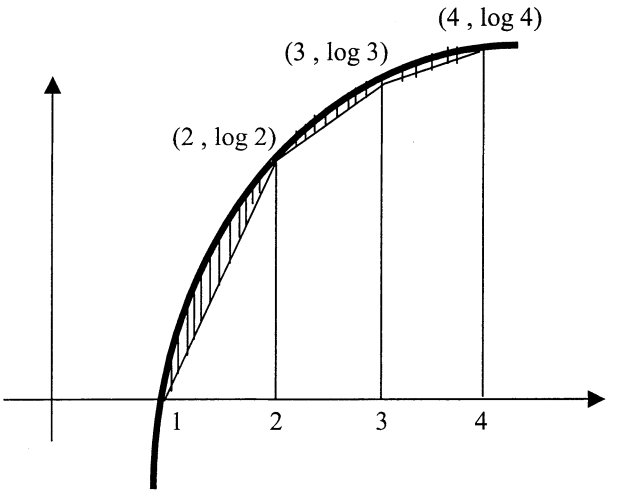


Figure A.1.7

For example, consider the area between $y = \log x$ and the polygonal approximation by vertices $(1, \log 1)$, $(2, \log 2)$, ..., (see the shaded area in Figure A.1.7). This region is unbounded, yet the area is finite. This finiteness of the shaded area is a corollary of the following (see [98])

Theorem A.1.5. *Let f be defined for $x \geq 1$ and suppose $f(x) \geq 0$, $f'(x) \geq 0$, $f''(x) \leq 0$, for all $x \geq 1$ (which is the case for the function $y = \log x$, which has been drawn in Figure A.1.7). For $n \in \mathbb{N}$, let*

$$T_n = \int_1^n f(t)\,dt - \sum_1^{n-1} (\textit{the area of the relevant trapezia}) ,$$

then $T_n \leq (f(2) - f(1))/2$.

Proof. Let $i-1$, i, $i+1$ be 3 consecutive integers and consider Figure A.1.8, which is the key to the proof. The line t is tangent to the graph of f at the point $B = (i, f(i))$. The area of the shaded region represents the errors of adjacent summands in the trapezoidal approximation.

Now translate the shaded region R between B and C, by one unit to the left, to become the region R' and note that no point of R' lies below t. Thus R' and the shaded region S between A and B are disjoint, except for B. The two shaded regions between $i-1$ and i, (viz. S and R') can now be similarly

shifted to the left by one unit (between $i-2$ and $i-1$) and so on till everything between 1 and n is fitted between 1 and 2.

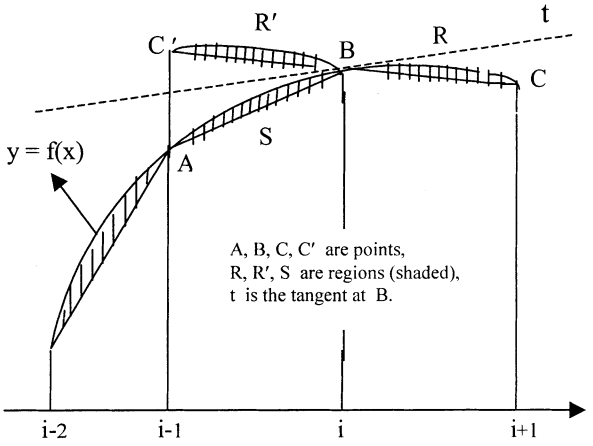


Figure A.1.8

Since the curve is increasing (because $f' \geq 0$), all the shaded regions between 1 and n are fitted in the triangle Δ, with vertices $(1, f(1))$, $(1, f(2))$, $(2, f(2))$ (see the figure for $n = 4$, i.e., Figure A.1.9). It follows that $T_n \leq$ Area of $\Delta = (f(2) - f(1))/2$, which completes the proof.

Letting $n \to \infty$, we get

Corollary A.1.6. $\sum_{n=1}^{\infty}(\int_n^{n+1} f(x)\,dx - (f(n) + f(n+1))/2) \leq (f(2) - f(1))/2$.

Proof. $\int_n^{n+1} f(x)\,dx - (f(n) + f(n+1))/2$ is the trapezoidal error between $[n, n+1]$ and the corollary follows.

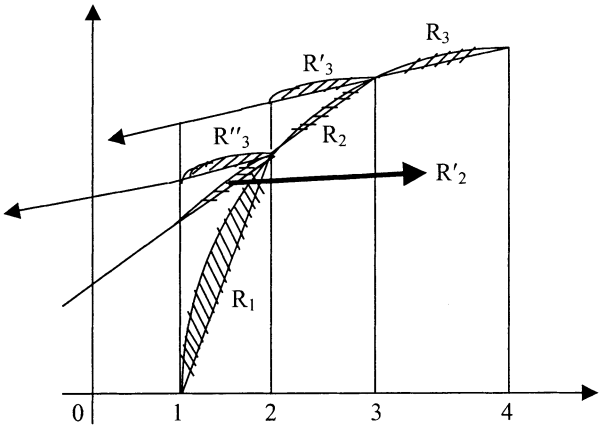


Figure A.1.9

This shows that the infinite series $\sum R_i$ (R_i are the shaded regions in the diagram) is convergent, with sum less than or equal to $(f(2) - f(1))/2$.

Using the proof of Theorem A.1.1 and the trapezoidal rule (Theorem A.1.2), we prove the following interesting

Corollary A.1.7. *The shaded region for the function $f(x) = \log x$, between 1 and ∞ (see Figure A.1.4) has area equal to $1 - \log\sqrt{(2\pi)}$.*

Proof. This shaded region has area precisely equal to (see proof of Corollary A.1.3)

$$\begin{aligned}\lim_{n\to\infty}\Big(\int_1^n \log t\,dt - \frac{1}{2}((\log 1+\log 2)+(\log 2+\log 3)+\cdots+(\log(n-1)+\log n))\Big)\\ &= \lim_{n\to\infty}(\log K_n + 1) \quad \text{(as proved in Corollary A.1.3)}\\ &= \log K + 1\\ &= \log(1/\sqrt{(2\pi)}) + 1 \quad \text{(by Stirling's formula)}\\ &= 1 - \log\sqrt{(2\pi)}\,,\end{aligned}$$

as required.

To provide yet another example of the trapezoidal rule, we derive the existence of Euler's constant γ, in the following:

Corollary A.1.8. $\gamma = \lim_{n\to\infty}(1 + 1/2 + \cdots + 1/n - \log n)$ *exists and satisfies*

$$2 + \zeta(3) < 6\gamma < 3 + \zeta(3)\,,$$

where, $\zeta(3) = 1 + 1/2^3 + 1/3^3 + \cdots$.

Proof. Take $f(t) = 1/t$ in Theorem A.1.2, so that $f'(t) = -1/t^2$, $f''(t) = 2/t^3$. The trapezoidal rule now gives

$$\int_{k-1}^{k} \frac{dt}{t} - (1/(k-1) + 1/k)/2 = -(1/12).f''(\xi_k)\,,$$

where $k - 1 < \xi_k < k$. We sum this equation from $k = 2$ to $k = n$, to get

$$\int_1^n \frac{dt}{t} - \frac{1}{2}((1 + 1/2) + (1/2 + 1/3) + \cdots + (1/(n-1) + 1/n)) = \sum_{k=2}^{n} -1/6\xi_k^3\,,$$

which gives, on simplification,

$$1 + 1/2 + \cdots + 1/n - \log n = \frac{1}{2}(1 + 1/n) + (1/6).\sum_{k=2}^{n} \xi_k^3\,.$$

Letting $n \to \infty$, this yields

$$1 + 1/2 + \cdots + 1/n - \log n \;\longrightarrow\; 1/2 + (1/6)\sum_{k=2}^{\infty} \xi_k^3\,.$$

Now use the estimate $k-1<\xi_k<k$ to get $1/k^3<1/\xi_k^3<1/(k-1)^3$; hence the above gives

$$\zeta(3)-1<\sum_{k=2}^{\infty}\xi_k^3<\zeta(3)\,.$$

Thus, we have proved that $\gamma=\lim_{n\to\infty}(1+1/2+\cdots+1/n-\log n)$ exists and satisfies $1/2+\zeta(3)/6-1/6<\gamma<1/2+\zeta(3)/6$, i.e., $2+\zeta(3)<6\gamma<3+\zeta(3)$, where, $\zeta(3)=1+1/2^3+1/3^3+\cdots$, as required.

§A.2. Schwarz differentiability

This is a rather specialized topic, one that offers excellent problems for non-routine examinations. Throughout our discussion, we suppose f to be a real-valued function defined on an open interval $\mathcal{I}$ of which $[a,b]$ is a closed subinterval (this is to avoid left hand limits at b and right hand limits at a).

Schwarz differentiability, also known as symmetric differentiability (s-differentiability; similarly s-differentiable, s-derivative for Schwarz derivative, etc) is defined as follows:

Definition A.2.1. For each $x\in[a,b]\subset\mathcal{I}$, if $\lim_{h\to0}\{(f(x+h)-f(x-h))/2h\}$ exists and is finite, we say f is s-differentiable in $[a,b]$ and the limit is called the s-derivative (i.e., the symmetric derivative or the Schwarz derivative) of f at x and is denoted by $f^{(s)}(x)$.

The following are some easy consequences of this definition:
(1) If $f'(x)$ (the usual derivative) exists at x, so does $f^{(s)}(x)$ and $f'(x)=f^{(s)}(x)$.
Proof. $(f(x+h)-f(x-h))/2h=(f(x+h)-f(x))/2h+(f(x)-f(x-h))/2h$ and taking limits as $h\to0$, the result follows.
(2). The converse to (1) is false; for example let $f(x)=|x|$; then at the point $x=0$, we have $f^{(s)}(0)=\lim(f(0+h)-f(0-h))/2h=\lim(f(0+h)-f(0-h))/2h=\lim(h-h)/2h=0$ exists, but $f'(0)$ does not exist.
(3) s-differentiability does not even imply continuity. Take, for example, the function

$$f(x)=\begin{cases}0 & \text{if } x\neq0,\\ 1 & \text{if } x=0\ .\end{cases}$$

Then at the point $x=0$, $\lim_{h\to0}(f(0+h)-f(0-h))/2h$ exists and equals 0, i.e., $f^{(s)}(0)=0$, but f is not continuous at $x=0$.

We see that the existence of $f^{(s)}(x)$ at the point x, is a much weaker requirement than the existence of $f'(x)$ at x. In view of this, all the basic elementary theorems regarding differentiabilty, such as Rolle's theorem, the first mean value theorem, etc. will have to be reformulated in terms of s-differentiability.

Thus, for example, we have
(4) It is not true that $f(x)$ is a constant when $f^{(s)}(x)=0$ in $[a,b]$, for example,

let

$$f(x) = \begin{cases} 0 & \text{if } x \neq 0 \,, \\ 1 & \text{if } x = 0 \,. \end{cases}$$

Then $f^{(s)}(x) = 0$ for all x but f is not a constant.

This failure is expected, since, we have
(5) Rolle's theorem too, is false for s-derivatives. For example, the function $f(x) = |x|$ in the interval $[-1, 1]$.

We shall deal with this problem after proving some easier results and the reformulated Rolle's theorem and the mean value theorem.
(6) It may happen that $f^{(s)}(a)$ exists and is finite but $f(a)$ does not exist (is not defined); for example, let $f(x) = 1/x^2$. Then $f^{(s)}(x) = 0$ for all x, including $x = 0$, but $f(0)$ is not defined.

Example A.2.2. For Dirichlet's function

$$f(x) = \begin{cases} 1 & \text{if } x \in \mathbb{Q} \,, \\ 0 & \text{otherwise} \,, \end{cases}$$

$f^{(s)}(x)$ exists for all $x \in \mathbb{Q}$ but not for $x \notin Q$.
(7) Even if f is continuous at $x = a$ and $f^{(s)}(a)$ exists, it does not follow that $f'(x)$ exists; for example take $f(x) = |x|$ and the point $x = 0$.
(8) Even if $f^{(s)}(x)$ is continuous in a neighbourhood of a and $f(x)$ is continuous at $x = a$, it does not follow that $f'(a)$ exists. For example, take $f(x) = 1/n$ if $x = \pm 1/n$, $(n \in \mathbb{N})$, $f(x) = 0$ otherwise. It is easily verified that $f^{(s)}(x) = 0$ for all x, so that $f^{(s)}(x)$ is continuous for all x. Further, $f(x)$ is continuous at $x = 0$. However, $f'(0)$ does not exist, because for $h = \pm 1/n$, $(f(0+h) - f(0)/h)$ equals ± 1, while for other values of h, this equals $0/h = 0$, so the limit of the quotient does not exist as $h \to 0$.

Theorem A.2.3. *If $f^{(s)}$ is continuous at $x = a$ and f is continuous in a neighbourhood of a, it does follow that $f'(a)$ exists.*

The proof of this is rather more involved than one would imagine at first glance and uses results analogous to Rolle's theorem and the mean value theorem, which we now develop.

Theorem A.2.4. *Let f be continuous in $[a, b]$, let $f^{(s)}$ exist in (a, b) and let $f(b) > f(a)$. Then there exists a number $c \in (a, b)$ such that $f^{(s)}(c) \geq 0$.*

Proof. Let k be such that $f(a) < k < f(b)$. The set $E = \{x \mid f(x) > k, a < x < b\}$ is bounded below by a. Let $c = \inf E$, then $c \neq a$ since $f(a)$ is not greater than or equal to k.

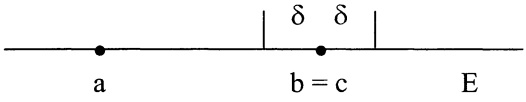


Figure A.1.10

Also $c \neq b$, for otherwise, since $f(x)$ is continuous at $x = c$, and $f(b) > k$, it would follow that there exists a $\delta > 0$ such that $f(x) > k$ for all $x \in (b-\delta, b+\delta)$ and so numbers less than c (e.g. $c - \delta/2$) would belong to E, contradicting the fact that c is the infimum of E. Furthermore, $f^{(s)}(x) \geq 0$, for, there exist points $x > c$ in every neighbourhood of c such that $f(x) > k$ while $f(x) \leq k$ for $x \in [a, c]$, so for $h > 0$, we have

$$(f(c+h) - f(c-h))/2h > (k-l)/2h > 0 \quad \text{where } l \leq k \ ,$$

giving $\lim_{h \to 0}[(f(c+h) - f(c-h))/2h] \geq 0$. But since $f^{(s)}(c)$ exists, so $f^{(s)}(c) \geq 0$.

A similar result holds with the inequality reversed, i.e.,

Theorem A.2.5. *Let f be continuous in $[a, b]$, let $f^{(s)}(x)$ exist in (a, b), and let $f(b) < f(a)$. Then there exists a number $c \in (a, b)$ such that $f^{(s)}(c) \leq 0$.*

Theorem A.2.6 (Analogue of Rolle's theorem). *Let f be continuous in $[a, b]$, let $f^{(s)}$ exists in (a, b) and let $f(a) = f(b) = 0$. Then there exists an $x_1 \in (a, b)$ such that $f^{(s)}(x_1) \geq 0$ and an $x_2 \in (a, b)$ such that $f^{(s)}(x_2) \leq 0$.*

Proof. If $f(x) \equiv 0$, the result is trivial. So let f be not identically zero; then there exists either a number c such that $f(c) > 0$ or a number d such that $f(d) < 0$ or both. By the above theorems, there exist x_1, x_2, with either $a < x_1 < c < x_2 < b$ or $a < x_2 < d < x_1 < b$ such that $f^{(s)}(x_1) \geq 0$ and $f^{(s)}(x_2) \leq 0$.

Theorem A.2.7 (Analogue of mean value theorem). *Let f be continuous in $[a, b]$ and let $f^{(s)}$ exist in (a,b). Then there exists points x_1, x_2, with $a < x_1, x_2 < b$ such that $f^{(s)}(x_2) \leq (f(b) - f(a))/(b-a) \leq f^{(s)}(x_1)$.*

Proof. This follows from Theorem A.2.6 in exactly the same way the mean value theorem for ordinary derivatives follows from Rolle's theorem (for ordinary derivative): just rotate the axes!

Theorem A.2.8. *Let f and $f^{(s)}$ be continuous in (a, b). Then $f'(x)$ exists and $f'(x) = f^{(s)}(x)$.*

Proof. Take h sufficiently small to ensure $a < x + h < b$. Then by Theorem A.2.7, there exist x_1, x_2 with $x < x_1, x_2 < x + h$ such that $f^{(s)}(x_2) \leq (f(x+h) - f(x))/h \leq f^{(s)}(x_1)$. If strict inequality holds at either end, we apply the intermediate value theorem for continuous functions. So there exists x_3 such that $f^{(s)}(x_3) = (f(x+h) - f(x))/h$, $x < x_3 < x + h$. Letting $h \to 0$

and noting that the left hand limit exists, we get $f^{(s)}(x) = f'(x)$.

We now give :

Proof of Theorem A.2.3. Given $\epsilon > 0$, there exist a neighbourhood N_a of a such that $f(x)$ is continuous in N_a and, since $f^{(s)}(x)$ is continuous at a, so

$$f^{(s)}(a) - \epsilon < f^{(s)}(x) < f^{(s)}(a) + \epsilon \,, \tag{1}$$

for all $x \in N_a$. By A.2.7, there exist x_1, x_2, strictly between a and $a+h$, such that

$$f^{(s)}(x_2) < (f(a+h) - f(a))/h < f^{(s)}(x_1) \,, \tag{2}$$

here $h \neq 0$ is such that $a + h \in N_a$. Now (1) and (2) imply that

$$f^{(s)}(a) - \epsilon < (f(a+h) - f(a))/h < f^{(s)}(a) + \epsilon \,. \tag{3}$$

Letting $h \to 0$, it follows that $f'(a)$ exists and equals $f^{(s)}(a)$.

We are now ready to prove a result analogous to the result: *if* $f'(x) = 0$, *then* $f(x)$ *is a constant*, in the Schwarz derivative set up. Indeed, we have the following :

Theorem A.2.9. *Suppose* f *is defined and is continuous in an open interval* I *of which* $[a, b]$ *is a closed subinterval and suppose* $f^{(s)}$ *exists in* I. *Then*

$$f^{(s)}(x) = 0 \ \textit{in} \ I \ \Rightarrow \ f(x) \ \textit{is a constant in} \ [a, b] \,.$$

Proof. Since $f^{(s)}(x) = 0$ in I, it is a continuous function and so by Theorem A.2.8, $f'(x)$ exists in $[a, b]$ and equals $f^{(s)}(x)$, i.e., equals 0. Thus $f'(x) = 0$ in $[a, b]$ giving $f(x) = k$.

Before we go on to uniform differentiability, we quickly introduce the second Schwarz derivative of $f(x)$, written as $f^{(ss)}(x)$. It is defined to be the Schwarz derivative of $f^{(s)}(x)$. Thus

$$\begin{aligned} f^{(ss)}(x) &= \lim_{h \to 0} [(f^{(s)}(x+h) - f^{(s)}(x-h))/2h] \quad \text{(provided the limit exists)} \\ &= \lim_{h \to 0} \frac{\lim\limits_{k \to 0} \frac{f(x+h+k) - f(x+h-k)}{2k} - \lim\limits_{k \to 0} \frac{f(x-h+k) - f(x-h-k)}{2k}}{2h} \,. \end{aligned}$$

But if the limit exists, as we suppose it does, then we may choose $k = h$; so that

$$\begin{aligned} f^{(ss)}(x) &= \lim_{h \to 0} \frac{[(f(x+2h) - f(x))/2h] - [(f(x) - f(x-2h))/2h]}{2h} \\ &= \lim_{h/2 \to 0} \frac{f(x+h) - 2f(x) + f(x-h)}{h^2} \quad \text{(writing } h/2 \text{ for } h) \\ &= \lim_{h \to 0} \frac{f(x+h) - 2f(x) + f(x-h)}{h^2} \quad \text{(as } h/2 \to 0, \text{ so does } h) \end{aligned}$$

Although, it looks difficult, the following result is easily proved:

Theorem A.2.10 (Schwarz, see [72]). *Suppose f is continuous in I and suppose $f^{(ss)}(x) = 0$ for all $x \in I$; then $f(x)$ is a linear function of x on any interval $[a, b] \subset I$.*

Proof. Let $\epsilon > 0$ and let

$$\phi(x) = f(x) - \left[f(a) + \left(\frac{f(b) - f(a)}{b - a}\right).(x - a)\right] + \epsilon(x - a)(x - b)\,,$$

so that $\phi(x)$ is continuous in $[a, b]$ and $\phi(a) = \phi(b) = 0$. Furthermore, using that $f^{(ss)}(x) = 0$ and the fact that if a function g is differentiable then $g'(x) = g^{(s)}(x)$, it may be easily checked that

$$\phi^{(ss)}(x) = 2\epsilon\,. \qquad (*)$$

We shall show first that

$$\phi(x) \leq 0 \quad \forall\, x \in [a, b]\,. \qquad (1)$$

Suppose not; then $\phi(x)$ assumes its maximum value $\phi(x_0)$ at an interior point $x_0 \in (a, b)$ (ϕ being continuous and so bounded and since ϕ attains its maximum). Then $(\phi(x_0 + h) - 2\phi(x_0) + \phi(x_0 - h))/2h \leq 0$ (since $\phi(x_0)$ is the maximum value). Letting $h \to 0$ gives $\phi^{(ss)}(x_0) \leq 0$, which contradicts (*).

Similarly the function

$$\psi(x) = -\left[f(x) - \left(f(a) + \left(\frac{f(b) - f(a)}{b - a}\right).(x - a)\right)\right] + \epsilon(x - a)(x - b)\,,$$

satisfies

$$\psi(x) \leq 0 \quad \forall\, x \in [a, b]\,. \qquad (2)$$

Combining (1) and (2) gives

$$\left|f(x) - \left(f(a) + \left(\frac{f(b) - f(a)}{b - a}\right).(x - a)\right)\right| \leq \epsilon|(x - a)(x - b)|$$

and since ϵ is arbitrary, it follows that the left hand side is 0, which gives

$$f(x) = f(a) + ((f(b) - f(a))/(b - a)).(x - a)\,,$$

a linear function of x, as required.

Remark A.2.11. We know that f' exists implies that $f^{(s)}$ exists and $f^{(s)} = f'$. It may be shown that f'' exists implies that $f^{(ss)}$ exists and $f^{(ss)} = f''$, for f'' exists $\Rightarrow$ $(f')'$ exists $\Rightarrow$ $(f^{(s)})^{(s)}$ exists and equals $(f^{(s)})'$ and hence $f^{(ss)}$ exists and equals $(f')' = f''$.

Alternatively, let $g(h) = f(x+h) + f(x-h)$, so that (by the Cauchy mean value theorem)

$$\begin{aligned}\frac{f(x+h) - 2f(x) + f(x-h)}{h^2} &= \frac{g(h) - g(0)}{h^2} \\ &= g'(\theta h)/2\theta h \quad (0 < \theta < 1) \\ &= (f'(x+\theta h) + f'(x-\theta h))/2\theta h \\ &\to f''(x) \text{ as } h \to 0\,,\end{aligned}$$

as required.

The converse is false, for example, let

$$f(x) = \int_0^x t \sin(1/t)\, dt\,;$$

then $f^{(ss)}$ exists at the point $x = 0$ but f'' does not.

We would now like to introduce the concept of *uniform Schwarz differentiability* in $[a, b]$.

Definition A.2.12. Let f be Schwarz differentiable at each point $x \in I$. We say that f is uniformly Schwarz differentiable in $[a, b]$ if given $\epsilon > 0$, there exists a number $\delta = \delta(\epsilon)$, δ independent of x, such that $|(f(x+h) - f(x-h))/2h - f^{(s)(x)}| < \epsilon$ if $|h| < \delta$ for all $x \in [a, b]$, $x \pm h \in I$.

According to Reich [85], the concept of uniform Schwarz differentiability is due to S.N. Mukhopadhyay.

Our aim is to show that $f'(x)$ exists, given the existence of $f^{(s)}(x)$ and certain other suitable conditions (cf. 7, 8 and Theorem A.2.3). We prove the following two simple results:

Theorem A.2.13 (Simeon Reich [91]). *Suppose $f^{(s)}$ is continuous at the point a and suppose that f is uniformly $s-$differentiable in a neighbourhood of a. Then $f'(a)$ exists and equals $f^{(s)}(a)$.*

Proof. Let $\epsilon > 0$ be given. By the uniform $s-$differentiability of f, there exist $\delta_1, \delta_2 > 0$, with $|(f(a+t+h) - f(a+t-h))/2h - f^{(s)}(a+t)| < \epsilon/2$ for all $|t| < \delta_1$, and all $0 < |h| < \delta_2$. By the continuity of $f^{(s)}(x)$ at a, $|f^{(s)}(a+t) - f^{(s)}(a)| < \epsilon/2$ for all $|t| < \delta_3$ ($\delta_3 > 0$). It follows that

$$\begin{aligned}&\left|\frac{f(a+t+h) - f(a+t-h)}{2h} - f^{(s)}(a)\right| \\ &\quad \le \left|\frac{f(a+t+h) - f(a+t-h)}{2h} - f^{(s)}(a+t)\right| + \left|f^{(s)}(a+t) - f^{(s)}(a)\right| \\ &\quad < \epsilon/2 + \epsilon/2 = \epsilon\,,\end{aligned}$$

if $0 < |t|$, $|h| < \delta/2$, where $\delta = \min\{\delta_1, \delta_2, \delta_3\}$. Putting $t = h$, we obtain $|(f(a+2h) - f(a))/2h - f^{(s)}(a)| < \epsilon$ if $0 < |h| < \delta/2$, or with $2h = h_1$,

$|(f(a+h_1)-f(a))/h_1 - f^{(s)}(a)| < \epsilon$ if $0 < h_1 < \delta$, i.e., $f'(a)$ exists and equals $f^{(s)}(a)$.

Theorem A.2.14. *Suppose $f^{(s)}$ is continuous in I and let*

$$F(x) = \int_a^x f^{(s)}(t)\,dt\ ,\quad x \in [a,b]\ .$$

Then $F^{(s)}(x)$ exists and equals $f^{(s)}(x)$ (cf. the fundamental theorem of Calculus).

Proof.

$$\begin{aligned} F(x+h)-F(x-h) &= \int_a^{x+h} f^{(s)}(t)\,dt - \int_a^{x-h} f^{(s)}(t)\,dt \\ &= \int_a^{x-h} f^{(s)}(t)\,dt + \int_{x-h}^{x+h} f^{(s)}(t)\,dt - \int_a^{x-h} f^{(s)}(t)\,dt \\ &= \int_{x-h}^{x+h} f^{(s)}(t)\,dt\ . \end{aligned}$$

Now

$$\begin{aligned} |f^{(s)}(t)| = |f^{(s)}(t) - f^{(s)}(x) + f^{(s)}(x)| &\le |f^{(s)}(t) - f^{(s)}(x)| + |f^{(s)}(x)| \\ &< \epsilon + |f^{(s)}(x)|\ , \end{aligned}$$

if $|t-x| \le h$, since $f^{(s)}(t)$ is continuous at x. So

$$\begin{aligned} |(F(x+h)-F(x-h))/2h| &\le (1/|2h|)\int_{x-h}^{x+h} |f^{(s)}(t)|\,dt \\ &\le (1/|2h|)\int_{x-h}^{x+h} (\epsilon + |f^{(s)}(x)|\,)\,dt \\ &= (1/|2h|)(\epsilon + |f^{(s)}(x)|\,).|2h| = |f^{(s)}(x)| + \epsilon\ . \end{aligned}$$

Letting $\epsilon \to 0$ and noting that then $h \to 0$, this gives the required result.

Theorem A.2.15. *Suppose f and $f^{(s)}$ are continuous in I, then f is uniformly s-differentiable in $[a,b]$.*

Proof. Let

$$\phi(x) = f(x) - \int_a^x f^{(s)}(t)\,dt\ \ (x \in I)\ .$$

Then $\phi(x)$ is continuous and s-differentiable in I and by Theorem A.2.14, $\phi^{(s)}(x) = f^{(s)}(x) - f^{(s)}(x) = 0$ (for $x \in [a,b]$). It follows, by Theorem A.2.9, that $\phi(x) = k$, a constant, i.e., that $f(x) = k + \int_a^x f^{(s)}(t)\,dt$ (for $x \in [a,b]$).

Hence

$$\begin{aligned}
&\left|\frac{f(x+h)-f(x-h)}{2h} - f^{(s)}(x)\right| \\
&\qquad = \left|\frac{1}{2h}\left(k + \int_a^{x+h} f^{(s)}(t)\,dt - k - \int_a^{x-h} f^{(s)}(t)dt\right) - f^{(s)}(x)\right| \\
&\qquad = \left|(1/2h)\left(\int_{x-h}^{x+h} f^{(s)}(t)\,dt\right) - f^{(s)}(x)\right| \\
&\qquad = |f^{(s)}(x+\theta h) - f^{(s)}(x)| \ , \ \theta \in [-1,1],
\end{aligned}$$

by the mean value theorem for integrals. Now $f^{(s)}(x)$ is continuous in $[a,b]$ and hence uniformly continuous. So given $\epsilon > 0$, there exists a $\delta > 0$ (independent of x) such that $|f^{(s)}(x+h) - f^{(s)}(x)| < \epsilon$ if $|h| < \delta$, $x \in [a,b]$. It follows that $|(f(x+h) - f(x-h))/2h - f^{(s)}(x)| < \epsilon$ if $|h| < \delta$, for all $x \in [a,b]$, i.e., $f(x)$ is uniformly s-differentiable in $[a,b]$.

Remark A.2.16. The continuity of f is necessary for the conclusion of Theorem A.2.15 to hold; for example, let $f(x) = 0$ if $x \neq 0$, and 1 if $x = 0$. Then $f^{(s)}(x) = 0$ for all x and so, is continuous everywhere, but f is not uniformly s-differentiable in any $[a,b]$ which contains the point 0.

The converse of the above theorem also holds, viz.

Theorem A.2.17 (Mukhopadhyay [69]). *If f is continuous and uniformly s-differentiable in I, then $f^{(s)}$ is continuous in $[a,b]$.*

Proof. Let $\xi \in [a,b]$. We have to show that $|f^{(s)}(\xi+h) - f^{(s)}(\xi)| < \epsilon$ if $|h| < \delta$.

Since $f^{(s)}$ is uniformly s-differentiable, so $-\epsilon/3 < (f(x+k) - f(x-k))/2k - f^{(s)}(x) < \epsilon/3$ if $|k| < \delta_1$ (for all x). Take $x = \xi + h$, to get

$$\frac{f(x+h+k)-f(x+h-k)}{2k} - \frac{\epsilon}{3} < f^{(s)}(\xi+h) < \frac{f(x+h+k)-f(x+h-k)}{2k} + \frac{\epsilon}{3} \quad (1)$$

Take $x = \xi$ and change sign throughout and rewrite to get

$$-\frac{f(\xi+k)-f(\xi-k)}{2k} - \frac{\epsilon}{3} < -f^{(s)}(\xi) < -\frac{f(\xi+k)-f(\xi-k)}{2k} + \frac{\epsilon}{3} \quad (2)$$

Adding (1) and (2) gives

$$\begin{aligned}
&(f(\xi+h+k) - f(\xi+h-k) - f(\xi+k) + f(\xi-k))/2k - 2\epsilon/3 < f^{(s)}(\xi+h) - f^{(s)}(\xi) \\
&\qquad < (f(\xi+h+k) - f(\xi+h-k) - f(\xi+k) + f(\xi-k))/2k + 2\epsilon/3 \ ,
\end{aligned}$$

if $|k| < \delta_1$, i.e., say for $A = f(\xi+h+k) - f(\xi+h-k) - f(\xi+k) + f(\xi-k)$,

$$A/2k - 2\epsilon/3 < |f^{(s)}(\xi+h) - f^{(s)}(\xi)| < A/2k + 2\epsilon/3 \ , \text{ if } |k| < \delta_1 \quad (3)$$

But now, since f is continuous at $x = \xi \pm k$, we have $f(\xi + k + h) - f(\xi + k) < 2k.(\epsilon/6)$ if $|h| < \delta_2$ and $f(\xi - k + h) - f(\xi - k) < 2k.(\epsilon/6)$ if $|h| < \delta_3$. Then

$$|A| = |f(\xi + h + k) - f(\xi + h - k) - f(\xi + k) + f(\xi - k)| < 2k.(\epsilon/6) + 2k.(\epsilon/6)$$

if $|h| < \min\{\delta_2, \delta_3\} = \delta$, say, i.e.,

$$-\epsilon/3 < A/2k < \epsilon/3 \text{ , if } |h| < \delta \text{ .} \tag{4}$$

Substituting in (3) gives

$$-\epsilon/3 - 2\epsilon/3 < |f^{(s)}(\xi + h) - f^{(s)}(\xi)| < \epsilon/3 + 2\epsilon/3 \text{ ,}$$

if $|h| < \delta$, $|k| < \delta_1$. Now making sure that $|k| < \delta_1$, this completes the proof.

Theorem A.2.18 (Mukhopadhyay [71]). *If f is uniformly $s-$differentiable in I, then*

(i) f is continuous in $[a, b]$ if $f^{(s)}$ is bounded in $[a, b]$,

(ii) $f^{(s)}$ is bounded in $[a, b]$ if f is continuous in I.

Proof. Suppose that f is discontinuous at the point $c \in [a, b]$. Then there exists $\eta > 0$, such that for each $\delta > 0$, there are points $x^* \in [a, b] \cap (c-\delta, c+\delta)$ such that

$$|f(x^*) - f(c)| > \eta \tag{1}$$

Further, since $f^{(s)}$ is bounded in $[a, b]$, there exists $M > 0$, such that

$$|f^{(s)}(x)| \leq M \text{ for all } x \in [a, b] \text{ .} \tag{2}$$

Hence, taking $x = (x^* + c)/2$, $h = (x^* - c)/2$ in the definition of uniform s-differentiability, we get

$$\begin{aligned}
&\left| \frac{f(x+h) - f(x-h)}{2h} - f^{(s)}(x) \right| \\
&\qquad = \left| \frac{f(\frac{x^*+c}{2} + \frac{x^*-c}{2}) - f(\frac{x^*+c}{2} - \frac{x^*-c}{2})}{2h} - f^{(s)}\left(\frac{x^*+c}{2}\right) \right| \\
&\qquad = |(f(x^*) - f(c))/(x^* - c) - f^{(s)}((x^* + c)/2)| \\
&\qquad \geq \eta/|x^* - c| - M \text{ (by (1) and (2)).}
\end{aligned}$$

Since x^* may be chosen as near to c as we want, this quantity above is greater than or equal to N, however large N be, and this contradicts the definition of uniform differentiability of f in I.

Conversely, by Theorem A.2.17, it follows that $f^{(s)}$ is continuous in $[a, b]$ and so bounded.

Theorem A.2.19. *If f is continuous at a and uniformly $s-$differentiable in a neighbourhood N of a, then $f'(x)$ exists at $x = a$.*

Proof. Since f is continuous at a, it is bounded, i.e., $|f(x)| \leq M$, in some neighbourhood N_a of a, which we may assume, is contained in N. Let $[c, d] \subset N_a \subset N$ $(c < a < d)$, so that f is uniformly s-differentiable in $[c, d]$.

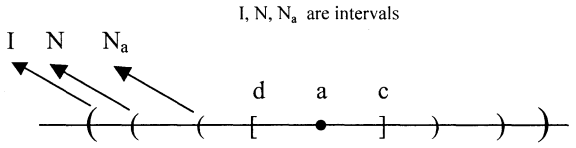


Figure A.1.11

Then there exists $h_0 > 0$ such that $|(f(x+h_0) - f(x-h_0))/2h_0 - f^{(s)}(x)| < \epsilon_0$ for $x \in [c, d]$, for some ϵ_0. Hence $|f^{(s)}(a)| < \epsilon_0 + |(f(x+h_0) - f(x-h_0))/2h_0| \leq \epsilon_0 + 2M/2h_0$, for all $x \in [c, d]$, i.e., $f^{(s)}$ is bounded in $[c, d]$ and so by Theorem A.2.18 (i), f is continuous in $[c, d]$ and hence further, by Theorem A.2.17, $f^{(s)}$ is continuous in some subinterval $[c_1, d_1]$, with $c_1 < a < d_1$, of $[c, d]$. It follows, by Theorem A.2.8, that $f'(a)$ exists.

Theorem A.2.20. *If f is continuous and uniformly s-differentiable in I, then f is uniformly differentiable in $[a, b]$.*

Proof. For $\epsilon > 0$, there exists a $\delta > 0$ such that

$$|(f(x+h) - f(x-h))/2h - f^{(s)}(x)| < \epsilon/2 \text{ , if } |h| < \delta_1, \text{ for all } x \in I. \tag{1}$$

Further, by Theorem A.2.17, $f^{(s)}$ is continuous in $[a, b]$ and so uniformly continuous; hence there exists a δ_2, such that

$$|f^{(s)}(x_1) - f^{(s)}(x_2)| < \epsilon/2 \text{ , if } |x_2 - x_1| < \delta_2, \text{ for all } x_1, x_2 \in [a, b]. \tag{2}$$

Let $\delta = \min\{\delta_1, \delta_2\}$ and choose h such that $|h| < \delta$. Now replace x by $x + h/2$ and h by $h/2$ in (1) to get (since $|h/2| < |h| < \delta_1$)

$$|(f(x+h) - f(x))/h - f^{(s)}(x+h/2)| < \epsilon/2 \,. \tag{3}$$

Hence

$$\left|\frac{f(x+h)-f(x)}{h} - f^{(s)}(x)\right| = \left|\frac{f(x+h)-f(x)}{h} - f^{(s)}(x+\frac{h}{2}) - f^{(s)}(x+\frac{h}{2}) - f^{(s)}(x)\right|$$
$$< \epsilon/2 + \epsilon/2 = \epsilon \,,$$

for all $x \in [a, b]$. This also shows that $f'(x) = f^{(s)}(x)$ in $[a, b]$.

Combining this with Theorem A.2.14, we obtain :

Theorem A.2.21. *If f is continuous and uniformly s-differentiable in I, then*

f' is continuous in $[a, b]$.

For most of the results given in this section see [3], [84], [72], [91], [71] and [69]; see also [70].

§A.3. Cauchy's functional equation: $f(x+y)=f(x)+f(y)$

Theorem A.3.1. *Let $f : \mathbb{R} \to \mathbb{R}$ be a function that satisfies the functional equation*

$$f(x+y) = f(x) + f(y) \qquad (*)$$

for all $x, y \in \mathbb{R}$. Then $f(rx) = r.f(x)$ for all $r \in \mathbb{Q}$, and indeed $f(x) = cx$ for all $x \in \mathbb{Q}$, for some $c \in \mathbb{R}$.

Proof. *Step 1.* We first show that $f((m/n).t) = (m/n).f(t)$ for all t and all positive rationals m/n. Now $(*)$ gives, by induction, $f(x_1 + x_2 + \cdots + x_n) = f(x_1) + f(x_2) + \cdots + f(x_n)$ and putting $x_k = x$ for each k, it follows that $f(nx) = nf(x)$ $(x \in \mathbb{R},\ n \in \mathbb{N})$.

Write $x = (m/n).(nx/m) = (m/n).t$ $(t = nx/m \in \mathbb{R},\ m, n \in \mathbb{N})$. Then $nx = mt$ and so $f(nx) = f(mt)$, i.e. $nf(x) = mf(t)$, i.e., $nf((m/n)t) = mf(t)$, i.e. $f((m/n)t) = (m/n)f(t)$.

Putting $t = 1$ and $f(1) = c$, this gives $f(m/n) = (m/n).c$, i.e., $f(x) = cx$ for all positive rationals x.

Step 2. We now show that $f(x) = cx$ for all rational numbers x. For $x = 0$, this is trivial, as on putting $x = y = 0$, $f(0) + f(0) = f(0)$, i.e., $f(0) = 0$. For $x < 0$, we proceed as follows:

First put $y = -x$ in $(*)$ to get $f(x - x) = f(x) + f(-x)$, i.e., $f(0) = f(x) + f(-x)$, i.e.,

$$f(-x) = -f(x) \ \ \forall\, x. \qquad (\dagger)$$

Then for $x < 0$, sat $x = -y$, $y > 0$, we have $f(y) = cy$ (since $y > 0$), i.e., $f(-x) = c(-x)$, i.e., $-f(x) = -cx$ (by $(\dagger)$), i.e., $f(x) = cx$, as required.

Theorem A.3.2. *Suppose f is continuous, then$(*)$ implies that $f(x) = cx$ for all $x \in \mathbb{R}$.*

Proof. Let $x \in \mathbb{R}$ and let x_n be a sequence of rational numbers that tends to x. Then $f(x_n) = cx_n$, for all n. Letting $n \to \infty$, we get $\lim_{n\to\infty} f(x_n) = cx$. But since f is continuous, $\lim_{n\to\infty} f(x_n) = f(x)$; hence $f(x) = cx$ for all $x \in \mathbb{R}$.

We also have:

Theorem A.3.3. *Suppose f satisfies $(*)$ and is continuous at a single point $x = a$; then $f(x) = cx$ for all $x \in \mathbb{R}$.*

Proof. It is enough to prove that f is continuous everywhere. To see this, let ξ be any point. Then we must show that $\lim_{x\to\xi} f(x) = f(\xi)$. We have

$\lim_{x\to\xi} f(x) = \lim_{x-\xi+a\to a} f(x-\xi+a+\xi-a)$ and writing $x-\xi+a=t$, this gives

$$\begin{aligned}\lim_{x\to\xi} f(x) &= \lim_{t\to a} f(t+\xi-a)\\ &= \lim_{t\to a}(f(t)+f(\xi-a))\\ &= \lim_{t\to a} f(t) + f(\xi-a)\\ &= f(a)+f(\xi-a) \qquad \text{(since } f \text{ is continuous at } t=a)\\ &= f(a+\xi-a) = f(\xi).\end{aligned}$$

The above was observed by G.Darboux in 1875.

Theorem A.3.4. *Suppose that f satisfies $(*)$ and $f(x)$ is nonnegative for all sufficiently small positive x; then $f(x) = cx$ for all $x \in \mathbb{R}$.*

Proof. Let $y > 0$ be sufficiently small, so that $f(y) \geq 0$ by hypothesis. Then

$$f(x+y) = f(x) + f(y) \geq f(x)\ ,$$

i.e., f is monotone increasing. Further $f(r) = cr$ for all $r \in \mathbb{Q}$.

Now let ξ be any real number and choose an increasing sequence r_n of rational numbers that tend to ξ and a decreasing sequence R_n of rational numbers that also tend to ξ. Then $r_n < \xi < R_n$ for all $n = 1, 2, \ldots$. Further, $cr_n = f(r_n) \leq f(\xi) \leq f(R_n) = cR_n$, since f is increasing. Here R_n and r_n are rational numbers. Letting $n \to \infty$ we get $c\xi \leq f(\xi) \leq c\xi$, as required.

Remark A.3.5. Suppose that f satisfies $(*)$ and is less than or equal to 0 for all sufficiently small positive numbers x, then in this case too $f(x) = cx$ for all $x \in \mathbb{R}$.

Theorem A.3.6. *Suppose that f satisfies $(*)$ and is bounded on an arbitrarily small interval (a, b); then $f(x) = cx$ for all $x \in \mathbb{R}$.*

Proof. Let $g(x) = f(x) - x.f(1)$. Then $g(x+y) = g(x) + g(y)$ (check) and g is bounded on (a, b). But if $r \in \mathbb{Q}$, then

$$\begin{aligned}g(r) &= f(r) - r.f(1)\\ &= r.f(1) - r.f(1) = 0 \quad \text{(by Theorem A.3.1).}\end{aligned}$$

Hence $g(x+r) = g(x) + g(r) = g(x)$ (for all $r \in \mathbb{Q}$). If now ξ is any real number, there exists a rational number r_ξ (in fact infinitely many such rational numbers) such that $\xi + r_\xi \in (a, b)$. Hence g is bounded everywhere. We claim that $g(x) \equiv 0$; for, if at some point p, $g(p) = A \neq 0$, then for $n \in \mathbb{N}$, $g(np) = n.g(p) = nA$ and so for sufficiently large n, g would assume arbitrarily large values, contradicting its boundedness. Thus $g \equiv 0$ and so $f(x) = x.f(1) = cx$, say.

These considerations lead to the following

Theorem A.3.7. *If Cauchy's functional equation*

$$f(x+y) = f(x) + f(y)$$

is satisfied for all $x, y \in \mathbb{R}$ and if f satisfies any one of the following conditions:

(i) f is continuous at a point $x = a$,

(ii) $f(x) \geq 0$ for $0 < x < \epsilon$, for some $\epsilon > 0$,

(ii)' $f(x) \leq 0$ for $0 < x < \epsilon$, for some $\epsilon > 0$,

(iii) $f(x)$ is bounded in (a, b), where (a, b) is some open interval,

then $f(x) = cx$, $c \in \mathbb{R}$, for all $x \in \mathbb{R}$.

A natural question that comes to mind is, *"Are there any functions $f : \mathbb{R} \to \mathbb{R}$, other than $f(x) = cx$, that satisfy Cauchy's functional equation?* If such a function could exist, then, by the above theorem, it would

(i) be totally discontinuous (i.e., discontinuous at each point),

(ii) neither be greater than or equal to 0 in $(0, \epsilon)$ nor be less than or equal to 0 in $(0, \epsilon)$, for any $\epsilon > 0$,

(iii) be unbounded in each interval (a, b).

Making use of the vector space basis of the (infinite dimensional) vector space $\mathbb{R}$ over $\mathbb{Q}$, it is easy to construct all functions that satisfy Cauchy's functional equation. Indeed, we have the following:

Theorem A.3.8. *The most general solution f, which satisfies Cauchy's functional equation, can be constructed as follows: Choose a basis $\mathbb{B}$ of $\mathbb{R}$ over $\mathbb{Q}$, so that if $x \in \mathbb{R}$ is any real number, then we can write uniquely*

$$x = r_1 b_1 + r_2 b_2 + \cdots + r_n b_n \ , \quad b_k \in \mathbb{B}, \ r_k \in \mathbb{R} \ .$$

Now define

$$f(x) = r_1 f(b_1) + r_2 f(b_2) + \cdots + r_n f(b_n) \ , \qquad (**)$$

where $f(b_1), f(b_2), \ldots, f(b_n)$ are arbitrarily chosen real numbers. Then f satisfies the required condition. Moreover, such a function is continuous if and only if there is a constant c such that $f(b_k) = c.b_k$, for all $k = 1, 2, \ldots, n$, and then $f(x) = cx$ for all x.

Proof. Let $x = r_1 b_1 + \cdots + r_n b_n$, $y = s_1 b_1 + \cdots + s_n b_n$, ($r_k$, s_k can be zero). Then $(**)$ gives

$$f(x+y) = f(x) + f(y) \ ,$$

for, $x + y = (r_1 + s_1)b_1 + \cdots + (r_n + s_n)b_n$ and therefore by $(**)$, $f(x + y) = (r_1 + s_1)f(b_1) + \cdots + (r_n + s_n)f(b_n)$, while

$$f(x)+f(y)=r_1f(b_1)+r_2f(b_2)+\cdots+r_nf(b_n)+s_1f(b_1)+s_2f(b_2)+\cdots+s_nf(b_n)$$
$$= (r_1 + s_1)f(b_1) + \cdots + (r_n + s_n)f(b_n) ,$$

i.e., Cauchy's equation is satisfied. Further, for $x = r_1b_1 + \cdots + r_nb_n$, by Theorem A.3.1, we have

$$\begin{aligned} f(x) &= f(r_1b_1 + \cdots + r_nb_n) \\ &= f(r_1b_1) + \cdots + f(r_nb_n) \text{ (since } f \text{ satisfies Cauchy's equation),} \\ &= r_1f(b_1) + \cdots + r_nf(b_n) \text{ (again by Theorem A.3.1).} \end{aligned}$$

We have thus proved that $f(x)$ has to be defined by $(**)$, i.e., that there is no other f. Such an f, if continuous, and only then, is given by $f(x) = cx$, by Theorem A.3.2. This completes the proof of the theorem.

Example A.3.9. Suppose $f : \mathbb{R} \to \mathbb{R}$, satisfies Cauchy's functional equation. If three vertices of a parallelogram lie on the graph Γ of the function f, then the fourth vertex does also lie on Γ.

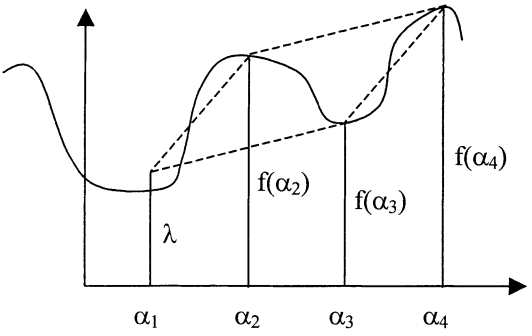


Figure A.1.12

With the notation of Figure A.1.12, it is enough to prove that $\lambda = f(\alpha_1)$. We have

(i) $\alpha_3 - \alpha_1 = \alpha_4 - \alpha_2$,

(ii) $f(\alpha_3) - \lambda = f(\alpha_4) - f(\alpha_2)$.

These conditions give

$$\begin{aligned} \lambda &= f(\alpha_2) + f(\alpha_3) - f(\alpha_4) \\ &= f(\alpha_2 + \alpha_3 - \alpha_4) \\ &= f(\alpha_1) , \end{aligned}$$

by (i) above.

Appendix - II

Hints and solutions to exercises

Exercise 1.1. Let $\mathcal{I}$ be the set of intervals $[r_1, r_2]$ with rational end points; let $\mathcal{T}$ be the set of isolated points of E and let $\mathcal{S}$ be the set of sparse points of E. First note that $\mathcal{I}$ is countable, for fix r_1, then $\{[r_1, r_2] \mid r_2 \in \mathbb{Q}\}$ is countable and there are only countably many $r_1 \in \mathbb{Q}$.

If $x \in \mathcal{T}$, find δ such that $E(x, \delta) = \{x\}$ and so for any $\delta_0 \leq \delta$, $E(x, \delta_0) = \{x\}$. Hence there exists a rational δ_0 such that $E(x, \delta_0) = \{x\}$. Now map $x \to [x - \delta_0, x + \delta_0]$. This map is one to one from $\mathcal{I}$ into $\mathcal{T}$. Hence $\mathcal{T}$ is countable.

Proceed likewise for S: Find a rational number δ_0 such that $E(x, \delta_0)$ is countable and the map $x \to [x - \delta_0, x + \delta_0]$, is a one-to-one map of $\mathcal{S}$ into $\mathcal{I}$.

Exercise 1.2. *(a)* $\Rightarrow$*(b)* : Let $\sup A = \alpha$. Construct an increasing sequence $\{a_n\}$ with $a_n \in A \cap [\alpha - 1/n, \alpha]$. This is possible since $\alpha = \sup A$. Then clearly $a_n \to \alpha$. But by (a), $\lim a_n \in E$, i.e., $\alpha \in E$. Similarly, $\inf A \in E$.
(b) $\Rightarrow$ *(c):* We have $\sup |x_n - x_m| < \epsilon$ if $m, n \geq N$, i.e.,

$$x_n < x_N + \epsilon \ , \ \text{if} \ n \geq N \ , \qquad (*)$$

i.e., $\{x_n\}$ is bounded. Let α be a limit point of $\{x_n\}$. As in the proof of (a) $\Rightarrow$ (b), choose a monotone subsequence $x_{n_1}, x_{n_2}, \ldots, x_{n_k}, \ldots$ of $\{x_n\}$ which tends to α. Then α is the supremum or the infimum of $\{x_{n_k}\}$. Hence, by (b), $\alpha \in E$ (with $A = \{x_{n_k}\} \subset E$). Now $|x_n - x_{n_{k_0}}| < \epsilon/2$, if n, n_{k_0} are large (by $(*)$) and $|x_{n_{k_0}} - \alpha| < \epsilon/2$ if n_{k_0} is large, because $x_{n_{k_0}}$ tends to α. Therefore

$$\begin{aligned} |x_n - \alpha| &= |x_n - x_{n_{k_0}} + x_{n_{k_0}} - \alpha| \\ &\leq |x_n - x_{n_{k_0}}| + |x_{n_{k_0}} - \alpha| \\ &< \epsilon/2 + \epsilon/2 = \epsilon \ , \end{aligned}$$

if n is large. Therefore the full sequence $\{x_n\} \to \alpha \in E$, as required.
(c) $\Rightarrow$ *(a):* Let $\{x_n\}$ be a bounded monotone sequence in E, which tends to α. We have to prove that $\alpha \in E$. Now for this sequence $\{x_n\}$, we have $|x_n - x_m| \leq |x_n - \alpha| + |x_m - \alpha| < \epsilon/2$, if $m, n \geq N$, say. Hence $\sup \{|x_n - x_m|\} < \epsilon$, if $m, n > N$. Therefore, this supremum tends to 0 as $N \to \infty$. Therefore by (c), $\{x_n\}$ tends to a limit which belongs to E, i.e., $\alpha \in E$.

Exercise 1.3. Let $a_1 = a_2 = 0$, where, for $n \geq 3$, a_n is equal to the length of a side of a regular polygon of n sides, inscribed in a unit circle, so that clearly $a_n \to 0$ as $n \to \infty$, but na_n is equal to the length of the complete polygon, which tends to the circumference of the circle, which is 2π, a transcendental number.

Exercise 2.1. $\delta(s,t) = 0$ implies that there exists x_n, y_n, $n = 1, 2, \ldots$, with $|x_n - y_n| < 1/n$, such that $f(x_n) = s$, $f(y_n) = t$. But $|x_n - y_n| < 1/n < \delta$ (if $n \geq N$) implies that $|f(x_n) - f(y_n)| < \epsilon$ (since f is continuous in $[a,b]$ and so uniformly continuous), i.e., $|s - t| < \epsilon$, which is false since $s \neq t$.

Next, choose $x_n < y_n$ with $f(x_n) = r$, $f(y_n) = t$, such that $|x_n - y_n| = \delta(r,t) + 1/n$. Since $r < s < t$, f takes the value s between x_n, y_n, say at w_n, i.e., $x_n < w_n < y_n$, such that $f(w_n) = s$. Then $|x_n - w_n| \leq |x_n - y_n| = \delta(r,t) + 1/n$ (for all n) and so is less than or equal to $\delta(r,t)$, i.e., $\delta(r,s) \leq \delta(r,t)$.

Finally, if $\delta(r,s) = \delta(r,t)$ for some $r < s < t$, then there exist $x_n < w_n < y_n$, with $f(x_n) = r$, $f(w_n) = s$, $f(y_n) = t$ and then $(w_n - x_n) - (y_n - x_n) \to 0$ as $n \to \infty$, i.e., $y_n - w_n \to 0$ as $n \to \infty$, and so $f(y_n) - f(w_n) \to 0$, by the continuity of f, i.e., $|t - s| < \epsilon$, which is false since $t \neq s$.

Exercise 2.2. (a). Let $\delta_1 > \delta_2$. Then $\phi(x, \delta_1) \geq \phi(x, \delta_2)$, for all x, since the left side is the oscillation of f in $(x - \delta_1, x + \delta_1)$ while the right side is the oscillation of f in $(x - \delta_2, x + \delta_2)$. Therefore, $\sup_x \phi(x, \delta_1) \geq \sup_x \phi(x, \delta_2)$, i.e. $\psi(\delta_1) \geq \psi(\delta_2)$.
(b) First, let f be bounded: $m \leq f(x) \leq M$. Then for all $x, y \in \mathbb{R}$, $|f(x) - f(y)| \leq M - m$, so $\phi(x, \delta) \leq M - m$ for all x, δ (> 0) and hence $\psi(\delta) \leq M - m$ for all $\delta > 0$, i.e., ψ is bounded. Conversely, let ψ be bounded, say $\sup \psi(\delta) = M$ (inf is greater than or equal to 0, of course). Then for all x and all $\delta > 0$, $\phi(x, \delta) \leq \psi(\delta) \leq M$. In this, take $x = 0$ (say) to get $\phi(0, \delta) \leq M$ for all $\delta > 0$ and so $|f(0) - f(y)| \leq M$, if $-\delta < y < \delta$ for all $\delta > 0$, i.e., $M - f(0) \leq f(y) \leq M + f(0)$ for all y (since δ can be any positive real number), i.e., f is bounded, as required.
(c). First, let f be uniformly continuous on $\mathbb{R}$. Then given $\epsilon > 0$, there exists an $\eta > 0$, such that $|f(x) - f(y)| < \epsilon$ if $|x - y| < \eta$ and so $\phi(x, \delta) \leq \epsilon$ if $\delta < \eta$ (for all $x \in \mathbb{R}$), and so $\psi(\delta) \leq \epsilon$ if $\delta < \eta$, i.e., $\psi(\delta) \to 0$ as $\delta \to 0$. Conversely, let $\psi(\delta) \to 0$ as $\delta \to 0$, and we just work backwards to get f is uniformly continuous on $\mathbb{R}$.

Exercise 2.3. $I \Rightarrow II$: Suppose II is false. Let $\epsilon > 0$ be given. Then there exists x as close to a as we want, such that

$$|f(x) - A| \geq \epsilon \,. \qquad (i)$$

Choose a sequence $\{x_n\}$ with $|x - a| < 1/n$ (so that for large n, x_n is sufficiently close to a) such that $|f(x_n) - A| \geq \epsilon$ (see (i)), i.e.,

$$f(x_n) \text{ does not tend to } A \,. \qquad (ii)$$

However, since $|x_n - a| < 1/n$ so $\{x_n\} \to a$ and so by (I), $f(x_n) \to A$, a contradiction to (ii).
$II \Rightarrow III$: By II, given $\epsilon > 0$, there exists a $\delta > 0$ such that $|f(x) - A| < \epsilon/2$ if $|x - a| < \delta$ and similarly $|f(y) - A| < \epsilon/2$ if $|y - a| < \delta$; and then $|f(x) - f(y)| \leq |f(x) - A| + |f(y) - A| < \epsilon/2 + \epsilon/2 = \epsilon$ if $0 < |x - a| < \delta$, $0 < |y - a| < \delta$, which is III.
$III \Rightarrow I$: Let $\{x_n\} \to a$ and we want to prove that $\{f(x_n)\}$ is convergent. By III, given $\epsilon > 0$, there exists a $\delta > 0$ such that $|f(x_n) - f(x_m)| < \epsilon$ if $|x_n - a| < \delta$, $|x_m - a| < \delta$, but since $x_n \to a$ so $|x_n - a| < \delta$ if $n \geq N$, i.e., $|x_n - a| < \delta$, $|x_N - a| < \delta$, if $n \geq N$ and so by III, we have $|f(x_n) - f(x_N)| < \epsilon$, i.e., $-\epsilon < f(x_n) - f(x_N) < \epsilon$, i.e., $f(x_n) < f(x_N) + \epsilon$ for all $n \geq N$, i.e., $f(x_n)$ is a bounded sequence and so contains a convergent subsequence, $f(x_{n_1}), f(x_{n_2}), \ldots, f(x_{n_k}), \ldots$, say, which tends to a limit L as $k \to \infty$. We claim that the entire sequence $\{f(x_n)\} \to L$. To see this, select n large and n_k (i.e., k) large so that $|x - a| < \delta$, $|x_{n_k} - a| < \delta$. Then by III, $|f(x_n) - f(x_{n_k})| < \epsilon$, but now $|f(x_n) - L| = |f(x_n) - f(x_{n_k}) + f(x_{n_k}) - L| \leq |f(x_n) - f(x_{n_k})| + |f(x_{n_k}) - L| < \epsilon + \epsilon$; hence $f(x_n) \to L$, as required.

Exercise 2.4. (i). Let $x_1 < x_2$ and we have to prove that $f(x_1) < f(x_2)$. We have

$$f(x_2) - f(x_1) = \sum_{n=1}^{\infty} (\phi_{x_2}(r_n) - \phi_{x_1}(r_n))/2^n \, ;$$

the rearrangement is allowed since the series is absolutely convergent. Here

$$(\phi_{x_2}(r_n) - \phi_{x_1}(r_n)) = \begin{cases} 1 - 1 & \text{if } r_n \leq x_1 \, , \\ 1 - 0 & \text{if } x_1 < r_n \leq x_2 \, , \\ 0 - 0 & \text{if } x_2 < r_n \, , \end{cases} = \begin{cases} 0 & \text{if } r_n \notin (x_1, x_2] \, , \\ 1 & \text{if } r_n \in (x_1, x_2] \, , \end{cases}$$

so the above sum is equal to $\sum_{r_n \in (x_1, x_2]} 1/2^n > 0$, since such r_n exist.
(ii) First let x_0 be irrational and let $x_0 < x$; then

$$f(x) - f(x_0) = \sum_{n=1}^{\infty} (\phi_x(r_n) - \phi_{x_0}(r_n))/2^n = \sum_{r_n \notin (x_0, x]} 1/2^n$$

and we can ensure that this is less than any ϵ, by making sure that the first few terms , say, $1/2, 1/2^2, \ldots, 1/2^j$ do not appear in this sum (note that $j = j(\epsilon)$), i.e., that $r_1, r_2, \ldots, r_j \notin (x_1, x_2]$, by choosing x close enough to x_0. Similarly for $x < x_0$. That proves continuity for the irrational x's.

Finally, if x_0 is rational, say equal to r_{n_0}, then for $x < x_0$ we have $|f(x) - f(x_0)| = \sum_{r_n \in (x, x_0]} 1/2^n$ and this sum contains the term $1/2^{n_0}$ since $r_{n_0} \in (x, x_0]$ and is not less than ϵ if $\epsilon < 1/2^{n_0}$. That completes the proof.

Exercise 2.5. f is continuous and so bounded and attains its bounds, so $S \neq \emptyset$.

Let $A = \sup S$ and we have to prove that $A \in S$. Suppose, to the contrary, that $A \notin S$, then $f(A)$ is not the supremum of $f(x)$ and so by the continuity of f, $f(x_0)$ is not the supremum of $f(x)$ for each $x_0 \in [a-\epsilon, A+\epsilon]$; but the interval $[A-\epsilon, A+\epsilon]$ contains points of S, i.e., points x_0 where $f(x_0)$ is the supremum of $f(x)$, which is a contradiction.

Suppose, to the contrary, that $g(x)$ is not identically 0 in $[0,1]$. Let $B = \sup g(x)$, and without loss of generality, assume that $B > 0$. Let S be the set of those points of $[0,1]$, where B is attained. By the first part $S \neq \emptyset$ and S contains $\sup S = A$, say. Then $A \neq 1$ (nor is A equal to 0) (since $g(A) = \sup g(x) = B > 0$, while $g(1) = 0$), so $0 < A < 1$. Therefore, there exists a k, $0 \leq A - k < A < a + k \leq 1$, with $g(A) = \frac{1}{2}\{g(A+k) + g(A-k)\}$. Here $g(A+k) \leq B$ (since $B = \sup g$) and $g(A-k) \leq B$, $g(A) = B$. Hence $g(A+k) = B$, i.e., $A + k \in S$, a contradiction since $A = \sup S$.

Exercise 2.6. Define $f_0(x) = x$, $f_1(x) = f(x)$, $f_2(x) = f(f(x))$, $\ldots$, $f_{n+1}(x) = f(f_n(x))$. Note that f is continuous (by the given condition). Suppose $f(x) > x$ for all x. This means $f_n(x)$ is a strictly increasing sequence for all x, i.e., in particular, $\{f_n(0)\}$ is a strictly increasing sequence. Now for each n,

$$|f_{n+1}(0) - f_n(0)| \leq k|f_n(0) - f_{n-1}(0)| \leq k^2|f_{n-1}(0) - f_{n-2}(0)| \leq \ldots \leq k^n|f(0) - 0|.$$

Hence for $m > n$,

$$\begin{aligned}
|f_m(0) - f_n(0)| &= |f_m(0) - f_{m-1}(0) + f_{m-1}(0) - f_{m-2}(0) + \cdots + f_{n+1}(0) - f_n(0)| \\
&\leq |f_m(0) - f_{m-1}(0)| + |f_{m-1}(0) - f_{m-2}(0)| + \cdots + \\
&\qquad + |f_{n+1}(0) - f_n(0)| \\
&\leq f(0)(k^{m-1} + k^{m-2} + \cdots + k^n) \\
&\leq f(0).k^n(1/(1-k)) \\
&\to 0 \quad \text{as} \quad n \to \infty.
\end{aligned}$$

Therefore, by the general principle of convergence, $\{f_n(0)\}$ is convergent, say to the limit α. But now $\{f(f_n(0))\}$ is the same sequence (check), and so tends to α, i.e., $f_{n+1}(0) \to \alpha$, but $f_{n+1}(0)$ also tends to α, which contradicts our supposition that $f(x) > x$ for all x. Similarly, " $f(x) < x$ for all x " is false. Hence either $f(x_0) = x_0$ for some x_0 as required and we are through, or

$$\exists \; x_1, x_2 \text{ such that } f(x_1) < x_1, f(x_2) > x_2 \,. \qquad (*)$$

Also, f is continuous, so $g(x) = f(x) - x$ is continuous and $g(x_1) < 0$, $g(x_2) > 0$ and so by the intermediate value property, there exists x_0 such that $g(x_0) = 0$, i.e., $f(x_0) = x_0$.

Now the uniqueness of x_0. If also $f(y_0) = y_0$, then $|f(y_0) - f(x_0)| \leq k|y_0 - x_0|$, but $f(y_0) = y_0$, $f(x_0) = x_0$ and so $|y_0 - x_0| \leq k|y_0 - x_0| < |y_0 - x_0|$, which is a contradiction unless $y_0 = x_0$. This completes the proof.

Finally, the last part. Consider the function $f(x) = \sqrt{(1+x^2)}$. We shall show that $|f(x) - f(y)| < |x - y|$. We have

$$|\sqrt{(1+x^2)} - \sqrt{(1+y^2)}|\,|\sqrt{(1+x^2)} + \sqrt{(1+y^2)}| = |1 + x^2 - 1 - y^2| = |x+y|\,|x-y|\,,$$

and therefore

$$\begin{aligned}|[\sqrt{(1+x^2)}-\sqrt{(1+y^2)}]/(x-y)| &= |x+y|/|\sqrt{(1+x^2)}+\sqrt{(1+y^2)}| \\ &\leq (|x|+|y|)/(\sqrt{(1+x^2)}+\sqrt{(1+y^2)}) \\ &< 1\end{aligned}$$

(since $|x| < \sqrt{(1+x^2)}$, $|y| < \sqrt{(1+y^2)}$). Hence $|\sqrt{(1+x^2)}-\sqrt{(1+y^2)}| < |x-y|$, as required.

But now $\sqrt{(1+x^2)}$ is never equal to x; indeed $\sqrt{(1+x^2)} > x$, always.

Exercise 2.7. Suppose, to the contrary, that for some $a < b$, we have $f(a) \neq f(b)$ and without loss of generality, let $f(a) < f(b)$. Choose an irrational ρ with $f(a) < \rho < f(b)$ and let $g(x) = f(x) - \rho$. Then g is continuous in $[a,b]$ and $g(a) < 0$, $g(b) > 0$, and so by the intermediate value property, there exists $c \in (a,b)$ such that $g(c) = 0$, i.e., $f(c) = \rho$. But f is rational valued and ρ is irrational–a contradiction.

Exercise 2.8. Suppose $f(x) \to A$ as $x \to \infty$. Let $\epsilon > 0$ be given; then $|f(x) - A| < \epsilon/2$ if $x \geq \Delta$. Now in the closed interval $[0,\Delta]$, since f is continuous, it is uniformly continuous. In the interval $[\Delta,\infty)$, we have

$$\begin{aligned}|f(x_1)-f(x_2)| &= |f(x_1)-A+A-f(x_2)| \\ &\leq |f(x_1)-A|+|f(x_2)-A| \\ &< \epsilon/2+\epsilon/2=\epsilon \quad , \text{ if } \quad x_1,x_2 \geq \Delta \ ,\end{aligned}$$

so f is uniformly continuous in $[\Delta,\infty)$.

The converse is false, for example, take $f(x) = x$, which is uniformly continuous in $[0,\infty]$, but f does not tend to a finite limit as x tends to ∞.

Exercise 2.9. Easy.

Exercise 2.10. The trick is to use simultaneous induction on n, i.e., our induction hypothesis is

(i) $P_{2n-1}(x) = 0$ has only one real root, *and*

(ii) $P_{2n}(x) = 0$ has no real root;

and we have to prove that

$(i)'$ $P_{2n+1}(x) = 0$ has only one real root, *and*

$(ii)'$ $P_{2n+2}(x) = 0$ has no real root.

Now $P'_{2n+1}(x) = P_{2n}(x)$, and by (ii), it has no real zero and $P_{2n}(0) = 1$, so being a continuous function, it stays above the x-axis for all x, i.e., retains positive sign for all x, i.e., $P'_{2n+1}(x) > 0$ for all x. It follows that

$$P_{2n+1}(x) \text{ is an increasing function of } x, \text{ for all } x. \qquad \text{(a)}$$

Further, for x large and negative, $P_{2n+1}(x) < 0$, while for x large and positive, $P_{2n+1}(x) > 0$; so $P_{2n+1}(x)$ is equal to 0 exactly once, proving $(i)'$ (note that (a) alone does not prove $(i)'$, for example e^x satisfies (a) but it is never 0).

Let α_{2n+1} be the unique real root of $P_{2n+1}(x) = 0$. Since $P_{2n+1}(x) = 1 + x + x^2/2! + \cdots$, we see that P_{2n+1} takes positive values for any positive x and so

$$\alpha_{2n+1} < 0 \,. \tag{b}$$

Now $P'_{2n+2}(x) = P_{2n+1}(x)$, which is $>, =, < 0$ according as x is $>, =, < \alpha_{2n+1}$. Thus the minimum of $P_{2n+2}(x)$ is at α_{2n+1}. We shall show that

$$P_{2n+2}(\alpha_{2n+1}) > 0 \,, \tag{c}$$

i.e., the minimum of P_{2n+2} is positive and so $P_{2n+2}(x) > 0$ for all x, proving $(ii)'$. To prove (c), note that

$$\begin{aligned} P_{2n+2}(\alpha_{2n+1}) &= P_{2n+1}(\alpha_{2n+1}) + \alpha^{2n+2}/(2n+2)! \\ &= 0 + \text{ a positive quantity} \\ &> 0 \,, \end{aligned}$$

as required. The last part of the question is almost trivial. By the uniform convergence of the series, the function $y = \exp(x) = \sum x^r/r!$ is a differentiable function of x for all $x \in \mathbb{R}$ and term by term differentiability is legal, i.e., $y' = (\exp(x))' = \exp(x)$, i.e., $dy/dx = y$. To prove that $\exp(x)$ is non-decreasing, it is enough to prove that $y' > 0$ for all x, i.e., that $\exp(x) > 0$ for all x. This is easily checked from the series for $\exp(x)$.

Exercise 2.11. First, suppose $\lim_{x \to a+} f(x) = \alpha$ and $\lim_{x \to b-} f(x) = \beta$ both exist. Define $f(a) = \alpha$, $f(b) = \beta$; then f is now a continuous function, defined on the closed interval $[a, b]$. Hence f is uniformly continuous on $[a, b]$ and so, on (a, b).

Conversely, suppose f is uniformly continuous on (a, b) and we have to prove that the two limits in question exist. Now $f(x) \to A$ as $x \to a+$ if and only if given ϵ, there exists a δ such that $|f(x_1) - f(x_2)| < \epsilon$ for all x_1, x_2 lying in a right neighbourhood of a. But by uniform continuity such a δ exists; indeed $|f(x_1) - f(x_2)| < \epsilon$ for all $x_1, x_2 \in (a, b)$ with $|x_1 - x_2| < \delta$. That completes the proof.

Exercise 2.12. (i) $f(0)$ is determined and equals 0. Indeed, being continuous at $x = 0$, we have $f(0) = \lim_{x \to 0+} f(x) = \lim_{x \to 0-} f(x)$, so $f(0)$ is determined. If $f(0)$ were positive, then, being continuous at $x = 0$, $f(x) > 0$ in a sufficiently small neighbourhood of 0, a contradiction to the hypothesis. Thus $f(x)$ is not greater than 0. Similarly, $f(0)$ is not less than 0. So $f(0) = 0$.
(ii) $f(0)$ is not determined by this condition; for example, let $f(x) = x.\sin(1/x)$, for $x \neq 0$. Then $|f(x)| = |x|\,|\sin(1/x)| \leq |x| < \epsilon$, if $0 < |x| < \delta$; so the condition

is satisfied but $f(0)$ is not defined.
(iii) Write $f(h) = a + \epsilon_h$, where $\epsilon_h \to 0$ as $h \to 0$. Then

$$\begin{aligned}(f(h) + f(-h) - 2f(0))/h &= (a + \epsilon_h + a + \epsilon_{-h} - 2f(0))/h \\ &= (2a + \epsilon_h + \epsilon_{-h} - 2f(0))/h \\ &\to 1 \text{ , as } h \to 0 \quad \text{(given)}.\end{aligned}$$

It follows that $2a = 2f(0)$, giving $f(0) = a$.

Exercise 2.13. For $x = \pm 1, \pm 1/2, \pm 1/3, \ldots$ and only these values of x, we have $\sin(\pi/x) = 0$. Further, it is enough to prove that f is integrable in $[0, 1]$, since the proof for $[-1, 0]$ is identical. So let $n > 0$. For $x \in (1/(n+1), 1/n)$, $\sin(\pi/x)$ is positive if n is even and is negative if n is odd. It follows that

$$\text{sign}(\sin(\pi/x)) = \begin{cases} 1 & \text{if } n \text{ is even} \\ -1 & \text{if } n \text{ is odd} . \end{cases},$$

Hence the graph of f is as shown in Figure A.2.1. This has countably many points of discontinuity; hence it is integrable.

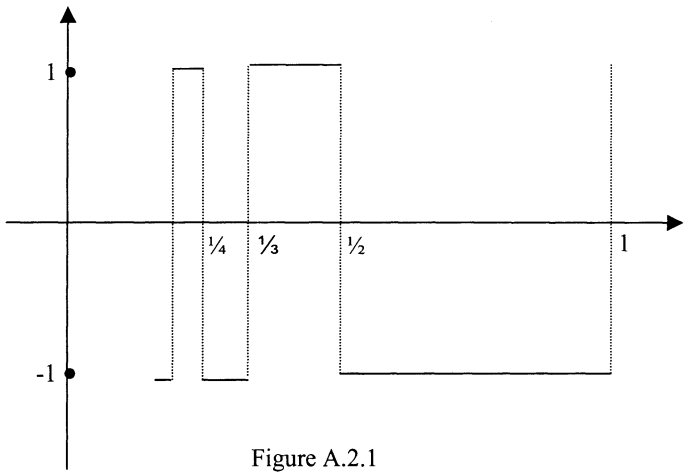


Figure A.2.1

Exercise 2.14. (i). The condition ensures that $f(x) \to 0$ as $x \to \infty$.

Indeed, $|f(x) - g(x)| < \epsilon$ if $x \geq \Delta$, i.e., $g(x) - \epsilon < f(x) < g(x) + \epsilon$ if $x \geq \Delta$. Here, the right side is less than ϵ, since $g(x) < 0$ and $-\epsilon < f(x)$, since $f(x) > 0$. Thus $-\epsilon < f(x) < \epsilon$ if $x \geq \Delta$, i.e., $|f(x)| < \epsilon$ if $x \geq \Delta$, which means $f(x) \to 0$ as $x \to \infty$.
(ii) If $\lambda \neq 0$, this does not ensure that $f(x)$ tends to a finite limit, for example, let

$$f(x) = \begin{cases} 1 \text{ , if } x \in [2n, 2n+1),\ n = 0, 1, 2, \ldots \\ -1 \text{ , if } x \in [2n+1, 2n+2),\ n = 0, 1, 2, \ldots \end{cases}$$

Then $f^2 = 1$, which tends to 1, but f does not tend to any limit as $x \to \infty$.
However, if $\lambda = 0$, then $f^2 \to 0 \Rightarrow f \to 0$ as $x \to \infty$.

Exercise 2.15. (i) Take $y = x$ to get $f(2x) = (f(x))^2 \geq 0$ for all x. Suppose $f(0) = 0$ for some a. Then $f(a + x) = f(a).f(x) = 0$ for all x, i.e., $f(x) \equiv 0$, a contradiction.
(ii) Let Δ be equal to $\sup f$ in $[-X, X]$, attained at ξ, i.e., $f(\xi) = \Delta$. Suppose, to the contrary, that $\inf f = 0$. Then, given $\epsilon > 0$, there exists η such that $0 < f(\eta) < \epsilon$. So $f(\eta + x) = f(\eta).f(x)$, i.e., $f(x) = (1/f(\eta)).f(\eta + x) > (1/\epsilon).f(\eta + x)$. Now take $x = \xi - \eta$; so $f(\xi - \eta) > (1/\epsilon).f(\xi) = (1/\epsilon).\Delta > \Delta$, if $\epsilon < 1$, a contradiction.

Exercise 2.16. The open intervals $(x - \delta_x, x + \delta_x)$ form an open cover of $[0, 1]$. By the Heine- Borel theorem, there exists a finite subcover, say, $\{O_i = (x_i - \delta_{x_i}, x_i + \delta_{x_i})\}$, $i = 1, 2, \ldots, n$, where, without loss of generality, assume that $O_i \cap O_{i+1} \neq \emptyset$ for each $i = 1, 2, \ldots, n - 1$.
Choose $p_i \in O_i \cap O_{i+1}$; then

(1) $f(0) - f(p_1) = \rho_1 \in \mathbb{Q}$,

(2) $f(x_2) - f(p_1) = \rho_2 \in \mathbb{Q}$,

(3) $f(x_2) - f(p_2) = \rho_3 \in \mathbb{Q}$,

(4) $f(x_3) - f(p_2) = \rho_4 \in \mathbb{Q}$,

$\ldots \quad \ldots \quad \ldots \quad \ldots$.

Now (1) $\Rightarrow f(p_1) \in \mathbb{Q}$, since $f(0) = 0$, and so (2) $\Rightarrow f(x_2) = f(p_1) + \rho_2 \in \mathbb{Q}$. Therefore, by (3), $f(p_2) \in \mathbb{Q}$, and so by (4), $f(x_3) \in \mathbb{Q}$, and so on. Thus $f(x_i) \in \mathbb{Q}$ for all i and so $f(x) \in \mathbb{Q}$, by the equation $f(x) - f(x') \in \mathbb{Q}$.

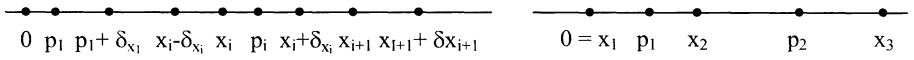


Figure A.2.2

The condition $f(0) = 0$ is essential, for, take $f(x) = \sqrt{2}$ for all $x \in [0, 1]$. This function satisfies the conditions of the theorem (except that $f(0) = 0$) but fails to satisfy the conclusion.

Exercise 2.17. *Hint:* Define $f(0) = 0$, if x is algebraic, and $f(x) = 1$ if x is not algebraic, and note that α and α^m are either both algebraic or both transcendental.

Exercise 2.18. *No:* For otherwise there would exist $x_1 \neq x_2$ such that $f(x_1) \neq f(x_2)$. Suppose, without loss of generality, that $x_1 < x_2$, say $x_1 + p = x_2$. Then $f(x_1 + p) = f(x_2)$, i.e., $f(x_1) = f(x_2)$, which is a contradiction.

Exercise 2.19. Let ϵ be given. We have to show that there exists a closed interval $[c,d] \subset [a,b]$ such that $f(x) < \epsilon$ for all $x \in [c,d]$. Consider the upper Riemann sum $S_\wp$ for the partition

$$\wp : a < a + (b-a)/n < a + 2(b-a)/n < \ldots < a + (n-1)(b-a)/n < b\,.$$

If the result is false, then

$$S_\wp > \epsilon/n + \epsilon/n + \cdots + \epsilon/n = \epsilon\,. \qquad (*)$$

By Darboux's theorem $S_\wp$ tends to the Riemann integral of f over $[a,b]$, which is given to be equal to 0. So $S_\wp$ ought to be less than ϵ if norm $\wp$ is small enough, which contradicts $(*)$ if n is large enough.

Now take $\epsilon = 1$ and find $[a_1, b_1]$ (the above $[c,d]$) contained in $[a,b]$ such that $f(x) < 1$ for all $x \in [a_1, b_1]$. Then $\int_{a_1}^{b_1} f = 0$ (for $0 = \int_a^b f = \int_a^{a_1} f + \int_{a_1}^{b_1} f + \int_{b_1}^b f$. Here each term is greater than or equal to 0, since $f \geq 0$, and so each term is equal to 0).

Now repeat the process with $\epsilon = 1/2$ to get $[a_2, b_2] \subset [a_1, b_1]$ such that $f(x) < 1/2$ for all $x \in [a_2, b_2]$, and so on.

Thus $f(x) < 1/r$ in $[a_r, b_r]$. By the nested interval property, there exists a ξ in $[a,b]$ which is in each $[a_r, b_r]$, so $0 \leq f(\xi) < 1/r$ for all $r = 1, 2, \ldots$; hence $0 \leq f(\xi) \leq 0$, giving $f(\xi) = 0$.

Exercise 2.20. The following graphs show examples of functions for which
(i) $(*)$ is satisfied, $(**)$ fails ;
(ii) $(*)$ fails, $(**)$ is satisfied.

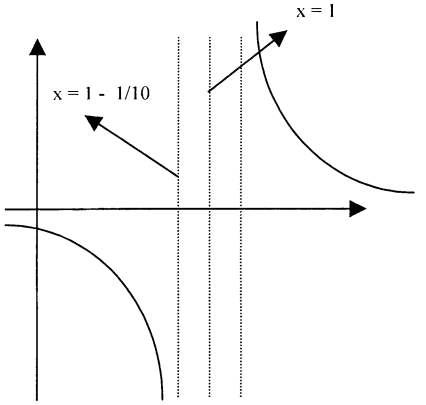


Figure A.2.3

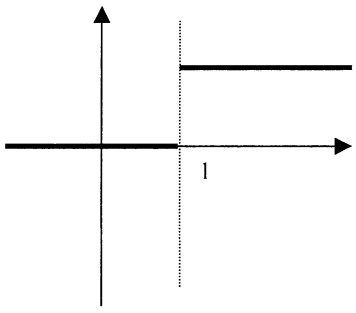


Figure A.2.4

$$(i)\ f(x) = \begin{cases} 0 & \text{if } 1 - \frac{1}{10} \leq x \leq 1 + \frac{1}{10}, \\ \frac{1}{x-(1+\frac{1}{10})} & \text{if } 1 + \frac{1}{10} < x, \\ \frac{-1}{x-(1-\frac{1}{10})} & \text{if } x < 1 - \frac{1}{10}. \end{cases}$$

$$(ii)\ f(x) = \begin{cases} 1 & \text{if } 1 \leq x, \\ 0 & \text{if } x < 1. \end{cases}$$

Exercise 2.21. First of all, since f is continuous, it is integrable and so, since $dg(x)/dx = f(x)$, we see that $g(x)$ has to be of the form $g(x) = \int_a^x f(t)\,dt + g(a)$. Further, $g(x+\pi) = \int_a^{x+\pi} f(t)\,dt + g(a) = \int_a^x f(t)\,dt + \int_x^{x+\pi} f(t)\,dt + g(a) = g(x) + \int_x^{x+\pi} f(t)\,dt$. However, since $g(x+\pi) = g(x)$, this implies $\int_x^{x+\pi} f(t)\,dt = 0$, i.e., $\int_0^{\pi} f(t)\,dt = 0$, which is the required necessary and sufficient condition.

Exercise 2.22. We modify the function

$$f(x) = \begin{cases} (-1)^1.x & \text{if } 1/2 \le x < 1, \\ (-1)^2.x & \text{if } 1/3 \le x < 1/2, \\ (-1)^3.x & \text{if } 1/4 \le x < 1/3, \\ \dots \quad \dots & \dots, \\ (-1)^n.x & \text{if } 1/(n+1) \le x < 1/n, \\ \dots \quad \dots & \dots, \end{cases}$$

whose graph is shown in Figure A.2.5, to make it continuous, without changing the value of the function at the points $x = 1, 1/2, 1/3, \dots$. The graph of the modified function is also drawn in Figure A.2.6.

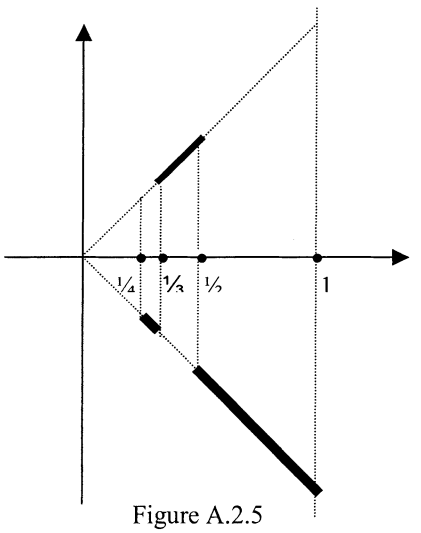


Figure A.2.5

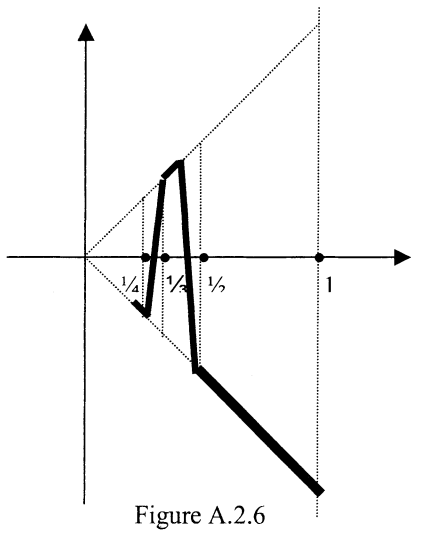


Figure A.2.6

Exercise 2.23. (i) The function f_n is given by

$$f_n(x) = \begin{cases} (n-1)x & \text{if } 0 \le x \le 1/n, \\ \frac{x}{n-1} + \frac{n-2}{n-1} & \text{if } 1/n \le x \le 1. \end{cases}$$

It follows that

$$g_n(x) = f_n^{-1}(x) = \begin{cases} x/(n-1) & \text{if } 0 \le x \le (n-1)/n, \\ (n-1)x - (n-2) & \text{if } (n-1)/n \le x \le 1. \end{cases}$$

$$= \begin{cases} x/(n-1) & \text{if } 0 \le x \le (n-1)/n, \\ (n-1)x - (n-2) & \text{if } (n-1)/n \le x < 1 \\ 1 & \text{if } x = 1. \end{cases} ,$$

Hence, as $n \to \infty$, we see that

$$f_n^{-1}(x) \longrightarrow \begin{cases} 0 \text{ if } 0 \le x < 1, \\ 1 \text{ if } x = 1 \ . \end{cases}$$

This, in any case, is clear from the graph of $f_n(x)$.
(ii) Let $x \in (0,1]$ be any point. Find m such that $2^m.x > 1/3$, but $2^{m-1}x \le 1/3$. Then $g(x) = 2x$, $g^2(x) = g(g(x)) = g(2x) = 4x$, ..., $g^m(x) = g(g^{m-1}(x)) = g(2^{m-1}.x) = 2^m x$, since $2^{m-1}x \le 1/3$, and $g^{m+1}(x) = g(g^m(x)) = g(2^m.x) = (2^m x + 1)/2$ (noted above), since $2^m.x > 1/3$. Here again, $(2^m.x + 1)/2 = 2^{m-1}.x + 1/2 > 1/2 > 1/3$, and therefore $g^{m+2}(x) = g((2^m.x+1)/2) = ((2^m.x+1)/2 + 1)/2 = (2^m.x + 3)/2$, and so on, up to $g^{m+n}(x) = (2^m.x + 2^n - 1)/2^n$, and letting $n \to \infty$, we see that this tends to 1. Thus

$$\lim_{n\to\infty} g^n(x) = 1 \text{ for all } x \in (0,1] .$$

Finally, if $x = 0$, then $g(0) = 0$ and so $g(g(0)) = 0$, and so on, i.e.,

$$g^n(0) \to 0 \text{ as } n \to \infty .$$

Exercise 2.24. $f(x) = p\sin x + qx\cos x + x^2$, and so $f(-x) = -p\sin x - qx\cos x + x^2$. Adding we get $f(x) + f(-x) = 2x^2$. Putting $x = 2$ gives $f(2) + f(-2) = 8$ or $f(-2) = 8 - f(2) = 5$.

Exercise 2.25. Since $x > \sin x$ for $x > 0$, it follows that $f(x) > 0$. Further, $f'(x) = (\sin^2 x - x^2\cos x)/x^2\sin^2 x$ and so it is enough to prove that $f'(x) > 0$, i.e., that $\sin^2 x > x^2\cos x$. Let $g(x) = x - \sin x(\cos x)^{-1/2}$. Then $g''(x) = (1/4)\sin x(\cos x)^{-5/2}(\cos^2 x - 3)$. Here obviously $g''(x) < 0$ if $x \in (0, \pi/2)$ and hence $g'(x)$ is decreasing in $(0, \pi/2)$, or $g'(x) < g'(0)$ (if $0 < x < \pi/2) = 0$ (check). Since $g'(x) < 0$, we see that $g(x)$ is decreasing, i.e., $g(x) < g(0)$ (if $0 < x < \pi/2) = 0$. Thus $g(x) < 0$ in $(0, \pi/2)$, i.e., $x < \sin x/\sqrt{\cos x}$ and since all the quantities involved are positive, we get $x^2\cos x < \sin^2 x$, as required.

Exercise 2.26. The graph of the function $y = \log x$ shows that it is concave in $[a, b]$. where $0 < a < b$. This implies (see Figure A.2.7) that the $y$coordinate of A is larger than the y-coordinate of B, i.e., that

$$\log(\lambda a + (1-\lambda)b) > \lambda\log a + (1-\lambda)\log b . \qquad (*)$$

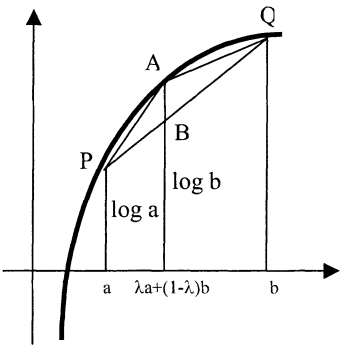


Figure A.2.7

Now take $a = \sin x$, $b = \sin x + \cos x$, $\lambda = \tan x$, so that for $0 < x < \pi/4$, $\lambda \in (0,1)$. Then $(*)$ gives $\log(\tan x.\sin x + (1 - \tan x)(\sin x + \cos x)) > \tan x.\log(\sin x) + (1 - \tan x).\log(\sin x + \cos x)$, i.e., $\log(\cos x) > \tan x.\log(\sin x) + (1 - \tan x).\log(\sin x + \cos x) > \tan x.\log(\sin x)$, (since in the second term, $1 - \tan x > 0$ and $\sin x + \cos x = \sqrt{2}.\cos(\pi/4 - x) > 1$), i.e. $\log(\cos x) > \log((\sin x)^{\tan x})$, i.e., $\cos x > (\sin x)^{\sin x/\cos x}$, i.e., $(\cos x)^{\cos x} > (\sin x)^{\sin x}$.

Exercise 2.27. (i) Let $a = \cos x$. Then $\sin a = \sin(\cos x)$. Since $\sin a < a$, we have

$$\sin(\cos x) < \cos(\sin x)\ . \tag{1}$$

But $\sin x < x$ and so

$$\cos(\sin x) < \cos x\ . \tag{2}$$

Now (1) and (2) give $\sin(\cos x) < \cos(\sin x)$.
Another method of doing (i): Without loss of generality assume that $-\pi < x \leq \pi$. We first show that $\cos(\sin x) \neq \sin(\cos x)$. For, otherwise, $\cos(\sin x) = \sin(\cos x) = \cos(\pi/2 - \cos x)$ and so $\sin x = \pm(\pi/2 - \cos x)$, since both $\sin x$ and $\pi/2 - \cos x \in (-\pi, \pi]$. Thus

$$\textit{either } \sin x + \cos x = \pi/2 \ \textit{ or } \ \sin x - \cos x = -\pi/2\ . \tag{$*$}$$

But $\sin x \pm \cos x = \sqrt{2}.\sin(x \pm \pi/4)$ and we get that

$$-\sqrt{2} \leq \sin x \pm \cos x \leq \sqrt{2}\ ,$$

or, since $-\pi/2 < -\sqrt{2}$ and $\sqrt{2} < \pi/2$,

$$-\pi/2 < \sin x \pm \cos x < \pi/2\ ,$$

contradicting $(*)$. Now, by continuity and the fact that $\cos(\sin 0) > \sin(\cos 0)$, it follows that $\cos(\sin x) > \sin(\cos x)$.
(ii) Let $\sin a = \cos x$; then $a = \sin^{-1}(\cos x)$. But $\sin a < a$, i.e.,

$$\cos x < \sin^{-1}(\cos x) \tag{1}$$

Also $x > \sin x$, and so $\sin^{-1} x > \sin^{-1}(\sin x) = x$ and therefore

$$\cos(\sin^{-1} x) < \cos x \,. \tag{2}$$

Now (1) and (2) imply $\cos(\sin^{-1}(x)) < \sin^{-1}(\cos x)$.
(iii)

$$\begin{aligned} |\sin(rx)| &= |\sin((r-1)x + x)| \\ &= |\sin((r-1)x)\cos x + \cos((r-1)x)\sin x| \\ &\leq |\sin((r-1)x)| + |\sin x| \,, \end{aligned}$$

and the result follows by induction.
Exercise 2.28 (Mathematics Magazine, **53** (1980), 245). The following function disproves the statement:

$$f(x) = \begin{cases} \pi - 2|\pi - x| & \text{if } |\pi - x| < \pi/2 \,, \\ 0 \quad \text{otherwise} \,. \end{cases}$$

The graph of this function is easily drawn. The function is zero outside of $(\pi/2, 3\pi/2)$. In $(\pi/2, \pi)$, we have $0 < \pi - x < \pi/2$ and so $f(x) = \pi - 2(\pi - x) = 2x - \pi$, while in $(\pi, 3\pi/2)$, we have $|\pi - x| = x - \pi$, since $\pi - x < 0$; so $f(x) = \pi - 2(x - \pi) = -2x + 3\pi$. We shall show that for each $x \in \mathbb{R}$, except $x = \pi$, $f^n(x) = 0$.

First of all, for $x = \pi$, $|x - \pi| = 0 < \pi/2$, so $f(\pi) = \pi$, hence $f(f(\pi)) = f(\pi) = \pi$, and so on, i.e., $f^n(\pi) = \pi \neq 0$ for any n.

For other values of x: If x lies outside the interval $(\pi/2, 3\pi/2)$, $f(x) = 0$ and so $f(f(x)) = f(0) = 0$, and so on. If x lies in the above interval, we proceed as follows: If x is such that $|x - \pi| \geq \pi/2^n$ for some $n \in \mathbb{N}$, then

$$f(x) = 0 \,. \tag{$*$}$$

Now for all x inside $[\pi/2, 3\pi/2]$, $|x - \pi|$ is greater than or equal to $\pi/2^n$ for some n (starting with $n = 1$), so for all these x too, $f^n(x) = 0$ for some n. Therefore now it remains to prove $(*)$. We use induction: For $n = 1$, $|x - \pi| \geq \pi/2$ implies $f(x) = 0$ (definition of f). Suppose $(*)$ holds for $n = k$ and let

$$|x - \pi| \geq \pi/2^{k+1} \,. \tag{$\sharp$}$$

Then $|f(x) - \pi| = |\pi - 2|\pi - x| - \pi| \geq \pi/2^k$ (by the definition of f, noting that $|x - \pi| < \pi/2$ and by $(\sharp)$). So by the induction hypothesis, $f^k(f(x)) = 0$, i.e., $f^{k+1}(x) = 0$, as required.

Exercise 4.1. First note that $|F(x)| \leq 3x^2$ for all $x \in \mathbb{R}$; for since $F(x) = F(-x)$ for all x and $F(0) = 0$, it is enough to prove this for $x > 0$. Now if $0 < y < x$, then

$$\int_y^x \sin(1/t)dt = \int_y^x (t^2 \sin(1/t)/t^2)dt = x^2 \cos(1/x) - y^2 \cos(1/y) - \int_y^x 2t \cos(1/t)dt.$$

Therefore

$$\left|\int_y^x \sin(1/t)dt\right| \leq x^2 + y^2 + \int_y^x 2t\,dt \leq 3x^2\,.$$

Hence

$$|F(x)| = \left|\int_y^x \sin(1/t)\,dt\right| = \left|\lim_{y\to 0}\int_y^x \sin(1/t)\,dt\right| = \lim_{y\to 0}\left|\int_y^x \sin(1/t)\,dt\right| \leq 3x^2\,,$$

as required. So now we have

$$\lim_{x\to 0}(F(x) - F(0))/(x-0) = \lim_{x\to 0}(F(x)/x) = X\,,$$

say, Then $\lim_{x\to 0} -3x^2/x \leq X \leq \lim_{x\to 0} 3x^2/x$ and it follows that $X = 0$.

Exercise 4.2. The required area $A = \int_\alpha^{\alpha+1} y\,dx$, and so $dA/dx = \Big[y\Big]_\alpha^{\alpha+1} = (e^{\alpha+1}/(\alpha+1)) - (e^\alpha/\alpha)$, and for the extrema, this is equal to 0, i.e., $e^\alpha.e/(\alpha+1) = e^\alpha/\alpha$, i.e. $\alpha = 1/(e-1)$.

Remark. Among all areas bounded by $y = f(x)$, the x-axis and two ordinates n units apart, the extreme one is that whose left ordinate is the solution for x of $f(x+n) = f(x)$. Note that no matter how irregular $y = f(x)$ may be between x and $x+n$ the two bounding ordinates are of equal length.

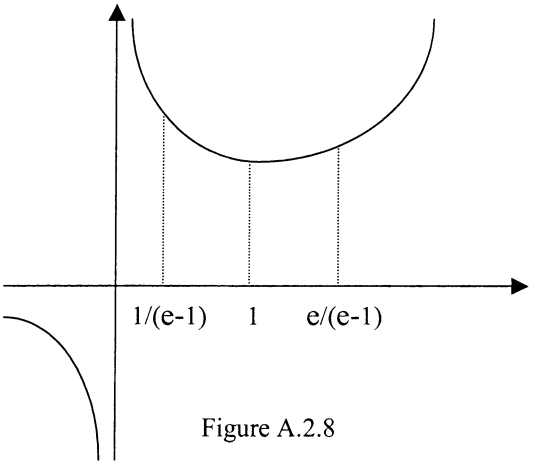


Figure A.2.8

Exercise 4.3. A routine exercise, try!

Exercise 4.4. Use induction on n. For $n = 1$, $x' = 0$, $x'' = 1$ is a solution, by (ii).

The induction step: By the horizontal chord theorem, there exists $\alpha, \alpha+1 \in [0,n]$ such that $f(\alpha) = f(\alpha+1)$. Define a function g on $[0,n-1]$ by

$$g(x) = \begin{cases} f(x) & \text{if } 0 \leq x \leq \alpha\,, \\ f(x+1) & \text{if } \alpha \leq x \leq n-1\,. \end{cases}$$

Then g is continuous in $[0, n-1]$ and $g(0) = g(n-1)$.

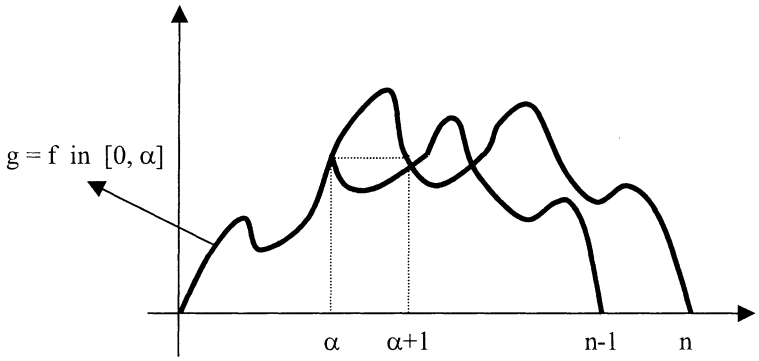


Figure A.2.9

Hence by the induction hypothesis, there exist $n-1$ solutions of $g(x') = g(x'')$, with $x'' - x' \in \mathbb{N}$, say, $(x'_1, x''_1), (x'_2, x''_2), \ldots, (x'_{n-1}, x''_{n-1})$. But now

$$0 = g(x''_i) - g(x'_i) = \begin{cases} f(x''_i) - f(x'_i) & \text{if } x''_i < \alpha\,, \\ f(x''_i + 1) - f(x'_i) & \text{if } x'_i \leq \alpha \leq x''_i\,, \\ f(x''_i + 1) - f(x'_i + 1) & \text{if } \alpha < x'_i\,. \end{cases}$$

So we have $n-1$ distinct solutions of $f(x') = f(x'')$ with $x'' - x' \in \mathbb{N}$. These $n-1$ solutions, together with the solution $(\alpha, \alpha+1)$ gives us n distinct solutions, as required.

Exercise 4.5 (See Amer. Math. Monthly, **82** (1975), 985-992). *Case 1:* $f(a) > a$. Consider the sequence of points $a, f(a), f^2(a), \ldots, f^{k-1}(a), f^k(a) = a$. Then there exists a point b in the sequence, say $b = f^i(a)$, such that $f(b) < b$ (i.e., $f(f^i(a)) < f^i(a)$), for otherwise the sequence would monotonically increase and could not return to a. Now let $g(x) = f(x) - x$. Then $g(b) = f(b) - b < 0$, but $g(a) = f(a) - a > 0$ (case 1). So by the intermediate value theorem, there exists a point c, between a and b, such that $g(c) = 0$, i.e., such that $f(c) = c$. *Case 2:* $f(a) < a$ may be similarly dealt with.

Exercise 4.6. Let x be any real number. We have

$$\begin{aligned} |f^n(x) - f^{n+1}(x)| &= |f(f^{n-1}(x)) - f(f^n(x))| \\ &\leq c(|f^{n-1}(x) - f^{n+1}(x))| + |f^n(x) - f^n(x)|) \\ &= c(|f^{n-1}(x) - f^{n+1}(x)|) \\ &= c(|f^{n-1}) - f^n(x) + f^n(x) - f^{n+1}(x)| \\ &\leq c(|f^{n-1}(x) - f^n(x)| + |f^n(x) - f^{n+1}(x)|) \end{aligned}$$

Thus, $|f^n(x) - f^{n+1}(x)|(1-c) \leq c|f^{n-1}(x) - f^n(x)|$, or $|f^n(x) - f^{n+1}(x)| \leq (c/(1-c)).|f^{n-1}(x) - f^n(x)|$, and repeated application of this shows that

$$\begin{aligned} |f^n(x) - f^{n+1}(x)| &\leq (c/(1-c))^2|f^{n-2}(x) - f^{n-1}(x)| \\ &\leq \cdots \quad \cdots \quad \cdots \\ &\leq (c/(1-c))^n|x - f(x)| \,. \end{aligned} \tag{$*$}$$

Hence, using $(*)$,

$$\begin{aligned} |f^n(x) - f^{n+r}(x)| &\leq |f^n(x) - f^{n+1}(x) + f^{n+1}(x) - f^{n+2}(x) + \cdots + \\ &\qquad + f^{n+r-1}(x) - f^{n+r}(x)| \\ &\leq |f^n(x) - f^{n+1}(x)| + |f^{n+1}(x) - f^{n+2}(x)| + \cdots + \\ &\qquad + |f^{n+r-1}(x) - f^{n+r}(x)| \\ &= (c/(1-c))^n|x - f(x)| + (c/(1-c))^{n+1}|x - f(x)| + \cdots + \\ &\qquad + (c/(1-c))^{n+r-1}|x - f(x)| \\ &= |x - f(x)|(c/(1-c))^n(1 + c/(1-c) + \cdots + (c/(1-c))^{r-1}) \\ &= |x - f(x)|(c/(1-c))^n((1 - c/(1-c)^r)/(1 - c/(1-c))) \\ &= |x - f(x)|(c/(1-c))^n(1-c).(1 - c/(1-c))^r/(1-2c) \\ &\leq |x - f(x)|(c/(1-c))^n.((1-c)/(1-2c)) \,, \end{aligned}$$

since $1 - c/(1-c) \leq 1$. But $c/(1-c) < 1$, and so the right hand side can be made smaller than any ϵ if $n \geq N$. Thus $\{f^n(x)\}$ is a Cauchy sequence and so convergent to a point ζ, say. Then we have

$$\begin{aligned} |\zeta - f(\zeta) &\leq |\zeta - f^n(x) + f^n(x) - f(\zeta)| \\ &\leq |\zeta - f^n(x)| + |f^n(x) - f(\zeta)| \\ &= |\zeta - f^n(x)| + |f(f^{n-1}(x)) - f(\zeta)| \,, \end{aligned}$$

which, by the hypothesis is less than or equal to $|\zeta - f^n(x)| + c(|f^{n-1}(x) - f(\zeta)| + |\zeta - f^n(x)|)$. Now let $x \to \infty$, then $f^n(x) \to \zeta$ and we get $|\zeta - f(\zeta)| \leq 0 + c(|\zeta - f(\zeta)| + |\zeta - \zeta|)$, i.e., $|\zeta - f(\zeta)| \leq c|\zeta - f(\zeta)|$, and since $c \neq 1$, we get $f(\zeta) = \zeta$, as required.

Exercise 5.1. (i) By the mean value theorem, $(f(x)f(a))/(x-a) = f'(c)$, for some $c \in (a, x)$. Choose a so large that $|f'(c)| < \epsilon$ (c is also large, being larger than a). Then $(f(x) - f(a))/(x-a) \to 0$ as $x \to \infty$, i.e., $f(x)/(x-a) - f(a)/(x-a) \to 0$ and so $f(x)/(x-a) \to 0$, i.e., $(f(x)/x)(x/(x-a)) \to 0$, but $x/(x-a) \to 1$, so $f(x)/x \to 0$.
(ii) No, for example, let $f(x) = (\sin x^2)/x$.
(iii) Again by the mean value theorem, $(g(2^{n+1}) - g(2^n))/(2^{n+1} - 2^n) = g'(c_n)$ $(2^n < c_n < 2^{n+1})$, i.e., $2^n g'(c_n) = g(2^{n+1}) - g(2^n)$. But now $|c_n g'(c_n)| < |2^{n+1}g'(c_n)| = 2(g(2^{n+1}) - g(2^n)) \to 2(l - l)$, where $g(x) \to l$ as $x \to \infty$, as required.

The example. Take $g(x) = \sin x/x$, then $x.g'(x) = \cos x - (\sin x)/x$ does not tend to any limit.

Exercise 5.2. The right hand side is equal to $\lim_{h\to 0}\{(f(x+h) - f(x))/h - (f(x) - f(x-h))/h\}/h = \lim_{h\to 0}\{f'(x+\theta h) - f'(x-h+\theta' h)\}/h$, (as by the mean value theorem $(f(b) - f(a))/(b-a) = f'(a + (b-a)\theta)$, with $a = x - h$, $b = x$ for the second term, where $\theta, \theta' \to 0$ as $h \to 0$), which on rewriting is equal to $\lim_{h\to 0}((f'(x+\theta h) - f'(x+\theta h + \theta' h - \theta h - h))/h(1 + \theta + \theta')).(1 + \theta - \theta') = \lim_{h\to 0} f''(x + \theta h - h(1 + \theta - \theta') + h(1 + \theta - \theta')\theta'').(1 + \theta - \theta')$, where $\theta, \theta', \theta'' \to 0$ as $h \to 0$ (by the mean value theorem, applied to f' with $a = x + \theta h - h(1 + \theta - \theta'), b = x + \theta h$), which then tends to $f''(x)$ by the continuity of $f''(x)$, as required.

Exercise 5.3. Let $\phi(x) = x.f((\log x)/\alpha)$, then $d\phi/dx = (1/\alpha)(f'((\log x)/\alpha) + \alpha f((\log x)/\alpha)) \to 0$, as $(\log x)/\alpha \to \infty$, by hypothesis, i.e., as $x \to \infty$.

Further, since $\phi'(x) \to 0$ as $x \to \infty$, we see by Exercise 5.1, that $\phi(x)/x \to 0$ as $x \to \infty$, i.e., $f((\log x)/\alpha) \to 0$ as $x \to \infty$, i.e., $f(x) \to 0$ as $x \to \infty$, as required.

Now by the mean value theorem, $(\phi(t_1) - \phi(t_2))/(t_1 - t_2) = \phi'(\xi) \to 0$ $(t_2 < \xi < t_1)$ as t_2 and therefore t_1, as well as $\xi \to \infty$ (proved above).

Thus if $t_1, t_2 > \Delta$, $|(\phi(t_1) - \phi(t_2))/(t_1 - t_2)| < \epsilon$. In this, replace t_1 by $e^{\alpha t_1}$ and t_2 by $e^{\alpha t_2}$. This then gives the required result.

Exercise 5.4. Let $\phi(t) = f'(t) + 2f(t) - 2$, then $\phi'(t) = f''(t) + 2f'(t)$, and so $\phi'(t) + \phi(t) = f''(t) + 3f'(t) + 2f(t) - 2 \to 2 - 2 = 0$. Hence by Exercise 5.3, $\phi(t) \to 0$ as $t \to \infty$, i.e.,

$$f'(t) + 2f(t) - 2 \to 0 \ , \ \text{as} \ t \to \infty \ .$$

Now let $\psi(t) = f(t) - 1$, then $\psi'(t) = f'(t)$. Therefore $\psi'(t) + 2\psi(t) = f'(t) + 2f(t) - 2 \to 0$ as $t \to \infty$ (by above). Therefore, again by Exercise 5.3, $\psi(t) \to 0$ as $t \to \infty$, i.e., $f(t) \to 1$ as $t \to \infty$.

Exercise 5.5. *Hint:* Use the first and the second mean value theorems, viz.

(i) $f(a+h) = f(a) + hf'(a + \theta h)$ (for some $0 < \theta < 1$),

(ii) $f(a+h) = f(a) + hf'(a) + (h^2/2)f''(a + \theta' h)$ (for some $0 < \theta' < 1$).

Exercise 5.6. Since $f'(x_0)$ exists, we have $(f(x_0 + h) - f(x_0))/h = f'(x_0) + \epsilon_h$, i.e., $f(x_0 + h) - f(x_0) = hf'(x_0) + h\epsilon_h$ and, by putting $-k$ for h, $f(x_0 - k) - f(x_0) = -kf'(x_0) - k\epsilon_{-k}$. Subtracting, we get $f(x_0 + h) - f(x_0) + f(x_0) - f(x_0 - k) = (h + k)f'(x_0) + h\epsilon_h + k\epsilon_{-k}$. Therefore the quotient $(f(x_0 + h) - f(x_0 - k))/(h + k) = f'(x_0) + (h\epsilon_h + k\epsilon_{-k})/(h + k)$. Here, since both $h\epsilon_h$ and $k\epsilon_{-k}$ tend to zero faster than h or k or $h + k$, we see that the latter term above tends to zero as h, k tend to 0, which proves the result.

For the function $f(x) = x^2 \sin(\pi/x)$, take $x_0 = 0$. Then

$$(f(h) - f(-k))/(h + k) = (h^2 \sin \pi/h + k^2 \sin \pi/k)/(h + k) \ .$$

Now choose $k = -h + h^2$, then the above quotient equals

$$(h^2 \sin(\pi/h) + (h^2 + h^4 - 2h^3) \sin(\pi/(h^2 - h)))/h = \sin(\pi/(h^2 - h)) + \text{a multiple of } h,$$

which does not tend to 0 as h tends to 0.

Exercise 5.7. (i) We have

$$\begin{aligned} f(x)/x &= ((f(x) - f(x'))/(x - x')).((x - x')/x) + f(x')/x \text{ , where } X < x' \leq x. \\ &= f'(c).(x - x')/x + f(x')/x \quad \text{, (by mean value theorem).} \end{aligned}$$

Now let $x \to \infty$ and see that $f(x)/x \to f'(c)$, $x' < c < \infty$. Letting $x' \to \infty$, it follows that $c \to \infty$, so the right hand side tends to λ (finite or infinite), as required.
(ii) Let $f(x) \to \lambda$ as $x \to \infty$, say $f(x) = \lambda + g(x)$, where $g(x) \to 0$ as $x \to \infty$. Then $(f(x+h) - f(x))/h = (\lambda + g(x+h) - \lambda - g(x))/h = (g(x+h) - g(x))/h$. Here keep h fixed and let $x \to \infty$ and we get $(f(x+h) - f(x))/h \to 0$ as $x \to \infty$, i.e., by the mean value theorem, $f'(x + \theta h) \to 0$ as $x \to \infty$ $(0 < \theta < 1)$. Letting $h \to 0$, this gives $f'(x) \to 0$ as $x \to \infty$.

Exercise 5.8. By the second mean value theorem, $f(a + h) = f(a) + h.f'(a) + (h^2/2).f''(a + \theta h)$, $(0 < \theta < 1)$. In this put $a = x, h = -x$ and we get

$$f(0) = f(x) + (-x)f'(x) + ((-x)^2/2).f''(x - \theta x) \text{ ,}$$

i.e., $f'(x) - f(x)/x = (x/2).f''(x(1 - \theta))$, as required, with $\xi = x(1 - \theta)$. Here $0 < \xi < x$, for we have to show that $0 < x(1 - \theta) < x$, i.e., since $0 \leq x$, that $0 < 1 - \theta < 1$, which is true, since $0 < \theta < 1$. If $f''(x) > 0$ for $x > 0$, then $f'(x)$ is an increasing function of x. It follows that the graph of f is as shown in Figure A.2.10, which then clearly shows that $f(x)/x$ too is an increasing function of x.

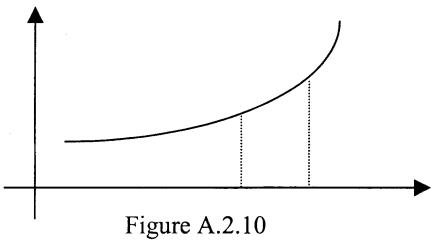


Figure A.2.10

Exercise 5.9. *Hint:* Let $g(x) = (x^a - 1).f(x)$, then $g(0) = g(1) = 0$. Now apply Rolle's theorem to $g(x)$ in $[0, 1]$.

Exercise 5.10. *Hint:* Let $h(x) = (f(x) - f(a))/(g(x) - g(a))/(g(b) - g(a)) - f(x) + f(a)$. Since $g'(x) \neq 0$ in $[a, b]$, it follows that $g(b) - g(a) \neq 0$. Check further that $h(a) = h(b) = 0$. Hence, by Rolle's theorem, there exists $c \in (a, b)$ such that $h'(c) = 0$, proving the result.

Exercise 5.11 (See American Math. Monthly, **87** (1980), 817). Put $a_0 = 0, a_n = 1$. Since f is differentiable on $[0,1]$ and so continuous, there exist a_k $(a_{k-1} < a_k \leq 1)$ such that $f(a_k) = k/n$. This is done successively for each k as follows:

First we want a_1 with $a_0 = 0 < a_1 < 1 = a_n$, such that $f(a_1) = 1/n$. This is possible because $f(a_0) = f(0) = 0$, $f(a_n) = f(1) = 1$. So there exist a_1 $(0 = a_0 < a_1 < a_n = 1)$, and $f(a_1) = 1/n$.

Next, $f(a_1) = 1/n$, $f(a_n) = 1$ and $1/n < 2/n < 1$, so there exists a_2 $(a_1 < a_2 < 1)$ such that $f(a_2) = 2/n$; and so on.

Hence by the mean value theorem, there exist x_k, $a_{k-1} < x_k < a_k$, such that

$$f'(x_k)=(f(a_k)-f(a_{k-1}))/(a_k-a_{k-1})=(k/n-(k-1)/n)/(a_k-a_{k-1})=1/n(a_k-a_{k-1}).$$

So now we have

$$\sum_{k=1}^{n} 1/f'(x_k) = \sum_{k=1}^{n} n(a_k - a_{k-1}) = n(a_n - a_0) = n(1-0) = n\,.$$

Since the a_k are strictly increasing and $a_{k-1} < x_k < a_k$, the points x_k are distinct, as required.

Another solution: The intermediate value theorem (IVT) is:

$$f \text{ continuous in } [a,b]\,,\ f(a) < c < f(b) \Rightarrow \exists\, \xi \in (a,b) \text{ such that } f(\xi) = c.$$

(i) Let $a_0 = 0$. Then $a_0 = f(0) < f(1) = 1 = a_n$. Hence by the IVT, there exists $a_1 \in (0,1)$ such that $f(a_1) = 1/n$.
(ii) Since $1/n < 2/n < 1$, i.e., $f(a_1) < 2/n < f(1)$, so by the IVT, there exists $a_2 \in (a_1, 1)$ such that $f(a_2) = 2/n$.
(iii) Since $2/n < 3/n < 1$, i.e., $f(a_2) < 3/n < f(1)$, so by the IVP, there exists $a_3 \in (a_2, 1)$ such that $f(a_3) = 3/n$; and so on; thus we have
(iv) $0 = a_0 < a_1 < \ldots < a_{n-1} < a_n = 1$, with $f(a_k) = k/n$, and so by IVT, there exists $\xi \in (a_{k-1}, a_k)$ such that

$$f'(\xi_k) = \frac{f(a_k) - f(a_{k-1})}{a_k - a_{k-1}} = \frac{k/n - (k-1)/n}{a_k - a_{k-1}} = \frac{1}{n(a_k - a_{k-1})}\,.$$

Therefore $\sum_{k=1}^{n} 1/f'(\xi_k) = \sum_{k=1}^{n} n(a_k - a_{k-1}) = n(a_n - a_0) = n(1-0) = n$, where, since the a_k are strictly increasing, the ξ_k are distinct.

Exercise 5.12. Let $\epsilon > 0$ and choose $\Delta > 0$ such that

$$|f(x) + f'(x)| < \epsilon \quad \text{for all } x \geq \Delta \qquad (*)$$

By Cauchy's mean value theorem, with $f(x).e^x$ for f and e^x for g, we have, for $x > \Delta$, a number $b \in (\Delta, x)$ such that

$$(f(x)e^x - f(\Delta)e^x)/(e^x - e^{\Delta x}) = (f(b)e^x + f'(b)e^x)/e^x = f(b) + f'(b)\,.$$

It follows that $|(f(x)e^x - f(\Delta)e^{\Delta})/(e^x - e^{x\Delta})| = |f(b) + f'(b)| < \epsilon$ (by $(*)$), if $x \geq \Delta$. Here the left hand side is equal to $|(e^x(f(x) - f(\Delta)e^{\Delta - x}))/e^x(1 - e^{\Delta - x})|$. So $|f(x) - f(\Delta)e^{\Delta - x}| < \epsilon|1 - e^{\Delta - x}|$ or $|f(x)| < |f(\Delta)|.e^{\Delta - x} + \epsilon|1 - e^{\Delta - x}| < 2\epsilon$, if x is large enough (do not forget this, for given ϵ, Δ is fixed). This completes the proof.

Exercise 5.13. For $a > 0$, the mean value theorem implies

$$f(a+1) = f(a) + f'(a) + (1/2)f''(a) + (1/6)f'''(\xi_1) \ , \quad \text{where } a < \xi_1 < a + 1 \ ,$$

and

$$f(a-1) = f(a) - f'(a) + (1/2)f''(a) - (1/6)f'''(\xi_2) \ , \quad \text{where } a - 1 < \xi_2 < a \ .$$

Subtract and add these to get

$$f(a+1) - f(a-1) = 2f'(a) + (1/6)f'''(\xi_1) + (1/6)f'''(\xi_2) \ ,$$

$$f(a+1) + f(a-1) = 2f(a) + f''(a) + (1/6)f'''(\xi_1) + (1/6)f'''(\xi_1) - (1/6)f'''(\xi_2) \ .$$

Let now $a \to \infty$. Then these become $c - c = \lim_{a \to \infty} 2f'(a) + 0 + 0$ and $c + c = 2c + \lim_{a \to \infty} f''(a) + 0 - 0$, proving the result.

Exercise 5.14. The second mean value theorem implies

$$f(a+h) = f(a) + (h/1!)f'(a) + (h^2/2!)f''(a + \theta h) \ , \quad 0 < \theta < 1 \ .$$

Take (i) $a = 2, h = -1$, (ii) $a = 3, h = -1$, (iii) $a = 2, h + 1$, to get
(i) $f(1) = f(2) - f'(2) + (1/2)f''(2 - \theta_1)$, i.e., since $f(1) = f(2) = 4$,

$$f'(2) = (1/2)f''(2 - \theta_1) > 0 \qquad (*)$$

(ii) $f(2) = f(3) - f'(3) + (1/2)f'(3 - \theta_2)$,
(iii) $f(3) = f(2) + f'(2) + (1/2)f''(2 + \theta_3)$.
By (ii), $f'(3) = f(3) + (1/2)f''(3 - \theta_2) - f(2)$, and substituting for $f(3)$ from (iii), this is equal to $f(2) + f'(2) + (1/2)f''(2 + \theta_3) + (1/2)f''(3 - \theta_2) - f(2)$, which is greater than 0 by (*) and the hypothesis.

Exercise 5.15. Let $f(x) = f_1(x).f_2(x) \ldots f_n(x)$, so that f is differentiable in $[a, b]$ and $f(a) = f(b) = 0$, $f(x) \neq 0$ for $x \in (a, b)$. So, by Rolle's theorem, there exists $c \in (a, b)$, such that

$$f'(c) = 0. \qquad (*)$$

Now, for $x \in (a, b)$, we have

$$f'(x) = \sum_{i=1}^{n} f_1(x) \ldots f_{i-1}(x).f_i'(x).f_{i+1}(x) \ldots f_n(x) = f(x).\sum_{i=1}^{n} f_i'(x)/f_i(x) \ .$$

Putting $x = c$, we get $0 = f'(c) = f(c).\sum_{i=1}^{n} f_i'(c)/f_i(c)$, (by (*)); but $c \in (a,b)$ and so $f(c) \neq 0$. It follows that $\sum_{i=1}^{n} f_i'(c)/f_i(c) = 0$.

Exercise 5.16. Let

$$f(x) = \begin{cases} 0 & \text{if } x \text{ is algebraic,} \\ 1 & \text{if } x \text{ is transcendental.} \end{cases}$$

This function is discontinuous everywhere and since x, x^m are either both algebraic or both transcendental, f has the desired property.

Exercise 5.17. Put $S(n) = f(1) + f(2) + \cdots + f(n)$. Then $S(n) - f(1) \leq \int_1^n f(x)dx \leq S(n)$. Dividing by $g(x)$ and noting that $f(1)/g(n)$ tends to 0 as n tends to infinity, we get $\lim_{n\to\infty} S(n)/g(n) = \lim_{n\to\infty}(\int_1^n f(x)\,dx/g(n))$. Here $\int_1^n f(x)\,dx = g(\infty) = \infty$, so by L'Hospial's rule, this equals $\lim_{n\to\infty} f(n)/g'(n)$.

As an application, we prove:

$$1 + 1/2 + 1/3 + \cdots + 1/n \approx \log n \text{ as } n \to \infty\,.$$

Indeed, take $f(x) = 1/x$, $g(x) = \log x$. These satisfy the hypothesis of the above exercise and hence also the conclusion, i.e., the required limit equals

$$\lim_{n\to\infty} (1/n)/(d(\log x)/dx) = \lim_{n\to\infty} ((1/n)/(1/n)) = 1\,.$$

Exercise 5.18 (Proof by N.R.C. Dockeray, Math. Mag., **15** (1931), 435). Let

$$\Phi(x) = \begin{vmatrix} f(x) & x^2 & x & 1 \\ f(a) & a^2 & a & 1 \\ f(b) & b^2 & b & 1 \\ f(c) & c^2 & c & 1 \end{vmatrix} \quad \text{so that} \quad \Phi'(x) = \begin{vmatrix} f'(x) & 2x & 1 & 0 \\ f(a) & a^2 & a & 1 \\ f(b) & b^2 & b & 1 \\ f(c) & c^2 & c & 1 \end{vmatrix}$$

(expand Φ by the first row and differentiate). Now $\Phi(a) = \Phi(b) = 0$, and so by Rolle's theorem, there exists a $\xi \in (a,b)$ such that $\Phi'(\xi) = 0$. Similarly, there exists an $\eta \in (b,c)$ such that $\Phi'(\eta) = 0$, i.e., $\Phi'(x)$ vanishes at $x = \xi, \eta$ and so there exists a $\zeta \in (\xi,\eta) \subset (a,c)$ such that $\Phi''(\zeta) = 0$. But again we have

$$\Phi''(x) = \begin{vmatrix} f''(x) & 2 & 0 & 0 \\ f(a) & a^2 & a & 1 \\ f(b) & b^2 & b & 1 \\ f(c) & c^2 & c & 1 \end{vmatrix} = -(b-c)(c-a)(a-b).f''(x) - 2\sum(b-c).f(a)\,,$$

i.e., there exists a $\zeta \in (a,c)$ such that

$$f(a)/(a-b)(a-c) + f(b)/(b-c)(b-a) + f(c)/(c-a)(c-b) = f''(\zeta)/2\,,$$

which is what we have to prove (check) but, if anything, this is a neater form!

Exercise 5.19. We have

$$\begin{aligned}\frac{d(x(f(x)+g(x)))}{dx} &= xf'(x)+f(x)+xg'(x)+g(x)\\ &= -g(x)+f(x)-f(x)+g(x)\\ &= 0\end{aligned}$$

Hence $x(f(x)+g(x))$ is a constant, say, $2A$. Similarly

$$\begin{aligned}\frac{d((f(x)-g(x))/x)}{dx} &= (x(f'(x)-g'(x))-(f(x)-g(x)))/x^2\\ &= (-g(x)+f(x)-f(x)+g(x))/x^2\\ &= 0\,.\end{aligned}$$

Hence, $(f(x)-g(x))/x$ is a constant, say, $2B$. These equations imply $f(x) = A/x + Bx$, $g(x) = A/x - Bx$ (A, B arbitrary constants).

Exercise 5.20. Use induction on n. For $n = 0$, let $g(x) = e^{-x}.f(x)$. Then $g(a) = g(b) = 0$. By Rolle's theorem there exists $c \in (a,b)$ such that $g'(c) = 0$, i.e., $e^{-c}f'(c) - f(c)e^{-c} = 0$, i.e., $f'(c) = f(c)$, as required.

For $n > 0$, let $h(x) = \sum_{i=0}^{n} f^i(x)$. Then $h(a) = h(b) = 0$ and $h(x) - h'(x) = f(x) - f^{(n+1)}(x)$. Apply the result for $n = 0$ to the function $h(x)$. So there exists $c \in (a,b)$ such that $0 = h(c) - h'(c) = f(c) - f^{(n+1)}(c)$, as required.

Exercise 5.21. By the mean value theorem, $(f(c+h) - f(c))/h = f'(\xi)$, for some ξ between c and $c+h$ (h may be negative). Letting h to go to 0, we get

$$f'(c) = \lim_{h\to 0} f'(\xi) = A\,.$$

Exercise 5.22. For $x \in [a,b]$, define $g(x) = f'(x) - (f(x) - f(a))/(b-a)$. We now consider cases:
Case 1: $f(c) > f(a)$. Choose d in $[a,c]$, such that $f(d)$ is a maximum for $f(x)$ in $[a,c]$ (so $f'(d) = 0$) and let $e \in (a,d)$ be such that

$$f'(e) = (f(d)-f(a))/(d-a) \quad \text{(first mean value theorem in } [a,d]) \qquad (*)$$

Then $g(e) = f'(e) - (f(e) - f(a))/(b-a) > 0$ (by $(*)$) and

$$g(d) = f'(d) - (f(d) - f(a))/(b-a) = -(f(d) - f(a))/(b-a) < 0\,,$$

since $f'(d) = 0$ and $f(d)$ is a maximum. Thus $g(e) > 0$, $g(d) < 0$ ans so, since $g(x)$ is continuous, by the intermediate value theorem, there exists a $\xi \in (e,d)$ such that $g(\xi) = 0$, as required.
Case 2: $f(c) < f(a)$. Proof is similar to that in case 1.
Case 3: $f(c) = f(a)$. Proof is again similar to that in case 1.

Exercise 5.23. We have $f(2) = f(x + (2-x)) = f(x) + (2-x)f'(x) + \{((2-$

$x)^2)/2\}f''(x+\theta_1(2-x))$, and $f(0) = f(x+(-x)) = f(x) + (-x)f'(x) + \{x^2/2\}f''(x+\theta_2(-x))$, where $0 < \theta_1, \theta_2 < 1$. Subtracting, we get $f(2) - f(0) = 2f'(x) + ((2-x)^2/2)f''(t) - (x^2/2)f''(t)$, say, where, since $0 < \theta_1, \theta_2 < 1$ and $0 \le x \le 2$, we get $0 \le t_1, t_2 \le 2$ (check). Hence

$$\begin{aligned} 2|f'(x)| &\le |f(2)| + |f(0)| + |((2-x)^2/2)f''(t_1)| + |(x^2/2)f''(t_2)| \\ &\le 1 + 1 + \frac{1}{2}((2-x)^2 + x^2) \text{ (since } |f| \le 1, |f''| \le 1) , \\ &= x^2 - 2x + 4 \\ &\le 4 \text{ (4 occurs at } x = 0, 2 \text{ in } [0,2]) , \end{aligned}$$

or $|f'(x)| \le 2$ in $[0,2]$, as required.

Exercise 5.24. Let $g(x) = x^2$. Then $g(0) = 0, g(1) = 1$, so the equation $g'(x) = cx$ has the unique solution $c = 2$ ($2x = cx$ and so $c = 2$). If f is any differentiable function on $[0,1]$, with $f(0) = 0, f(1) = 1$, then the function $h(x) = f(x) - g(x)$ is differentiable on $[0,1]$ and $h(0) = h(1) = 0$. Hence, by Rolle's theorem, there exists a point $\xi \in (0,1)$ such that $h'(\xi) = 0$, i.e., $f'(\xi) - g'(\xi) = 0$, i.e., $f'(\xi) = 2\xi$, i.e., $c = 2$ works and no other c works, as required.

Exercise 5.25. Differentiate the relation $dx/dy = 1/(dy/dx)$ with respect to y, to give

$$\begin{aligned} d^2x/dy^2 = d(1/(dy/dx))/dy &= (d(1/(dy/dx))/dx).(dx/dy) \\ &= ((-1/(dy/dx)^2).(d^2y/dx^2)).(1/(dy/dx)) \\ &= -d^2y/dx^2/(dy/dx)^3 . \end{aligned}$$

On cross multiplying, this gives $d^2y/dx^2 = -(dy/dx)^3.(d^2x/dy^2) = -(dy/dx)^3.(1/(d^2y/dx^2))$, (since we are assuming that $d^2x/dy^2 = 1/(d^2y/dx^2)$), which gives $(d(dy/dx)/dx)^2 = -(dy/dx)^3$.

Let now $dy/dx = \phi(x)$. Then this becomes $d\phi/dx = (-\phi)^{3/2}$, i.e., $(-\phi)^{-3/2}.d(-\phi) = -dx$, which gives, on integration, $(-\phi)^{-1/2}/(-1/2) = -x - a$, say, or $2/\sqrt{-\phi} = x+a$, i.e., $dy/dx = -4(x+a)^{-2}$, giving $y = 4/(x+a) - b$, say, i.e., *the family of translates of the rectangular hyperbola* $xy = 4$, *i.e.*, $(x+a)(y+b) = 4$.

Exercise 5.26. (i) Since $y = \cos x.\cos(x+2) - \cos^2(x+1)$,

$$\begin{aligned} dy/dx &= -\cos x.\sin(x+2) - \cos(x+2).\sin x + 2\cos(x+1).\sin(x+1) \\ &= -(\sin(x+x+2)) + \sin(2(x+1)) \\ &= 0 . \end{aligned}$$

It follows that $y = K$, a constant. Taking $x = \pi/2$, we find $y = -\cos^2(\pi/2 + 1) = -\sin^2 1 = K$, i.e., the graph is a straight line, parallel to the x-axis, through the point $(\pi/2, -\sin^2 1)$.

(ii) To sketch the graph of $|x-1|+|y|=1$, first let $x \geq 1$, i.e., $x-1 \geq 0$ and our function becomes

$$y = \begin{cases} 2-x & \text{if } y > 0 \quad (1) \\ x-2 & \text{if } y < 0\,. \quad (2) \end{cases}$$

If now $x-1 \leq 0$, i.e., $x \leq 1$, our function becomes

$$y = \begin{cases} x & \text{if } y > 0 \quad (3) \\ -x & \text{if } y < 0\,. \quad (4) \end{cases}$$

Thus the graph is shown in Figure A.2.11.

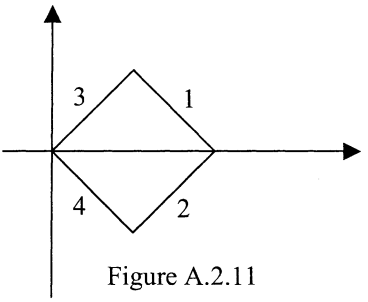


Figure A.2.11

Exercise 5.27. $f'(c) = \lim_{h\to 0}(f(c+h)-f(c))/h = \lim_{h\to 0} hf'(c+h\theta(h))/h$. Now $\theta(h)$ is continuous, so $\theta(h)$ is bounded in a neighbourhood of $h=0$ and hence $h.\theta(h) \to 0$ as $h \to 0$. So the above equals the limit of $f'(c+q)$ as q tends to 0. It follows that $f'(x)$ is continuous at $x=c$.

Exercise 5.28. It does not follow that $f'(1)$ exists, for example take $f(x) = |x-1|$.

Exercise 5.29. Differentiate (1) and (2) with respect to y:

$$f'(x+y) = f(x)f'(y) - g(x)g'(y)\,,$$

$$g'(x+y) = f(x)g'(y) + g(x)f'(y)\,.$$

In this put $y=0$ and we get $f'(x) = -g(x)g'(0)$ and $g'(x) = f(x)g'(0)$. Eliminating $g'(0)$, we get $f(x)f'(x)+g(x)g'(x)=0$. Integrating, we get $f^2(x)+g^2(x)=K$. Here, if $K=0$, then $f^2=0$ and $g^2=0$, i.e., $f \equiv 0$, and $g \equiv 0$, which is not the case since we are given that f and g are not constants.

Now $f^2(x+y)+g^2(x+y) = (f^2(x)+g^2(x))(f^2(y)+g^2(y))$, so $K = K.K$. Since $K \neq 0$, it follows that $K=1$, as required.

Exercise 5.30. Consider the curve $\Gamma : y = \log x$. We have $dy/dx = 1/x$. At the point $P(\alpha,\beta)$, let the tangent go through $(0,0)$. It is easy to check that then $(\alpha,\beta) = (e,1)$. Thus for $x_0 < e$, the line joining $(0,0)$ and $(x_0, \log x_0)$ cuts Γ in a further point $(y_0, \log y_0)$, as required.

Next, $\log y - \log x = \int_x^y dt/t$. Since the graph of $f(t) = 1/t$ is concave upwards, this integral is less than its trapezoidal approximation (see §A.1), i.e., $\log y - \log x < (y - x)(1/x + 1/y)/2$, i.e.,

$$2(\log y - \log x)/(y - x) < 1/x + 1/y \qquad (*)$$

Now the line through the three points $(0,0), (x, \log x), (y, \log y)$ (collinear, since $x \log y = y \log x$) has slope $(\log y - \log x)/(y - x) = \log x/x = \log y/y$ and so

$$\log x/x + \log y/y = 2((\log y - \log x)/(y - x)) ,$$

which, by $(*)$ above, is less than $1/x + 1/y$.

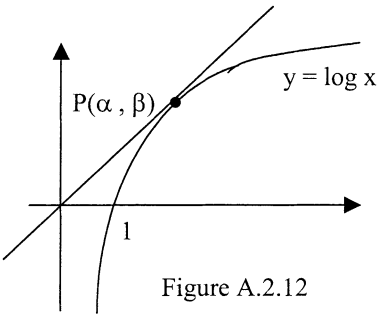


Figure A.2.12

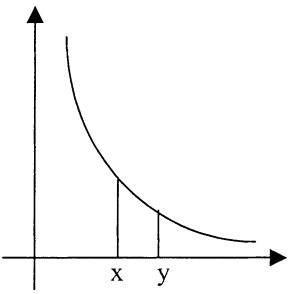


Figure A.2.13

Exercise 5.31. By symmetry, we need to find the point $P = (a, b)$ on $y = e^x$ that is closest to the line $y = x$. This point has its tangent parallel to $y = x$ (clearly). Hence the point satisfies $(d(e^x)/dx)_P = 1$, i.e., $(e^x)_{(a,b)} = 1$, i.e., $e^a = 1$, i.e., $a = 0$, which gives $b = e^a = e^0 = 1$, i.e., $P = (0, 1)$. It follows that the two required "closest" points are

$$(0,1) \text{ on } y = e^x \text{ and } (1,0) \text{ on } y = \log x .$$

Exercise 5.32. Let $P = (p, p)$ be an intersection of $y = \log_a x$ and $y = a^x$ (by symmetry, it lies on the line $y = x$). For the function $y = \log_a x$, we have $dy/dx = 1/x . \log_e a$ and so at P, since we want $(dy/dx)_P = 1$, we have $1 = 1/(p . \log_e a)$, which gives

$$p = 1/\log_e a . \qquad (1)$$

Similarly, for the function $y = a^x$, we have $dy/dx = a^x . \log_e a$ and so at P, $1 = a^p . \log_e a$, giving

$$1/\log_e a = a^p \qquad (2)$$

Now (1) and (2) imply $a^p = p$ and taking logarithms to base a, this implies

$$p = \log_a p \qquad (3)$$

Hence $1/\log_e a = p$ (by (1)) $= \log_a p$ (by (3)) $= \log_e p/\log_e a$, which implies $\log_e p = 1$, i.e., $p = e$.

So now, since $P = (p,p)$ lies on the curve $y = a^x$, we get, $p = a^p$, i.e., $e = a^e$, i.e., $e.\log_e a = 1$, i.e., $\log_e a = 1/e$, i.e., $a = e^{1/e}$. thus the two curves $y = (e^{1/e})^x$ and $y = \log_{e^{1/e}} x$ meet at just one (real) point (see Figure A.2.14).

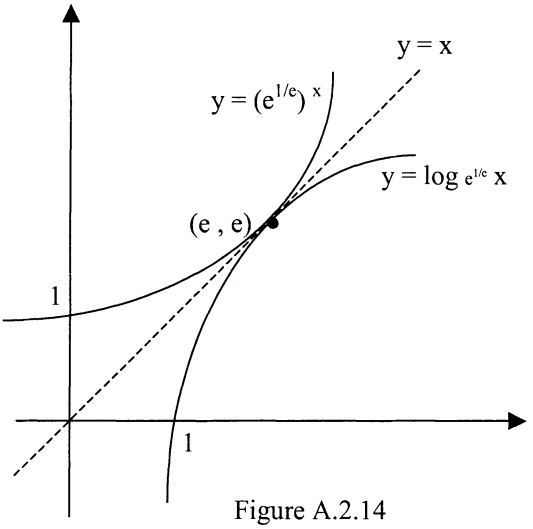


Figure A.2.14

Exercise 5.33. Write $t = \log x$, then $(*)$ holds if and only if

$$k.\log t = t, \text{ i.e. }, 1/k = \log t/t = F(t),$$

say, so that $F(t) \to 0$ as $t \to \infty$.

Now $F'(t) = (1 - \log t)/t^2$, which is greater than 0 if $t \in (0,e)$ and is less than 0 if $t \in (e,\infty)$; so $F(t)$ is increasing in (e,∞) and $F(t)$ has a maximum at $t = e$. But $F(1) = 0$, $F(e) = 1/e > 1/3$; so there exist two solutions of the equation $(*)$.

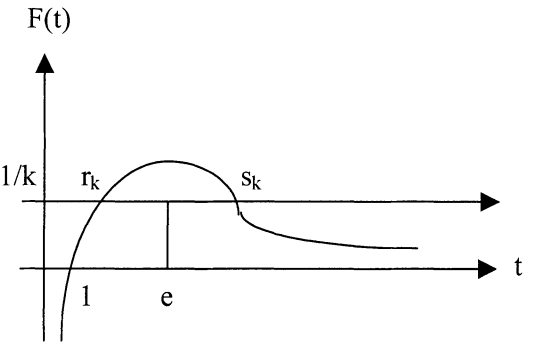


Figure A.2.15

Further, as $k \to \infty$, $1/k \to \infty$, so the two roots tend to 1 and ∞. These are the limits of the roots of the equation $\log t/t = 1/k$ as $k \to \infty$; an equation got from $(*)$ by taking logarithms, so the roots of $(*)$ tend to e and ∞ as $k \to \infty$.

Exercise 5.34. We have

$$f'(x) = \frac{\log(\frac{x+1}{x})(\frac{1}{x+2} - \frac{1}{x+1}) - \log(\frac{x+2}{x+1})(\frac{1}{x+1} - \frac{1}{x})}{(\log(x+1) - \log x)^2} ,$$

so it is enough to prove that $f'(x) > 0$, i.e., that

$$\log((x+1)/x).(1/(x+2) - 1/(x+1)) > \log((x+2)/(x+1)).(1/(x+1) - 1/x),$$

or, simplifying, it is enough to prove that

$$\log((x+1)/x).(1/(x+2)) < \log((x+2)/(x+1)).(1/x) \qquad (*)$$

Now

$$\log(x+2) - \log(x+1) = \int_{x+1}^{x+2} dt/t > 1.(1/(x+2)) ,$$

since $t \leq x+2$, so $1/t \geq 1/(x+2)$, i.e.,

$$\log((x+2)/(x+1)).(1/x) > (1/(x+2)).(1/x) \qquad (1)$$

Similarly

$$(\log(x+1)/x)(1/(x+2)) = (1/(x+2))\int_{x}^{x+1} dt/t < 1/(x+2) = (1/(x+2)).1.(1/x),$$

since $x \leq t \leq x+1$, so $1/x \geq 1/t \geq 1/(x+1)$, i.e., $1/t \leq 1/x$; hence,

$$(\log(x+1)/x)(1/(x+2)) < (1/(x+2))(1/x) . \qquad (2)$$

Now (1) and (2) give $(*)$.

Exercise 5.35. We prove that $f(n+h) > f(n-h)$ $(0 < h < n)$. For, this is the same thing as showing that $((n+h)^n e^{-n-h})/((n-h)^n e^{-n+h}) > 1$ (on taking logarithms), which is the same thing as showing that $\log((n+h)/(n-h)) > 2h/n$. Here the left side is the area H below the curve $y = 1/x$, above the x-axis, between the two ordinates $x = n-h$ and $x = n = h$, while the right hand side is the area T of the trapezium formed by the tangent to $y = f(x)$ at $x = n$, the x-axis, and the above two ordinates.

Exercise 5.36. (i) Since $\mathbb{R}$ is connected and f is continuous, $f(\mathbb{R})$ $(\subseteq \mathbb{R})$ is connected and so $f(\mathbb{R})$ is an interval, I, say, i.e., $I = (a,b)$, $[a,b]$, $[a,b)$, $(a,b]$, (∞,a), $(-\infty,a]$, (a,∞), $[a,\infty)$, $(\infty,-\infty)$. We shall show that $I = [a,b], [a,\infty), (-\infty,a], (-\infty,\infty)$, i.e., I is closed. Let a be, say, the left end point of I. First suppose a is finite (a may or may not belong to I). The case

$I = \{a\}$ is trivial, as then $f(x) = a$, for all x. So suppose I has positive length, i.e., points to the right of a, close enough to a, are in I. Noting that $f(\alpha) = \alpha$ for all $\alpha \in I$ (by $(*)$), we have $f(a) = f(\lim_{x\to a+} x) = \lim_{x\to a+} f(x)$ (since f is continuous) $= \lim_{x\to a+} x$ (by above) $= a$, i.e., $f(a) = a$ and so $a \in I$, as required. If a is infinite, say $-\infty$, then the right neighbourhood of a is in I trivially. Thus in all cases I is closed (possibly unbounded), as claimed.

This now gives us all the required continuous functions that satisfy $(*)$ as follows: First let $I = [a, b] = f(\mathbb{R})$ be bounded. Then the graph of $y = f(x)$ lies between the two horizontal lines $y = a$ and $y = b$ (see Figure A.2.16).

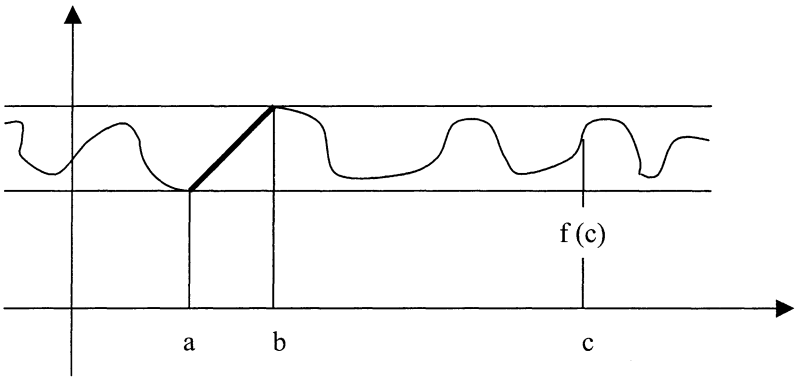


Figure A.2.16

Further, f restricted to $[a, b]$ ($[a, b]$ on the x-axis now) is the identity function. Thus all the required functions are any continuous continuations of the bold line (shown in Figure A.2.16) on either side, that keep within the limits $y = a$ and $y = b$. Thus for example, if c is any point on the x-axis, $f(c) \in [a, b]$, and so $f(f(c)) = c$, as required.

If I is a ray, say $[a, \infty)$, Figure A.2.16 will help in completing the proof.
(ii) This will follow if we show that I, the range of f, is either equal to $\mathbb{R}$, or to a singleton point $\{K\}$. Then in the former case, since $f(x) = x$ for all $x \in I$, we see that $f(x) = x$ for all x, while in the latter case, $f(x) \in I = \{K\}$ for all x, i.e., $f(x) = K$.

If $I = \mathbb{R}$, we are through, so let I be one of $[a, b]$, $[a, \infty)$, $(-\infty, b]$. It is enough to deal with one case, say $I = [a, \infty)$ (other cases being similar).

Take x near a, to the right of a; then $f(x) = x$ and so

$$\lim_{x\to a+} (f(x) - f(a))/(x - a) = \lim_{x\to a+} (x - a)/(x - a) = 1 .$$

However, for x to the left of a,

$$x - a < 0, \text{ but } f(x) \geq a \text{ (since } f(x) \in I \text{ for all } x) \qquad (**)$$

and so

$$\lim_{x\to a-}(f(x)-f(a))/(x-a)=\lim_{x\to a-}(f(x)-a)/(x-a)\le 0 \quad \text{(by (**))}$$

(the limit exists, since f is differentiable). This shows that the left and the right derivatives of $f(x)$ at $x=a$ can not be equal, so $f(x)$ is not differentiable at $x=a$. It follows that $I=\{a\}$, as required.

Exercise 5.37. I (F.D. Hammer, American Math. Monthly, **82** (1975), 415-416). Let $f(x)=\sum_{n=1}^{\infty} g(n!\,x)/(n!)^2$, where $g(y)$ is the periodic function of period 1, defined in $[-1/2,1/2]$, by $g(y)=y(1-4y^2)$, so that $g(n)=0$ for all $n\in\mathbb{Z}$ and $g(n+1/2)=0$ for all $n\in\mathbb{Z}$, and $g(x)$ has a continuous derivative for all x and this derivative is equal to 1 at all integers, so (i) is satisfied. A rough graph of the function $g(x)$ is drawn in Figure A.2.17.

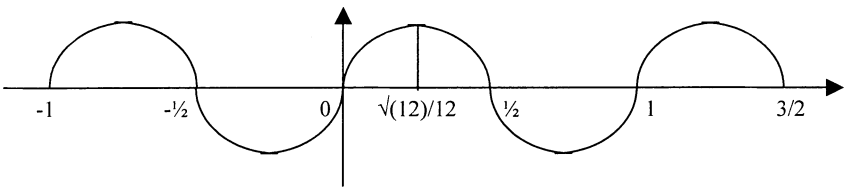


Figure A.2.17

Further, for any $x\in\mathbb{Q}$, $f(x)$ has at most finitely many nonzero terms and they are rational, i.e., (ii) is satisfied.

Finally, the formal derivative of $f(x)$ converges uniformly and absolutely and therefore converges to the derivative of $f(x)$. For $x\in\mathbb{Q}$, the derivative of $f(x)$ is the same as the series for $e=\sum 1/n!$, except for at most finitely many terms, which are rational. Thus, for $x\in\mathbb{Q}$, $f'(x)=e+\alpha$, $\alpha\in\mathbb{Q}$, i.e., (iii) holds.

II. Let

$$f(x)=\begin{cases}1/2^{2n-1} & \text{if } x=k/2^n, \text{ where } k \text{ is an odd integer less than } 2^n,\\ 0 & \text{otherwise.}\end{cases}$$

Since the number of odd k that are less than 2^n is 2^{n-1}, we see that $f(x)$ equals $1/2^{2n-1}$ for 2^{n-1} values of x, i.e., 2^{n-1} times. Hence the variation of $f(x)$ in $[0,1]$ is $\sum_{n=1}^{\infty} 2^{n-1}/2^{2n-1}=\sum_{n=1}^{\infty} 1/2^n=1$, i.e., $f(x)$ is of bounded variation and therefore can be written as a difference of two monotone increasing functions, i.e., $f=f_1-f_2$. It follows that f is differentiable almost everywhere, i.e., on a dense set. However, f is discontinuous for $x=k/2^n$, which forms a dense subset of $[0,1]$.

Another example: Let

$$f(x)=\begin{cases}q^{-q} & \text{if } x=p/q,\ (p,q)=1,\\ 0 & \text{if } x \text{ is irrational.}\end{cases}$$

Then $f(x)$ is discontinuous at each rational, but differentiable at each algebraic irrational.

III. If such a function exists, then by the second mean value theorem, for each $n = 1, 2, \ldots$, there exists a $\xi_n \in (0, 1/n)$, such that

$$f(1/n) = f(0) + f'(0)/n + f''(\xi_n)/2n^2 \,. \qquad (*)$$

Now $\sum f(1/n)$ is convergent, so $f(1/n) \to 0$ as $n \to \infty$ and since $f(x)$ is continuous at $x = 0$, $f(0) = 0$. Further, $f''(x)$ is continuous, and so bounded, say $|f''(x)| \leq M$. Then $\sum |f''(\xi_n)/2n^2| \leq \sum M/2n^2$ and so $f''(\xi_n)/2n^2$ is absolutely convergent. Now by $(*)$, noting that $f(0) = 0$, we have

$$\sum_{n=1}^{N} f(1/n) = \sum_{n=1}^{N} f'(0)/n + \sum_{n=1}^{N} f''(\xi_n)/2n^2 \,.$$

Here, the left hand side is convergent and the second term on the right hand side is convergent, but the first term on the right hand side is divergent, which is a contradiction, unless $f'(0) = 0$ and $(*)$ gives

$$\sum |f(1/n)| = \sum |f''(\xi_n)/2n^2| \,.$$

Here the right side is convergent while the left side is divergent–a contradiction. So such a function can not exist.

Exercise 6.1. (i) is true, for the hypothesis implies that $a_n^4 \to 0$ and so $a_n^4 < 1$ if $n > N$, therefore $a_n^5 < a_n^4$ if $n > N$ (even if $a_n < 0$), and so $\sum a_n^5$ is convergent.
(ii) is false, for example take

$$a_n = \begin{cases} 1/n^2 & \text{if } n \text{ is not a square,} \\ 1/n & \text{if } n \text{ is a square.} \end{cases}$$

(iii) is false, for example take $a_n = (-1)^n/\log n$.
(iv) does not ensure convergence, for example take

$$a_n = \begin{cases} n & \text{if } n \text{ is a prime,} \\ 0 & \text{otherwise.} \end{cases}$$

(v) ensures convergence, for suppose $|a_{n+1}/a_n| \leq k$ for $n \geq N$. Then for $M > N$, we have

$$|a_M/a_N| = |(a_M/a_{M-1}).(a_{M-1}/a_{M-2}) \ldots (a_{N+1}/a_N)| \leq k^{M-N} \,,$$

i.e., $|a_m| \leq k^{m-n}|a_N| \to 0$ as $M \to \infty$, since $k < 1$, i.e., $a_n \to 0$.
(vi) does not ensure convergence, for example take the sequence

$$\frac{1}{2}, \frac{0}{2}; \frac{1}{3}, \frac{2}{3}, \frac{3}{3}; \frac{3}{4}, \frac{2}{4}, \frac{1}{4}, \frac{0}{4}; \frac{1}{5}, \frac{2}{5}, \frac{3}{5}, \frac{4}{5}, \frac{5}{5}; \frac{5}{6}, \frac{4}{6}, \frac{3}{6}, \frac{2}{6}, \frac{1}{6}, \frac{0}{6}; \ldots$$

Then $a_{n-1} - a_n \to 0$ as $n \to \infty$, a_n are bounded but a_n oscillate between 0 and 1.
(vii) $\sum a_n$ convergent implies that $a_n < 1$ for all $n > N$ and hence $a_n^2 < a_n$ if $n > N$ so that $\sum a_n^2$ is convergent.
(viii) $a_n/(1-a_n) < 2a_n + 2a_n^2$ if $n > N$ (check), so $\sum a_n/(1-a_n)$ is convergent, since $\sum a_n$ and $\sum a_n^2$ are convergent.
(ix) $\sum a_n^2$ is not necessarily convergent, for example, take $a_n = (-1)^n/\sqrt{n}$.
(x) $\sum a_n/(1-a_n)$ not necessarily convergent, for example, let

$$\sum a_n = \underbrace{\frac{1}{2} - \frac{1}{2} + \frac{1}{2} - \frac{1}{2} + \frac{1}{2} - \frac{1}{2}}_{(2^2-1 \text{ pairs } 1/2-1/2)} + \underbrace{\frac{1}{3} - \frac{1}{3} + \frac{1}{3} - \frac{1}{3} + \cdots + \frac{1}{3} - \frac{1}{3}}_{(3^2-1 \text{ pairs } 1/3\text{-}1/3)} + \cdots +$$
$$+ \underbrace{\frac{1}{n} - \frac{1}{n} + \frac{1}{n} - \frac{1}{n} + \cdots + \frac{1}{n} - \frac{1}{n}}_{((n^2-1) \text{ pairs } 1/n-1/n)} + \cdots$$

Here, $\sum a_n$ is convergent with sum 0 clearly, since the series is alternating and monotone decreasing and $a_n \to 0$; but

$$\sum \frac{a_n}{1-a_n} = \underbrace{1 - \frac{1}{3} + 1 - \frac{1}{3} + 1 - \frac{1}{3}}_{(2^2-1 \text{ pairs } 1-1/3)} + \underbrace{\frac{1}{2} - \frac{1}{4} + \frac{1}{2} - \frac{1}{4} + \cdots + \frac{1}{2} - \frac{1}{4}}_{(3^2-1 \text{ pairs } 1/2\text{-}1/4)} + \cdots +$$
$$+ \underbrace{\frac{1}{n-1} - \frac{1}{n+1} + \frac{1}{n-1} - \frac{1}{n+1} + \cdots + \frac{1}{n-1} - \frac{1}{n+1}}_{((n^2-1) \text{ pairs } 1/(n-1)-1/(n+1))} + \cdots$$

Here $1/(n-1) - 1/(n+1) = 2/(n^2-1)$, and so each block of n^2-1 pairs sums to 2 and so our series under consideration is divergent.

Exercise 6.2. We have $b_n/a_n = 1 + \epsilon_n$, where $\epsilon_n \to 0$, and so

$$\begin{aligned} b_n &= a_n + \epsilon_n a_n \text{ , since } a_n > 0 \\ &< a_n + a_n \text{ , } \quad \text{if } n > N, \text{since } \epsilon_n < 1 \\ &= 2a_n \text{ .} \end{aligned}$$

So $\sum b_n$ is convergent.

Exercise 6.3. Let $\epsilon > 0$ (and $k > 0$) be given. For N arbitrarily large, select an n $(n > m > N)$ such that $a_n \geq k/n$ and then

$$\begin{aligned} |a_{m+1} + \cdots + a_n| &> (n-m)a_n && \text{(since } a_n \text{ is decreasing)} \\ &\geq (n-m)k/n && \text{(since } a_n \geq k/n) \\ &= k - mk/n \\ &> \epsilon \text{ ,} \end{aligned}$$

if n is large enough, i.e., we can not make $|a_{m+1} + \cdots + a_n|$ arbitrarily small and so $\sum a_n$ is divergent, by the general principle of convergence.

Exercise 6.4. Observe that $\sum(c_n + id_n)$ convergent means $\sum c_n$ and $\sum d_n$ are convergent. The length λ_N of the segment $S_{N-1}S_N$ is the distance between the points $\sum_{n=1}^{N-1}(c_n + id_n)$ and $\sum_{n=1}^{N}(c_n + id_n)$, which is equal to $\sqrt{(c_N^2 + d_N^2)} \leq c_N + d_N$, since c_N and d_N are both greater than or equal to 0. Hence the length

$$\sum_{N=1}^{M} \lambda_N \leq \sum_{N=1}^{M} c_N + \sum_{N=1}^{M} d_N$$

which stays bounded as $M \to \infty$, since $\sum c_N$ and $\sum d_N$ are convergent.

The result is false if the condition $c_n, d_n \geq 0$ is replaced by $c_n + d_n \geq 0$. For example, let $c_n = (-1)^{n+1}/n$, $d_n = (-1)^n/n$. Then $\lambda = \sqrt{(c_N^2 + d_N^2)} = \sqrt{(1/N^2 + 1/N^2)} = \sqrt{2}/N$; therefore $\sum_{N=1}^{M} \lambda_N = \sqrt{2}.\sum_{N=1}^{M}(1/N)$, which is unbounded, since $\sum 1/N$ is divergent.

Exercise 6.5. Let $\Lambda = \limsup d_n$, $\lambda = \liminf d_n$. Then Λ, λ both exist and are finite, for if say Λ were infinite, i.e., d_n unbounded, then $d_n + d_n^{-1} \geq d_n$ would be unbounded, which can not be the case since $\{d_n + d_n^{-1}\}$ is convergent.

Suppose that $\Lambda > \lambda$ (≥ 1, since $d_n \geq 1$). Choose subsequences

$$d_{n_1}, d_{n_2}, \ldots \to \Delta\,, \quad 1/d_{m_1}, 1/d_{m_2}, \ldots \to \lambda\,.$$

Then $1/d_{n_1}, 1/d_{n_2}, \ldots \to 1/\Lambda$ and $1/d_{m_1}, 1/d_{m_2}, \ldots \to 1/\lambda$. Therefore $d_{n_1} + 1/d_{n_1}, d_{n_2} + 1/d_{n_2}, \ldots \to \Lambda + 1/\Lambda$ and $d_{m_1} + 1/d_{m_1}, d_{m_2} + 1/d_{m_2}, \ldots \to \lambda + 1/\lambda$. But the sequence $\{d_n + 1/d_n\}$ is convergent to S, say; so any subsequence is also convergent to S. It follows that $S = \Lambda + 1/\Lambda = \lambda + 1/\lambda$. This gives $(\Lambda - \lambda)(\Lambda\lambda - 1)/\Lambda\lambda = 0$, i.e., either $\Lambda = \lambda$ or $\Lambda\lambda = 1$ and both are impossible since $\Lambda > \lambda \geq 1$.

The example: Let $0 < k < 1$ and let $d_n = k, 1/k, k, 1/k, \ldots$ (then all terms are greater than or equal to k), so $1/d_n = 1/k, k, 1/k, k, \ldots$ and hence $d_n + 1/d_n = k + 1/k \to k + 1/k$, but d_n does not tend to any limit.

Exercise 6.6. We have $P_{n+1}/P_n = 1 + a_{n+1} > 1$, so $P_{n+1} > P_n$, i.e., P_n increases, but

$$P_n = (1 + a_1)(1 + a_2)\ldots(1 + a_n) < e^{a_1}.e^{a_2}.\ldots.e^{a_n} = e^{a_1 + a_2 + \cdots + a_n} = e^{\lambda}\,,$$

where $\lambda = \sum_{i=1}^{\infty} a_i$ remains bounded since $\sum a_i$ is convergent.

If now $\sum a_n/R_n$ were convergent, then $Q_n = (1 + a_1/R_1)(1 + a_2/R_2)\ldots(1 +$

a_n/R_n) would tend to a limit, by the first part,

$$\text{i.e., } ((a_1 + R_1)/R_1)((a_2 + R_2)/R_2)\ldots((a_n + R_n)/R_n) \to \mu, \text{ say },$$

$$\text{i.e., } \left(\frac{a_1 + a_2 + \cdots}{a_2 + \cdots}\right).\left(\frac{a_2 + \cdots}{a_3 + \cdots}\right)\ldots\left(\frac{a_n + a_{n+1} + \cdots}{a_{n+1} + \cdots}\right) \to \mu ,$$

$$\text{i.e., } (a_1 + a_2 + \cdots)/(a_{n+1} + \cdots) \to \mu ,$$

$$\text{i.e., } \lambda/(a_{n+1} + \cdots) \to \text{a finite limit },$$

which can not be the case, since $a_{n+1} + \cdots \to 0$, by the general principle of convergence.

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